The Cheat Code: Essential Math and Statistics for Social Scientists

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Contents

Preface

This text is based on the idea that new students can benefit from learning the foundations of mathematics and statistics relatively thoroughly with a social science framework. This book is my attempt to write the introduction I wish I had received when I began my university studies.

Why yet another textbook?

This work overlaps with various others. But while most books, possibly for good reasons, choose to focus on one thing at a time, I try here to give the reader a feel for a lot of things simultaneously. I hope this combination will make it easier to see how these things are connected.

This book focuses somewhat on causal relationships - how and why things are the way they are. To do this we need to learn about the methods and conditions for causal analysis. Within social science, the methods for causal analysis have changed for the better in recent decades, and my hope is that this text will contribute to this development.

What you will learn

Part I presents the foundations of mathematics, from basic arithmetic through functions, logarithms, equations, derivatives, optimization, integration, and linear algebra. The part also contains examples of how mathematics can be used to describe theories and phenomena in social science.

Part II introduces probability theory and statistical inference. The part covers discrete and continuous probability distributions and hypothesis testing.

Part III introduces how we can study variation and covariation, the least squares method and regression analysis, with applications in social science. It also introduces counterfactual analysis and how regression analysis can be used to study causal relationships.

Part IV introduces three analytical tools commonly used in social science: Microsoft Excel, Stata, and the programming language R.

Part V contains a final chapter with some common misconceptions and suggestions for further reading.

Thanks!

This book is a development and translation (from Swedish) of an earlier book by me. I am still grateful to those who helped me create that book. Among those, a special thanks to Jakob Lundberg for help with the Laffer curve and to Fredrik Söderqvist for help with the section on monopsony.

The author, Årsta 2026

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Part I

Mathematics Basics

Part I: Mathematics Basics

Introduction to some of the most basic rules of mathematics, as well as some examples on how we may describe theories using math.

Chapter 1

Why math?

This first chapter introduces some ideas on why math is useful.

1.1 Mathematics in Social Science

This book is about mathematics, statistics, and some common misconceptions. Many seem to believe that math requires a special talent or genetic defect to learn. But I, who wrote this, am not particularly smart. I'm probably not particularly dumb either. I hope that I am like most people are. If I can learn math, so can you.

Another misconception is that mathematics for social scientists only concerns economists. The inherent logic of mathematical language is the same regardless of what we study. There is therefore nothing in mathematics that makes it more or less suitable for use within the behavioral science we call economics compared to the behavioral sciences we call sociology, political science, gender studies, history, etc.

Scientific research is a fancy expression for studying things carefully. There is no single definition that everyone agrees upon for exactly what is and is not science. Analytical work within social science consists of studying people. Advanced analytical work is performed both at universities, government agencies, and private companies.

You, the reader, are not expected to have any higher education in mathematics, statistics, or social science. In several of the chapters below, examples are given of how one can use mathematics to describe ideas and theories. The ambition with these examples is primarily to illustrate and inspire. All are simplified to facilitate pedagogy.

1.2 Measure, theorize, compare

Simply put, mathematics and statistics are used for three things in social science:

1. measure things
2. describe relationships and theories
3. compare ideas against reality

This section briefly goes through what these three things can entail.

1.2.1 Measure things

We often put numbers on things we observe around us, such as the number of eggs in the fridge or the amount of money in your bank account. This is called quantifying, measuring the quantity of something. In analytical work, information about reality is called data. Data can consist of all types of information and all forms of information can in some sense and to some extent be quantified. Data that is not measured in numbers are called qualitative or categorical data. For example, a person's gender, name, and country of birth.

Even qualitative data can in many situations, in various ways, be rewritten as numbers. In analytical work, an important aspect is often to note whether a phenomenon exists or not, yes or no. Suppose we want to investigate whether someone living in town A had an income of over 100 million USD last year. The answer is either yes or no. This information can be saved as 0 for no and 1 for yes.

When we collect information for analytical work, we must be careful. If the goal is to investigate how high salaries are in town A, it is important that we have a clear understanding of what the concept "salary" means in this particular measurement and perhaps which residents of town A we are interested in. If the goal is to compare salaries in town A this year with salaries from 10 years ago, the information must be collected at both times in a way that makes such a comparison as accurate as possible.

An important part of analytical work is how we classify information, how we define things. Say we work at a company and want to conduct a survey among our customers, where we are to divide customers into "new customers" and "loyal customers." Or say we work for a department at a hospital and need to divide patients into those who need urgent care and those who need care later. In both these situations, we need to establish clear rules for how we sort our information.

Are there things which cannot be measured? While we can often find ways to approximate or indirectly measure difficult phenomena, some things remain practically impossible to measure in any meaningful sense. For instance, many things are vague or poorly defined and therefore impossible to measure. For example, "the culture of a country," my love for chocolate, or how boring it is to fish.

A common method for collecting information about things that are hard or impossible to measure is to redefine the phenomenon or question. For instance, to study 'job satisfaction,' we might measure it indirectly through survey responses, employee turnover rates, or productivity metrics. To study the love for chocolate, instead of measuring the feeling of eating chocolate, we can send out a survey and let people rate how much they like chocolate. In that case, we measure the answer to the question and not the feeling itself. Another method could be to observe what proportion of their income people spend on chocolate. Or measure what happens in the body when eating chocolate.

There are also phenomena that are clearly defined but practically impossible to measure for practical reasons, or because it would be too expensive to collect the information. For example, "how many carbohydrates did all residents of Europe eat between 10 and 11 o'clock on August 14, 1996?"

A related problem is that it can be difficult to capture what we are actually seeking. Say we wish to measure how good the police are. One measure of successful police work is to see how many crimes the police solve. Another measure might be the absence of crime. But how much crime is (not) committed depends in turn on what is considered illegal in society and how much crime is detected. There is no simple answer to how we solve these things. In some cases, acceptable solutions are possible but the world is full of difficult subjects where numbers will only give us superficial guidance.

In social science, we often measure relatively abstract phenomena. For example, how many people feel something or hold an opinion. A feeling or an opinion can be defined in several ways and often what we measure is perhaps more an expression of this feeling or opinion. We are therefore often forced to use an approximate measure of the phenomenon we are to investigate. Say, for example, that we are to study the following:

- How happy are people today?

To answer this question accurately, we must first specify several things:

1. Definition and measurement: What do we mean by the word "happiness"? Is our goal to measure the feeling of happiness or an expression of happiness? How we define happiness also affects how we should collect information. Should we scan brain activity, or ask several people in a survey that we send out via email?

2. Population and time: We also need to think about whose happiness we should study, which people should be included in the investigation. "People" can include both children and adults, citizens in the country you study, and/or people who are just on vacation over the weekend. Do we mean how people feel right now "today," this specific date? Or how happy people have felt over the past year?

3. Purpose: All the above questions can in turn be affected by the study's more overarching purpose and what types of conclusions we hope to be able to draw in a broader perspective. Do we want to compare happiness between countries? Do we wish to compare happiness over time?

Different collection methods have their advantages and disadvantages in turn. A well-known challenge in these contexts is that those who answer the questions can be influenced by the collection method, the form and sequence of questions, or other information. Should we collect answers using surveys or physical interviews? Can the respondent be anonymous? What type of background information should participants receive? Should we ask about economics, religion, and death?

1.2.2 Describe relationships and theories

In social science, we also use mathematics to describe relationships and theories between different phenomena. In social science, we often want to formulate theories about causal relationships. Another word for causal relationship is causality or causal connection. Two phenomena can covary even if no causal relationship exists between them. A theory about causal relationships can, for example, sound like this:

- Phenomenon A affects phenomenon B.

Or:

- A change in phenomenon X causes a change in phenomenon Y.

In our daily lives, we naturally relate to causal relationships, for example:

- I have a headache because of the weather.
- If I drink water, I will feel less thirsty.
- If I learn math, others will think I am smart.

A common misconception is that mathematics means our description of relationships and theories becomes simpler than what is the case if we only use words. All social science theories are ideas about human behavior and must therefore, by definition, be simplified descriptions of reality.

Even if we only use words, without mathematics, to describe our theory, we will always be forced to simplify. Reality is simply too complex. Mathematics helps us understand when we simplify and when we are precise. Social scientists often use difficult words to describe even relatively simple relations and connections.

Let us say we are interested in a phenomenon that for some reason is not measurable. We can neither collect data nor find a suitable substitute. Mathematics can even in this situation be useful for describing our ideas. Mathematics allows us to be more precise, helps us understand when we simplify, and thereby contributes to clarifying reasoning. Consider the following statement as an example:

- The higher income people have, the happier they are.

Whether this is a clear statement or not can be discussed. For example, it is not clear whether this is a theory about a causal relationship or whether we just expect a relation between these phenomena. We may for example, have reason to believe that income and happiness covary without therefore meaning that one causes the other. If we mean that there is a causal relationship between income and happiness, it is not clear from the sentence in what way income, for example, causes happiness. This theory, described in this way, also says nothing about how large the actual effect might be, for example how much happier I become if my salary increases by 10 percent compared to if my salary doubles.

Our thesis is formulated as higher income being connected to a happier life. To know if something is “higher” or “more” than something, we must be able to estimate at least two comparable values. Regardless of whether we can collect information about these phenomena, we can have an understanding of them. With mathematics, we can, for example, describe our statement in the following way:

$$y = a * x \tag{1.1}$$

This expression can be read as y equals a times x. The symbol $*$ means multiplication (times). The three letters symbolize the following:

-y= a person’s happiness

-x= a person’s income

- The letter a indicates how large an effect a change in a person's income has on the same person's happiness. Right now, the value of a is unknown.

As equation (1.1) is written with y to the left of the equals sign and x to the right, the equation describes how variations in income (x) bring about, or explain, variations in happiness (y). This equation suggests a linear relationship where happiness increases proportionally with income. However, this is likely an oversimplification - real relationships are rarely this straightforward.

With the help of mathematics, it is easy to discuss further precision of our theory. We might want to change how a and x are connected. We can add more variables besides x and describe a more complex idea. We might want to describe how a change in income at age 50 affects happiness differently than a change in income at age 20. We might actually mean that higher income brings about several other things that can make life happier, such as more leisure time and better food, which we might also want to write into the equation. Whether these changes are necessary depends on what we are going to use our reasoning for, what we hope to be able to discuss or show.

The three functions of mathematics in social science — measuring, describing relationships, and comparing ideas against reality — work together in the research process. We first need to measure phenomena to collect data. Then we use mathematical tools to describe potential relationships and formulate theories about how these phenomena might be connected. Finally, we test whether our theoretical descriptions actually match what we observe in the real world. This brings us to the third major function of mathematics in social science.

1.2.3 Compare ideas against reality

The third area that mathematics is used for in social science is to compare statements and theories with reality. This can be done in many different ways, both simple and advanced. Part II introduces causal relationships and how we can calculate covariation between two or more phenomena. The stronger statements we make, the more careful we must be with our methods, analyses, and conclusions.

As a rule, one can say that it is relatively easy to find covariation between different phenomena, but it is very difficult to find good results that show causal relationships, which is emphasized and illustrated in several examples.

Another central part of analytical work consists of calculating probabilities. When we study information and trace patterns in it, there is often a theoretical possibility that our results could have arisen by chance. Our possibilities to estimate probabilities are a central part of analytical work. We return to probability in Part III .

Analysis of causal relationships, covariation, and probabilities is used in both social science and natural science. People have calculated whether scientific theories match reality for hundreds of years. Here are some examples to show that mathematics and statistics concern you regardless of what you are interested in.

- Does money make you happy? One of humanity's perhaps oldest questions is how we should become happy or at least as glad and satisfied as possible. Many have wondered and calculated whether wealth makes us happy, including Charles Jones and Peter Klenow who published an academic article about this in 2016 that received much attention: "Beyond GDP? Welfare across Countries and Time" Jones & Klenow (2016) . In the article, they present calculations that show that the average income in a country covaries with various factors that might explain people's well-being: consumption, leisure, mortality, and economic inequality. A central question in their work is how we can measure these things and whether this is actually a causal relationship.
- How do we know that Einstein was right? Albert Einstein is perhaps best known for his work with the theories of relativity, which are partly based on the observation that the speed of light in vacuum is constant, regardless of how the light's source moves. Einstein's work contributed to overturning several theories about how our reality fundamentally works. As an example, many previously believed that light traveled in a substance called ether, which is the same word used to talk about radio transmissions. The American physicist Dayton Miller presented results in 1933 from various experiments that he claimed showed that Einstein was wrong Miller (1933) . Miller's experiments were already discredited by the 1920s through the Michelson-Morley experiment and subsequent research, though he continued to claim contradictory results into the 1930s. Miller used

interferometers to measure ether and thereby perform experiments. Based on the data he collected, he calculated the probability that the observations were a coincidence. It was partly based on discussions about mathematics and statistics that the research community concluded that Miller was wrong and Einstein was right.

- Does smoking cause cancer? Nowadays it is considered an indisputable fact that smoking causes cancer. This has not always been the case. Research results regarding tobacco and cancer were presented as early as the end of the 18th century, but as late as the 1950s, the debate was still sometimes intense around this. This discussion also revolved around how one can observe causal relationships and whether the mathematics behind the calculations was correct. Ronald Fisher, one of the 20th century's most influential statisticians, strongly questioned many of the research results of that time and considered the studies to be substandard.
- Do humans affect the climate? Nowadays, virtually all researchers working with this agree that the average temperature on Earth is rising and that this can largely be traced to human activity. But this has been subject to discussions, both regarding the actual calculations as well as the mathematics behind them. During the 1970s, some studies suggested that aerosol pollution might have a cooling effect, but the scientific consensus was already moving toward understanding human-caused warming. This discussion also largely concerns whether the calculations are correct and what type of conclusions can be drawn from them IPCC (2007) .

1.3 Why Calculate

Mathematics is no guarantee that our reasoning about the world becomes better. But mathematics gives us an opportunity to improve our reasoning in a way that is impossible without it. This section briefly goes through some arguments for why the absence of mathematics risks leading thought astray.

1.3.1 We lie to both ourselves and others

Humans see patterns in all sorts of things. This is central to our survival but often leads to erroneous conclusions. For example, there are plenty of stories about success and predictions: "Here are 10 things successful people do before breakfast" and "Why politician A will win the next election." A large amount of such literature is carried forward by magical thinking. A lot of such stories may of course be enjoyable either way, but many people adopt the same habits as successful celebrities without the results even being of a similar nature. Many confident statements about politics and the future are pure fantasies.

Another example of misleading conclusions is those we often see in various types of graphical illustrations, diagrams or charts. Diagrams are excellent to illustrate information and create an overview. But any graphical illustration can also often mislead us. Companies use graphical illustrations of data and statistics to convince us to buy their particular products. Politicians and political activists want to convince us that they are right.

It is also easy to deceive yourself with graphics. One way to illustrate how easily things go wrong is given in figure 1.1 . In the figure, eight arrowheads can be seen with four distances between each pair. Many people perceive the distances between the arrowheads as differing. But this is incorrect. All four distances between respective pairs are equally long.

This illusion demonstrates how our perceptions can be systematically biased, which is why we need mathematical tools to avoid being misled by our intuitions when analyzing data.

In this example, you as a reader can check the distances between the arrowheads by counting the number of squares. But many illustrations of information can contain much more complicated optical illusions that are harder to detect. Mathematics can here help us make it harder for us to lie to ourselves and others. This type of optical illusion is called the Müller-Lyer illusion, after a German sociologist who researched the phenomenon at the end of the 19th century.

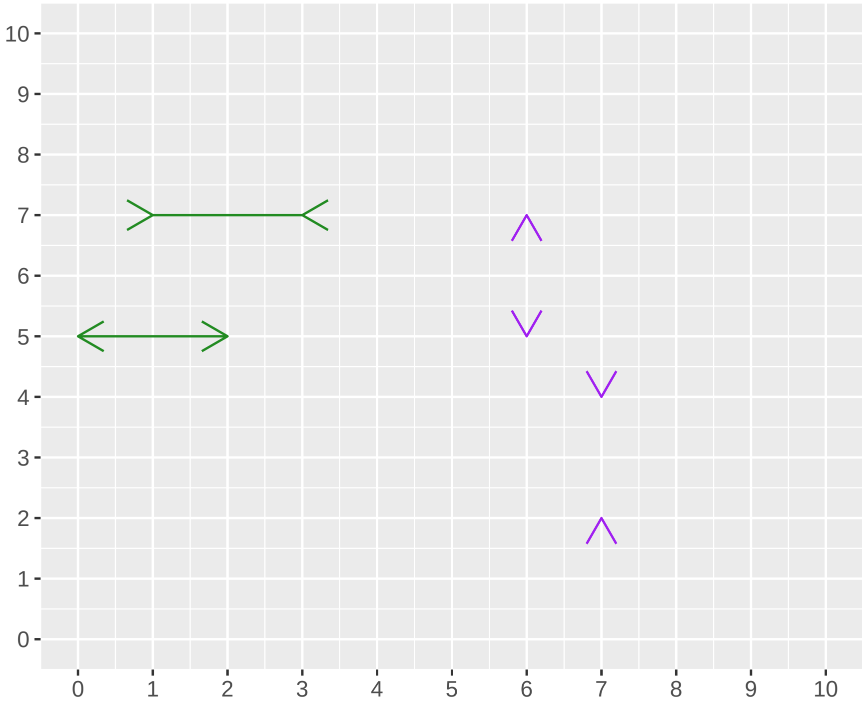

Figure 1.1: How far is it between the arrowheads?

1.3.2 Two dimensions are too few

The diagram in figure 1.1 has two dimensions: height and width. A cube has three dimensions: height, width, and depth. In pictures, it is often difficult to illustrate more than two dimensions. But reality is full of relationships that are more complex than what can be shown in diagrams. In our everyday lives, it is easy to find many examples that are the result of three or more factors.

The effect of one phenomenon can also affect the effect of other phenomena. This type of relationship rarely can be illustrated simply and clearly in pictures or diagrams. In these cases, the simple illustrations instead become misleading. In later chapters, we go through the advantages and disadvantages of graphical illustrations and how mathematics can be used to compare relationships between several phenomena simultaneously.

1.3.3 Science is needed

In social science, we often work with questions close to ideology and values. It is sometimes difficult for analysts to keep separate what we know about the world and what we wish we knew. Sometimes it can seem impossible to reach out with new knowledge when everyone seems to have decided what they believe about the world. It can then be easy to despair about the use of more technically advanced methods.

Consider how many flawed proposals that might have gained traction in the past can now be quickly identified and dismissed using mathematical analysis Looking at the history of human societies, large parts of modern social science are young. Keep it up.

1.4 Tips for your studies

Here follow some general advice for your studies.

Mathematics as language. Mathematics can be described as a language or an addition to our usual language. If a mathematical expression feels confusing, it can help to describe the same thing with your own words.

Mathematics as patterns. Mathematics can also be described as a method for seeing patterns. We categorize things to be able to count: 2 red apples, 3 green pears. In social science, we often trace patterns in human behavior to understand how and why people do things.

Repeat with patience. Mathematics consists of a large number of rules. To remember large amounts of information often requires much repetition. It is natural that you have to go back and read the same thing again several times. Learning mathematics can be compared to training strength or fitness. No one expects to become strong in one day. Confusion is a natural element in much learning.

Why do you want to learn math? Many experience that their work becomes more rewarding if they have a more or less clear goal in front of them. It can help to think about what you want to be able to use mathematics for. You may have whatever motive you want for your studies. An interesting job. A higher salary so you can buy an expensive car. Or maybe you just think it's fun to calculate.

Try different methods. There are different ways to absorb information. Try different approaches to find what works best for you. To learn mathematics, you need to calculate yourself. Many then benefit from using pen and paper. It can also help to illustrate things in pictures. Everything can be explained and described in different ways.

Turn the page. If you get stuck on a task, you can try jumping ahead to the next section or chapter to go back later. Don't get stuck on something you don't understand. Take a break or read something else. There can be many explanations for why you get stuck. Sometimes the brain needs rest.

Go to the gym. Regular breaks and exercise are central to all work. The brain works best when the rest of the body works best. Your body is a physical apparatus that needs to move, whether you want to or not. Physical training counteracts stress and helps the brain process information. The best training is that which actually happens, and even simpler exercises are better than none at all.

Ask AI. AI chat-bots can often offer insightful explanations to both technical and more theoretical aspects of math, statistics and scientific methods.

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Chapter 2

Basics

This chapter goes through basic mathematics. A challenge with introductions is that you want everything to come first. Some concepts may seem abstract or lack clear motivation initially, but will hopefully become clearer further along.

2.1 Different types of numbers

In mathematics, there are different types of numbers. In this book, we work almost exclusively with real numbers, which are all "ordinary" numbers:

- Positive numbers = All numbers above 0. For example: 5 and 0.00123.
- Negative numbers = All numbers below 0. For example: -1.
- Integers = All numbers that can be written without decimals, for example the number 0 or the number 23. Examples of numbers that are not integers are 3.14 and 5.1.
- Rational numbers = All numbers that can be expressed as a quotient (fraction) of two other integers. Example: $\frac{7}{9}$, which can be read as "7 divided by 9".
- Irrational numbers = All numbers that cannot be expressed as a quotient of two integers. Irrational numbers have an infinite number of decimals and the decimals do not end in a repeating pattern. A well-known example of an irrational number is the number Pi, which is approximately 3.141592. Pi is a circle's circumference divided by its diameter (the distance from edge to edge). Pi is written with the Greek letter lowercase pi, π, and is often rounded to two decimals: $\pi = 3.14$.

Real numbers include, among others, the following values:

-2	0	0.00123	2.19	5	27

In English periods are used as decimal separators. Many countries instead use a comma as decimal separator. All integers can be written with decimals if desired. The integer 17 can, for example, be written as 17.0 or 17.00, or 17.000 and so on. As a rule, we omit the decimals when all decimals (all decimal places) are 0 in a number. The zeros after the decimal point do not provide the reader with any interesting information. The same applies if we have, for example, the number 6.8. It does not help the reader to write this as 6.80 or 6.8000.

Real numbers can be described as being located on a number line with larger values to the right and smaller values to the left. Numbers above 0 are called positive numbers. Numbers below 0 are called negative numbers. If there is no sign in front of a digit, the number is positive. If there is a minus sign in front, the number is negative. Figure 2.1 shows a number line with some integers around 0.

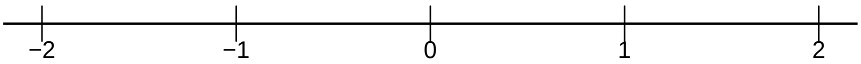


Figure 2.1: The number line

A number's absolute value is the number's distance from 0 on the number line. The absolute value of the number 5 is 5. The absolute value of -5 is also 5. Absolute value is written with vertical bars. Here are three examples:

$$\begin{aligned} |5| &= 5 \\ |-5| &= 5 \\ |4-18| &= |-14| = 14 \end{aligned} \tag{2.1}$$

Numbers that when divided by 2 result in a new integer are called even numbers. For example, the number 12 is even because $12/2 = 6$. Since 6 is an integer, 12 is even. All even numbers multiplied by themselves result in a new even number. If the result of division by 2 becomes a number with decimals, the number being divided is an odd number. For example, the number 11 is an odd number because $11/2 = 5.5$. All odd numbers multiplied by themselves always result in a new odd number. For example:

$$3 \times 3 = 9 \tag{2.2}$$

Both the number 3 and 9 are odd numbers, which is easy to see since $3/2 = 1.5$ and $9/2 = 4.5$. Two other examples: $5 \times 5 = 25$ and $7 \times 7 = 49$. The numbers 5, 7, 25 and 49 are odd numbers. Here are two examples:

$$\begin{aligned} 2 \times 2 &= 4 \\ 6 \times 6 &= 36 \end{aligned} \tag{2.3}$$

The numbers 2, 4, 6 and 36 are even numbers. Sometimes we find it useful to rewrite a fraction as a number with decimals, for example $1/2 = 0.5$, or $\frac{1}{4} = 0.25$. Some fractions cannot be written exactly with decimals. One such example is $1/3 \approx 0.33333...$. Numbers that can be expressed exactly as a fraction with two integers are called rational numbers. Numbers that cannot be expressed as a fraction with two integers are called irrational numbers. A well-known example of an irrational number is pi, $\pi \approx 3.14$. All real numbers can be written in the form of a fraction where we divide the number by 1. All real numbers can also be expressed as the number multiplied by 1. As an example, the number 123 is the same as $1 * 123$ and $123/1$:

$$123 = 1 * 123 = \frac{123}{1} \tag{2.4}$$

The number 1 can be written as a fraction with the same numerator and denominator:

$$1 = \frac{123}{123} = \frac{5}{5} = \frac{17.3}{17.3} \tag{2.5}$$

2.2 The four arithmetic operations

We have four arithmetic operations:

- Addition $(+)$
- Subtraction $(-)$
- Multiplication (written as $*$ or $\times$ or $\cdot$)

- Division (written as / or $\div$)

The equals sign ($=$) means that the result to the left of the sign becomes the same as the result to the right of the sign. The equals sign is also called the equality sign. Example: $1+2=3$. The left side of the equals sign is sometimes called the left-hand side, while the right side is called the right-hand side. When two or more numbers are added or subtracted, these numbers are called terms. The result of addition is called the sum:

$$\underbrace{1}_{\text{Term 1}} + \underbrace{2}_{\text{Term 2}} = \underbrace{3}_{\text{Sum}} \tag{2.6}$$

The result from subtraction is called the difference:

$$\underbrace{1}_{\text{Term 1}} - \underbrace{2}_{\text{Term 2}} = \underbrace{-1}_{\text{Difference}} \tag{2.7}$$

When two or more numbers are multiplied, the numbers are called factors. The result of multiplication is called the product:

$$\underbrace{1}_{\text{Factor 1}} \times \underbrace{2}_{\text{Factor 2}} = \underbrace{2}_{\text{Product}} \tag{2.8}$$

Multiplication can be written with any of the three symbols $(*, \times, \cdot)$. There are several different symbols for the same arithmetic operation, but they mean exactly the same thing. For example:

$$2 \times 4 * 2 \cdot 2 = 4 * 2 * 2 * 2 = 2 * 2 * 2 * 4 = 32 \tag{2.9}$$

This is perhaps a good example of how mathematics can be described as a language, a way of writing. The main thing is that the reader understands what you write. In division we have a number that we call the numerator, which is divided by another number, which we call the denominator. The result is called the quotient:

$$\text{Quotient} = \frac{\text{Numerator}}{\text{Denominator}} \tag{2.10}$$

Division can also be written in several ways, for example $\frac{1}{2}$ or $1/2$, which have the same meaning. The expressions $\frac{1}{2}$ and $1/2$ can be read as “one divided by two” or “one over two”. We must also take into account parentheses and powers. A power is a mathematical expression where a number, the base, is raised to another number, the exponent:

$$\text{Power} = \text{Base}^{\text{Exponent}} \tag{2.11}$$

An example of a power expression is 10^2, which can be read as “10 to the power of 2”. 10 is the base and 2 is the exponent. The exponent indicates how many times the base should be multiplied by itself. For example:

$$\begin{aligned} 10^2 &= 10 * 10 = 100 \\ 2^3 &= 2 * 2 * 2 = 8 \\ 3^4 &= 3 * 3 * 3 * 3 = 81 \end{aligned} \tag{2.12}$$

2.3 Order of operations

There is a specific order for how we should perform the different arithmetic operations:

1. Parentheses
2. Powers
3. Multiplication and division, from left to right in each expression
4. Addition and subtraction, from left to right in each expression

We will now calculate an expression that contains multiplication, addition and subtraction. We therefore start with all multiplication signs, regardless of where in the expression they appear:

$$4 + 1 * 2 - 3 = 4 + 2 - 3 \tag{2.13}$$

This expression should be read as four plus one times two minus three equals four plus two minus three. The words "equals" mean that the result on both sides of the equals sign (=) has the same value and gives the same result. In the next step we calculate addition and subtraction from left to right. In this case it doesn't matter if we first calculate addition and then subtraction, or vice versa:

$$4 + 2 - 3 = 4 - 1 = 6 - 3 = 3 \tag{2.14}$$

For plus and minus it doesn't matter in which order the numbers appear:

$$-1 + 2 + 3 + 4 + 5 = 13 \tag{2.15}$$

Even if we change the order of the terms to the left of the equals sign we get the same result, as long as the mathematical operator in front of the number (plus or minus) follows:

$$2 - 1 + 3 + 4 + 5 = 13 \tag{2.16}$$

Note that when a positive number, above 0, appears first in the expression we don't need to write out the plus sign. In multiplication it doesn't matter which of the numbers being multiplied comes first or last. In the order of operations, in the introduction to this section, points 3 and 4 describe how we should first calculate multiplication and division. This should also be calculated from left to right. Here is an example when this rule is important:

$$1/2/2/2 \tag{2.17}$$

This expression can be read as 1 divided by 2, divided by 2, divided by 2. If we read the numbers from the left, which is correct, we get:

$$\begin{aligned} &\textbf{Correct!}\text{Solving from the left:} \\ (1/2)/2/2 &= ((1/2)/2)/2 \\ &= (1/4)/2 \\ &= 1/8 \\ &= 0.125 \end{aligned} \tag{2.18}$$

If we instead solve the equation from the right, which is incorrect, we get:

$$
\begin{aligned}
&\textbf{Incorrect! } \text{Solving from the right:}\\
1/2/\left(2/2\right) &= 1/2/\left(1\right) \\
&= 1/\left(2/1\right) \\
&= 1/2 \\
&= 0.5
\end{aligned}
\tag{2.19}
$$

In row 2 we see the result from the parentheses, 2 divided by 2, which gives the result 1. On the second-to-last row we have 1 divided by 2, which equals 0.5. That a calculation continues on the next line has no mathematical meaning. It is merely a way of writing to make reading easier. Here in the book we also use it so that the mathematical expressions fit on the pages. Let us take another example:

$$2 \times 3 \div 4 \times 5 \tag{2.20}$$

This calculation contains both multiplication ($\times$) and division ($\div$). The calculation rules say that these two should be calculated simultaneously but from left to right. Let us calculate the answer from both left and right to compare:

$$
\begin{array}{c|c}
\textbf{Correct! From the left.} & \textbf{Incorrect! From the right.} \\
2 \times 3 \div 4 \times 5 = & 2 \times 3 \div 4 \times 5 = \\
= 6 \div 4 \times 5 = & = 2 \times 3 \div 20 = \\
= \frac{3}{2} \times 5 = & = 2 \times \frac{3}{20} = \\
= \frac{15}{2} = & = \frac{6}{20} = \\
= 7.5 & = 0.3
\end{array}
\tag{2.21}
$$

The order of operations says that we should first calculate parentheses, powers and then multiplication and division, as well as addition and subtraction. Consider the following expression as an example:

$$\left(1/2/2\right)^2 \times \left(1+2\right)^3 \tag{2.22}$$

We start by calculating the contents of the parentheses. In the left parenthesis we calculate from left to right:

$$\left(\frac{1}{4}\right)^2 \times \left(3\right)^3 = \left(\frac{1}{4} * \frac{1}{4}\right) \times 3^3 = \frac{1}{16} \times 9 = \frac{9}{16} \tag{2.23}$$

When two parentheses are multiplied by each other, it is common for the multiplication sign to be omitted:

$$\left(\frac{1}{4}\right) \times \left(3\right) = \left(\frac{1}{4}\right)\left(3\right) \tag{2.24}$$

The rest of this chapter goes through in more detail how to calculate with parentheses (section 2.4), division (section 2.6) and powers (section 2.8).

2.4 Parentheses

Parentheses can be used to make extensive mathematical expressions easier to read. There are special calculation rules for parentheses. If there is a minus sign in front of a parenthesis, it affects all terms inside the parenthesis. If we remove the parenthesis, all signs change from minus to plus and from plus to minus. For example:

$$1 - \left(-2+3\right) = 1+2-3 = 0 \tag{2.25}$$

A plus sign in front of the parenthesis does not change any signs in the parenthesis:

$$1 + (-2 + 3) = 1 - 2 + 3 = 2 \tag{2.26}$$

When multiplying numbers in parentheses, corresponding calculation rules apply. Multiplication of two negative numbers becomes a positive number. Multiplication of a negative and a positive number becomes a negative number. Examples:

$$-1 * (-2 + 3) = 2 - 3 \tag{2.27}$$

Or we could take:

$$1 * (-2 + 3) = -2 + 3 \tag{2.28}$$

Multiplication of a parenthesis affects all values in the parenthesis:

$$-1 \times (2 + 3 + 4 + 5) = -14$$

All real numbers multiplied by 0 become 0:

$$0 \times -1 \times (1 - 2 - 3 - 4 - 5) = 0$$

Here follow some illustrations of calculation rules for parentheses where we use the letters a, b, c, d and e to symbolize arbitrary real numbers. We use letters to describe general rules. The reader may try replacing the letters with numbers. We start by multiplying a by the parenthesis $(b + c)$, where a is multiplied by each letter in the parenthesis:

$$a\,(b + c) = ab + ac \tag{2.29}$$

It is also possible to illustrate the same thing with numbers:

$$2\,(3 + 4) = 2 * 3 + 2 * 4 = 6 + 8 = 14 \tag{2.30}$$

It doesn't matter whether the number we multiply the parenthesis by is to the left or right of the parenthesis. Here is an example where we rewrite an expression in different ways, but all different writings give the same result:

$$\begin{aligned} c\,(a + b)\,d &= cd\,(a + b) \\ &= (a + b)\,cd \\ &= (ca + cb)\,d \\ &= cda + cdb \end{aligned} \tag{2.31}$$

A parenthesis may also be multiplied with another parenthesis. Each number in each parenthesis is then multiplied by each number in the other parenthesis. Let us take an example again where we rewrite an expression in different ways. All these different ways of writing give the same result:

$$\begin{aligned} (a + b)\,(c + d) &= ac + ad + b\,(c + d) \\ &= ac + ad + bc + bd \\ &= a\,(c + d) + b\,(c + d) \\ &= c\,(a + b) + d\,(a + b) \end{aligned} \tag{2.32}$$

Many times our work can be made easier by rewriting an expression so that it becomes a bit neater. When we write multiplication with letters, it is common to omit the multiplication sign. It also doesn't matter in which order the numbers appear:

$$a * b = ab = ba = b * a \tag{2.33}$$

The same thing if we have an expression where we combine letters and numbers. Here is an example where the letter x symbolizes an arbitrary number:

$$x * 100 = x100 = 100x \tag{2.34}$$

The multiplication sign may be omitted when we multiply parentheses:

$$4 * (3 - 2) = 4\,(3 - 2) \tag{2.35}$$

2.5 Equality and inequality

If we change something on one side of the equals sign (=) we must make corresponding changes on the other side of the equals sign. Otherwise the meaning of the equals sign becomes incorrect. If we take the expression $1 + 2 = 3$ for example, then we can subtract 2 from both sides of the equals sign:

$$\begin{aligned} 1 + 2 - 2 &= 3 - 2 \\ 1 &= 1 \end{aligned} \tag{2.36}$$

Or we can multiply both sides by 3:

$$\begin{aligned} 2 * (1 + 2) &= 2 * (3) \\ 2 * 3 &= 2 * 3 \\ 6 &= 6 \end{aligned} \tag{2.37}$$

There are also symbols for inequalities, so-called inequality signs. Here are some examples of inequality signs:

$>$ Greater than
$<$ Less than
$\geq$ Greater than or equal to
$\leq$ LEss than or equal to
$\neq$ Not equal to

Let us go through some examples:

$$\begin{aligned} a &> 4 \\ b &< 5 \\ c &\geq 7 \\ d &\leq 8 \\ e &\neq 6 \end{aligned} \tag{2.38}$$

The first line $a > 4$ means that the letter a symbolizes a value that can be anything greater than 4. The second line $b < 5$ means that b symbolizes an arbitrary value that is less than 5. The third line $c \geq 7$ means that c is an arbitrary value that is equal to or greater than 7. The fourth line $d \leq 8$ means that d is a number that is less than or equal to 8. In the fifth line $e \neq 6$ the letter e is a value that is not equal to 6. This means that the letter e can symbolize any value as long as it is not 6. For example 5.99 or 452.

If we have an expression with an inequality sign and change something on one side, we sometimes must also change on the other side for the inequality sign to still apply. Alternatively, we must change the inequality sign. When we multiply or divide by negative numbers, this can result in the inequality having to be changed to be correct. Example:

$$\begin{gathered} -4 > -5 \\ -4 \times (-1) > -5 \times (-1) \\ \textbf{Incorrect! } 4 > 5 \end{gathered} \tag{2.39}$$

The first and second lines are correctly executed but the third line is incorrect. In this situation when we use multiplication by a negative value (-1) we must reverse the inequality sign:

$$\begin{gathered} -4 \times (-1) > -5 \times (-1) \\ 4 < 5 \end{gathered} \tag{2.40}$$

Now it's correct. There may be multiple inequality signs in the same equation:

$$0 < a < 1 \tag{2.41}$$

This means that a symbolizes a value that is greater than 0 but less than 1. Both inequalities in the expression must apply simultaneously. Since the inequality signs do not include the values 0 and 1, a can be very close to 0 or 1, but not exactly 0 or 1.

2.6 Fractions

In the introduction to this chapter we described how when dividing two numbers a numerator is divided by a denominator and that the result is called a quotient. Another word for this is fraction:

$$\text{Fraction} = \text{Quotient} = \frac{\text{Numerator}}{\text{Denominator}} = \text{Numerator/Denominator} \tag{2.42}$$

The words "numerator" and "denominator" refer to the placement above or below the fraction line. The fraction line is also called the division sign. This can be read as numerator divided by denominator or numerator divided by denominator. Fractions are often used to describe a portion of something: "three-quarters full." It has no mathematical meaning whether we write the fraction vertically, $\frac{3}{4}$, or on one line and separate the numerator and denominator with a slash /, such as 3/4. We can also write division as $3 \div 4$. All three ways of writing mean the same thing:

$$\frac{3}{4} = 3/4 = 3 \div 4 = 0.75 \tag{2.43}$$

We can read this as three divided by four or three quarters or 0.75. If we multiply a fraction by a number, only the numerator is multiplied:

$$\frac{1}{2} \times 3 = \frac{1 \times 3}{2} = \frac{3}{2} \tag{2.44}$$

When a fraction is multiplied by another fraction, we multiply the numerators together and the denominators together:

$$\frac{1}{2} \times 3 = \frac{1}{2} \times \frac{3}{1} = \frac{1 \times 3}{2 \times 1} = \frac{3}{2} \tag{2.45}$$

The same principle applies to fractions with different numerators and denominators. Here we have two fractions that are multiplied by each other:

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d} = \frac{ac}{bd} \tag{2.46}$$

where the letters a, b, c and d symbolize different arbitrary numbers. The letters are only meant to describe that these rules are general, regardless of which real numbers the letters represent. Another example:

$$\frac{1}{3} * \frac{4}{6} = \frac{1 * 4}{3 * 6} = \frac{4}{18} \tag{2.47}$$

If both the numerator and denominator are multiplied by the same number, we can cancel this number. Here is an example where we cancel the letter b:

$$\frac{a}{b} * \frac{b}{c} = \frac{ab}{cb} = \frac{a\cancel{b}}{c\cancel{b}} = \frac{a}{c} \tag{2.48}$$

Or the same thing, but with some numbers:

$$\frac{1}{3} \times \frac{3}{2} = \frac{1 \times \cancel{3}}{\cancel{3} \times 2} = \frac{1}{2} \tag{2.49}$$

Another example:

$$\frac{1 * 2 * 3 * 4}{1 * 2 * 3} = \frac{24}{6} = 4 \tag{2.50}$$

Another way to describe this calculation is to cancel factors that appear in both the numerator and denominator:

$$\frac{\cancel{1 * 2 * 3} * 4}{\cancel{1 * 2 * 3}} = 4 \tag{2.51}$$

Fractions can be positive and negative. The fraction $-\frac{1}{2}$ can be read as minus one half. If there is no sign in front of the fraction, the number is positive. If only one of the numerator and denominator is negative, the entire fraction becomes negative:

$$\frac{-1}{3} = -\frac{1}{3} = \frac{1}{-3} \tag{2.52}$$

Or like this:

$$\frac{1 + 2 + 3 + 4}{-1} + 1 = -\frac{10}{1} + 1 = -9 \tag{2.53}$$

If both the numerator and denominator are negative numbers, the fraction becomes positive:

$$\frac{-2}{-3} = \frac{\cancel{(-1)}2}{\cancel{(-1)}3} = \frac{2}{3} \tag{2.54}$$

Often it can help to convert the numbers so that two fractions get the same denominator, that is, the same value below the division line. We can do this by multiplying or dividing both the numerator and denominator in a fraction by the same number. As long as we perform multiplication or division by the same value on the entire numerator and the entire denominator, the value of the fraction does not change:

$$\frac{2}{4} = \frac{2/2}{4/2} = \frac{1}{2} = 0.5 \tag{2.55}$$

The same thing but we multiply both the numerator and denominator by the number 3:

$$\frac{2*3}{4*3} = \frac{6}{12} = \frac{1}{2} = 0.5 \tag{2.56}$$

The fraction 6/12 gives exactly the same result as the fraction 2/4, the fraction 1/2 and the fraction $\frac{2*3}{4*3}$. Another way to see this is to multiply both the numerator and denominator by an arbitrary number that we can call a:

$$\frac{2*a}{4*a} = \frac{2}{4} * \frac{\cancel{a}}{\cancel{a}} = 0.5 \tag{2.57}$$

We cancel a from the numerator and denominator since we multiply a by the entire numerator and the entire denominator. Fractions can be divided by other fractions. This is often useful. Here is an example where each new expression gives the same result:

$$\frac{0.5}{0.5} = \frac{\left(\frac{10}{20}\right)}{\left(\frac{1}{2}\right)} = \frac{2*\frac{1}{2}}{1} = \frac{2/2}{1} = \frac{1}{1} \tag{2.58}$$

One more example:

$$\frac{1/2}{3/4} = \frac{4*\left(\frac{1}{2}\right)}{3} = \frac{2}{3} \tag{2.59}$$

2.7 Least common denominator

Now we will add the following two fractions:

$$\frac{1}{3} + \frac{4}{6} \tag{2.60}$$

To do this we need to find the least common denominator of the two fractions, where the denominator is the value below the division line. The least common denominator for two fractions is the lowest value we can use as the denominator in both fractions and still write both fractions as integers. To find the least common denominator we convert both fractions so that they get the same denominator. We do this by converting the first fraction so that its denominator becomes 6. To find the number we need to use for this we first divide the denominator in fraction no. 2 by the denominator in fraction no. 1:

$$\frac{6}{3} = 2 \tag{2.61}$$

Then we multiply only the first fraction by the number 2:

$$\frac{1}{3} * 2 = \frac{2}{6} \tag{2.62}$$

Let us describe this entire calculation based on our original expression:

$$2 * \left(\frac{1}{3}\right) + \frac{4}{6} = \frac{2}{6} + \frac{4}{6} = \frac{6}{6} = 1 \tag{2.63}$$

Another method for finding common denominators is to multiply each numerator and denominator by the denominators from all other fractions. Example:

$$\begin{aligned}
&\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \\
=&\frac{1*(3*4*5)}{2*(3*4*5)} + \frac{1*(2*4*5)}{3*(2*4*5)} + \frac{1*(2*3*5)}{4*(2*3*5)} + \frac{1*(2*3*4)}{5*(2*3*4)} = \\
=&\frac{60+40+30+24}{120} = \\
=&\frac{154}{120} = \\
=&\frac{77}{60}
\end{aligned} \tag{2.64}$$

Let us take another example where we mix letters and numbers. The letters symbolize arbitrary numbers and it doesn't matter which numbers they are:

$$\begin{aligned}
&\frac{a}{2} + \frac{b}{3} + \frac{c}{4} = \\
=&\frac{a*3*4}{2*3*4} + \frac{b*2*4}{3*2*4} + \frac{c*2*3}{4*2*3} = \\
=&\frac{12a+8b+6c}{24}
\end{aligned} \tag{2.65}$$

As long as we don't know more about what the letters a, b and c symbolize, there is also no way to solve this expression. If we separate the numbers so that this becomes separate fractions we easily get back to the original expression or something similar:

$$\frac{12a+8b+6c}{24} = \frac{12}{24}a + \frac{8}{24}b + \frac{6}{24}c = \frac{a}{2} + \frac{b}{3} + \frac{c}{4} \tag{2.66}$$

A type of notation that is good to be familiar with is the following:

$$\frac{a-b}{b} = \frac{a}{b} - \frac{b}{b} = \frac{a}{b} - 1 \tag{2.67}$$

We simplify the first fraction $\frac{a-b}{b}$ by dividing it into the two fractions $\frac{a}{b} - \frac{b}{b}$. This works when we have terms in the numerator that are separated by plus and minus. We can also divide fractions with multiple terms in the numerator:

$$\frac{a+b*c-d}{fg} = \frac{a}{fg} + \frac{b}{fg} * \frac{c}{fg} - \frac{d}{fg} \tag{2.68}$$

In this context, however, it is important to point out a common mistake. While we can split the numerators and keep the denominator, we generally cannot split the denominator and keep the numerator. As an example, the fraction $\frac{b}{a-b}$ generally cannot be split into two separate fractions. This probably becomes easier to see if we use numbers instead. The left side in the following example is not the same as the right side:

$$\begin{aligned} \frac{6}{3-2} &\neq \frac{6}{3} - \frac{6}{2} \\ \frac{6}{1} &\neq 2-3 \\ 6 &\neq -1 \end{aligned} \tag{2.69}$$

Above we went through how we can rewrite two fractions so that they get the same denominator by multiplying different fractions by each other's denominators. Here is another way to describe this method:

$$\frac{a}{b} * \frac{d}{d} + \frac{c}{d} * \frac{b}{b} = \frac{ad}{bd} + \frac{cb}{db} = \frac{ad+cb}{bd} \tag{2.70}$$

The first fraction's numerator (a) and denominator (b) are multiplied by d, since this is the right fraction's denominator. This can be described as multiplying the first fraction by a fraction consisting of d/d. The right fraction is instead multiplied by the fraction b/b, since b is the left fraction's denominator. The same method can also be used if we have more than two fractions with different denominators. Here is an example with three fractions:

$$\begin{aligned} &\frac{a}{b} + \frac{c}{d} + \frac{e}{f} = \\ &= \frac{a}{b}\left(\frac{d}{d} * \frac{f}{f}\right) + \frac{c}{d}\left(\frac{b}{b} * \frac{f}{f}\right) + \frac{e}{f}\left(\frac{b}{b} * \frac{d}{d}\right) = \\ &= \left(\frac{adf}{bdf}\right) + \left(\frac{cbf}{dbf}\right) + \left(\frac{ebd}{fbd}\right) = \\ &= \frac{adf+cbf+ebd}{bdf} \end{aligned} \tag{2.71}$$

In row two, each fraction's numerator and denominator are multiplied by the numerators from the other two fractions. This results in the three fractions getting one and the same denominator and can thereby be written as one large fraction.

2.8 Powers

In the introduction to this chapter we defined power expressions (exponential expressions) as Power = Base$^{\text{Exponent}}$. Powers mean, among other things, that we can express large numbers in compressed form, for example:

$$\begin{aligned} 5^3 &= 5*5*5 = 125 \\ 5^9 &= 9,765,625 \end{aligned} \tag{2.72}$$

If we have the base b with exponent 2, this is called b squared . We can think of it as a figure with two dimensions: height and width, where all sides of the figure are equally long, that is, a square where all four sides have length b. The number 1^2 can be thought of as a square where height and width are equal to 1. We can write this as $1^2 = 1 \times 1 = 1$. This can be read as the square where both width and height are equal to 1 has an area that is 1 area unit. Which unit of measurement is used doesn't matter. It can be centimeters or miles. If the square's sides are 1 cm long, the area is 1 square centimeter.

The number 2^2 can be thought of as a square where width multiplied by height becomes $2 \times 2 = 2^2 = 4$. The number $3^2 =$ a 3×3 square. Figure 2.2 illustrates this with three squares. The smallest square has height and width = a with area $a^2 = a \times a$. The middle square has height = width = b and area = b^2. The largest square has height = width = c and area = c^2.

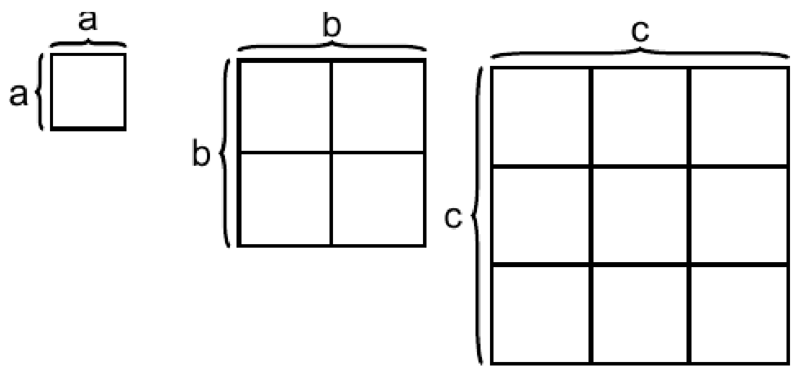


Figure 2.2: Three squares

A square whose area is calculated as 2^2 can be described as being 4 area units. Exactly what the word area units represents depends on which unit the square's length and width are given in. A square whose width and height are 2 centimeters has an area of 4 square centimeters. If the exponent is instead 3, such as 2^3, it is called cubed. The number 2^3 can be thought of as a cube with three dimensions: width, height and length. If this cube's width, height and length are 2 centimeters, the cube's volume is 8 cubic centimeters.

Illustrating mathematics in pictures is often helpful. But illustrations also have their limitations. For example, no advanced method is required to calculate 2^4. But there is no simple way in which a four-dimensional figure can be illustrated. Even parentheses can have powers and then the same calculation rules for parentheses apply as otherwise. For example, if we multiply two parentheses with a negative number, the result becomes a positive number:

$$(-2)^3 = (-2) * (-2) * (-2) = (4) * (-2) = -8 \tag{2.73}$$

If two power expressions are multiplied with each other and they have the same base, the exponents can be added. For example:

$$2^2 * 2^3 = 2^{2+3} = 2^5 \tag{2.74}$$

This can also be seen by writing out all numbers:

$$2^2 * 2^3 = (2 * 2) * (2 * 2 * 2) = 2 * 2 * 2 * 2 * 2 = 2^5 \tag{2.75}$$

The expression with the letters a and b for arbitrary real numbers:

$$a\,(a + b) = a^2 + ab \tag{2.76}$$

First the number a outside the parentheses is multiplied with the first letter inside the parentheses, which is also an a. We then get $a * a = a^2$. Then a outside the parentheses is multiplied with the second letter in the parentheses, which is b. We then get $a * b = ab$. Let us show the same thing but with numbers

$$2\,(2+3) = 2^2 + 2*3 = 4+6 = 10 \tag{2.77}$$

When dividing powers with the same base, the exponents can be subtracted:

$$\frac{3^4}{3^2} = 3^{4-2} = 3^2 = 9 \tag{2.78}$$

Another way to write the same thing:

$$\frac{3^4}{3^2} = \frac{3*3*\cancel{3*3}}{\cancel{3*3}} = \frac{3*3}{1} = 3^2 = 9 \tag{2.79}$$

This also means that when a power stands as denominator in a fraction, the power can be written as numerator with the opposite sign in front of the exponent:

$$3^{-2} = \frac{1}{3^2} = \frac{1}{9} \tag{2.80}$$

Or for example:

$$4^5 = \frac{1}{4^{-5}} \tag{2.81}$$

Here is another example where we move the denominator to the nominator, and vice versa:

$$\frac{2^{-3}}{3^{-5}} = 2^{-3} * 3^5 = \frac{3^5}{2^3} \tag{2.82}$$

A power consists of base and exponent. The entire power expression can in turn have an additional exponent. In that case the two exponents can be multiplied:

$$\left(4^2\right)^3 = 4^2 * 4^2 * 4^2 = 4^{2*3} = 4^6 \tag{2.83}$$

Powers do not need to be whole numbers. A form of powers that is often used is numbers between 0 and 1. For example $9^{\frac{1}{2}}$, the number 9 raised to one half. Another common way to write the same thing is the following:

$$9^{\frac{1}{2}} = \sqrt[2]{9} = \sqrt{9} \tag{2.84}$$

This expression is called the square root of 9, or taking the root of the number 9. The square root of a number a is the number that multiplied by itself becomes the number a. If we take the square root of 9 we get $9^{1/2} = \pm 3$ since both $3^2 = 9$ and $\left(-3\right)^2 = 9$. Suppose we call the number that multiplied by itself becomes a the letter b. The calculation rule for square root can then be summarized:

$$a^{\frac{1}{2}} = \pm b \tag{2.85}$$
$$\text{where } \left(\pm b\right)^2 = a$$

The expression square root comes from the fact that it is precisely the number 2 we use. If we are only interested in the positive b-value, this is called seeking the positive square root. Often when we are to calculate the square root of a number, it is precisely the positive square root we seek. The square root is so commonly occurring that this is often described only with the symbol $\sqrt{}$, without the number 2. If the exponent is one third, that is $\sqrt[3]{}$, this is called the cube root. The cube root of the number c is the number that multiplied by itself 3 times becomes c:

$$c^{\frac{1}{3}} = \sqrt[3]{c} = \pm b \qquad (2.86)$$
$$\text{where } (\pm b)^3 = c$$

The cube root of the number 8 is 2 since $2^3 = 2*2*2 = 8$. Any real number can be an exponent: positive, negative, rational and irrational numbers. The rules for roots follow from how powers are defined. As an example:

$$8^{\frac{1}{3}} * 8^{\frac{1}{3}} * 8^{\frac{1}{3}} = 8^{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}} = 8^1 = 8 \qquad (2.87)$$

Another way to write the same thing:

$$8^{\frac{1}{3}} * 8^{\frac{1}{3}} * 8^{\frac{1}{3}} = 2 * 2 * 2 = 2^3 = 8 \qquad (2.88)$$

Table 2.1: Exponent properties

Rule	Formula
1	$a^b * a^c = a^{b+c}$
2	$\frac{1}{a^b} = a^{-b}$
3	$\frac{a^b}{a^c} = a^{b-c}$
4	$\left(a^b\right)^c = a^{bc}$
5	$\left(\frac{a}{b}\right)^c = \frac{a^c}{b^c} = a^c * b^{-c}$
6	$(a * b)^c = a^c * b^c$
7	$a^1 = a$
8	$a^0 = 1$

Table 2.1 summarize some useful properties and rules for exponents and powers. The letters a, b and c represent arbitrary real numbers. The first rule means that if we multiply two power expressions with the same base, the exponents can be added as follows:

$$5 * 5^2 = 5^1 * 5^2 = 5^{1+2} = 5^3 = 125 \qquad (2.89)$$

where we also use the power rule $a^1 = a$(row 7 in the table), which describes how all real numbers can be written as power expressions with exponent 1. Row 2 in table 2.1 means that the sign in front of the exponent changes if the power is moved between numerator and denominator:

$$a^b \left(\frac{c+d}{e-f}\right) = \frac{1}{a^{-b}} \frac{(c+d)}{(e-f)} = \frac{c+d}{a^{-b}e - a^{-b}f} \qquad (2.90)$$

The calculation rule on row 8 means that all real numbers raised to the power of 0 equal 1. Sometimes we may encounter fractions where the numerator or denominator also consists of a fraction. Then it can help to rewrite the values as power expressions:

$$\frac{2}{\frac{1}{2}} = \frac{2^1}{\frac{1^1}{2^1}} = \frac{2^1}{1^1 * 2^{-1}} = \frac{2^1 * 2^1}{1} = 2 * 2 = 4 \tag{2.91}$$

Note how the exponent for the number 2 changes sign from 2^1 to 2^{-1} and back to 2^1 as we move the number between numerator and denominator.

2.9 Parentheses with powers

Parentheses can also have powers. For example:

$$\begin{aligned}(2+3)^2 &= (2+3)(2+3) \\ &= 2*2+2*3+3*2+3*3 \\ &= 4+6+6+9 \\ &= 25\end{aligned} \tag{2.92}$$

Or like this:

$$(1+2+3)^2 = 6^2 = 6*6 = 36 \tag{2.93}$$

If we put a minus sign in front of this parenthesis, the result becomes negative:

$$\begin{aligned}-(1+2+3)^2 &= -\left((1+2+3)^2\right) \\ &= -\left((6)^2\right) \\ &= -(6^2) \\ &= -(36) \\ &= -36\end{aligned} \tag{2.94}$$

Here is an example with negative numbers in a parenthesis squared:

$$(-1-2-3)^2 = (-6)^2 = (-6)(-6) = 36 \tag{2.95}$$

In this example, the letters a and b represent arbitrary numbers:

$$(a+b)^2 = (a+b)(a+b) = a^2 + ab + ba + b^2 \tag{2.96}$$

Since $ab = ba$ we may simplify this expression a bit to:

$$\begin{aligned}(a+b)^2 &= a^2 + ab + ba + b^2 \\ &= a^2 + ab + ab + b^2 \\ &= a^2 + 2ab + b^2\end{aligned} \tag{2.97}$$

The calculation in equation (2.96) and (2.98) is sometimes called notable products, meaning that they are common patterns which may be handy to learn by hand. The second notable product shows how a parenthesis with subtraction squared can be calculated:

$$\begin{aligned}(a-b)^2 &= (a-b)(a-b) \\ &= a^2 - ab - ba + b^2 \\ &= a^2 - 2ab + b^2\end{aligned} \tag{2.98}$$

For multiplication of two parentheses with addition in one and subtraction in the other we get:

$$(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2 \tag{2.99}$$

To summarize we have here the following notable products:

$$\begin{aligned}\text{Square of the sum: } (a+b)^2 &= a^2 + 2ab + b^2 \\ \text{Square of the difference: } (a-b)^{\,2} &= a^2 - 2ab + b^2 \\ \text{Difference of two squares: } (a+b)(a-b) &= a^2 + ab - ab - b^2 = a^2 - b^2\end{aligned} \tag{2.100}$$

Notice that the same rules apply if we replace the letters in the equations with more comprehensive mathematical expressions. Suppose we have $c = (a+b)$ and $b = 2a$ and want to calculate c^2. We then get:

$$\begin{aligned}c^2 &= (a+b)^2 \\ &= (a+2a)^2 \\ &= a^2 + 2a(2a) + (2a)^2 \\ &= a^2 + 4a^2 + 4^2 \\ &= 9a^2\end{aligned} \tag{2.101}$$

2.10 Factorization and divisibility

To be able to simplify expressions, it is also good to be comfortable with factorization. Factorization means that we factor out a common denominator, a factor, that can be multiplied with the affected terms. We start with an example where we factor out the number 5 from the following expression:

$$5x + 10 = 5(x+2) \tag{2.102}$$

Just as before, the letter x here represents an arbitrary number. That we use a different letter has no mathematical significance. Let us now factorize the expression 25+125. In this case, we can factor out 25:

$$25 + 125 = 25(1+5) \tag{2.103}$$

We may also use the notable products in equation (2.101) . Suppose we have the expression $x^2 - 25$. This we may rewrite in the following way:

$$x^2 - 25 = (x+5)(x-5) = x^2 - 5x + 5x - 25 = x^2 - 25 \tag{2.104}$$

When we work with powers, it is often useful to think about how one may rewrite the power expression as factors with multiplication. For example:

$$\left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1*1*1}{2*2*2} = \frac{1}{8} \tag{2.105}$$

If we have a parenthesis with an exponent and multiple expressions in the parenthesis, the entire parenthesis should be multiplied by itself. For example:

$$\left(\frac{1}{4}+\frac{2}{5}\right)^2 = \left(\frac{5+8}{20}\right)^2 = \left(\frac{13}{20}\right)\left(\frac{13}{20}\right) = \frac{169}{400} \tag{2.106}$$

This can in turn be compared with if we multiply two different parentheses:

$$\left(3+\frac{2}{3}\right)(2-3) = \left(\frac{11}{3}\right)(-1) = -\frac{11}{3} \tag{2.107}$$

Suppose we have a fraction $\frac{a}{b}$ where a and b are arbitrary integers, not equal to 0. If $\frac{a}{b}$ results in a new integer that we call c where c is yet another integer, this is called a being divisible by b. The number b is in that case called a divisor of a. The number b can be described as a factor in a.

If we have $\frac{5}{4} = 1.25$, then 4 is not a divisor of 5 since the result 1.25 is not an integer. The decimal value from division is also called a remainder. For example: $\frac{5}{4} = 1 + \frac{1}{4} = 1$ remainder $\frac{1}{4}$. Suppose we now have the fraction $\frac{a}{b} = c$. We can now multiply the left and right sides by b and divide both sides by c. In this way we swap places between b and c(see section 2.6):

$$\begin{aligned} \frac{a}{b} &= c \\ \frac{a}{c} &= b \end{aligned} \tag{2.108}$$

Since a, b and c are integers, equation (**??**) illustrates that both b and c are divisors of a. Consider the following example with numbers. The number 2 is a divisor of 6:

$$\frac{6}{2} = 3 \tag{2.109}$$

The number 3 is an integer, which is a requirement for calling 2 a divisor. The number 3 is also a divisor of 6 since $\frac{6}{3} = 2$. Factorization instead means that we break up the number into two or more factors. For the number 6, the numbers 3 and 2 are factors:

$$2 \times 3 = 6 \tag{2.110}$$

Prime numbers are the positive integers greater than 1 that are only divisible by 1 and themselves. The number 6 is not a prime number since this number has the divisors 1, 2, 3 and 6. The first prime numbers are 2, 3, 5, 7 and 11. Prime numbers can, among other things, be used to find all divisors of a number by writing a number as a factor of prime numbers. This is called prime factorization. Since prime numbers are only divisible by 1 and themselves, there is only one way to prime factorize each integer.

The numbers 2 and 3 are prime numbers and therefore $2 * 3 = 6$ is a prime factorization of the number 6. Prime factorization of $9 = 3 * 3 = 3^2$. Let us prime factorize the number 121. One way to do this is to test the first prime numbers. This is often a good method since many numbers are divisible by precisely these prime numbers. We try to divide 121 by 2, 3, 5, 7 and 11 and get:

$$\frac{121}{11} = 11 \tag{2.111}$$

This means that $11^2 = 121$ and 11 is thus a divisor of 121. Based on the unique prime factorization, it is now possible to find all possible divisors of the number by combining the prime numbers with each other. Let us take the number 24 as an example. Prime factorization gives

$$24 = 2^3 * 3 = 2 * 2 * 2 * 3 \tag{2.112}$$

We get the divisors $2 * 2 = 4$, $2 * 2 * 2 = 8$ and $2 * 3 = 6$. This we can in turn see by taking:

$$\begin{aligned} \frac{24}{2} &= 12 \\ \frac{24}{3} &= 8 \\ \frac{24}{4} &= 6 \\ \frac{24}{6} &= 4 \\ \frac{24}{8} &= 3 \end{aligned} \tag{2.113}$$

2.11 Sum

The sum of a collection of numbers can be written with the symbol $\sum$ which is the Greek letter capital sigma. The use of $\sum$ is so common that this is called the summation symbol or summation sign. Say for example that we have a set consisting of the three numbers 4, 5 and 6. The set (not the sum) with these numbers we call X and each individual number we call x_i where i indicates which of the numbers it is: $x_1 = 4$, $x_2 = 5$ and $x_3 = 6$. In total we have n amount of numbers where $n = 3$. The sum of the numbers in X can be described like this:

$$\sum_{i=1}^{n=3} x_i = x_1 + x_2 + x_3 = 4 + 5 + 6 = 15 \tag{2.114}$$

If we write $\sum_{i=1}^{n}$ the symbols $i = 1$ and n mean that we should sum from observation number $i = 1$ up to observation number $i = n$. Since n indicates the number of observations, this means that we should sum all values. The summation symbol is thus an abbreviation for addition. This can also be combined with other arithmetic operations. For example

$$\sum_{i=1}^{3} \frac{x_i}{2} = \frac{4}{2} + \frac{5}{2} + \frac{6}{2} = 7,5 \tag{2.115}$$

The notation $\sum_{i=1}^{3}$ with $i = 1$ and 3 above means that we sum the values $i = 1$ to $i = 3$ for all numbers x_i. Since division by 2 is not affected by the summation $\frac{1}{2}$ can be moved to the left of the summation:

$$\sum_{i=1}^{3} \frac{x_i}{2} = \frac{1}{2} * \sum_{i=1}^{3} x_i \tag{2.116}$$

where the value $\frac{1}{2}$ is constant, it is the same regardless of which value x_i has. That these two expressions give the same result as equation (2.114) and (2.115) . Here is another example where we factor out multiplication by 3:

$$\sum_{i=1}^{3} 3x_i = 3*4+3*5+3*6 = 45 \tag{2.117}$$
$$3\sum_{i=1}^{3} x_i = 3*15 = 45$$

If instead of addition we want to multiply a collection of values and calculate the sum of this product, this is called calculating the product sum, which is written with Π(the Greek letter capital Pi)

$$\prod_{i=1}^{n=3} x_i = x_1 * x_2 * x_3 = 4*5*6 = 120 \tag{2.118}$$

2.12 Median

Median is the middle value in a collection of values. Mathematically this can be described as having a collection of numbers that we sort from lowest to highest. We call the collection of numbers for X. Each number is numbered as $x_1, x_2, ..., x_n$, where n is the amount of numbers. If we have the numbers 4, 5, 6 then $x_1 = 4, x_2 = 5$ and $x_3 = 6$, and the amount of numbers $n = 3$. The median depends on whether n is an even or odd number:

$$\begin{aligned} \text{Median } &= x_{\frac{n+1}{2}}, \text{ if } n \text{ is an odd number} \\ \text{Median } &= \frac{x_{\frac{n}{2}} + x_{\frac{n+2}{2}}}{2}, \text{ if } n \text{ is an even number} \end{aligned} \tag{2.119}$$

This might look more complicated than it is. We have the three numbers 4, 5, 6 and $n = 3$. Since n is an odd number, we use the upper definition from equation (2.120) which gives us:

$$x_{\frac{n+1}{2}} = x_{\frac{3+1}{2}} = x_2 = 5 \tag{2.120}$$

where x_2 is the second number in our collection. Let us take an example where we seek the median in the following collection of values:

x_i	1	1	1	4	5	6	6	6	7	8	8	9	9	9	9	9
i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

The upper row x_i are the values and the lower row i are the order numbers. We have here $n = 16$, which is why we use the median's second definition in equation (2.120) :

$$\frac{x_{\frac{n}{2}} + x_{\frac{n+2}{2}}}{2} = \frac{x_{\frac{16}{2}} + x_{\frac{16+2}{2}}}{2} = \frac{6+7}{2} = 6,5 \tag{2.121}$$

The median is between the eighth and ninth values, x_8 and x_9.

2.13 Mean

Another useful measure for average value is the mean, which is defined in the following way:

$$\text{Mean} = \frac{\text{Total of the numbers we want to calculate the mean for}}{\text{Amount of numbers}} \tag{2.122}$$

Let us calculate the mean for the three numbers 4, 5 and 6. The sum of the numbers $(4+5+6)$ is divided by the amount of numbers (3):

$$\text{Mean} = \frac{4+5+6}{3} = \frac{15}{3} = 5 \tag{2.123}$$

The mean is often written with a bar over it, which is why the mean for all the values in X can be written $\bar{X}$. The calculation of this mean can then be described:

$$\bar{X} = \frac{\sum_{i=1}^{n} x_i}{n} \tag{2.124}$$

Now we shall calculate the mean of the numbers 3, 5, 4, 6 and 7. Each value we call x_i where $x_1 = 3, x_2 = 5, x_3 = 4, x_4 = 6$ and $x_5 = 7$. The entire collection of the five numbers we call X. The mean $\bar{X}$ can be calculated as:

$$\bar{X} = \frac{\sum_{i=1}^{5} x_i}{5} = \frac{3+5+4+6+7}{5} = 5 \tag{2.125}$$

If instead of $\sum_{i=1}^{5}$ we write $\sum_{i=2}^{4}$ we sum the second, third and fourth values in the number series. Since we now take the mean of three numbers, the denominator in the expression will change from 5 to 3:

$$\bar{X} = \frac{\sum_{i=2}^{4} x_i}{3} = \frac{5+4+6}{3} = 5 \tag{2.126}$$

2.14 Weighted mean

A type of mean that is often useful is what is called weighted mean. As the name suggests, this is a mean where each value is weighted. Suppose we have two school classes, A and B, with 10 and 20 students in each class respectively. On the latest math test, the students in the two classes had on average 32 and 42 points respectively. If we just take the mean of these two values we get:

$$\text{Mean: } \frac{32+42}{2} = 37 \tag{2.127}$$

But this mean takes no account of the fact that one school class is twice as large and therefore does not give a correct picture of the students' average result. The weighted mean is calculated by taking the sum of each value multiplied by weight and dividing by the sum of the weights:

$$\text{Weighted mean: } \bar{X}_{\text{Weighted}} = \frac{\sum_i^n x_i v_i}{\sum v_i} \tag{2.128}$$

where n is the number of observations, x_i is the value for observation i and v_i is the weight that belongs to each value. Let us use this equation to calculate a weighted mean where we use the number of students in each class as weight. We multiply the two test scores by the number of students in the classes:

$$\bar{X}_{\text{viktat}} = \frac{\sum_i^2 x_i v_i}{\sum_i^2 v_i} = \frac{32*10+42*20}{10+20} \approx 38.67 \tag{2.129}$$

The weighted mean becomes approximately 38.67. This gives a more representative picture of the students' test results, compared to the unweighted mean. The denominator in the equation is the sum of the weights. The weights in this case are the number of school students in each class: $\sum_i^2 v_i = 10 + 20$.

Consider another example. An unweighted mean of the three values 1, 2 and 3 is $6/3 = 2$. Now we shall redo this calculation but let value 3 have twice as much weight as values 1 and 2. We therefore give value 3 the weight 50% and values 1 and 2 the weight 25% each:

$$\bar{X}_{\text{weighted}} = \frac{\sum \text{obs}_i \text{weight}_i}{\sum \text{weight}_i} = \frac{1 * 0.25 + 2 * 0.25 + 3 * 0.5}{0.25 + 0.25 + 0.5} = 2.25 \tag{2.130}$$

2.15 Percent and per mille

Fractions can be used to calculate a portion of something, for example percent (%) or per mille (‰). The word percent means per hundredth and per mille means per thousand. Another way to describe per hundredth and per thousandth is as fractions divided by one hundred and one thousand respectively:

$$\begin{aligned} \text{One percent} &= \frac{1}{100} = 1\% \\ \text{One per mille} &= \frac{1}{1,000} = 1‰ \end{aligned} \tag{2.131}$$

Suppose we have 100,000 apples and shall throw away 13% of these. To calculate how many apples we shall throw away we take:

$$100,000 \times \frac{13}{100} = \frac{100,000 \times 13}{100} = \frac{1,300,0\cancel{00}}{1\cancel{00}} = 13,000 \tag{2.132}$$

Sometimes the word percentage rate is used to describe a proportion in percent. In this example, the percentage rate of apples that we shall throw away is 13%. When calculating with percent, one sometimes wants to calculate percentage change as well as change in number of percentage points. Say for example that you have 500 and receive 25, so your total wealth has increased from 500 to 525. To calculate how large the increase is calculated in percent, we take the increase as a percentage share of the original amount, 25/500, and divide both numerator and denominator by 5, so that the numerator becomes 100:

$$\frac{25}{500} = \frac{25/5}{500/5} = \frac{5}{100} = 0.05 = 5\% \tag{2.133}$$

Say now that you receive an additional 25. This increase in percent becomes:

$$\frac{25}{525} \approx 0.0476 = 4.76\% \tag{2.134}$$

where $\approx$ means approximately equal to. The number 0.0476 is 4.76 percent. To know how many percent your wealth has increased from the original sum 500, we divide the total increase 50 by the original sum:

$$\frac{50}{500} = \frac{50/5}{500/5} = \frac{10}{100} = 10\% \tag{2.135}$$

The International Labour Organization (ILO), an agency under the organization United Nations, UN, estimates that in 2016 on average 40 million people in the world lived under slavery-like conditions. The same year, the earth's population was 7.426 billion. Let us calculate what proportion of the earth's population lived under slavery-like conditions, calculated as percent and per mille:

$$\text{Percent: } \frac{40}{7,426} * 100 = 0.00539 * 100 = 0.539\% \tag{2.136}$$
$$\text{Per mille: } \frac{40}{7,426} * 1000 = 0.00539 * 1,000 = 5.39‰$$

These figures suggest that approximately half a percent, or 5.4 per mille, of the earth's population live under slavery-like conditions.

Suppose we have a tax that is 32% on average. In fraction form this is $\frac{32}{100}$ or with decimals 0.32. If a person has 20,000 USD that shall be taxed at this percentage rate. To calculate 32% of 20,000 USD:

$$\frac{32}{100} * 20,000 = \frac{32 * 20,000}{100} = 6,400 \tag{2.137}$$

Note that the number 20,000 can also be written as a fraction:

$$\frac{32}{100} * \frac{20,000}{1} = \frac{32 * 20,000}{100 * 1} = \frac{640,000}{100} = 6,400 \tag{2.138}$$

Table 2.2: The same value described in different ways

Decimal number	Fraction	Power	Percent	Per mille
0.01	$\frac{1}{100}$	10^{-2}	1%	10‰
0.05	$\frac{5}{100}$	$5 * 10^{-2}$	5%	50‰
0.1	$\frac{1}{10}$	10^{-1}	10%	100‰
0.1011	$\frac{1,011}{10,000}$	$1,011 * 10^{-4}$	10.11%	101.1‰
0.5	$\frac{1}{2}$	2^{-1}	50%	500‰
0.75	$\frac{3}{4}$	$3 * 4^{-1} = 3 * 2^{-2}$	75%	750‰
1	$\frac{1}{1}$	$1^1 * 1^{-1}$	100%	1,000‰
2.5	$\frac{5}{2}$	$5 * 2^{-1}$	250%	2,500‰
10.3	$\frac{103}{10}$	$103 * 10^{-1}$	1,030%	10,300‰

We have now covered several key mathematical concepts including decimal numbers, fractions, percent, per mille, and powers. In mathematics, the same numerical value can often be expressed in multiple equivalent forms. Being able to convert between these different representations while maintaining the same value is a fundamental skill that helps solve problems more efficiently. Table 2.2 demonstrates various ways to express identical values using decimal numbers, fractions, powers, percentages, and per mille.

2.16 Combinatorics

Combinatorics is a part of mathematics that among other things can be used to calculate the number of possible combinations of two phenomena. Consider an example with more characters and possible combinations. Suppose we shall choose one letter and one number. The Swedish alphabet has 29 letters and we have 10 different numbers to choose from 0–9. Among the possible combinations we have for example $a1, a2, a3, ..., b7, b8, b9, ..., k5, k6$ and so on. The number of possible combinations is given by the number of possible choices multiplied with each other:

$$\text{Number of letters in the alphabet} \times \text{Number of digits} = 29 \times 10 = 290 \tag{2.139}$$

Erik has three t-shirts and two pairs of jeans. From these he can choose $3 \times 2 = 6$ combinations when he shall go out. Now we shall select two of the letters a, b and c and combine these. We may choose the same letter twice. We then have nine possible combinations: *aa*, *bb*, *cc*, *ab*, *ba*, *ac*, *ca*, *bc* and *cb*. One way to calculate this is to take the number of possible characters (3) raised to the power of the number of characters to be chosen (2):

$$3^2 = 3 * 3 = 9 \tag{2.140}$$

The same thing applies if the number of possible combinations increases. Say that you shall choose a password for your email and it must be at least 12 characters long. Each character can consist of a letter from the Swedish alphabet, one of the numbers 0 to 9 or one of the following 10 special characters: !, “, #, ¤, %, &, /, (,) and =. To calculate the number of possible combinations, we add the possible alternatives for each character:

$$29 \text{ letters} + 10 \text{ digits} + 10 \text{ special characters} = 49 \tag{2.141}$$

The number of possible passwords can then be calculated to:

$$49^{12} = 191,581,231,380,566,000,000 \tag{2.142}$$

This number can be read as approximately 191.6 trillion, where 1 trillion is written with a one and 18 zeros. This enormous number of possible combinations is exactly why strong passwords are crucial for cybersecurity, using long passwords with a mix of letters, numbers and symbols etc. When hackers attempt to break into accounts, they often simply try different password combinations until they find the correct one. With $49^{12} \approx 191.6$ trillion possible combinations, even a powerful computer trying millions of passwords per second would need an impractically long time to test every possibility.

Now we shall calculate how many ways the letters can be arranged $\{a, b, c\}$ without using any letter twice in the same combination. In total we have 6 different ordered combinations: abc, acb, cab, bac, bca and cba.

Ordered combinations are called permutations. To calculate the number of permutations, the number of ordered combinations, we take the number of alternatives and multiply this with each integer down to 1:

$$3! = 3 * 2 * 1 = 6 \tag{2.143}$$

The expression 3! can be read as three factorial. For eight different alternatives we get:

$$8! = 8 * 7 * 6 * 5 * 4 * 3 * 2 * 1 = 40,320 \tag{2.144}$$

Generally, it holds that 0!=1. To understand this we may imagine a scenario where we shall choose to put on some nice clothes when we shall go to a party and want to calculate the number of possible combinations. We realize that we don’t have any clothes we like and therefore have 0 alternatives. But not choosing any clothes is also a type of combination.

The total number of possible combinations therefore becomes 1. Factorial is according to the same logic only defined for non-negative integers, the natural numbers: 0, 1, 2,... and so on. A general way to describe the above permutations is that if n is an arbitrary natural number then it holds for all natural numbers that:

$$n! = n * (n-1) * (n-2) * ... * 3 * 2 * 1 \tag{2.145}$$

If $n = 13$ then 13! means that we get:

$$13! = 13 * 12 * 11 * ... * 3 * 2 * 1 \tag{2.146}$$

Say now that we again shall choose among the letters $\{a, b, c\}$ but we shall only arrange 2 of these. When we choose the first letter we have three to choose from. When we then shall choose the next letter we have two left to choose from. If we as the first letter chose c we have left a or b to choose from as the second letter. The number of possible combinations is in this case $3 * 2 = 6$. To describe this more generally we may rewrite equation (2.145) to:

$$n * (n-1) * ... * (n-k+1) \tag{2.147}$$

where n is still an arbitrary number and k is the number that we shall choose. The number k must by definition here be a number between 0 and n. So in the case where we have three letters and shall select two we have that $n = 3$ and $k = 2$. The number of possible alternatives can then be described as:

$$3 * (3 - 2 + 1) = 3 * 2 = 6 \tag{2.148}$$

This equation can also be written like this:

$$n * (n-1) * ... * (n-k+1) = \frac{n!}{(n-k)!} \tag{2.149}$$

Let us use this new equation to calculate the number of possible combinations of the three letters $\{a, b, c\}$ if we shall select two of them:

$$\frac{n!}{(n-k)!} = \frac{3!}{(3-2)!} = \frac{3*2*1}{1!} = \frac{6}{1} = 6 \tag{2.150}$$

Let us take another example. We find ourselves now in the future. You have become a mathematical genius and rich from your many brilliant inventions. But you feel bored and decide to sort your luxury cars. You have a total of 12 cars and now want to calculate how many different ways you can combine two of your cars on the driveway to your enormous house. We have $n = 12$ and $k = 2$. The total number of possible arrangements is given by:

$$\frac{n!}{(n-k)!} = \frac{12!}{(12-2)!} = \frac{12*11*10*9*8*7*6*5*4*3*2*1}{10*9*8*7*6*5*4*3*2*1} \tag{2.151}$$

Since all numbers in the denominator are also found in the numerator the fraction can be simplified by deleting the same multipliers in the numerator as in the denominator. We mark the numbers that shall be crossed out with parentheses:

$$\frac{12*11*(10*9*8*7*6*5*4*3*2*1)}{(10*9*8*7*6*5*4*3*2*1)} = 12*11 = 132 \tag{2.152}$$

For the first car you have 12 to choose from. When you shall choose the second you have 11 to choose from. Equation (2.149) can also be used for the first examples in this section. To calculate $n!$ for all possible combinations of all available alternatives, that is $k = n$, we get:

$$n! = \frac{n!}{(n-k)!} = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n! \tag{2.153}$$

Even the absence of alternatives is a possible alternative. If we have n possible alternatives but choose not to choose any of these, $k = 0$, we get:

$$\frac{n!}{(n-k)!} = \frac{n!}{(n-0)!} = \frac{n!}{n!} = 1 \tag{2.154}$$

So far our selected values have been placed in a specific order, for example acb or bca. If we are uninterested in the order, the alternatives abc, acb, bca, bac, cab and cba count as only one alternative.

When we are not interested in which order the values stand in, this is called a combination. If we have n possible alternatives, select k number from these and want to know the number of possible combinations, it can be calculated in the following way:

$$\binom{n}{k} = \frac{n!}{(n-k)! * k!} \tag{2.155}$$

The notation $\binom{n}{k}$ can be read as "n over k". If we select two ($k = 2$) from three alternatives ($n = 3$):

$$\binom{n}{k} = \frac{n!}{(n-k)! * k!} = \frac{3!}{(3-2)!2!} = \frac{3 * \cancel{2 * 1}}{1 * \cancel{2 * 1}} = 3 \tag{2.156}$$

Let the three alternatives be the letters $\{a, b, c\}$. The possible alternatives are then ab, ac, bc. Let us take the example with the 12 luxury cars again, $n = 12$ and $k = 2$, but without internal order:

$$\begin{aligned} \binom{n}{k} &= \frac{n!}{(n-k)! * k!} \\ &= \frac{12!}{(12-2)!2!} \\ &= \frac{12 * 11 * (10 * 9 * 8 * 7 * 6 * 5 * 4 * 3 * 2 * 1)}{(10 * 9 * 8 * 7 * 6 * 5 * 4 * 3 * 2 * 1) * 2 * 1} \\ &= \frac{12 * 11}{2} \\ &= 6 * 11 \\ &= 66 \end{aligned} \tag{2.157}$$

Another useful trick is to think symmetrically about combinatorial problems. Say for example that we have the letters $\{a, b, c\}$ and shall choose two of these. This can either be set up as choosing 2 letters, alternatively choosing 1 letter that shall remain:

$$\begin{aligned} \binom{3}{2} &= \binom{3}{1} \\ \frac{3!}{(3-2)!2!} &= \frac{3!}{(3-1)!1!} \\ \frac{3 * \cancel{2 * 1}}{1 * \cancel{2 * 1}} &= \frac{3 * \cancel{2 * 1}}{\cancel{2 * 1} * 1} \\ \frac{3}{1} &= \frac{3}{1} \end{aligned} \tag{2.158}$$

This can generally be described as:

$$\binom{n}{k} = \binom{n}{n-k} \tag{2.159}$$

To see this one can substitute the right side of this expression into equation (2.155) and then in the fraction replace k with $n - k$:

$$\binom{n}{n-k} = \frac{n!}{(n-(n-k))!\,(n-k)!} = \binom{n}{k} \tag{2.160}$$

2.17 Chapter summary

- Real numbers: all values on the number line, integers, rational and irrational numbers. The four arithmetic operations: addition, subtraction, multiplication and division. One may also use powers, parentheses and fractions. Addition: terms and sum. Subtraction: terms and difference. Multiplication: factors and product. Division: numerator, denominator and quotient. Order of operations: parentheses, powers, multiplication, division, addition and subtraction, from left to right.
- Absolute value describes a number's distance from 0 on the number line. Example: $|-1| = 1$. Equality (=) and inequality (for example > and <) describe the relation between values. Mathematical operations on one side of the equals sign must be matched by operations on the other side. Percent means hundredth, $1/100 = 1\%$. Per mille means thousandth, $1/1,000 = 1‰$. Factorization means that we factor out a common factor. Prime numbers are the positive integers greater than 1 that are only divisible by 1 and themselves. Example: 2, 3, 5, 7, 11.
- Power = base$^{\text{exponent}}$. Example: $2^3 = 2*2*2 = 8$. For $a^{\frac{1}{2}} = b$, where a is an arbitrary number, b is the number that multiplied by itself becomes a. That is: $b^2 = a$. $a^{\frac{1}{2}} = \sqrt{a}$ = square root of a. $a^{\frac{1}{3}}$ = cube root of a.
- Notable products include $(a+b)^2 = a^2+2ab+b^2$, and $(a-b)^2 = a^2-2ab+b^2$, and: $(a+b)(a-b) = a^2-b^2$.
- Mean: $\frac{\sum_{i=1}^{n=3} x_i}{n} = \frac{x_1+x_2+x_3}{3}$. Weighted mean: $\frac{\sum_i^n x_i v_i}{\sum v_i}$ where v_i is weight for each value x_i. Median is the middle value. Product sum: $\prod_{i=1}^{n=3} x_i = x_1 * x_2 * x_3$.

-$3^2 = 9$: Number of combinations with repetition when we choose 2 values from a set with 3 values. Three factorial: $3! = 3*2*1 = 6$= number of permutations, possible combinations without repetition with order. Number of permutations when we choose k alternatives from n values: $\frac{n!}{(n-k)!}$. Possible combinations k from n alternatives without order: $\binom{n}{k} = \frac{n!}{(n-k)!k!}$.

2.18 Exercises

1. Calculate:

(a) (3+4times2-1)

(b) ((3+4)times2-1)

(c) (3+4times(2-1))

Answer: a) $3+8-1 = 10$, b) $7 \times 2 - 1 = 14 - 1 = 13$, c) $3 + 4 \times 1 = 3 + 4 = 7$.

2. Find lowest common denominator and calculate the sum:

(a) (frac{1}{3}+frac{2}{5})

(b) (frac{3}{4}-frac{1}{6})

(c) (frac{2}{3}+frac{5}{6}-frac{1}{2})

Answer: a) $\frac{5}{15} + \frac{6}{15} = \frac{11}{15}$, b) $\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$, c) $\frac{4}{6} + \frac{5}{6} - \frac{3}{6} = \frac{6}{6} = 1$

3. Simplify:

(a) ($2^{\{3\}\text{times}2}$\{4\})

(b) (frac{$5^{\{6\}\}\{5}$\{2\}\})

(c) (($3^{\{2\})}$\{3\})

Answer: a) $2^{3+4} = 2^7 = 128$, b) $5^{6-2} = 5^4 = 625$, c) $3^{2\times3} = 3^6 = 729$

4. Calculate:

(a) (|7-12|)

(b) (|-3|+|5|)

(c) (|4-9|-|-2|)

Answer: a) $|-5| = 5$, b) $3 + 5 = 8$, c) $5 - 2 = 3$

5. Factorize:

(a) (6x+12)

(b) (x^{2}-9)

(c) (15x^{2}+10x)

Answer: a) $6(x + 2)$, b) $(x + 3)(x - 3)$, c) $5x(3x + 2)$

6. Calculate

(a) (sqrt{64})

(b) (sqrt[3]{27})

(c) (16^{1/2})

Answer: a) 8 (positive root), b) 3 , c) 4

7. A company has 1,250 employees. If 32% of employees work part-time, how many work:

(a) Part-time?

(b) Full time?

(c) If 15% of part-time employees move up to full-time, how many part-time employees are left?

Answer: a) $1,250\times0,32 = 400$ persons, b) $1,250-400 = 850$ persons, c) $400-(400\times0,15) = 400-60 = 340$ persons

8. Develop the following parenthetical expressions:

(a) ((x+3)^{2})

(b) ((2x-5)^{2})

(c) ((x+4)(x-4))

Answer: a) $x^2 + 6x + 9$, b) $4x^2 - 20x + 25$, c) $x^2 - 16$

9. Solve the following:

(a) (frac{2/3}{1/2})

(b) (frac{3/4+1/2}{1/3})

(c) (frac{5}{2/7})

Answer: a) $\frac{2/3}{1/2} = \frac{2}{3} \times \frac{2}{1} = \frac{4}{3}$, b) $\frac{3/4+2/4}{1/3} = \frac{5/4}{1/3} = \frac{5}{4} \times 3 = \frac{15}{4}$, c) $5 \times \frac{7}{2} = \frac{35}{2}$

10. We have the following numbers: 2, 5, 8, 8, 9, 11, 14. Calculate:

(a) Median

(b) Mean

(c) Sum

Answer: a) Median = 8, b) Mean = $\frac{2+5+8+8+9+11+14}{7} = \frac{57}{7} \approx 8.14$, c) Sum = 57

11. Two classes have written the same exam. Class A with 18 students had a mean score of 72. Class B with 24 students had a mean score of 68. Calculate:

(a) The weighted mean of both classes,

(b) Total number of points for class A

(c) Total number of points for class B

Answer: a) Weighted mean $= \frac{18\times72+24\times68}{18+24} = \frac{1296+1632}{42} = \frac{2928}{42} \approx 69.7$, b) $18 \times 72 = 1,296$ points, c) $24 \times 68 = 1,632$ points

12. Calculate

(a) (4!)

(b) (binom{5}{2})

(c) In how many ways can 3 people line up?

Answer: a) $4! = 4 \times 3 \times 2 \times 1 = 24$, b) $\binom{5}{2} = \frac{5!}{2!3!} = \frac{5\times4}{2\times1} = 10$, c) $3! = 3 * 2 * 1 = 6$ ways

13. Write the following numbers in different forms:

(a) Write 0.125 as a fraction and percent

(b) Write (frac{3}{8}) as a decimal number and percent

(c) Write 45% as a fraction and decimal number

Answer: a) $\frac{1}{8}$ and 12,5%, b) 0.375 and 37.5%, c) $\frac{9}{20}$ and 0.45

14. Prime factor the following numbers and enter all divisors:

(a) 36

(b) 45

(c) 60

Answer: a) $36 = 2^2 \times 3^2$, divisors: 1, 2, 3, 4, 6, 9, 12, 18, 36, b) $45 = 3^2 \times 5$, divisors: 1, 3, 5, 9, 15, 45, c) $60 = 2^2 \times 3 \times 5$, divisors: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

15. Calculate the sum:

(a) (sum_{i=1}^{5}2i)

(b) (sum_{i=2}$^{\{4\}(i}${2}-1))

(c) (prod_{i=1}^{3}(i+1))

Answer: a) $2 + 4 + 6 + 8 + 10 = 30$, b) $(4 - 1) + (9 - 1) + (16 - 1) = 3 + 8 + 15 = 26$, c) $2 \times 3 \times 4 = 24$

Chapter 3

Examples with Applications

This chapter provides examples of how mathematics can be applied in issues related to social science.

3.1 Interest

In section 2.15 we went through percent, hundredth, and per mille, thousandth. Now we will calculate how much the interest will be if we deposit money in a bank account with two percent interest per year. Since we deposit money in the bank, this may be described as lending money to the bank and the interest we receive on our account is the compensation that the bank gives us.

Two percent can be written with numbers as either 2 % or as 0.02. To calculate the interest we multiply this value with the amount of money we deposit in the account. Suppose we deposit 100 USD. The interest then becomes $100 * 0.02 = 2$ USD. After one year we have 100 USD + the interest that the bank gives us: $100 + 2 = 102$.

Another way to calculate this is to multiply $100 \times 1.02 = 102$. We multiply our savings with 1 plus the interest rate 2%: $1 + 0.02 = 1.02$. This shows how much money we have after the interest is added to our savings amount. If we let the money remain in the account for another year, we will next year receive interest on the amount 102. The result after year 2 is given by $102 \times 1.02 = 104.04$. Another way to describe this:

$$\begin{aligned} 102 * 1.02 &= (100 * 1.02) * 1.02 \qquad (3.1) \\ &= 100 * 1.02 * 1.02 \\ &= 100 * (1.02)^2 \\ &= 104.04 \end{aligned}$$

Note that all rows in the equation give exactly the same result. On the third row we write together $1.02 * 1.02$ as $(1.02)^2$. This is read as 1.02 raised to the power of two and is a power expression, see section 2.8 .

3.2 Interest on interest

When interest is added to money that has already increased due to interest, this is called interest on interest, or compound interest or cumulative interest. Suppose we deposit 5,000 USD in a savings account where the annual interest is two percent. We let the money remain in the account for three years. The first year we receive the interest that is generated on our 5,000 USD. Our savings amount after the first year then becomes:

$$\begin{aligned}\text{Savings after year 1} &= 5,000 * 1.02 \\ &= 5,000 * 1 + 5,000 * 0.02 \\ &= 5,000 + 100 \\ &= 5,100\end{aligned} \tag{3.2}$$

We here multiply the original amount with 1 plus the interest rate 0.02 since we shall calculate the total amount that we will have after the interest is paid out. The two following years we receive interest both on our original savings plus interest on the money we have received in interest previous years. Let us write it with one equation per year. After year 1 we have:

$$5,000 * 1 + 5,000 * 0.02 = 5,000 + 100 = 5,100 \tag{3.3}$$

After year 2:

$$5,100 * 1.02 = 5,100 + 102 = 5,202 \tag{3.4}$$

After year 3:

$$5,202 * 1.02 = 5,306.04 \tag{3.5}$$

These three calculations can also be written in one and the same equation as:

$$5,000 * 1.02 * 1.02 * 1.02 = 5,306.04 \tag{3.6}$$

Just as before, powers can be used to make this calculation even more compact. We set 1.02 as the base in a power expression and define the exponent equal to the number of years: 3. Our total amount after three years:

$$5,000 * 1.02^3 = 5,306.04 \tag{3.7}$$

When we take a loan we instead have to pay interest on interest. Say for example that you use a credit card with 12% annual interest. A credit card allows you to borrow money from the bank when making purchases and pay interest on the loan. The interest is in this case added per month. To calculate the monthly interest we take the annual interest, 12%, and divide by the number of months in a year, 12:

$$\text{Interest per month: } 0.12/12 = 0.01 = 1\% \tag{3.8}$$

Say now that you use the credit card to shop for 20,000 and wait three months to pay. How much will the interest be? The interest starts in this example being added to your loan directly when you shop and is calculated at the beginning of each month on the debt that then applies. New interest is added to old interest for each month that the loan is not repaid. After one month:

$$20,000 * 1.01 = 20,200 \tag{3.9}$$

After two months:

$$20,200 * 1.01 = 20,402 \tag{3.10}$$

After three months:

$$20,402 * 1.01 = 20,606.02 \tag{3.11}$$

Let us write this using a power expression. After three months your loan has grown to the following amount:

$$20,000 * 1.01^3 = 20,606.02 \tag{3.12}$$

The interest cost for borrowing for three months on the credit card becomes in this case $20,606.02–20,000 = 606.02$ USD. This phenomenon is called interest on interest or cumulative interest. The word cumulative refers here to that the interest is added and forms a sum, whereupon interest is added to that sum and so on. A related concept is cumulative sum, which describes a sum that is calculated per value in a series, including all values in the series up to that point. Table 3.1 shows some numbers and a column with a cumulative sum and a column with a regular sum.

Table 3.1: Illustration of cumulative sum.

Observation number	Value	Cumulative sum
1	23	23
2	32	$23 + 32 = 55$
3	11	$23 + 32 + 11 = 66$
4	14	$23 + 32 + 11 + 14 = 80$
Sum: 80		

3.3 Effective interest

According to the consumer credit law, lenders are obligated to inform about a loan's effective interest rate. Effective interest rate is the total cost of a loan including any additional fees, calculated as annual percentage interest rate on the loan amount. Many loans come with additional fees, for example a fee for creating the loan (setup fee) or a fee for the lender to send bills (billing fee). In practice this means that when it says in an advertisement that you can buy something and pay "interest-free" later, you often in practice pay an effective interest rate despite it not being called interest.

A store offers you to buy a TV for 10,000 USD and postpone the payment without interest for 24 months. For this the store charges a fee of 700 USD and an administrative fee of 40 USD every month, totaling $40 \times 24 = 960$. Total amount of extra costs:

$$700 + 40 \times 24 = 1,660 \tag{3.13}$$

Expressed as a percentage of the loan (10,000 USD) this gives the effective interest rate:

$$\frac{1,660}{10,000} = 16.6\% \tag{3.14}$$

Let us take another example. Various companies offer payday loans, paid out directly. A company offers you to borrow 10,000 USD at 40% nominal interest rate for 12 months. "Nominal interest rate" refers to the interest rate that is written out, for example in an advertisement. To find out what we really have to pay we must however calculate with all fees that are added. Nominal interest rate + any fees = effective interest rate. To the nominal interest rate is added 700 USD in setup fee, 700 USD in extension fee and 420 USD per month in billing fee. One twelfth of the nominal interest rate is added to the loan each month. We have nominal interest rate + setup fee + extension fee + the billing fees. Total amount to repay for a loan of 10,000 USD is then:

$$\begin{aligned}
&10,000 \times \left(1 + \frac{0.4}{12}\right)^{12} + 700 + 700 + 420 \times 12 = \\
\approx &10,000 \times 1.4821264 + 700 + 700 + 5,040 = \\
= &14,821.26 + 700 + 700 + 5,040 = \\
= &21,261.26
\end{aligned} \tag{3.15}$$

To get the effective interest rate we subtract the loan amount 10,000 from this sum and calculate how much the remainder is as a percentage of the loan amount:

$$\frac{21,261.26 - 10,000}{10,000} = \frac{11,261.26}{10,000} \approx 112.6\% \tag{3.16}$$

The effective interest rate for this loan is thus 112.6%. Some loans can have over 1,000 percent in effective interest rate.

3.4 Index

An index is a method to compare the development of a quantity. A common way to calculate an index is to compare values against a base year or a base level. We may choose whichever year we want to use as base year. Often one calculates index so that the index value for the base year is 100. The number 100 is arbitrarily chosen.

Table: Basis for index

Time series A

Year	2007	2008	2009	2010	2011	2012
Data	123	234	336	207	252	199
Time series B	Year	2010	2011	2012	2013	2014
—	—	—	—	—	—	—
Data	2	4	5	4	7	6.3

By comparing two or more indexes (indices) it is easy to see relative differences. In table 3.4 we have the two time series A and B. For each year t an index can be calculated by dividing the value for this year with the value for the base year. Index year t is then:

$$\text{Index}_t = \frac{\text{Value}_t}{\text{Value}_{\text{base year}}} * 100 \tag{3.17}$$

Now we shall create one index per time series, using the first year in each series as the base year. The index value for time series A year 2007 is:

$$\text{Index A}_{2007} = 100 * \frac{\text{Time series A}_{2007}}{\text{Time series A}_{2007}} = 100 * \frac{123}{123} = 21.7 \tag{3.18}$$

For the year 2010 we get:

$$\text{Index A}_{2010} = 100 * \frac{\text{Time series A}_{2010}}{\text{Time series A}_{2007}} = 100 * \frac{207}{123} \approx 168.3 \tag{3.19}$$

The last year for time series A is 2012:

$$\text{Index A}_{2012} = 100 * \frac{199}{123} \approx 161.8 \tag{3.20}$$

For time series B the first year is 2010:

$$\text{Index B}_{2010} = 100 * \frac{\text{Time series B}_{2010}}{\text{Time series B}_{2010}} = 100 * \frac{2}{2} = 100 \tag{3.21}$$

The index value year 2015 for time series B:

$$\text{Index B}_{2015} = 100 * \frac{6.3}{2} = 315 \tag{3.22}$$

Figure 3.1 illustrate the two time-series A and B in two plots. In the plot to the left we show the original values, same as in table 3.4 . In the plot to the right we see the indexed versions of the two series. The first observation in the graph to the right is the base year for each time series. Try yourself to calculate index for the two time series using other base years.

Figure 3.1: Time series index

3.5 Population index

Indices can be used in many situations. Table 3.2 shows the size of the population for USA every tenth year 1950 – 2020. We will now calculate a population index with 1950 as base year. For 1950 we get the following index value:

$$\text{Index}_{1950} = \frac{154}{154} * 100 = 1 * 100 = 100 \tag{3.23}$$

For 1960 we get:

$$\text{Index}_{1960} = \frac{180}{154} * 100 = 1.169 * 100 = 116.9 \tag{3.24}$$

In table 3.3 the results for all years are shown. Please check for yourself that the results are correct. Between 1950 and 2020 the US population increased by circa 120%: 220.1 – 100 = 120.1. In 2000 the population had increased by 82.5%, compared to 1950.

Table 3.2: US population

Year	1950	1960	1970	1980	1990	2000	2010	2020
Population, millions	154	180	208	230	253	281	311	339
Source: Our World In Data								

Table 3.3: US population index

Year	1950	1960	1970	1980	1990	2000	2010	2019
Population index 1960 = 100	100,0	116.9	134.8	149.1	164.3	182.5	201.7	220.1

Let us also calculate the percentage change of the population size between each year. This will show the growth rate per ten-year period (per decade). We do this by calculating the difference between the two values for two different time points and dividing by the value for the earlier time point. We start with the difference in population size 1960 and 1950:

$$\frac{180 - 154}{154} = \frac{180}{154} - \frac{154}{154} \approx 1.169 - 1 = 0.169 \tag{3.25}$$

The results for all years are shown in table 3.4 . If we were to calculate percentage change for the index in table 3.3 we would get the same result. We can see this by inserting the definition of index from equation (3.17) into our equation for percentage change:

$$\text{Percent change between year } t \text{ and year } t - 1 = \frac{\text{value}_t}{\text{value}_{t-1}} - 1 \tag{3.26}$$

We insert the definition of index for the two values:

$$\begin{aligned} \frac{\left(\frac{\text{Value}_t}{\text{Value}_{\text{base year}}}\right)100}{\left(\frac{\text{Value}_{t-1}}{\text{Value}_{\text{base year}}}\right)100} - 1 &= \frac{\text{Value}_t * \cancel{\text{Value}^{-1}_{\text{base year}} * 100}}{\text{Value}_{t-1} * \cancel{\text{Value}^{-1}_{\text{base year}} * 100}} - 1 \\ &= \frac{\text{Value}_t}{\text{Value}_{t-1}} - 1 \end{aligned} \tag{3.27}$$

which is the same thing as percentage change without indexing.

Table 3.4: Population, percentage change per decade

Year	1950	1960	1970	1980	1990	2000	2010	2020
Population, percent change	-	0.169	0.153	0.106	0.102	0.111	0.105	0.0912

3.6 Income index

Now we will compare income decile groups, which means that we divide the US population into ten groups. Decile group one is the tenth of the population that has the lowest earnings. Decile group ten is the tenth with the highest earnings. Many people change income group over the years, for example because they go from unemployment to work, or from work to retirement. Table 3.5 shows the threshold income after tax for decile group 1 and 9, among US households 1970–2020. The values for decile group 1 represents the income level, below which 10 percent of the households fall. The values for group 9 is the income levels, below which 90 percent of the households falls. Above the group 9 values is the 10 percent riches household each year.

Table 3.5: Income per decile group

Decile group 1	Decile group 9	1970
10,514	53,081	1980
11,615	53,943	1990
11,803	63,236	2000
13,465	70,966	2010
13,243	74,392	2020
16,987	94,904	Note: Data from Our World in Data and the Luxembourg Income Study (2025). Income is measured in 2017 dollars, adjusted for inflation.

Now we shall create two indices for the two decile groups with 1970 as base year. The index value for year 1970 will for each group be equal to 100. The results are shown in table 3.6 . For decile group 1 earnings increased by 61.56%. For decile group 9 earnings increased by 178.79%. If we choose another base year the results look different.

Table 3.6: Income index per decile group

Decile group 1	Decile group 9	1970
$\frac{10{,}514}{10{,}514} * 100 = 100$	$\frac{53{,}081}{53{,}081} * 100 = 100$	1980
110.47	101.62	1990
112.26	119.13	2000
128.07	133.69	2010
125.95	140.15	2020
$\frac{16{,}987}{10{,}514} * 100 \approx 161.56$	$\frac{94{,}904}{53{,}081} * 100 \approx 178.79$	

3.7 Adjust prices and wages for inflation

A price index shows the indexed price development for one or several goods or services. One such index that is often used is the consumer price index (CPI) which is created by collecting prices on a large number of different goods and services and calculating an average for a hypothetical basket of goods. The basket of goods is usually based on information about what people spend money on. If some other characteristic of the goods and services changes, for example changed quality, then the calculation is adjusted for this. Ideally, CPI therefore measures only pure price changes.

When we compare price development on individual goods and services we often benefit from adjusting for the general price development in society (inflation), for example by dividing with CPI. This is called to adjust for inflation, inflation-adjusting or deflating prices.

By adjusting for nominal price changes we get a measure of how much the value of money has changed. This in turn makes it easier to compare prices over time. This can be illustrated with a hypothetical example. Say that a good costs 100 USD year 1 and we earn 100 USD per hour. The year after all prices and wages have doubled. Year 2 the good costs 200 USD and we earn 200 USD per hour.

To say that the prices of goods has doubled is not particularly informative if we do not take into account that our earnings also have doubled. And in the same way, the new higher earnings does not tell us anything about our standard of living, as long as we do not account for the price development.

A price that is not deflated is called a nominal price. A nominal price is the price that is stated on the price tag, the normal price. A price that is deflated is called real price. Compare the description of nominal and effective interest rate in section 3.3 . The price index that is used to divide the prices is called a deflator. In this example we shall use the consumer price index, CPI, . One can also use other types of price indices to deflate. The definition of a real price for time point t:

$$\text{Real price}_t = p_t \left(\frac{\text{CPI}_{\text{base}}}{\text{CPI}_t} \right) \tag{3.28}$$

where p_t is the nominal price we shall deflate, CPI_{base} is the CPI value for the year we want to use as base year and CPI_t is the CPI value for the same year as the current nominal price we are deflating. If the price of a good increases more slowly than the price of other things, this good has become cheaper relative to other things. Its real price has decreased. CPI may also be used to measure the purchasing power of wages. If the price of the goods and services we buy increases more slowly than our wages we say that our real purchasing power, and thereby our real wages, has increased.

One of the first handheld mobile phones that was sold to the public, Motorola DynaTAC 8000X, used to cost 3,995 USD in 1984. The simpler models of Motorola's mobile phones that are for sale today, in 2025, cost around 200 USD. The smartphones that Motorola sells today have many more functions than what DynaTAC even came close to, but let us nevertheless calculate the difference between these two prices. We divide the 2025 price by the 1984 price (the base year price) and multiply by 100, which gives us an index value:

$$\frac{200}{3,995} * 100 \approx 5.006 \tag{3.29}$$

The price for one of Motorola's cheapest mobile phones in 2025 costs approximately 5% of what their model cost 40 years ago.

Now we shall deflate the price of Motorola's mobile phones. To deflate a price from a specific year we divide this price by the CPI value for the same year and multiply the result by the index value for the base year (the year we want to measure the prices in). We want to calculate the 1984 price for Motorola DynaTAC 8000X in 2025 prices. A long time series on the US consumer price index is available from the Federal Reserve Bank of Minneapolis website . If CPI in 1984 is set to 100, the CPI in 2025 is approximately 310. The real price of the Motorola DynaTAC 8000X in 2025 prices is therefore:

$$\text{Real price } = \text{price}_{1984} \left(\frac{\text{CPI}_{2025}}{\text{CPI}_{1984}} \right) \approx 3,995 \left(\frac{310}{100} \right) = 12,384.5 \tag{3.30}$$

The real price for Motorola DynaTAC 8000X in year 1984 was $12,384.5, calculatedintoday's monetaryvalue.Letuscalculate$
200. Since the price \$200 for the modern phone is already in 2025 prices we do not need to recalculate it:

$$\frac{200}{12,384.5} * 100 \approx 1.6 \tag{3.31}$$

This means that if we adjust for the average price development between 1984 and 2025, Motorola's modern phone models cost only 1.6% of the price of their flagship model in 1984. Real prices may fall for different reasons, such as technological progress and new production methods.

3.8 Economic growth

Gross domestic product per capita (GDP per capita) for a given year is calculated by summing the value of all goods and services sold that year and dividing by the number of inhabitants. GDP is often measured per country and year, but could also be estimated for a local region or a group of countries. Government agencies are often responsible for collecting information and presenting official estimates of GDP. GDP is often used as an approximation of the amount of income and production in a country.

To compare GDP over time, values are often adjusted for price changes (inflation), to real GDP (instead of nominal GDP). To compare GDP between countries we exchange local currency GDP to a common currency, such as US dollars. When comparing countries it is often useful to adjust real GDP for price differences between those countries, to adjust for purchasing power. This inflation and purchasing power adjusted measure is called USD PPP (Purchasing Power Parities).

Now we will compare GDP growth for the US in the years 2010–2019. During these years, US GDP per capita increased from $48,453 to$ 56,469, see table 3.7 .

We begin by calculating the percentage change of GDP per capita per year. If we denote GDP per capita in year t as y_t, the annual percentage rate of change is:

$$\text{GDP growt in year } t = \frac{y_t - y_{t-1}}{y_{t-1}} \tag{3.32}$$

where y_t is GDP per capita some year t and y_{t-1} is GDP per capita the year before t. For year 2010 we get:

$$\frac{y_{2010} - y_{2009}}{y_{2009}} = \frac{49,267 - 48,453}{48,453} = 1.68\% \tag{3.33}$$

This and the results for the remaining years are shown in table 3.8 . As can be seen in the table, the growth varies somewhat between the different years. By calculating an average, we smooth out the changes.

Table 3.7: GDP per capita, USA

Year	GDP per capita	Year	GDP per capita	2009
48,453	2015	52,808	2010	49,267
2016	53,301	2011	49,675	2017
54,152	2012	50,436	2018	55,455
2013	51,011	2019	56,469	2014
51,797	Note: Data from Maddison Project via Our World In Data			

One way to compare different time periods is to calculate average GDP growth for five years at a time. An average for GDP growth in the years 2010–2014 and an average for the years 2015–2019. We denote these two averages as $\Delta\bar{y}_1$, for the period 2010–2014 and $\Delta\bar{y}_2$, for the period 2015–2019. These are calculated in the following way:

$$\bar{\Delta y_1} = \frac{1.68\% + 0.83\% + 1.53\% + 1.14\% + 1.54\%}{5} = 1.34\% \tag{3.34}$$
$$\bar{\Delta y_2} = \frac{1.95\% + 0.93\% + 1.60\% + 2.40\% + 1.83\%}{5} = 1.74\%$$

Another way to compare development over time while simultaneously smoothing out short-term variations is to use what is called a moving average. With a moving average we calculate an average of the values closest to the current time point. This can be done in several ways. Our moving average for the GDP change in year t, we call $gm\,(y_t)$. To calculate a three-year moving average for year t we summarize the three values year $t-1$, year t and year $t+1$ and divide by 3:

$$gm\,(\Delta y_t) = \frac{y_{t-1} + y_t + y_{t+1}}{3} \tag{3.35}$$

To calculate $gm\,(y_{t+1})$ we instead take:

$$gm\,(\Delta y_{t+1}) = \frac{y_t + y_{t+1} + y_{t+2}}{3} \tag{3.36}$$

Table 3.8 describes percentage change per year for GDP per capita 2010–2019. In table 3.9 the moving average for the same values is shown. The overall development is the same but the growth rate is now more even over time.

Table 3.8: GDP per capita, percent change per year

Year	GDP per capita, percent change	Year	GDP per capita, percent change	2010
1.68%	2015	1.95%	2011	0.83%
2016	0.93%	2012	1.53%	2017
1.60%	2013	1.14%	2018	2.40%
2014	1.54%	2019	1.83%	

Table 3.9: GDP per capita, three year moving average

Year	GDP change	Year	GDP change	2010–2012
1.35%	2014–2016	1.48%	2011–2013	1.17%
2015–2017	1.49%	2012–2014	1.40%	2016–2018
1.65%	2013–2015	1.54%	2017–2019	1.94%

3.9 Child mortality

In recent decades, child mortality throughout the world has declined sharply. However, it still varies greatly among the world's countries. Table 3.10 describes the average child mortality per continent in 1950 and 2016. Both in 1950 and 2016, child mortality was highest in Africa.

However, child mortality has also decreased the most in the African countries, from approximately 300 per 1,000 children in 1950, to being down to 62.7 deceased children per 1,000 under 5 years of age in 2016. During the same period, child mortality in the countries in Europe decreased from 86.7 in 1950 to 4.9 in 2016.

Table 3.10: Number of deceased children per 1,000 under 5 years of age

Year 1950	Year 2016	Africa
302	62.7	Asia
234	22.8	Europe
86.7	4.9	North America
152	15.9	Oceania
140	22.7	South America
181	17.8	Source: Gapminder, OurWorldInData.org.

Since the beginning of the 1800s, the average child mortality in the world has decreased from on average over 400 deaths per 1,000 children (40%), down to approximately 20% in 1950 and to approximately 4% in 2016. That is, a decrease of 16 percentage points:

$$4 - 20 = -16\,\text{percentage points} \tag{3.37}$$

The countries in Africa have during the period seen child mortality decrease from 302 to 62.7 per 1,000 children. Calculated in percentage points this becomes:

$$\frac{62.7 - 300}{1,000} = \frac{-237.3}{1,000} = -0.2373 = -23.73\% \tag{3.38}$$

In Europe the average child mortality decreased from 86.7 to 4.9 per 1,000 children. Expressed in percentage points:

$$\frac{4.9 - 86.7}{1,000} = \frac{-81.8}{1,000} = -0.0818 = -8.18\% \tag{3.39}$$

3.10 Chapter summary

- Interest on a loan is often stated in percent. If you take a loan of 5,000 with 20% interest, the interest is 5,000 * 20/100 = 1,000.
- Interest on interest, or compound interest or cumulative interest, describes how the interest for the next period is added to both the principal amount and the interest that has been added. Example: a savings amount in the first time period increases with the interest that is added to the amount. The next time period, interest is given on both the savings amount and the first period's interest. The same phenomenon applies to loans but in reverse, interest is added to the interest as the loan is not repaid.
- Effective interest rate is the actual total cost of a loan. Besides ordinary interest, this can also include fees that the lender charges, such as for example "setup fee" or "notification fee".
- An index can be calculated by taking $100 * (\text{value year } t) / (\text{value base year})$ where the letter t symbolizes year. The base year here gets the value 100 but can have any value whatsoever.
- To calculate percentage change of a sum a between time point 1 and 2, we take $(a_2 - a_1)/a_1$ where a_1 and a_2 are the values that the sum has at two different time points.
- To calculate real prices and wages, we adjust for nominal price changes. This is called deflating, which can be done with a price index. Real price = nominal price * (price index base year / price index). A commonly occurring price index is the consumer price index (CPI).
- One way to calculate the growth rate for something is to calculate the percentage change per time period. A time period can for example be one year. Example: $(x_t - x_{t-1})/x_{t-1}$ where x is the phenomenon we are to calculate the change for and the letter t symbolizes time period and $t-1$ the previous time period. A moving average is calculated on values from time points simultaneously: 2010–2012, 2011–2013 and so on.

3.11 Exercises

1. You deposit \$8,000 in a savings account with an annual interest rate of 3.5%.

(a) Calculate the interest earned after one year.

(b) Calculate the total amount in the account after one year.

(c) If the interest rate changes to 4.2% in the second year, what will be the total amount after two years?

Answer: (a) Interest $= 8{,}000 \times 0.035 = 280$, (b) Total $= 8{,}000 + 280 = 8{,}280$ or $8{,}000 \times 1.035 = 8{,}280$, (c) After year 2: $8{,}280 \times 1.042 = 8{,}627.76$

2. A credit card has a nominal annual interest rate of 18%. Interest is compounded monthly.

(a) What is the monthly interest rate?

(b) If you borrow \$5,000 and don't make any payments for 6 months, how much will you owe?

(c) Calculate the effective annual interest rate when interest is compounded monthly.

Answer: (a) Monthly rate $= 0.18/12 = 0.015 = 1.5\%$, (b) Amount owed $= 5{,}000 \times 1.015^6 = 5{,}467.45$, (c) Effective annual rate $= (1.015)^{12} - 1 = 1.1956 - 1 = 0.1956 = 19.56\%$

3. You take a loan of \$15,000 with a nominal interest rate of 8% per year. The bank charges a setup fee of \$450 and a monthly billing fee of \$25 for 24 months.

(a) Calculate the total amount of fees (excluding the nominal interest).

(b) Calculate the total cost of the loan including all fees and interest over 24 months.

(c) What is the effective interest rate for this loan?

Answer: (a) Total fees $= 450+(25\times24) = 450+600 = 1{,}050$, (b) Interest over 24 months $= 15{,}000\times0.08\times 2 = 2{,}400$; Total cost $= 15{,}000+2{,}400+1{,}050 = 18{,}450$, (c) Effective interest $= (18{,}450-15{,}000)/15{,}000 = 3{,}450/15{,}000 = 0.23 = 23\%$

4. The following data shows quarterly sales (in thousands) for a company: Q1 2023: 450, Q2 2023: 520, Q3 2023: 480, Q4 2023: 610, Q1 2024: 490, Q2 2024: 585.

(a) Create an index with Q1 2023 as the base period (=100).

(b) Calculate the percentage change in sales from Q3 2023 to Q4 2023.

(c) What is the average quarterly growth rate for the entire period?

Answer: (a) Q1 2023: 100, Q2 2023: 115.6, Q3 2023: 106.7, Q4 2023: 135.6, Q1 2024: 108.9, Q2 2024: 130.0, (b) $(610 - 480)/480 = 130/480 = 0.271 = 27.1\%$, c) Average growth $= (585/450)^{1/5} - 1 = 1.3^{0.2} - 1 = 0.0539 = 5.39\%$ per quarter

5. The population of a city was 250,000 in 2010, 275,000 in 2015, and 308,000 in 2020.

(a) a) Create a population index with 2010 as the base year.
(b) b) Calculate the percentage growth from 2010 to 2020.
(c) c) What was the average annual growth rate between 2015 and 2020?

Answer: (a) 2010: 100, 2015: 110, 2020: 123.2, (b) $(308,000 - 250,000)/250,000 = 58,000/250,000 = 0.232 = 23.2\%$, c) Annual rate $= (308,000/275,000)^{1/5} - 1 = 1.12^{0.2} - 1 = 0.0229 = 2.29\%$

6. Income data for two groups (in dollars): Group A had median income of $35,000 in 2015 and$ 42,500 in 2023. Group B had median income of $65,000 in 2015 and$ 72,000 in 2023.

(a) a) Create income indices for both groups with 2015 as the base year.
(b) b) Which group experienced higher percentage growth?
(c) c) Calculate the ratio of Group B's income to Group A's income for both years. Has inequality increased or decreased?

Answer: (a) Group A: 2015: 100, 2023: 121.4; Group B: 2015: 100, 2023: 110.8, (b) Group A: 21.4% growth, Group B: 10.8% growth; Group A experienced higher growth, c) 2015:
$)65,000/$

$(35,000 = 1.86$; 2023:
$)72,000/$
$(42,500 = 1.69$; Inequality has decreased

7. A laptop cost $2,400 in 2018. The CPI was 251.1 in 2018 and 304.7 in 2024.

(a) Calculate the real price of the 2018 laptop in 2024 dollars.

(b) If a similar laptop costs$ 1,800 in 2024, has the real price increased or decreased?

(c) By what percentage has the real price changed?

Answer: (a) Real price in 2024 dollars = 2,400 $\times$ (304.7/251.1) = 2,400 $\times$ 1.213 = 2,911.20, (b) Real price has decreased (nominal 1,800 $<$ real equivalent 2,911.20), (c) Percentage change = (1,800 $-$ 2,911.20)/2,911.20 = $-0.382 = -38.2\%$

8. GDP per capita for a country was: 2019: $45,200, 2020:$ 43,800, 2021: $44,900, 2022:$ 46,100, 2023: $47,500.

(a) Calculate the annual growth rate for each year.

(b) What was the average annual growth rate for the period 2020-2023?

(c) Calculate a 3-year moving average for the growth rates (for years where it's possible).

Answer: (a) 2020: -3.1% , 2021: 2.5% , 2022: 2.7 %, 2023: 3.0%, (b) Average 2020-2023 = $(47,500/43,800)^{1/3} - 1] = [1.084^{0.333} - 1] = 0.027 = 2.7\%$, c) 2021: 0.7%, 2022: 2.73%

9. Child mortality data shows: Country X had 185 deaths per 1,000 children under 5 in 1990 and 28 per 1,000 in 2020. Country Y had 45 per 1,000 in 1990 and 8 per 1,000 in 2020.

(a) Calculate the change in percentage points for each country.

(b) Calculate the percentage reduction in child mortality for each country.

(c) Which country achieved a greater relative improvement?

Answer: (a) Country X: -15.7 percentage points, Country Y: -3.7 percentage points, (b) Country X: $(185 - 28)/185 = 84.9\%$ reduction, Country Y: $(45 - 8)/45 = 82.2\%$ reduction, c) Country X achieved greater relative improvement (84.9% vs 82.2%)

10. An investment of $20,000 earns 2.5% interest for the first 3 years, then 3.8% for the next 2 years (compound interest applies).

(a) Calculate the value after 3 years.

(b) Calculate the final value after 5 years.

(c) What is the average annual growth rate over the entire 5-year period?

Answer: (a) After 3 years:
$)20,000 \times 1.025^3 =$
$(21,550.63$, (b) After 5 years:
$)21,550.63 \times 1.038^2 =$
$(23,223.84$, c) Average annual rate = (
$)23,223.84/$
$(20,000)^{1/5} - 1 = 1.1612^{0.2} - 1 = 0.0303 = 3.03\%$

Chapter 4

Functions and graphs

This chapter introduces mathematical functions and graphs. Functions are central to a large amount of mathematical work and can within social science be used both to discuss both theory and empirical data. In later chapters we use mathematical functions in many examples. A graph is some kind of illustration of quantitative values, which can be used to illustrate both theory and data. Sometimes the words diagram, plot or figure are used with a similar meaning.

4.1 Mathematical functions

A mathematical function takes an input value and connects this to an output value. Example of a mathematical function: Take each input value x and multiply by 2. We call the result y:

$$y = 2x \tag{4.1}$$

When $x = 1$ then $y = 2$. When $x = 52$ then $y = 104$, and so on. The letters y and x are called variables here. Variable y is a function of x, which we see because y stands to the left of the equals sign and everything else, including variable x, stands to the right of the equals sign. Variable x is the input value and variable y is the output value. This can also be described as the variable y being explained by the variable x. Variable x is the explanatory, or independent, variable. Variable y is the explained, or dependent, variable.

Which input values that x can take we may also determine and describe. This is called defining for which values of x equation (4.1) applies. It can for example be any real number or all numbers within an interval. If x is any number between 1 and 100, the minimum value for y is:

$$y = 2 * 1 = 2 \tag{4.2}$$

and the largest value for y becomes $y = 2 * 100 = 200$. In equation (4.1) the number 2 indicates how much y changes when the variable x increases by 1. The number 2 represents a constant that is always 2, regardless of whether $x = 1$ or $x = 14$ and so on. For $x = 5$ we get $y = 2 * 5 = 10$. For $x = 6$ we get $y = 2 * 6 = 12$. When x increases from 5 to 6, y increases by 2. In the equation, y and x are called variables as mentioned. The number 2 in the equation is called a coefficient or parameter.

A function can also be described in the following way:

$$y = f(x) \tag{4.3}$$

This expression can be read as y is a function of x, where we now call the mathematical function f. The letter f doesn't mean anything particular other than that a function exists in some form. The letter f is arbitrarily chosen. A mathematical function that we define ourselves we may also name whatever we want. The function f can be both simple and advanced. The only thing we can discern from $f(x)$ is that it uses the variable x as input value and results in variable y. Suppose we have a function $g(x)$ where we only know that the function uses the variable x. As long as we have not said anything more about the functions' form, f and g can be the same function. Functions don't have to do anything spectacular. A function can for example be: $y = h(x) = x$, where each value for y is the same value as x.

Functions can be designed in many (infinite) different ways and can contain any number of variables and coefficients (parameters, constants). Here is an example where a variable y is a function of the two variables x and z. We call the function v:

$$y = v(x, z) = 2x + 3z \tag{4.4}$$

Table 4.1: Some values for $y = v(x, z) = 2x + 3z$

x	z	$y = 2x + 3z$
1	1	$2*1+3*1=5$
2	2	$2*2+3*2=10$
3	3	15
4	4	20
5	5	25

Table 4.1 shows some values for x and z, as well as some values for y calculated with function $v(x, z)$. In chapter 2 we went through absolute values, a number's distance from 0 on the number line. Absolute values can also be used when we work with functions. Consider the following function $p(x)$ as an example:

$$p(x) = 5 + 2x \tag{4.5}$$

The absolute value of $p(x = 5)$ becomes then:

$$|p(5)| = |5 + 2 * 5| = 15 \tag{4.6}$$

The absolute value of $p(x = -5)$ becomes:

$$|p(-5)| = |5 - 2 * (-5)| = 5 \tag{4.7}$$

4.2 Function as theory

Mathematical functions can among other things be used to describe how we think about relationships between different phenomena in reality, or to describe theories about causal relationships. A causal relationship can for example be formulated: Variations in phenomenon A cause variations in phenomenon B. Or: An increase in X will lead to a decrease in Y. These types of formulations do not necessarily mean that one phenomenon explains everything that happens in the other phenomenon. Variations in one phenomenon may only explain a small part of the variation in the other phenomenon.

Suppose we have a theory that higher income for some reason causes people to live longer. One way to describe causal relationships is with arrows: $X \rightarrow Y$, where X symbolizes annual income calculated in thousands of USD and Y stands for life expectancy in years. The arrow goes from income X to life expectancy Y, which symbolizes how we believe the relationship works: changes in X cause changes in Y. With the help of mathematics one may specify the theory, for example in the form of the following equation:

$$Y = a * X \tag{4.8}$$

The letters Y and X are just designations. If we prefer we may write out the words, life expectancy $=$ $a *$ income, but equations often become hard to read if we write out long words. The coefficient a is an unknown value, which symbolizes how much Y changes when X changes by 1. In this case a is a constant, a number that doesn't change even if X changes. We don't know what value a has, only that this number is the same for all values of X. This means that regardless of whether X changes from 3 to 5, or if X changes from 543 to 600, a will have the same value.

Purely mathematically the coefficient a can be both positive, negative or 0. If $a = 0$ it means that Y doesn't change when X changes, which in that case means that there is no relationship between the variables. As equation (4.8) is formulated we mean that a person with 0 USD in income will have a life expectancy of 0 years:

$$Y = a * 0 = 0 \tag{4.9}$$

That doesn't sound like a credible theory. Let us assume that there is a lower limit, so that even if one doesn't earn any money one can manage to get by somehow. We call this theoretically expected life expectancy b and add it to our equation:

$$Y = aX + b \tag{4.10}$$

A person with zero income can now, according to our theory, be expected to live for b years:

$$Y = a * 0 + b = b \tag{4.11}$$

What is a variable and coefficient should always be clear from the context, but can sometimes be hard to keep apart. There is no rule for which letters one uses for what.

Let us assume that we have our theory about income and life expectancy formulated as in equation (4.8) and that we have arrived at $a = 0.01$. That is, for every 1,000 USD a person's annual income increases, the person lives 0.01 years longer, regardless of at what income or age this occurs.

Regardless of whether any such relationship exists or not, this theoretical model is a clumsy description of reality. But with mathematics we can more easily discuss the details. There might be parts of the theory that still seem to match reality while other parts need to be rewritten. With mathematics it also becomes easier to be more exact in which parts of the theory seem to match better or worse with reality.

4.3 Coordinate system and graphs

Functions of the type $y = f(x)$ can be drawn in graphs. This is useful for many different reasons. Often we draw graphs with two axes, one for height and one for width. Another word for this is dimensions: a vertical dimension for the height, and a horizontal dimension for the width. By tradition the horizontal axis, the width, is called the x-axis while the vertical axis, the height, is called the y-axis. These names of the axes in the graph have no mathematical meaning however. It is just a common way to describe them. We may call the axes in the graph whatever we want.

A graph with x- and y-axis illustrates combined values of the variables x and y. The values for the variable x are placed in the graph with regard to the horizontal x-axis while the values for y are placed with regard to the vertical y-axis. Consider the following function as an example:

$$y = 10x \tag{4.12}$$

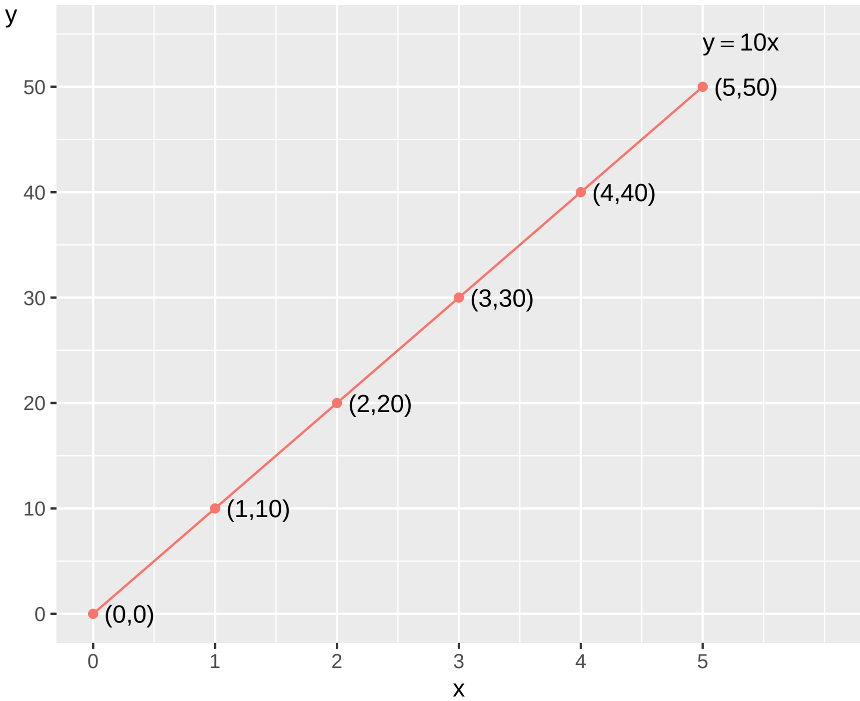


Figure 4.1: Illustration of the function $y = 10x$

Now we are going to draw this function in a graph for the values of x that are between 0 and 5. We use the x-values to calculate the values for y. The combined values of x and y are shown in figure 4.1 . The points along the black line show the combined values of x(the horizontal axis) and y(the vertical axis) at the whole numbers 0 to 5. Two combined values of the variables x and y can be written with parentheses (x, y), which represents a point in the graph. The point $(x, y) = (2, 20)$ has the values $x = 2$ and $y = 20$. The point where $x = y = 0$ can be written $(x, y) = (0, 0)$ and is called the zero point or origin.

The black line is leaning to the upper right corner of the graph. The coefficient 10 in the equation indicates how much variable y changes each time variable x increases by 1. When x increase from 1 to 2, y increase from 10 to 20. In the graph we may see this by following the black line and comparing the values on the x- and y-axis.

A graph may have lines from multiple functions for example the following functions $k(x)$ and $u(x)$:

$$
\begin{aligned}
y &= k(x) = 5 - 0.1x \\
y &= u(x) = x - 3
\end{aligned}
\tag{4.13}
$$

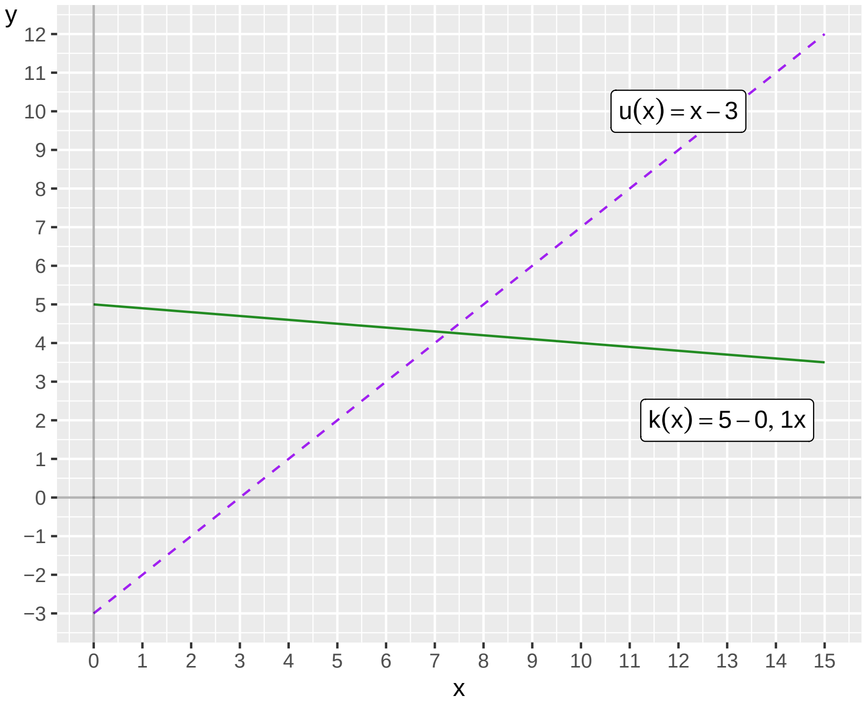


Figure 4.2: Two functions in the same graph

Figure 4.2 illustrates the functions $k(x)$ and $u(x)$. The line for the function $f(x)$ in figure 4.1 slopes upward to the right in the graph. This is called that the line have a positive slope. The function $u(x)$ also has a positive slope. If the line instead slopes downward to the right in the picture, it has a negative slope, higher values for x mean smaller values for y. The function $k(x)$ results in a negative slope.

If the line is completely horizontal it has no slope. Figure 4.3 shows three lines as examples of negative, positive and no slope.

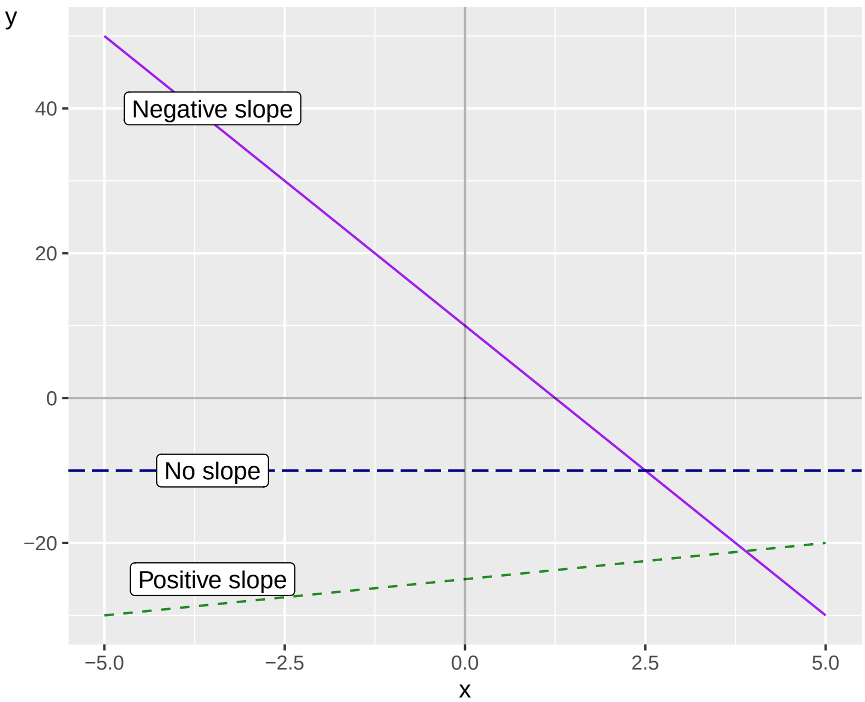


Figure 4.3: Positive slope, negative slope and no slope

The functions we work with in this book are defined so that for each value of x that a function uses, this x will only connect to at most one value of $y = f(x)$. In a two-dimensional graph this means that the line for the function in the graph only coincides with one point on the x-axis and never goes over or under itself, since the current x-value would then be connected to several values on the y-axis. This also means that when you see a line in a graph with unique y-values for each x, this line can always be drawn with a mathematical function, regardless of how the line looks. A function that connects a unique value of x to several values of y is called a multi-valued function but is not described here.

4.4 Equation of a straight line

A particularly important type of equation is the equation of a straight line. The most basic form of this equation can be written in the following way:

$$y = a + bx \tag{4.14}$$

where y and x are variables and the letters a and b are constant coefficients or parameters. Every straight line in a graph can be drawn with an equation of this form. The letter b indicates how much y changes when x increases by 1. The letter a indicates the value for y when $x = 0$, which we can check by substituting $x = 0$ into the equation:

$$y = a + bx = a + b * 0 = a \tag{4.15}$$

The first coefficient, in this case called a, is also called the y-intercept, or just intercept. This is since a is the y-value where the line intersects the y-axis when $x = 0$. Coefficient b is called the slope coefficient, since it indicates the slope of the line (the change in y when x increases by 1). In section 4.2 we described a theory about the relationship between income and life expectancy with the function $Y = aX$. This is also a form of the equation of a straight line but the coefficient that indicates the y-intercept is in this case equal to 0.

If we use equation (4.14) and set $a = 0$ we get:

$$y = a + bx = 0 + bx = bx \tag{4.16}$$

So we have one function $y = bx$ and from earlier the function $Y = aX$. The two functions can describe exactly the same relationship between the two variables. That we use different letters in the two functions has no mathematical significance. The important thing when we use letters is that we explain what the letters symbolize.

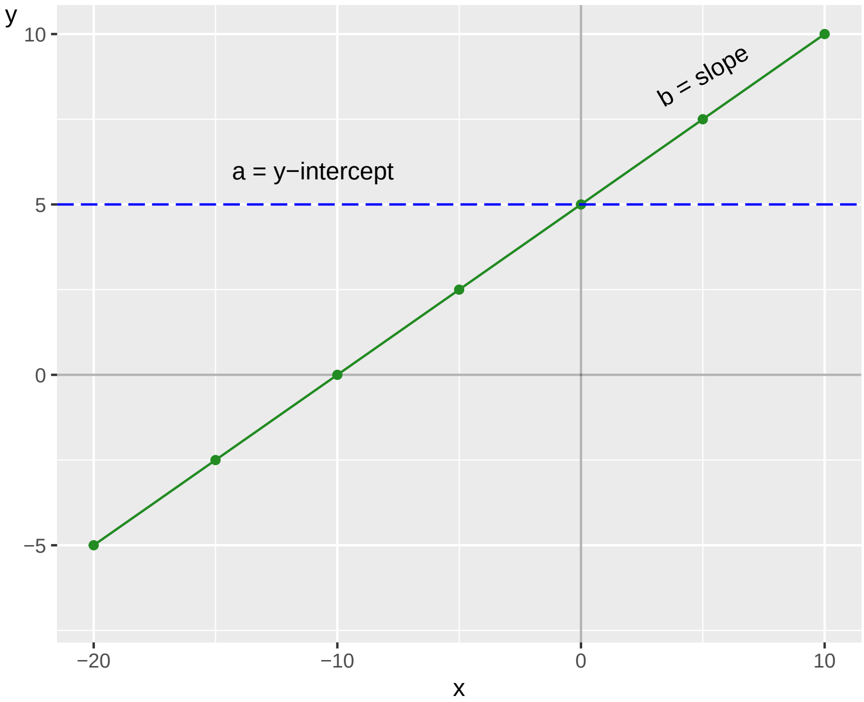


Figure 4.4: A straight line

In figure 4.4 an example of a line in a graph is shown. We don't know yet what equation the line has. But since the line is straight we know that it can be represented by an equation with the corresponding form as $y = a + bx$. Now we are going to calculate the values for the two constant coefficients a and b. The coefficients a and b can be calculated from the graph. We find coefficient a by comparing where the line intersects the y-axis when $x = 0$, which is at $y = 5$. We substitute the values $x = 0$ and $y = 5$ into our equation $y = a + bx$ to get a:

$$\begin{aligned} y &= a + bx \\ 5 &= a + b * 0 \\ 5 &= a \end{aligned} \tag{4.17}$$

Now we are going to calculate the slope coefficient b. Two points to the left of the y-axis we see that the line intersects the x-axis at the point where $x = -10$. Here follow two methods for calculating b, the slope of the line.

Method 1: The point on the line where $x = -10$ is $y = 0$. We know that $a = 5$. We substitute these values into our equation to get b:

$$\begin{aligned} y &= a + bx \\ 0 &= 5 + b(-10) \\ 10b &= 5 \\ b &= \frac{5}{10} = \frac{1}{2} = 0.5 \end{aligned} \tag{4.18}$$

Method 2: Coefficient b is the slope of the line, how much y changes when x increases by 1. We have noted two points for the line in the graph: point 1 where $(x_1, y_1) = (-10, 0)$ and point 2 where $(x_2, y_2) = (0, 5)$. From these the slope of the line can be calculated by comparing how much y changes between y_1 and y_2, and how much x changes between x_1 and x_2:

$$b = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 0}{0 - (-10)} = \frac{5}{10} = \frac{1}{2} = 0.5 \tag{4.19}$$

Sometimes the equation of a straight line is written with other letters, for example:

$$\gamma = \alpha_1 + \alpha_2 \Gamma \tag{4.20}$$

The important thing is what the letters symbolize and that this is clearly evident. In this case the Greek letters γ and Γ(small and capital gamma) are two different variables, while α_1 and α_2(alpha 1 and 2) are the equation's constant coefficients. That we add the subscript numbers 1 and 2 to the constants α_1 and α_2 is only a way of writing. The subscript numbers indicate that α_1 is a different coefficient than α_2. Just as above, α_1 indicates the line's y-intercept (the value for γ when $\Gamma = 0$) while coefficient α_2 indicates the line's slope.

In figure 4.5 some plots are shown where we have drawn different examples of functions of the form $y = a + bx$. Go through carefully and make sure that you are certain that you understand how the line in each plot relates to its equation. Check that the values for a and b match what you see in the plot. Here we have the following four different functions:

$$\begin{aligned} &\text{(a) } y = 0.5x \\ &\text{(b) } y = -0.5x \\ &\text{(c) } y = 15 - 2x \\ &\text{(d) } y = 5 + \frac{1}{4}x \end{aligned} \tag{4.21}$$

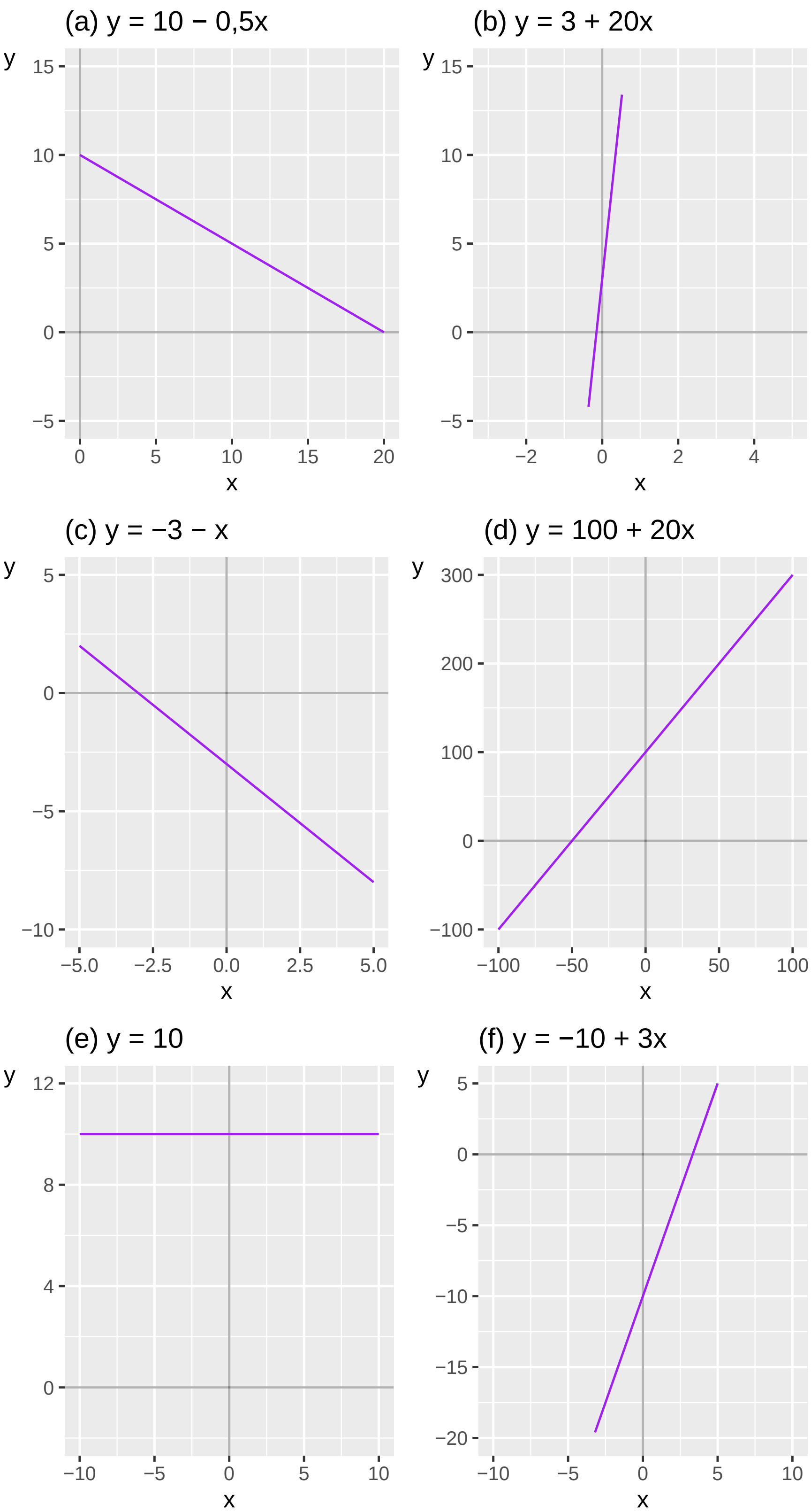


Figure 4.5: Examples of the linear function

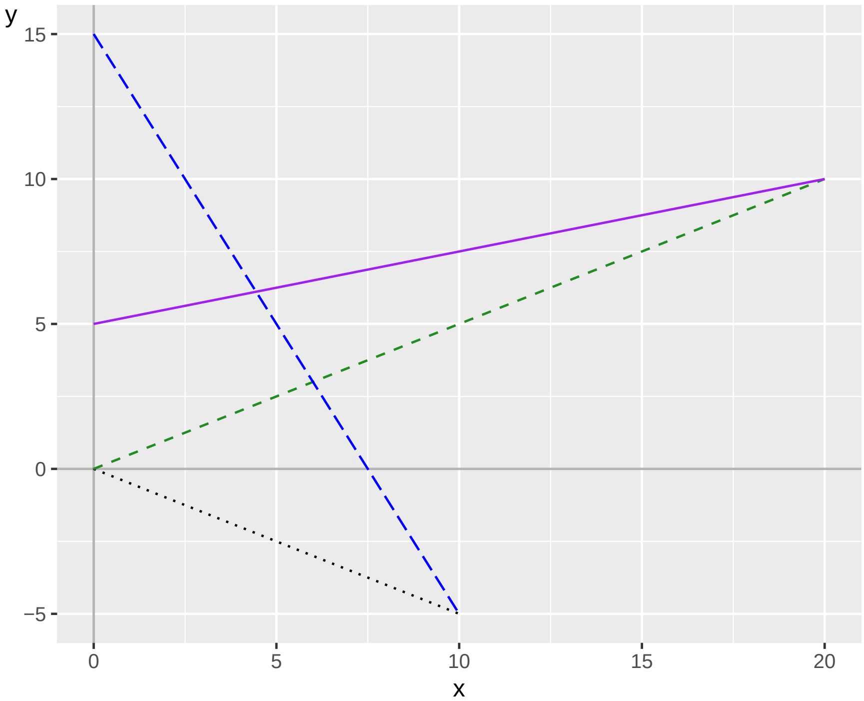


Figure 4.6: Four lines in the same graph

where each function describes a relation between the two variables y and x. In figure 4.6 a line is drawn for each respective function. Let us go through how we can see which line belongs to which function. In the graph two lines pass through the origin: $(x, y) = (0, 0)$. Both functions (a) and (b) have the value $y = 0$ when $x = 0$, which we may see since these functions lack a constant for the y-intercept, a term that is not multiplied by x. One of these lines has a positive slope, y increases as x increases, while the other line has a negative slope, y decreases as x increases. Function (a) draws a line with positive slope because x is multiplied by the positive value 0.5. The line for function (b) has negative slope since x is multiplied by -0.5.

The two remaining lines also have a positive and negative slope respectively, which is sufficient to determine which of functions (c) and (d) belongs to which line. Function (c) has the negative constant -2 in front of x, while function (d) has the positive constant $\frac{1}{4}$ in front of x.

Function (c) is the line in the graph that meets the y-axis at $y = 15$, which corresponds to the constant 15 in the function. Function (d) meets the y-axis at $y = 5$, which corresponds to the constant 5 in the function.

The equation of a straight line is central to a large amount of analytical work and one of the red threads through most of this book. Therefore, make sure to take a little extra time to ensure that you are following along this far. Feel free to go back and review the text. If something feels unclear or strange, it can sometimes help to read other descriptions of the same phenomena online or in other books.

4.5 Solve for x

In many situations we want to solve for a variable from an equation, for example variable x. This means that we rewrite our equation so that x stands on one side of the equality sign and everything else in the equation stands on the other side. Here is an example where we solve for x by dividing both sides by 4:

$$\begin{aligned} 4x &= 5 \\ x &= \frac{5}{4} \\ x &= 1.25 \end{aligned} \tag{4.22}$$

Since we have found a solution for x, we are done with the task. Consider another example where we shall solve for x:

$$3 + 2x = 10 \tag{4.23}$$

To solve for x we change both the left and right side of the equality sign simultaneously. We start by subtracting 3 from both sides:

$$\begin{aligned} 3 + 2x &= 10 \\ 3 + 2x - 3 &= 10 - 3 \\ 2x &= 7 \end{aligned} \tag{4.24}$$

Then we divide both sides by 2:

$$\begin{aligned} 2x &= 7 \\ \frac{\cancel{2}x}{\cancel{2}} &= \frac{7}{2} \\ x &= \frac{7}{2} \end{aligned} \tag{4.25}$$

To solve for a variable often means that we create a new equation. Let us take the following expression as an example:

$$a + bx = c \tag{4.26}$$

The letter x is our variable that we shall solve for. The letters a, b and c now symbolize different constants, any real numbers. Even though we don't know what numbers the letters represent, we can still solve for x. The result becomes a new equation:

$$\begin{aligned} a + bx &= c \\ bx &= c - a \\ x &= \frac{c - a}{b} \end{aligned} \tag{4.27}$$

Numbers and different letters can also be combined in the same function. For example:

$$z = h\left(p\right) = 3 + 5a + 8p + ab \tag{4.28}$$

where z and p are variables and a and b are coefficients. We solve for p:

$$\begin{aligned} z &= 3 + 5a + 8p + ab \\ &= 3 + a\left(5 + b\right) + 8p \\ -8p &= 3 + a\left(5 + b\right) - z \\ p &= \frac{-3 - a\left(5 + b\right) + z}{8} \end{aligned} \tag{4.29}$$

In the next example we shall solve for x:

$$\begin{aligned} x\left(5+58\right) &= 3+4 \\ x &= \frac{7}{63} \\ x &= \frac{1}{9} \end{aligned} \tag{4.30}$$

We now have a combination of numbers, letters that represents numbers and the variable x. We solve for x:

$$\begin{aligned} 9x\left(a+b\right) &= a+b \\ x &= \frac{1}{9}\left(\frac{a+b}{a+b}\right) \\ x &= \frac{1}{9} \end{aligned} \tag{4.31}$$

Often we encounter situations when the variable we shall solve for stands on both sides of the equality sign. Here is an example with an equation where we shall solve for variable z. We move all parts where z appears to the left side:

$$\begin{aligned} z\left(\frac{25}{80+20}\right) &= 4\left(z-2\right) \\ z\frac{25}{100} &= 4z-8 \\ z\frac{1}{4}-z4 &= -8 \end{aligned}$$

Now we have all terms where z appears on the left side of the equality sign. But the equation can be simplified further to solve for an expression for z. We therefore move everything else in the equation to the right side:

$$\begin{aligned} z\frac{1}{4}-z4 &= -8 \\ z\left(\frac{1}{4}-4\right) &= -8 \\ z &= \frac{-8}{-15/4} \\ z &= \frac{32}{15} \end{aligned} \tag{4.32}$$

Here follows another example where we shall now solve for x. Initially we now have several terms where the variable x appears. In the equation, besides x, there are also numbers and the letter y. What y represents is not crucial for this example. It can be another variable or only a constant written as a letter. To solve for x we move all terms which do not include any x to the right hand side:

$$\begin{aligned} -17x-14y-\frac{60x}{30} &= -5y \\ -17x-\frac{60x}{30} &= -5y+14y \\ -17x-2x &= 9y \\ 19x &= -9y \\ x &= -\frac{9}{19}y \end{aligned} \tag{4.33}$$

Here is an example where we have a larger equation with the two variables x and y, with several parameters, constants. Let us simplify this a bit:

$$\begin{aligned} 17x + 14y - \frac{60x}{30} &= 5y \\ 17x - 2x &= 5y - 14y \\ 15x &= -9y \end{aligned} \tag{4.34}$$

This looks a bit more comfortable. But we may also continue to simplify from here, for example to solve for x:

$$x = -\frac{9}{15}y \tag{4.35}$$

We can also solve for y:

$$\begin{aligned} x &= -\frac{9}{15}y \\ -9y &= 15x \\ 9y &= -15x \\ y &= -\frac{15}{9}x \end{aligned} \tag{4.36}$$

It is difficult to do much more with this equation unless we get more information about x and y. Even if we rewrite it, we need to know something more to be able to find a unique solution for the variables.

4.6 Linear equations

The equation of a straight line, $y = a + bx$, can be used to draw a straight line in a graph with two axes, one horizontal and one vertical. More generally, a function $f(x)$ is defined as linear if the following two conditions are fulfilled:

$$\begin{aligned} \text{Condition 1: } & f(x_1 + x_2) = f(x_1) + f(x_2) \text{ for all } x_1 \text{ and } x_2 \\ \text{Condtion 2: } & f(ax) = af(x) \text{ for all } a \end{aligned} \tag{4.37}$$

where x_1 and x_2 are two different input values on the explanatory variable x, which are used in function f. Let us illustrate with the function $y = f(x) = 0.25x$. We try the values $x_1 = 5$ and $x_2 = 10$. Condition 1:

$$\begin{aligned} f(5 + 10) &= f(5) + f(10) \\ 0.25 * 15 &= 0.25 * 5 + 0.25 * 10 \\ 3.75 &= 3.75 \end{aligned} \tag{4.38}$$

We try condition 2 with the values $a = 5$ and $x = 3$:

$$\begin{aligned} f(5 * 3) &= 5f(3) \\ 0.25 * 15 &= 5 * 0.25 * 3 \\ 3.75 &= 3.75 \end{aligned} \tag{4.39}$$

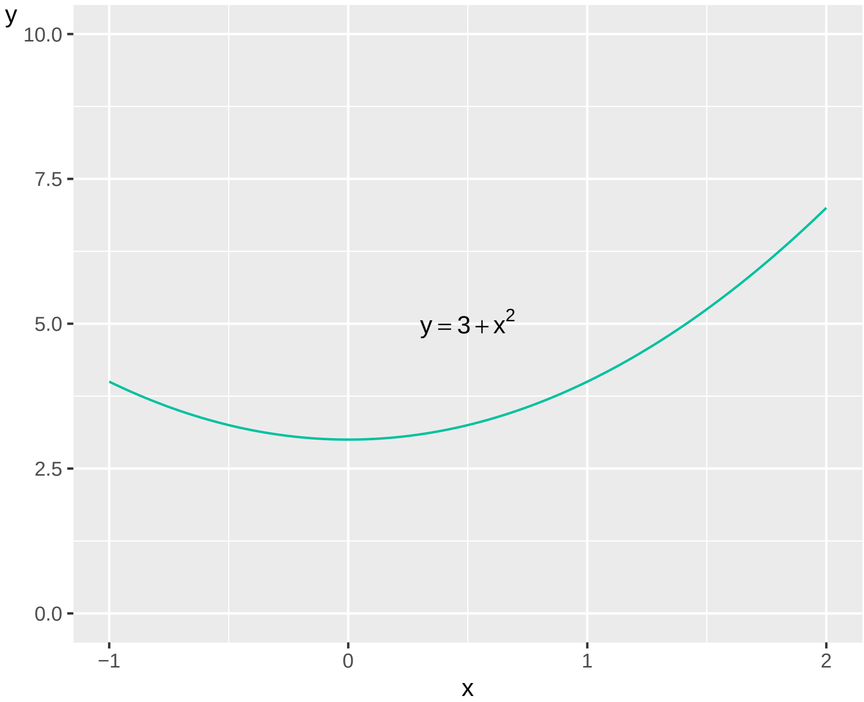


Figure 4.7: The line for the function $y = 3 + x^2$

This means that the function $y = f(x) = 0.25x$ is linear. Let us now check if the following function is linear: $y = 3 + x^2$. For condition 1 we try $x_1 = 5$ and $x_2 = 10$:

$$\begin{aligned} f(5+10) &= f(5) + f(10) \\ 3 + (15)^2 &= (3+5^2) + (3+10^2) \\ 3 + 225 &\neq 28 + 103 \end{aligned} \tag{4.40}$$

The function does not fulfill the condition and is therefore not linear. Due to the term x^2 we also see that the equation is not drawn as a straight line in a graph. Figure 4.7 illustrates this, where the line is curved.

4.7 A bit of set theory

A collection of numbers is called a set or number set. A set can consist of many numbers or a few. Here is an example of a number set $A = \{2, 6, 13\}$ which is called A and contains the three numbers 2, 6 and 13. In a number set the order of the numbers has no significance, unless this is explicitly stated. For example the set $A = \{2, 6, 13\}$ can also be described as $A = \{13, 2, 6\}$ or as $A = \{6, 2, 13\}$. In the introduction to the chapter we described the integers, which are usually symbolized by the letter $\mathbb{Z}$:

$$\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\} \tag{4.41}$$

The integers, $\mathbb{Z}$, are an example of a number set. There are infinitely many integers and all are included in the set $\mathbb{Z}$. We also work with the rational numbers, which are symbolized by $\mathbb{Q}$, where all numbers are included that can be expressed as a fraction with two integers where the denominator is not 0. For example:

$$\frac{1}{13}, \frac{78}{84}, \frac{3}{2}, \frac{140}{3} \tag{4.42}$$

We also work with the irrational numbers, which are the numbers that cannot be written as a fraction of two integers. Here we have for example $2^{1/2}$(the square root of 2), π (the number pi) and e(Euler's number, approximately 2.718, discussed more below). The real numbers are denoted $\mathbb{R}$. One way to describe a set that includes all irrational numbers is $\mathbb{R}\backslash\mathbb{Q}$, where the backslash is called set difference. The expression $\mathbb{R}\backslash\mathbb{Q}$ can be read as "the real numbers, excluding the rational numbers". Another useful symbol is $\in$, which means "in" or "belongs to", such as for example:

$$3 \in \mathbb{R} \tag{4.43}$$

This means that the number 3 belongs to the real numbers. Above we described the following:

$$\mathbb{Z} \in \mathbb{Q} \in \mathbb{R}, \tag{4.44}$$

This means that the integers $\mathbb{Z}$ are included in the rational numbers $\mathbb{Q}$, which are included in the real numbers $\mathbb{R}$. Since all integers are included in the rational numbers $\mathbb{Z}$ is a subset of $\mathbb{Q}$. The condition for something to be described as a subset is that every value is included in the larger set. And since all $\mathbb{Q}$ are included in $\mathbb{R}$, $\mathbb{Q}$ is also a subset of $\mathbb{R}$. Subset can be written with the symbol $\subseteq$:

$$\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \tag{4.45}$$

Conversely $\mathbb{R}$ is a superset of $\mathbb{Q}$, which is a superset of $\mathbb{Z}$. Superset can be written with the symbol $\supseteq$:

$$\mathbb{R} \supseteq \mathbb{Q} \supseteq \mathbb{Z} \tag{4.46}$$

A common way to define number sets is with parentheses and brackets: $(-1, 1)$ or $[-1, 1]$. Both these expressions describe an interval between -1 and 1. Parentheses, (), mean that the number set includes all values between the interval but not the endpoints, the values -1 and 1. This is called the number set being open. Brackets, [], mean that the set also includes the endpoint values -1 and 1, which is called the number set being closed. Say for example that we have a letter, a, which represents a value between 1 and 5. This can be written:

$$a \in [1, 5] \tag{4.47}$$

This means that a is one of all possible values between 1 and 5. The letter a can symbolize the number 1 or the number 2.43. To define an interval where the boundary values 1 and 5 respectively are not included, we write instead:

$$a \in (1, 5) \tag{4.48}$$

In this case the number a can be any of all values between 1 and 5, but not exactly 1 or 5. Parentheses mean that all values are included in the intervals except the boundary values 1 and 5. For example 1.000001 and 4.999999967 and all other decimal values that come infinitely close to 1 and 5 are included, but not the integers themselves.

Now we have described how the small letter a can assume a value between 1 and 5. If we instead want to define a set of numbers that contains all values in an interval, for example 1 to and including 5, we may write:

$$A = \{x | x \in [1, 5]\} \tag{4.49}$$

The letter A now symbolizes the set of values. Each value in set A is represented by the letter x, where x is in the interval $[1, 5]$. The complement set to a set A is the set with all values that are not included in A. The complement set to A is written $A^{\complement}$:

$$A^{\complement} = \{x | x \in (-\infty, 1) \text{ och } x \in (5, \infty)\} \tag{4.50}$$

The values in $A^{\complement}$ consist of all real numbers from negative infinity up to 1 as well as from 5 up to positive infinity. Positive and negative infinity is not real values and therefore not included in $\mathbb{R}$ either. If a set contains only the number 0 this can be written as $\{0\}$ and is called the null set.

If a set is empty and contains no value at all, not even the number 0, this is called the empty set, which can be written as $\emptyset$ or $\{\}$. The empty set is by definition a subset of every other set. For example, if we have the set $(14, 17)$, the values 14 to 17, then $\emptyset \in (14, 17)$ as well. Suppose we have the two sets A and B respectively and shall create a new set C that consists of all unique values in A and B together. This can be written with the symbol $\cup$, which is called union:

$$C = A \cup B \tag{4.51}$$

Note that $\cup$ is not the same thing as addition. Say for example that we have the following two sets:

$$A = \{1, 2, 3, 4, 5\}, \quad B = \{3, 4, 5, 6, 7\} \tag{4.52}$$

If we take the union of these two sets we get:

$$C = A \cup B = \{1, 2, 3, 4, 5, 6, 7\} \tag{4.53}$$

If we instead want the new set C to contain the values in A that are not found in B we can remove these with the operator $\setminus$(set difference), which we also mentioned above:

$$C = A \setminus B = \{1, 2\} \tag{4.54}$$

Say now that the new set C should contain the values from A and B that they have in common. This we can achieve with the operator $\cap$, which is called the intersection:

$$C = A \cap B = \{3, 4, 5\} \tag{4.55}$$

If all decimal values are included in an interval this is called a continuous interval. If we have a continuous interval, all decimals between 0 and 1, we by definition always have an infinite set of values, since the number of decimals can always increase. If we instead have a set that only contains the integers 1 to 3 the set is not continuous since the values between the integers are not included in the set. Instead this is a discrete set, where the word discrete means that the number of values included in the set can be counted. Let us now take the following function:

$$y = f(x) = \frac{1}{x}, \forall x \in (-\infty, 0) \cup (0, \infty) \tag{4.56}$$

The symbol $\forall$ means "for all". The expression $\forall x \in$ therefore means "for all x in". The entire expression means that the function $f(x)$ is defined for all x except the following values: negative infinity, 0 and positive infinity. The reason why $1/x$ is not defined for $x = 0$ here is that $f(x)$ then goes to infinity, not a real value. The reason for this is, somewhat simplified, the following. If we multiply $1/x$ with x we get $x * (1/x) = (x/x) = 1$. But there is no real number that can be multiplied with $1/0$ to get 1, since any number multiplied with 0 equals 0.

A function that is defined over a collection of real values and does not have any interruptions or "sudden" changes between two adjacent values is called continuous. Function f in equation (4.56) is continuous over the intervals $(-\infty, 0)$ and $(0, \infty)$. Since f is not defined for $x = 0$ the function is not continuous for intervals that include the value 0. Let us now take the following example:

$$y = h(x) = \begin{cases} 1 + x, & \forall x \in (-\infty, 5) \\ 20 - x & \forall x \in [5, \infty) \end{cases} \tag{4.57}$$

Function h is defined as $h(x) = 1 + x$ for all real x(negative infinity $-\infty$ is not a real number) up to just below $x = 5$. From $x = 5$ up to just below positive infinity (∞) the function is defined as $h(x) = 20 - x$. For $x = 4.99$ and $x = 5$ we get:

$$h(x = 4.99) = 1 + 4.99 = 5.99 \tag{4.58}$$
$$h(x = 5) = 20 - 5 = 15$$

Function $h(x)$ is not continuous, which is illustrated in figure 4.8 . The small ring symbolizes that $h(x)$ is defined infinitely close to $x = 5$. At the value $x = 5$ the function's value is instead the filled point $(x, y) = (5, 15)$. The slope of the lines differ in the manner described in equation (4.57) .

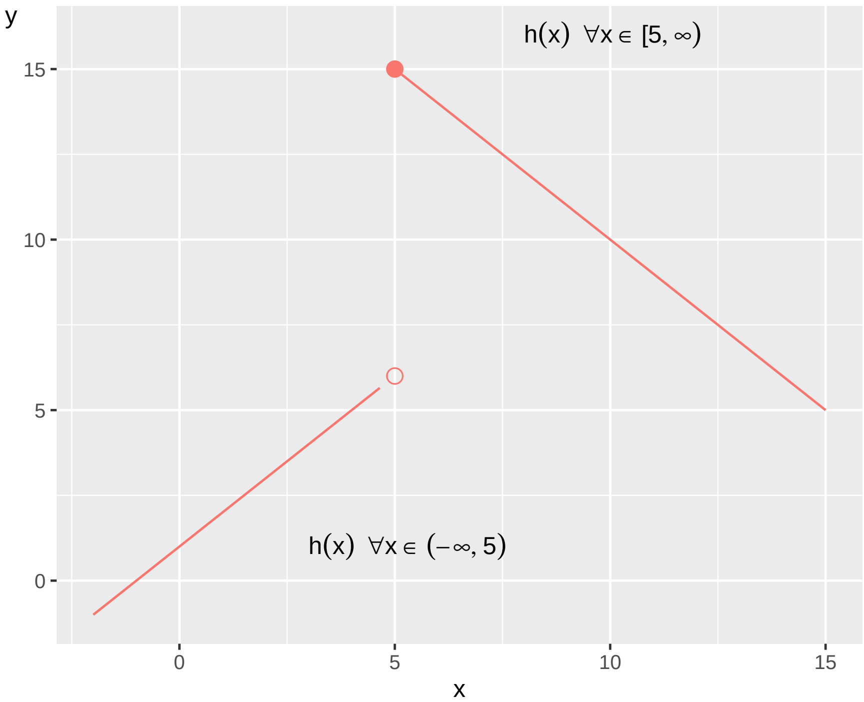


Figure 4.8: A non-continuous function

4.8 Domain for a function

Let us again take our theory about the relationship between income and life expectancy that we used in section 4.2 : $y = f(x) = ax$, where y is life expectancy, x is income and coefficient a indicates how much the life expectancy (y) changes when the income (x) increases by one unit. In this case x and y cannot assume just any values. Neither x nor y can in this example be negative, below 0. We therefore say that for $x < 0$ the function $f(x)$ is not defined. Another way to describe this is:

$$y = f(x) = ax, \ \forall x \geq 0 \tag{4.59}$$

The entire expression can be read as y is a function of x for all x $(\forall x)$ that is equal to or above zero (≥ 0). The values that a function can result in, the values that y can take, are called the codomain. The input values of x, for which our function is defined, are called the domain.

Let us take another example:

$$y = f(x) = \frac{1}{x}, \ \forall x \neq 0 \tag{4.60}$$

This new function f is defined for all values of x that are not equal to 0. Another way to write the same thing is:

$$y = f(x) = \frac{1}{x},\ x \in \mathbb{R}^{+} \cup \mathbb{R}^{-} \tag{4.61}$$

where the symbols $x \in \mathbb{R}^{+} \cup \mathbb{R}^{-}$ mean that x is in ($\in$) the number set the positive real numbers ($\mathbb{R}^{+}$) as well as (union, $\cup$) the negative real numbers ($\mathbb{R}^{-}$), that is the positive and negative real numbers but not 0. This can also be written:

$$y = f(x) = \frac{1}{x},\ x \in \mathbb{R} \setminus 0 \tag{4.62}$$

where the symbols $x \in \mathbb{R}\setminus 0$ mean that x is included in ($\in$) the number set the real numbers ($\mathbb{R}$), excluding ($\setminus$) the number zero 0.

Let us now look at the following function that has the two input variables x and z:

$$y = h(x, z) = z + 3x \tag{4.63}$$

Variable z is defined for the positive integers, the natural numbers excluding zero, $z \in \mathbb{N}\setminus 0$, and variable x is defined for the positive real numbers, $x \in \mathbb{R}^{+}$. Suppose we instead have a variable $y = g(x, z)$ as a function of x and z. Both input variables are defined over all real numbers, which can be written as $(x, z) \in \mathbb{R}^{2}$.

The exponent 2 on $\mathbb{R}$ refers to the fact that we have two number lines: one number line for variable x and one for z. This in turn creates a two-dimensional field, a plane surface. The variable $y = g(x, z)$ is also defined for all real numbers, $y \in \mathbb{R}$. Function g thus has the domain $\mathbb{R}^{2}$ and the codomain $\mathbb{R}$:

$$f : \mathbb{R}^{2} \to \mathbb{R} \tag{4.64}$$

4.9 Combining functions with operations

Given that two or more functions have overlapping domains (the values a function is defined for) one can use addition, subtraction, multiplication and division on these. Suppose we have the two functions $f(x)$ and $g(x)$, defined over the same interval for x. Then the following applies:

$$\begin{aligned}
(f + g)(x) &= f(x) + g(x) \\
(f - g)(x) &= f(x) - g(x) \\
(f * g)(x) &= f(x) * g(x) \\
(f/g)(x) &= \frac{f(x)}{g(x)}
\end{aligned} \tag{4.65}$$

Let us apply this to the following example:

$$\begin{aligned}
f(x) &= 3 + 2x \\
g(x) &= x^{3} - 4
\end{aligned} \tag{4.66}$$

If we add the functions we get:

$$(f + g)(x) = f(x) + g(x) = 3 + 2x + x^{3} - 4 \tag{4.67}$$

Subtraction with functions:

$$(f-g)(x) = f(x) - g(x) = 3 + 2x - (x^3 - 4) = -x^3 + 2x + 7 \tag{4.68}$$

Multiplication of the two functions:

$$\begin{aligned}(f * g)(x) &= f(x) * g(x) \\ &= (3 + 2x)(x^3 - 4) \\ &= 3x^3 - 12 + 2x^4 - 8x \\ &= 2x^4 + 3x^3 - 8x - 12\end{aligned} \tag{4.69}$$

Division:

$$(f/g)(x) = \frac{f(x)}{g(x)} = \frac{3 + 2x}{x^3 - 4} \tag{4.70}$$

Another way to combine functions is to create a composite function, which means that we put one function inside another function. Say for example that we have two separate functions, f and g, where $y = g(x)$, where the variable y is a function of the variable x:

$$y = g(x) = 2x \tag{4.71}$$

We also have z as a function of the variable y through function f:

$$z = f(y) = y^2 \tag{4.72}$$

In the parentheses of function f we may now write $g(x)$ instead of y, so that we get $f(g(x))$. Another way to write this is $f \circ g(x)$:

$$z = f(g(x)) = f \circ g(x) \tag{4.73}$$

The definition of $g(x)$ we get from equation (4.71) :

$$z = f(g(x)) = f(2x) = (2x)^2 \tag{4.74}$$

This can be read as g of f of x. Or expressed differently, the variable z is a function of y, which is a function of x. This is then calculated as usual, so if $x = 3$ we get:

$$f(g(3)) = (2 * 3)^2 = 36 \tag{4.75}$$

Say now that we have the functions t and h:

$$\begin{aligned}t(x) &= \frac{3x}{2} \\ h(x) &= x + 5\end{aligned} \tag{4.76}$$

We are now looking for $t(h(x))$. The fact that we now call the variable in both functions by the same letter x has no mathematical significance. We can still work with the functions as a composite function:

$$t\left(h\left(x\right)\right)=t\left(x+5\right)=\frac{3\left(x+5\right)}{2} \tag{4.77}$$

For $x=3$ we get:

$$t\left(h\left(3\right)\right)=t\left(3+5\right)=\frac{3\left(3+5\right)}{2}=12 \tag{4.78}$$

4.10 Chapter summary

- A mathematical function $y=f\left(x\right)$ takes the input value, input, x and connects this to an output value, output, y. Written this way we do not know what function f looks like. Mathematical functions can among other things be used to describe social science theories.
- The values that the output value y can take are called the codomain. The values that the input value x can take, for which the function is defined, are called the domain. A function that is defined over all real values over an interval without exceptions or sudden interruptions is called continuous.
- Functions can be illustrated in graphs with two axes: height and width. The height, the vertical axis, is often called the y-axis. The width, the horizontal axis, is often called the x-axis.
- The equation of a straight line can be written $y=a+bx$, where a is called the y-intercept and is a constant that indicates the value for y when $x=0$. b is the slope coefficient and indicates how much y changes when x increases by 1. $b>0$: the function's line in the graph has positive slope. $b<0$: negative slope. $b=0$: the line is horizontal.
- A function is linear if the following two conditions are fulfilled: (1) $f\left(x_1+x_2\right)=f\left(x_1\right)+f\left(x_2\right)$, for the two different x-values x_1 and x_2; and (2) $f\left(ax\right)=af\left(x\right)$ for all a.
- Elasticity = how much one variable changes in percentage terms with respect to percentage change in another variable.
- A collection of numbers is called a set, for example $A=\{1,2,3\}$. The expression $117\in\mathbb{R}$ means that the number 117 belongs to the real numbers. The complement set to set A is written $A^{\complement}$ and includes all values that are not included in A. The null set contains only the value 0 $\{0\}$. The empty set contains no value and is written $\emptyset$. For the two sets A and B, where $A\neq B$, $A\cup B=C$ where all values that are included in A and B respectively are included. $A\backslash B$ is the set difference, the values in A that are not found in B. $A\cap B=C$ includes the values that are common to sets A and B.
- A closed interval is written with brackets, for example $[0,1]$, which includes all values between 0 and 1 including the boundary values 0 and 1. An open interval can be written $(0,1)$, which includes all values between 0 and 1 without including the boundary values 0 and 1.
- Given that the two functions $f\left(x\right)$ and $g\left(x\right)$ have overlapping domains, these two functions can be combined with addition: $\left(f+g\right)\left(x\right)=f\left(x\right)+g\left(x\right)$; subtraction: $\left(f-g\right)\left(x\right)=f\left(x\right)-g\left(x\right)$; multiplication: $\left(f*g\right)\left(x\right)=f\left(x\right)*g\left(x\right)$; division: $\left(f/g\right)\left(x\right)=f\left(x\right)/g\left(x\right)$.
- A composite function consists of a function whose input value is the result of another function. For example $f\left(g\left(x\right)\right)$, where function f uses the result of function g, which has the input variable x.

4.11 Exercises

1. Given the function $f(x)=3x-5$, calculate: (a) $f(2)$, (b) $f(-1)$, (c) $f(0)$

Answer: (a) $f(2)=3(2)-5=6-5=1$, (b) $f(-1)=3(-1)-5=-3-5=-8$, (c) $f(0)=3(0)-5=0-5=-5$

2. Consider the function $g(x,y)=2x+4y-1$. Find:

(a) (g(1,2))

(b) (g(3,-1))

(c) (g(0,0))

Answer: (a) $g(1,2) = 2(1) + 4(2) - 1 = 2 + 8 - 1 = 9$, (b) $g(3,-1) = 2(3) + 4(-1) - 1 = 6 - 4 - 1 = 1$, (c) $g(0,0) = 2(0) + 4(0) - 1 = 0 + 0 - 1 = -1$

3. For the linear function $y = 4 + 2x$:

(a) What is the y-intercept?

(b) What is the slope?

(c) Find the value of (y) when (x=3) .

(d) Find the value of (x) when (y=10)

Answer: (a) y-intercept = 4, (b) slope = 2, (c) $y = 4 + 2(3) = 4 + 6 = 10$, (d) $10 = 4 + 2x$, so $6 = 2x$, therefore $x = 3$

4. Two points on a straight line are $(2,5)$ and $(6,13)$.

(a) Calculate the slope of the line.

(b) Find the equation of the line in the form (y=a+bx) .

(c) What is the y-intercept?

Answer: (a) slope $= \frac{13-5}{6-2} = \frac{8}{4} = 2$, (b) Using point $(2,5)$: $5 = a + 2(2)$, so $a = 1$. Equation: $y = 1 + 2x$, (c) y-intercept = 1

5. Solve for x in the following equations:

(a) (5x+3=18)

(b) (2x-7=3x+2)

(c) (frac{x}{4}+2=5)

Answer: (a) $5x = 15$, so $x = 3$, (b) $-7 - 2 = 3x - 2x$, so $-9 = x$, (c) $\frac{x}{4} = 3$, so $x = 12$

6. Determine whether the following functions are linear using the two conditions for linearity:

(a) (f(x)=3x)

(b) (g(x)=x^{2}+1)

(c) (h(x)=-2x)

Answer: (a) Linear: $f(x_1 + x_2) = 3(x_1 + x_2) = 3x_1 + 3x_2 = f(x_1) + f(x_2)$ and $f(ax) = 3(ax) = a(3x) = af(x)$, (b) Not linear: $g(x_1 + x_2) = (x_1 + x_2)^2 + 1 \neq x_1^2 + 1 + x_2^2 + 1$, (c) Linear: satisfies both conditions

7. Given sets $A = \{1,3,5,7\}$ and $B = \{3,4,5,6\}$, find:

(a) (Acup B)

(b) (Acap B)

(c) (Asetminus B)

(d) (Bsetminus A)

Answer: (a) $A \cup B = \{1,3,4,5,6,7\}$, (b) $A \cap B = \{3,5\}$, (c) $A \setminus B = \{1,7\}$, (d) $B \setminus A = \{4,6\}$

8. For the function $f(x) = \frac{1}{x-2}$:

(a) What is the domain of the function?

(b) Why is the function not defined at (x=2) ?

(c) Is the function continuous over the interval ([3,5]) ?

Answer: (a) Domain: $x \in \mathbb{R} \setminus \{2\}$ or $(-\infty, 2) \cup (2, \infty)$, (b) Division by zero is undefined when $x = 2$, (c) Yes, the function is continuous over $[3,5]$ since 2 is not in this interval

9. Given $f(x) = x + 3$ and $g(x) = 2x - 1$, find:

(a) ((f+g)(x))

(b) ((f-g)(x))

(c) ((fcdot g)(x))

(d) ((f/g)(x))

Answer: (a) $(f+g)(x) = x+3+2x-1 = 3x+2$, (b) $(f-g)(x) = x+3-(2x-1) = -x+4$, (c) $(f \cdot g)(x) = (x+3)(2x-1) = 2x^2+5x-3$, (d) $(f/g)(x) = \frac{x+3}{2x-1}$

10. Given $f(x) = x^2$ and $g(x) = x-4$, find:

(a) (f(g(x)))

(b) (g(f(x)))

(c) (f(g(2)))

(d) (g(f(3)))

Answer: (a) $f(g(x)) = f(x-4) = (x-4)^2 = x^2-8x+16$, (b) $g(f(x)) = g(x^2) = x^2-4$, (c) $f(g(2)) = f(-2) = 4$, (d) $g(f(3)) = g(9) = 5$

11. Draw the graph of $y = -2x+6$ and identify:

(a) The point where the line crosses the y-axis

(b) The point where the line crosses the x-axis

(c) Whether the line has positive or negative slope

Answer: (a) y-axis crossing: $(0,6)$, (b) x-axis crossing: set $y = 0 : 0 = -2x+6$, so $x = 3$, point is $(3,0)$, (c) Negative slope (coefficient of x is -2)

12. A theory suggests that happiness (H) is related to income (I) by the equation $H = 0.002I + 50$.

(a) What is the baseline happiness when income is zero?

(b) How much does happiness increase when income increases by $1000 ?

(c) What income level corresponds to a happiness score of 60?

Answer: (a) Baseline happiness = 50, (b) Happiness increases by $0.002 \times 1000 = 2$ units, (c) $60 = 0.002I + 50$, so $10 = 0.002I$, therefore $I = 5000$

Chapter 5

Logarithms

This chapter introduces logarithms, which among other things can be used to facilitate calculations with powers. Logarithms can be used to solve equations with unknown exponents, compare percentage changes, linearize exponential relationships, and analyze exponential growth such as economic development over long periods.

5.1 Base, exponent, logarithm

In chapter 2 we described powers with a base and an exponent: power = base$^{\text{exponent}}$. Logarithms are another way to write power expressions. The logarithm of the number a is the exponent x that the base b must be raised to to get the value a:

$$\begin{aligned} a &= b^x \\ \log_b a &= x \end{aligned} \tag{5.1}$$

If we for example take the logarithm with base 3 of the number 9, this is equal to the value that the base 3 must be raised to for us to get 9:

$$\log_3(9) = 2 \text{ since } 3^2 = 9 \tag{5.2}$$

Logarithmic calculation is a designation for the computational exercise we do when we search for a specific exponent in a power expression. Logarithmic calculation is a form of mathematical function called $\log()$: the logarithm function. Sometimes logarithms are written with parentheses $\log(a)$ and sometimes without parentheses $\log a$, but the parentheses do not have any specific mathematical function. Sometimes one can see the notation ${}^b\log a = x$, with the base b written to the left of the word log.

Logarithms are not defined for negative values, such as $\log -9$, since there is no base b which can be raised to any exponent and get -9, or any other negative value.

Now we have the following expression:

$$\log_b a = x \tag{5.3}$$

and want to solve for a. We do this by putting $\log_b a$ as an exponent to the base b. At the same time we use the right-hand side also as an exponent to base b:

$$\begin{aligned} \log_b a &= x \\ b^{\log_b a} &= b^x \\ a &= b^x \end{aligned} \tag{5.4}$$

Here is the same example but instead of b we use base 3:

$$\begin{aligned} \log_3 9 &= 2 \\ 3^{\log_3 9} &= 3^2 \\ 9 &= 3^2 \end{aligned} \tag{5.5}$$

5.2 The common logarithm

A common base for logarithmic calculation is 10. It is so common that when one sees an equation where the function $\log()$ appears but no base is defined, it is usually logarithmic calculation with base 10 that is meant, which is called the common logarithm. Just as above, the common logarithm means that $\log_{10} a$ is equal to the value that 10 must be raised to for us to get the value a, where a is some number. Here are some examples of logarithms with base 10:

$$\begin{gathered} \log_{10} 10 = 1 \text{ since } 10^1 = 10 \\ \log_{10} 100 = 2 \text{ since } 10^2 = 100 \\ \log_{10} 1,000 = 3 \text{ since } 10^3 = 1,000 \end{gathered} \tag{5.6}$$

The common logarithm means that for the number a we have:

$$a = 10^{\log_{10}(a)} \tag{5.7}$$

Logarithmic calculation is useful for several things. Say for example that we want to solve for x from the following function

$$1,000,000 = 100^x \tag{5.8}$$

For this we take the common logarithm of the numbers on both sides:

$$\begin{aligned} \log_{10}(1,000,000) &= x \log_{10}(100) \\ 6 &= x * 2 \\ x &= 3 \end{aligned} \tag{5.9}$$

Let us go through the equation above step by step. In the left side we get the logarithm of 1,000,000 with base 10, which is equal to the number that 10 must be raised to for us to get 1,000,000. The answer is 6. In the right side, logarithmic calculation means that one can move down the exponent x from 100^x and write this in front of $\log(100)$. Generally it applies for exponents that $(a^b)^c = a^{b*c}$ and we know that the common logarithm of 100 is 2, since $10^2 = 100$. Therefore we get:

$$100^3 = (10^2)^3 = 10^{2*3} = 10^6 \tag{5.10}$$

If we take the logarithm of 10^6 we may write:

$$\log_{10}(10^6) = \log_{10}(10^2)^3 = 3 * \log_{10}(10^2) = 3 * 2 = 6 \tag{5.11}$$

Since $\log 100 = 2$ we can also write:

$$100 = 10^{\log 100} = 10^2 \tag{5.12}$$

If the logarithm function appears in an equation we must remember that it is the result of the function that should be used in the calculation and nothing else. Consider the following equation: $\log 1,000 + 42$. To calculate this we take the result of the first term, $\log 1,000 = 3$, and add 42:

$$\log_{10} 1,000 + 42 = 3 + 42 = 45 \tag{5.13}$$

Same thing if we have several logarithmic expressions in the same equation. Here is an example with two terms:

$$\log_{10} 100 + \log_{10} 10 = 2 + 1 = 3 \tag{5.14}$$

5.3 Properties of logarithms

Logarithms have the following properties:

$$\begin{aligned} \textbf{Product rule:}\ & \log(a * b) = \log(a) + \log(b) \\ \textbf{Quotient rule:}\ & \log\left(\frac{a}{b}\right) = \log(a) - \log(b) \\ \textbf{Power rule:}\ & \log(x^a) = a\log(x) \end{aligned} \tag{5.15}$$

To see what these three rules mean we can take some simple examples. Let us start with multiplication, the product rule:

$$\begin{aligned} \log 1,000 &= \log(100 * 10) \\ &= \log 100 + \log 10 \\ &= 2 + 1 \\ &= 3 \end{aligned} \tag{5.16}$$

Note the difference between the following two expressions:

$$\begin{aligned} \log a * \log b &\neq \log a + \log b \\ \log ab &= \log a + \log b \end{aligned} \tag{5.17}$$

To see the meaning of this difference we substitute a and b with numbers and use the common logarithm:

$$\begin{aligned} \log_{10} 5 + \log_{10} 2 &= \log_{10}(5 * 2) = \log_{10} 10 = 1 \\ \log_{10} 5 * \log_{10} 2 &\approx 0.699 * 0.3 \approx 0.21 \end{aligned} \tag{5.18}$$

Let us now illustrate the Quotient rule, using the following example:

$$\log 100 = \log \frac{1,000}{10} = \log 1,000 - \log 10 = 3 - 1 = 2 \tag{5.19}$$

When we work with logarithms and powers one can use the Power rule:

$$\log 10,000 = \log 100^2 = 2\log 100 = 2 * 2 = 4 \tag{5.20}$$

The Power rule states that $\log x^a = a \log x$. This can also be written:

$$a \log x = \log\left(x^a\right) = \log\left(10^{\log x}\right)^a \tag{5.21}$$

5.4 Natural logarithm

We have previously mentioned the constant π, which is approximately 3.14. Another commonly occurring constant is Euler's number e, which is approximately 2.718. Just as the number 10, the number e(approximately 2.718) is a common base for logarithmic calculation, which is called the natural logarithm and is often written as $\ln(x)$:

$$\ln x = \text{the number we need to raise e to obtain the value x} \tag{5.22}$$

Another way to write the same thing:

$$\begin{aligned} y &= \ln(x) \\ \text{where } e^y &= x \end{aligned} \tag{5.23}$$

Just as for other logarithms it applies that if $x = 1$ then $\ln(x) = \ln(1) = 0$, since all numbers raised to 0 equal 1. Above we went through how $100 = 10^{\log_{10} 100} = 10^2 = 100$. When we have $\ln()$ we instead have the base e. This means that e raised to the natural logarithm of the number a equals just a. Since the natural logarithm uses the number e as base, the following applies:

$$a = e^{\ln(a)} \tag{5.24}$$

In equation (5.5) we went through how by using the logarithm on a power expression one can isolate the exponent. If we have a power expression where Euler's number e is the base we may do the same with the natural logarithm:

$$\begin{aligned} e^y &= x \\ \ln e^y &= \ln x \\ y &= \ln x \end{aligned} \tag{5.25}$$

The antilogarithm of the natural logarithm has a specific function called the natural exponential function and is denoted $\exp()$:

$$\begin{aligned} y &= \ln x \\ \exp(y) &= \exp(\ln x) \\ e^y &= e^{\ln x} \\ e^y &= x \end{aligned} \tag{5.26}$$

We shall now calculate:

$$e^{\ln 8 + \ln 2} + 4 \tag{5.27}$$

We start by summing the exponent in the first term, where we use the logarithm law for addition, $\ln a + \ln b = \ln ab$:

$$e^{\ln 8 + \ln 2} + 4 = e^{\ln 16} + 4 \tag{5.28}$$

The first term is now e raised to the natural logarithm of 16. Since $\ln(16)$ by definition is the number that e must be raised to in order to get 16, we have that $e^{\ln 16} = 16$. Based on this, we now have the following:

$$e^{\ln 16} + 4 = 16 + 4 = 20 \tag{5.29}$$

5.5 Change of base for logarithms

We have so far worked with logarithmic calculation where the logarithmic terms in the expressions have the same base. Sometimes we may however encounter expressions where we have logarithms with different bases in the same expression. Then it can be useful to know how we can change the base of a logarithmic expression. This also facilitates the understanding of logarithmic calculation generally. Suppose we have the expression:

$$x = a^{\log_a x} \tag{5.30}$$

Note how the right side only says that a raised to $\log_a x$, with a as base, is the same thing as x. We now take the natural logarithm on both sides. Since we shall describe base change of logarithms we now write the natural logarithm as $\log_e()$ instead of $\ln()$. We now get:

$$\begin{aligned} \log_e x &= \log_e \left(a^{\log_a x}\right) \\ \log_e x &= \left(\log_a x\right)\left(\log_e a\right) \\ \log_a x &= \frac{\log_e x}{\log_e a} \end{aligned} \tag{5.31}$$

The last line describes the calculation rule for how to change the base from the logarithm with base a to any other base. Here we used the natural logarithm, where we wrote $\log_e$ instead of ln. But the same principle applies for any other base we would want to change to. As an example:

$$\log_a x = \frac{\log_e x}{\log_e a} = \frac{\ln x}{\ln a} = \frac{\log_5 x}{\log_5 a} \tag{5.32}$$

In the first term we have the logarithm of x with base a. In the second and third terms we have the natural logarithm in numerator and denominator. In the fourth term we have the logarithm with base 5 in numerator and denominator. This calculation rule can also be illustrated with an example. We have:

$$100 = 10^{\log_{10} 100} \tag{5.33}$$

Now we shall rewrite this expression to a form where we instead use the natural logarithm. We may then take the natural logarithm of both sides:

$$\ln(100) = \ln\left(10^{\log_{10} 100}\right) = \left(\log_{10} 100\right)\ln(10) \tag{5.34}$$

We move the parts with natural logarithm to one side so that we get a new definition for $\log_{10} 100$:

$$\log_{10} 100 = \frac{\ln 100}{\ln 10} \tag{5.35}$$

We know that $\log_{10} 100 = 2$ since $10^2 = 100$. That the right side with natural logarithm is also equal to 2 we can see by taking:

$$\frac{\ln 100}{\ln 10} = \frac{\ln 10^2}{\ln 10} = \frac{2 \ln 10}{\ln 10} = 2 \tag{5.36}$$

5.6 Using natural log to calculate percentage change

The natural logarithm has several useful properties. One of these is that for small changes, the difference between the natural logarithm of two values is approximately the same as the percentage difference. That is, for small a the following applies:

$$\ln(1 + a) \approx a \tag{5.37}$$

which means that if we raise Eulers number e to the small value a we get $1 + a$. For instance, the value 102 is 2 percent higher than 100: $102/100 - 1 = 0.02 = 2\%$. If we take the natural log of $102/100 = 1.02$ we get approximately 2 percent:

$$\begin{aligned} \ln\left(\frac{102}{100}\right) &= \ln(102) - \ln(100) \\ &\approx 4.62497 - 4.60517 \\ &\approx 0.0198 \\ &\approx 2\% \end{aligned} \tag{5.38}$$

This may seem like a complicated way to measure percent. But this is very useful in a lot of statistical analysis, which we will introduce in part III.

Notice that it is only natural logarithms that work as an approximation for small percent change. With another log base we get other results. For instance, $\log_{10}(105/100) \approx 0.02$, which is of course wrong.

For small differences the natural logarithm gives an acceptable approximation of percentage differences. But when the difference between the values increases, this no longer applies. For example:

$$\ln\left(\frac{30}{15}\right) \approx 0.69 = 69\% \tag{5.39}$$

which is wrong, since a change from 15 to 30 is 100%. If we know in advance how much the percentage change is, it is possible to create a correct measure by changing the base of the logarithms. The details are not included here, but explained pedagogically by Huntington-Klein (2021) .

5.7 Logarithms for linearity

A lot of analysis is based on more or less linear relationships, such as if we increase (decrease) A this will result in an increase (decrease) of B. There is a lot of important associations that are not linear. But some of them we may want to recalculate so that they become more linear. One such example is exponential growth, where something is increasing at an increasing rate. If we take logarithms.

To see how logarithmic transformation can create a more linear relationship between different values we look at the four graphs in figure 5.1 . The figure shows four different combinations of values for the variables $x, x^2, \ln x$ and $\ln x^2$, which is also described in table 5.1 . Of the four graphs (a), (b), (c) and (d), it is only in graph (d) we have a linear relationship between the variables. In graph (d) the values for $\ln(x^2)$ are shown on the y-axis and values for $\ln x$ on the x-axis. The line in the graph is straight, which means that it by definition must be able to be drawn with a variant of the straight line equation:

$$y = a + bx \tag{5.40}$$
$$\ln(x^2) = a + b(\ln x)$$

In graph (a) we have plotted combined values of x and x^2. The line grows exponentially and is therefore not linear. In graph (b) combined values of x and $\ln(x^2)$ are shown. The line is curved and the rate of increase decreases with higher values for x, which is not a linear relationship. Graph (c) shows combined values of x^2 and $\ln(x)$, which also does not result in a linear function.

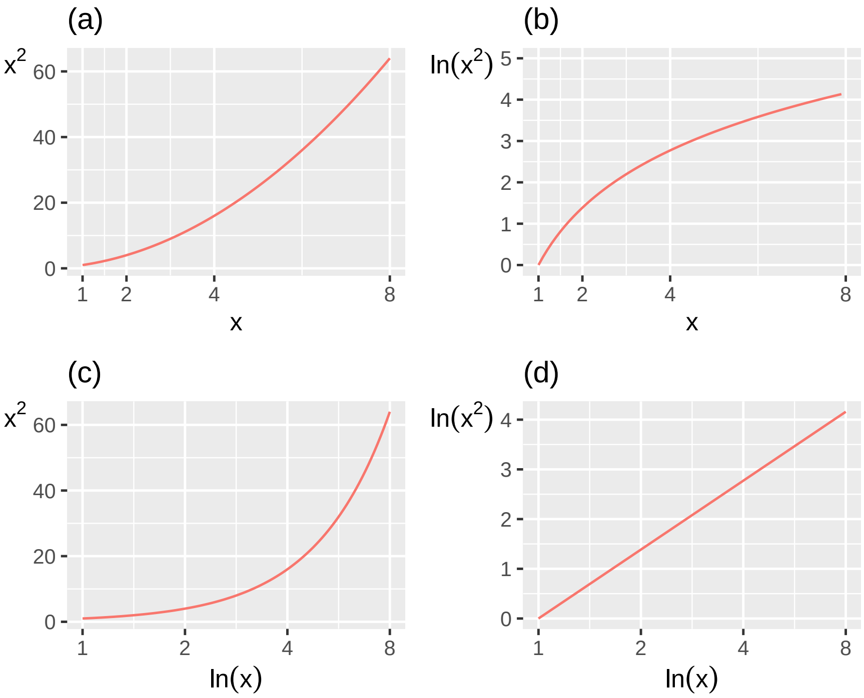


Figure 5.1: Illustration of x, x^2, $\ln x$ and $\ln x^2$

Table 5.1: Rounded values for x, x^2, $\ln x$ and $\ln x^2$

x	$y = x^2$	$\ln x$	$y = \ln x^2$
1	1	0.0	0.0
2	4	0.7	2.1
3	9	1.1	3.3
4	16	1.4	4.2
5	25	1.6	4.8
6	36	1.8	5.4
7	49	1.9	5.8
8	64	2.1	6.2
9	81	2.2	6.6
10	100	2.3	6.9

In section 4.4 we defined the slope of the straight line as:

$$b = \frac{y_2 - y_1}{x_2 - x_1} \tag{5.41}$$

where we compare the distance between two points where the first point has coordinates (x_1, y_1) and the second point has coordinates (x_2, y_2). Now we will use this again to calculate the slope of the function that draws the straight line in graph (d) in figure 5.1 . Let us use the point $(x_1, y_1) = (2, 4)$ and the point $(x_2, y_2) = (3, 9)$:

$$\begin{aligned} b &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{(\ln x_2^2) - (\ln x_1^2)}{(\ln x)_2 - (\ln x)_1} \\ &= \frac{\ln (x_2^2 / x_1^2)}{\ln (x_2 / x_1)} \\ &= \frac{\ln (9/4)}{\ln (3/2)} \\ &= 2 \end{aligned} \tag{5.42}$$

The result 2 is the same thing as the exponent in the term x^2. For the logarithm law about powers in equation (5.16) we saw that $\ln (x^a) = a \ln x$. In this case we have a function of type $y = x^2$. If we take logarithms of both sides of $y = x^2$ we get:

$$\begin{aligned} \ln y &= \ln (x^2) \\ \ln y &= 2 \ln x \end{aligned} \tag{5.43}$$

This is a linear function, a function of the same form as the straight line equation. This might become even easier to see if we substitute $\ln y = z$ and $\ln x = q$. We then get:

$$z = 2q \tag{5.44}$$

where z and q are now our variables and 2 is a constant. This equation can be described as a variant of the straight line equation $y = a + bx$ when $a = 0$.

5.8 Taking logarithms to get x

In section 5.3 we introduced the logarithm law for powers and saw that $\log (a^b) = b \log (a)$. Sometimes we want to use this to solve for x in an equation. Consider the following example:

$$3^x = 21 \tag{5.45}$$

We take logarithms of both sides:

$$\begin{aligned} 3^x &= 21 \\ \log (3^x) &= \log (21) \\ x \log 3 &= \log 21 \\ x &= \frac{\log 21}{\log 3} \approx 2.77 \end{aligned} \tag{5.46}$$

Let us go through this calculation step by step. We start by taking logarithms of both sides of the equation, which allows us to move down x from exponent to multiplier for $\log 3$. The result becomes the same thing regardless of which base we use when we take logarithms, as long as we use the same base on both sides of the equals sign.

Suppose now that we wish to know what makes people happy and satisfied. Within social science the concept utility function is often used in these contexts. The concept "utility" is used here to describe preferences and what a person finds useful depends on their preferences. Say now that we have a theory that a person's utility is a result of two things: work and leisure. On a somewhat abstract level utility = percent of the year that we feel happy, work = working hours per year while leisure = all time that is not used for work. We describe our theory with the following equation:

$$N = A^z F^{1-z} \tag{5.47}$$

where N stands for utility, A for work and F for leisure. The exponent z is a constant that describes what significance the two explanatory variables have for the person's utility. While A and F are included in the equation based on our general theory of how the world works, the value for z can be calculated based on information about A and F(more about this later). We feel satisfied with the theory itself but we want to rewrite the equation to a linear function. This can be done with logarithmic transformation:

$$\ln N = \ln \left(A^z F^{1-z}\right) \tag{5.48}$$

The next step is to use the Product law for logarithms (equation (5.16)):

$$\ln N = \ln \left(A^z\right) + \ln \left(F^{1-z}\right) = z \ln A + (1 - z) \ln F \tag{5.49}$$

Let us substitute some symbols to make it even clearer:

$$n = \beta_1 a + \beta_2 f \tag{5.50}$$

where small letters n, a and f symbolize logarithmic versions of the variables N, A and F, while β_1 and β_2 symbolize our slope coefficients, which were previously z and $(1 - z)$. Why would we want to rewrite our equation? Partly it becomes easier to illustrate the relationship between the variables, partly the linear equation is easier to work with if we shall compare our theory against reality, which we return to in chapter 21

5.9 Log GDP

In this section we shall again work with gross national product, GDP (see section 3.8). Table 5.2 shows nominal GDP for the US for every ten years 1800–2000. The third column contains the natural logarithm of GDP. The natural logarithm is here rounded to one decimal place. Year 1800 GDP was 0.486 billions USD and $\ln(0.486) \approx -0.7$, which means that $e^{-0.7} \approx -0.486$.

In the table we see how an equally large relative change in x corresponds to the same absolute change of the natural logarithm. Every time GDP doubles, regardless of from which level, the change in $\ln(GDP)$ is approximately 0.7. Every time GDP triples, $\ln(GDP)$ increases by approximately 1.1. When GDP increases tenfold, $\ln(GDP)$ increases by approximately 2.3.

Table 5.2: US GDP 1800–2000 and the natural log of GDP

Year	GDP billions USD	ln GDP
1800	0.486	−0.7
1810	0.714	−0.3
1820	0.718	−0.3
1830	1.032	0
1840	1.59	0.5
1850	2.656	1
1860	4.41	1.5
1870	7.899	2.1

Year	GDP billions USD	ln GDP
1880	10.592	2.4
1890	15.607	2.7
1900	21.197	3.1
Note: Data from Measuring Worth		

Year	GDP billions USD	ln GDP
1910	33.746	3.5
1920	89.246	4.5
1930	92.16	4.5
1940	102.899	4.6
1950	299.827	5.7
1960	542.382	6.3
1970	1,073.303	7
1980	2,857.307	8
1990	5,963.144	8.7
2000	10,250.952	9.2
.	.	.

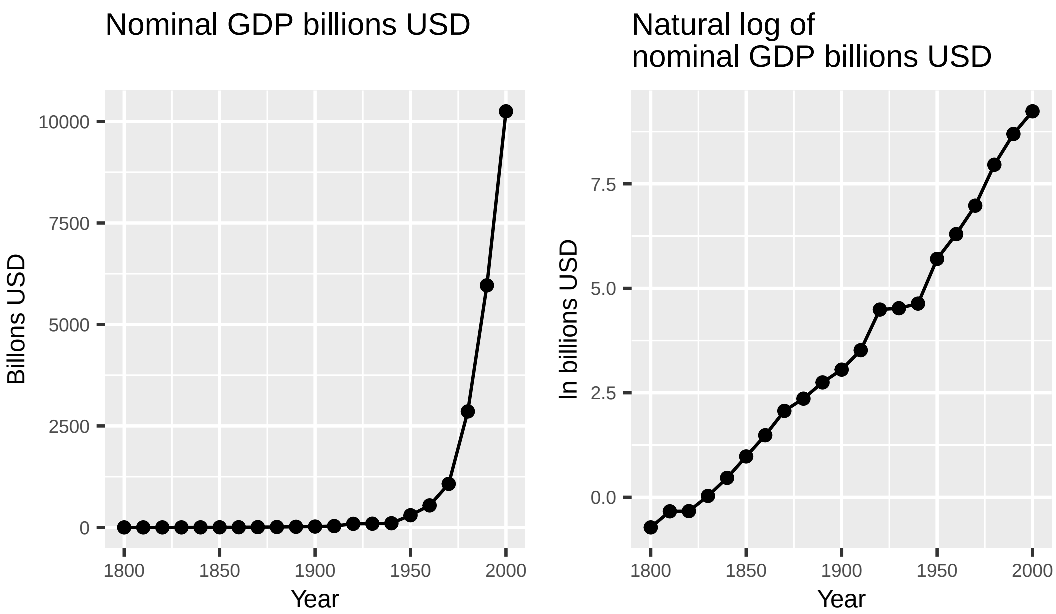


Figure 5.2: US GDP per capita, 1,000 USD PPP. Data from Our World in Data

The same thing is illustrated in figure 5.2 , where the numbers from table 5.2 are shown in two graphs: in the left graph a line for GDP in USA between the years 1800–2000 is shown, in billions of dollars. The line increases exponentially. The greater GDP becomes, the greater is the rate of increase. In the right graph the curve for ln(GDP) is shown. The line now increases at a more even pace and is almost straight. This effect occurs because of the logarithmization of the nominal values, which makes the graph's scale to show relative changes. Note that this does not necessarily mean that ln(GDP) is a better measure than nominal GDP. Which measure you should use depends on what you want to measure, compare or show.

5.10 Using logarithms to discuss theory

There is a well-known covariation between a country's residents' subjectively estimated satisfaction with their lives and the country's average income level, which simply can be described as the covariation between "happiness" and income. This is illustrated in figure 5.3 where we see data for happiness and income in 2017 in all the world's countries. Each dot represents a combined value of happiness and gross national product per inhabitant, GDP per capita, that is average amount of total income per inhabitant.

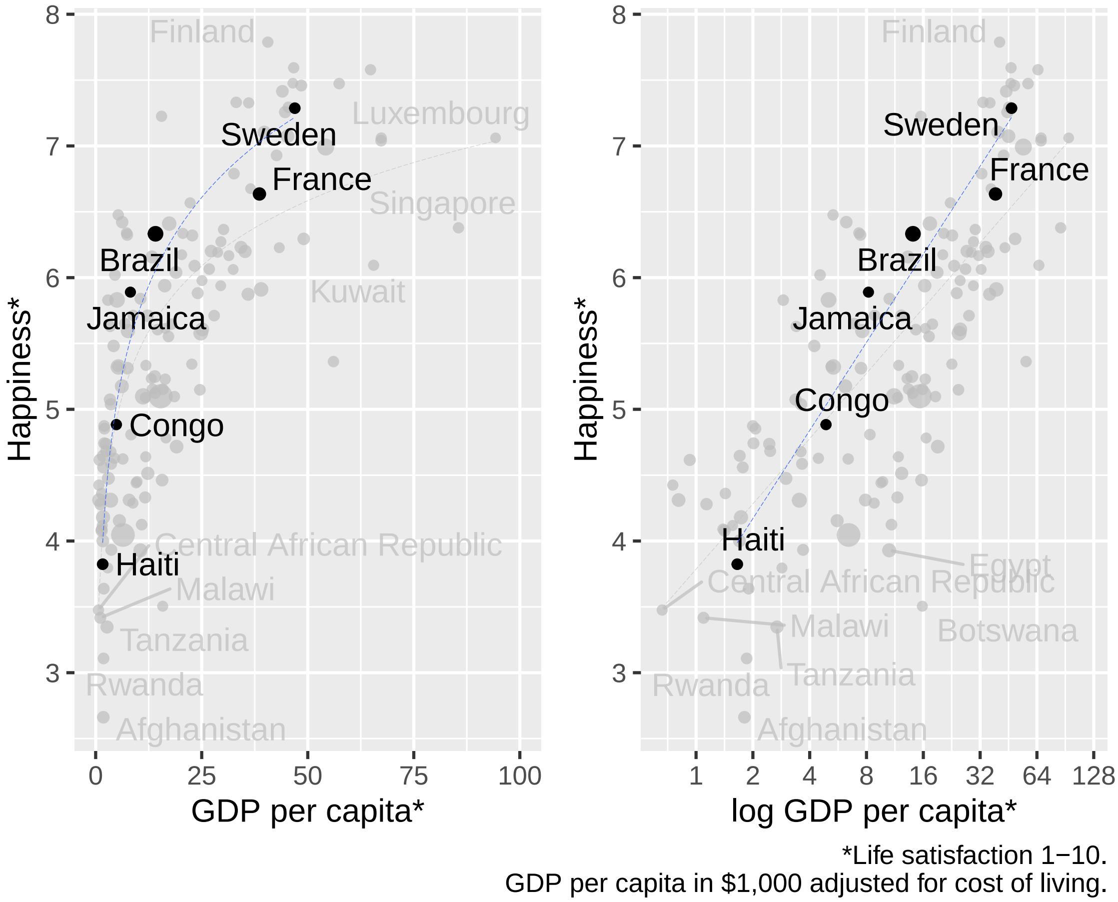


Figure 5.3: Happiness and GDP. Source: www.ourworldindata.org

Happiness is measured here as a survey response on a scale from 1 to 10 where 10 is the highest level of satisfaction with life. The measure of GDP we use here is so-called purchasing power adjusted GDP per capita calculated in thousands of American dollars. This is a rough measure of average income adjusted for price differences between countries. Some dots represent groups of countries, for example all countries that are members of the organization OECD or all countries in Africa south of the Sahara.

In the graph to the left, inhabitants in richer countries are on average happier than in poorer countries. But happiness is only higher with higher GDP up to a certain level. For the richest countries there does not seem to be any particularly clear covariation between happiness and GDP. In the right graph we compare happiness and logarithmic values for GDP. Now the average happiness among inhabitants seems on average to be higher with higher GDP, both among countries with low and high GDP. A large part of the countries lie almost as on a straight line from the graph's lower left corner up to the right corner. In both graphs we have also drawn a trend line so that the pattern shows more clearly. The black line is the trend for the marked countries. The less visible gray line is the trend line for all the world's countries.

To understand technically why the pattern changes we compare the highlighted countries Sweden, France, Brazil, Jamaica, Congo, Kenya and Haiti. Table 5.3 shows the values for these countries, sorted by GDP. To understand the right graph we look particularly at the difference between Congo (happiness 4.9), Brazil (happiness 6.3) and Sweden (happiness 7.3). If we compare GDP the differences appear much larger than happiness, where Congo has a GDP per capita of 4.9, Brazil 14.1 and Sweden 46.9.

Table 5.3: Happiness and GDP in six countries

Country	Happiness	GDP per capita	$\log_{10}$ (GDP per capita)
Haiti	3.8	1.7	0.2
Congo	4.9	4.9	0.7
Jamaica	5.9	8.2	0.9
Brazil	6.3	14.1	1.1
France	6.6	38.6	1.6
Sweden	7.3	46.9	1.7
Source: Our World in Data, www.ourworldindata.org.			

If we compare relative differences in income Sweden has a little more than three times as high average income as Brazil, which has almost three times as high average income as Congo. This is also reflected in the logarithmic values, where Sweden's $\log_{10}$ (GDP) is 1.7, Brazil's 1.1 and Congo's 0.7. Logarithmic transformation facilitates relative comparisons since the difference between logarithmic values is the same for corresponding relative differences in absolute values.

The relative differences in income covariate relatively well with life satisfaction. Exactly why this is the case is impossible to say, especially based on only the information we use here. An important part of the explanation might be that an income increase of a fixed sum, for example 100 USD, usually is not as valuable for a rich person as it is for a poor. However, a relative income increase, such as 10%, are useful no matter if you are poor or rich to begin with. That is, life satisfaction might increase even among very high income earners if their income increase sufficiently much.

The right graph has a striking pattern. But note that not even this in itself is any proof of a direct causal relationship between happiness and income. High income countries are usually very different from poor countries in many aspects, and these things also might affect life satisfaction. It is therefore not correct to say that we here see a direct causal relation from just income. Therefore we cannot draw any conclusions about how much happiness in a population would increase if incomes would increase. To understand this, just think about the different ways that income could increase in theory, while other parts of life could get worse at the same time. Part II introduce methods to study causal relationships.

5.11 Chapter summary

- Logarithms are another way to write powers. In a power expression b^x, b is called the base and x the exponent. Given the base b, $\log_b x =$ the number that b must be raised to obtain the value x. Thus: if base $= b = 10$ then $\log_{10} 100 = 2$ since $10^2 = 100$.
- The function $\log()$ often uses base 10. This is so common that if it only says $\log()$ this usually means that the base is 10. The function $\ln()$ is log with Euler's number e(approximately 2.718) as base. For all bases it applies that $\log 1 = 0$ since all numbers, all bases, raised to 0 become 1.
- The logarithm laws:
  - Multiplication: $\log(ab) = \log a + \log b$.
  - Division: $\log(a/b) = \log a - \log b$.
  - Powers: $\log x^a = a \log x$.
- To change base for the logarithm it applies that $\log_b x = \frac{\log_a x}{\log_a b} = \frac{\ln x}{\ln b}$, where b, a and x are real numbers.
- For small differences the difference between the natural logarithm of two values is approximately equal to the percentage difference. Example: $\ln(102/100) = \ln 102 - \ln 100 \approx 0.02$, approximately 2%.
- Logarithmic transformation can among other things be used to compare relative differences, create linear equations and solve expressions with powers.

5.12 Exercises

1. Calculate:

(a) (log_{2}64)

(b) (log_{3}81)

(c) (log_{5}125)

Answer: (a) $\log_2 64 = 6$ since $2^6 = 64$, (b) $\log_3 81 = 4$ since $3^4 = 81$, (c) $\log_5 125 = 3$ since $5^3 = 125$

2. Write each power expression as a logarithm:

(a) (2^{8}=256)

(b) (10^{-2}=0.01)

(c) (e^{1}=e)

Answer: (a) $\log_2 256 = 8$, (b) $\log_{10} 0.01 = -2$, (c) $\ln e = 1$

3. Calculate using the common logarithm (base 10):

(a) (log_{10}10,000)

(b) (log_{10}0.001)

(c) (log_{10}10^{7})

Answer: (a) $\log_{10} 10,000 = 4$, (b) $\log_{10} 0.001 = -3$, (c) $\log_{10} 10^7 = 7$

4. Use the product rule to simplify:

(a) (log(20)+log(5))

(b) (log_{2}(8)+log_{2}(4))

(c) (log_{3}(9)+log_{3}(27))

Answer: (a) $\log(20 \times 5) = \log 100 = 2$, (b) $\log_2(8 \times 4) = \log_2 32 = 5$, (c) $\log_3(9 \times 27) = \log_3 243 = 5$

5. Use the quotient rule to calculate:

(a) (log(1,000,000)-log(1,000))

(b) (log_{3}(81)-log_{3}(3))

(c) (log_{2}(64)-log_{2}(4))

Answer: (a) $\log(1,000,000/1,000) = \log 1,000 = 3$, (b) $\log_3(81/3) = \log_3 27 = 3$, (c) $\log_2(64/4) = \log_2 16 = 4$

6. Use the power rule to simplify:

(a) (log(1,000^{2}))

(b) (log_{2}(4^{3}))

(c) (ln(e^{5}))

Answer: (a) $2 \log 1,000 = 2 \times 3 = 6$, (b) $3 \log_2 4 = 3 \times 2 = 6$, (c) $\ln(e^5) = 5$

7. Simplify using logarithm laws:

(a) (2log(10)+log(100))

(b) (2log_{3}(9)+log_{3}(3))

(c) (logsqrt{10,000})

Answer: (a) $2 \times 1 + 2 = 4$, (b) $2 \times 2 + 1 = 5$, (c) $\frac{1}{2} \log 10,000 = \frac{1}{2} \times 4 = 2$

8. Calculate using the natural logarithm:

(a) (ln(e^{4}))

(b) (e^{ln5})

(c) (e^{ln3+ln4})

Answer: (a) $\ln(e^4) = 4$, (b) $e^{\ln 5} = 5$, (c) $e^{\ln 3+\ln 4} = e^{\ln 12} = 12$

9. Solve for x :

(a) (log_{3}x=4)

(b) (log x=5) (base 10)

(c) (10^{x}=500)

Answer: (a) $x = 3^4 = 81$, (b) $x = 10^5 = 100,000$, (c) $x = \log 500 = \log(5 \times 100) = \log 5 + 2 \approx 0.699 + 2 = 2.699$

10. Use the change of base formula $\log_a x = \frac{\ln x}{\ln a}$:

(a) Calculate (log_{8}(64)) by rewriting using natural logarithms.

(b) Calculate (log_{4}(32)) .

(c) Express (log_{2}(100)) using common logarithms (base 10) and calculate.

Answer: (a) $\frac{\ln 64}{\ln 8} = \frac{6\ln 2}{3\ln 2} = 2$, (b) $\frac{\ln 32}{\ln 4} = \frac{5\ln 2}{2\ln 2} = 2.5$, (c) $\frac{\log 100}{\log 2} = \frac{2}{0.301} \approx 6.64$

11. Use the natural logarithm to approximate percentage changes:

(a) Monthly wages increased from 40,000 to 40,800. Calculate (ln(40,800)-ln(40,000)) and compare with the actual percentage change.

(b) GDP per capita in country A is 8,000 USD and in country B 12,000 USD. Calculate (ln(12,000)-ln(8,000)) . Is this a good approximation of the percentage difference?

(c) A price index rose from 100 to 103. Use ln to approximate the percentage change and compare with the exact value.

Answer: (a) $\ln(40,800/40,000) = \ln(1.02) \approx 0.0198 \approx 2\%$; actual: $800/40,000 = 2\%$ — good approximation, (b) $\ln(12,000/8,000) = \ln(1.5) \approx 0.405 \approx 40.5\%$; actual: 50% — poor approximation for large differences, (c) $\ln(103/100) = \ln(1.03) \approx 0.030 \approx 3\%$; actual: 3% — good approximation

Chapter 6

Polynomial Equations

This chapter introduces polynomials. A polynomial is an equation with variables and constants combined with addition, subtraction and multiplication and with a positive integer exponent. This is useful for a lot of things in analytical work, for instance describing different theories about how humans behave.

6.1 Quadratic Equations

The following expression is an example of a polynomial:

$$x^2 + 3 \tag{6.1}$$

The following expression is not a polynomial because the exponent -2 is negative:

$$x^{-2} + 3 \tag{6.2}$$

The degree of the polynomial is given by the largest positive integer exponent in the expression. Let us now take the following polynomial:

$$x^2 + x + 2 \tag{6.3}$$

Since the exponent in the term x^2 is 2, this is called a second-degree polynomial (quadratic equation). If the largest exponent in the expression had been 3, for example x^3, this would have been a third-degree polynomial (cubic equation). If the largest exponent is 4, it is called a fourth-degree equation, and so on. Examples:

$$\begin{aligned} \text{Quadratic polynomial: } & x^2 - 25x + 321 \\ \text{Cubic polynomial: } & x^3 + x^2 - 11x + 20 \end{aligned} \tag{6.4}$$

Often we want to find an equation's zeros: the x-values for which the entire equation equals 0. This is also called the real solutions to the equation. A polynomial's degree determines the number of possible solutions. A quadratic equation can have at most two zeros: two, one or no solution. A cubic equation can have at most three solutions: three, two, one or none.

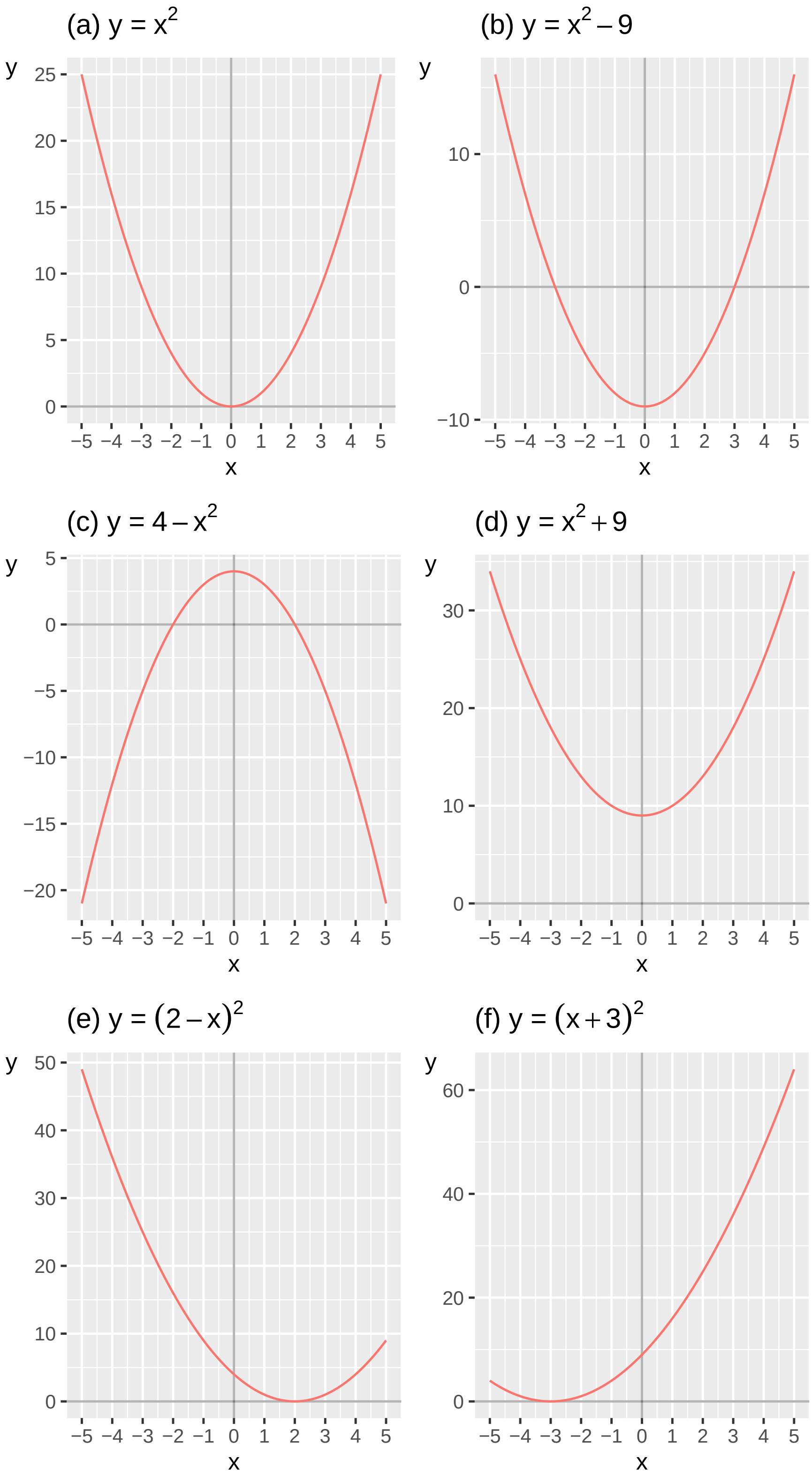


Figure 6.1: Examples of quadratic equations

To find the solutions, zeros, of an equation we set the equation equal to 0 and solve for x. Sometimes we may also find the solutions using graphs. In figure 6.1 six graphs are shown with one quadratic equation each, written as functions of the form $y = f(x)$. The solution for each function is the point where the function's line crosses the x-axis. Let us go through graphs (a) to (f) in order. Graph (a) illustrates the function:

$$y = f(x) = x^2 \tag{6.5}$$

The term x^2 indicates that this is a quadratic equation and it has at most two solutions. In this case there is only one x-value that gives the result $f(x) = 0$ and that is $x = 0$:

$$f(0) = 0^2 = 0 \tag{6.6}$$

We write the solution as $x^* = 0$. Graph (b) shows the function:

$$y = x^2 - 9 \tag{6.7}$$

The function's line crosses the x-axis at two points, where the function, $y = f(x)$, equals 0: $x = -3$ and $x = 3$. These are the function's solutions. When we have more than one solution we can write $x_1^* = -3$ and $x_2^* = 3$. We may also see this if we set the function equal to 0 and solve for x:

$$\begin{aligned} f(x) = x^2 - 9 &= 0 \\ x^2 &= 9 \\ \left(x^2\right)^{\frac{1}{2}} &= 9^{\frac{1}{2}} \\ x^{2*\frac{1}{2}} &= 9^{\frac{1}{2}} \\ x &= \pm 3 \end{aligned} \tag{6.8}$$

In row three we take the square root of both sides, that is exponent 1/2(see section 2.8). We get the solutions $x_1^* = 3$ and $x_2^* = -3$ and can also test these by substituting them into equation (6.7) :

$$\begin{aligned} y =& x^2 - 9 \\ x_1^* = 3 \Rightarrow y =& 3^2 - 9 = 0 \\ x_2^* = -3 \Rightarrow y =& (-3)^2 - 9 = 0 \end{aligned} \tag{6.9}$$

Graph (c) illustrates the function:

$$y = 4 - x^2 \tag{6.10}$$

The line crosses the x-axis at $x_1^* = -2$ and $x_2^* = 2$. We can also solve for the function's zeros:

$$\begin{aligned} 4 - x^2 &= 0 \\ x^2 &= 4 \\ x^* &= \pm 2 \end{aligned} \tag{6.11}$$

Graph (d) illustrates the function:

$$y = x^2 + 9 \tag{6.12}$$

In the graph the function's line does not cross the x-axis. This means that the function lacks a solution that is a real number, the function lacks a real solution. If we calculate the function's solutions:

$$
\begin{aligned}
x^2 + 9 &= 0 \\
x &= (-9)^{1/2} \\
&= \pm\sqrt{-9}
\end{aligned}
\tag{6.13}
$$

To calculate x we must here take the square root of a negative number, such as $\sqrt{-9}$, which is not defined for any real numbers. We can however instead use the imaginary unit, which is denoted i. The imaginary unit i has the following properties:

$$
\begin{aligned}
i &= (-1)^{1/2} \\
i^2 &= -1
\end{aligned}
\tag{6.14}
$$

We now get:

$$
x = \pm\sqrt{-9} = \pm\sqrt{-1 * 9} = \pm\sqrt{i^2 * 3^2} = \pm 3i \tag{6.15}
$$

The equation's solutions are $x_1^* = 3i$ and $x_2^* = -3i$. Graph (e) illustrates the function:

$$
y = (2 - x)^2 \tag{6.16}
$$

In the graph the line passes through the x-axis at $x^* = 2$. For equation (6.16) to equal zero, x must be 2. All other values for x result in $y > 0$. Graph (f) shows the function:

$$
y = (x + 3)^2 \tag{6.17}
$$

For $y = 0$ we need $x + 3 = 0$, and thus $x^* = -3$, which can also be seen in the graph.

6.2 Factoring Equations

This section and the two following go through methods for solving polynomials. Section 2.10 introduced factorization of integers. An equation can also be factorized. We illustrate this with the following equation:

$$
(x + 1)(x + 2) = 0 \tag{6.18}
$$

The entire expression on the left side must equal 0. The two parentheses are multiplied together, making each parenthesis a factor. For the entire expression to equal 0, one or both parentheses must equal 0. This gives us the solutions $x_1^* = -1$ and $x_2^* = -2$. If we multiply the parentheses in equation (6.18) , this results in a quadratic polynomial:

$$
\begin{aligned}
(x + 1)(x + 2) &= 0 \\
x^2 + 3x + 2 &= 0
\end{aligned}
\tag{6.19}
$$

If we have a quadratic polynomial of this type, it can facilitate our work if we succeed in factorizing so that we get the two parentheses. To be sure that we have found the correct solutions, we substitute our x-values into the quadratic polynomial and check:

$$
\begin{aligned}
x^2 + 3x + 2 &= 0 \\
x = -1 \Rightarrow (1)^2 + 3(-1) + 2 &= 0 \\
x = -2 \Rightarrow (-2)^2 + 3(-2) + 2 &= 0
\end{aligned}
\tag{6.20}
$$

Both values result in the quadratic polynomial becoming 0, which confirms that these x-values $x_1^* = -1$ and $x_2^* = -2$ are solutions to the equation. Now we shall find the possible solutions to the following quadratic equation:

$$x^2 + 5x = 0 \tag{6.21}$$

The variable x appears in both terms on the left-hand side. We factor out x and put the rest of the expression in parentheses:

$$x(x + 5) = 0 \tag{6.22}$$

For the left-hand side to equal 0, we must have either $x = 0$ or $x + 5 = 0$, which in that case requires that $x = -5$. This gives us the solutions $x_1^* = 0$ and $x_2^* = -5$, which we can test by substitution in equation (6.21) :

$$
\begin{aligned}
x^2 + 5x &= 0 \\
x_1^* = 0 \Rightarrow (0)^2 + 5 * 0 &= 0 \\
x_2^* = -5 \Rightarrow (-5)^2 + 5(-5) &= 25 - 25 = 0
\end{aligned}
\tag{6.23}
$$

Both these x-values result in the equation becoming 0, which confirms that these are solutions to our equation. Since $x(x + 5) = x^2 + 5x$ is a quadratic equation we seek at most two solutions.

It can be difficult to find expressions that can be factorized from an equation. Sometimes one must try different approaches. Or draw the equation in a graph to find a first solution and then try from there. Even if a polynomial at first glance can look complicated, we can often make it more manageable with simple means. Consider the following quadratic polynomial as an example:

$$3x^2 + 3x - 168 = 0 \tag{6.24}$$

We want to find the zeros and start by dividing by 3:

$$
\begin{aligned}
\frac{3x^2}{3} + \frac{3x}{3} - \frac{168}{3} &= 0 \\
x^2 + x - 56 &= 0
\end{aligned}
\tag{6.25}
$$

This expression is easier to factorize. We seek two factors that result in the product 56. Since we shall factor out two factors we also use the fact that the second term should become $+x$, that is $(+1)x$. We can here try different approaches until we succeed in rewriting the expression:

$$
\begin{aligned}
x^2 + x - 56 &= 0 \\
(x + 8)(x - 7) &= 0
\end{aligned}
\tag{6.26}
$$

This gives us two proposed solutions: $x_1^* = -8$ respectively $x_2^* = 7$. We test these proposed solutions by substituting the x-values into the quadratic equation:

$$\begin{aligned} x^2 + x - 56 &= 0 \\ x = -8 \Rightarrow (-8)^2 + (-8) - 56 &= 0 \\ x = 7 \Rightarrow 7^2 + 7 - 56 &= 0 \end{aligned} \tag{6.27}$$

Both these x-values result in the equation becoming 0, which means that they are solutions, thus $x_1^* = -8$ and $x_2^* = 7$. Let us now find the solutions (zeros) of the following polynomial:

$$x^2 + 2x - 8 = 0 \tag{6.28}$$

We factorize this expression and get:

$$\begin{aligned} x^2 + 2x - 8 &= 0 \\ (x + 4)(x - 2) &= 0 \end{aligned} \tag{6.29}$$

This gives us the solutions $x_1^* = -4$ and $x_2^* = 2$. Feel free to test the solutions yourself using substitution.

6.3 Rational Roots

Another method for finding an equation's solutions is to search for its rational roots. We illustrate this with the following quadratic equation:

$$2x^2 + x - 3 \tag{6.30}$$

The method we will now go through is based on the polynomial having a rational root, which in that case is also a solution to the equation. If the equation has a root, this is found among the numbers we can put together as the fraction p/q. The letter p is an integer factor to the equation's constant, where the constant in this case refers to the number 3. The letter q is an integer factor to the number that is multiplied with the power with the largest integer exponent. In this case $2x^2$ is the term in the equation with the largest exponent, and the number that is multiplied with the power is thus 2.

For p: Potential factors for the number 3 are ± 1 and ± 3, where the symbol $\pm$ is called the plus-minus sign and indicates that the values 1 respectively 3 can be both positive (plus) and negative (minus). Potential candidates for p are $-3, -1, 1$ and 3.

For q: The number 2 has the potential integer factors ± 1 and ± 2. For q we get the candidates $-2, -1, 1$ and 2. We combine the possible integer factors in the fraction p/q and thereby get the equation's possible rational roots:

$$\frac{p}{q} = \pm 1, \pm \frac{1}{2}, \pm 3, \pm \frac{3}{2} \tag{6.31}$$

These roots we test in equation (6.30) and check if the result becomes 0, which in that case indicates that the value is a solution. For $x = 1$ we get:

$$x = 1 \Rightarrow 2(1)^2 + 1 - 3 = 0 \tag{6.32}$$

For $x = -\frac{3}{2}$ we get:

$$\begin{aligned} x = -\frac{3}{2} \Rightarrow & 2\left(-\frac{3}{2}\right)^2 - \frac{3}{2} - 3 \\ &= 2\left(\frac{9}{4}\right) - \frac{3}{2} - 3 \\ &= \frac{9-3}{2} - \frac{6}{2} = 0 \end{aligned} \tag{6.33}$$

These two values are our solutions: $x_1^* = 1$ and $x_2^* = -\frac{3}{2}$. Let us now take the following quadratic equation:

$$x^2 + 4x - 5 \tag{6.34}$$

To study its solutions we start by checking the equation's potential rational roots. The constant 5 gives that $p = 1$ and 5, while $q = 1$. Potential rational roots are in this case:

$$\frac{p}{q} = \pm 1, \pm 5 \tag{6.35}$$

Through substitution we can see that:

$$\begin{gathered} x^2 + 4x - 5 \\ x = 1 \Rightarrow 1^2 + 4 * 1 - 5 = 0 \\ x = -5 \Rightarrow (-5)^2 + 4(-5) - 5 = 0 \end{gathered} \tag{6.36}$$

This gives us the solutions $x_1^* = 1$ and $x_2^* = -5$.

6.4 The Quadratic Formula

A third method for solving quadratic equations is the quadratic formula. Suppose we have the following quadratic equation:

$$ax^2 + bx + c = 0 \tag{6.37}$$

The letters a, b and c are constants and $c \neq 0$. The solutions for this equation can be calculated with the quadratic formula:

$$\begin{gathered} x^* = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q} \\ \text{where } p = \frac{b}{a}, \text{ och } q = \frac{c}{a} \end{gathered} \tag{6.38}$$

Table 6.1: Derivation of the quadratic formula

In general	**Our example**
$ax^2 + bx + c = 0$, where a, b, c are constants and $c \neq 0$	$2x^2 + 3x + \frac{1}{4} = 0$
$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$	$x^2 + \frac{3}{2}x + \frac{1}{8} = 0$

In general	**Our example**
$x^2 + px + q = 0$	$x^2 + \frac{3}{2}x + \frac{1}{8} = 0$
$x^2 + px + q - q = 0 - q$	$x^2 + \frac{3}{2}x + \frac{1}{8} - \frac{1}{8} = 0$
$x^2 + px = -q$	$x^2 + \frac{3}{2}x = -\frac{1}{8}$
$x^2 + px + \left(\frac{p}{2}\right)^2 = -q + \left(\frac{p}{2}\right)^2$	$x^2 + \frac{3}{2}x + \left(\frac{3}{2}\right)^2 = -\frac{1}{8} + \left(\frac{3}{2}\right)^2$
$\left(x + \frac{p}{2}\right)^2 = -q + \left(\frac{p}{2}\right)^2$	$\left(x + \frac{3}{2}\right)^2 = -\frac{1}{8} + \left(\frac{3}{2}\right)^2$
$x + \frac{p}{2} = \pm\sqrt{-q + \left(\frac{p}{2}\right)^2}$	$x^* + \frac{3}{2} = \pm\sqrt{-\frac{1}{8} + \left(\frac{3}{2}\right)^2}$
$x = -\frac{p}{2} \pm \sqrt{-q + \left(\frac{p}{2}\right)^2}$	$x^* = -\frac{3}{2} \pm \sqrt{-\frac{1}{8} + \left(\frac{3}{2}\right)^2}$

Table 6.1 shows how the quadratic formula can be derived. In the left column a general example with letters is given step by step. In the right column we go through the same thing for the following equation, where we solve for x:

$$2x^2 + 3x + \frac{1}{4} = 0 \tag{6.39}$$

In equation (2.96) we went through the squaring rule for parentheses:

$$(x + d)^2 = x^2 + 2dx + d^2 \tag{6.40}$$

From equation (6.39) we start by moving $\frac{1}{4}$ to the right side and dividing everything by 2:

$$\begin{aligned} 2x^2 + 3x &= -\frac{1}{4} \\ x^2 + \frac{3}{2}x &= -\frac{1}{8} \end{aligned} \tag{6.41}$$

In this equation the fraction $\frac{3}{2}$ now corresponds to the term $2d$ in equation (6.39) . The expression d^2 is therefore $\left(\frac{3}{2}\right)^2$. We add this to both sides and rewrite:

$$\begin{aligned} x^2 + \frac{3}{2}x + \left(\frac{3}{2}\right)^2 &= -\frac{1}{8} + \left(\frac{3}{2}\right)^2 \\ \left(x + \frac{3}{2}\right)^2 &= -\frac{1}{8} + \left(\frac{3}{2}\right)^2 \end{aligned} \tag{6.42}$$

We take the square root of both sides and move everything except x to the right side:

$$x^* = -\frac{3}{2} \pm \sqrt{-\frac{1}{8} + \left(\frac{3}{2}\right)^2} \tag{6.43}$$

The square root of a number is given by both a negative and a positive value, which is symbolized by the plus-minus sign $\pm$. This gives us two solutions to equation (6.39) :

$$\begin{aligned} x_1^* &= -\frac{3}{2} + \sqrt{-\frac{1}{8} + \left(\frac{3}{2}\right)^2} \\ x_2^* &= -\frac{3}{2} - \sqrt{-\frac{1}{8} + \left(\frac{3}{2}\right)^2} \end{aligned} \tag{6.44}$$

6.5 Third degree and Higher

Polynomials of higher degree can also be illustrated in graphs. In figure 6.2 the following two functions are illustrated::

$$
\begin{aligned}
y &= 2x^3 - x^2 + 7 \\
y &= 6x^2 + 3x^3 - x^4
\end{aligned} \tag{6.45}
$$

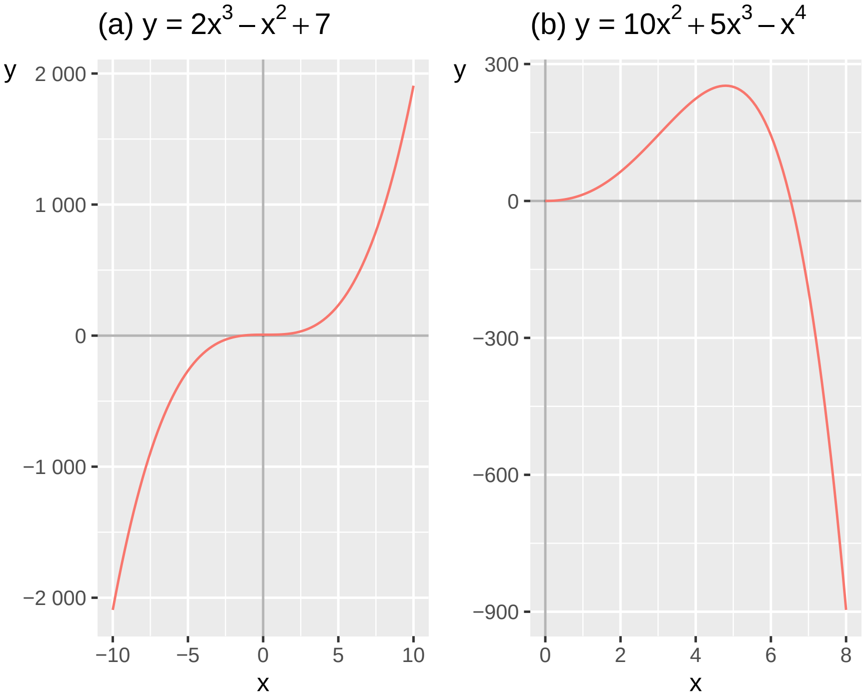


Figure 6.2: Examples of third- and fourth-degree polynomials

In the first row we have a third-degree polynomial, since we have x^3. On the second row we have a fourth-degree polynomial, which we see on the term x^4. Even though we have several terms where x appears in each function we still have only two variables, x and y. The first function in equation (6.46) in graph (a) gets increasingly higher y-values at higher x-values, the line curves upward further to the right in the graph.

For the second equation, graph (b), y first increases slightly and then decreases with higher x-values. In both cases this is controlled by which sign stands in front of the term that has the largest positive integer exponent. In graph (a) the increase is controlled by x^3 being positive in the first equation. In graph (b) the negative development is controlled by x^4 being negative. Let us now look at the following four functions, which are illustrated in figure 6.3 in their respective graphs:

$$
\begin{aligned}
\text{(a)}: \; y &= 75 - x^2 \\
\text{(b)}: \; y &= -10 + 10x^2 - x^3 \\
\text{(c)}: \; y &= 100x - 100x^2 + x^3 \\
\text{(d)}: \; y &= 10x^2 + 10x^3 - x^4
\end{aligned} \tag{6.46}
$$

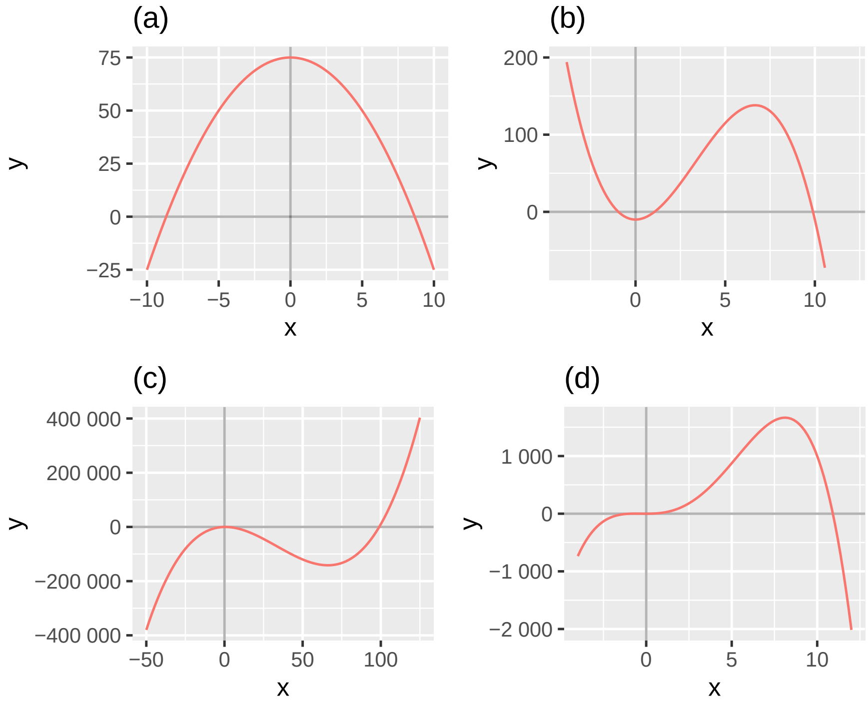


Figure 6.3: More examples of polynomials

Graph (a) shows the function $y = 75 - x^2$, which we can see since the line meets the y-axis at $y = 75$, when $x = 0$. Graph (b) shows the line for the function $y = -10 + 10x^2 - x^3$. The line meets the y-axis at $y = -10$ when $x = 0$. The negative term $-x^3$ dominates the line's development further to the right in the graph where the line curves downward below the x-axis.

Graph (c) shows the line for function $y = 100x - 100x^2 + x^3$ which meets the y-axis at $y = 0$ when $x = 0$. The positive term $+x^3$ dominates the line's development upward for higher values of x, further to the right in the graph. Graph (d) shows the function $y = 10x^2 + 10x^3 - x^4$ which also meets the y-axis at $y = 0$ when $x = 0$. The negative term $-x^4$ causes the line to curve downward for higher values of x, further to the right in the graph.

6.6 Factorizing Polynomials of Higher Degree

Polynomials of higher degree can be more difficult to solve. Sometimes we can rewrite equations so that we first succeed in solving a quadratic equation and solve this with one of the methods above, like the quadratic formula. Consider the following cubic equation:

$$x^3 + 2x^2 - 5x - 6 \tag{6.47}$$

We seek the zeros of this equation. When one should factorize it can be difficult to know how to start and sometimes one has to try different approaches. We set the equation equal to 0 and see after some puzzling that we can write:

$$\begin{aligned} x^3 + 2x^2 - 5x - 6 &= 0 \\ x^3 + 3x^2 - x^2 - 3x - 2x - 6 &= 0 \\ \left(x^2 - x - 2\right)(x + 3) &= 0 \\ (x + 1)(x - 2)(x + 3) &= 0 \end{aligned} \tag{6.48}$$

Now that we have factorized out the three parentheses in the last row we can more easily find proposals for the equation's zeros: $x_1 = -1$, $x_2 = 2$ and $x_3 = -3$. We test our solutions by substituting these x-values into the third-degree polynomial on the first line:

$$\begin{aligned} x^3 + 2x^2 - 5x - 6 = 0 \\ x = -1 \Rightarrow (-1)^3 + 2(-1)^2 + 5 - 6 = 0 \\ x = 2 \Rightarrow 2^3 + 2 * 2^2 - 5 * 2 - 6 = 0 \\ x = -3 \Rightarrow (-3)^3 + 2(-3)^2 - 5(-3) - 6 = 0 \end{aligned} \tag{6.49}$$

All these three x-values are solutions to the polynomial: $x_1^* = -1$, $x_2^* = 2$ and $x_3^* = -3$. Let us now study if the following cubic equation with variable x has any solutions:

$$x^3 - 2x^2 - x \tag{6.50}$$

We set the equation equal to 0 and factor out x:

$$\begin{aligned} x^3 - 2x^2 - x = 0 \\ x\left(x^2 - 2x - 1\right) = 0 \end{aligned} \tag{6.51}$$

This gives us our first proposed solution, $x_1^* = 0$, which we see since we now have two factors: x respectively the parenthesis $(x^2 - 2x - 1)$. At least one of these factors must be 0. The expression in the parenthesis is a quadratic equation that we can solve with for example the quadratic formula. This gives us two new proposed solutions: $x_2^* = 1 - 2^{1/2}$ and $x_3^* = 1 + 2^{1/2}$. Test these solutions yourself by substituting these values for x in equation (6.50) .

Let us now take the following cubic equation:

$$2x^3 + 3x^2 + \frac{1}{4}x = 0 \tag{6.52}$$

Since all terms contain x we can factor this out:

$$x\left(2x^2 + 3x + \frac{1}{4}\right) = 0 \tag{6.53}$$

This gives us our first solution in $x_1^* = 0$, which we test by putting it into equation (6.52) :

$$x = 0 \Rightarrow 2 * 0^3 + 3 * 0^2 + \frac{1}{4} * 0 = 0 \tag{6.54}$$

Now it remains to find the remaining solutions to equation (6.52) by calculating what is left in the parenthesis in equation (6.53) . This we did above when we derived the quadratic formula, in equation (6.39) .

6.7 Divisions of Polynomials of Higher Degree

Another useful method for solving polynomials of higher degree is polynomial division. The method is based simply on having a polynomial of higher degree and having found one of the solutions which we use to factorize. To find the remaining solutions we can divide the remaining polynomial by our factor and thereby solve for new factors. Consider the following fourth-degree polynomial:

$$x^4 - 2x^3 - 13x^2 + 14x + 24 \tag{6.55}$$

The constant 24 indicates that the polynomial's roots should be found among the numbers $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12, \pm 24$. To proceed we try substituting these numbers in our equation and see if the result becomes 0. In that case we have found a root. We start by testing $x = -2$. For $x = -2$ we get:

$$x^4 - 2x^3 - 13x^2 + 14x + 24 \tag{6.56}$$
$$x = -2 \Rightarrow (-2)^4 - 2(-2)^3 - 13(-2)^2 + 14(-2) + 24 \neq 0$$

For $x = -2$ the equation is not equal to 0, which is why this is not a solution or root. Let us try $x = 2$ instead:

$$x = 2 \Rightarrow (2)^4 - 2(2)^3 - 13(2)^2 + 14(2) + 24 = 0 \tag{6.57}$$

Since $x_1^* = 2$ is a solution to equation (6.55) we know that $x - 2$ is a factor in the equation. To find the remaining solutions to the polynomial we shall now divide the polynomial by the factor $x - 2$. For this we use polynomial division with the help of long division:

x^4	$-2x^3$	$-13x^2$	$+14x$	$+24$	$(x-2)$

We start by dividing x in our parenthesis $(x - 2)$ into the term with the largest exponent, x^4. The result, x^3, we write above this term:

x^3	x^4	$-2x^3$	$-13x^2$	$+14x$	$+24$
$(x-2)$					

Then we multiply x^3 by our denominator, the parenthesis. The results of this, $(x^4 - 2x^3)$ we write under the polynomial. Then we subtract this from the polynomial and save the remainder in the next row:

x^3	x^4	$-2x^3$	$-13x^2$	$+14x$	$+24$
$(x-2)$	$-(x^4-2x^3)$	$-13x^2$	$+14x$	$+24$	

Now we again divide x from our denominator $(x - 2)$ into the term with the largest exponent in this new equation $(-13x^2 + 14x + 24)$, the result is multiplied by the denominator and subtracted from the new equation:

x^3	$-13x$	x^4	$-2x^3$	$-13x^2$	$+14x$
$+24$	$(x-2)$	$-(x^4-2x^3)$	$-13x^2$	$+14x$	$+24$
$-(-13x^2+26x)$	$-12x$	$+24$			

We repeat the same procedure once more:

x^3	$-13x$	-12	x^4	$-2x^3$	$-13x^2$
$+14x$	$+24$	$(x-2)$	$-(x^4-2x^3)$	$-13x^2$	$+14x$
$+24$	$-(-13x^2+26x)$	$-12x$	$+24$	$-(-12x+24)$	0

On the bottom row there is now 0. The equation has divided evenly and what now stands on the top row is the expression that we can multiply our denominator $(x - 2)$ by to get our original fourth-degree polynomial:

$$\begin{aligned} x^4 - 2x^3 - 13x^2 + 14x + 24 = \\ = (x-2)\left(x^3 - 13x - 12\right) \end{aligned} \tag{6.58}$$

Let us now do the same thing also with the longer factor, the third-degree polynomial $(x^3 - 13x - 12)$. But now we again need to find a new root to divide with. The constant 12 indicates the following potential roots: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8$ or ± 12. By testing these in our equation we find our second root: $x_2^* = -1$:

$$\begin{gathered} x^3 - 13x - 12 \\ x = -1 \Rightarrow (-1)^3 - 13(-1) - 12 = 0 \end{gathered} \tag{6.59}$$

We therefore divide the third-degree polynomial $x^3 - 13x - 12$ by the expression $x + 1$:

x^2	$-x$	-12	x^3	$-13x$
-12	$(x+1)$	$-\left(x^3+x^2\right)$	$-x^2$	$-13x$
-12	$-\left(-x^2-x\right)$	$-12x$	$-(-12x-12)$	0

Again the equation divides evenly and we have gotten a new polynomial:

$$x^3 - 13x - 12 = (x+1)\left(x^2 - x - 12\right) \tag{6.60}$$

The quadratic polynomial in the second parenthesis we can solve with the quadratic formula or factorization. We can also continue to search for roots with the help of the constant and then polynomial divide the result thereof as well. The constant 12 indicates that the potential roots to this quadratic polynomial can be $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$. By trying our way forward we find the third solution $x_3^* = -3$:

$$x = -3 \Rightarrow 9 + 3 - 12 = 0 \tag{6.61}$$

From this we now make the following division:

x	-4	x^2	$-x$
-12	$(x+3)$	$-\left(x^2+3x\right)$	$-4x$
-12	$-(-4x-12)$	0	

The division divides evenly and we get 0 on the bottom row. On the top row we have now found the last factor:

$$x - 4 = 0 \;\Rightarrow\; x_4^* = 4 \tag{6.62}$$

We have now found all four roots to our fourth-degree polynomial and can factorize out four terms:

$$x^4 - 2x^3 - 13x^2 + 14x + 24 = (x+1)(x-2)(x+3)(x-4) \tag{6.63}$$

Feel free to test the solutions by substituting them into the fourth-degree polynomial.

6.8 Chapter summary

- Equations that consist of variables and constants with a positive integer exponent and where the different parts are combined with plus, minus and multiplication, are called polynomials or polynomial equations. Example of polynomial: $x^2 + 3$. The following is not a polynomial: $x^{-2} + 3$, since the exponent -2 is negative.
- A polynomial's degree is determined by the largest positive integer exponent in the expression. Example of a quadratic polynomial: $x^2 + 3$. Example of a cubic polynomial: $x^3 + 3$.
- To find the solution to a polynomial we set the equation equal to 0 and solve for possible values of the variable or variables. Example: $x^2 - 9 = 0$ has the solutions $x = \pm\sqrt{9} = \pm 3$. In a graph this can be illustrated with the points when the function y=0.
- One method for finding the equation's solutions is to factorize. Example: $x^2 - x - 6 = (x+2)(x-3) = 0$ has the solutions $x_1^* = -2$ and $x_2^* = 3$.
- If a polynomial has a rational root we can also try p/q as a potential solution. The letter p is an integer factor to the equation's constant and q is an integer factor to the number that is multiplied with the power in the equation that has the largest exponent.
- A third method is the quadratic formula: given $ax^2 + bx + c = 0$, where a, b, c are constants and $c \neq 0$, the solutions are given by $x^* = -p/2 \pm \sqrt{(p/2)^2 - q}$, where $p = b/a$, $q = c/a$.
- Polynomials of higher degree can also be solved with factorization. Example: the cubic polynomial $(x+1)(x-2)(x+3) = 0$ has the solutions $x_1^* = -1, x_2^* = 2, x_3^* = -3$.
- Another method for solving polynomials of higher degree is polynomial division. If we have found a solution to a polynomial, $x_1^* = 1$, we can divide the polynomial by the equation $x - 1$. Thereby we can then find more factors and solutions.

6.9 Exercises

1. For each expression, state whether it is a polynomial. If yes, state its degree:

(a) (x^{3}-2x+5)

(b) (x^{-1}+3)

(c) (4x^{2}-x+7)

Answer: (a) Yes, degree 3, (b) No (negative exponent), (c) Yes, degree 2

2. Solve:

(a) (x^{2}=25)

(b) (x^{2}-16=0)

(c) (4x^{2}=36)

Answer: (a) $x = \pm 5$, (b) $x = \pm 4$, (c) $x = \pm 3$

3. Factorize and solve:

(a) (x^{2}-5x+6=0)

(b) (x^{2}+2x-8=0)

(c) (x^{2}-7x+12=0)

Answer: (a) $(x-2)(x-3) = 0$, $x_1^* = 2, x_2^* = 3$, (b) $(x+4)(x-2) = 0$, $x_1^* = -4, x_2^* = 2$, (c) $(x-3)(x-4) = 0$, $x_1^* = 3, x_2^* = 4$

4. Factorize using special identities:

(a) (x^{2}-9)

(b) (x^{2}+6x+9)

(c) (x^{2}-10x+25)

Answer: (a) $(x+3)(x-3)$, (b) $(x+3)^2$, (c) $(x-5)^2$

5. Use the rational root theorem to find candidates, then solve:

(a) (x^{2}-x-6=0)

(b) (x^{2}+x-12=0)

(c) (x^{2}-3x-10=0)

Answer: (a) $x_1^* = 3,\ x_2^* = -2$, (b) $x_1^* = 3,\ x_2^* = -4$, (c) $x_1^* = 5,\ x_2^* = -2$

6. Solve using the quadratic formula $x^* = -p/2 \pm \sqrt{(p/2)^2 - q}$ where $p = b/a$ and $q = c/a$:

(a) (x^{2}-5x+4=0)

(b) (x^{2}+3x-10=0)

(c) (2x^{2}-7x+3=0)

Answer: (a) $p = -5,\ q = 4 :\ x^* = 5/2 \pm 3/2$, $x_1^* = 4,\ x_2^* = 1$, (b) $p = 3,\ q = -10 :\ x^* = -3/2 \pm 7/2$, $x_1^* = 2,\ x_2^* = -5$, (c) $p = -7/2,\ q = 3/2 :\ x^* = 7/4 \pm 5/4$, $x_1^* = 3,\ x_2^* = 1/2$

7. For each equation, determine whether it has two, one, or no real solutions by calculating $(p/2)^2 - q$:

(a) (x^{2}-4x+5=0)

(b) (x^{2}-4x+4=0)

(c) (x^{2}-4x+3=0)

Answer: (a) $(p/2)^2 - q = 4 - 5 = -10 :$ two solutions, $x_1^* = 3,\ x_2^* = 1$

8. For the cubic polynomial $x^3 - 6x^2 + 11x - 6 = 0$:

(a) Test (x=1) as a potential root.

(b) Divide (x$^{\{3\}\text{-6x}}${2}+11x-6) by ((x-1)) and write down the resulting quadratic polynomial.

(c) Solve the quadratic polynomial from (b) to find all three solutions.

Answer: (a) $1 - 6 + 11 - 6 = 0$, confirmed root, (b) $x^3 - 6x^2 + 11x - 6 = (x-1)(x^2 - 5x + 6)$, (c) $(x-2)(x-3) = 0 :\ x_1^* = 1,\ x_2^* = 2,\ x_3^* = 3$

9. Factorize and find all solutions:

(a) (x^{3}-x=0)

(b) (x$^{\{3\}\text{+3x}}${2}-4x=0)

(c) (x$^{\{3\}\text{-4x}}${2}+4x=0)

Answer: (a) $x(x+1)(x-1) = 0 :\ x_1^* = 0,\ x_2^* = -1,\ x_3^* = 1$, (b) $x(x+4)(x-1) = 0 :\ x_1^* = 0,\ x_2^* = -4,\ x_3^* = 1$, (c) $x(x-2)^2 = 0 :\ x_1^* = 0,\ x_2^* = 2$ (double root)

10. Solve using any appropriate method:

(a) (3x^{2}-12=0)

(b) (x^{2}-2x-15=0)

(c) (x$^{\{3\}\text{-2x}}${2}-x+2=0) (hint: test (x=1))

Answer: (a) $x^2 = 4,\ x = \pm 2$, (b) $(x-5)(x+3) = 0 :\ x_1^* = 5,\ x_2^* = -3$, (c) $(x-1)(x-2)(x+1) = 0 :\ x_1^* = 1,\ x_2^* = 2,\ x_3^* = -1$

Chapter 7

Systems of Linear Equations

So far we have worked with functions that we solve individually. Several linear functions can also be connected and apply simultaneously. This is called equations belonging to a system of equations. This is a central part of many analytical methods and is widely used in both empirical and theoretical work.

7.1 Linear system

A linear system of equations consists of two or more linear equations. The word "system" means that all the equations apply simultaneously. We begin with a simple example with two equations and two variables, y and x:

$$\begin{cases} y &= 2x + 3 \\ x &= y - 1 \end{cases} \tag{7.1}$$

Now we shall investigate if there exists a unique solution for this system. A unique solution for the system would mean that there exist unique values for variables x and y that fit in the equations simultaneously. A linear system of equations of this type can have either a unique solution, infinitely many solutions or no solution, see next section.

If we are to investigate a system of equations' solutions, a useful method is what is called substitution. The idea is based on that we replace, substitute, parts of one equation with parts from the other. For example to use the definition of a variable from one equation and insert this into the other equation. We begin by taking the definition of x from the lower equation: $x = y - 1$. This definition of x we insert at the place x stands in the upper equation. The upper equation now becomes:

$$\begin{aligned} y &= 2x + 3 \\ y &= 2\,(y - 1) + 3 \\ y &= 2y - 2 + 3 \\ y^* &= -1 \end{aligned} \tag{7.2}$$

We have now obtained a new definition of the variable y. We mark this solution candidate with an asterisk, y^*, since we need to verify it satisfies both equations. We substitute y^* into the lower equation and thereby also obtain a candidate solution for x:

$$\begin{aligned} x &= y^* - 1 \\ x^* &= (-1) - 1 = -2 \end{aligned} \tag{7.3}$$

We must always test substituting our solutions into the original system of equations to see that they really are correct. We check x^* by substituting this into the first equation:

$$\begin{aligned} y &= 2x^* + 3 \\ y &= 2\,(-2) + 3 = -1 \end{aligned} \tag{7.4}$$

Since this gives the same result this confirms that we have found the unique solutions for our two variables: $x^* = -2$ and $y^* = -1$. The variables x and y can assume other values, but no other values will result in the two equations being satisfied simultaneously. Therefore, these other values are not solutions to the system. Let us now study the following system of equations:

$$\begin{cases} x &= 1 + y \\ x &= 2 - y \end{cases} \tag{7.5}$$

The equations are now formulated as two functions for x. We can also here use substitution and therefore replace x in the upper equation with the definition of x from the lower equation:

$$\begin{aligned} 2 - y &= 1 + y \\ 1 &= 2y \\ y^* &= \frac{1}{2} \end{aligned} \tag{7.6}$$

This gives us the following:

$$\begin{aligned} x &= 2 - y^* \\ x^* &= 2 - \frac{1}{2} = \frac{3}{2} \end{aligned} \tag{7.7}$$

We test substituting this solution into the upper equation:

$$\begin{aligned} x^* &= 1 + y \\ \frac{3}{2} &= 1 + y \\ y^* &= \frac{1}{2} \end{aligned} \tag{7.8}$$

This means that we have found the solution for both x and y. Previously we went through how we may illustrate one or more functions with two variables in graphs with two axes. Let us look at the first system of equations we went through above:

$$\begin{cases} y &= 2x + 3 \\ x &= y - 1 \end{cases} \tag{7.9}$$

To more easily be able to see how this looks in a graph we can rewrite the lower equation, so that we get another expression for y:

$$\begin{cases} y &= 2x + 3 \\ y &= x + 1 \end{cases} \tag{7.10}$$

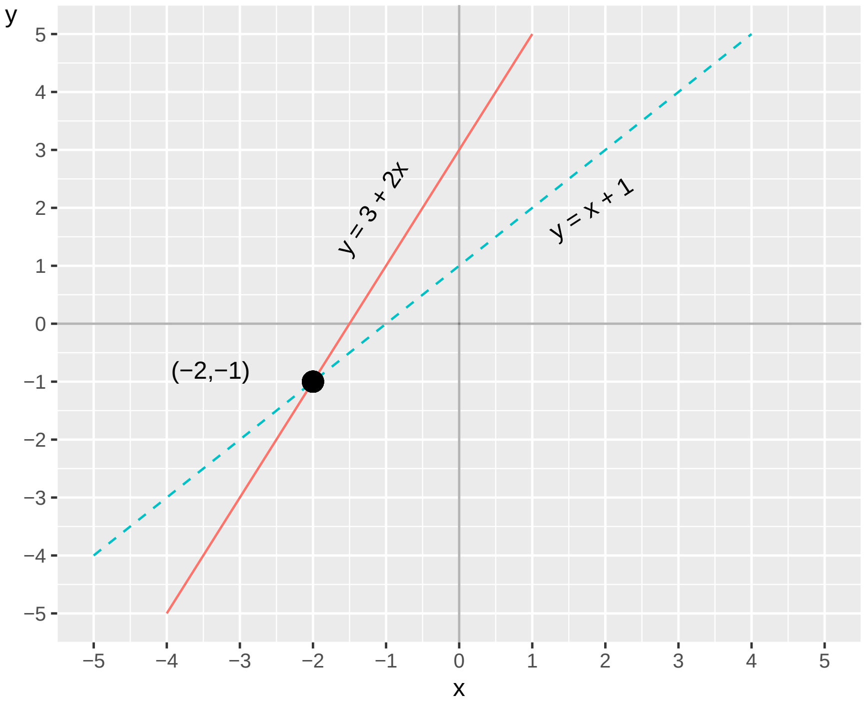


Figure 7.1: Two equations in one graph

Now we have two functions for y that we can draw in a graph, which is shown in figure 7.1 . One line per function. The two lines meet at the point $(x, y) = (-2, -1)$. These values for x and y are the only possible solution to our system of equations. This is also the only point where the two lines meet. We know this because we are here working with linear equations. Even if we were to extend the lines far beyond the points that are visible in the graph the lines will never bend and meet at any other point.

7.2 Number of solutions in the system

In a system of equations with two straight lines (linear functions) we can either have a unique solution, no solution or infinitely many solutions. Figure 7.2 shows three plot with illustrations of the possibilities for number of solutions. Each plot illustrate two lines, drawn with one equation each. In the left graph the lines meet at one point and this system of equations has one unique solution (compare figure 7.1). In the middle graph the lines are parallel and the system therefore lacks a solution. In the right graph the lines go along the same points and therefore meet an infinite number of times. The system of equations for the right graph therefore has an infinite number of solutions.

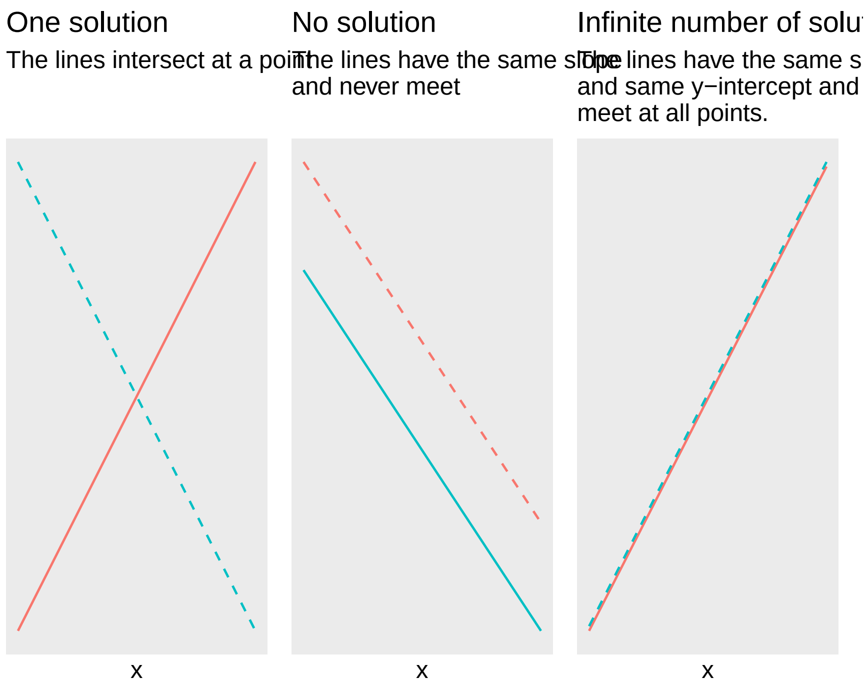


Figure 7.2: Possible number of solutions for linear systems of equations

A linear system of equations lacks a solution if the two straight lines never meet at any point. This applies if the slope of the lines is the same, which means that the lines are parallel. In that case there are no values for x and y that satisfy the system of equations in its entirety. If we for example have the following system of equations:

$$\begin{cases} y & = 5x - 4 \\ 10x & = 2y \end{cases} \tag{7.11}$$

we can from the second, lower, equation solve for $y = 5x$. If we replace y in the first equation with this we get:

$$\begin{aligned} y &= 5x - 4 \\ 5x &= 5x - 4 \\ 5x - 5x &\neq -4 \end{aligned} \tag{7.12}$$

The last line leads to a paradox. The expression $5x - 5x$ equals 0 and not -4. The system lacks a solution. We have infinitely many solutions if the equations instead can be described with the same slope and the same value for the y-intercept. For example:

$$\begin{cases} y & = 3x \\ 13 + 12x & = 4y + 6 + 7 \end{cases} \tag{7.13}$$

We rewrite the second equation:

$$\begin{aligned} 13 + 12x &= 4y + 6 + 7 \\ 12x &= 4y + 6 + 7 - 13 \\ 3x &= y \end{aligned} \tag{7.14}$$

This means that the system of equations can be written:

$$\begin{cases} y & = 3x \\ 3x & = y \end{cases} \tag{7.15}$$

Since the two equations is the same the has an infinite number of solutions. Another situation that leads to an infinite number of solutions is if we have a linear system of equations with more variables than equations, since we then cannot find any unique definitions for the variables in the system.

7.3 Row Reduction

Another useful method that can be used to find the solution for linear systems of equations is elimination, also called row reduction or Gaussian elimination. Suppose we have the following system:

$$\begin{cases} x = 6 + 2y \\ 2x = 3 - 8y \end{cases} \tag{7.16}$$

When we use row elimination we can both add and subtract an entire row with the values from another row, and multiply individual rows with a constant. These two operations can also be combined. To find the solution to the system we can therefore multiply row 1 by the number 2:

$$\begin{cases} (x = 6 + 2y) \times 2 \\ 2x = 3 - 8y \end{cases} \tag{7.17}$$

We then get the system of equations:

$$\begin{cases} 2x = 12 + 4y \\ 2x = 3 - 8y \end{cases} \tag{7.18}$$

We then take row 2 minus row 1 (we subtract row 1 from row 2):

$$\begin{array}{r} -\,(2x = 12 + 4y) \\ \underline{(2x = 3 - 8y)} \\ 0 = -9 - 12y \end{array} \tag{7.19}$$

The third row, below the line, shows the difference between row 2 minus row 1. From the bottom row we can solve for y and then get the solution y^*:

$$\begin{aligned} -9 - 12y &= 0 \\ y^* &= -\frac{3}{4} \end{aligned} \tag{7.20}$$

This value we can then substitute for y in the original row 1 in the system of equations. This gives us the solution for x:

$$\begin{aligned} x &= 6 + 2y^* \\ x^* &= 6 + 2\left(-\frac{3}{4}\right) = 9/2 \end{aligned} \tag{7.21}$$

Let us check our solution by substituting x^* into row 2 and calculate y^*:

$$\begin{aligned} 2\left(\frac{9}{2}\right) &= 3 - 8y \\ y^* &= \frac{1}{8}(3-9) = -\frac{3}{4} \end{aligned} \tag{7.22}$$

This seems to be correct. The result becomes the same as when we used substitution. We can see this if we for example take the definition $x = 6 + 2y$ from row 1 and substitute into row 2:

$$\begin{aligned} 2x &= 3 - 8y \\ 2(6+2y) &= 3 - 8y \\ y^* &= -\frac{3}{4} \end{aligned} \tag{7.23}$$

7.4 Taking logarithms in systems of equations

Even when we have systems of equations we can benefit from being able to add or remove logarithms. Here is a linear system of equations with two equations and an equal number of variables, x and y. Other letters symbolize constants:

$$\begin{cases} y + 12 & = 3 \\ 5^x + 12y & = 8 \end{cases} \tag{7.24}$$

The upper equation gives us the solution for $y^* = -9$. This we can substitute into the lower equation. Thereafter we can rewrite in logarithmic form to solve for x:

$$\begin{aligned} 5^x + 12y^* &= 8 \\ 5^x + 12 * (-9) &= 8 \\ 5^x &= 8 + 108 \\ \ln 5^x &= \ln 116 \\ x \ln 5 &= \ln 116 \\ x^* &= \frac{\ln 116}{\ln 5} \approx 2.95 \end{aligned} \tag{7.25}$$

Consider the following example of logarithmic functions in a system of equations:

$$\begin{cases} \log x + 2\log y & = 4 \\ \log y - 3 & = \log x \end{cases} \tag{7.26}$$

We therefore use $\log x = \log y - 3$ in the upper equation:

$$\begin{aligned} \log x + 2\log y &= 4 \\ (\log y - 3) + 2\log y &= 4 \\ 3\log y &= 7 \\ y^3 &= 10^7 \\ y^* &= 10^{\frac{7}{3}} \end{aligned} \tag{7.27}$$

Now we have a definition for y, which we substitute into the lower equation in the system of equations (7.26) :

$$\begin{aligned} \log y^* - 3 &= \log x \\ \log\left(10^{\frac{7}{3}}\right) - 3 &= \log x \\ \frac{7}{3} - 3 &= \log x \\ -\frac{2}{3} &= \log x \\ x &= 10^{-\frac{2}{3}} \end{aligned} \tag{7.28}$$

Now we have the solutions $y^* = 10^{\frac{7}{3}}$ and $x^* = 10^{-\frac{2}{3}}$. Verify these solutions by substitution.

7.5 Three or more variables

So far we have only gone through systems of equations with two variables but systems of equations can also have several variables. Let us look at the following system as an example, where we have the three variables x, y and z:

$$\begin{cases} x &= 5 + y \\ y &= 13 - z \\ z &= x + y \end{cases} \tag{7.29}$$

We shall now investigate if the system has one or infinitely many solutions. The definition of z from row 3 we can substitute into row 2:

$$\begin{aligned} y &= 13 - z \\ y &= 13 - (x + y) \end{aligned} \tag{7.30}$$

From this we solve for a new definition for x:

$$y = 13 - x - y \tag{7.31}$$

$$x = 13 - 2y \tag{7.32}$$

This definition of x we substitute into row 1 in equation (7.29) , which gives the solution for y:

$$\begin{aligned} x &= 5 + y \\ (13 - 2y) &= 5 + y \\ 3y &= 8 \\ y^* &= \frac{8}{3} \end{aligned} \tag{7.33}$$

From this we can in turn find solutions for x and z. We begin by substituting y^* into equation (7.32) :

$$\begin{aligned} x &= 13 - 2y^* \\ x^* &= 13 - 2 * \left(\frac{8}{3}\right) = \frac{23}{3} \end{aligned} \tag{7.34}$$

By using the solutions for x^* and y^* in the third equation in the system we get the solution for z:

$$z^* = \frac{23+8}{3} = \frac{31}{3} \tag{7.35}$$

Let us check this by substituting z^* into row 2:

$$y^* = 13 - \frac{31}{3} = \frac{8}{3} \tag{7.36}$$

That seems to be correct. We can also use row elimination, but to facilitate understanding in this case we arrange all terms in the same order on all rows. We start from the system:

$$\begin{cases} x & = 5 + y \\ y & = 13 - z \\ z & = x + y \end{cases} \tag{7.37}$$

We rewrite:

$$\begin{cases} x & = 5 + y \\ 0 & = 13 - z - y \\ x & = z - y \end{cases} \tag{7.38}$$

We add row 3 to row 2::

$$\begin{cases} x & = 5 + y \\ x & = 13 - 2y \\ x & = z - y \end{cases} \tag{7.39}$$

We subtract row 1 from row 2:

$$\begin{cases} x & = 5 + y \\ 0 & = 8 - 3y \\ x & = z - y \end{cases} \tag{7.40}$$

From row two we now get the solution $y^* = \frac{8}{3}$. This we can substitute into row 1, whereupon we find the solution $x^* = \frac{23}{3}$. These two solutions we can then substitute into row 3 and then find $z^* = \frac{23}{3} + \frac{8}{3} = \frac{31}{3}$.

7.6 Some more examples

We shall now investigate the solution to the following system of equations:

$$\begin{cases} y = 1 + x \\ x = 2 - y \end{cases} \tag{7.41}$$

In this case we can relatively easily create two definitions for y and set these equal to each other and solve for x^*. We begin by rewriting the second equation in the system so that we get $y = 2 - x$. We set these equal to each other and thereby find the solution for x:

$$\begin{aligned} 2 - x &= 1 + x \\ x^* &= \frac{1}{2} \end{aligned} \tag{7.42}$$

We substitute this into the first equation and get the solution for y:

$$\begin{aligned} y &= 1 + x^* \\ y^* &= 1 + \frac{1}{2} = \frac{3}{2} \end{aligned} \tag{7.43}$$

Verify these solutions by substitution. Consider a similar example but with fractions and test whether we can find a solution for x and y:

$$\begin{cases} y = -\frac{3}{4} - \frac{5}{6}x \\ x = -1 + y \end{cases} \tag{7.44}$$

The lower equation in the system we can rewrite to $y = x + 1$. We set the two definitions for y equal to each other and solve for x^*:

$$\begin{aligned} x + 1 &= -\frac{3}{4} - \frac{5}{6}x \\ \frac{11}{6}x &= -\frac{7}{4} \\ x^* &= -\frac{21}{22} \end{aligned}$$

This gives us the solution for y:

$$y^* = -\frac{3}{4} - \frac{5}{6}\left(-\frac{21}{22}\right) = \frac{1}{22} \tag{7.45}$$

Try yourself to substitute the solutions into the original system of equations to check that it is correct. Here we have a system of equations with the variables x, y and z:

$$\begin{cases} x = y + 2z \\ y = 5 - x + 3z \\ z = 1 \end{cases} \tag{7.46}$$

Already from equation (7.46) we see that $c^* = 1$. This we can use in row 2:

$$\begin{aligned} y &= 2 - x + 3z^* \\ y &= 5 - x + 3 \end{aligned} \tag{7.47}$$

This definition of y as well as row 3, $z = 1$, we can substitute into row 1:

$$\begin{aligned} x &= y + 2z \\ x &= (5 - x + 3) + 2 \\ x^* &= 5 \end{aligned} \tag{7.48}$$

Row 3, $z = 1$, and $x^* = 5$ we can substitute into row 2:

$$\begin{aligned} y &= 5 - x + 3z \\ y^* &= 5 - 5 + 3 = 3 \end{aligned} \tag{7.49}$$

We have now found the three solutions we are looking for the system: $x^* = 5$, $y^* = 3$ and $z^* = 1$. Verify these solutions by substitution. Consider the following expression with the same letters, but new numbers and equations:

$$\begin{cases} x = 10 - 0.3z \\ y = x + 5 \\ z = x - y \end{cases} \tag{7.50}$$

Row 2 and 3 in combination give:

$$z^* = x - x - 5 = -5 \tag{7.51}$$

We substitute z^* into row 1 and get:

$$x^* = 10 - 0.3 * (-5) = 11.5 \tag{7.52}$$

This we can substitute into row 2:

$$y^* = 11,5 + 5 = 16,5 \tag{7.53}$$

Let us now take the following system of equations as an example:

$$\begin{cases} x = 23 - 5y + z \\ x = 2 + 8z \end{cases} \tag{7.54}$$

Since this system of equations has more variables (three: x, y and z) than equations (two), the system has an infinite number of solutions.

7.7 Beyond the third dimension

Note that an ordinary graph with two axes has two dimensions: height and width, the vertical and the horizontal axis, which are usually called y and x. As we went through above there is no limitation for how many variables that can be included in an equation. However, it becomes considerably more complicated to illustrate relationships between more than two variables. We can draw two variables in a graph with two axes. If we have three variables we can draw this in a three-dimensional graph as a cube.

We can also use other tricks to add additional dimensions, for example let a variable determine the size of the dots in a graph, or illustrate one or more variables with the help of colors. But there is still a limitation of how much information that can be illustrated in one and the same graph. Already three variables in the same graph often becomes difficult to overview.

Does this mean that it is meaningless to use more than three variables simultaneously? No. Suppose we have a theory that a person's life expectancy depends on their income and that this can be written in the form of the equation for a straight line:

$$y = a + bx \tag{7.55}$$

where y and x are variables and a and b are coefficients. The coefficient a is a constant that indicates the value that y takes when $x = 0$, the line's y-intercept. Coefficient b is the constant slope coefficient that indicates how much y changes when x increases by one unit.

Say now that we in addition to income also think that life expectancy in various ways depends on an additional five variables: (1) the person's gender, (2) genetic conditions, (3) eating habits, (4) alcohol consumption and (5) number of kilometers the person jogs each week. These other variables we can in that case describe as additional dimensions in our theoretical model. We now have six explanatory variables and can write these into our linear equation as $x_1,\ x_2,\ x_3,\ x_4,\ x_5$ and x_6. Each variable gets its own slope coefficient, which we call $b_1,\ b_2,\ b_3,\ b_4,\ b_5$ and b_6. Each slope coefficient describes how y changes when the respective explanatory variable $(x_1, ..., x_6)$ changes:

$$y = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5 + b_6x_6 \tag{7.56}$$

We are thus often interested in relationships, covariations and correlations that are difficult to illustrate in graphs and pictures.

In this model with several variables have a longer equation compared to what we have worked with earlier. But despite this we still describe a relatively "simple" linear relationship between y and the each explanatory variable, x_1 to x_6. To see why this is a rather simple approach, we can for example imagine that the different explanatory variables $(x_1, ..., x_6)$ also might affect each other and also affect each other's slope coefficients $(b_1, ..., b_6)$. Parts of the relationship between the variables might only apply under certain conditions, and so on. It is not difficult to reason about several changes to the theory that would require more complex mathematics to be described.

In our everyday life we often consider complex relationships and correlations between different phenomena. We can easily understand that our lives are affected by many different types of phenomena that in many cases also affect each other. Certain phenomena we additionally have limited or no influence over. Despite this we seem to have surprisingly easy to become convinced by simple arguments about complex and significant causal relationships through banal arguments within politics or in advertising, how one phenomenon causes another phenomenon. It is difficult to be human.

7.8 Chapter summary

- A system of equations consists of several equations that are valid simultaneously. Linear systems of equations can have a unique solution, infinitely many solutions or lack a solution. For a system of equations with two equations a unique solution means that the equations' lines in a graph meet at one point. An infinite number of solutions means that the lines have the same x- and y-values at all points. If the system lacks a solution the lines never meet.
- To find solutions to a system of equations we can use substitution or row reduction.
- A graph with two axes is a good way to illustrate how two variables, for example x and y, relate to each other. It is usually difficult to illustrate the relationship between three or more variables.

7.9 Exercises

1. Solve using substitution:

(a) (displaystyle begin{cases} y=3x-1 y=x+5 end{cases})

(b) (displaystyle begin{cases} x=2+y x=8-2y end{cases})

(c) (displaystyle begin{cases} y=4x 2x+y=18 end{cases})

Answer: (a) $3x - 1 = x + 5 \Rightarrow x^* = 3,\ y^* = 8$, (b) $2 + y = 8 - 2y \Rightarrow y^* = 2,\ x^* = 4$, (c) $2x + 4x = 18 \Rightarrow x^* = 3,\ y^* = 12$

2. For each system, determine whether it has a unique solution, infinitely many solutions, or no solution:

(a) (displaystyle begin{cases} y=2x+1 y=2x+4 end{cases})

(b) (displaystyle begin{cases} y=3x-2 2y=6x-4 end{cases})

(c) (displaystyle begin{cases} y=x+1 y=-x+5 end{cases})

Answer: (a) Same slope, different intercept: no solution, (b) Second equation is $y = 3x - 2$, same as first: infinitely many solutions, (c) Unique solution: $x^* = 2,\ y^* = 3$

3. Solve using row reduction:

(a) (displaystyle begin{cases} x+y=7 2x-y=2 end{cases})

(b) (displaystyle begin{cases} 2x+y=9 x-y=0 end{cases})

(c) (displaystyle begin{cases} x+2y=8 3x-y=3 end{cases})

Answer: (a) Add rows: $3x = 9 \Rightarrow x^* = 3, y^* = 4$, (b) Add rows: $3x = 9 \Rightarrow x^* = 3, y^* = 3$, (c) Row 2 $\times 2$ then add to row 1: $7x = 14 \Rightarrow x^* = 2, y^* = 3$

4. In a market for a good, demand is given by $p = 100 - 2q$ and supply by $p = 10 + 3q$, where p is price and q is quantity.

(a) Set up the system of equations.

(b) Find the equilibrium price and quantity.

(c) At (q=15) , calculate demand price and supply price. Is there excess supply or excess demand?

Answer: (a)

$$\begin{cases} p = 100 - 2q \\ p = 10 + 3q \end{cases}$$

, (b) $100 - 2q = 10 + 3q \Rightarrow q^* = 18, p^* = 64$, (c) $p_D(15) = 70, p_S(15) = 55$: excess demand (demand price exceeds supply price)

5. Solve using logarithms where needed:

(a) (displaystyle begin{cases} y+7=4 2^{x}-3y=13 end{cases})

(b) (displaystyle begin{cases} log x=2 log x+log y=5 end{cases})

(c) (displaystyle begin{cases} 5^{x}=25 y=2x-1 end{cases})

Answer: (a) $y^* = -3$; $2^x - 3(-3) = 13 \Rightarrow 2^x = 4 \Rightarrow x^* = 2$, (b) $x^* = 100$; $\log 100 + \log y = 5 \Rightarrow \log y = 3 \Rightarrow y^* = 1000$, (c) $5^x = 5^2 \Rightarrow x^* = 2, y^* = 3$

6. Solve the following systems with three variables:

(a) (displaystyle begin{cases} z=2 y=z+3 x=y+1 end{cases})

(b) (displaystyle begin{cases} x+y+z=12 x=3y z=2y end{cases})

(c) (displaystyle begin{cases} x+y=7 y+z=8 x+z=9 end{cases})

Answer: (a) $z^* = 2, y^* = 5, x^* = 6$, (b) $3y + y + 2y = 12 \Rightarrow y^* = 2, x^* = 6, z^* = 4$, (c) $x + y + z = 12 \Rightarrow z^* = 5, x^* = 4, y^* = 3$

7. Check whether the given values are solutions to each system:

(a) (x$^{\{*\}=2,,y}$\{*\}=3) for(displaystyle begin{cases} y=x+1 2x+y=7 end{cases})

(b) (x$^{\{*\}=1,,y}$\{*\}=2) for(displaystyle begin{cases} 3x+y=4 x-y=0 end{cases})

(c) (x$^{\{*\}=4,,y}$\{*\}=-1) for(displaystyle begin{cases} x+2y=2 3x-y=13 end{cases})

Answer: (a) $3 = 2+1$ and $4+3 = 7$: yes, (b) $3(1)+2 = 5 \neq 4$: no, (c) $4+2(-1) = 2$ and $12-(-1) = 13$: yes

8. Solve using row reduction or substitution:

(a) (displaystyle begin{cases} x+y+z=6 x-y+z=2 x+y-z=2 end{cases})

(b) (displaystyle begin{cases} x=3 y=x+2 z=x+y end{cases})

(c) (displaystyle begin{cases} 2x+y=10 y+z=7 x+z=6 end{cases})

Answer: (a) Row 2 $-$ Row 1: $y^* = 2$; Row 3 $-$ Row 1: $z^* = 2$; $x^* = 2$, (b) $x^* = 3, y^* = 5, z^* = 8$, (c) $x^* = 3, y^* = 4, z^* = 3$

9. Three friends A, B and C together spent 240 on a dinner. A spent twice as much as B. C spent 40 more than B.

(a) Set up a system of equations.

(b) Solve for what each person paid.

(c) What percentage share of the total did each person pay?

Answer: (a)

$$\begin{cases} A + B + C = 240 \\ A = 2B \\ C = B + 40 \end{cases}$$

, (b) $2B + B + B + 40 = 240 \Rightarrow B^* = 50,\ A^* = 100,\ C^* = 90$, (c) A: $100/240 \approx 41.7\%$, B: $50/240 \approx 20.8\%$, C: $90/240 = 37.5\%$

10. In a simple labor market, labor demand is $L_D = 200 - 5w$ and labor supply is $L_S = 20 + 3w$, where w is the wage.

(a) Find the equilibrium wage and employment level.

(b) A minimum wage of 25 is introduced. Calculate (L_{D}) and (L_{S}) at this wage.

(c) Is there excess supply or excess demand at the minimum wage, and by how many workers?

Answer: (a) $200 - 5w = 20 + 3w \Rightarrow w^* = 22.5,\ L^* = 87.5$, (b) $L_D(25) = 75,\ L_S(25) = 95$, (c) Excess supply (unemployment) of $95 - 75 = 20$ workers

Chapter 8

Theories of life, death and dating

In this chapter we will go through how we can use mathematics to describe social science theories. The examples are greatly simplified and intended only as illustration. In this part of the work we are only superficially interested in reality. Later on, in Part II and III , we will go through how we can use mathematics to more carefully compare our theories against information about reality.

8.1 Income and life expectancy

Suppose we have a theory that higher income extends life for people. We describe our theory in the following way:

$$Y = a * X \tag{8.1}$$

where Y is the number of years a person lives, X is the number of USD the person has in income during their adult life and a is a coefficient that describes in what way X affects Y. On an overall level it seems reasonable that people with higher incomes live longer. Many people with higher incomes have better opportunity to choose healthy alternatives, greater opportunities to rest and so on. But our model feels a bit too simple so we could think about whether we can improve it.

As our model is written right now the relationship is linear, for each USD of higher income life is extended by a. But that might not be so credible. The large effect of higher income might arise primarily at low incomes, for example when going from 0 USD to a small income. In that case we now have a theory that income X has a diminishing effect on life expectancy Y. One way to describe this mathematically is the following:

$$Y = X^b,\ 0 < b < 1 \tag{8.2}$$

The exponent b is between 0 and 1. Exactly what value b has we are a bit uncertain about. We might plan to study it by collecting data on people's life expectancy and incomes, but not right now. For our theoretical discussion this information is sufficient as it is now.

That a single variable, income, could give a complete explanation to such a complex phenomenon as life expectancy also doesn't seem so credible. Let us therefore also add a variable that summarizes all hereditary biological conditions, for example risks for certain diseases. We call this new variable G. Our theoretical model now becomes:

$$Y = X^b + G,\ 0 < b < 1 \tag{8.3}$$

Our theoretical model is still very limited and if we wish we could add more variables. There are no mathematical limitations for what we add. What should be included in the model simply depends on what we are going to use the mathematics for.

8.2 Mathematical adventures in national income

We have earlier compared data on GDP. Alongside reality we can also describe GDP theoretically, for example like this:

$$Y = C + G + I + X - M \tag{8.4}$$

where Y is GDP, C is private households' consumption of goods and services, G is public sector consumption, I is investments in machines and factories, X is export of goods and services and M is import. This is a common way to describe GDP and you can find real data on these parts online. But here we only discuss these things as abstract concepts.

Theoretically there is nothing that prevents us from adding more letters. Say for example that we want to distinguish investments I into two parts: investments in manufacturing industries that create things and investments in service industries that create services. We call these two components I_t and I_s.

We also want to distinguish household consumption with a variable for rich households and a variable for poor households. We call these variables C_{rich} and C_{poor}. We need not define precisely where to draw the line for rich and poor respectively. We now get the following definition:

$$Y = C_{\text{rich}} + C_{\text{poor}} + I_t + I_s + X - M \tag{8.5}$$

We can continue to build on our description if we feel like it. We might for example also want to add letters that symbolize natural resources and divide this into more or less environmentally harmful production. We might want to have a measure of export and import that occurs in contact with certain types of countries.

But now we are mostly adding things because we can. What should be included in the equation depends primarily on what the purpose is, what we are going to use the mathematics for and what we want to discuss or illustrate. We can for example use such an equation to show how different things in the economy affect each other or how important they are for our economic prosperity.

8.3 A matter of class

A large amount of social science is interested in how the population can be categorized into social classes, which in turn can mean slightly different things. For example working class and capitalist class. If we are to work analytically with this we need a precise definition and then mathematics can help us. Say for example that we make the following division into three classes:

- The working class in society consists of those individuals whose total income during a year comes more than 50% from wage income, earned through employment.
- The capitalist class in society consists of those individuals whose income during a year comes more than 50% from capital income.
- Other individuals in society, whom we call "others".

We disregard all possible conceivable problems with this division, for example ambiguities in the definitions, difficulties in collecting data and so on. Would this division work in practice for analytical work? Again, just as in the preceding examples, it depends on what we hope to be able to discuss, illustrate and examine. There are certainly arguments and phenomena where this division can serve a purpose. And there are guaranteed questions where this division is far too crude or just meaningless in general.

Another discussion about class with a long tradition is that about differences in lifestyles. Say for example that we have a theory about people's relationship to opera performances and that this relationship depends partly on what income the person's household has and partly on the number of years the person spent on university education. We formulate this mathematically:

$$O_i = C_i^a + U_i^b,\ 0 < a < 1,\ 1 < b \leq 2 \tag{8.6}$$

where O_i is interest in opera for person i. C_i^a is income for person i's household. The letter a is a coefficient that describes what significance C has for O. We here define a condition for this: $0 < a < 1$, which means that income indeed increases the person's interest in opera but that the effect is diminishing. U_i^b is the person's time in university education and b is a coefficient with the following condition: $1 < b \leq 2$. Since b is over 1 this means that the longer the person studies, the more the interest increases and the rate of increase also increases.

8.4 Supply and demand

Within social science, linear equation systems are used among other things to reason about how we think that supply and demand work for a good or service, and how this contributes to determining the price for a good or service. The word "supply" refers to how much goods or services the sellers in a market want to offer at a given price. The word "demand" refers to how large a quantity of goods or services the buyers want to pay for at a given price. The word "equilibrium" refers in this case to the system having a solution for our variables. Mathematics is only one way to describe how we think reality can work.

We will now give an example of a linear equation system for supply and demand. You can yourself think of what good or service this might be thought to describe. The simplest is probably to think of a clearly bounded market with a physical good that is sold and demanded by a limited group of producers and consumers, perhaps in the style of some competing vegetable vendors at a market square. In reality, markets can be very difficult to clearly bound. Supply and demand can moreover work in several complex ways. In this case we have two linear equations for supply and demand, which we can write as an equation system:

$$\begin{cases} q_{\text{supply}} & = 3p \\ q_{\text{demand}} & = 20 - 2p \end{cases} \tag{8.7}$$

where q and p are our variables. The other numbers are coefficients. In this case our equation system consists of two definitions of q: one equation for supply and one for demand in the market. Our coefficients describe how we think theoretically that supply and demand work for a good or service. The two equations in (8.7) describe precisely this, how supply and demand of the production quantity q relate to the price p.

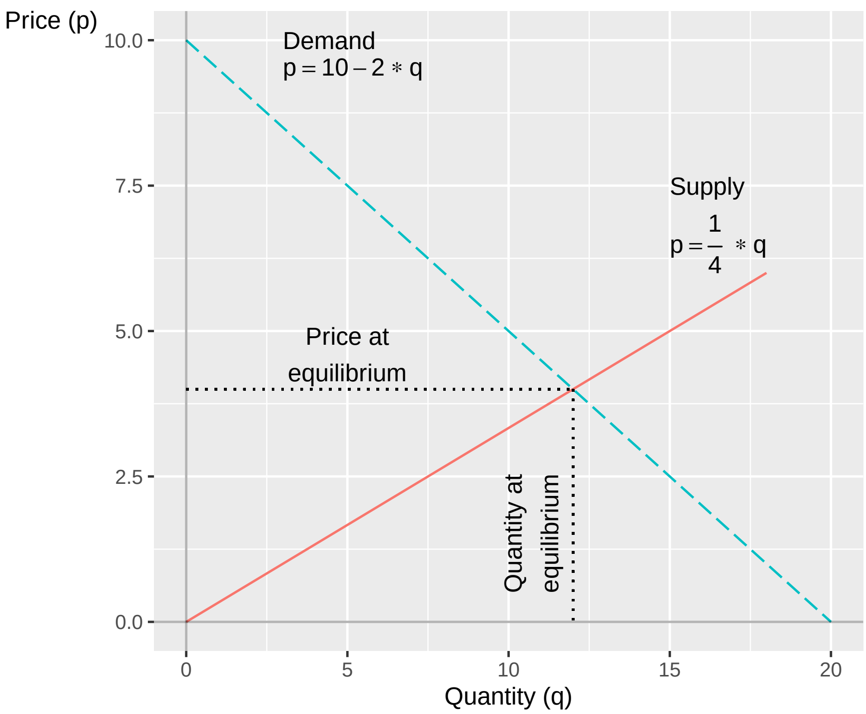


Figure 8.1: Supply and demand based on two linear equations

What we are looking for is the point where $q_{\text{supply}} = q_{\text{demand}}$. In this situation we think that the market is in equilibrium, when supply and demand are equal. We can start by drawing up the system in a graph, see figure 8.1 . In the graph the lines meet at a unique point, which defines the equilibrium price p^* and the equilibrium quantity q^*. To calculate the equilibrium price we set q_{supply} and q_{demand} equal to each other and solve for p:

$$\begin{aligned} q_{\text{supply}} &= q_{\text{demand}} \\ 3p &= 20 - 2p \\ p^* &= 20/5 = 4 \end{aligned} \tag{8.8}$$

This p^* we can substitute into the equation system and solve for q for supply and demand in equilibrium:

$$\begin{aligned} q_{\text{supply}} =& 3p = 12 \\ q_{\text{demand}} =& 20 - 2p = 12 \end{aligned} \tag{8.9}$$

Both equations give that $q^* = 12$, whereupon we have the system's unique solution. We can check that this is correct with the help of substitution.

8.5 Two types of cookies

Linear equations can also be used to describe alternatives in a choice situation. For example: ahead of this year's big cake sale the local scout organization should choose strategy in the big bake. One alternative is that the scouts bake real cakes with expensive ingredients. The cakes will in that case become exquisite, but expensive to make. The profit per sold package will be 20 USD. We can describe this in the equation:

$$\text{profit}_1 = 20 * \text{cookies}_1 \tag{8.10}$$

Another alternative is to bake the cakes with water and that big sack of old flour they found in the container behind the clubhouse last summer. No other ingredients. Since these cakes are super cheap to make, the profit will now be 50 USD per cake tin. The disadvantage is that the organization will incur fines of 1,000 USD for having poisoned some of their customers. The expected profit from this alternative can be described in the following equation:

$$\text{profit}_2 = 50 * \text{cookies}_2 - 1,000 \tag{8.11}$$

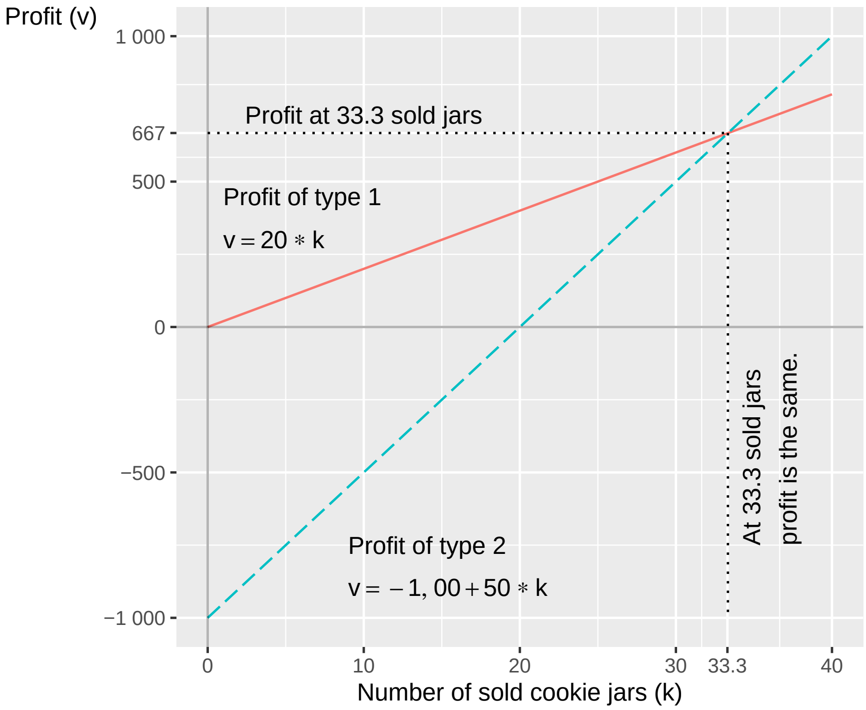


Figure 8.2: Profit from two different types of cookies

How many cake tins must the scouts sell for the expected profit in alternatives 1 and 2 to become equivalent? An equivalent profit means that the two equations should have the same value. We can thus calculate the number of cake tins by setting the two equations equal to each other. This means that we have a linear equation system:

$$\begin{cases} \text{profit}_1 & = 20 * \text{cookies}_1 \\ \text{profit}_2 & = 50 * \text{cookies}_2 - 1,000 \end{cases} \tag{8.12}$$

We now define $\text{cookies}_1 = \text{cookies}_2 = \text{cookies}$ and calculate:

$$\begin{aligned} 50 * \text{cookies} - 1,000 &= 20 * \text{cookies} \\ 30 * \text{cookies} &= 1,000 \\ \text{cookies}^* &= 100/3 \approx 33.33 \end{aligned} \tag{8.13}$$

At 33.3 sold cake tins the two alternatives are equivalent. The profit then becomes:

$$\begin{aligned} &\text{Alternative 1: profit} = 20 * 33.3 = 667 \\ &\text{Alterantive 2: profit} = 50 * 33.3 - 1,000 = 667 \end{aligned} \tag{8.14}$$

This is illustrated in figure 8.2 , where the profit for the two alternatives is described with two lines. The lines meet at the solution we have calculated: 33.3 sold cake tins with 667 USD profit.

8.6 Finding the one

When we calculate data about reality we like to use material that is easy to quantify, that easily allows itself to be measured. One such thing is money, which is moreover a unit of measurement that many people understand. The disadvantage is that many decisions in reality are more complex. As mentioned earlier, however, mathematics can help us discuss reality even when we are interested in phenomena that can be harder to put numbers on.

Let us now take an example where we say that the numbers instead measure how fun something is. We begin by describing this example with numbers, which possibly can be easier to understand. After that we will go through how we can work with the same type of theory without numbers, which many times is at least as useful and sometimes better. The point of this is to illustrate how mathematics can help us reason, even if we cannot calculate exact answers.

In this example we use a simplified happiness scale from 0-100 as a pedagogical tool. While it is difficult (to say the least) to measure happiness in reality, numbers help us understand the mathematical principles. What matters is not the exact values, but how we can use systems of equations to compare strategies.

Erik wants to get married as soon as possible and is considering different strategic alternatives for meeting the right one. Erik doesn't intend to jump between strategies and the marriage goal must happen this year. It is thus about choosing a strategy and sticking to it. Erik estimates their happiness today at 80 on a scale from 0 to 100, and hopes to be able to increase their happiness to 100 by meeting a partner. Erik doesn't like dating however and each date brings down Erik's happiness a little.

Strategy 1: Using a free dating app. Just installing the app and starting to go through the profiles lowers Erik's happiness by 50 percentage points. For each date Erik has to go on, Erik's happiness drops by an additional two percentage points. Let us describe this in an equation:

$$\text{happiness} = 80 - 2 * date_{\text{app}} - 50 \tag{8.15}$$

Strategy 2: Erik's best friend has chosen four candidates who are ready for marriage. Since the social effort is now a bit higher, each date lowers Erik's happiness by 22 percentage points:

$$\text{happiness} = 80 - 22 * date_{\text{friend's friend}} \tag{8.16}$$

This gives us a linear equation system with two equations. Now we wonder at what number of dates the two strategies can be considered equivalent, as well as what level of (un)happiness this would entail. Let us denote the variables $d_1 = date_{\text{app}}$ and $d_2 = date_{\text{friend's friend}}$ and calculate how many dates are required for the two strategies to be considered equivalent:

$$\begin{cases} \text{happiness} & = 80 - 2d_1 - 50 \\ \text{happiness} & = 80 - 22d_2 \end{cases} \tag{8.17}$$

To calculate the number of dates when both strategies give the same expected result we define $d_1 = d_2 = d$ and set the two equations equal to each other:

$$\begin{aligned} 80 - 22d &= 80 - 2d - 50 \\ 50 &= 20d \\ 2,5 &= d^* \end{aligned} \tag{8.18}$$

At 2.5 dates the two strategies can be considered equivalent. We can also see this by trying to substitute the result for d^* into our system:

$$\begin{aligned} \text{Strategy 1: } \text{happiness} &= 80 - 2 * 2,5 - 50 = 25 \\ \text{Strategy 2: } \text{happiness} &= 80 - 22 * 2,5 = 25 \end{aligned} \tag{8.19}$$

The solution for the system is $d^* = 2.5$ and $\text{happiness}^* = 25$. Figure 8.3 illustrates the two strategies as straight lines in the graph and the equilibrium $(d^*, \text{happiness}^*) = (2.5,\ 25)$ as the point where the two lines meet.

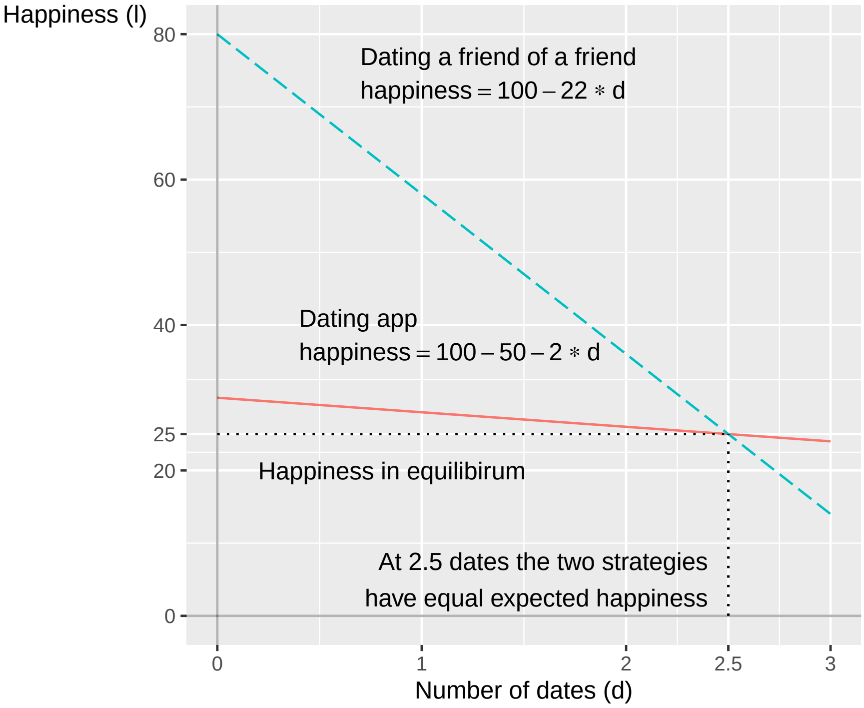


Figure 8.3: Two different strategies for dating

Measuring happiness on a scale from 0 to 100 is a bit clumsy. But by setting up equations we can more clearly discuss what conditions must be fulfilled for one or the other relationship to hold. If we take our example with dating we could write out the equations but use letters instead of numbers for our coefficients. For example, in the case of Erik's dating the equation system could be written like this instead:

$$\begin{cases} \text{happiness} & = m - ad_1 - b \\ \text{happiness} & = m - cd_2 \end{cases} \tag{8.20}$$

The letter m is current level of happiness, the two types of dates are symbolized by d_1(the app) and d_2(friend's friend), a and c are the slope coefficients (the cost) for going on different types of dates and b is the emotional cost of starting the dating app. From this we can solve out the following definitions for the variables d(number of dates) and happiness:

$$\begin{aligned} d^* &= \frac{b}{c-a} \\ \text{happiness}^* &= \frac{mc - ma - cb}{c-a} = \frac{m\,(c-a) - cb}{c-a} \end{aligned} \tag{8.21}$$

This allows us to describe how the different variables are connected and what is required for d^* to assume a positive value. Given that $b > 0$, then $c > a$ must hold. That is, given that happiness decreases when Erik opens the dating app ($b > 0$), then the happiness decrease from going on a date with the friend's friend (c) must be greater than the happiness decrease per date via the app (a).

Now let us explore an alternative scenario where Erik's happiness increases from going on dates. While there's no single correct way to model this experience mathematically, a straightforward modification is to change the coefficients a and c from minus to plus:

$$\begin{cases} \text{happiness} & = m + ad_1 - b \\ \text{happiness} & = m + cd_2 \end{cases} \tag{8.22}$$

Erik's happiness still decreases from opening the app, which we see through the fact that the term b is negative. Whether Erik in this situation prefers to date friends' friends or use the dating app depends on what values the coefficients a, b and c have.

8.7 A theory about work

In this example we will describe the labor market with a linear equation system. The theories are simplified to make the mathematical description easier. While this example to some degree illustrate how we can think about real labor markets, the models (just as in all theoretical examples in this book) make several unrealistic assumptions.

Our equation system consists of two equations that give a simplified picture of how we can imagine that wages (W) and prices (P) are determined:

$$\begin{cases} \text{Wage-setters: } \frac{W}{P} = \frac{a}{U^b}, & b > 0 \\ \text{Price-setters: } \frac{P}{W} = \frac{c}{U^d}, & c \geq 0 \end{cases} \tag{8.23}$$

The first equation describes how wages W are set with a markup, coefficient a, over prices P. So if workers face higher prices they also demand higher wages. The second equation describes how prices P are set with a markup c on wages W. If companies face higher wages they set higher prices. Unemployment U has a negative effect on both wages and prices, and this effect is determined by coefficients b and d.

The letters a, b, c and d are constants that summarize the phenomena that affect wages and prices. Exactly what this symbolizes we don't worry about here. It could for example be taxes, legislation or how employees and employers negotiate wages and conditions via their trade unions and employer organizations. The first equation (W/P) is also called the wage-setting curve, which can be described as a supply curve for labor. The second equation (P/W) is also called the price-setting curve and can be described as a demand curve for labor.

We now seek a solution for the variable W/P, real wage (see section 3.7), and U, percent unemployment. We begin by rewriting the second equation and solving for $W/P = U^d/c$. We substitute this into the first equation:

$$\begin{aligned} \frac{W}{P} &= \frac{a}{U^b} \\ \frac{U^d}{c} &= \frac{a}{U^b} \\ U^{d+b} &= ac \\ U^* &= (ac)^{\frac{1}{b+d}} \end{aligned} \tag{8.24}$$

The solution for U^* we can then use to solve out $(W/P)^*$. We substitute U^* into the first equation in the system (8.23) :

$$\begin{aligned} \left(\frac{W}{P}\right)^* &= \frac{a}{(U^*)^b} \\ &= \frac{a}{(ac)^{\frac{b}{b+d}}} \\ &= \frac{a^{1-\frac{b}{b+d}}}{c^{\frac{b}{b+d}}} \\ &= a^{\frac{d+b-b}{b+d}} c^{\frac{-b}{b+d}} \\ &= a^{\frac{d}{b+d}} c^{\frac{-b}{b+d}} \end{aligned} \tag{8.25}$$

Or we can start from the left side of the equation (8.25) and substitute U^* there:

$$\left(\frac{W}{P}\right)^{*} = \frac{(U^{*})^{d}}{c} = \frac{(ac)^{\frac{d}{b+d}}}{c} = a^{\frac{d}{b+d}} c^{\frac{-b}{b+d}} \tag{8.26}$$

Now we have the solution for the two variables U and W/P:

$$\left(U^{*}, (W/P)^{*}\right) = \left((ac)^{\frac{1}{b+d}}, a^{\frac{d}{b+d}} c^{\frac{-b}{b+d}}\right) \tag{8.27}$$

The parenthesis in the right side describes the solutions for the respective variable. U^{*} is a definition of what in social sciences is called equilibrium unemployment. This is not a linear equation system but with the help of logarithmization we can make it linear. We therefore take the logarithm of both sides of the respective equation in the system (8.23) :

$$\begin{cases} \log\left(\frac{W}{P}\right) = \log\left(\frac{a}{U^{b}}\right), & b > 0 \\ \log\left(\frac{P}{W}\right) = \log\left(\frac{c}{U^{d}}\right), & c \geq 0 \end{cases} \tag{8.28}$$

$$\begin{cases} \log W - \log P = \log a - \log U^{b} \\ \log P - \log W = \log c - \log U^{d} \end{cases}$$

$$\begin{cases} w - p = \log a - bu \\ p - w = \log c - du \end{cases}$$

where $w - p = \log\left(\frac{W}{P}\right)$ and $bu = b \log U$. The letter u is the logarithm of percent unemployment and a, b, c and d are coefficients that define how our variables are connected. Just like above we rewrite the second equation, set the two definitions of $w - p$ equal to each other and solve for u. To save space we do not write out log():

$$\begin{aligned} du - c &= a - bu \\ u\,(b + d) &= a + c \\ u^{*} &= \frac{a + c}{b + d} \end{aligned} \tag{8.29}$$

The definition of u^{*} is the logarithmized version of the solution in equation (8.27) . To convert to unemployment in percent we take the exponent:

$$\begin{aligned} \exp\left(u^{*}\right) &= \frac{1}{b + d} \exp\left(a + c\right) \\ U^{*} &= ac^{\frac{1}{b+d}g} \end{aligned} \tag{8.30}$$

Figure 8.4 shows how this model can be illustrated in a graph. The relationship between W/P and U is described with the two functions for supply and demand, also called the wage- and price-setting curve. If we read the x-axis from left to right the axis measures percent of the workforce that has work, employment rate. Employment rate can be defined as $1 - U$, where U is percent unemployed. The solution for the variables U^{*} and $(W/P)^{*}$ is the point where the lines meet.

In the diagram to the left is shown how the lines look in ordinary form, as in equation (8.23) . In the diagram to the right are shown the logarithmized functions, as in equation (8.29) .

Figure 8.4: Supply and demand for labor

8.8 External effects

In section 8.4 we described an example where supply and demand determine price and quantity for a good or service in a free market. We can also use this mathematics to describe unintended effects. Let us illustrate with an example. Suppose we have a completely unregulated market, without taxes or fees, for a product where supply and demand can be described with the equation system:

$$\begin{cases} \text{Supply:} & q = 2 + p \\ \text{Demand:} & q = 20 - p \end{cases} \tag{8.31}$$

where q is quantity of the product and p is price. The solution to this equation system is $(q^*, p^*) = (11, 9)$. The problem now is that this quantity, 11 units, does not take into account the total cost for society, the social cost. The production creates emissions that cause long-term damage to the climate. The costs for these damages are not included in the price. In social sciences this phenomenon is called an external effect, or externality. An external effect is a consequence that is not intended, something that neither the consumers nor the producers want to happen, and therefore is not reflected in the price.

Now let's say researchers have calculated what the total cost for society is and believe that the following equation system better reflects both the private and the social cost:

$$\begin{cases} \text{Supply, incl. social cost} & q = 10 + p \\ \text{Demand:} & q = 20 - p \end{cases} \tag{8.32}$$

The solution to this new system is $(q^*, p^*) = (5, 15)$. The demand curve is the same in both examples. The price, now taking into account the social cost, becomes 15 instead of 9. This is not reflected in the actual price in the market. One way to raise the market price to reflect also the social cost could be to introduce a tax on the climate-damaging emissions.

External effects don't have to be negative. If we imagine that supply and demand for education in mathematics for social scientists in a completely free market could be described with the following equation system:

$$\begin{cases} \text{Supply} & q = 1 + p \\ \text{Demand:} & q = 7 - 2p \end{cases} \tag{8.33}$$

would result in equilibrium levels $(q^*, p^*) = (3, 2)$. But now some social scientists have calculated that the social benefit of this type of education is higher than what market demand reflects. The social benefit is instead captured with the following equation system:

$$\begin{cases} \text{Supply:} & q = 1 + p \\ \text{Demand incl social cost:} & q = 16 - 2p \end{cases} \tag{8.34}$$

This would instead mean that $(q^*, p^*) = (6, 5)$. The optimal amount of education in mathematics for social scientists would in that case be double, compared to the market equilibrium. Since the supply equation is the same in both cases, the equilibrium price would be higher in this situation. This example could in that case be an argument for why the government should subsidize this production by paying the difference between the price in a free market ($p = 2$) and the optimal price with regard to social benefits ($p = 5$). In reality, political measures are always more complex than simple hypothetical examples.

8.9 Linear relationships in real life

Many times we want to reformulate our mathematical problems so that these can be described with linear equations, since this type of mathematics often simplifies our analytical work. A challenge with this is that many things in reality are not at all linear. Erik's happiness is probably affected differently by date no. 1 compared to date no. 15, thus a nonlinear relation between happiness and dates. Either Erik is by that point so dulled that happiness barely changes at all. Or maybe the opposite: Erik has kept up their spirits until then but now loses hope, why happiness starts to fall drastically.

The same thing applies to the example concerning market trading. With small price changes it is possibly reasonable to assume that supply in the market square increases by the same small amount. But with more dramatic price increases the market square would start to attract many new sellers and supply would shoot up.

Sometimes linear equations fit better and sometimes worse. Many times it can be that even phenomena that are not linear in their entirety can be linear within a sufficiently large interval that it can still be reasonable to use linear equations. But other times it can be precisely the non-linear part of a relationship between two or more phenomena that is the interesting part. There is no simple answer to when one or the other is more appropriate. It depends on the study's purpose, research question, theory and method. Mathematics can help us analyze things but does not automatically ensure that our analysis becomes smart or correct. Later on we will go into a bit more about how we can handle these phenomena.

8.10 Chapter summary

- Equation systems can among other things be used to illustrate supply and demand in a market, where supply and demand are described in the form of one equation each. Where the equations are equal to each other the lines meet in a graph, which gives equilibrium price and equilibrium quantity of the good or service.
- Equation systems can also be used to describe the trade-off between two different strategies and their payoff, for example in the form of a scout association's planning of cookie sales.

8.11 Exercises

1. The supply and demand for a good can be described with the equation system $q_{\text{supply}} = 3p$

(a) Find the equilibrium price (p^{*})

(b) Suppose supply changes to (q_{text{supply}}=5p) Find the new equilibrium.

(c) With the original supply (q_{text{supply}}=3p) Find the new equilibrium.

Answer: (a) $3p = 20 - 2p \Rightarrow p^* = 4$, (b) $5p = 20 - 2p \Rightarrow p^* = 20/7 \approx 2.86$, (c) $3p = 30 - 2p \Rightarrow p^* = 6$

2. A scout organization has two strategies for cookie sales. Strategy 1 gives a profit of 20 USD per tin: $\text{profit}_1 = 20 \times \text{cookies}$ Strategy 2 gives 50 USD per tin but incurs a fixed cost of 1,000 USD: $\text{profit}_2 = 50 \times \text{cookies} - 1,000$

(a) At how many sold tins are the two strategies equivalent?

(b) What is the profit at that number of tins?

(c) If the scouts expect to sell 40 tins, which strategy gives higher profit?

Answer: (a) $20c = 50c - 1,000 \Rightarrow c^* = 100/3 \approx 33.3$, (b) profit ≈ 667 , (c) Strategy 1: 800 USD, Strategy 2: 1,000 USD; Strategy 2 gives higher profit.

3. Erik is choosing between two dating strategies. Strategy 1 (dating app): $\text{happiness} = 30 - 2d$ Strategy 2 (friend's friend): $\text{happiness} = 80 - 22d$

(a) At how many dates (d^{*})

(b) What is the happiness level at (d^{*})

(c) If Erik goes on 1 date, which strategy gives higher happiness?

Answer: (a) $30 - 2d = 80 - 22d \Rightarrow d^* = 2.5$, (b) happiness $= 25$, (c) Strategy 1: 28, Strategy 2: 58; Strategy 2 gives higher happiness.

4. An unregulated market has supply $q = 2 + p$

(a) Find the free-market equilibrium (left(q^{*},p{*}right))

(b) When social costs are included, supply changes to (q=-4+p) Find the new equilibrium.

(c) By how much does the price change when social costs are included?

Answer: (a) $2 + p = 20 - p \Rightarrow p^* = 9$, (b) $-4 + p = 20 - p \Rightarrow p^* = 12$, (c) Price increases by 3

5. In a free market, supply and demand for mathematics education are $q = 1 + p$

(a) Find the free-market equilibrium (left(q^{*},p{*}right))

(b) With social benefit included, demand changes to (q=16-2p) Find the new equilibrium.

(c) How does equilibrium quantity change between (a) and (b)?

Answer: (a) $1 + p = 7 - 2p \Rightarrow p^* = 2$, (b) $1 + p = 16 - 2p \Rightarrow p^* = 5$, (c) Quantity increases from 3 to 6.

6. Consider the GDP model $Y = C + G + I + X - M$

(a) Calculate (Y)

(b) If investment (I)

(c) Using the original values, if import (M)

Answer: (a) $Y = 100 + 80 + 50 + 40 - 30 = 240$, (b) $Y = 250$, (c) $Y = 220$

Chapter 9

Non-linear equations in systems

Non-linear equations can also be included in systems together. This can help us describe theories and reason about different phenomena. Non-linear equation systems can be more difficult to work with compared to linear equation systems. But sometimes this type of equations can also be solved in similar ways to linear ones. This chapter gives a brief introduction with some examples from social science.

9.1 Solutions with non-linear systems

As described above, for linear equation systems we can know in advance how many solutions we can expect given certain conditions: none, one or infinitely many solutions. With non-linear equation systems there are no such rules. If the lines in a graph are not straight they can meet 0, 1, 2, 3 or however many times up to infinity. This can be illustrated by comparing the following equations:

$$\begin{cases} f\left(x\right) & = 2x^3 - x^2 + 7 \\ g\left(x\right) & = 5x \\ h\left(x\right) & = 10 + x \end{cases} \tag{9.1}$$

and plot these in figure 9.1 . There is no x-value for which all three functions meet, so a system with these three functions thus lacks solutions. The first function, $f\left(x\right)$, is the solid line. The second function, $g\left(x\right)$, is the dotted line and the third function, $h\left(x\right)$, is the dashed line.

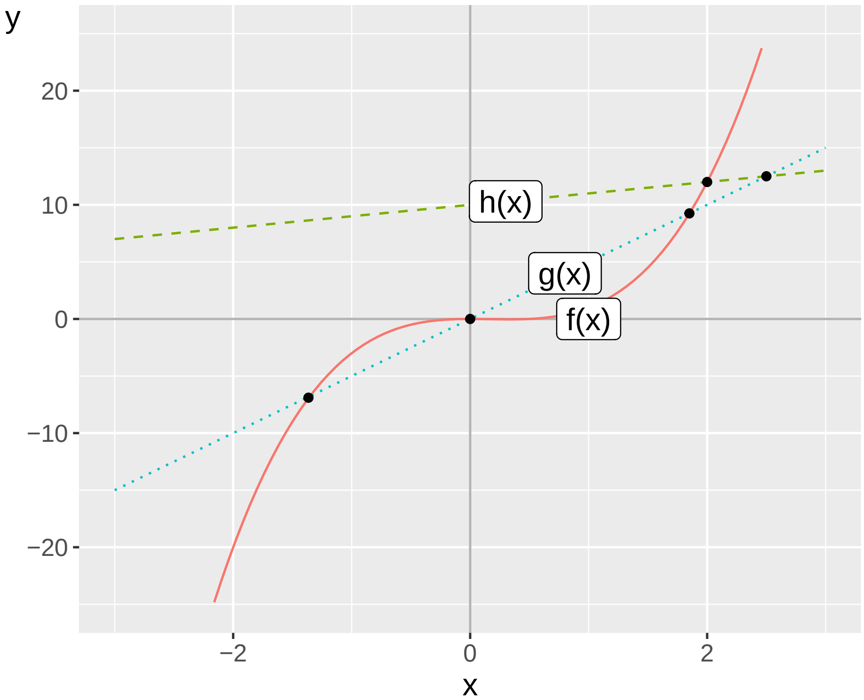


Figure 9.1: Example of a non-linear equation system

Function $f(x)$ is non-linear while the other two are linear. Since function $g(x)$ only has one slope coefficient, the number 5, and no coefficient for y-intercept, the line intersects the y-axis at $x = y = 0$. At this point, $(x, y) = (0, 0)$, function $f(x)$ also intersects the y-axis, which is also seen in the figure.

The lines for $f(x)$ and $g(x)$ meet at three points. A system with only these two functions would therefore have three solutions. The lines for $f(x)$ and $h(x)$ meet at one point and a system with only these two functions would therefore have a unique solution. The lines for $g(x)$ and $h(x)$ meet at the point $x = 10/4 = 2.5$.

9.2 Substitution

There are several basic methods that we can use to solve non-linear equation systems. One of these is substitution. Suppose we have a system with the following two equations and the two variables y and x:

$$\begin{cases} y = x \\ y = (x-2)^2 \end{cases} \tag{9.2}$$

The first equation is linear and the second is non-linear. This system we can solve by taking the definition of y from the first equation and substituting into the lower equation:

$$\begin{aligned} x &= (x-2)^2 \\ x &= x^2 - 4x + 4 \\ 0 &= x^2 - 5x + 4 \\ 0 &= (x-4)(x-1) \end{aligned} \tag{9.3}$$

The equation is 0 for $x_1^* = 4$ and $x_2^* = 1$ respectively. Let us test these solutions by substituting the values into the system. For $x = 4$ we get

$$\begin{cases} y = 4 \\ y = (4-2)^2 = 4 \end{cases} \tag{9.4}$$

For $x = 1$ we get:

$$\begin{cases} y = 1 \\ y = (1-2)^2 = 1 \end{cases} \tag{9.5}$$

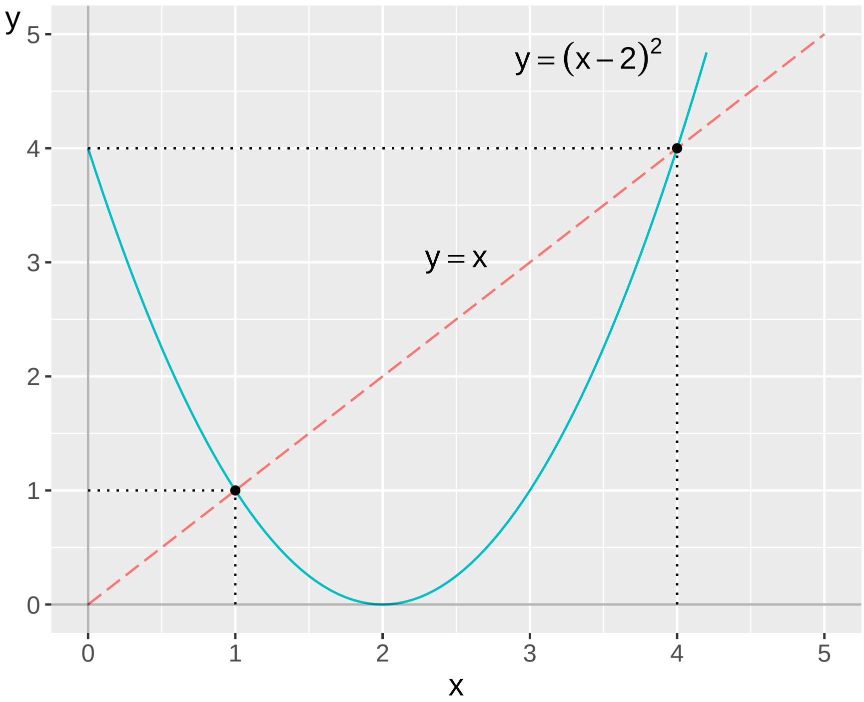


Figure 9.2: Two equations with two solutions

We get the solutions $(x^*, y^*) = (1, 1)$ and $(4, 4)$. Figure 9.2 illustrates the two equations and the two points where the equations meet. Let us now examine the following equation system:

$$\begin{cases} y = f(x) = \left(2 - \frac{1}{x^2}\right)^{\frac{1}{2}} \\ y = g(x) = x^2 \end{cases}, \forall x > 0 \tag{9.6}$$

We have two definitions of y that we can set equal to each other:

$$\begin{aligned} \left(1 - \frac{1}{x^2}\right)^{1/2} &= x^2 \\ x^4 + \frac{1}{x^2} - 2 &= 0 \end{aligned} \tag{9.7}$$

One way to simplify our work now is to define a new variable, $t = x^2$:

$$\begin{aligned} x^4 + \frac{1}{x^2} - 2 &= 0 \\ t^2 + \frac{1}{t} - 2 &= 0 \end{aligned} \tag{9.8}$$

This expression we can rewrite:

$$t^3 - 2t + 1 = 0 \tag{9.9}$$

To find candidate solutions to this third-degree polynomial we try $t = 0$ and $t = 1$:

$$\begin{aligned} t = 0 &\Rightarrow (0)^3 - 2 * 0 + 1 \neq 0 \\ t = 1 &\Rightarrow 1^3 - 2 * 1 + 1 = 0 \end{aligned} \tag{9.10}$$

This means that $t_1 = 1$ is a solution and that $t - 1$ is thereby a factor of the expression $t^3 - 2t + 1$. To find more roots we can use polynomial long division (see section 6.5):

t^2	$+t$	-1	t^3	$-2t$	$+1$
t	-1	$-(t^3$	$-t^2)$	t^2	$-2t$
$+1$	$-(t^2$	$-t)$	$-t$	$+1$	$-(-t$
$+1)$	0				

Since the division worked out evenly, we got 0 at the bottom, this confirms that $t - 1$ is a factor of $t^3 - 2t + 1$. On the top row we get the quotient $t^2 + t - 1$, which is a second-degree polynomial that we can solve with the quadratic formula. This gives us two additional solutions for t:

$$\begin{aligned} t &= -\frac{1}{2} \pm \sqrt{1 + \left(\frac{1}{2}\right)^2} \\ t_2 &= -0.5 - \sqrt{1.25} \\ t_3 &= -0.5 + \sqrt{1.25} \approx 0.618 \end{aligned} \tag{9.11}$$

Now we have three solutions for t that we can use in $t = x^2$ to solve out $x = \pm\sqrt{t}$. Since the square root is not defined for negative real numbers we only have use of $t > 0$. Our first solution was $t_1 = 1$, which gives two candidate solutions for x:

$$x_{1,2} = \pm\sqrt{1} \tag{9.12}$$

Since the equation system is only defined for $x > 0$, $x_1 = 1$ is a possible solution. We test by substitution in the system in equation (9.6) :

$$\begin{cases} y = f(1) = \left(2 - \frac{1}{1}\right)^{\frac{1}{2}} = 1 \\ y = g(1) = 1 \end{cases} \tag{9.13}$$

That is correct. Of t_2 and t_3 we are only interested in t_3:

$$\begin{aligned} x &= \pm\sqrt{t_3} \\ x &= \pm\sqrt{0.618} \\ \Rightarrow x_2 &= 0.786 \end{aligned} \tag{9.14}$$

We are only interested in the positive square root, which gives us a second solution for x. We test this by substitution in our equation system:

$$\begin{cases} y = f(0.786) &= \left(2 - \frac{1}{(0.786)^2}\right)^{\frac{1}{2}} = 0.618 \\ y = g(0.786) &= (0.786)^2 = 0.618 \end{cases} \tag{9.15}$$

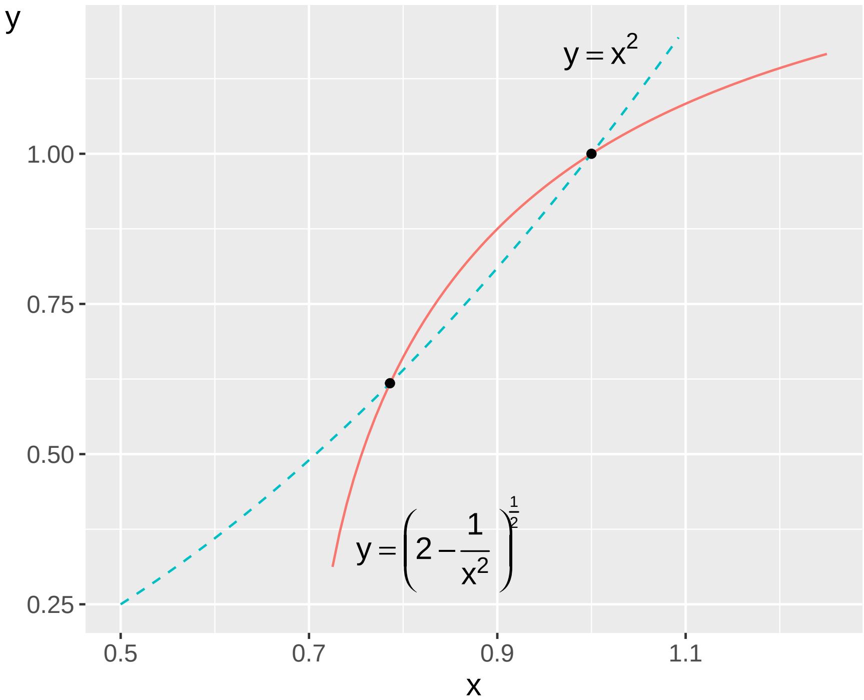


Figure 9.3: Two equilibria in a non-linear system

We have found four roots for the fourth-degree polynomial we started with in equation (9.8) . The system's two solutions $x_1 = 1$ and $x_2 = 0.786$ as well as two negative roots. The equation system and its solutions are illustrated in figure 9.3 .

9.3 Row reduction with non-linear equations

Another useful method for solving non-linear equation systems is row reduction, or elimination, in a similar way as we did for linear equation systems in section 7.3 . The principle is based on that we can add and subtract equations in the same system with each other and thereby eliminate terms so that we get new definitions of the variables. To illustrate this method we can check if the following equation system has any solutions:

$$\begin{cases} y = 20 - 2x^2 \\ y = x^2 - 10 \end{cases} \tag{9.16}$$

We subtract row 2 from row 1:

$$\begin{cases} 0 = 30 - 3x^2 \\ y = x^2 - 10 \end{cases} \tag{9.17}$$

From row 1 we can now find the solutions for x:

$$\begin{aligned} x^2 &= 10 \\ x^* &= \pm 10^{\frac{1}{2}} \end{aligned} \tag{9.18}$$

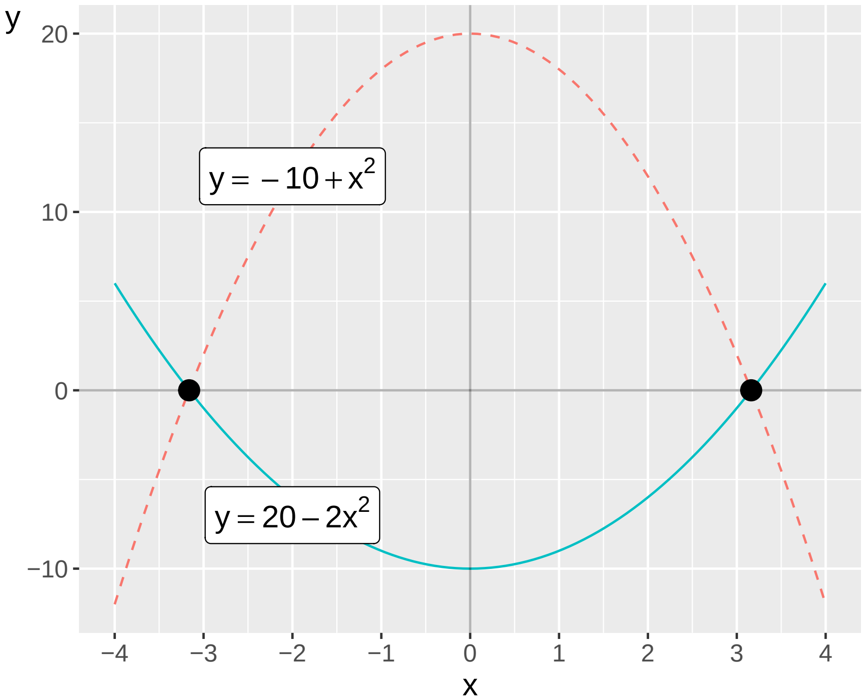


Figure 9.4: The non-linear equation system in equationeqrefeq:ickelinjart-ekvations-sys-2

In this case we can without difficulty also use substitution and reach the same result, for example by taking the definition of y in row 2 and substituting this into row 1. From this we can solve out x:

$$\begin{aligned} x^2 - 10 &= 20 - 2x^2 \\ x &= \pm 10^{1/2} \end{aligned} \tag{9.19}$$

The equation system in equation (9.16) is illustrated in figure 9.4 , where the two equilibria, where the functions' lines meet, are marked with points.

9.4 Example 1: Two friends

Earlier we used linear equation systems to describe some examples of social science discussions. Non-linear equations and equation systems also occur abundantly in social science. Here follow some simplified examples that have drawn inspiration from more comprehensive academic works and research articles. In the first example we have two friends, Erik and Maria, who do everything together. Both are considering whether they should start smoking. Erik's relationship to smoking can be described with the following function:

$$\text{Erik's smoking} = y = x^2 \tag{9.20}$$

The letter x is a variable that describes how much Maria smokes. Maria's smoking can in turn be described with the function:

$$\text{Maria's smoking} = x = y^2 \tag{9.21}$$

We now have the following equation system:

$$\begin{cases} y = x^2 & \forall x \in [0,1] \\ x = y^2 & \forall y \in [0,1] \end{cases} \tag{9.22}$$

The system describes how Erik's smoking is a positive function of Maria's smoking, which in turn is a positive function of Erik's smoking. Both equations are only defined for values of x and y within the interval $[0, 1]$. One way to find the system's solutions is to rewrite equation two to $y = x^{\frac{1}{2}}$ and then use this in the first equation:

$$\begin{aligned} x^2 &= x^{\frac{1}{2}} \\ x^2 - x^{\frac{1}{2}} &= 0 \\ x^{\frac{1}{2}}\left(x^{\frac{3}{2}} - 1\right) &= 0 \end{aligned} \tag{9.23}$$

Now we can see that the system's solutions are given by $x^{\frac{1}{2}} = 0$ and $\left(x^{\frac{3}{2}} - 1\right) = 0$. For $x^{\frac{1}{2}} = 0$, x must be 0, which is the first solution: $x_1^* = 0$. For $\left(x^{\frac{3}{2}} - 1\right)$ we get a second solution:

$$\begin{aligned} x^{\frac{3}{2}} - 1 &= 0 \\ \left(x^3\right)^{\frac{1}{2}} &= 1 \\ x_2^* &= 1 \end{aligned} \tag{9.24}$$

We have the solutions $x_1^* = 0$ and $x_2^* = 1$, which gives us the solutions for y:

$$\begin{aligned} y_1^*\left(x_1^*\right) &= 0^2 = 0 \\ y_2^*\left(x_2^*\right) &= 1^2 = 1 \end{aligned} \tag{9.25}$$

Solution 1 gives the point $(x_1^*, y_1^*) = (0, 0)$ and solution 2 gives the point $(x_2^*, y_2^*) = (1, 1)$. These solutions mean that either none of the friends smoke $(0, 0)$ or both smoke all the time $(1, 1)$, where the value 1 can be described as much as possible.

These solutions can be described as the system's stable equilibria. If one of the friends tries smoking with less than 1 pack per day, such as smoking half a pack, the equation system moves toward the solution $(0, 0)$. Say for example that $x = 0.5$. This gives $y = (0.5)^2 = 0.25$. That is, if one smokes half a pack then the other only smokes a quarter pack. Since both friends influence each other, the next day we get $x = (0.25)^2 = 0.0625$, whereupon $y = (0.0625)^2 \approx 0.0039$, approaching closer to 0 to infinity. This process continues indefinitely, with both smoking levels decreasing toward zero—making $(0, 0)$ a stable equilibrium. $(1, 1)$ is a stable equilibrium if any of the two friends start there, since $y = x^2 = (1)^2 = 1$. In this theoretical example they cannot smoke more than the value 1.

The two functions and the two solutions (the two stable equilibria) are illustrated in figure 9.5 . The lines meet at the two points $(0, 0)$ and $(1, 1)$.

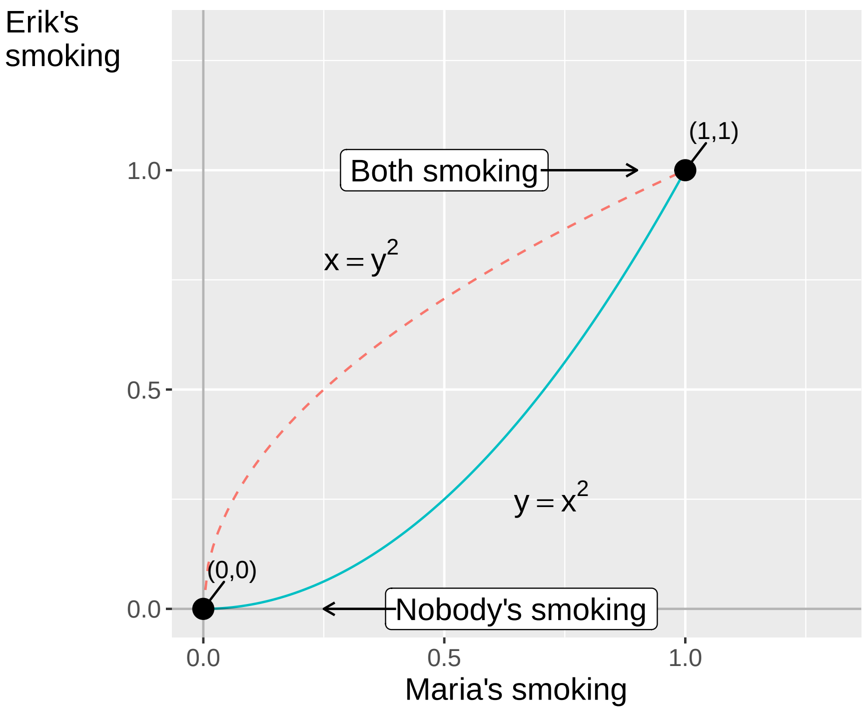


Figure 9.5: The smoking habits of two friends

9.5 Example 2: More smoking

Let us continue calculating Erik and Maria's smoking but based on other equations. The purpose of this example is to illustrate how we by changing a little in the equations can illustrate different types of theories and patterns. Suppose we now have the following equation system:

$$\begin{cases} y = x^2 + \frac{1}{4} & \forall x \in [0,1] \\ x = y^2 + \frac{1}{4} & \forall y \in [0,1] \end{cases} \tag{9.26}$$

where x and y still symbolize the two friends' smoking and how smoking is affected by the other's smoking. In both equations we have added the constant $\frac{1}{4}$. This constant can be interpreted as both liking to smoke a little bit but that the other's smoking can also affect one's own smoking. To solve out the system's solutions we start by taking the definition of y from the upper equation and substituting this into the lower equation:

$$\begin{aligned} x &= \left(x^2 + \frac{1}{4}\right)^2 + \frac{1}{4} \\ &= x^4 + \frac{x^2}{2} + \frac{1}{16} + \frac{1}{4} \\ 0 &= x^4 + \frac{x^2}{2} - x + \frac{5}{16} \end{aligned} \tag{9.27}$$

Now we have a fourth-degree equation from which we can proceed and solve out x. We first rewrite the equation so that we get:

$$16x^4 + 8x^2 - 16x + 5 = 0 \tag{9.28}$$

From this it is easier to see the equation's potential rational roots, see section 6.3 . We seek p/q where p is an integer factor of the constant 5 and q is an integer factor of 16, the multiplier of the power expression in the equation that has the largest exponent. Possible values for p are ± 1 and ± 5. Possible values for q are $\pm 1, \pm 2, \pm 4, \pm 8, \pm 16$. We now get the following potential roots:

$$\frac{p}{q} = \pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}, \pm\frac{1}{8}, \pm\frac{1}{16}, \pm\frac{5}{2}, \pm\frac{5}{4}, \pm\frac{5}{8}, \pm\frac{5}{16}, \pm 5 \tag{9.29}$$

By testing these roots in equation (9.28) we can see that $x = \frac{p}{q} = \frac{1}{2}$ is a root for the equation. This value for x happens to also be the system's only solution. This results in the corresponding solution for y:

$$y^* = (x^*)^2 + \frac{1}{4} = \left(\frac{1}{2}\right)^2 + \frac{1}{4} = \frac{1}{2} \tag{9.30}$$

The system's unique solution is thus $(x^*, y^*) = \left(\frac{1}{2}, \frac{1}{2}\right)$. This equation system thus results in Erik and Maria smoking half a pack of cigarettes per day. Suppose now that the two friends' smoking can be described with the following equation system:

$$\begin{cases} y = 1 - x^2 & \forall x \in [0,1] \\ x = 1 - y^2 & \forall y \in [0,1] \end{cases} \tag{9.31}$$

Let us start by interpreting the equations in ordinary words. In both equations the respective variable is a negative function of the other variable. This is an illustration that both smoke, given that the other does not smoke. The system has three solutions. Let us look at the first two: $(x^*, y^*) = (1, 0)$ and $(x^*, y^*) = (0, 1)$. Both of these solutions mean that one person smokes one pack per day while the other does not smoke at all. There is also a third solution:

$$(x^*, y^*) = \left(\frac{5^{1/2} - 1}{2}, \frac{5^{1/2} - 1}{2}\right) \approx (0.62,\ 0.62) \tag{9.32}$$

This can be interpreted as both being able to consider smoking simultaneously, but in that case they will smoke less than they would have done if they smoked separately. They will both smoke somewhat half-heartedly.

This simple example illustrates how we with the help of mathematics can describe social relationships, or at least parts of these. We can of course rewrite or expand the equations to thereby describe in more detail how we imagine people behave in different situations or how the process could develop over time. We can also use several variables to describe how several people in a group affect each other.

9.6 Example 3: They call it love

In section 8.6 we looked at an example concerning Erik's dating strategies. Say now that Erik has found a partner and wants to work to make their relationship function. But Erik doesn't want to put more work into the relationship than what Erik's partner does. The partner feels the same and also doesn't want to engage unless Erik does it. We can describe with the help of a non-linear equation system in a similar way as the example with smoking:

$$\begin{cases} K = P^2 + 0.1 \\ P = K^2 + 0.1 \end{cases} \tag{9.33}$$

where K and P symbolize Erik and Erik's partner and 0.1 is a constant that symbolizes how much work they are prepared to put in regardless of how much the other one exerts themselves. From the lower equation we solve an expression for K:

$$K = (P - 0.1)^{1/2} \tag{9.34}$$

Now we have two definitions of K that we can set equal to each other:

$$\begin{aligned} (P-0.1)^{1/2} &= P^2+0.1 \\ P-0.1 &= \left(P^2+0.1\right)^2 \\ P-0.1 &= P^4+2*0.1*P^2+(0.1)^2 \end{aligned} \tag{9.35}$$

On the right-hand side on the bottom row we now have the following fourth-degree polynomial:

$$P^4+0.2P^2-P+0.11=0 \tag{9.36}$$

From here we can continue by testing rational roots (see section 6.3) or different values for P and see if we can find the first root. After that we can work further with polynomial division. We test $P=0, \frac{1}{2}$ and 1:

$$\begin{aligned} \text{For } P=0: &\quad 0+0-0+0.11=0.11\neq 0 \\ \text{For } P=\frac{1}{2}: &\quad \left(\frac{1}{2}\right)^4+0.2\left(\frac{1}{2}\right)-\frac{1}{2}+0.11=-0.2775\neq 0 \\ \text{For } P=1: &\quad 1+0.2-1+0.11=0.31\neq 0 \end{aligned} \tag{9.37}$$

None of these were a solution, but note how the results for the polynomial are positive for $P=0$ and $P=1$, but negative for $P=1/2$. There should be one or more values for P between these values that result in the polynomial being equal to 0, meaning there is a solution for P between 0 and 1/2, as well as another solution between 1/2 and 1. Another method to find these solutions is to plot a graph for the functions in question, either by hand or with the help of a computer.

Figure 9.6 illustrates the line for the fourth-degree polynomial in equation (9.36) . The line passes 0 where P is just over 0.1 and just under 0.9. This is the system's two stable equilibria, and the solutions to the equations.

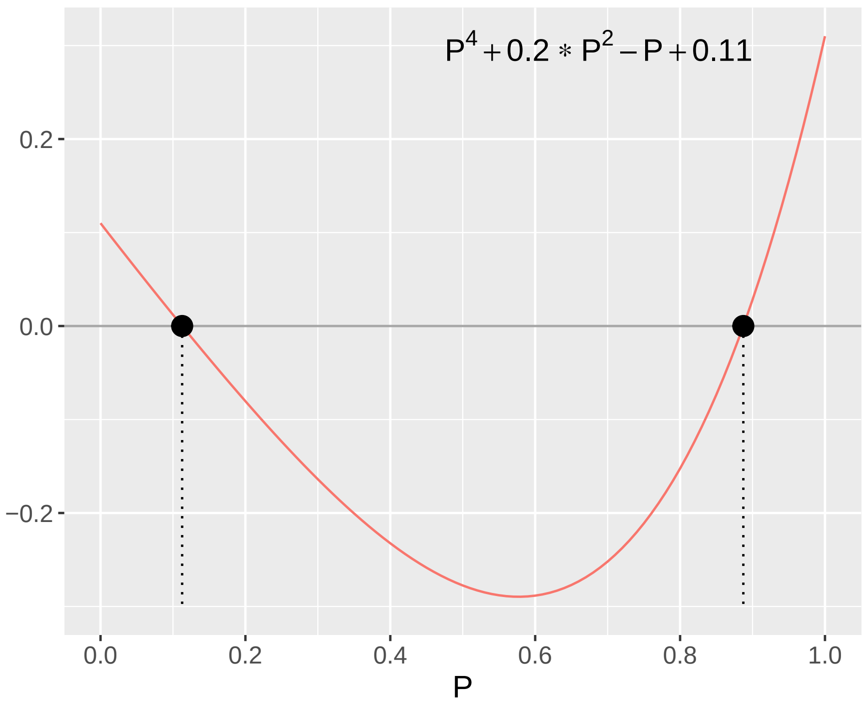


Figure 9.6: Erik's polynomial

The same solutions are seen when the two equations in system (9.33) are illustrated together. There are two solutions to the system: one solution with relatively low values where both Erik and the partner exert themselves relatively little to make the relationship work, and one solution where both work hard to make things work. The solutions to the system are $K_1 = P_1 \approx 0.11$ and $K_2 = P_2 \approx 0.89$.

9.7 Example 4: Another theory about work

In section 8.7 we went through an equation system that described a theory for work. That system was non-linear in its first form but with the help of logarithmization we got linear equations. In that example, linearity did not affect the number of solutions to the system. We calculated a unique solution for the two variables in both cases. However, there are discussions around labor market and social economy where we can also benefit from non-linear equation systems with multiple equilibria. We will therefore look at yet another simplified model that can be used to reason around the labor market and the equilibrium levels for employment, unemployment and real wage.

In this example we have two non-linear functions, one for wage-setting and one for price-setting:

$$\begin{cases} \frac{W}{P} = \frac{a}{U} & , b > 0 \\ \frac{P}{W} = \frac{c}{d(1-U)^{1/2}} & , d > 0 \end{cases} \tag{9.38}$$

The letters W and P symbolize wages and prices. The expression W/P is real wage, and U is percent unemployment. The parenthesis $(1 - U)$ is the share of the workforce that is employed. The letters a, b, c and d are positive coefficients. The first equation in the system is the wage-setting curve and the second equation is the price-setting curve.

Unlike in section 8.7 we now assume that the price-setting curve also has a positive slope. This means that the curves can meet more than once and that we thereby get at least two solutions. Now we will check the conditions for this. We start by rewriting the second equation so that this gives us an expression for W/P. We set this new expression equal to the first equation's definition of W/P, which gives:

$$\begin{aligned} \frac{d\,(1-U)^{1/2}}{c} &= \frac{a}{U} \\ U\,(1-U)^{1/2} &= \frac{ac}{d} \\ U^2 - U &= \left(\frac{ac}{d}\right)^2 \\ U^2 - U - h &= 0 \end{aligned} \tag{9.39}$$

where h in the last row is a designation for $\left(\frac{ac}{d}\right)^2$. We now have a second-degree polynomial that we can solve with for example the quadratic formula (see equation (6.38)):

$$\begin{aligned} U &= -\left(\frac{-1}{2}\right) \pm \sqrt[2]{\left(\frac{-1}{2}\right)^2 - (-h)} \\ U_1 &= \frac{1}{2} - \sqrt{\frac{1}{4} + h} \\ U_2 &= \frac{1}{2} + \sqrt{\frac{1}{4} + h} \end{aligned} \tag{9.40}$$

These solutions we can substitute into the system's equations and solve for the solutions of the variable W/P. For the first equation this gives two solutions:

$$\left(\frac{W}{P}\right)_1 = \frac{a}{U} = \frac{a}{\frac{1}{2} + \sqrt{\frac{1}{4} + h}} \tag{9.41}$$
$$\left(\frac{W}{P}\right)_2 = \frac{a}{U} = \frac{a}{\frac{1}{2} - \sqrt{\frac{1}{4} + h}}$$

Figure 9.7 illustrates the two equations and solutions. At the point (solution) down to the left in the figure the amount of jobs and wages are relatively smaller, compared to the point (solution) up to the right.

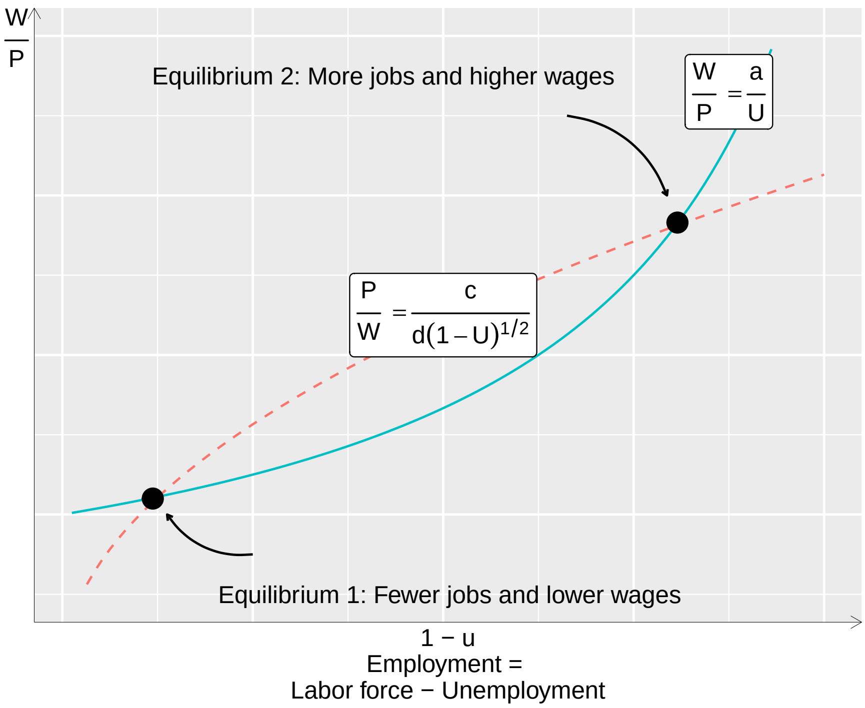


Figure 9.7: Two equilibria for wage and labor

As mentioned above, a central assumption in this example is that the curves slope in the same direction. If the curves slope in different directions they can only meet once, in the way we went through in section 8.7 . Exactly why the curves according to this theory would slope in the same direction is not covered in this review. We simply accept it as a theoretical assumption. Another approach would be that we start the reasoning on a more fundamental level and describe how workers and employers behave (Manning, 1990).

9.8 Chapter summary

- Non-linear equations can be included in equation systems. For non-linear equations there are no general rules for number of solutions. Non-linear equations can meet anywhere between 0 and infinitely many times.
- To find solutions to a non-linear equation system we can use row reduction and substitution in similar ways as we use it for linear equation systems.
- Another way to find solutions for a non-linear equation system is to use polynomial division.
- Non-linear equation systems can among other things be used to describe theoretical questions where we have more than one solution to the system, more than one equilibrium. This can for example be used to describe macroeconomic perspectives on wage and price formation or social dynamics between friends or partners.

9.9 Exercises

1. Solve the equation system

$$\begin{cases} y = x \\ y = (x-2)^2 \end{cases}$$

(a) Substitute the first equation into the second and write the resulting equation for (x)

(b) Find all solutions (x^{*})

(c) Write the complete solutions (left(x$^{\{*\},y}$\{*\}right))

Answer: (a) $x = (x-2)^2 = x^2 - 4x + 4 \Rightarrow x^2 - 5x + 4 = 0$, (b) $x_1^* = 1$, $x_2^* = 4$, (c) $(1,1)$ and $(4,4)$

2. Use row reduction to solve the equation system

$$\begin{cases} y = 20 - 2x^2 \\ y = x^2 - 10 \end{cases}$$

(a) Subtract row 2 from row 1 to eliminate (y) What equation for (x)

(b) Find all solutions (x^{*})

(c) Find the corresponding values (y^{*})

Answer: (a) $0 = 30 - 3x^2$, (b) $x^* = \pm\sqrt{10}$, (c) $y^* = x^2 - 10 = 10 - 10 = 0$

3. Solve the equation system

$$\begin{cases} y = 10 - x^2 \\ y = 2x^2 - 5 \end{cases}$$

(a) Eliminate (y)

(b) Find the corresponding (y^{*})

(c) Write the complete solutions (left(x$^{\{*\},y}$\{*\}right))

Answer: (a) $0 = 15 - 3x^2 \Rightarrow x^* = \pm\sqrt{5}$, (b) $y^* = 10 - 5 = 5$, (c) $\left(\sqrt{5}, 5\right)$

4. Erik and Maria's smoking can be described by

$$\begin{cases} y = x^2 \\ x = y^2 \end{cases}$$

(a) Show that (left(0,0right))

(b) Show that (left(1,1right))

(c) Starting from (x=0.8) Is the system moving toward (left(0,0right))

Answer: (a) $y = 0^2 = 0$, (b) $y = 1^2 = 1$, (c) $y = (0.8)^2 = 0.64$

5. A modified version of the smoking model has the equation system

$$\begin{cases} y = x^2 + \frac{1}{4} \\ x = y^2 + \frac{1}{4} \end{cases}$$

(a) Show that (x^{*}=frac{1}{2})

(b) Calculate the corresponding (y^{*})

(c) How many solutions does this system have compared to the system in exercise 4?

Answer: (a) $y = \left(\frac{1}{2}\right)^2 + \frac{1}{4} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, (b) $y^* = \frac{1}{2}$, (c) This system has one solution; exercise 4 has two solutions.

6. Erik and a partner's effort in a relationship is described by

$$\begin{cases} K = P^2 + 0.1 \\ P = K^2 + 0.1 \end{cases}$$

The solutions satisfy the polynomial $P^4 + 0.2P^2 - P + 0.11 = 0$

(a) Evaluate the polynomial at (P=0) Is the result positive or negative?

(b) Evaluate the polynomial at (P=0.5) Is the result positive or negative?

(c) What does the sign change between (a) and (b) tell us about the number of solutions?

Answer: (a) $0^4 + 0.2 \cdot 0^2 - 0 + 0.11 = 0.11 > 0$, (b) $(0.5)^4 + 0.2\,(0.5)^2 - 0.5 + 0.11 \approx -0.28 < 0$, (c) The sign change means there is at least one solution between $P = 0$ and $P = 0.5$.

7. Find the intersection points of the system

$$\begin{cases} y = x^2 - 3 \\ y = 2x \end{cases}$$

(a) Set up the equation (x^{2}-3=2x) and rearrange.

(b) Find all solutions (x^{*}) .

(c) Write the complete solutions (left(x$^{\{*\},\mathrm{y}}${*}right)) .

Answer: (a) $x^2 - 2x - 3 = 0$, (b) $x^* = 3$ or $x^* = -1$, (c) $(3, 6)$ and $(-1, -2)$.

8. In a market, demand is $P = 40 - Q^2$ and supply is $P = 3Q$.

(a) Set up the equation (40-Q^{2}=3Q) and rearrange.

(b) Find the equilibrium quantity (Q^{*}) (positive solution only).

(c) Find the equilibrium price (P^{*}) .

Answer: (a) $Q^2 + 3Q - 40 = 0$, (b) $Q^* = 5$, (c) $P^* = 3 \cdot 5 = 15$.

9. Consider the system

$$\begin{cases} y = x^2 + 2 \\ y = x - 1 \end{cases}$$

(a) Set up the equation (x^{2}+2=x-1) and rearrange.

(b) Calculate the discriminant (Delta=b^{2}-4ac) .

(c) What does (Delta<0) tell us about the number of intersections?

Answer: (a) $x^2 - x + 3 = 0$, (b) $\Delta = 1 - 12 = -11 < 0$, (c) $\Delta < 0$: the line and parabola do not intersect.

10. Find the intersection points of the system

$$\begin{cases} y = x^2 + 1 \\ y = 3x - 1 \end{cases}$$

(a) Set up and solve the resulting quadratic equation.

(b) Find the solutions (left(x$^{\{*\},\mathrm{y}}${*}right)) .

(c) How many intersection points do the parabola and the line have?

Answer: (a) $x^2 - 3x + 2 = (x - 1)\,(x - 2) = 0$, (b) $(1, 2)$ and $(2, 5)$, (c) two intersection points.

11. Show that the line $y = 4x - 4$ is tangent to the parabola $y = x^2$.

(a) Set up the equation (x^{2}=4x-4) and rearrange.

(b) Calculate the discriminant and interpret the result.

(c) Find the tangent point (left(x$^{\{*\},\mathrm{y}}${*}right)) .

Answer: (a) $x^2 - 4x + 4 = 0$, (b) $\Delta = 16 - 16 = 0:$ exactly one solution, confirming tangency, (c) $x^* = 2$, $y^* = 4$.

12. In a labor market, labor supply is $L_s = w^2$ and labor demand is $L_d = 12 - w$, where w is the wage.

(a) Set up the equation (L_{s}=L_{d}) .

(b) Rearrange to a polynomial equation and solve for (w^{*}) (positive solution only).

(c) Find the equilibrium labor (L^{*}) .

Answer: (a) $w^2 = 12 - w$, (b) $w^2 + w - 12 = (w + 4)(w - 3) = 0 \Rightarrow w^* = 3$, (c) $L^* = 12 - 3 = 9$.

Chapter 10

Derivative

This chapter introduces the derivative. The derivative is a function for calculating the slope of a line at a specific point. The derivative can be used to find a function's maximum and minimum values, which is central to a large amount of analytical work.

10.1 Limit

Before we get to derivatives, we shall introduce limits. The limit is the value that a function approaches at a specific point. Limits can be used to handle expressions that are not defined for all numbers. Suppose we have the function:

$$y = f(x) = \frac{1}{x^2} \tag{10.1}$$

where x is a real number. The real numbers are not defined for division by 0. To understand why, see what happens with $f(x)$ when x approaches 0:

$$\begin{aligned} \text{For } x = 0.1 \text{ is } \frac{1}{x^2} &= \frac{1}{0.01} = 100 \\ \text{For } x = 0.01 \text{ is } \frac{1}{x^2} &= \frac{1}{0.0001} = 10,000 \\ \text{For } x = 0.0001 \text{ is } \frac{1}{x^2} &= \frac{1}{0.0000001} = 100,000,000 \end{aligned} \tag{10.2}$$

When x approaches 0, the expression $1/x^2$ approaches infinity. Infinity is written with the symbol ∞, which is not a real number. This means that the function $y = f(x) = 1/x^2$ is not defined for $x = 0$. To describe that $f(x)$ is defined for all x except 0 one may write:

$$y = f(x) = \frac{1}{x^2},\ \forall x \neq 0 \tag{10.3}$$

The variable y is a function of x for all x that are not equal to 0. That equation (**??**) approaches infinity as x approaches 0 can be described in the following way:

$$\lim_{x \to 0} f(x) = \frac{1}{x^2} = \infty \tag{10.4}$$

The expression $\lim_{x \to 0}$ stands for limes, which means limit. To describe how function f behaves when x goes toward positive infinity, we write:

$$\lim_{x \to \infty} f(x) = \frac{1}{x^2} = 0 \tag{10.5}$$

When x goes toward positive infinity, the limit for $f(x) = 1/x = 0$. The function goes toward the value 0. For many expressions, it's possible to find the limit by calculating the expression for values close to the limit, as in equation (**??**) . Suppose we seek the limit for the expression $x - 1$ when $x \to 1$:

$$\lim_{x \to 1} x - 1 \tag{10.6}$$

To find the limit, we calculate values close to $x = 1$:

x	$x - 1$
0.9	0.1
0.99	0.01
0.999	0.001

The closer x gets to 1, the closer $x - 1$ gets to the value 0. This is the limit for $x - 1$:

$$\lim_{x \to 1} x - 1 = 0 \tag{10.7}$$

We also calculate $x - 1$ for values just above 1:

x	$x - 1$
1.1	0.1
1.01	0.01
1.001	0.001

We get the same limit regardless of whether we approach x from above or from below. The limit for an expression can however differ depending on whether we approach the limit from above or from below. Take for example the function:

$$y = \frac{1}{x} \; \forall x \neq 0 \tag{10.8}$$

The function $y = \frac{1}{x}$ is defined for all x that are not equal to 0. For $x > 0$, the same limit still applies as for $y = f(x) = 1/x^2$. When x approaches 0 from values above 0, the expression $1/x$ goes toward positive infinity. But if we instead look at how x approaches 0 for values below 0, we get the following result instead:

$$\begin{aligned} \text{For } x = -1 \text{ is } y &= \frac{1}{x} = \frac{1}{-1} = -1 \\ \text{For } x = -0.2 \text{ is } y &= \frac{1}{x} = \frac{1}{-0.2} = -5 \\ \text{For } x = -0.0001 \text{ is } y &= \frac{1}{x} = \frac{1}{-0.0001} = -10,000 \end{aligned} \tag{10.9}$$

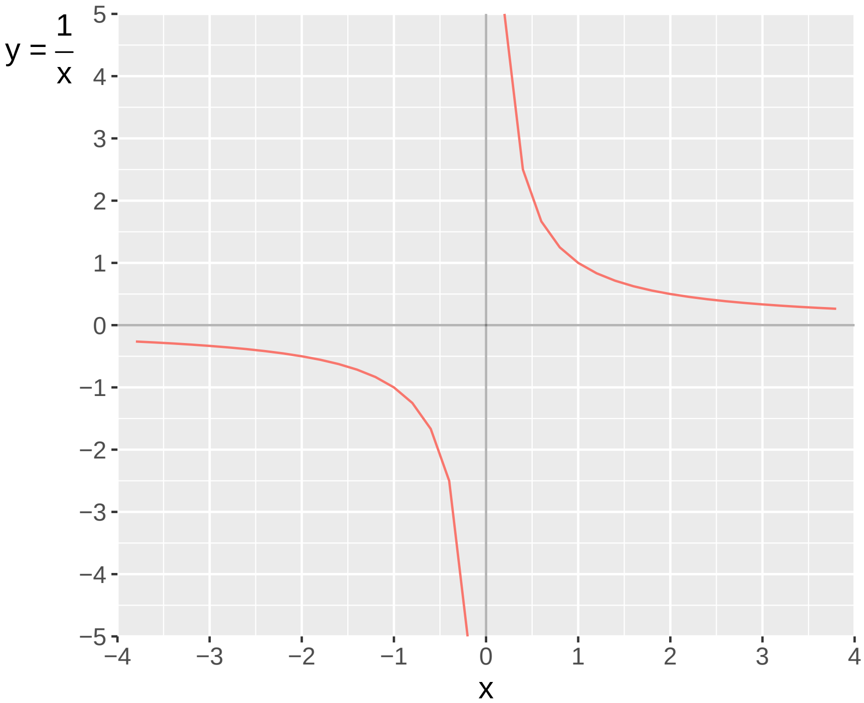


Figure 10.1: The function $y = f(x) = 1/x$

Since x approaches 0 from values below 0, $1/x$ goes toward negative infinity, which can be written $-\infty$. This is illustrated in figure 10.1 where function $f(x) = 1/x$ goes toward positive infinity $+\infty$ as x approaches 0 for positive numbers and toward negative infinity $-\infty$ for negative numbers:

$$\begin{aligned}\lim_{x\to 0^+} f(x) &= \frac{1}{x} = +\infty \\ \lim_{x\to 0^-} f(x) &= \frac{1}{x} = -\infty\end{aligned} \tag{10.10}$$

The notation $x \to 0^+$ means that x goes toward 0 from the positive side of the input value ($x > 0$). This is also called a right-hand limit, because x approaches from the right on the number line. The expression $x \to 0^-$ means that x approaches 0 from smaller values ($x < 0$), which is called a left-hand limit.

10.2 Simplify first

We often benefit from simplifying the function we are working with as far as possible. This also applies when we seek limits. Consider now the following equation:

$$y = g(x) = \frac{x^3 - 2x^2 + 4x - 8}{x - 2} \tag{10.11}$$

If we substitute $x = 2$, we get:

$$g(2) = \frac{2^3 - 2 * 2^2 + 4 * 2 - 8}{2 - 2} = \frac{0}{0} \tag{10.12}$$

For $x = 2$ the nominator and denominator is zero, which is undefined for real values. It is however possible to calculate the limit as x approaches 2 and thereby get a value for $g(2)$. But first we factorize the numerator so we get:

$$y = g(x) = \frac{x^3 - 3x^2 + 3x - 9}{x - 2} = \frac{(x^2 + 4)(x - 2)}{x - 2} \tag{10.13}$$

Since we have the parenthesis $(x - 2)$ in both numerator and denominator, we can cancel this and get:

$$g(x) = \frac{(x^2 + 4)\cancel{(x - 2)}}{\cancel{(x - 2)}} = x^2 + 4 \tag{10.14}$$

In this simplified form the function is defined for $x = 2$:

$$y = g(2) = 2^2 + 4 = 8 \tag{10.15}$$

Since this is a simplified version of the function we call this value 8 the function's limit for $x = 2$, which we write as:

$$\lim_{x \to 2} g(x) = \frac{x^3 - 2x^2 + 4x - 8}{x - 2} = 8 \tag{10.16}$$

Suppose now that we have the following function:

$$y = f(x) = \frac{(x^2 + 3)(x - 3)}{x - 2} \tag{10.17}$$

for $x = 2$ the denominator is zero and therefore undefined. First we can expand the numerator: $(x^2 + 3)(x - 3) = x^3 - 3x^2 + 3x - 9$. We cannot simplify the expression more. In this case we have two limits. As x approaches 2 from the left $(x < 2)$ the function $f(x)$ approaches positive infinity $+\infty$. As x approaches 2 from the right $(x > 2)$ function $f(x)$ approaches negative infinity $-\infty$:

$$\begin{aligned} \lim_{x \to 2^+} f(x) &= \frac{x^3 - 3x^2 + 3x - 9}{x - 2} = -\infty \\ \lim_{x \to 2^-} f(x) &= \frac{x^3 - 3x^2 + 3x - 9}{x - 2} = +\infty \end{aligned} \tag{10.18}$$

While limits help us understand function behavior near problematic points, they also enable us to solve a fundamental question: how can we measure the exact slope of a curved line at a single point? This leads us to the derivative.

10.3 Differentiating

The derivative can be used to calculate the slope of a line at a specific point. In section 4.4 we went through the equation of the straight line. In a function of the form $y = a + bx$, b is the slope of the line and a is the value y takes when $x = 0$. b is a measure of how much y changes when x increases by 1. Since the line is straight, it has the same slope regardless of which value x has. The slope between the points (x_1, y_1) and (x_2, y_2) we may calculate as:

$$b = \frac{y_2 - y_1}{x_2 - x_1} \tag{10.19}$$

If we instead have a line that is not straight over all values of x, one may use the derivative to calculate the slope at a specific value of x. Let us illustrate with the function

$$y = x^2 \tag{10.20}$$

Figure 10.2 illustrates how the slope of the line for x^2 varies depending on the value for x. The solid curved line is drawn with x^2 and the dashed straight lines show the slope at two points on the curved line. At the point $(x, y) = (0, 0)$ the solid line has slope 0 and is horizontal. At the point $(x, y) = (2, 4)$ the slope is instead positive, up toward the graph's right corner.

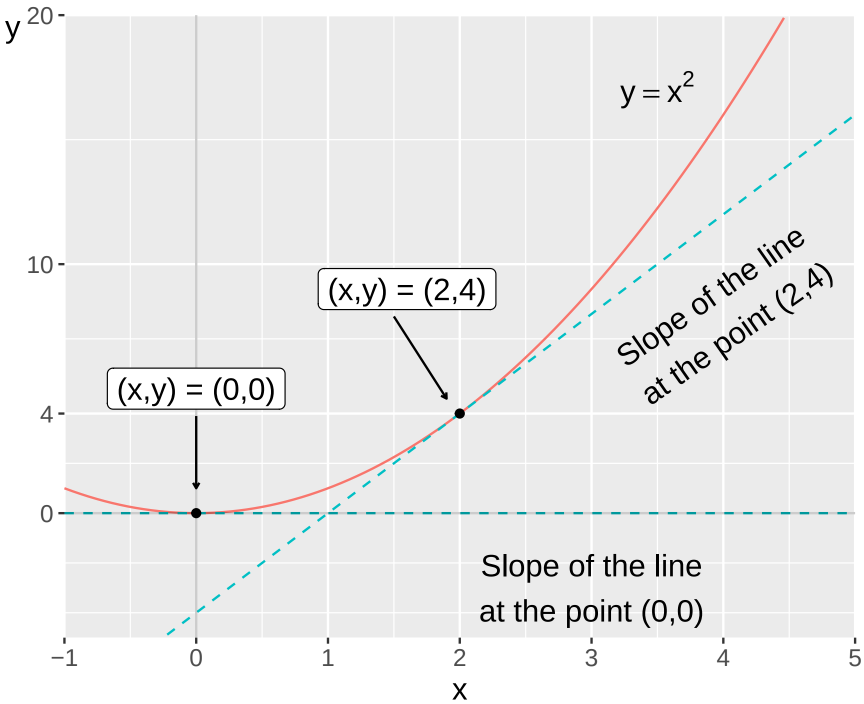


Figure 10.2: The function $y = x^2$

The derivative indicates the limit of the line's slope. The limit here refers to how much the function $f(x)$ changes at a small change in x. This is described by the derivative's h-definition, where the letter h describes the distance between two values in the variable x

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{10.21}$$

Equation (10.21) can be read as the derivative of function $f(x)$, which we write as $f'(x)$, is given by the function value at point x+h minus the function value at point x, as h goes toward 0, divided by h. The derivative's h-definition therefore describes the limit when the distance between the two points x and $x+h$ shrinks toward 0. For function $f(x) = x^2$ we derive this by substituting this function into equation (10.21) :

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{(x+h)(x+h) - x^2}{h} \\ &= \lim_{h \to 0} \frac{h^2 + 2xh + x^2 - x^2}{h} \\ &= \lim_{h \to 0} \frac{h^2 + 2xh}{h} \end{aligned} \tag{10.22}$$

All terms in the numerator and denominator contain h, which we may simplify:

$$f'(x) = \lim_{h \to 0} \frac{h \times \cancel{h} + 2x\cancel{h}}{\cancel{h}} = \lim_{h \to 0} h + 2x = 2x \tag{10.23}$$

Now we have shown that the function $y = f(x) = x^2$ has the derivative:

$$y' = f'(x) = 2x \tag{10.24}$$

We now have two functions that both use the variable x: on one hand $f(x) = x^2$ and on the other hand its derivative, $f'(x) = 2x$. Let us also figure out what slope the line has in figure 10.2 . The derivative of our function is $f'(x) = 2x$. To get the slope of the line at a specific point, that is how much y changes when x increases by 1, we substitute the value for x at the current point. In the figure, the points at $x = 2$ and $x = 4$ are marked with dashed lines. If we substitute these values for x into the derivative's function, we get:

$$\begin{aligned} f'(2) &= 2 * 2 = 4 \\ f'(4) &= 2 * 4 = 8 \end{aligned} \tag{10.25}$$

which shows how the line for the function $y = f(x) = x^2$ slopes at these points.

The dashed lines in the figure are straight lines, which we may calculate backwards in the following way. We know that the derivative is $f'(x) = 2x$ and that at $x = 2$, $f'(2) = 2 * 2 = 4$. The last is the value for y at this point. To get the dashed straight line's equation, we therefore substitute the values we know into such an equation. We seek the following function:

$$y = a + bx \tag{10.26}$$

We already know the values for y, b and x:

$$4 = a + 4 * 2 \tag{10.27}$$

This gives the following value for a:

$$a = 4 - 8 = -4 \tag{10.28}$$

The equation for the dashed line at $x = 2$ in figure 10.2 is therefore:

$$y = -4 + 4x \tag{10.29}$$

The above walkthrough was relatively elaborate. Fortunately, there are several calculation rules that we may use to avoid doing this process every time we need to use derivatives (see section 10.9 below). The calculation rule for the above example can be formulated as if we have the following function:

$$f(x) = x^a \tag{10.30}$$

its derivative is:

$$f'(x) = ax^{a-1} \tag{10.31}$$

10.4 The next derivative

For the derived version of $y = f(x)$ the following notations say exactly the same thing:

$$y' = f'(x) = \frac{df(x)}{dx} = \frac{dy}{dx} = \frac{\partial y}{\partial x} \tag{10.32}$$

The symbol ∂ here have the same meaning as the letter d. The expression dy/dx can be read as a small change of y divided by a small change of x. This notation is called Leibniz notation, after Wilhelm Leibniz (1646–1716).

So far we have only worked with the first derivative. If we take the derivative of the first derivative, we get the second derivative. This is calculated and written in a similar way as the first derivative. For example:

$$\begin{aligned} \text{Function:} y &= f(x) = x^3 \\ \text{First derivative: } \frac{\partial y}{\partial x} &= f'(x) = 3x^2 \\ \text{Second derivative } \frac{\partial}{\partial x}\left(\frac{\partial y}{\partial x}\right) &= f''(x) = 2 * 3x = 6x \end{aligned} \tag{10.33}$$

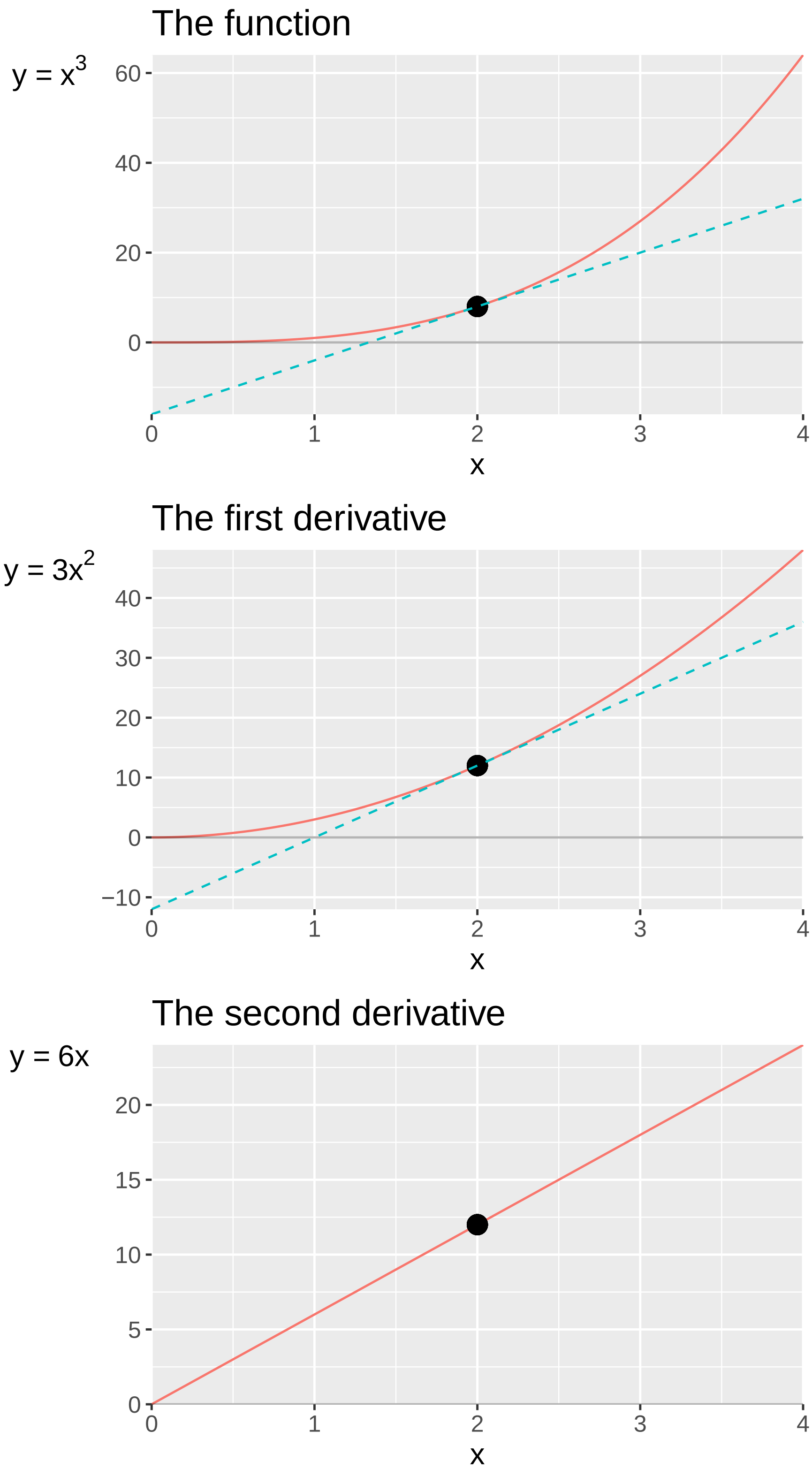


Figure 10.3: The function $y = f(x) = x^3$ and its first and second derivative

These are three different mathematical expressions: the function, the first derivative and the second derivative. All three expressions are different mathematical functions that use the variable x as input value, which is evident from the parenthesis with x in the name of each function: $f(x)$, $f'(x)$ and $f''(x)$. Figure 10.3 illustrate all these three functions in one graph each, with variable y on the vertical axis and variable x on the horizontal axis. In all three graphs, the slope of the line is marked at $x = 2$. Let us calculate the value for each function in equation (10.34) for $x = 2$:

$$
\begin{aligned}
\text{Function: } f(2) &= 2^3 = 8 \\
\text{First derivative: } f'(2) &= 3 * 2^2 = 12 \\
\text{Second derivative: } f''(2) &= 6 * 2 = 12
\end{aligned}
\tag{10.34}
$$

If we take the derivative of the second derivative, we get what is called the third derivative. The derivative after that is called the fourth derivative and so on. If we start here from the function $y = f(x) = x^3$, then we have:

$$
\begin{aligned}
y' &= f'(x) = 3x^2 \\
y'' &= f''(x) = 6x \\
y''' &= f'''(x) = 6 \\
y'''' &= f''''(x) = 0
\end{aligned}
\tag{10.35}
$$

where rows 3 and 4 are the third and fourth derivatives of the function $y = x^3$. The fourth derivative is the slope of the slope of the slope of the line's slope for $y = x^3$, which is 0 for all x. We can see this already in the third derivative, which is defined as $y''' = 6$ for all values of x, which in a graph becomes a horizontal line at $y = 6$.

10.5 Exponent, logarithm and *e*

We have so far worked with functions where the variable x stands as the base in a power expression, like $f(x) = x^a$, where exponent a is a coefficient (see section 2.8). In an exponential function, the variable is instead the exponent. For example:

$$
y = f(x) = b^x \tag{10.36}
$$

where the base b is a coefficient and the exponent x is a variable. Sometimes the expression the exponential function is used in definite form, which then refers to $y = e^x$ with the number e as base. This is sometimes also written as $y = \exp(x) = e^x$. Let us derive the derivative for the function $y = b^x$ using the derivative's h-definition (see equation (10.21)):

$$
\begin{aligned}
y'_x &= \lim_{h \to 0} \frac{b^{x+h} - b^x}{h} \\
&= \lim_{h \to 0} \frac{b^x \left(b^h - 1\right)}{h} \\
&= b^x \left(\lim_{h \to 0} \frac{b^h - 1}{h} \right)
\end{aligned}
\tag{10.37}
$$

In the last step, we factor out b^x since this is independent of h and therefore is not affected by h going toward 0. The limit for the expression is dependent on the base b, which is illustrated in figure 10.4 . The solid line shows values for the limit $\lim_{h \to 0} \left(b^h - 1\right)/h$ for different b. When $b = 1$, the limit is 0.

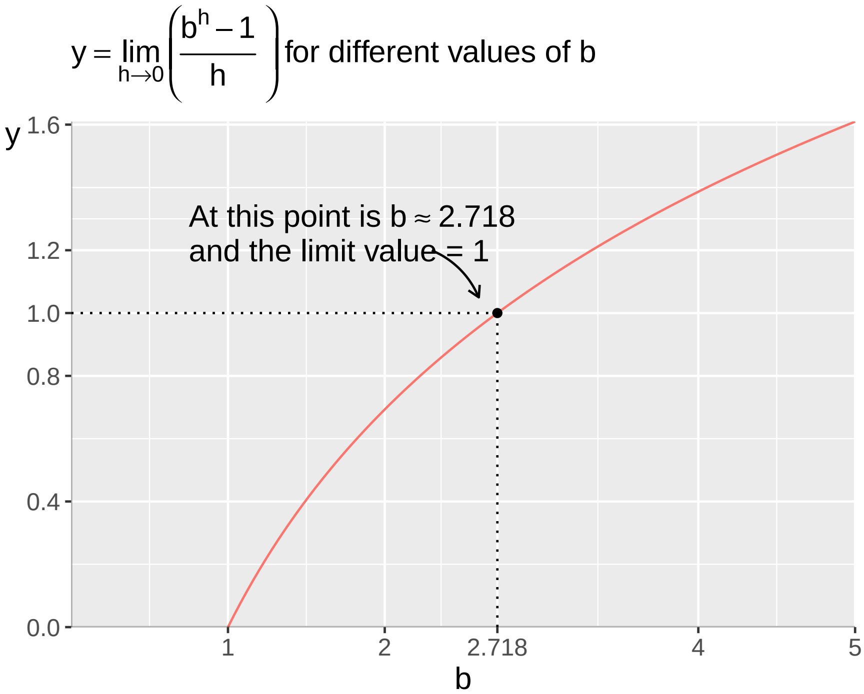


Figure 10.4: The importance of the base b

The limit $\lim_{h \to 0} (b^h - 1)/h$ from equation (**??**) is equal to 1 when b is equal to approximately 2.718, which is also marked in figure 10.4 . This number is so commonly occurring that it has received its own name, Euler's number, and is abbreviated e. We have mentioned this number before. The natural logarithm of the base b in equation (**??**) is the number that e(approximately 2.718) must be raised to in order to get b. This means that the general result for the equation can be described as the natural logarithm of b. Say now that we have a function with the following form:

$$f(x) = b^x \tag{10.38}$$

Then its derivative

$$f'_x(x) = b^x \left(\lim_{h \to 0} \frac{b^h - 1}{h} \right) = b^x \log_e b = b^x \ln b \tag{10.39}$$

If the logarithm's base b is equal to the number e, we get:

$$b^x \log_e b = e^x \ln e = 1 * e^x = e^x \tag{10.40}$$

This means that if we have a function of the type $y = f(x) = e^x$, then its derivative is:

$$y'_x = e^x \tag{10.41}$$

That is, the derivative of function $y = e^x$ is also equal to e^x. The second derivative of this function is similarly $y''_{xx} = e^x$. Let us now derive the derivative of the following function:

$$y = h(x) = e^{ax} \tag{10.42}$$

The variable y is a function of the variable x according to the function $h(x)$. The letter a is a constant coefficient and e is Euler's number. To calculate the derivative of this function, we again use the derivative's h-definition:

$$y'_x = \lim_{h\to 0}\frac{e^{a(x+h)} - e^{ax}}{h} = \lim_{h\to 0}\frac{e^{ax}\left(e^{ah}-1\right)}{h} = e^{ax}\left(\lim_{h\to 0}\frac{e^{ah}-1}{h}\right) \tag{10.43}$$

In the third step, we factor out e^{ax} from the parenthesis, which is possible since this factor is not affected by $h \to 0$. Moving out e^{ax} thus does not affect the limit. To proceed from here, we may multiply both numerator and denominator in the parenthesis by the constant a, an arbitrary real number. Since a in the numerator also does not affect the limit, we move this out of the parenthesis. The constant a in the denominator we leave in place:

$$\begin{aligned} y'_x &= e^{ax}\left(\lim_{h\to 0}\frac{a}{a}\times\frac{e^{ah}-1}{h}\right) \\ &= ae^{ax}\left(\lim_{h\to 0}\frac{e^{ah}-1}{ah}\right) \\ &= ae^{ax}\left(\lim_{h\to 0}\frac{e^{ah}-1}{ah}\right) \\ &= ae^{ax} \end{aligned} \tag{10.44}$$

When we move a out of the parenthesis, both numerator and denominator go toward 0, which is why the limit goes toward 1. In the last line we therefore remove the limit and only have $y'_x = ae^{ax}$ left. This expression has the same form as what we derived in equation (10.40) . We may summarize this as a general calculation rule. If we have a function with the form:

$$y = f(x) = e^{g(x)} \tag{10.45}$$

its derivative is:

$$y'_x = g'_x e^{g(x)} \tag{10.46}$$

These examples also make it easy to continue with other types of functions, for example the following power function:

$$y = f(x) = b^{ax} \tag{10.47}$$

where the letters b and a are coefficients and y and x are variables. One way to derive the derivative of equation (10.47) is to rewrite the expression as a power with base e, Euler's number:

$$y = f(x) = b^{ax} = e^{\ln b^{ax}} = e^{ax\ln b} \tag{10.48}$$

We here take the natural logarithm of our function b^{ax}, which gives $\ln b^{ax} = ax\ln b$. We use this as an exponent to the base e, which gives us an expression that we can differentiate:

$$f'_x(x) = (a\ln b)\, e^{ax\ln b} = (a\ln b)\, b^{ax} \tag{10.49}$$

10.6 The derivative of $\log x$

Let us now differentiate the following function:

$$y = \log x \tag{10.50}$$

where $\log x$ is the logarithm of x with arbitrary base b, where $b > 0$ and $b \neq 1$. We use the derivative's h-definition:

$$y' = (\log x)' = \lim_{h\to 0} \frac{\log(x+h) - \log(x)}{h} \tag{10.51}$$

Since $\log a - \log b = \log(a/b)$ we write:

$$y' = \lim_{h\to 0} \frac{\log\left(1 + \frac{h}{x}\right)}{h} \tag{10.52}$$

We factor out the numerator h^{-1}:

$$y' = \lim_{h\to 0} \frac{1}{h} \log\left(1 + \frac{h}{x}\right) \tag{10.53}$$

Let us now define $z = h/x$ so $1/h = \frac{1}{z}\frac{1}{x}$:

$$y' = \lim_{h\to 0} \frac{1}{x}\frac{1}{z} \log(1+z) \tag{10.54}$$

Note that as $h \to 0$, $z \to 0$ as well. The multiplier $1/z$ in front of the logarithm we move up to the exponent:

$$y' = \frac{1}{x} \lim_{h\to 0} \log(1+z)^{1/z} \tag{10.55}$$

We shall now look more closely at the part of the expression that consists of $(1+z)^{1/z}$. Let us temporarily rewrite this to the following function:

$$y = (1+z)^{1/z} \tag{10.56}$$

and look at some values in a graph. Figure 10.5 shows the line for this function over some values for z between -1 and 2. Due to the exponent, this function is not defined for $z = 0$.

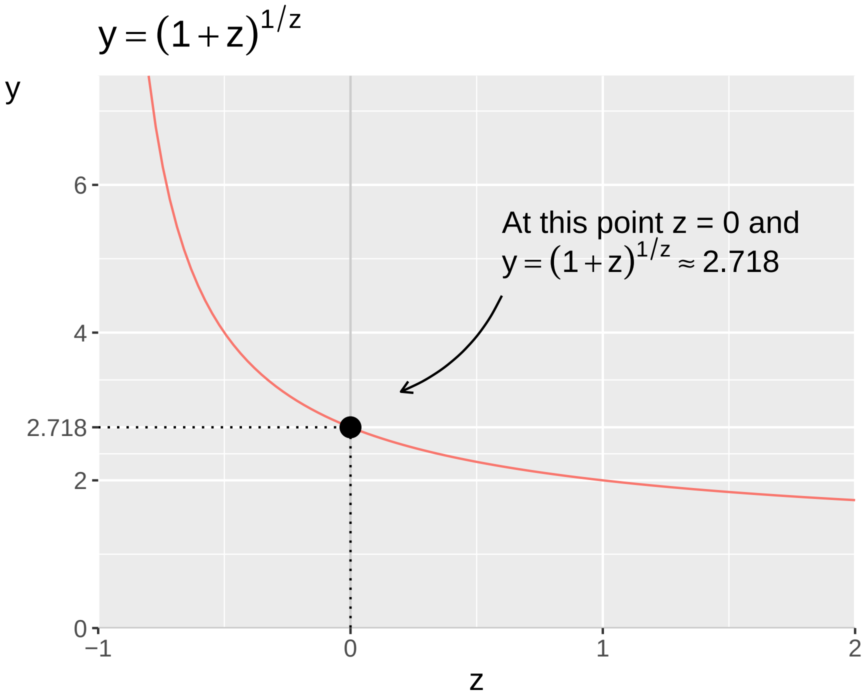


Figure 10.5: The function $y = (1+z)^{1/z}$

Note how the limit for this function approaches the value 2.718 as z approaches 0. The number 2.718 we recognize already as Euler's number, e. The limit for the expression in equation (10.55) is equal to e:

$$y' = \frac{1}{x}\left(\lim_{z\to 0}\log(1+z)^{1/z}\right) = \frac{1}{x}\log e \tag{10.57}$$

As mentioned at the beginning of the example, the logarithm here has base b. Let us now change base to e, so that we get the natural logarithm. We went through this in section 5.5 where in equation (**??**) we had the general rule for changing base between logarithms:

$$\log_a x = \frac{\log_b x}{\log_b a} \tag{10.58}$$

In this case this means:

$$\log_b e = \frac{\ln e}{\ln b} = \frac{1}{\ln b} \tag{10.59}$$

The derivative of $\log x$ then becomes:

$$y'_x(x) = (\log_b x)'_x = \frac{1}{x\ln b} \tag{10.60}$$

When we have base $b = e$ the derivative is:

$$y'_x = (\ln x)'_x = \frac{1}{x} \tag{10.61}$$

10.7 L'Hôpital's rule

Here follows a simplified description of L'Hôpital's rule (also called Bernoulli's rule), which is a useful calculation rule for derivatives. Suppose we have the functions $f(x)$ and $g(x)$ and these go toward 0 when x goes toward the limit a. Alternatively that the function g approaches infinity. We calculate a constant derivative sufficiently close to the limit a, in such a way that we get the following:

$$\lim_{x \to a} \frac{f'}{g'} \tag{10.62}$$

For these functions the following then applies:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'_x(x)}{g'_x(x)} \tag{10.63}$$

This is called L'Hôpital's rule and means that the limit for function f divided by function g as x goes toward a is the same as the limit for the derivative of f divided by the derivative of g as x goes toward a. We illustrate the usefulness of L'Hôpital's rule with the function:

$$y = k(x) = \frac{x^2+3}{x-2} \tag{10.64}$$

Function $k(x)$ is not defined for $x = 2$, since the denominator then becomes 0. To facilitate the calculation of the limit for $x = 2$, we then use L'Hôpital's rule. We define the numerator as $f(x) = x^2 + 3$ and the denominator as $g(x) = x - 2$. We have the derivatives $f'_x = 2x$ and $g'_x = 1$. Now we calculate the limit as $x \to 2$:

$$\lim_{x \to 2} k(x) = \lim_{x \to 2} \frac{2x}{1} = 4 \tag{10.65}$$

The number 4 comes from $x \to 2$ and $2x = 2 * 2 = 4$. The limit for the derivative of $k(x)$ with respect to x as $x \to 2$ is thus 4. According to L'Hôpital's rule, this is also the limit for the quotient $\frac{f(x)}{g(x)}$ as $x \to 2$.

10.8 Partial derivative

When we took the derivative of $f(x)$ above, we wrote $f'(x)$ and since we only have one variable in the function, we understand that this refers to taking the derivative with respect to the variable x. We may also write to our variable to be extra clear:

$$f'_x(x) = \text{the derivative of f with respect to variable x} \tag{10.66}$$

If we have several variables, such as $f(x, y)$, we may calculate the derivative for each variable. This is called the partial derivative. For the derivative of function f with respect to x and y respectively, we write f'_x and f'_y. If we differentiate f with respect to one variable x we do not differentiate with respect to other variables in the same function. Other terms in the function are treated as constants. Take the following function:

$$f(x, y) = x^2 + y^2 \tag{10.67}$$

We shall differentiate this with respect to x and y, and thereafter differentiate the results with respect to x and y again:

$$
\begin{aligned}
f'_x &= 2x \\
f'_y &= 2y \\
f''_{xx} &= 2 \\
f''_{yy} &= 2
\end{aligned}
\tag{10.68}
$$

We may also differentiate f first with respect to first x and get $f'_x = 2x$, and then differentiate this function with respect to y:

$$f''_{xy} = 0 \tag{10.69}$$

The same thing if we instead differentiate f first with respect to y and then x:

$$f''_{yx} = 0 \tag{10.70}$$

Let us take another example with the following function:

$$g(x, y) = x^2 y^2 \tag{10.71}$$

The first and second derivatives of function g with respect to x and y:

$$
\begin{aligned}
g'_x &= 2xy^2 \\
g''_{xx} &= 2y^2 \\
g'_y &= 2yx^2 \\
g''_{yy} &= 2x^2 \\
g''_{xy} &= 4xy \\
g''_{yx} &= 4yx
\end{aligned}
\tag{10.72}
$$

10.9 Differentiation rules

Table 10.1 describes with general equations some examples of rules for differentiation, where x is the variable and the letters a, b and c are constants. The letter e is Euler's number. $f(x)$ refers to function f of x and f'_x refers to the first derivative of this function, with respect to x.

Table 10.1: Differentiation rules

	Function	**First derivative**
1	$f(x) = bx^a + c$	$f'_x(x) = abx^{a-1}$
2	$f(x) = \frac{1}{x} = x^{-1}$	$f'_x(x) = -1 \cdot x^{-2} = -1/x^2$
3	$f(x) = \sqrt{x} = x^{\frac{1}{2}}$	$f'_x(x) = \frac{1}{2}x^{-\frac{1}{2}} = 1/(2\sqrt{x})$
4	$\frac{1}{f(x)} = (f(x))^{-1}$	$\left(\frac{1}{f(x)}\right)'_x = -\frac{f'_x(x)}{[f(x)]^2}$
5	$f(x) = b^{ax}$	$f'_x(x) = (a \ln b)\, b^{ax}$
6	$f(x) = e^{g(x)}$	$f'_x(x) = g'_x e^{g(x)}$
7	$f(x) = e^{ax}$	$f'_x(x) = ae^{ax}$
8	$f(x) = \log x$, $g(x) = \ln x$	$f'_x(x) = \frac{1}{x \ln b}$, där b är log bas., $g'_x(x) = \frac{1}{x}$

Table 10.2: Differentiation rules for two combined functions

	Combination	**First derivative**
1	$(f+g)(x) = f(x) + g(x)$	$(f+g)'_x = f'_x + g'_x$
2	$(f-g)(x) = f(x) - g(x)$	$(f-g)'_x = f'_x - g'_x$
3	$(f \times g)(x) = f(x) \times g(x)$	$(f \times g)_x = f'_x(x)\,g(x) + f(x)\,g'_x(x)$
4	$(f/g)(x) = \frac{f(x)}{g(x)}$	$\left(\frac{f}{g}\right)'_x = \frac{f'_x(x)g(x)+f(x)g'_x(x)}{[g(x)]^2}$
5	$f \circ g = f(g(x))$	$f'_x(g(x))\,g'_x(x)$

In section 4.9 we went through how we may use addition, subtraction, multiplication and division on functions, like $(f+g)(x) = f(x) + g(x)$. In the same section we introduced composite functions, such as $y = f(g(x))$. Table 10.2 summarizes calculation rules for derivatives where two functions are added (row 1), subtracted (row 2), multiplied (row 3), divided (row 4) or composed (row 5). We illustrate these calculation rules with the functions:

$$\begin{aligned} f(x) &= x^2 \\ g(x) &= 3 - 2x \end{aligned} \tag{10.73}$$

The first derivative of each function with respect to x is:

$$\begin{aligned} f'_x(x) &= 2x \\ g'_x(x) &= -2 \end{aligned} \tag{10.74}$$

If the functions are added, we get:

$$\begin{aligned} f + g &= x^2 + 3 - 2x \\ (f+g)'_x &= f' + g' = 2x - 2 \end{aligned} \tag{10.75}$$

With subtraction:

$$\begin{aligned} f - g &= x^2 - 3 + 2x \\ (f-g)'_x &= f' - g' = 2x + 2 \end{aligned} \tag{10.76}$$

Row three in table 10.2 describes what is called the product rule:

$$\begin{aligned} f \times g &= x^2(3 - 2x) \\ (f \times g)'_x &= f'g + fg' = 2x(3-2x) - 2x^2 \end{aligned} \tag{10.77}$$

Row four in table 10.2 describes the quotient rule. This shows how to calculate the derivative when one function is divided by another function:

$$\begin{aligned} \frac{f}{g} &= \frac{x^2}{3-2x} \\ \left(\frac{f}{g}\right)'_x &= \frac{f'g + fg'}{g^2} = \frac{2x(3-2x) - 2x^2}{(3-2x)^2} \end{aligned} \tag{10.78}$$

Row five in table 10.2 describes the chain rule, which shows how to calculate the derivative for composite functions. For example:

$$f(g(x)) = (3-2x)^2 \tag{10.79}$$

We also have that $g'_x = -2$ and $f'_{g(x)} = 2(3-2x)$. The derivative of $f(g(x))$ with respect to x is then:

$$f(g(x))'_x = f'_g(g)\, g'_x = 2(3-2x)(-2) = -4(3-2x) \tag{10.80}$$

10.10 Elasticity

A measure that is often used in social science is elasticity. This indicates how much one variable changes in percentage terms at a percentage change in another variable. Say for example that q is the number of sold units of this good (the market's demand), and p is the price for this good.

Change between two price levels can be written as $\Delta p = p_2 - p_1$, which we read as the difference between price level 1 and price level 2. The letter Δ(the Greek letter capital delta) symbolizes change here. Change between two quantities: $\Delta q = q_2 - q_1$. The price elasticity can then be defined as:

$$\text{price elasticity: } \frac{\frac{\Delta q}{q}}{\frac{\Delta p}{p}} = \frac{\frac{q_2 - q_1}{q_1}}{\frac{p_2 - p_1}{p_1}} \tag{10.81}$$

This is percentage change of q divided by percentage change of p. If demand q changes a lot when p changes, this means that the good has high price elasticity, customers are very price-sensitive. If demand q barely changes at all when p changes, price elasticity is low and customers are relatively price-insensitive.

In equation (10.32) we described the derivative using the letter d and the symbol ∂. The derivative gives us the slope of the line at a specific point. Similarly, we may compare the elasticity at a specific point, which is called point elasticity. Say for example that a large store sells milk for 10 USD per package. Every week 100 packages are sold. In January, the store decides to raise the price by 20%, to 12 USD per package. In the long run, sales drop to 97 packages per week. The price elasticity of demand for milk is then:

$$\frac{\partial q/q}{\partial p/p} = \frac{-0.03}{0.2} = -0.15 \tag{10.82}$$

When the price increases by 20%, demand decreases by 3%, which is why the price elasticity is -0.15%. We do not know if this elasticity applies to all price levels. Studying price elasticity in reality is more complex than this example, see chapter 27 .

10.11 Chapter summary

- The limit, limes, for a function describes the value the function approaches when it comes infinitely close to a point. Example: $\lim_{x\to 0} \frac{1}{x} = \infty$ and $\lim_{x\to\infty} \frac{1}{x} = 0$.
- The derivative is a function that describes the limit for a line's slope at a point. The derivative's h-definition: $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$, where h indicates the distance between two values in the variable x.
- For the function $f(x)$ we may write the first derivative with respect to x as f'_x, the second derivative as f''_{xx} and the third derivative as f'''_{xxx}. Example: $y = f(x) = x^3$, $f'_x = 3x^2$, $f''_{xx} = 6x$ and $f'''_{xxx} = 6$.
- The function $y = f(x) = a + bx$ has the slope b for all values of x that the function is defined for. The derivative can be written $f' = b$.

- Partial derivative: differentiate a function with several variables with respect to one variable at a time. Example: $y = f(x,z) = x^2 + z^2$ has first derivative $f'_x = 2x$ and $f'_z = 2z$.
- Examples of differentiation rules:

  -$f(x) = bx^a + c$ gives $f'_x = abx^{a-1}$.

  -$g(x) = b^{ax}$ gives $g'_x(x) = (a \ln b)\, b^{ax}$.

  -$h(x) = \log x$ gives $h'_x(x) = 1/(x \ln b)$.
- • Examples of differentiation with combined and composite functions

  -$(f+g)(x)'_x = f'_x + g'_x$

  -$(f-g)(x)'_x = f'_x - g'_x$

  -$\left(\frac{1}{f(x)}\right)'_x = \left(f(x)^{-1}\right)'_x = -f'_x f^{-2} = \frac{-f'_x}{(f(x))^{-2}}$
  - Product rule: $(f(x)\,g(x))'_x = ((f \times g)(x))'_x = f'g + g'f$
  - Quotient rule: $\left(\frac{f(x)}{g(x)}\right)'_x = \frac{f'_x g + f g'_x}{g}$
  - Chain rule: $f(g(x))'_x = g' f'$.
- L'Hôpital's rule: $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'_x(x)}{g'_x(x)}$.

10.12 Exercises

1. Consider the function $f(x) = \frac{1}{x}$

(a) What is (lim_{xrightarrowinfty}fleft(xright))

(b) What is (lim_{xrightarrow0^{+}}fleft(xright))

(c) What is (lim_{xrightarrow0^{-}}fleft(xright))

Answer: (a) $\lim_{x\to\infty} \frac{1}{x} = 0$, (b) $\lim_{x\to 0^+} \frac{1}{x} = +\infty$, (c) $\lim_{x\to 0^-} \frac{1}{x} = -\infty$.

2. Consider the function $f(x) = \frac{x^2-4}{x-2}$

(a) Show that directly substituting (x=2)

(b) Factorize the numerator and simplify (fleft(xright))

(c) Calculate (lim_{xrightarrow2}fleft(xright))

Answer: (a) $f(2) = \frac{4-4}{2-2} = \frac{0}{0}$, (b) $x^2 - 4 = (x-2)(x+2)$, (c) $\lim_{x\to 2}(x+2) = 4$.

3. Use the power rule $(x^a)' = ax^{a-1}$

(a) (fleft(xright)=x^{4})

(b) (fleft(xright)=3x^{2}+2x+5)

(c) Calculate the slope of (fleft(xright)=x^{2})

Answer: (a) $f'(x) = 4x^3$, (b) $f'(x) = 6x + 2$, (c) $f'(3) = 2 \cdot 3 = 6$.

4. Given $f(x) = x^4$

(a) Find the first derivative (f'left(xright))

(b) Find the second derivative (f' 'left(xright))

(c) Find the third derivative (f' ' 'left(xright))

Answer: (a) $f'(x) = 4x^3$, (b) $f''(x) = 12x^2$, (c) $f'''(x) = 24x$.

5. Differentiate the following exponential and logarithmic functions.

(a) (fleft(xright)=e^{3x})

(b) (fleft(xright)=ln x)

(c) (fleft(xright)=2^{x})

Answer: (a) $f'(x) = 3e^{3x}$, (b) $f'(x) = \frac{1}{x}$, (c) $f'(x) = \ln(2) \cdot 2^x$.

6. Consider the function $f(x,y) = x^3 + 2x^2y + y^2$

(a) Find the partial derivative (f’_{x})

(b) Find the partial derivative (f’_{y})

(c) Find the second partial derivative (f’ ’_{xx})

Answer: (a) $f'_x = 3x^2 + 4xy$, (b) $f'_y = 2x^2 + 2y$, (c) $f''_{xx} = 6x + 4y$.

7. Use the product rule $(fg)' = f'g + fg'$ to differentiate the following.

(a) (fleft(xright)=x^{2}e{x})

(b) (fleft(xright)=xln x)

(c) Find the (x) -value where (fleft(xright)=xln x) has a horizontal tangent ((f’left(xright)=0)).

Answer: (a) $f'(x) = xe^x(x+2)$, (b) $f'(x) = \ln x + 1$, (c) $\ln x + 1 = 0 \Rightarrow x = e^{-1} = \frac{1}{e}$.

8. Use the quotient rule $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ to differentiate the following.

(a) (fleft(xright)=frac{x}{x+1})

(b) (fleft(xright)=frac{x^{2}}{e}{x}})

(c) Find (f’left(0right)) for (fleft(xright)=frac{x^{2}}{e}{x}}) .

Answer: (a) $f'(x) = \frac{1}{(x+1)^2}$, (b) $f'(x) = \frac{x(2-x)}{e^x}$, (c) $f'(0) = 0$.

9. Use the chain rule $(f(g(x)))' = f'(g(x)) \cdot g'(x)$ to differentiate the following.

(a) (fleft(xright)=left(3x+2right)^{4})

(b) (fleft(xright)=e^{x}{2}})

(c) (fleft(xright)=lnleft(2x+1right))

Answer: (a) $f'(x) = 12(3x+2)^3$, (b) $f'(x) = 2xe^{x^2}$, (c) $f'(x) = \frac{2}{2x+1}$.

10. The demand function for a product is $Q(P) = \frac{100}{P^2}$.

(a) Find (Q’left(Pright)=frac{dQ}{dP}) .

(b) The price elasticity is (varepsilon=frac{dQ}{dP}cdotfrac{P}{Q}) . Calculate (varepsilon) .

(c) Is demand elastic (left(left|varepsilonright|>1right)) or inelastic (left(left|varepsilonright|<1right)) ?

Answer: (a) $Q'(P) = -\frac{200}{P^3}$, (b) $\varepsilon = -2$, (c) $|\varepsilon| = 2 > 1 :$ demand is elastic.

11. A firm’s total revenue is $TR(Q) = 20Q - Q^2$.

(a) Find marginal revenue (MRleft(Qright)=TR’left(Qright)) .

(b) At what quantity (Q^{*}) is (MR=0) ?

(c) Find (TR’ ’left(Qright)) and interpret the result.

Answer: (a) $MR(Q) = 20 - 2Q$, (b) $Q^* = 10$, (c) $TR''(Q) = -2 < 0 :$ marginal revenue is decreasing.

12. Consider $f(x,y) = 3x^2y + 2xy^3$.

(a) Find (f’*{x}) and (f’*{y}) .

(b) Find the cross-partial derivative (f’ ’_{xy}) .

(c) Verify Young’s theorem: (f’ ’*{xy}=f’ ’*{yx}) .

Answer: (a) $f'_x = 6xy + 2y^3$, $f'_y = 3x^2 + 6xy^2$, (b) $f''_{xy} = 6x + 6y^2$, (c) $f''_{yx} = 6x + 6y^2 = f''_{xy}$.

Chapter 11

Optimization with functions

This chapter introduces optimization, which we shall use here to find a desired solution based on a function. In social science, we may for example use this to describe how a company can maximize its profit or describe how a person can maximize their utility and happiness or minimize their unhappiness.

11.1 Extreme values

Figure 11.1 shows a line drawn with an unknown function where two points are marked. The lowest point on the line is the function's minimum value and the highest point is the maximum value. Both of these two are examples of what are called extreme values, also called extreme points or critical points. Consider the following function:

$$y = f(x) = x^2, \forall 1 \leq x \leq 10 \tag{11.1}$$

The symbol $\forall$ means "for all". The symbol $\leq$ means "less than or equal to". The expression $\forall 1 \leq x \leq 10$ should be read as "for all x that are greater than or equal to 1 and less than or equal to 10". This means that this function is defined over the interval 1 to 10 on the variable x.

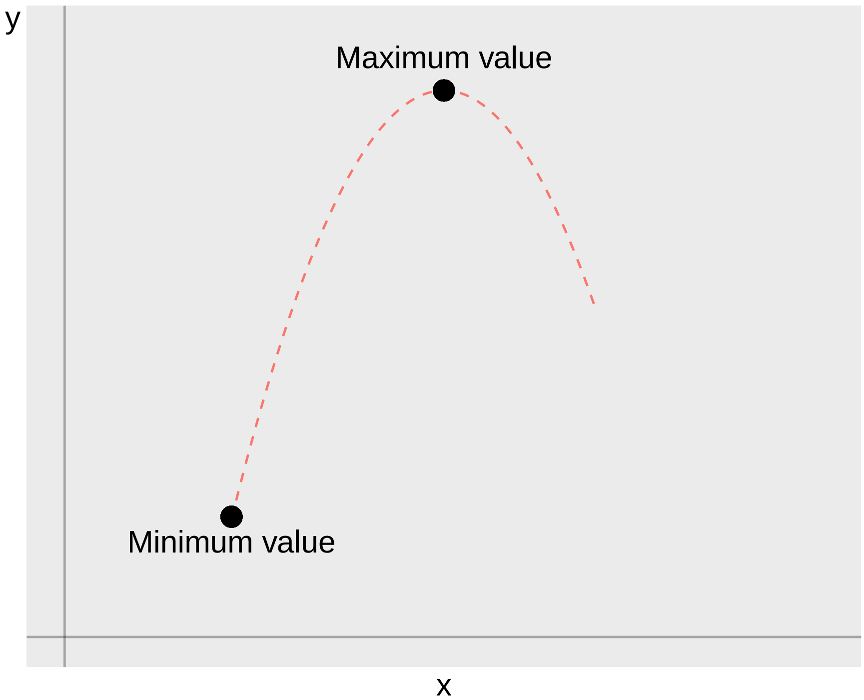


Figure 11.1: A function with a maximum and minimum value

Since x here can only take a value between 1 and 10, the function $f(x)$ and the variable y will only assume values between $1^2 = 1$ and $10^2 = 100$. Our function is relatively simple and we may therefore see that the function's smallest and largest value in this case are given by these input values:

$$\begin{aligned} \text{Smallest value} = \text{minimum value: } f(1) = 1^2 = 1 \\ \text{Greatest value} = \text{maximum value: } f(10) = 10^2 = 100 \end{aligned} \tag{11.2}$$

All other values for x between 1 and 10 result in values for y that are located between these two results. Even a function that is defined over a set can lack extreme values. We can illustrate this with the function:

$$y = x, \; \forall x \in (0, 3) \tag{11.3}$$

In a graph, this function would create a straight line. Since x is defined over an open interval, the points $(x, y) = (0, 0)$ and $(3, 3)$ are not included. The function is defined infinitely close to the values $x = 0$ and $x = 3$. But for every decimal close to these values, there will be a decimal that is even closer. The function therefore lacks minimum and maximum values.

11.2 Differentiate to find extreme values

The derivative with respect to a variable x gives us the slope of the line at a specific point. This is a measure of how much the function or the dependent variable, for example y, changes when variable x increases by 1. We can use this information for different purposes. Suppose we have the following function:

$$y = f(x) = -x^2, \quad \forall x \in [-\infty, \infty] \tag{11.4}$$

Function f describes a desirable phenomenon and we therefore seek the value for x where f is maximized. We seek the function's maximum value, which the derivative can help us with. The function is defined over a bounded interval and is illustrated in figure 11.2 . If the first derivative is equal to 0 at a value on the variable and the line at the point is therefore horizontal and lacks slope, this is an extreme point, minimum or maximum.

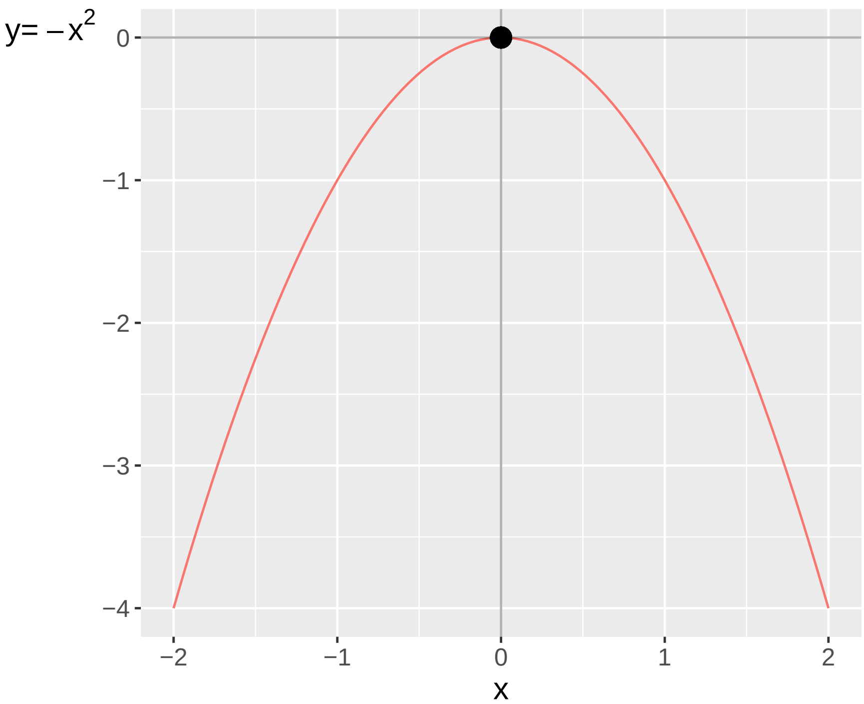


Figure 11.2: The function $y = f(x) = x^2$

If we know the function's second derivative, the slope of the slope, this can also give us more information about what type of extreme point it is. If the second derivative is negative at an extreme point, it means that the points adjacent to the extreme value are smaller than the point and we then know that the extreme value is a maximum point. If the second derivative is positive, the slope of the slope increases, then the extreme value is a minimum point.

To find the function's extreme points, we calculate the first derivative and set the result equal to 0. This gives us what is called a first-order condition. A first-order condition is a mathematical expression that describes the condition that at the point that is maximum, the first derivative must be equal to 0. From this we can solve for x. From equation (11.4) we get the following first derivative which we set equal to 0:

$$f'(x) = -2x = 0 \tag{11.5}$$

This first derivative is equal to 0 at $x = 0$, which is why we from $f'(x) = 0$ can solve for $x^* = 0$. This is the function's only extreme value. We check the function's second derivative to see if this is a maximum or minimum point:

$$f''(x) = -2 \tag{11.6}$$

Since the second derivative is negative, $f'' < 0$, for all x, $f(x = 0)$ must be a maximum point. The derivative is not the only way to find a function's extreme values. Function f is defined for all $x \in [-\infty, \infty]$. The function's minimum is therefore given by the two points $x = -\infty$ and $x = \infty$, negative and positive infinity, where the function goes toward negative infinity. Note that this is based on the function being defined for infinity. If the domain instead is $x \in (-\infty, \infty)$, where negative and positive infinity are excluded from the domain, then the function lacks a minimum point. There will always be an x-value that results in lower $f(x)$. The value $x = 0$ is however still the function's maximum point. Consider now the following function:

$$y = h(k) = k^2 - k + 3, \quad \forall x \in [-\infty, \infty] \tag{11.7}$$

where h is the name of the function and k is the variable. To find out for which values of k the function has its highest and lowest values, we take the first and second derivative:

$$\begin{aligned} h'_k(k) &= 2k - 1 \\ h''_{kk}(k) &= 2 \end{aligned} \tag{11.8}$$

From the first derivative we see that $k^* = \frac{1}{2}$. Since the second derivative is $h''_{kk} > 0$, we assume that this is a minimum point. At $k^* = \frac{1}{2}$ we have:

$$h\left(k^* = \frac{1}{2}\right) = k^2 - k + 3 = \left(\frac{1}{2}\right)^2 - \frac{1}{2} + 3 = 2 + \frac{3}{4} \tag{11.9}$$

The function's line is illustrated in figure 11.3 where the minimum point $(k, y) = \left(\frac{1}{2}, 2.75\right)$ is marked. The function's maximum is given by the two points $x = -\infty$ and $x = \infty$, where function h goes toward positive infinity.

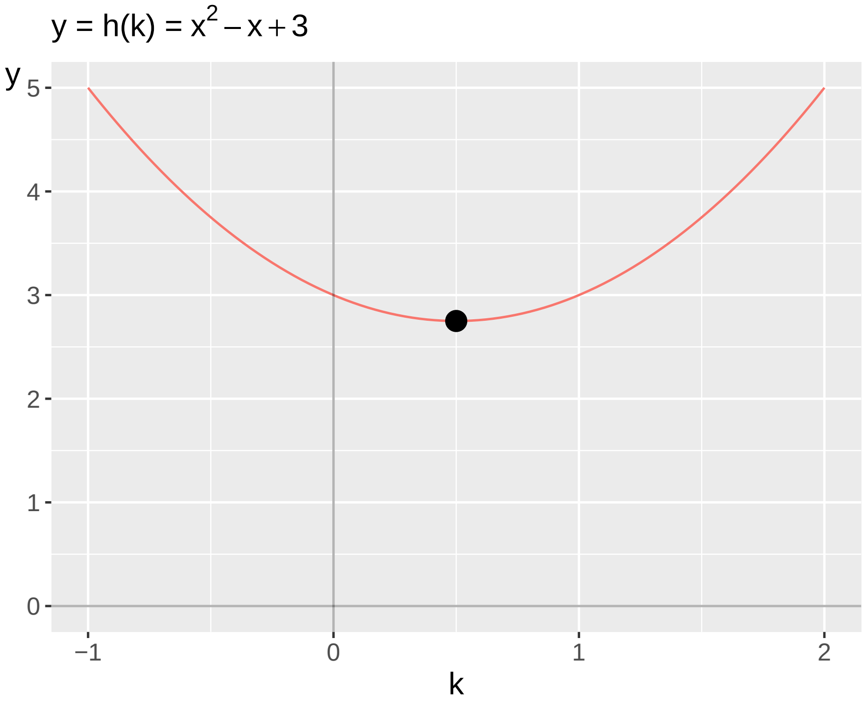


Figure 11.3: The line for $h\left(k\right)=k^{2}-k+3$ and its minimum point

11.3 Global and local

Maximum and minimum points can be global and local. A local maximum value is the point within an interval where a function assumes its largest value. A global maximum value is the point that gives the function's largest value over all values that the function is defined for. The same thing for minimum values. Let us illustrate with the following function:

$$y=f\left(x\right)=x^{3}-x^{2}-x+2,\quad\forall x\in\left[-10,10\right]\tag{11.10}$$

This is a third-degree function defined for all x from -10 to plus 10. The first and second derivatives give:

$$\begin{aligned}f_{x}^{\prime}&=3x^{2}-2x-1\\f_{xx}^{\prime\prime}&=6x-2\end{aligned}\tag{11.11}$$

From the first derivative we solve for two zero points, two values for x where the first derivative $f_{x}^{\prime}$ is equal to 0:

$$3x^{2}-2x-1=0\tag{11.12}$$

From the quadratic formula we see that:

$$x^{*}=-\left(-\frac{2/3}{2}\right)\pm\sqrt{\left(-\frac{2/3}{2}\right)^{2}-\left(-1/3\right)}\tag{11.13}$$

This gives the solutions $x_{1}=-\frac{1}{3}$ and $x_{2}=1$. At these two points $f^{\prime}=0$. We check the solutions by substituting the values into the second derivative:

$$f''_{xx}\left(-\frac{1}{3}\right) = 6\left(-\frac{1}{3}\right) - 2 = -4 \qquad (11.14)$$
$$f''_{xx}(1) = 6 * 1 - 2 = 4$$

The first row means that the second derivative is negative at $x = -\frac{1}{3}$. This indicates that this is a maximum point. The second row gives that the second derivative is positive at $x = 1$, which suggests that this is a minimum point. Let us also calculate y at these points:

$$\begin{aligned} y = f\left(x = -\frac{1}{3}\right) &= x^3 - x^2 - x + 2 \\ &= \left(-\frac{1}{3}\right)^3 - \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) + 2 \\ &= -\frac{1}{27} - \frac{1}{9} + \frac{1}{3} + 2 \\ &= 2 + \frac{5}{27} \end{aligned}$$

For $x = 1$ we have:

$$y = f(x = 1) = x^3 - x^2 - x + 2 = (1)^3 - (1)^2 - (1) + 2 = 1 \qquad (11.15)$$

The function is defined for all $x \in [-10, 10]$. Let us also calculate y for the endpoints (the outer values) -10 and 10. For $x = -10$:

$$\begin{aligned} y = f(x = -10) &= x^3 - x^2 - x + 2 \\ &= (-10)^3 - (-10)^2 - (-10) + 2 \\ &= -1,000 - 100 + 10 + 2 \\ &= -1,088 \end{aligned}$$

For $x = 10$:

$$\begin{aligned} y = f(x = 10) &= x^3 - x^2 - x + 2 \\ &= (10)^3 - (10)^2 - (10) + 2 \\ &= 1,000 - 100 - 10 + 2 \\ &= 892 \end{aligned} \qquad (11.16)$$

The function's value at x=-10 is smaller than the value at $x = 1$, since $-1,088 < 1$ and the function's value at $x = 10$ is larger than at $x = -\frac{1}{3}$ since $892 > 2 + \frac{5}{27}$. We therefore conclude that the function's global minimum is at $(x, y) = (-10, -1,088)$ and its global maximum is at $(x, y) = (10, 892)$. Additionally, the function has a local minimum at the point $(x, y) = (1, 1)$ and a local maximum at $(x, y) = \left(-\frac{1}{3}, 2 + \frac{5}{27}\right)$.

The next question is how local these extreme values are, within which intervals of x are these points maximum and minimum? We start by looking at this in a graph, which is illustrated in figure 11.4 . The two local extreme values are marked along the line with points. The global extreme values at $x = -10$ and $x = 10$ are outside the picture and marked with arrows.

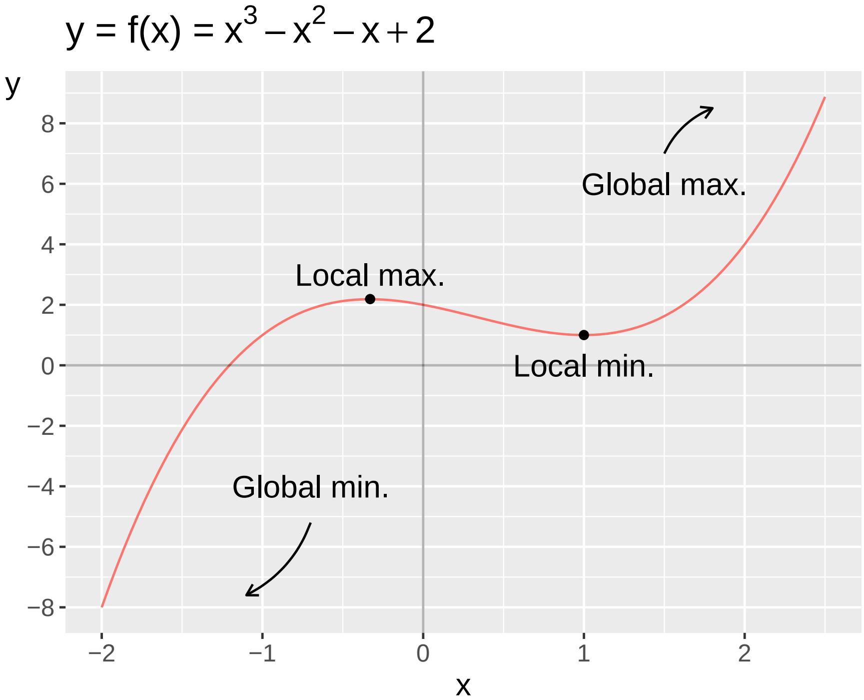


Figure 11.4: Local and global minimum and maximum points

From the picture we see that the point $(x, y) = (1, 1)$ is a minimum for the function for x-values from -1 and upward. For all values for x above -1 up to 10 (the function's domain), the point $(1, 1)$ is the one that gives the lowest value. For the point $(x, y) = \left(-\frac{1}{3}, 2 + \frac{5}{27}\right)$ we see that this is the function's maximum for all x from approximately 1.6 and downward.

11.4 More examples

Let us find the extreme values for some functions. We seek the largest and smallest value, maximum and minimum value. Let us start with the function:

$$y = f(x) = x, \quad \forall x \leq 0 \tag{11.17}$$

Since f is defined for $x \leq 0, y = f(0) = 0$ is the function's largest value, its global maximum value. Since f is defined for all negative values, y goes toward negative infinity, $-\infty$, when x does the same, which is the function's global minimum value. If we instead have the function:

$$y = f(x) = x, \quad \forall x < 0 \tag{11.18}$$

where the value 0 is excluded from the domain, then the function lacks a global maximum value. We only come infinitely close to 0. The minimum value is also here negative infinity, $-\infty$. Let us now take the function:

$$y = f(x) = 3x, \quad \forall x \in (2, 12) \tag{11.19}$$

The domain $(2, 12)$ excludes the values 2 and 12. The function therefore lacks global minimum and maximum values. If the function is instead defined for $x \in [2, 12]$, the function's global minimum value is:

$$y = f(2) = 6 \tag{11.20}$$

The function's global maximum value:

$$y = f(12) = 36 \tag{11.21}$$

Let us now calculate the global minimum and maximum value for the following function:

$$y = f(x) = x^2 - x, \quad \forall x \in \mathbb{R} \tag{11.22}$$

We take the first and second derivatives of y with respect to x:

$$\begin{aligned} y'_x &= 2x - 1 \\ y''_{xx} &= 2 \end{aligned} \tag{11.23}$$

From the first derivative we see that $x = \frac{1}{2}$. This gives the following value for y:

$$y = f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 - \frac{1}{2} = -\frac{1}{4} \tag{11.24}$$

The second derivative is $y''_{xx} > 0$ which means that this is a global minimum value. Since the function is defined over the real numbers $\mathbb{R}$, which does not include infinity, the function lacks a global maximum value.

11.5 Terrace

Let us now look at the line in figure 11.5 . At the two marked points in the diagram, the first derivative is equal to 0. In the figure, the point $(x, y) = (-0.5, -1.7)$ is a minimum point. At the point $(1, 0)$, the second derivative is equal to 0, which is why this is a terrace point. At the terrace point, our function increases for larger values of x and decreases for smaller values of x. Let us summarize what applies for the second derivative for points where the first derivative is equal to 0. We have the following rules:

$$\begin{aligned} \text{Maximimum point: } & f'' < 0 \\ \text{Minimimum point: } & f'' > 0 \\ \text{Terrace point: } & f'' = 0 \end{aligned} \tag{11.25}$$

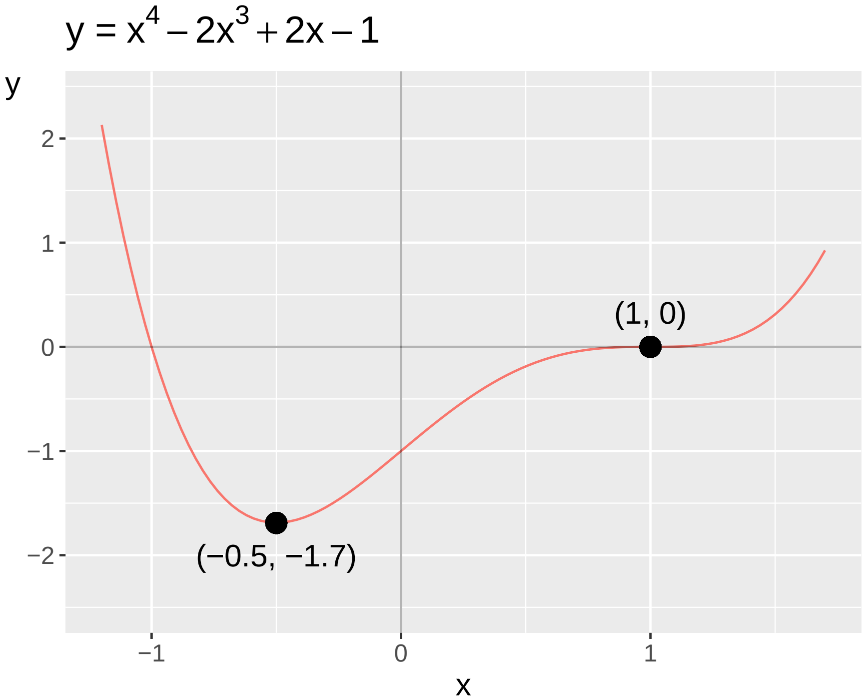


Figure 11.5: Function with minimum and terrace point

The function for the line in figure 11.5 is:

$$y = f(x) = x^4 - 2x^3 + 2x - 1 \tag{11.26}$$

The first derivative with respect to x:

$$f'_x(x) = 4x^3 - 6x^2 + 2 \tag{11.27}$$

The second derivative with respect to x:

$$f''_{xx}(x) = 12x^2 - 12x \tag{11.28}$$

Try yourself to find these functions' solutions.

11.6 Optimization with constraints

So far we have worked with optimization to find extreme values in functions without any particular form of limitation, other than the function itself. Often, however, we have use of being able to add constraints to our calculations. We often have use of this in social science, for example in reasoning about how people and organizations try to achieve their goals based on limited means or opportunities. Constraints can however make it more difficult to find the solution to optimization problems.

A common method for finding extreme points with constraints is the Lagrange method. We begin by describing the method on a brief overall level. Suppose we shall optimize function $f(x)$ with respect to a constraint that we formulate generally as function $g(x) = 0$. We begin by just writing this as a function:

$$\max_x f(x), \quad \text{w.r.t. } g(x) = 0 \tag{11.29}$$

Function f shall be maximized with respect to variable x so that function g is equal to 0. The abbreviation “w.r.t” stands for with respect to . Sometimes the abbreviation “s.t.” is used which stands for subject to . Then we set up the Lagrange function:

$$\mathcal{L}(x,\lambda) = f(x) + \lambda g(x) \tag{11.30}$$

where the Greek letter small lambda λ symbolizes the Lagrange multiplier, which in the Lagrange function is a new variable that is connected with the Lagrange method. We shall return to the interpretation of λ. For the functions f and g to be able to be set up in this way, we may need to rewrite them. To find the extreme points, we differentiate the Lagrange function with respect to the two variables x and λ:

$$\begin{aligned} \mathcal{L}'_x &= f'_x + \lambda g'_x \\ \mathcal{L}'_\lambda &= g \end{aligned} \tag{11.31}$$

If we set these expressions equal to 0, we get the first-order conditions and can solve for any solutions for x and λ. The Lagrange multiplier λ has a specific interpretation depending on the maximization problem itself and can for example indicate changes with respect to if the constraint, which λ is multiplied by, changes by one unit. Let us illustrate by maximizing the following function:

$$f(x) = x^2 \tag{11.32}$$

While we maximize $f(x)$, we shall fulfill the following constraint: $x^2 = 2$. We call this expression function $g(x)$ and move all terms in the function to one side of the equals sign so that we get $g(x) = 0$:

$$g(x) = x^2 - 2 = 0$$

Now we set up the Lagrange function in the following way:

$$\begin{aligned} \mathcal{L}(x,\lambda) &= f(x) + \lambda g(x) \\ \mathcal{L}(x,\lambda) &= x^2 + \lambda\left(x^2 - 2\right) \end{aligned} \tag{11.33}$$

To find the function's solutions, we take the derivative of $\mathcal{L}$ with respect to x and λ. By setting these respective derivatives equal to 0, we get the first-order conditions:

$$\begin{aligned} \mathcal{L}'_x &= 2x + 2x\lambda = 0 \\ \mathcal{L}'_\lambda &= x^2 - 2 = 0 \end{aligned} \tag{11.34}$$

From $\mathcal{L}'_x$ we get the solution for λ:

$$\lambda^* = -1 \tag{11.35}$$

From $\mathcal{L}'_\lambda$ we get the solution for x:

$$x^* = \sqrt{2} \tag{11.36}$$

This means that if we shall maximize the function $f(x) = x^2$ so that the constraint $x^2 - 2 = 0$ is fulfilled, then $x^* = 2^{1/2} \approx 1,41$. Let us now take the following maximization problem with constraints:

$$\begin{gathered} \text{Maximize } f(x,y) = x^2 y \\ \text{under the condition } g(x,y) = x + y = 10 \end{gathered} \tag{11.37}$$

We solve this by setting up the Lagrange function:

$$\mathcal{L}(x, y, \lambda) = x^2 y + \lambda(x + y - 10) \tag{11.38}$$

From this we derive the partial derivatives and set these equal to 0 to get the first-order conditions:

$$\begin{aligned} \mathcal{L}'_x &: 2xy + \lambda = 0 \\ \mathcal{L}'_y &: x^2 + \lambda = 0 \\ \mathcal{L}'_\lambda &: x + y - 10 = 0 \end{aligned} \tag{11.39}$$

Since $\mathcal{L}'_x = \mathcal{L}'_y = 0$, we create respective definitions of λ:

$$\begin{aligned} \mathcal{L}'_x &: \lambda = 2xy \\ \mathcal{L}'_y &: \lambda = x^2 \end{aligned} \tag{11.40}$$

By setting the definitions of λ equal to each other, we get a new constraint:

$$x^2 = 2xy \tag{11.41}$$

We simplify this by dividing both sides by x:

$$x = 2y \tag{11.42}$$

We also have the constraint left:

$$x + y - 10 = 0 \tag{11.43}$$

Based on this we now solve for x and y. We have $x = 2y$, which we substitute into equation (11.43) :

$$\begin{aligned} x + y - 10 &= 0 \\ 2y + y - 10 &= 0 \end{aligned} \tag{11.44}$$

This gives us the solution $y^* = \frac{10}{3}$. This y we use in equation (11.42) , $x = 2y$, and solve for x:

$$x^* = \frac{20}{3} \tag{11.45}$$

These are the values for x and y that optimize $f(x, y) = x^2 y$, while the constraint $x + y = 10$ is fulfilled:

$$f(x^*, y^*) = f\left(\frac{20}{3}, \frac{10}{3}\right) = (20/3)^2 (10/3) = \frac{4,000}{27} \tag{11.46}$$

11.7 Chapter summary

- A function can have maximum and minimum points where the function's value is maximized or minimized. These are called extreme points.
- One way to find extreme points is to use derivatives and calculate a line's slope at a specific point. The first derivative gives the line's slope, the second derivative gives the slope of the slope, and so on. To calculate the derivative we use limits, limes.
- Example: the function $y = f(x) = x^2$ has first derivative $f' = 2x$ and second derivative $f'' = 2$.
- If the first derivative is $f' = 0$ at a point, this is a maximum point if $f'' < 0$, minimum point if $f'' > 0$ and terrace point if $f'' = 0$.
- The Lagrange method is a method for solving maximization or minimization problems with constraints. The general form for the Lagrange function is $\mathcal{L}(x, \lambda) = f(x) + \lambda g(x)$ where λ is the Lagrange multiplier and g is the function for the constraint.

11.8 Exercises

1. Find the extreme point of $f(x) = -x^2 + 4x$

(a) Set (f'left(xright)=0)

(b) Use the second derivative to determine whether this is a maximum or minimum.

(c) Calculate (fleft(x^{*}right))

Answer: (a) $-2x + 4 = 0 \Rightarrow x^* = 2$, (b) $f''(x) = -2 < 0$: maximum, (c) $f(2) = 4$.

2. Find the extreme point of $h(k) = k^2 - k + 3$

(a) Set (h'left(kright)=0)

(b) Use the second derivative to determine whether this is a maximum or minimum.

(c) Calculate (hleft(k^{*}right))

Answer: (a) $2k - 1 = 0 \Rightarrow k^* = \frac{1}{2}$, (b) $h''(k) = 2 > 0$: minimum, (c) $h\left(\frac{1}{2}\right) = \frac{11}{4} = 2.75$.

3. Given $f(x) = x^3 - x^2 - x + 2$

(a) Find the critical points by setting (f'left(xright)=0)

(b) Determine whether each critical point is a local maximum or local minimum.

(c) State the function's global maximum and global minimum over the interval.

Answer: (a) $3x^2 - 2x - 1 = 0 \Rightarrow x_1^* = -\frac{1}{3}, x_2^* = 1$, (b) $f''\left(-\frac{1}{3}\right) = -40$ (local min), (c) global extrema determined by boundary values.

4. Find the extreme point of $f(x) = x^2 - x$

(a) Set (f'left(xright)=0)

(b) Is this a maximum or minimum?

(c) Calculate the extreme value (fleft(x^{*}right))

Answer: (a) $2x - 1 = 0 \Rightarrow x^* = \frac{1}{2}$, (b) $f''(x) = 2 > 0$: minimum, (c) $f\left(\frac{1}{2}\right) = -\frac{1}{4}$.

5. Use the Lagrange method to maximize $f(x) = x^2$

(a) Write the Lagrange function (mathcal{L}left(x,lambdaright))

(b) Write the two first-order conditions.

(c) Find (x^{*})

Answer: (a) $\mathcal{L} = x^2 + \lambda(x^2 - 8)$, (b) $\mathcal{L}'_x = 2x + 2x\lambda = 0$, (c) $x^* = \sqrt{8} = 2\sqrt{2}$.

6. Use the Lagrange method to maximize $f(x, y) = xy$ subject to $x + y = 6$.

(a) Write the Lagrange function (mathcal{L}left(x,y,lambdaright)) .

(b) Write the three first-order conditions.

(c) Find (x^{}) and (y^{}) .

Answer: (a) $\mathcal{L} = xy + \lambda\left(x + y - 6\right)$, (b) $\mathcal{L}'_x = y + \lambda = 0$, (c) $x^* = y^* = 3$.

7. A firm has profit function $\pi\left(Q\right) = 100Q - Q^2 - 500$.

(a) Find the profit-maximizing quantity (Q^{*}) by setting (pi'left(Qright)=0) .

(b) Use the second derivative to confirm this is a maximum.

(c) Calculate the maximum profit (pileft(Q^{*}right)) .

Answer: (a) $\pi'\left(Q\right) = 100 - 2Q = 0 \Rightarrow Q^* = 50$, (b) $\pi''\left(Q\right) = -2 < 0 :$ maximum, (c) $\pi\left(50\right) = 5000 - 2500 - 500 = 2000$.

8. A firm's total cost is $TC\left(q\right) = q^3 - 6q^2 + 15q + 10$.

(a) Find the marginal cost (MCleft(qright)=TC'left(qright)) .

(b) Find the quantity (q^{*}) that minimizes (MC) .

(c) Verify it is a minimum using the second derivative.

Answer: (a) $MC\left(q\right) = 3q^2 - 12q + 15$, (b) $MC'\left(q\right) = 6q - 12 = 0 \Rightarrow q^* = 2$, (c) $MC''\left(q\right) = 6 > 0 :$ minimum.

9. Maximize $U\left(x,y\right) = x^{1/2}y^{1/2}$ subject to $2x + y = 12$.

(a) Write the Lagrange function (mathcal{L}left(x,y,lambdaright)) .

(b) Write the three first-order conditions.

(c) Find (x^{}) , (y^{}) , and (Uleft(x{*},y{*}right)) .

Answer: (a) $\mathcal{L} = x^{1/2}y^{1/2} + \lambda\left(12 - 2x - y\right)$, (b) FOCs: $\frac{1}{2}x^{-1/2}y^{1/2} = 2\lambda$; $\frac{1}{2}x^{1/2}y^{-1/2} = \lambda$; $2x + y = 12$, (c) $y = 2x \Rightarrow x^* = 3, y^* = 6, U = 3\sqrt{2}$.

10. Find the extreme point of $f\left(x,y\right) = -x^2 + 2x - y^2 + 4y + 5$.

(a) Find (f'{x}) and (f'{y}) , and the critical point (left(x{*},y{*}right)) .

(b) Calculate (D=f' '{xx}f' '{yy}-left(f' '_{xy}right)^{2}) and classify the critical point.

(c) Calculate (fleft(x{*},y{*}right)) .

Answer: (a) $f'_x = -2x + 2 = 0 \Rightarrow x^* = 1$; $f'_y = -2y + 4 = 0 \Rightarrow y^* = 2$, (b) $D = \left(-2\right)\left(-2\right) - 0 = 4 > 0$, $f''_{xx} < 0 :$ local maximum, (c) $f\left(1,2\right) = 10$.

11. Consider $f\left(x,y\right) = x^2 - y^2$.

(a) Find the critical point.

(b) Calculate (D=f' '{xx}f' '{yy}-left(f' '_{xy}right)^{2}) .

(c) Interpret: what type of critical point is it?

Answer: (a) $\left(0,0\right)$, (b) $D = 2\cdot\left(-2\right) - 0^2 = -4 < 0$, (c) $D < 0 :$ saddle point (neither maximum nor minimum).

12. Maximize $U\left(x,y\right) = xy^2$ subject to $x + 2y = 6$.

(a) Write the Lagrange function.

(b) From the first-order conditions, find the relationship between (x) and (y) .

(c) Find (x^{}) , (y^{}) , and (Uleft(x{*},y{*}right)) .

Answer: (a) $\mathcal{L} = xy^2 + \lambda\left(6 - x - 2y\right)$, (b) $y^2 = \lambda$; $2xy = 2\lambda \Rightarrow x = y$, (c) $x^* = y^* = 2, U\left(2,2\right) = 8$.

Chapter 12

Theories about cake and monopoly

In the previous chapter, we went through how we can use derivatives to search for a function's extreme values, for example which input value in a variable gives the lowest or highest value in a function. This chapter shows examples of how we can use this type of methods to discuss theories about the world. The examples are simplified but inspired by more advanced theories within social science.

12.1 The optimal amount of cake

Cake is delicious but we get nauseous if we eat too much. Let us describe this with mathematics. We think that cake makes us happy and call the variable happiness y. Suppose we measure happiness on a scale from 1 to 5, where 5 is the greatest possible conceivable happiness. Happiness, y, is in turn a function, f, of how many cakes we eat. We denote the number of eaten cakes as the variable x. For each cake we consume, our happiness increases by 3 units, which we may describe with the function:

$$\text{Happiness} = y = f(x) = 3x \tag{12.1}$$

The variable x is continuous and for each bite of cake we eat we become happier. But we also become more full and eventually our happiness start to decrease if we eat more. This negative effect on happiness we can insert into our function by adding the term $-x^2$.

The term $-x^2$ is quadratic, raised to the power of 2. This means that for low values, for example 0.5, the negative effect will be relatively small. For example: $-(0.5)^2 = -0.25$ or $-(0.1)^2 = -0.01$. But for higher values of x the negative effect will instead become increasingly larger, such as for example $-(2)^2 = -4$ or $-(5)^2 = -25$. The entire function is now:

$$y = f(x) = 3x - x^2 \tag{12.2}$$

This is a maximization problem where we choose the amount of cake to eat, x, to maximize our happiness, y:

$$\max_{\text{w.r.t. } x} f(x) = 3x - x^2 \tag{12.3}$$

To calculate which amount of cake leads to the greatest possible happiness we calculate the first and second derivatives of function f with respect to x:

$$\begin{aligned} f'_x &= 3 - 2x \\ f''_{xx} &= -2 \end{aligned} \tag{12.4}$$

To find the happiness-maximizing amount of cake we set the first derivative f'_x equal to 0 and solve for x:

$$x^* = \frac{3}{2} = 1,5 \tag{12.5}$$

The second derivative f''_{xx} in equation (**??**) is negative. This indicates that $x = 1.5$ is a maximum point. The amount of cake that leads to the greatest possible happiness is one and a half cakes. To calculate the amount of happiness at $x^* = 1.5$ we take:

$$f\left(x^* = 1,5\right) = 3 * 1,5 - \left(1,5\right)^2 = 4,5 - 2,25 = 2,25 \tag{12.6}$$

One and a half cakes gives 2.25 in happiness, where happiness is measured on a scale from 1 to 5. However, exactly how much happiness is achieved through cake eating is not central in this example. Figure 12.1 illustrates this example and its solution. To the left and right of $x = 1.5$ we see that happiness is less than 2.25.

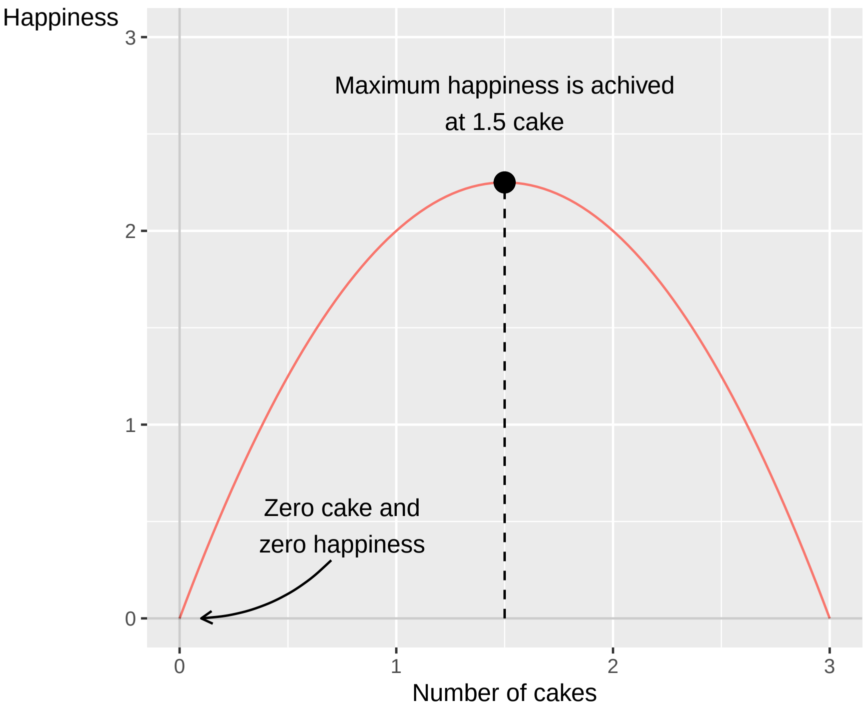


Figure 12.1: How cake may affect one's happiness

Another way to describe this phenomenon is with the help of the concept of marginal effect. Generally, the concept of marginal effect refers to how a result changes with small changes in a variable. In this case, how much the total amount of happiness changes with small changes in the amount of cake.

We seek the amount of cake where marginal happiness equals 0, where the first derivative of function f is 0, $f'_x = 0$. The value of x where $f'_x = 0$ is an extreme point. If the second derivative is negative, small changes in x at x^* mean that happiness decreases. In that case, the extreme point is a local maximum point. Since equation (12.2) only has one maximum point, this is also a global maximum point, that is, the point where the happiness given by cake eating is the greatest possible.

The example is trivial but introduces the fundamental logical starting points in a large number of theories that occur abundantly within social science. Much social science theory can be formulated as people trying to achieve one or several goals, by choosing between different action alternatives. Many theories about human behavior are often formulated directly or indirectly as maximization or minimization problems.

Regardless of how ingenious a theoretical model may be, it says nothing in itself about how someone thinks when they are going to eat cake, or make any other type of decision. To know something about reality, we need information about reality. In Part II and III , commonly occurring methods for how we can study information about reality and compare against theory are introduced.

12.2 To work or not to work

The example with cake in the previous section is an abstract illustration of how an individual must choose the right level to get the best possible result from a situation. Now we shall look at a similar example but with a clearer connection to social science. Say that Erik shall maximize his utility, N, by choosing between the following:

1. Erik's possibility to buy goods and services is financed through working and getting wages. This entire phenomenon we call C.
2. Erik's leisure time, which we denote L.

Erik's utility maximization thus depends on a choice between work and leisure. What effect C and L have on Erik's utility we describe with the following function:

$$N = u(C, L) = C^{1/2}L^{1/2} \ , \forall C, L > 0 \tag{12.7}$$

where the utility N is explained by the function $u()$ with the included variables C and L. Both variables can only assume positive values, since Erik cannot have negative consumption or leisure. The exponents for C and L, the fraction $1/2$, describe how much N increases when the respective variable changes by one unit, for example one USD and one clock hour respectively. The question is now at which amounts of C and L Erik's utility is maximized.

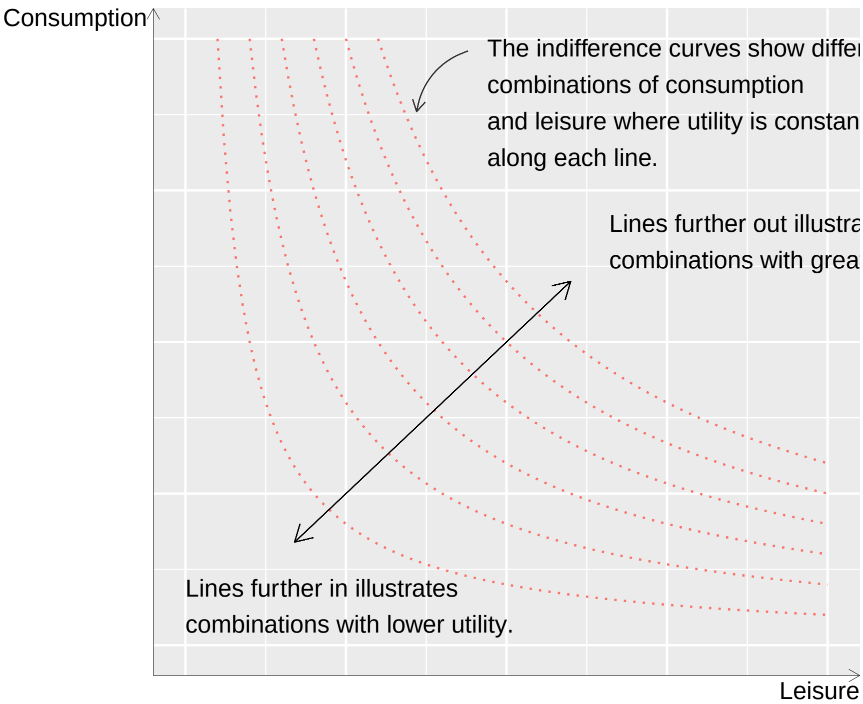


Figure 12.2: The relationship between consumption and leisure with indifference curves

To better understand the utility function we start by illustrating the theoretical relationship, which is shown in figure 12.2 . The two-dimensional graph shows three variables: C, L and N, where each curved line shows a value for N. The two axes in the graph, horizontal and vertical axis, measure consumption and leisure. Higher up along the vertical axis = more consumption. Further to the right along the horizontal axis = more leisure. The curved lines in the graph are called indifference curves. Along one and the same indifference curve, utility N is the same for different combinations of C and L. This theory means that Erik is indifferent to exactly which combination of C and L it becomes, as long as we remain on the same line.

Erik can accept less leisure as long as he can compensate with greater consumption, up to the left in the graph, and less consumption if he can compensate with more leisure. The lines further away from the origin represent combinations of consumption and leisure that together result in greater utility. Erik's consumption and leisure are also limited by Erik's budget, which is determined partly by how much he chooses to work and partly by his wage. This condition can be described with the following budget function:

$$C = w\,(T - L) = wT - wL \tag{12.8}$$

where w is wage, T is total available waking time and L is still leisure time. The parenthesis $(T - L)$ describes how much leisure time Erik gives up and instead spends on work and thereby gets wages. The higher the wage, the less leisure needs to be sacrificed to consume the same amount of goods and services. The budget function can be used to draw a straight line in the graph with the indifference curves. Figure 12.3 illustrates the budget line and indifference curves. The two axes are consumption (C) and leisure (L).

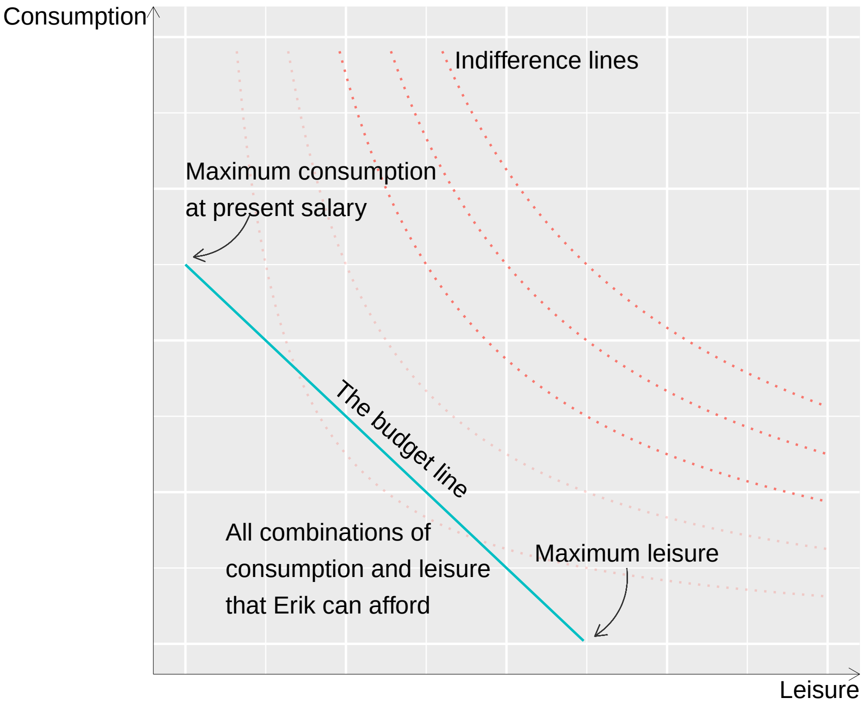


Figure 12.3: The budget line and different indifference curves

If Erik spent all available time on work, $L = 0$, then $C = wT$, a point along the vertical axis, consumption. This is the maximum amount of consumption that Erik can achieve, given the current wage. The line slopes negatively down toward the horizontal axis and the slope is determined by w. The budget function can therefore be written out as a straight diagonal line. Higher wage gives a steeper line and enables greater consumption higher up on the vertical axis. The further away from the origin that the budget line lies, the more Erik can consume. All combinations of consumption and leisure, C and L, within the budget line are possible to choose.

Given that Erik wants to maximize his utility and this works in the way that it is described in the utility function, Erik will choose a combination of consumption and leisure where one of the indifference curves just touches the budget line. This is illustrated in figure 12.4 with the budget line and three examples of indifference curves. As long as Erik has the same income, same budget line, and Erik's utility can be described as it is done in the utility function, same shape of the indifference curves, this is the only logical result.

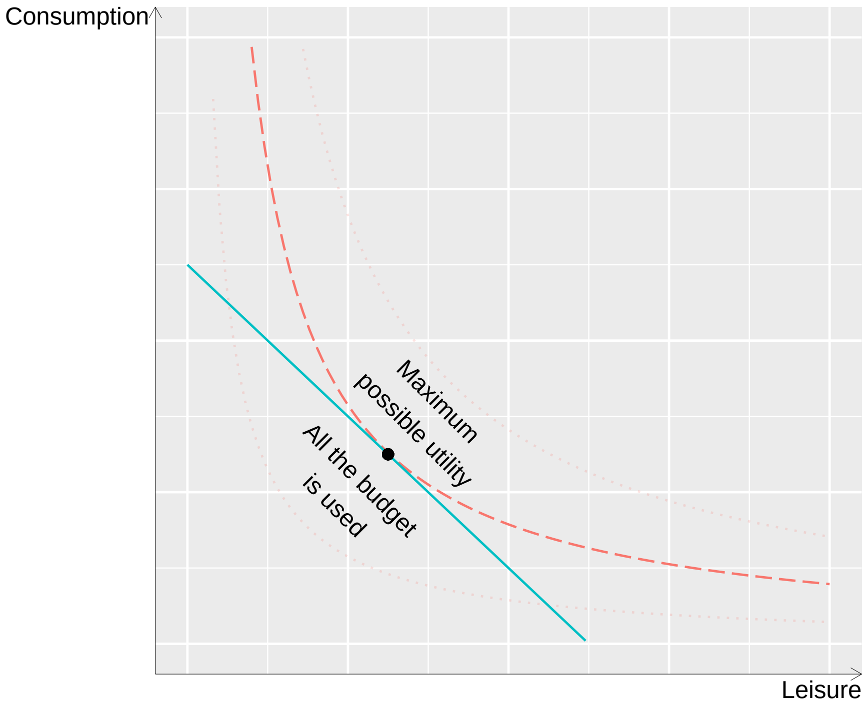


Figure 12.4: Maximum utility and budget

Why doesn't he choose a point within the budget line? At every point within the budget line, the gray area in figure 12.3 , there is another point further away from the origin where Erik can get a greater amount of utility.

Why doesn't he choose another point along the budget line? At every point on the budget line closer to the vertical or horizontal axis, Erik's utility will instead be defined by another indifference curve closer to the origin, which thus means a smaller amount of total utility. Let us now set up Erik's maximization problem:

$$\max_{\text{w.r.t. } C,L} C^{1/2}L^{1/2} \tag{12.9}$$
$$\text{under condition } C \leq w\left(T-L\right)$$

This we solve by setting up the Lagrange function:

$$\mathcal{L}\left(C,L\right): C^{1/2}L^{1/2} + \lambda\left(wT - wL - C\right) \tag{12.10}$$

We take the derivative of $\mathcal{L}$ with respect to C and L respectively:

$$\mathcal{L}'_C : \frac{1}{2}C^{-1/2}L^{1/2} - \lambda \tag{12.11}$$
$$\mathcal{L}'_L : \frac{1}{2}C^{1/2}L^{-1/2} - \lambda w$$

From $\mathcal{L}'_L$ we solve for λ:

$$\lambda = \frac{1}{w}\frac{1}{2}C^{1/2}L^{-1/2} \tag{12.12}$$

This definition of λ we put into the first-order condition we got from $\mathcal{L}'_C$:

$$\frac{1}{2}C^{-1/2}L^{1/2} = \frac{1}{w}\frac{1}{2}C^{1/2}L^{-1/2} \tag{12.13}$$
$$w = \frac{\frac{1}{2}C^{1/2}L^{-1/2}}{\frac{1}{2}C^{-1/2}L^{1/2}}$$

This equation shows how wage w relates to Erik's valuation of consumption C and leisure L at the point that maximizes Erik's utility. In the numerator we have the first derivative of the utility function with respect to L, u'_L. In the denominator we have the first derivative of the utility function with respect to C, which we write as u'_C. We therefore rewrite this equation to:

$$w = \frac{u'_L}{u'_C} \tag{12.14}$$

This means exactly what we saw in figure 12.4 , where the utility-maximizing point on the budget line is the one where one of the curved indifference curves touches the budget line. At this point, the slope of the budget line which is determined by w, is the same as the slope of the indifference curve that touches the budget line, which is determined by the form of the utility function.

12.3 Maximum possible tax level

A large amount of social science deals with taxes and how they affect people's lives. A well-known example is the so-called Laffer curve, named after the American economist Arthur Laffer. Corresponding theories within social science can be traced back to at least the 1400s.

We start by imagining that we have a society without taxes. People work and produce goods and services that are bought and sold. The total income in society we call y, whose value is not decisive for the example. Now the government wants to introduce a tax that shall be set as a percentage rate on all incomes and is called t. The tax t can be a value between 0 and 1, between 0 and 100%. The state's total tax revenue, i, can then be calculated to:

$$i = ty \tag{12.15}$$

If all incomes in society sum to 1,000 USD and the tax is 5%, the state's tax revenue becomes $i = ty = 1000 * 0.05 = 50$ USD. The incomes that are taxed, y, are also called the tax base. However, the tax also affects people's willingness to work and pay tax. The higher the tax becomes, the less people want to work, which in turn brings down the state's tax revenue. This mechanism we want to take into account in our equation over tax revenue. We therefore rewrite equation (12.15) so that we instead get the following expression:

$$i = ty\left(1-t\right)^b \tag{12.16}$$

The letter b indicates how the people in society react to a tax increase, which is also called tax base elasticity. If b is 0, people work exactly as much regardless of how high the tax is. A higher positive value of b means that tax increases have a greater negative effect on people's willingness to work. A negative value $b < 0$ means that people work more at higher tax.

Equation (12.16) is an example of the Laffer curve. From this equation we see that if the tax is set to 100% then $t = 1$ and in that case the tax revenue becomes $i = 0$. So if the state wants to maximize its tax revenue, the tax must be set somewhere between 0 and 100%. To find the tax level that gives the greatest possible tax revenue we set this up as a maximization problem where we shall calculate the maximum of tax revenue i with respect to tax level t:

$$\max_{\text{w.r.t. } t} i = ty\left(1-t\right)^b \tag{12.17}$$

To solve the maximization problem we take the derivative of i with respect to t:

$$i_t' = y(1-t)^b - by(1-t)^{b-1}t \tag{12.18}$$

where we use the chain rule for derivatives (table 10.2). We then set the first derivative equal to 0 and solve for t using logarithms:

$$\begin{aligned}
y(1-t)^b - by(1-t)^{b-1}t &= 0 \\
\ln y + b\ln(1-t) - \ln b - \ln y - \ln t - (b-1)\ln(1-t) &= 0 \\
\ln(1-t) - \ln t - \ln b &= 0 \\
\ln\left(\frac{1-t}{t}\right) &= \ln b \\
\frac{1}{t} - 1 &= b \\
t^* &= \frac{1}{1+b}
\end{aligned} \tag{12.19}$$

The definition of t^* in the last line gives us an expression for the greatest possible tax revenue, that is, the value for t that maximizes the variable i. The greatest possible tax revenue depends on the tax base elasticity b.

The government in our fictitious society wants to introduce tax but is uncertain about the value of b. To get more clarity they compare with two other countries. Figure 12.5 describes the Laffer curves for these two societies that represent two different types of society. In the figure, both lines go toward 0 tax revenue as the tax level approaches 100%, where we imagine that no one wants to pay tax.

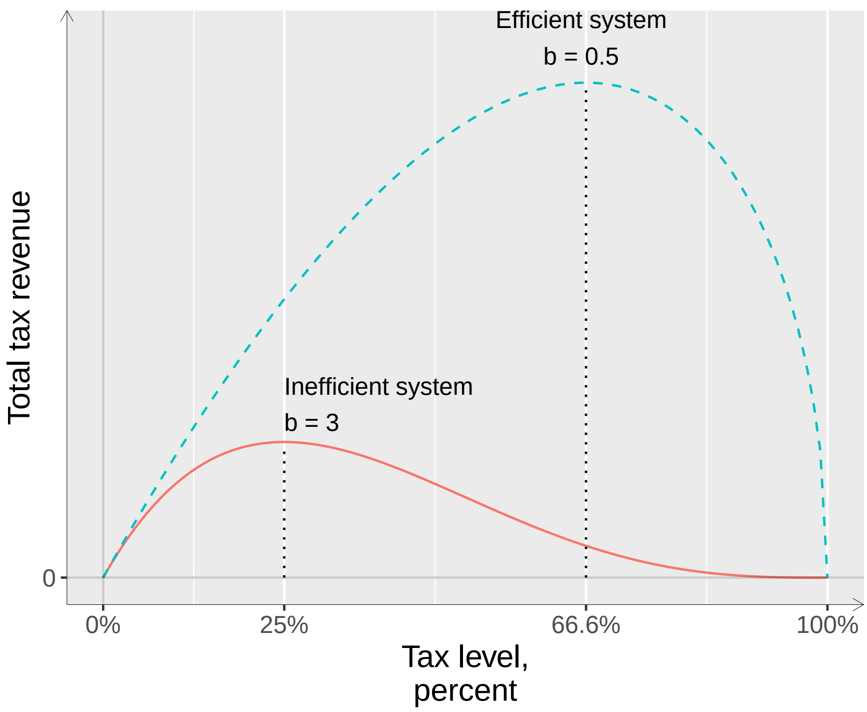


Figure 12.5: Two Laffer curves

In the first country, the public sector functions inefficiently. People have low trust in the political system and the taxes that the state collects are not used for anything that people benefit from. The taxes are designed in a complicated and inefficient way. The tax base elasticity is in this case $b = 3$. The greatest possible tax revenue is achieved in this case at a tax level of:

$$t(b=3) = \frac{1}{1+3} = 0.25 \tag{12.20}$$

This means that if the state wants to maximize tax collection it must set the tax at 25%. All other tax levels result in lower total tax revenue. In the graph this is illustrated with the solid line at "Inefficient system". The curve's highest point is at $t = 0.25$. This means that already at a relatively low tax level, further increases in tax will cause total tax revenue to decrease.

In the second society, the public sector functions relatively well. The inhabitants get back for their tax money a large amount of services of good quality and have high trust in the political system. The tax system is moreover efficiently designed and easy to follow for those who want to. The tax base elasticity is in this case $b = 0.5$ and the greatest possible tax revenue is reached at the tax level:

$$t\,(b = 0.5) = \frac{1}{1 + 0.5} = \frac{2}{3} \tag{12.21}$$

Expressed as a percentage, that corresponds to approximately 66.7%. In the graph this society is described with the dashed line, marked with "Efficient system". This line's highest point is at $t = 0.667$. This means that the government can continue to raise taxes to relatively high percentages before a further increase causes tax revenue to start decreasing.

12.4 Consumption and saving

A popular topic within social science is people's relation to consumption and saving. If we think of consumption as the money people spend on goods and services today, we regard saving as consumption that is postponed to some time in the future. To the extent that we have the possibility to save something for the future, a common challenge is how much we should consume today and in the future respectively to become as satisfied as possible. Here follows a simple example of how we can use mathematics to describe this phenomenon.

The theoretical description here is a simplified version of what is called Fisher's model for intertemporal choice, or intertemporal consumption. The word "intertemporal" refers to the theory describing how people make decisions in different time periods that affect later time periods. Irving Fisher was one of the early 1900s' most famous economists and presented these ideas in his book Theory of Interest 1930, which can be read for free online. Let us assume that we divide the entire life into two time periods: 1 (today) and 2 (the future). We shall now use the following notations:

-c_1, c_2= consumption during time period 1 and 2 respectively.

-y_1, y_2= income during period 1 and 2.

-s_1= saving during time period 1.

-r= real interest rate plus 1. If the interest rate is 5%, then r=1.05. Real interest rate is nominal interest rate adjusted for price changes, see section 3.7 .

We also assume for simplicity's sake that we consume all our income at the latest during time period 2. So if we consume our entire income during each respective time period we get $c_1 = y_1$ and $c_2 = y_2$, where the amount of consumption corresponds to the amount of income.

If we save a little during the first time period and consume the rest, we describe this as $c_1 = y_1 - s_1$. In time period 2, the income will then be $y_2 + s_1 * r$, where we multiply saving by interest rate to describe the interest income we received on our savings from time period 1. We therefore have two budget constraints, one for each time period:

$$\begin{aligned} c_1 + s_1 &= y_1 \\ c_2 &= y_2 + s_1 * r \end{aligned} \tag{12.22}$$

We solve for $s_1 = y_1 - c_1$ from the first line and substitute into the second:

$$\begin{aligned} c_2 &= y_2 + (y_1 - c_1)\,r \\ c_1 * r + c_2 &= y_1 * r + y_2 \end{aligned} \tag{12.23}$$

We can think of the interest rate r here as the price for consuming today, during time period 1. The interest rate r is therefore multiplied by c_1. If we instead save for the future, the income today (time period 1) will have increased by the interest rate tomorrow (time period 2). The interest rate r is therefore also multiplied by y_1. Let us now divide both sides by r:

$$c_1 + \frac{c_2}{r} = y_1 + \frac{y_2}{r} \tag{12.24}$$

This version of the budget can be described as the value of consumption and income in period 2, adjusted for the interest rate. That is, c_2/r is the relative value of consumption in period 2, relative to consumption in period 1. Let us now also assume that increased consumption increases our utility. But we content ourselves with describing this only in abstract terms. Let us assume that there exists a mathematical function U, which in some form describes what utility we get during life from our consumption during both time periods:

$$\text{Utility} = U\left(c_1, c_2\right) \tag{12.25}$$

We imagine that our total utility function has the following properties:

$$\begin{gathered} U\left(c_1, c_2\right)'_{c_1} > 0,\, U\left(c_1, c_2\right)'_{c_2} > 0 \\ U\left(c_1, c_2\right)''_{c_1 c_1} < 0,\, U\left(c_1, c_2\right)''_{c_2 c_2} < 0 \end{gathered} \tag{12.26}$$

where the first line describes how the first derivative is positive for both c_1 and c_2(more consumption, more utility). The second line describes how the second derivative is negative in both cases, which means that the more we consume, the less utility increases with additional consumption. The question is now how much we should consume during each respective time period. If the interest rate r is high, it may be wise to save a little to thereby be able to consume even more in time period 2, and vice versa. This we formulate as a maximization problem:

$$\begin{gathered} \max_{\text{w.r.t.}c_1, c_2} U\left(c_1, c_2\right) \\ \text{under condition } c_2 = y_2 + \left(y_1 - c_1\right) r \end{gathered} \tag{12.27}$$

We have a utility function that shall be maximized subject to a budget constraint. We set up the Lagrange function (see section 11.6):

$$\mathcal{L} : U\left(c_1, c_2\right) + \lambda\left(y_2 + \left(y_1 - c_1\right) r - c_2\right) \tag{12.28}$$

where λ is the Lagrange multiplier. The first-order conditions we get by differentiating $\mathcal{L}$ with respect to c_1 and c_2 and setting the results equal to 0:

$$\begin{gathered} \mathcal{L}'_{c_1} : U'_{c_1} + \lambda\left(-r\right) = 0 \\ \mathcal{L}'_{c_2} : U'_{c_2} + \lambda\left(-1\right) = 0 \end{gathered} \tag{12.29}$$

From this we solve for different definitions of λ from both conditions. The first first-order condition gives:

$$\lambda = \frac{U'_{c_1}}{r} \tag{12.30}$$

From the second first-order condition we get:

$$\lambda = U'_{c_2} \tag{12.31}$$

We set the two definitions for λ equal to each other and rewrite:

$$\frac{U'_{c_1}}{U'_{c_2}} = \frac{1}{r} \tag{12.32}$$

This expression describes how the marginal utility of consumption in time period 1, divided by the marginal utility of consumption in time period 2 will be equal to 1 divided by the real interest rate. That is, if the interest rate rises we have greater utility from consumption in period 2. This is logical in the sense that the consumption we postpone becomes worth more. The result is illustrated in figure 12.6 . The two straight lines describe two different interest rate levels: one line if the interest rate is "low" and one line if the interest rate is "high". Both these lines are drawn from the budget constraint: $c_2 = y_2 + (y_1 - c_1)\, r$. The line's slope is given by r, which is illustrated in the figure. The line's y-intercept is given by the incomes y_1 and y_2 as well as saving in period 1: $y_1 - c_1$. The curved line, the indifference curve, describes our preferences for consumption today and tomorrow respectively and is given by the utility function $U\,(c_1, c_2)$.

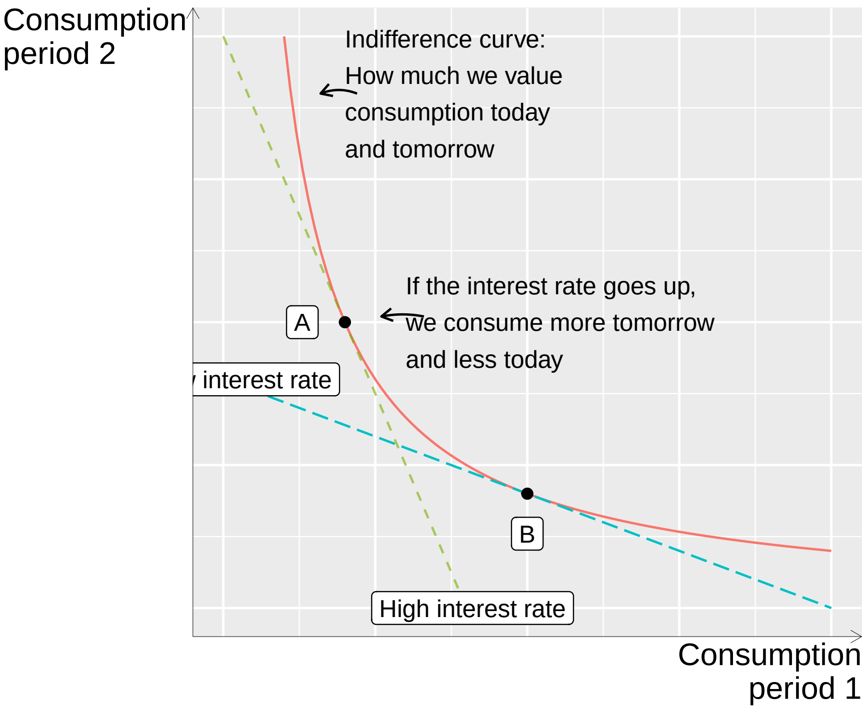


Figure 12.6: Consume today or tomorrow

The lines in this figure can be compared with those in figure 12.4 . There may be many straight lines depending on our budget. If our incomes increase, the straight lines' y-intercept moves up to the right in the graph, away from the origin. The curved line shows where the budget lines meet our preferences. The curved line touches the budget line since we use our entire budget and consume all our income during period 1 and 2. Now we shall develop the theory a little further by adding impatience. Say that our total utility function in equation (12.25) can be divided into two separate functions, one for each time period:

$$\text{Utility } = U\,(c_1, c_2) = U\,(c_1) + \beta U\,(c_2) \tag{12.33}$$

We have the same relation to consumption in both time periods. The function U therefore also has the same form in both periods, but different input values: c_1 and c_2 respectively. The multiplier β(Greek letter beta) describes how much we value future consumption, the consumption that takes place in time period 2. Or expressed differently, how impatient we are. High β= we are patient. Low β= we are impatient. Say now that the first derivative of function U for each respective variable only contains the variable that we differentiate with respect to:

$$\begin{aligned} U\left(c_1 c_2\right)'_{c_1} &= U'_{c_1}\left(c_1\right) \\ U\left(c_1 c_2\right)'_{c_2} &= \beta U'_{c_2}\left(c_2\right) \end{aligned} \tag{12.34}$$

In that case we may rewrite our result in equation (12.32) to the following:

$$\frac{U'_{c_1}\left(c_1\right)}{\beta U'_{c_2}\left(c_2\right)} = \frac{1}{r} \tag{12.35}$$

This is almost the same thing as our first result. The difference now is that our valuation of consumption in period 1 and 2 respectively also depends on how highly we value future consumption, that is, the value of β. A smaller value for β means that we are more impatient and value consumption today (period 1) higher than consumption in the future (period 2).

The example in this section illustrates how we can use maximization problems to reason about decisions that affect the outcome both today and later. Similar mathematics can be used to reason about all possible types of decisions that have more or less long-term effects.

12.5 A profit-maximizing company

A company manufactures the two products X and Y and wants to maximize its profit. The amount of produced units of the two products we denote with x and y. The selling prices for the two goods are controlled by the market, which the company cannot influence. The company can, however, choose how large an amount of the products should be produced.

For product x we have price $p_x = 10$ and for product y we have $p_y = 20$. The company's revenue can be described as a function of price multiplied by quantity. The function for total revenue we call TR after total revenue:

$$\text{Total revenue} = TR\left(x, y\right) = p_x x + p_y y = 10x + 20y \tag{12.36}$$

From equation (12.36) we solve for y:

$$TR = 10x + 20y \tag{12.37}$$

Different combinations of the quantities x and y can result in equally large revenues. Say that the company wants the revenue to be equal to 2,000 USD. This can be achieved by producing 100 of y and 0 of x, or producing 0 of y and 200 of x. Or 80 of y and 40 of x, and so on. This we may describe in the following way:

$$y = 2,000 - \frac{1}{2}x,\ \forall x, y \geq 0 \tag{12.38}$$

In the same way we set up this equation for other sums in total revenue and illustrate these as lines in a diagram. Figure 12.7 shows this for the following equations:

$$\begin{aligned} y &= 2,000 - 0.5x \\ y &= 1,600 - 0.5x \\ y &= 1,200 - 0.5x \\ y &= 800 - 0.5x \\ y &= 400 - 0.5x \end{aligned} \tag{12.39}$$

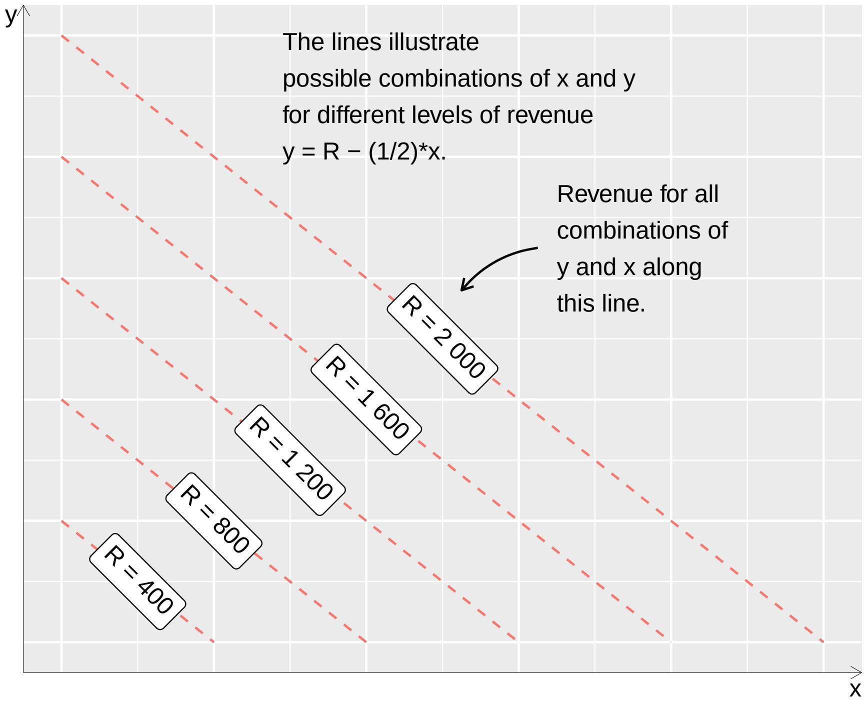


Figure 12.7: Company revenue from product x and y

The total production costs consist of fixed expenses of 10 USD and variable costs for manufacturing products X and Y:

$$\text{Total cost} = TC\,(x, y) = x^2 + 2y^2 + 10 \tag{12.40}$$

where TC is the name of the function. The company's profit is equal to revenue minus costs, $TR - TC$, which we may write as a function, V:

$$V\,(x, y) = TR - TC = 10x + 20y - \left(x^2 + 2y^2 + 10\right) \tag{12.41}$$

The profit is a function of the quantities x and y, which can be described as the company wanting to maximize its profit by choosing the amount to produce of x and y:

$$\max_{\text{w.r.t. } x,y} V\,(x, y) = 10x + 20y - \left(x^2 + 2y^2 + 10\right) \tag{12.42}$$

This problem we solve by taking the derivative of V with respect to x and y. We do this because we are looking for a maximum point where the first derivative is equal to 0 and the second derivative is negative. The first derivative of $V\,(x, y)$ with respect to x and y respectively:

$$\begin{aligned} V'_x &= 10 - 2x \\ V'_y &= 20 - 4y \end{aligned} \tag{12.43}$$

We now have two expressions that describe how V changes when x and y respectively increase by one unit. We set the expressions in equation (12.44) equal to 0, which gives us the first-order conditions, and solve for x and y. These values for x and y are the solutions we are looking for, which we write as x^* and y^*:

$$\begin{aligned} x^* &= 10/2 = 5 \\ y^* &= 20/4 = 5 \end{aligned} \tag{12.44}$$

We also see that the second derivative of V with respect to x and y is negative by differentiating the expressions in equation (12.44) once more:

$$\begin{aligned} V''_{xx} &= -2 \\ V''_{yy} &= -4 \end{aligned} \tag{12.45}$$

Another way to describe what we just did is marginal effect and marginal profit. The marginal effect refers here to how much V changes with small changes in the variables x and y.

The profit is maximized when the marginal profit is equal to 0, that is, when the first derivative of V is $V'_x = 0$ and $V'_y = 0$. Let us calculate how much profit the company will make by substituting these values for x^* and y^* into $V(x^*, y^*) = V(5,5)$:

$$V(5,5) = 10*5 + 20*5 - \left(5^2 + 2*5^2 + 10\right) = 65 \tag{12.46}$$

Given the prices $p_x = 10$ and $p_y = 20$ and current production costs, the company can maximize its profit by producing 5 units of product A and 5 units of product B, and then makes a profit of 65 USD. The company's total costs are given by equation (12.40) . If we take the first derivative with respect to x and y of this function we get the company's marginal cost, MC, marginal cost:

$$\begin{aligned} MC_x &= C'_x = 2x \\ MC_y &= C'_y = 2y \end{aligned} \tag{12.47}$$

The profit is optimized at $(x^*, y^*) = (5,5)$, where the marginal costs are:

$$\begin{aligned} MC_x &= C'_x(x^*) = 2*5 = 10 \\ MC_y &= C'_y(y^*) = 2*5 = 10 \end{aligned} \tag{12.48}$$

If the company produces the quantities 5 of x and 5 of y, the marginal costs, at exactly this production quantity, are equal to 10 with respect to x and 10 with respect to y. The company thus maximizes its profit when the marginal cost is equal to the price for the products, given that the price is determined by the market:

$$P^* = MC \tag{12.49}$$

If we imagine that perfect competition prevails, all companies have the same cost and revenue functions. All companies will in that case produce in this way, so that $P^* = MC$. At the production quantities $y = 5$ and $x = 5$, profit in the company is maximized. If the company produces more or less of product X or Y than these amounts, the profit will be smaller. Figure 12.8 shows an illustration of this. For each given production quantity of y, the line in the diagram shows the relationship between profit and produced quantity x.

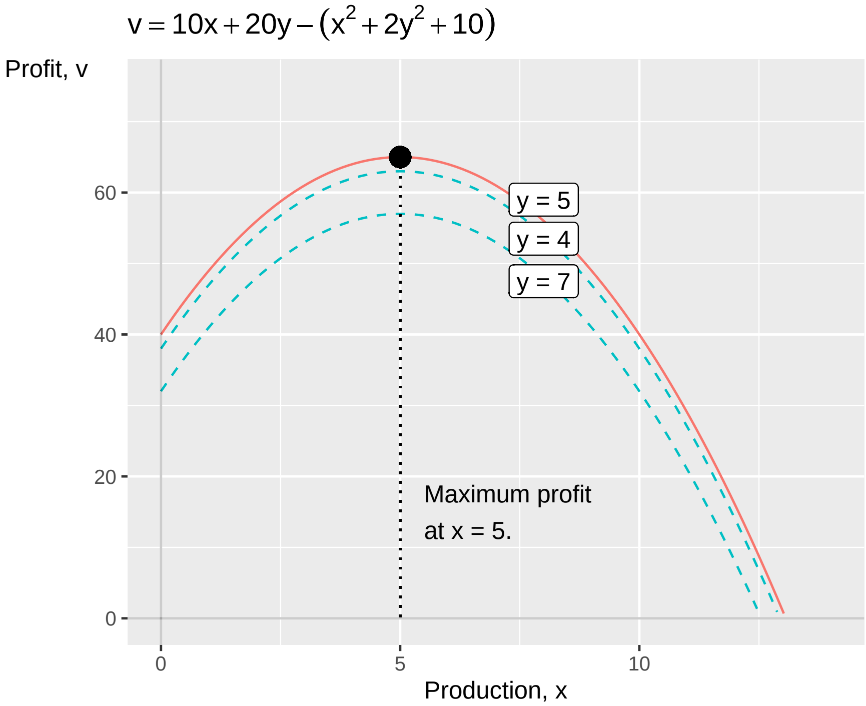


Figure 12.8: Profit maximization

The profit can still be above 0 for other values of x and y but since other values also result in other costs, the profit will then be smaller. The dashed lines below the solid line in the figure illustrate that for both larger and smaller values for y, the total profit will be smaller for all other x-values.

12.6 Profit maximization with constraints

Suppose now that we have a profit-maximizing company, but this time we add a constraint in the form of the company's production. The company sells a good where the price, p, is determined by the market. The company's costs for production are given by the function:

$$TC = 10 + q^2 + wL \tag{12.50}$$

The letter q is the amount of production that the company delivers, w is the wage for the company's employees and L the number of full-time employees, or full-time positions. The wages, w, are also determined by the outside world, in this case the labor market.

The company can, however, choose how much it should produce (q) and how many full-time positions it should have (L). The company's production is determined by the function:

$$q(L) = 2L \tag{12.51}$$

This is a simplified example of what in economics is called a production function. The production function describes the relationship between the company's inputs and its production, q. In this case, the company's only input is labor, L. The demand for the company's products is given by the function:

$$q = 100 - 4p \tag{12.52}$$

From this equation we solve for an expression for p:

$$p = 25 - \frac{q}{4} \tag{12.53}$$

Total revenue is given by the function:

$$TR = pq = \left(25 - \frac{q}{4}\right) q = 25q - \frac{q^2}{4} \tag{12.54}$$

The company's profit is now given by the function:

$$V = TR - TC = 25q - \frac{q^2}{4} - (10 + q^2 + wL) \tag{12.55}$$

The wage on the labor market for the company's employees is $w = 2$:

$$V = 25q - \frac{q^2}{4} - (10 + q^2 + 2L) \tag{12.56}$$

We must also take into account the constraint, which is a result of the company's production conditions (the production function):

$$2L \geq q \tag{12.57}$$

This means that q must be less than or equal to 5 times the number of full-time positions, L. This is important because the company maximizes its profit, V, by choosing production quantity, q, and number of full-time employees, L. All this we set up as a maximization problem:

$$\max_{\text{w.r.t. } q,L} V = 25q - \frac{q^2}{4} - \left(10 + q^2 + 2L\right), \text{ s.t. } 2L \geq q \tag{12.58}$$

To solve this we set up the Lagrange function:

$$\mathcal{L}(q, L) = 25q - \frac{q^2}{4} - 10 - q^2 - 2L + \lambda (2L - q) \tag{12.59}$$

where λ is the Lagrange multiplier. We differentiate with respect to q and L, which leads us to the first-order conditions:

$$\begin{aligned} \mathcal{L}'_q &= 25 - \frac{2}{4}q - 2q - \lambda \\ \mathcal{L}'_L &= -2 + \lambda 2 \end{aligned} \tag{12.60}$$

We put $\mathcal{L}'_L = 0$ and solve for λ:

$$\lambda^* = \frac{2}{2} = 1 \tag{12.61}$$

We put λ^* in $\mathcal{L}'_q$ and solve for q^*:

$$\begin{aligned} 25 - \frac{2}{4}q - 2q - 1 = 0 \\ 24 = q\left(2 + \frac{1}{2}\right) \\ q = \frac{24}{5/2} \\ q^* = 48/5 = 9.6 \end{aligned} \tag{12.62}$$

From the production function we see that:

$$\begin{aligned} q^* &= 2L \\ 9,6 &= 2L \\ L^* &= 4.8 \end{aligned} \tag{12.63}$$

This results in the following profit:

$$V\left(q^*, L^*\right) = 25q - \frac{q^2}{4} - \left(10 + q^2 + 2L\right) = 124.4 \tag{12.64}$$

12.7 Monopoly

Let us take another example with a company that sells goods or services on a market, but this time we assume that the company has a monopoly on the market. Monopoly means that the company is the sole seller and thereby has power to influence both price and quantity. In section 12.5 above we went through an example where companies in competition had to accept the price that the market determines for its goods or services. The monopoly seller can instead choose to produce a smaller quantity at a higher price and thereby make greater profit. Let us illustrate with an example. A company produces the quantity Q of a good, which entails total costs corresponding to:

$$TC = 10 + Q^2 \tag{12.65}$$

TC is abbreviation for Total Cost. The company's customers demand goods in relation to price P in a way that we can describe with the function:

$$\begin{aligned} \text{Demand: } Q &= 20 - \frac{P}{5} \\ \Rightarrow P &= 100 - 2Q \end{aligned} \tag{12.66}$$

Total revenue is given by the function TR which consists of price multiplied by the quantity of production Q:

$$TR = PQ = \left(100 - 2Q\right)Q = 100Q - 2Q^2 \tag{12.67}$$

The monopolist wants to maximize its profit V, which is a result of revenue minus costs:

$$\begin{aligned} \max_{Q,P} V &= TR - TC \\ &= PQ - TC \\ &= Q\left(100 - 2Q\right) - \left(10 + Q^2\right) \\ &= 100Q - 2Q^2 - 10 - Q^2 \\ &= 100Q - 3Q^2 - 10 \end{aligned} \tag{12.68}$$

To get the first-order condition we differentiate the function V with respect to the variable Q:

$$V'_Q = 100 - 6Q \tag{12.69}$$

We set this expression equal to 0 and solve for the profit-maximizing production quantity:

$$\begin{aligned} 100 - 6Q &= 0 \\ Q &= \frac{100}{6} \\ Q^* &= 16 + \frac{2}{3} \end{aligned} \tag{12.70}$$

The monopoly company maximizes its profit at $Q^* = 16 + \frac{2}{3}$, whereupon the selling price is given by customers' demand, equation (12.66) :

$$P^*(Q^*) = 100 - 2Q^* = 100 - 2 * \left(16 + \frac{2}{3}\right) = 66 + \frac{2}{3} \tag{12.71}$$

The company's total profit is given by the function for V, equation (12.68) :

$$\begin{aligned} V(Q^*) &= 100Q - 3Q^2 - 10 \\ &= 100\left(16 + \frac{2}{3}\right) - 3\left(16 + \frac{2}{3}\right)^2 - 10 \\ &= 823 + \frac{1}{3} \end{aligned} \tag{12.72}$$

On a hypothetical market with so-called perfect competition, the equilibrium price on the market will approach the marginal cost. Perfect competition is a hypothetical state that rarely or never occurs in reality. Regardless, we may describe this price level in the following way:

$$P^* = MC \tag{12.73}$$

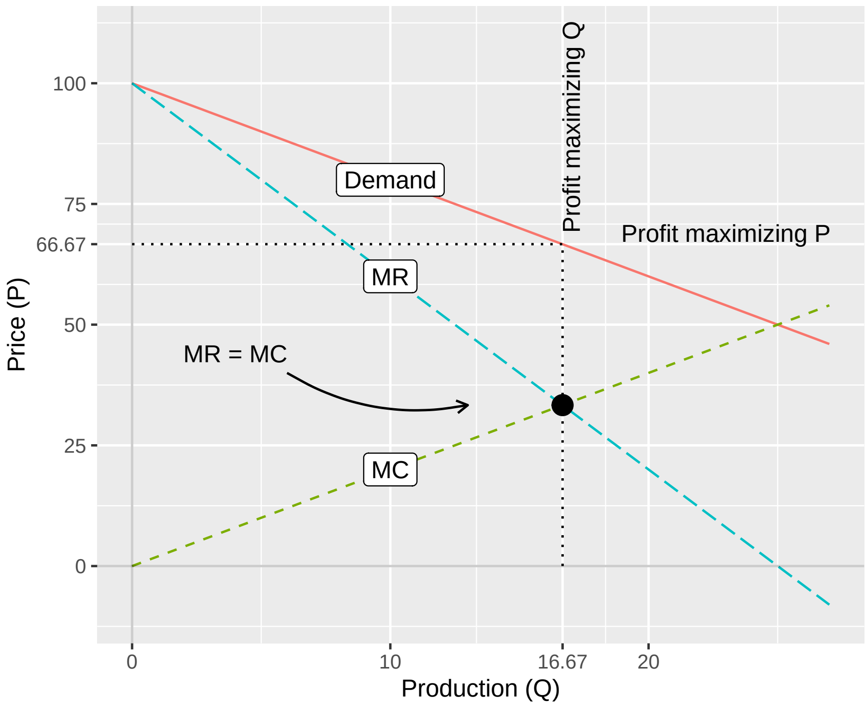


Figure 12.9: Monopoly

This we saw in the example in section 12.5 . As seen in equation (12.67) , total revenue TR is a function of the variable Q. If we take the first derivative of this function we get the function for the monopoly company's marginal revenue, MR:

$$MR = TR'_Q = 100 - 4Q \tag{12.74}$$

When the company produces the profit-maximizing quantity $Q^* = 16 + 2/3$ this gives marginal revenue:

$$MR\left(Q^*\right) = 100 - 4Q = 100 - 4\left(16 + 2/3\right) = 33 + \frac{1}{3} \tag{12.75}$$

The function for total costs, equation (12.65) , is also a function of the variable Q. If we take the first derivative of this we get the function for the company's marginal costs, MC:

$$MC = TC'_Q = 2Q \tag{12.76}$$

which at the profit-maximizing production quantity $Q^* = 16 + 2/3$ is:

$$MC\left(Q^*\right) = 2Q = 2\left(16 + 2/3\right) = 33 + \frac{1}{3} \tag{12.77}$$

This is the same result as we found for $MR\left(Q^*\right)$. A profit-maximizing monopoly company produces quantity Q where $MR = MC$, while profit-maximizing companies in perfect competition produce a quantity so that $MC = P^*$. Figure 12.9 illustrates the monopoly company's pricing, where the equilibrium quantity is determined by the point where the curves for MR and MC meet. The purpose of this exercise is primarily to see how the behavior of companies in competition can differ from companies with monopoly. The reasoning is simplifications of reality and builds on specific assumptions.

12.8 Monopsony

Monopsony describes a situation when instead of a single seller (monopoly) we have a single buyer, which in turn means that the buyer has power to influence the conditions for their purchases, both quantity and price. In reality, elements of monopsony occur in, for example, geographically delimited areas and in markets that are subject to special laws. Monopsony can also occur in the labor market, for example if there is only one or a few companies that employ a certain type of professional group. Let us illustrate with an example. Assume that a company buys quantity q of a good at price p. The supply for the good on the market is given by the function:

$$\text{Supply: } q = 2p_{\text{purchases}} \tag{12.78}$$

We start by solving for a definition of the purchase price p:

$$p_{\text{purchases}} = \frac{q}{2} \tag{12.79}$$

The company in turn uses the purchased good to produce its own goods. Total manufacturing costs, tc, are given by price multiplied by quantity:

$$tc = p_{\text{purchases}} q \tag{12.80}$$

We substitute the definition of $p_{\text{inköp}}$ from equation (12.79) into the equation for tc:

$$tc = p_{\text{purchases}} q = \left(\frac{q}{2}\right) q = \frac{q^2}{2} \tag{12.81}$$

The first derivative of tc with respect to q gives us the marginal cost mc:

$$mc = tc'_q = q \tag{12.82}$$

The company produces goods, whose demand is determined by the function:

$$\text{Demand: } q = 10 - p_{\text{sales}} \tag{12.83}$$

We simplify a little and use the designation q also for this good. We may think of this example as the company purchasing an input-good to its own production. From this function for demand we solve for the company's selling price for its own goods:

$$p_{\text{sales}} = 10 - q \tag{12.84}$$

The company's total revenue, tr:

$$tr = p_{\text{sales}} q \tag{12.85}$$

We take p_{sales} from equation (12.84) and substitute into the equation for tr:

$$tr = (10 - q)\, q = 10q - q^2 \tag{12.86}$$

From this expression for total revenue we take the first derivative with respect to q, which gives us marginal revenue mr:

$$mr = tr'_q = 10 - 2q \tag{12.87}$$

The company's profit v is total revenue tr minus total cost tc:

$$v = tr - tc \tag{12.88}$$

We replace tc and tr with the definitions from equation (12.80) and (12.86) :

$$v\,(q) = \left(10q - q^2\right) - \frac{q^2}{2} = 10q - \frac{3}{2}q^2 \tag{12.89}$$

The function for the company's profit $v\,(q)$ is determined by what quantity q the company chooses to purchase and produce. Based on this we formulate the company's maximization problem:

$$\max_q v\,(q) = 10q - \frac{3}{2}q^2 \tag{12.90}$$

To solve the maximization problem we take the first derivative of the profit function $v\,(q)$ with respect to the variable q:

$$v_q' = 10 - 3q \tag{12.91}$$

We set this expression equal to 0 to get the first-order condition:

$$v_q' : \ 10 - 3q = 0 \tag{12.92}$$

From this we may solve for q, the quantity that maximizes the company's profit:

$$q^* = \frac{10}{3} \approx 3,33 \tag{12.93}$$

As before, we mark this quantity with an asterisk, q^*. The purchase price for the good that the company is the sole buyer of is given by the supply function, equation (12.78) :

$$p_{\text{purchases}}\left(q^*\right) = \frac{q^*}{2} = \frac{10/3}{2} = \frac{5}{3} \tag{12.94}$$

The selling price for the company's own production is given by equation (12.84) :

$$p_{\text{sales}}\left(q^*\right) = 10 - q^* = 10 - \frac{10}{3} = \frac{20}{3} \tag{12.95}$$

The company's marginal cost mc at q^* is given by equation (12.82) :

$$mc\left(q^*\right) = q^* = \frac{10}{3} \tag{12.96}$$

and the company's marginal revenue mr is given by equation (12.87) :

$$mr\left(q^*\right) = 10 - 2q^* = 10 - 2 * \left(\frac{10}{3}\right) = \frac{10}{3} \tag{12.97}$$

The company will maximize its profit by choosing such a quantity so that the company's marginal revenue is equal to its marginal costs, $mr = mc$. When we went through examples with companies in markets with perfect competition we could instead see that the company's production occurs where the marginal costs are equal to the market price $P^* = MC$, equation (12.49) .

Figure 12.10 illustrates profit maximization for monopsony according to the model we have now gone through. The lines show the company's marginal cost mc, marginal revenue mr, demand for the company's products from equation (12.83) as well as supply of the purchased good, from equation (12.79) .

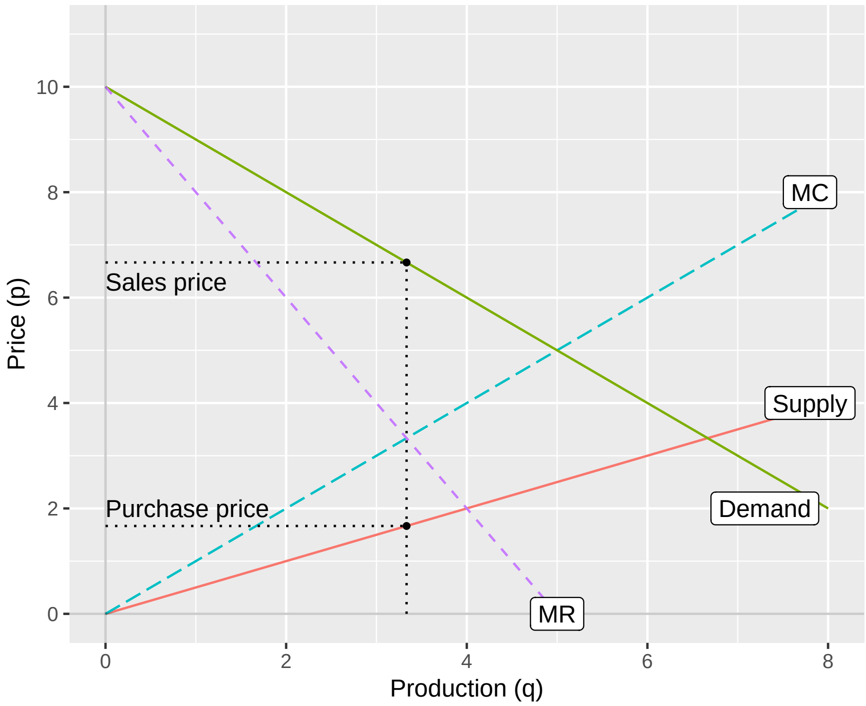


Figure 12.10: Monopsony

12.9 If we do not optimize

In this chapter we have used different examples to illustrate how people maximize their utility or profit. The world is also full of examples that cannot easily be described as utility-maximizing. In section 9.4 we went through an example with two friends, Erik and Maria who make decisions about whether they should start smoking, based on what their friend does. The decisions were summarized in the following system of equations:

$$\begin{cases} y = x^2 \\ x = y^2 \end{cases} \tag{12.98}$$

where x and y describe Erik's and Maria's smoking. The system has two solutions: both smoke or no one smokes. We call smoking the variable r and that $r = 0$ means no one smokes and $r = 1$ that they smoke. Now we let Erik and Maria have r as input variable in their utility function $U(r)$. Expressed as an optimization problem we may describe it as wanting to optimize utility U by choosing a value of r:

$$\max_r U(r), \forall r \geq 0 \tag{12.99}$$

Variable r must be greater than or equal to 0 since the amount of smoking cannot be negative. In our system of equations (12.98) smoking r is not included, other than indirectly. The primary decision factor is the other person's smoking, which is what the variables x and y describe. This means that in the system of equations we have maximized utility (which is not written out) both in the situation when no one smokes ($r = 0$) and when both smoke ($r = 1$).

If all possible alternatives of a phenomenon can result in utility maximization, it is not particularly helpful to describe this as a utility maximization problem for this utility (smoking, the variable r) as in equation (12.99) . The example we have gone through here concerns whether the persons can be described as completely rational, which is a concept with special meaning within social science and which is discussed among other things within the part of economics called microeconomics. Here we content ourselves with noting that such examples exist and among other things illustrate that it can be difficult to draw conclusions regarding all types of behaviors we observe in the world.

12.10 Chapter summary

- Optimization problems can be used for many different social science questions. For example, to find the optimal amount of cake a person should eat or what is the optimal balance between work and leisure.
- We can use optimization problems both to describe phenomena that we often think of as economic, for example a company's decision to produce or not produce different goods or services. We can also use this method for phenomena that we think of as more social, for example how people optimize their utility or happiness.
- Even if we may describe an optimization problem where a company should maximize its profit or an individual should maximize their utility, it is not given what the result will be. Seemingly simple problems can have several solutions that are not always easy to interpret.

12.11 Exercises

1. Consider the happiness function $f(x) = 4x - 2x^2$, where x is the number of cakes eaten.

(a) Calculate the first derivative (f'(x)) .

(b) Find the amount of cake (x^{*}) that maximizes happiness.

(c) Calculate the maximum happiness (f(x^{*})) .

Answer: (a) $f'(x) = 4 - 4x$, (b) $x^* = 1$, (c) $f(1) = 2$.

2. Erik has utility function $N = C^{1/2}L^{1/2}$, total time $T = 12$, and wage $w = 4$.

(a) Write out the budget function (C) as a function of (L) .

(b) Set up the Lagrange function for the constrained maximization.

(c) From the first-order conditions, find (L^{*}) and (C^{*}) .

Answer: (a) $C = 4(12 - L) = 48 - 4L$, (b) $\mathcal{L} = C^{1/2}L^{1/2} + \lambda(4(12 - L) - C)$, (c) FOCs give $C = 4L \Rightarrow L^* = 6, C^* = 24$.

3. The state's tax revenue is $i = ty(1 - t)^b$.

(a) Calculate (t^{*}) that maximizes tax revenue when (b=3) .

(b) Calculate (t^{*}) that maximizes tax revenue when (b=0.5) .

(c) If total income (y=1000) and (b=1) , calculate the maximum tax revenue at (t^{*}) .

Answer: (a) $t^* = \frac{1}{1+3} = 0.25$, (b) $t^* = \frac{1}{1+0.5} = \frac{2}{3} \approx 0.667$, (c) $t^* = \frac{1}{2}$, $i = \frac{1}{2} \cdot 1000 \cdot \frac{1}{2} = 250$.

4. Anna earns $y_1 = 200$ in period 1 and $y_2 = 100$ in period 2. The interest rate is $r = 1.1$.

(a) Write out the intertemporal budget constraint with the given values.

(b) If Anna saves (s_{1}=50) in period 1, what is her consumption (c_{2}) in period 2?

(c) If the interest rate rises to (r=1.2) , does the budget line become steeper or flatter?

Answer: (a) $c_1 + c_2/1.1 = 200 + 100/1.1 \approx 290.9$, (b) $c_2 = 100 + 50 \cdot 1.1 = 155$, (c) steeper: larger r increases the opportunity cost of current consumption.

5. A company produces goods x and y with prices $p_x = 8$ and $p_y = 12$. Total costs are $TC(x, y) = x^2 + y^2 + 5$.

(a) Write out the profit function (V(x,y)) .

(b) Find the first-order conditions (V'{x}=0) and (V'{y}=0) .

(c) Find the profit-maximizing quantities (x^{*}) and (y^{*}) , and the maximum profit (V(x^{*},y^{*})) .

Answer: (a) $V = 8x + 12y - x^2 - y^2 - 5$, (b) $V'_x = 8 - 2x = 0$; $V'_y = 12 - 2y = 0$, (c) $x^* = 4, y^* = 6, V(4,6) = 47$.

6. A monopoly company has total costs $TC = 5+Q^2$ and faces demand $Q = 15-P/2$, giving $P = 30-2Q$.

(a) Write out the total revenue function (TR(Q)=PQ) .

(b) Write out the profit function (V(Q)=TR-TC) .

(c) Find the profit-maximizing quantity (Q^{*}) and the monopoly price (P^{*}) .

Answer: (a) $TR = (30-2Q)Q = 30Q - 2Q^2$, (b) $V = 30Q - 3Q^2 - 5$, (c) $Q^* = 5, P^* = 20, V(5) = 70$.

7. A competitive firm has total revenue $TR = 5Q$ and total cost $TC = Q^2 + Q$.

(a) Write the profit function (pileft(Qright)) .

(b) Find the profit-maximizing quantity (Q^{*}) .

(c) Calculate the maximum profit (pileft(Q^{*}right)) .

Answer: (a) $\pi(Q) = 5Q - Q^2 - Q = 4Q - Q^2$, (b) $\pi' = 4 - 2Q = 0 \Rightarrow Q^* = 2$, (c) $\pi(2) = 4$.

8. A consumer maximizes utility $U(x,y) = 2\sqrt{x} + y$ subject to the budget constraint $x + y = 10$.

(a) Set up the Lagrange function.

(b) From the first-order conditions, find (x^{*}) .

(c) Find (y^{*}) and (Uleft(x^{*},y^{*}right)) .

Answer: (a) $\mathcal{L} = 2\sqrt{x} + y + \lambda(10 - x - y)$, (b) FOC: $1/\sqrt{x} = \lambda = 1 \Rightarrow x^* = 1$, (c) $y^* = 9, U(1,9) = 11$.

9. A monopolist faces demand $P = 50 - 2Q$ and has total cost $TC = 2Q$.

(a) Write the profit function (pileft(Qright)=TR-TC) .

(b) Find the profit-maximizing quantity (Q^{*}) and price (P^{*}) .

(c) Calculate the maximum profit.

Answer: (a) $\pi(Q) = 48Q - 2Q^2$, (b) $Q^* = 12, P^* = 26$, (c) $\pi(12) = 288$.

10. Maria earns $y_1 = 300$ in period 1 and nothing in period 2. The interest rate is $r = 0.5$. She maximizes $U = \ln(c_1) + \ln(c_2)$.

(a) Write the intertemporal budget constraint (c_{1}+c_{2}/left(1+rright)=y_{1}) .

(b) Use the first-order conditions to find (c_{1}^{*}) and (c_{2}^{*}) .

(c) How much does Maria save in period 1?

Answer: (a) $c_1 + c_2/1.5 = 300$, (b) FOCs: $c_2 = 1.5c_1 \Rightarrow c_1^* = 150, c_2^* = 225$, (c) $s_1 = 150$.

11. A government maximizes social welfare $W = \ln(x_1) + 2\ln(x_2)$ subject to $x_1 + x_2 = 100$.

(a) Set up the Lagrange function.

(b) Find the optimal allocation (left(x_{1}^{*},x_{2}^{*}right)) .

(c) Interpret: which group receives more, and why?

Answer: (a) $\mathcal{L} = \ln x_1 + 2\ln x_2 + \lambda(100 - x_1 - x_2)$, (b) $x_2 = 2x_1 \Rightarrow x_1^* \approx 33.3, x_2^* \approx 66.7$, (c) group 2 receives twice as much due to its double welfare weight.

12. A monopolist sells in two markets with demand $P_1 = 20 - Q_1$ and $P_2 = 14 - Q_2$. Marginal cost is constant at $MC = 2$.

(a) Write the profit function (pi=pi_{1}+pi_{2}) .

(b) Find (Q_{1}^{*}) and (Q_{2}^{*}) by setting (MR_{i}=MC) in each market.

(c) Find prices (P_{1}^{*}) , (P_{2}^{*}) and total profit.

Answer: (a) $\pi = 18Q_1 - Q_1^2 + 12Q_2 - Q_2^2$, (b) $Q_1^* = 9, Q_2^* = 6$, (c) $P_1^* = 11, P_2^* = 8, \pi^* = 117$.

Chapter 13

Integration

This chapter introduces integration, also called integrating. Integration can be described as finding a function's "anti-derivative". Integrals can among other things be used to calculate the area between two lines in a graph, or the volume and mass of an object. This we can in turn use for describing theories and calculate probability, which is a central aspect of all empirical analysis.

13.1 The primitive function

Suppose we have a function f'_x which is the derivative of function F with respect to the variable x. With integration we seek a function F whose derivative gives f'_x. The function we then get through integration is called integral. Function F is the integral to function f, and function f is the integrand. Consider the following function F of x:

$$F(x) = x^a + b \tag{13.1}$$

where a and b are constants. Taking the derivative of F with respect to x:

$$\frac{d}{dx}F(x) = f'_x(x) = ax^{a-1} \tag{13.2}$$

Function F is the primitive function to f'_x. Another expression for primitive function is indefinite integral, which is described more thoroughly below. The primitive function is the function whose derivative gives us the function f'_x. If we are to calculate the integral of function $F(x)$ in equation (13.1) we instead seek a primitive function to function F. The derivative of the primitive function we in that case seek should result in function F.

Integration is often described with two symbols: $\int$ and the letters dx, where x is the variable that the integration should be performed with respect to. The expression dx can be read as "with respect to variable x". The description of the integral of a function $f(x)$ can be written in different ways:

$$\int (f(x))\, dx = \int f(x)\, dx = \int dx f(x) \tag{13.3}$$

The mathematical meaning in these three ways of writing is the same. The three expressions can all be read as "the integral of function $f(x)$, with respect to variable x". We may also write out the function $f(x)$ to clarify which expression we seek an integral for. Taking the definition of $f(x)$ from equation (13.1) and insert between the symbols $\int$ and dx:

$$\int f(x)\, dx = \int (x^a + b)\, dx \tag{13.4}$$

In function $f(x)$ we have two terms that should be integrated: x^a and b. When we differentiate we subtract 1 from the exponent. When we integrate we instead add 1. For the first term, x^a, we integrate by adding 1 to the exponent: x^{a+1}. We also divide the same term by the new exponent $(a+1)$. The integral of x^a is:

$$\int (x^a)\, dx = \frac{x^{a+1}}{a+1} \tag{13.5}$$

That we divide by $a+1$ means that when differentiating the integral we will get back x^a as result. Let us check this by taking the derivative of the expression $\frac{x^{a+1}}{a+1}$ with respect to x:

$$\left(\frac{x^{a+1}}{a+1}\right)_x' = \frac{(a+1)\, x^{a+1-1}}{a+1} = x^a \tag{13.6}$$

The exponent moves down from x in front of the expression. Since the expression is divided by the same exponent we cancel these. The next term in equation (13.1) is the constant b. When we integrate we add x to terms that consist only of constants. The next term in our integral therefore becomes bx:

$$\int (b)\, dx = bx \tag{13.7}$$

Last in the integral we add another constant that we call C. This because when differentiating we cancel the terms where the current variable is not included. The constant C thus represents an arbitrary number, for example all conceivable constants that could be included in a primitive function to $f(x)$. We now get the following primitive function that we call F:

$$F(x) = \int (f(x))\, dx = \int (x^a + b)\, dx = \frac{x^{a+1}}{a+1} + bx + C \tag{13.8}$$

$F(x)$ is a primitive function, the integral, to $f(x)$ in equation (13.1) . We check the integral by differentiating $F(x)$ with respect to x:

$$F(x)_x' = \frac{\partial}{\partial x}\left[\frac{x^{a+1}}{a+1} + bx + C\right] = \frac{a+1}{a+1}x^a + b = x^a + b \tag{13.9}$$

This results in the same expression as function $f(x)$ in equation (13.1) , which indicates that our integral was correct. With the same method we can also integrate longer equations with several terms. As long as each term, separated by addition and subtraction, has similar form as above we can use the same method for each term separately.

Let us now take the function:

$$h(x) = x^3 + 2x^2 + 4x \tag{13.10}$$

A primitive function for $h(x)$ is $H(x)$:

$$\begin{aligned} H(x) &= \int h(x)\, dx \\ &= \int (x^3 + 2x^2 + 4x)\, dx \\ &= \frac{x^4}{4} + \frac{2x^3}{3} + \frac{4x^2}{2} + C \end{aligned} \tag{13.11}$$

The derivative of function H with respect to x becomes function h in equation (13.10) :

$$H'_x(x) = x^3 + 2x^2 + 4x = h(x) \tag{13.12}$$

13.2 The meaning of constant C

By definition there are infinitely many primitive functions to a function $f(x)$. To see this we take the following function as an example:

$$y = f(x) = 2x \tag{13.13}$$

A primitive function to f can look like this:

$$F_1(x) = x^2 \tag{13.14}$$

Or like this:

$$F_2(x) = x^2 + 5 \tag{13.15}$$

Or like this:

$$F_3(x) = x^2 + 123 + a^4 \tag{13.16}$$

where a is a constant coefficient (or a variable) that we do not differentiate over. If we differentiate the functions $F_1(x)$, $F_2(x)$ and $F_3(x)$, the result in all three cases becomes $y = 2x$. To handle this when we calculate integrals we always assume that a constant that we call C is included in all primitive functions. Let us now take the following function as an example:

$$g(x) = x^3 + 4 \tag{13.17}$$

The integral of this function, which we call $G(x)$:

$$G(x) = \int g(x)\,dx = \int \left[x^3 + 4\right] dx = \frac{x^4}{4} + 4x + C \tag{13.18}$$

The derivative of function G with respect to x again gives function $g(x)$:

$$G'_x = x^3 + 4 \tag{13.19}$$

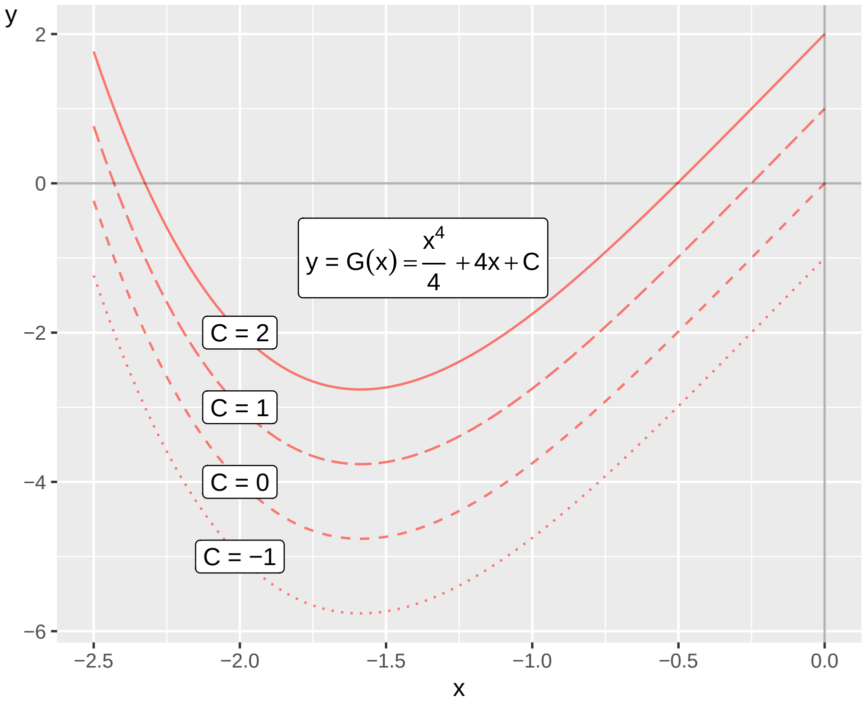


Figure 13.1: The primitive function $G(x)$ with different constants C

In function $G(x)$, which is the integral to $g(x)$, a constant C is again included. Since C can be any value the integral $F(x)$ can assume infinitely many different definitions, depending on the value for C. Figure 13.1 illustrates some examples of the function $G(x)$ and the meaning of C. In the graph four lines are drawn with the function:

$$y = G(x) = \frac{x^4}{4} + 4x + C \tag{13.20}$$

The four lines have different values for the constant $C : -1, 0, 1$ and 2 respectively.

13.3 Primitive function with conditions

Primitive functions can also be combined with conditions. Say for example that we have the function:

$$t(x) = 3x + 5 \tag{13.21}$$

and seek an expression for its primitive function given the condition that $T(0) = 2$, which means that the primitive function $T(x)$ equals 2 when $x = 0$. Another way to say the same thing is that $T(x)$ passes through the point $(0, 2)$. The primitive function to $t(x)$ is:

$$T(x) = \frac{3x^2}{2} + 5x + C \tag{13.22}$$

The constant C can be any number but in this case there is only one version of function $T(x)$ that fulfills the condition $T(0) = 2$:

$$T(x)_1 = \frac{3x^2}{2} + 5x + 2 \tag{13.23}$$

In this function, which fulfills the condition, $C = 2$. That $T_1(x)$ fulfills the condition $T(0) = 2$ we can see by calculating $T_1(x = 0)$:

$$T(0)_1 = \frac{3(0)^2}{2} + 5(0) + 2 = 2 \tag{13.24}$$

13.4 Primitive function for an exponent

In section 10.5 we went through how we may differentiate exponential functions such as:

$$[e^x]'_x = e^x, \qquad \frac{\partial}{\partial x}[e^{ax}] = ae^{ax} \tag{13.25}$$

where x is the included variable, e is Euler's number and a is constant. We see among other things that the function $f(x) = e^x$ has derivative $f'_x(x) = e^x$. With this in mind we shall now go through how we can find a primitive function when we have x as exponent in a power expression. Let us start with the function:

$$f(x) = e^x \tag{13.26}$$

We shall now derive a primitive function for this:

$$\int f(x)\,dx = F(x) = e^x + C \tag{13.27}$$

The letter C is a constant. If we take the derivative of F with respect to x we get:

$$F'_x(x) = e^x \tag{13.28}$$

This is the same thing as $f(x)$. Let us also take the integral of a function of the type $g(x) = e^{ax}$. The integral to this function we call $G(x)$:

$$G(x) = \int g(x)\,dx = \int e^{ax}dx = \frac{e^{ax}}{a} + C \tag{13.29}$$

Just as in the examples in the previous section we add the denominator a to the term e^{ax} so that the derivative of the primitive function $G(x)$ becomes $g(x)$. Let us differentiate the function $G(x)$ with respect to x:

$$G'_x = a * \frac{e^{ax}}{a} = e^{ax} \tag{13.30}$$

This is the same thing as $g(x)$. In section 10.5 we also went through the derivative of a^x where a is an arbitrary constant and x is variable:

$$[a^x]'_x = a^x \ln a \tag{13.31}$$

If the base $a = e$(Euler's number) and we have function $h(x) = a^x$, the derivative is:

$$h(x)'_x = a^x \ln a = a^x = e^x \tag{13.32}$$

Let us now derive a primitive function for the function $h(x) = a^x$. According to the same logic as above we now add a denominator and a constant C and then get the following primitive function:

$$H(x) = \int h(x)\,dx = \int (a^x)\,dx = \frac{a^x}{\ln a} + C \tag{13.33}$$

Let us illustrate these rules with two examples where we write out numbers instead of letters. Suppose we have the following function:

$$q(x) = e^{12x} + 5 \tag{13.34}$$

Now we seek a primitive function for $q(x)$, which we call $Q(x)$:

$$Q(x) = \frac{e^{12x}}{12} + 5x + C \tag{13.35}$$

Since C is a general term this is called that we have determined all primitive functions for $q(x)$. Let us take another example with several terms. This time we seek a primitive function for the function:

$$v(x) = 3^x + e^{2x} + 4 + \frac{x^3}{2} \tag{13.36}$$

The integral to the function $v(x)$ we call $V(x)$:

$$\begin{aligned} V(x) &= \int v(x)\,dx \\ &= \int \left[3^x + e^{2x} + 4 + \frac{x^3}{2}\right] dx \\ &= \frac{3^x}{\ln(3)} + \frac{e^{2x}}{2} + 4x + \frac{x^4}{8} + C \end{aligned} \tag{13.37}$$

Note how we in this example on the third line take the opportunity to use several different rules that we have introduced earlier.

13.5 Calculate area with integral

Integrals can among other things be used to calculate an area, for example the area between the line for a function and the horizontal x-axis, or the area between the lines in a graph. Let us start with how we calculate the area in a graph without integral for the following function:

$$y = f(x) = 4, \forall 0 \leq x \leq 4 \tag{13.38}$$

The function $f(x)$ equals 4 for all values of x between 0 and 4. The line for this function is horizontal. The area between this line and the x-axis:

$$\begin{aligned} \text{width } \times \text{ height} &= \text{area} \\ 4 \times 4 &= 16 \text{ area units} \end{aligned} \tag{13.39}$$

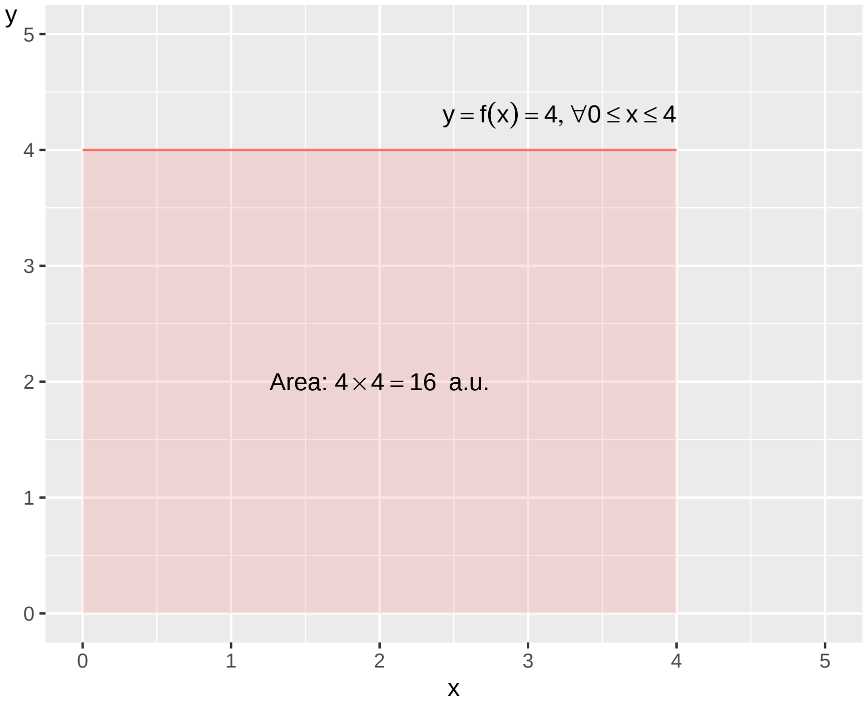


Figure 13.2: The area under the function $y = f(x) = 4, \forall 0 \leq x \leq 4$

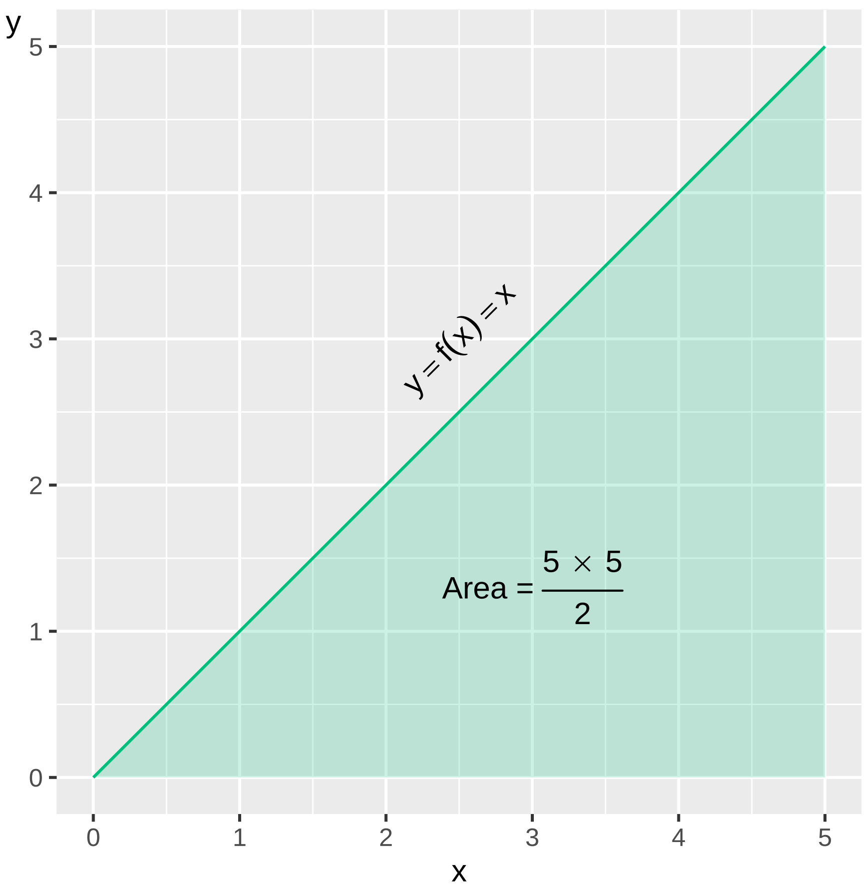


Figure 13.3: The function $y = f(x) = x$ and the area under the line

The expression *area units*, a.u., is a general concept that we use since we have not specified any unit of measurement for x and y. Area units can for example be square centimeters or square meters. The function in equation (13.38) and the area under the line is illustrated in figure 13.2 . If we instead have the function:

$$y = h(x) = x, \forall 0 \le x \le 5 \tag{13.40}$$

this is, in a graph, a straight line between $(0,0)$ and $(5,5)$. The area for the surface under the line down to the x-axis is then:

$$\text{Area} = \frac{\text{height} \times \text{width}}{2} = \frac{5 \times 5}{2} = 12.5 \text{ a.u.} \tag{13.41}$$

The function and the area are illustrated in figure 13.3 . Let us now calculate the area for these two examples with the help of integrals. We start with the function $y = f(x) = 4$. The integral for function f:

$$F(x) = 4x + C \tag{13.42}$$

where C is an arbitrary constant. To calculate the area we derive the primitive function over the interval 0 to 4. We calculate the area by taking the integral for $f(x)$ over the values a and b, 0 to 4. This integral is written $\int_a^b$:

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a) \tag{13.43}$$

We calculate the integral from the smaller to the larger value. The expression $[F(x)]_a^b$ means that we calculate the primitive function for a and b, which is then set up as $F(b)$ minus $F(a)$. We set up the equation with the integral with the higher value as input value, minus the integral with the lower value. In this case we have $a = 0$ and $b = 4$:

$$\begin{aligned} \int_0^4 f(x)\,dx &= F(4) - F(0) \\ &= (4(4) + C) - (4(0) + C) \\ &= 16 \end{aligned} \tag{13.44}$$

This is the same result as we found in equation (**??**) . The constant C is written out in both equations but since we get a positive and a negative C both can be canceled. Let us now take the integral to the function $y = h(x) = x$:

$$H(x) = \int h(x)\,dx = \int [x]\,dx = \frac{x^2}{2} + C \tag{13.45}$$

To calculate the area:

$$\int_a^b h(x)\,dx = H(b) - H(a) \tag{13.46}$$

We now have $b = 5$ and $a = 0$:

$$\begin{aligned}\int_0^5 h(x)\,dx &= \int_0^5 [x]\,dx \\ &= H(5) - H(0) \\ &= \left(\frac{(5)^2}{2} + C\right) - \left(\frac{(0)^2}{2} + C\right) \\ &= 12.5\end{aligned} \tag{13.47}$$

This is the same result as the answer we got above. As long as the integral lacks integration limits, that is an interval for x, this is called an indefinite integral. When we specify the interval for x, as in this case $\int_0^5$, this is called a definite integral.

Let us now take the function:

$$y = g(x) = x^2 \tag{13.48}$$

Its integral is:

$$G(x) = \int (x^2)\,dx = \frac{x^3}{3} + C \tag{13.49}$$

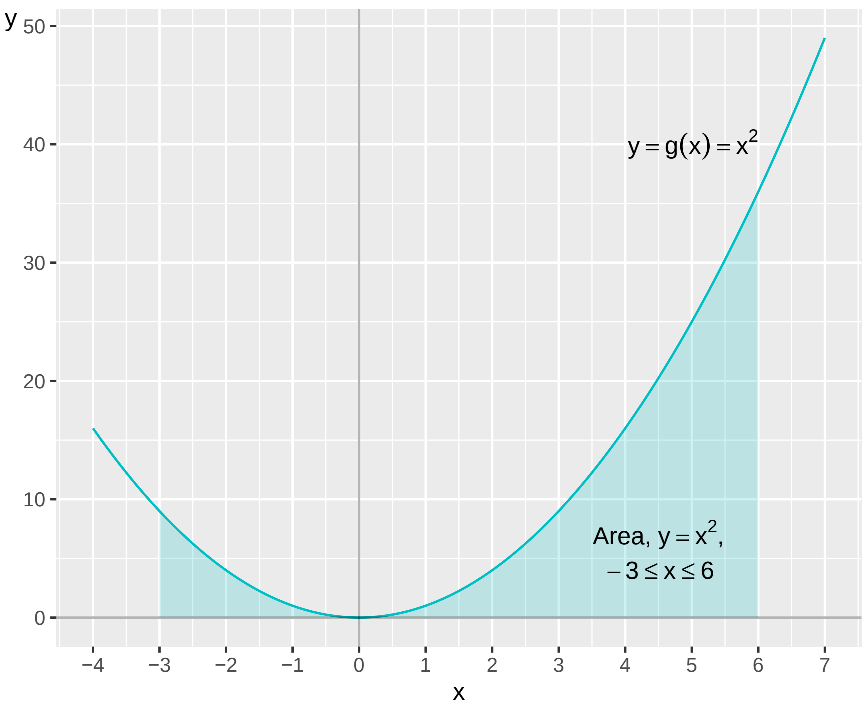


Figure 13.4: The function $y = f(x) = x^2$

We shall now calculate the area for $G(x)$ for the values $-3 \le x \le 6$:

$$G(x) = \int_{-3}^{6} g(x)\,dx \tag{13.50}$$

Our integral has the boundary values $x = -3$ and $x = 6$. Since the integral has a specified interval for x this means that this is a definite integral:

$$G(x) = \int_{-3}^{6} g(x)\,dx = [G(x)]_{-3}^{6} = G(6) - G(-3) \tag{13.51}$$

$G(6)$ and $G(-3)$ is the primitive functions for $x = 6$ and $x = -3$:

$$\begin{aligned}\int_{-3}^{6} g(x)\,dx &= G(6) - G(-3) \\ &= \left(\frac{1}{3}(6)^3 + C\right) - \left(\frac{1}{3}(-3)^3 + C\right) \\ &= \frac{216}{3} + C - \frac{(-27)}{3} - C \\ &= 63 \text{ a.u.}\end{aligned} \tag{13.52}$$

In row 3 we set the function $G(x = 6)$ as positive term and $G(x = -3)$ as negative term. Figure 13.4 illustrates the line to the function $y = x^2$ and the area between the horizontal x-axis and the function's line for $-3 \leq x \leq 6$.

13.6 The area under the x-axis

So far we have only calculated the area for lines above the x-axis. But as we have seen in previous examples functions can also have negative y-values, whereupon the lines pass below the x-axis. The form for definite integrals that we used above is defined in such a way that it calculates the area between a line and the horizontal x-axis:

$$\int_{b}^{a} f(x)\,dx = F(b) - F(a) \tag{13.53}$$

The area between the x-axis and a line above the x-axis becomes a positive value. The area between the x-axis and a line below the x-axis becomes a negative value. If we calculate the area based on a function that both passes above and below the x-axis we get the net area. For example:

$$\begin{aligned}\int_{0}^{5} [-2 + x]\,dx &= F(5) - F(0) \\ &= \left(-2(5) + \frac{(5)^2}{2}\right) - \left(-2(0) + \frac{(0)^2}{2}\right) \\ &= (-10 + 12.5) \\ &= 2.5\end{aligned}$$

The function and the two surfaces are illustrated in figure 13.5 where the first area (A_1) is the surface below the x-axis, which equals -2. The second area (A_2) is the surface above the x-axis, which equals 4.5. The net result is thus the area for the part that is above x minus the surface that is below x.

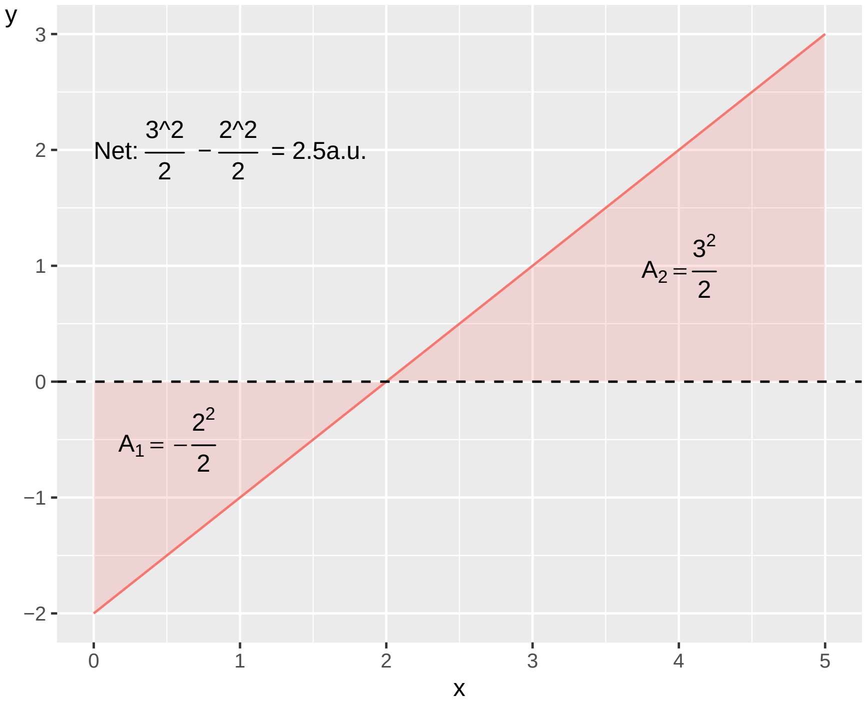


Figure 13.5: Area above and under the x -axis

Suppose we instead want to calculate all surface between the function's line and the x-axis, regardless of whether the line is above or below the x-axis, and sum this area to one value. In that case we divide the integral and put a minus sign for the part that is below the x-axis. We divide the integral at the value for x where the line intersects the x-axis, which is where $y = f(x) = 0$. We start by calculating the x-value we seek:

$$\begin{aligned} -2 + x &= 0 \\ x^* &= 2 \end{aligned} \tag{13.54}$$

Then we divide the integral into two parts and put a minus sign in front of the part that is below the x-axis:

$$\text{Total area:} -\int_0^2 f(x)\,dx + \int_2^5 f(x)\,dx \tag{13.55}$$

Now we calculate the two integrals and sum:

$$\begin{aligned} -\int_0^2 f(x)\,dx + \int_2^5 f(x)\,dx &= -\left[-2x + \frac{x^2}{2}\right]_0^2 + \left[-2x + \frac{x^2}{2}\right]_2^5 \\ &= -[-4+2] + [(-10+12,5) - (-4+2)] \\ &= 6,5 \end{aligned} \tag{13.56}$$

The total area between the line and the x-axis is 6.5 area units.

13.7 The area between two lines

Integrals can also be used to calculate the area between two curves. Suppose we have the following two continuous functions $g(x)$ and $h(x)$:

$$g(x) = -x + x^3 - x^2 + 10 \tag{13.57}$$
$$h(x) = 2 + 2x - \frac{1}{2}x^2$$

Over an interval of x-values $[a, b]$ is $g(x) \geq h(x)$. The area A is bounded for this interval upward by $g(x)$ and downward by $h(x)$ and can be calculated as:

$$A = \int_a^b [g(x) - h(x)]\, dx \tag{13.58}$$

Lines and the area are illustrated in figure 13.6 . We shall now calculate the area between these for $0 \leq x \leq 3$ and therefore set up equation (13.58) with the functions from equation (**??**) and calculate the integral:

$$\begin{aligned}
\int_0^3 g(x) - h(x)\, dx &= \int_0^3 \left[-x + x^3 - x^2 + 10 - \left(2 + 2x - \frac{1}{2}x^2\right)\right] dx \qquad (13.59)\\
&= \int_0^3 \left[x^3 - \frac{1}{2}x^2 - 3x + 8\right] dx\\
&= \left[\frac{x^4}{4} - \frac{x^3}{6} - \frac{3x^2}{2} + 8x\right]_0^3\\
&= \left(\frac{3^4}{4} - \frac{3^3}{6} - \frac{3(3)^2}{2} + 8(3)\right) - \left(\frac{0^4}{4} - \frac{0^3}{6} - \frac{3 * 0^2}{2} + 8 * 0\right)\\
&= 20.25 + 4.5 - 13.5 + 24\\
&= 44.25 \text{ a.u.}
\end{aligned}$$

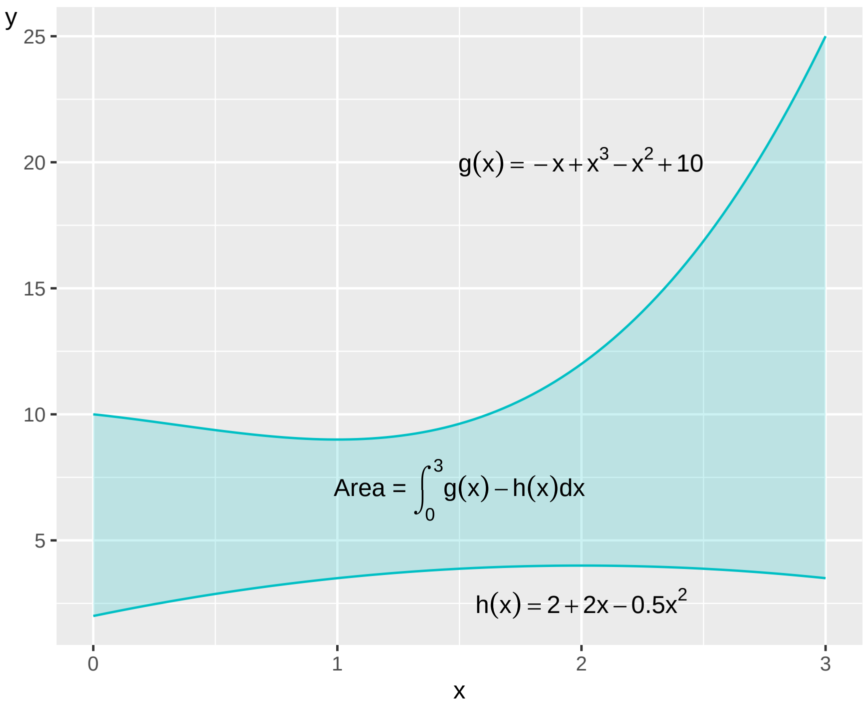


Figure 13.6: The area between two functions

13.8 Integration rules

Table 13.1 describes integrals for some commonly occurring functions, several of which we have gone through above. The calculation rule in the fourth row that proceeds from function $f(x) = \frac{1}{x}$ describes how the derivative of $\log_b x = \frac{1}{x \ln b}$(section10.5). When the base b = Euler's number (e), the derivative of $\ln x = 1/x$(see equation (10.60) and (10.61)). Now we shall use the latter to calculate the primitive function of $1/x$. The integral of $1/x$ equals the natural logarithm of the absolute value of x, plus the constant C:

$$\int \frac{1}{x} dx = \ln|x| + C \tag{13.60}$$

Table 13.1: Integrals for some common functions

	Function $f(x)$	**Integral** $\int f(x)\,dx$
1	x^a, $a \neq -1$	$\frac{x^{a+1}}{a+1} + C$
2	$x^{-2} = \frac{1}{x^2}$	$\frac{x^{-1}}{-1} + C$
3	$\frac{1}{x}$	$\ln|x| + C$
4	$\frac{1}{ax+b}$	$\frac{1}{a}\ln|ax+b| + C$
5	e^x	$e^x + C$
6	e^{kx}	$\frac{e^{kx}}{k} + C$
7	a^x	$\frac{a^x}{\ln a} + C$
8	$\ln x$	$x \ln x - x + C$

The reason for the absolute value is that $1/x$ holds for all real numbers except $x = 0$, while $\ln x$ is only defined for $x > 0$, see chapter 5 . The fourth calculation rule describes the integral:

$$\int \left(\frac{1}{ax+b}\right) dx = \frac{1}{a}\ln|ax+b| + C \tag{13.61}$$

where a and b are constants. This we see from the chain rule for derivative of composite functions, which gives that $(f \circ g)'_x = g'(f' \circ g)$, see table 10.2 . Let us consider the natural logarithm as function f and $ax + b$ as function g. Now we shall calculate the derivative of the following expression:

$$\frac{1}{a}\ln(ax+b) \tag{13.62}$$

We rewrite this in the following way:

$$\begin{aligned}
\left(\frac{1}{a}\ln(ax+b)\right)'_x &= \frac{1}{a}(f \circ g)'_x \\
&= \frac{1}{a} g'_x (f'_x \circ g) \\
&= \frac{1}{a}(ax+b)'_x (\ln(ax+b))'_x \\
&= \frac{1}{a}(a)\left(\frac{1}{ax+b}\right) + 0 \\
&= \frac{1}{ax+b}
\end{aligned} \tag{13.63}$$

Table 13.2: Integration rules

	Integral	Can also be written as
1	$\int af(x)\,dx$, where a is a constant	$a\int f(x)\,dx$
2	$\int (f+g)\,dx$	$\int f(x)\,dx + \int g(x)\,dx$
3	$\int (f-g)\,dx$	$\int f(x)\,dx - \int g(x)\,dx$

Table 13.2 describes some common rules for integrals. The first rule describes how we can factor out the constant a from the integral $\int f(x)\,dx$, since a is not affected by the integration. We illustrate with the function:

$$f(x) = 4x^2 + 16x \tag{13.64}$$

The integral of f:

$$F(x) = \frac{4x^3}{3} + 8x^2 + C \tag{13.65}$$

Now we rewrite f so we get $4\left(x^2+4x\right)$ and call the content in the parentheses $g(x)$. If we take the integral of this we get:

$$G(x) = \frac{x^3}{3} + 2x^2 + C \tag{13.66}$$

From here we see that $4G(x) = F(x)$, which is what the first calculation rule in table 13.2 describes. The other two integration rules can also be illustrated with function $f(x) = 4x^2 + 16x$. Suppose we divide this function into the two functions:

$$h(x) = 4x^2, \quad k(x) = 16x \tag{13.67}$$

Their respective integrals are then:

$$H(x) = \frac{4x^3}{3} + C, \quad K(x) = 8x^2 + C \tag{13.68}$$

From this we see that $H + K = F$. If we sum the two constants C and C in the functions H and K we get a new arbitrary constant, which has the same meaning for the integrals and is not affected by the integration. This is what the second calculation rule in table 13.2 describes.

The third calculation rule is a variant of the second calculation rule. In a similar way as above we check that the following holds::

$$\int (h-k)\,dx = \int \left(4x^2 - 16x\right)dx = \frac{4x^3}{3} - 8x^2 + C = H - K \tag{13.69}$$

which confirms the third rule. For definite integrals we may also describe some additional rules. Given that f is a continuous function over an interval where the values a, b, c are included the following holds for definite integrals:

$$\begin{aligned}
&\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx \\
&\int_a^a f(x)\,dx = 0 \\
&\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx
\end{aligned} \tag{13.70}$$

The first of these rules says that if we take the integral over an interval from a to b, it corresponds to the same value but negative if we take the integral from b to a instead. If we take the following function as an example:

$$y = f(x) = x^3 + 4 \tag{13.71}$$

We calculate its integral:

$$\int_1^2 f(x)\,dx = \left[\frac{x^4}{4} + 4x\right]_1^2 = (4+8) - \left(\frac{1}{4} + 4\right) = 7 + \frac{3}{4} \tag{13.72}$$

This means that:

$$\begin{aligned} -\int_2^1 f(x)\,dx &= -\left[\frac{x^4}{4} + 4x\right]_2^1 \\ &= -\left(\left(\frac{1}{4} + 4\right) - (4+8)\right) \\ &= 7 + \frac{3}{4} \end{aligned} \tag{13.73}$$

The second calculation rule means:

$$\int_a^a f(x)\,dx = F(a) - F(a) = 0 \tag{13.74}$$

The third calculation rule means that if we have a continuous function where the values a,b,c are included within an interval the integral can be divided into two integrals and give the same result as if we had taken the integral over the entire interval directly. We will illustrate this with the following example where the integral to the left sums the variable x over the interval $[1, 3]$ and the two integrals in the right side sum x the interval $[1, 2]$ as well as the interval $[2, 3]$:

$$\begin{aligned} \int_1^3 x^2 dx &= \int_1^2 x^2 dx + \int_2^3 x^2 dx \\ \left[\frac{x^3}{3}\right]_1^3 &= \left[\frac{x^3}{3}\right]_1^2 + \left[\frac{x^3}{3}\right]_2^3 \\ 9 - \frac{1}{3} &= \left(\frac{8}{3} - \frac{1}{3}\right) + \left(9 - \frac{8}{3}\right) \\ 8 + \frac{2}{3} &= 8 + \frac{2}{3} \end{aligned} \tag{13.75}$$

Both sides give the same final result.

13.9 More examples

We shall now calculate the primitive function for the following function:

$$f(x) = x + 12 \tag{13.76}$$

The primitive function for f is:

$$\begin{aligned} F(x) &= \int (x+12)\,dx \\ &= \frac{1}{2}x^{1+1} + 12 * x + C \\ &= \frac{x^2}{2} + 12x + C \end{aligned} \tag{13.77}$$

If we instead have the following function:

$$f(x) = 4x^2 + 3y \tag{13.78}$$

the primitive function fort his is:

$$F(x) = \frac{4x^3}{3} + 3yx + C$$

Extensive equations can at first glance look more complicated than they actually are. For example the function:

$$f(x) = a + 2bx + 3c + 4x^5 \tag{13.79}$$

With the help of similar methods as in previous examples we see that the primitive function for this function f is:

$$\begin{aligned} F(x) &= ax + bx^2 + 3cx + \frac{4x^6}{6} + C \\ &= \frac{2}{3}x^6 + bx^2 + (a+3c)\,x + C \end{aligned} \tag{13.80}$$

In the second row we rearrange a little and put the term with the highest exponent of x first in the expression. Sometimes we also benefit from rewriting the equation before we calculate the integral. For example:

$$g(k) = k^5 ak^{-2} \tag{13.81}$$

This function g can be written:

$$g(k) = ak^3 \tag{13.82}$$

The primitive function for function g is:

$$G(k) = \frac{ak^4}{4} + C \tag{13.83}$$

Let us now take the following function $k(T)$:

$$k(T) = T^{-1} + 36T - \frac{1}{T^2} \tag{13.84}$$

The primitive function for this function k is:

$$K(T) = \ln T + 18T^2 + T^{-1} + C$$

13.10 Multiple integral

If we have several variables in the same function, for example $f(x, y)$ or $g(x, y, z)$, we can use a multiple integral. Consider now the following function:

$$z = f(x, y) = x^2 + y \tag{13.85}$$

The variable z is a function of the variables x and y, where $(x, y) \in \mathbb{R}^2$. Now we shall calculate an integral with respect to both x and then y, over the interval 0 to 1 for both variables:

$$\int_0^1 \int_0^1 \left(x^2 + y\right) dxdy \tag{13.86}$$

To calculate this integral we solve one integral at a time, in the same way as when we calculated integrals above. We start by taking the integral with respect to x and calculate the area between 0 and 1:

$$\int_0^1 \int_0^1 \left(x^2 + y\right) dxdy = \int_0^1 \left[\frac{x^3}{3} + xy\right]_0^1 dy = \int_0^1 \left(\frac{1}{3} + y\right) dy \tag{13.87}$$

Then we take the integral with respect to y and calculate this:

$$\int_0^1 \left(\frac{1}{3} + y\right) dy = \left[\frac{y}{3} + \frac{y^2}{2}\right]_0^1 = \frac{5}{6} \tag{13.88}$$

The result is $\frac{5}{6}$ volume units.

When we calculated the integral for a function with one variable we calculated among other things the area for the surface under a line in a graph. Since we now calculate the area under a two-dimensional surface (x, y)-plane, the result becomes a three-dimensional volume. In this example we have the dimensions z, x and y. When we have two variables this is called a double integral or surface integral. Integration over three variables is called triple integral and can be calculated in a corresponding way.

13.11 Partial integration

Partial integration, or integration by parts, is a method for finding the solution to integrals that are the product of two or more functions. The need for this method arises when the integral of a product is not equal to the product of the factors' integrals. Suppose we have the function:

$$y = h(x) = x^2 a^{2x} \tag{13.89}$$

where x is the included variable and a is a constant. The function $h(x)$ consists of the two factors x^2 and a^{2x}, which we describe as separate functions, f and g respectively:

$$h(x) = x^2 a^{2x} = f(x)\, g(x) \tag{13.90}$$

If we take the integral for the functions f and g we get the primitive functions F and G:

$$F = \int x^2 dx = \frac{x^3}{3}, \qquad G = \int a^{2x} dx = \frac{a^{2x}}{2 \ln a} \tag{13.91}$$

The derivative of F and G respectively with respect to x results in the functions f and g. But the derivative of the product of $F \times G$ does not result in the product of $f \times g = x^2 a^{2x}$. The derivative of $(F \times G)$ with respect to x becomes instead:

$$[F(x) \times G(x)]'_x = \frac{\partial}{\partial x}\left[\frac{x^3}{3} * \frac{a^{2x}}{2}\right] = \frac{a^{2x}\ln(a) * x^3}{3} + \frac{a^{2x}x^2}{2} \tag{13.92}$$

where we use the differentiation rules from section 10.9 . To calculate the integral of $h(x)$ we must instead use partial integration. The general form for this, given that we divide function $h(x)$ into the two functions $f(x)$ and $g(x)$, looks as follows:

$$\textbf{Partial integration:} \int fg'dx = fg - \int f'gdx \tag{13.93}$$

where $f' = f'_x(x)$ and $g' = g'_x(x)$. To use this method we seek to begin with an expression that can be formulated as the left side in this equation, $\int f(x)\, g'_x(x)\, dx$.

We now choose which of the two expressions x^2 and a^{2x} from function h we should define as f and g'_x respectively. This choice in turn affects how the rest of the calculation. Suppose we define $f = x^2$ and $g' = a^{2x}$. We call the primitive function we seek H:

$$H = \int fg'dx = fg - \int f'gdx = x^2\frac{a^{2x}}{2\ln a} - \int 2x\frac{a^{2x}}{2\ln a}dx \tag{13.94}$$

where $f' = 2x$ and $g = \frac{a^{2x}}{2\ln a}$. The second term consists of a new integral of a product. In this term we can cancel the number 2 in the numerator and denominator and factor out the denominator, since this does not contain the variable x and therefore is not affected by the integration:

$$\int 2x\frac{a^{2x}}{2\ln a}dx = \frac{1}{\ln a}\int xa^{2x}dx \tag{13.95}$$

To solve this integral we again use partial integration based on equation (13.93) : $\int fg'dx = fg - \int f'gdx$. We now choose to define $f = x$ and $g' = a^{2x}$, which gives $f' = 1$ and $g = \frac{a^{2x}}{2\ln a}$. The primitive function H now becomes:

$$\begin{aligned}
H &= \int fg'dx \\
&= x^2\frac{a^{2x}}{2\ln a} - \frac{1}{\ln a}\left(x\frac{a^{2x}}{2\ln a} - \int\frac{a^{2x}}{2\ln a}dx\right) \\
&= x^2\frac{a^{2x}}{2\ln a} - \frac{1}{2(\ln a)^2}\left(xa^{2x} - \int a^{2x}dx\right)
\end{aligned} \tag{13.96}$$

At the far right in the equation we again get a new integral. But this time it is only the expression a^{2x} that should be integrated. This we can solve without partial integration:

$$
\begin{aligned}
H &= \int fg'dx \qquad (13.97)\\
&= x^2\frac{a^{2x}}{2\ln a} - \frac{1}{2\left(\ln a\right)^2}\left(xa^{2x} - \int a^{2x}dx\right)\\
&= x^2\frac{a^{2x}}{2\ln a} - \frac{1}{2\left(\ln a\right)^2}\left(xa^{2x} - \frac{a^{2x}}{2\ln a}\right)\\
&= \frac{x^2a^{2x}}{2\ln a} - \frac{xa^{2x}}{2\left(\ln a\right)^2} + \frac{a^{2x}}{4\left(\ln a\right)^3} + C\\
&= \frac{2\left(\ln a\right)^2 x^2a^{2x} - 2\ln\left(a\right)xa^{2x} + a^{2x}}{4\left(\ln a\right)^3} + C\\
&= \frac{a^{2x}\left(2\ln\left(a\right)x\left(\ln\left(a\right)x - 1\right) + 1\right)}{4\left(\ln a\right)^3} + C
\end{aligned}
$$

We are finished and we have now found the primitive function H to the function $h\left(x\right) = x^2a^{2x}$. If we take the first derivative of function $H\left(x\right)$ with respect to x it certainly requires a somewhat cumbersome calculation but the result becomes again function $h\left(x\right)$. Let us take another example. We now seek the integral to the following function:

$$
v\left(x\right) = x^2e^{2x} \qquad (13.98)
$$

This also requires partial integration. We define $f = x^2$ and $g' = e^{2x}$, which gives $f' = 2x$ and $e^{2x}/2$:

$$
\begin{aligned}
\int v\left(x\right)dx &= fg - \int f'gdx \qquad (13.99)\\
&= x^2\frac{e^{2x}}{2} - \int 2x\frac{e^{2x}}{2}dx\\
&= \frac{x^2e^{2x}}{2} - \int xe^{2x}dx
\end{aligned}
$$

We define $f = x$ and $g' = e^{2x}$ and therefore have that $f' = 1$ and $g = e^{2x}/2$. This gives the following integral, which we call $V\left(x\right)$:

$$
\begin{aligned}
V\left(x\right) &= \int fg'dx \qquad (13.100)\\
&= \frac{x^2e^{2x}}{2} - \int xe^{2x}dx\\
&= \frac{x^2e^{2x}}{2} - \left(\frac{xe^{2x}}{2} - \int\frac{e^{2x}}{2}dx\right)\\
&= \frac{x^2e^{2x}}{2} - \frac{1}{2}\left(xe^{2x} - \int e^{2x}dx\right)\\
&= \frac{x^2e^{2x}}{2} - \frac{1}{2}\left(xe^{2x} - \frac{e^{2x}}{2}\right)\\
&= \frac{e^{2x}\left(2x^2 - 2x + 1\right)}{4} + C
\end{aligned}
$$

Thus we have found a primitive function to function $v\left(x\right)$ in equation (13.98) .

Let us try to solve the same function again but this time define the temporary functions f and g the opposite way compared to what we just did. We thus have the same function: $v\left(x\right) = x^2e^{2x}$ but this time we set $f = e^{2x}$ and $g' = x^2$, which gives $f' = 2e^{2x}$ and $g = x^3/3$. Partial integration gives:

$$\begin{aligned}\int v(x)\,dx &= fg - \int f'g dx \\ &= e^{2x}\frac{x^3}{3} - \int 2e^{2x}\frac{x^3}{3}dx \\ &= e^{2x}\frac{x^3}{3} - \frac{2}{3}\left(\int e^{2x}x^3dx\right)\end{aligned} \tag{13.101}$$

We now define $f = e^{2x}$ and $g' = x^3$. This gives $f' = 2e^{2x}$ and $g = x^4/4$:

$$\begin{aligned}\int v(x)\,dx &= e^{2x}\frac{x^3}{3} - \frac{2}{3}\left(\int e^{2x}x^3dx\right) \\ &= e^{2x}\frac{x^3}{3} - \frac{2}{3}\left(e^{2x}\frac{x^4}{4} - \int 2e^{2x}\frac{x^4}{4}dx\right)\end{aligned} \tag{13.102}$$

Now unfortunately a pattern begins to emerge. We continue to solve integrals but the new integrals become rather more complicated than the previous ones, which counteracts the purpose. This illustrates that we can benefit greatly from setting up our problem in as manageable a way as possible.

13.12 Income distribution with integrals

Here follows an example of how we may use integrals for social science theory. In part IV we shall also introduce how we may use integrals when we calculate probability.

Say that a society consists of four people and each one has the same income, 1,000 USD, so that the total incomes in the society are 4,000 USD. Table 13.3 describes the population and its incomes. In the two columns furthest to the right in the table the cumulative amount of people and the incomes are summed as percentage shares of the total.

Table 13.3: Four persons with the same income

Person	Income for this person	Cumulative income for all	Cumulative share of persons	Cumulative share of all income
1	1,000	1,000	25 %	25 %
2	1,000	2,000	50 %	50 %
3	1,000	3,000	75 %	75 %
4	1,000	4,000	100 %	100 %

Let's say that person no. 4 instead of 1,000 USD has 2,000 USD in income. The total incomes in the society become 5,000. Person no. 4 will have $2,000/5,000 = 40\%$ of society's total income. Person 1, 2 and 3 will now have one fifth, 20%, of total income each. This group of people and their incomes is described in table 13.4 .

Table 13.4: Four persons where one person has twice as high income as the others

Person	Income for each person	Cumulative income for all	Cumulative share of people	Cumulative share of all income
1	1,000	1,000	25 %	20 %
2	1,000	2,000	50 %	40 %
3	1,000	3,000	75 %	60 %
4	2,000	5,000	100 %	100 %

Figure 13.7 illustrates both distributions. The diagonal line, 45 degrees, represents the distribution where all the people have the same income. 75% of the people, person 1 to 3, have 75% of the incomes. The lower line represents the distribution where person no. 4 has twice as high income as the others. 75% of the people now have 60% of the incomes. This type of distribution curves are called Lorenz curves .

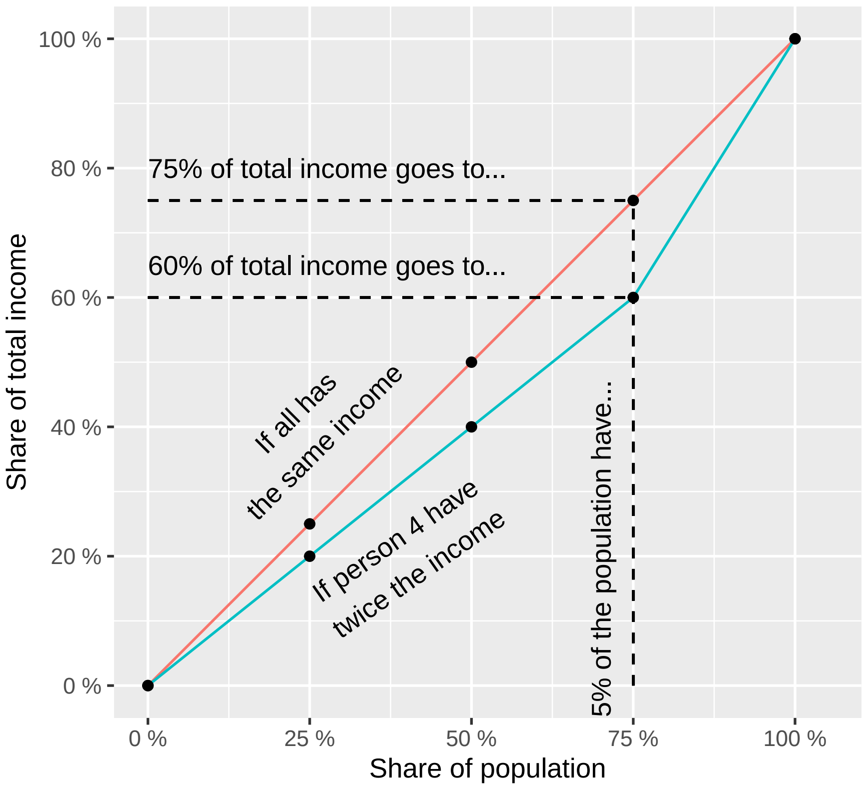


Figure 13.7: Example with a Lorenz curve

Now we instead want to use mathematical functions to describe the income distribution. This may be useful if we want to reason theoretically. Let the following function describe what percentage share of the society's incomes, y, goes to a percentage share of the population, x:

$$y = L(x) = x^a, \ a \geq 1 \tag{13.103}$$

where a is a constant coefficient. If $a = 1$ the function becomes $y = x$ and the inhabitants have the same income. The percentage cumulative amount of the population increases at the same pace as the incomes. Say now that we have a society where the income distribution can instead be described with the function:

$$y_1 = L_1(x) = x^3 \text{ (Distribution 1)} \tag{13.104}$$

Now something happens that causes the income distribution to change. It could for example be some form of technological change, a change in demography between one generation and the next or that the government changes taxes and transfers. Regardless of the reason the new income distribution can instead be described with the function:

$$y_2 = x^2 \text{ (Distribution 2)} \tag{13.105}$$

Now we shall find out how much the total income distribution has changed. This we can calculate with the help of the so-called Gini coefficient, named after the Italian statistician and sociologist Corrado Gini. The Gini coefficient indicates what share of the incomes must be redistributed for everyone to have the same income.

Say for example that we have two people where one person has 0 in income and the other has 100 USD in income. In that case 50% of the incomes must be redistributed if both people shall have the same income, 50 USD. If B is the area under the Lorenz curve and A is the area between the Lorenz curve and the diagonal line we calculate the Gini coefficient, G in the following way:

$$G = \frac{A}{A+B} \tag{13.106}$$

Since we by definition know that $A + B = 1/2$:

$$G = \frac{A}{A+B} = \frac{A}{\frac{1}{2}} = 2A \tag{13.107}$$

From this we also see that:

$$\begin{aligned} A + B &= \frac{1}{2} \\ 2A &= 1 - 2B = G \end{aligned} \tag{13.108}$$

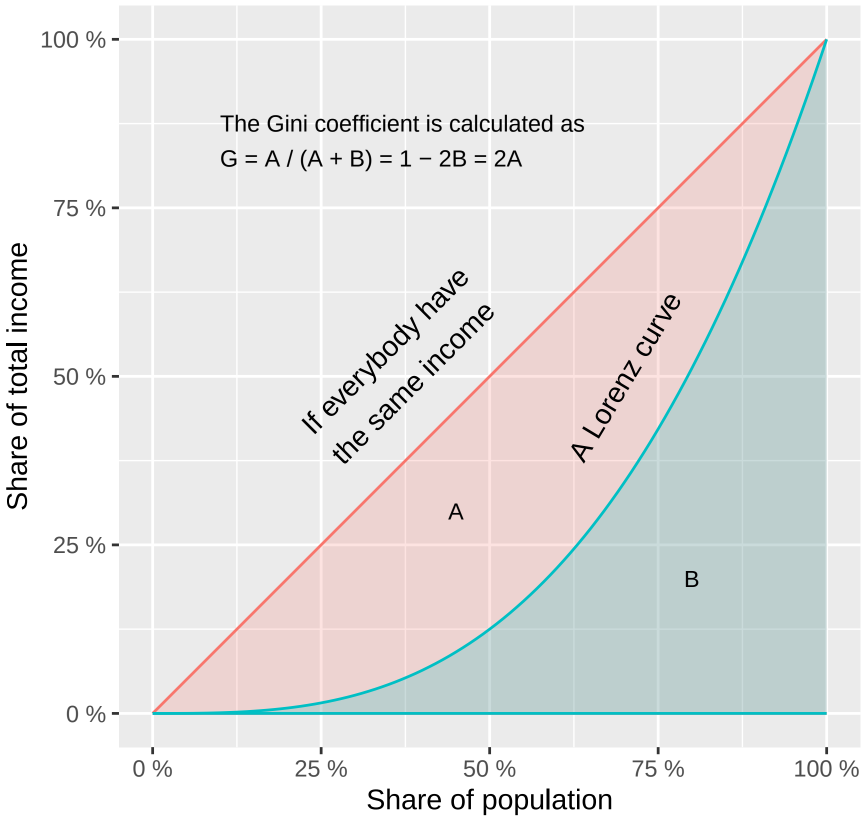


Figure 13.8: The Gini coefficient

This is illustrated in figure 13.8 generally with some lines as examples. The vertical axis shows cumulative percentage share of the population and the horizontal axis shows cumulative percentage share of all incomes in the society. The Lorenz curve describes an example of a distribution where some inhabitants have more income than others. The entire area under the diagonal line down to the horizontal axis constitutes half of the total area in the graph. This is easy to see since the line divides the graph in the middle from the lower left corner to the upper right. We can also calculate the same thing with the help of the integral. Let us take the function for the diagonal line:

$$y = x \tag{13.109}$$

The integral from 0 to 1:

$$\int_0^1 x dx = \frac{1^2}{2} = \frac{1}{2} \tag{13.110}$$

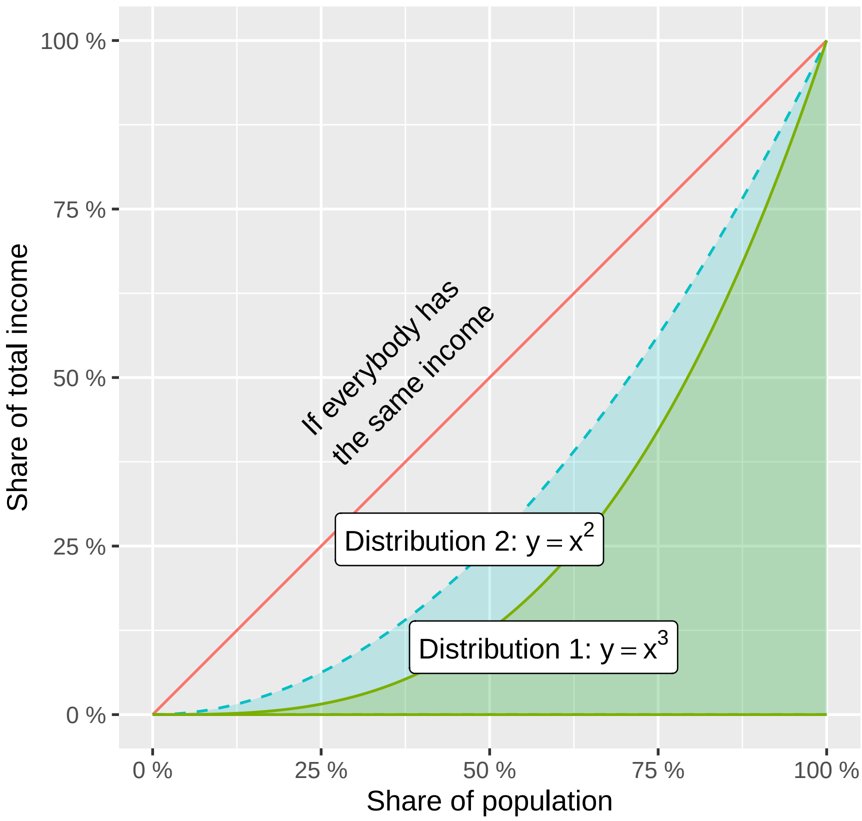


Figure 13.9: Two measures of income distribution

The Gini coefficient for the income distribution before tax can be calculated as $G = 1 - 2B$, where B is the area under a Lorenz curve. In this case we have two Lorenz curves, which describe two examples of distributions: $y = x^3$(Gini before) and $y = x^2$(Gini after) respectively. To see how the income distribution has changed we calculate the area under each curve and compare. Gini before:

$$G_1 = 1 - 2\int_0^1 x^3 dx = 1 - 2\left(\frac{1}{4}\right) = \frac{1}{2} = 50\% \tag{13.111}$$

Gini after:

$$G_2 = 1 - 2\int_0^1 x^2 dx = 1 - 2\left(\frac{1}{3}\right) = \frac{1}{3} = 33,3\% \tag{13.112}$$

The income spread has according to this measure decreased from 50% to 33.3%. The difference is illustrated in figure 13.9 where the dashed line shows income distribution no. 2. The diagonal line shows, as before, how the income distribution would have looked if everyone had had the same income.

13.13 Chapter summary

- Integration can be described as "anti-derivative". This gives us the integral for a function. Example: the function $f(x) = x^2$ has the integral $F(x) = x^3/3 + C$. Function F is the integral and is also called the primitive function. Function f is the integrand. In these functions x is a variable and C an arbitrary constant. Since C can be any number, f has an infinite number of primitive functions.
- Primitive functions can also be calculated based on conditions. Example: we have the function $f(x) = x^2$ and seek the primitive function so that the condition $F(0) = 2$ is fulfilled. This gives the integral $F(x) = x^3/3 + C$ where $C = 2$.
- Integrals can among other things be used to calculate the area under a curve down to the x-axis. Example: to calculate the area between the line for $y = x^2$ for all $x \in [0, 4]$ we take $\int_0^4 f(x)\,dx = F(4) - F(0) = (4^3/3 + C) - (0^3/3 + C) = 21 + 1/3$ area units. To calculate the area below and above the x-axis we divide the integral into two parts, one for the region above and one for the region below the x-axis.
- The area between two curves is given by $\int_a^b [g(x) - h(x)]\,dx$, where $g(x) \geq h(x)$ over an interval of x-values and the area is bounded upward by $g(x)$ and downward by $h(x)$.
- Examples of integration rules: The function $f(x) = x^a$ where a is a constant coefficient, gives integral $F(x) = x^{a+1}/(a+1) + C$. The function $f(x) = e^{kx}$ where k is a constant coefficient and e is Euler's number, gives $F(x) = e^{kx}/k + C$. The function $f(x) = a^x$ gives $F(x) = a^x/\ln a + C$. The function $f(x) = \ln x$ gives $F(x) = x\ln x - x + C$.
- Within social science integrals can among other things be used to discuss theoretical distributions of variables, for example distribution of incomes in a population.

13.14 Exercises

1. Find the primitive function for $h(x) = 2x^3 + 4x$.

(a) Find (H(x)=int h(x)dx) .

(b) Verify the result by differentiating (H(x)) .

(c) Find the specific primitive function (H(x)) such that (H(0)=3) .

Answer: (a) $H(x) = \frac{x^4}{2} + 2x^2 + C$ (b) $H'(x) = 2x^3 + 4x = h(x)$ (c) $H(0) = C = 3$

2. Find primitive functions for the following exponential expressions.

(a) (int e^{3x}dx)

(b) (int2^{x}dx)

(c) (intleft(e$^{\{3x\}+2}$\{x\}right)dx)

Answer: (a) $\frac{e^{3x}}{3} + C$ (b) $\frac{2^x}{\ln 2} + C$ (c) $\frac{e^{3x}}{3} + \frac{2^x}{\ln 2} + C$

3. Calculate the definite integral $\int_0^3 x^2 dx$.

(a) Find the primitive function (G(x)=int x^{2}dx) .

(b) Calculate (G(3)-G(0)) .

(c) What does the result represent geometrically?

Answer: (a) $G(x) = \frac{x^3}{3} + C$ (b) $G(3) - G(0) = \frac{27}{3} - 0 = 9$ (c) The area under the curve $y = x^2$

4. Consider the function $f(x) = x - 2$ for $0 \leq x \leq 5$.

(a) Find the value (x^{*}) where (f(x)=0) .

(b) Calculate the net integral (int_{0}^{5}(x-2)dx) .

(c) Calculate the total area between the curve and the x-axis (treating the area below the x-axis as positive).

Answer: (a) $x^* = 2$ (b) $\int_0^5 (x-2)dx = \left[\frac{x^2}{2} - 2x\right]_0^5 = 2.5$ (c) $-\int_0^2 (x-2)dx + \int_2^5 (x-2)dx = 2 + 4.5 = 6.5$

5. Consider the functions $g(x) = x + 3$ and $h(x) = x^2 + 1$ on the interval $[0, 2]$.

(a) Verify that (g(x)geq h(x)) on ([0,2]) by evaluating at (x=0) and (x=1) .

(b) Set up the integral for the area between the two curves.

(c) Calculate the area.

Answer: (a) $g(0) = 3 > h(0) = 1$ (b) $A = \int_0^2 (-x^2 + x + 2)dx$ (c) $\left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x\right]_0^2 = \frac{10}{3} \approx 3.33$

6. Consider the function $s(x) = 3/x + 1$.

(a) Find the primitive function (S(x)) .

(b) Find the specific primitive function (S(x)) such that (S(1)=5) .

(c) Calculate (int_{1}^{e}s(x)dx) .

Answer: (a) $S(x) = 3\ln|x| + x + C$ (b) $S(1) = 0 + 1 + C = 5 \Rightarrow C = 4$ (c) $S(e) - S(1) = (3 + e + C) - (1 + C) = 2 + e \approx 4.72$

7. Find the primitive function $H(x) = \int (6x^2 - 4x + 1)dx$.

(a) Find (H(x)) .

(b) Find the specific primitive function such that (H(0)=2) .

(c) Calculate the definite integral (int_{0}$^{\{2\}(6x}${2}-4x+1)dx) .

Answer: (a) $H(x) = 2x^3 - 2x^2 + x + C$ (b) $H(0) = C = 2$ (c) $\left[2x^3 - 2x^2 + x\right]_0^2 = 16 - 8 + 2 = 10$

8. A firm's marginal cost is $MC(q) = 3q^2 - 4q + 5$ and fixed cost is $FC = 20$.

(a) Find the total cost function (TC(q)) by integrating (MC(q)) and adding (FC) .

(b) Calculate (TC(4)) .

(c) Calculate the total variable cost when output increases from (q=1) to (q=3) .

Answer: (a) $TC(q) = q^3 - 2q^2 + 5q + 20$ (b) $TC(4) = 64 - 32 + 20 + 20 = 72$ (c) $\left[q^3 - 2q^2 + 5q\right]_1^3 = 24 - 4 = 20$

9. The demand function for a good is $p(q) = 18 - 3q$ and the market price is $p^* = 6$.

(a) Find the equilibrium quantity (q^{*}) where (p(q$^{\{*\})=p}${}) .*

(b) Set up the integral for the consumer surplus (CS=int_{0}$^{\{q}${}}*left(p(q)-p^{}*right)dq) .

(c) Calculate (CS) .

Answer: (a) $18 - 3q^* = 6 \Rightarrow q^* = 4$ (b) $CS = \int_0^4 (12 - 3q)dq$ (c) $\left[12q - \frac{3q^2}{2}\right]_0^4 = 48 - 24 = 24$

10. Consider the function $k(x) = 2e^x + \frac{3}{x}$ for $x > 0$.

(a) Find the primitive function (K(x)=int k(x)dx) .

(b) Find the specific primitive function such that (K(1)=4) .

(c) Calculate (int_{1}^{2}k(x)dx) .

Answer: (a) $K(x) = 2e^x + 3\ln x + C$ (b) $K(1) = 2e + C = 4 \Rightarrow C = 4 - 2e$ (c) $\left[2e^x + 3\ln x\right]_1^2 = 2e^2 + 3\ln 2 - 2e \approx 11.4$

11. Consider $f(x) = x^2 - 2x$ on the interval $[0, 3]$.

(a) Find the values where (f(x)=0) .

(b) Calculate the net integral (int_{0}$^{\{3\}(x}${2}-2x)dx) .

(c) Calculate the total area between (f(x)) and the (x) -axis (treating all area as positive).

Answer: (a) $x = 0$ (b) $\left[\frac{x^3}{3} - x^2\right]_0^3 = 9 - 9 = 0$ (c) $\int_0^2 (2x - x^2)dx = \frac{4}{3}$

12. The Lorenz curve $L(x) = x^3$ describes the income distribution in a population, where $x \in [0, 1]$ is the cumulative population share and $L(x)$ is the corresponding income share.

(a) What income share does the poorest 50% receive? Calculate (L(0.5)) .

(b) Calculate (int_{0}^{1}L(x)dx) .

(c) The Gini coefficient is (G=1-2int_{0}^{1}L(x)dx) . Calculate (G) and interpret ((G=0) is perfect equality, (G=1) maximum inequality).

Answer: (a) $L(0.5) = 0.125$ (b) $\int_0^1 x^3 dx = \frac{1}{4}$ (c) $G = 1 - 2 \cdot \frac{1}{4} = 0.5$

Chapter 14

Linear Algebra: Vectors and Matrices

This chapter introduces linear algebra, specifically how to use vectors and matrices. These methods can be used, among other things, to calculate covariation and to study networks. Linear algebra is a central piece of empirical analysis, which is introduced more in detail in part III to V.

14.1 Vectors

If we have a mathematical quantity that only has one type of value, for example length, weight, temperature or income, this can be called a scalar. Often when we say scalar we mean a real number, even though the word scalar also can refer to other types of numbers. Quantities with multiple values are called vectors. Suppose we have a vector $\vec{p}$, where the arrow over p indicates that this is a vector, with information about a person's weight and height:

$$\text{Person } 1 = \vec{p}_1 = (\text{weight}, \text{height}) \tag{14.1}$$

Elements can also be written vertically:

$$\vec{p} = \begin{pmatrix} \text{weight} \\ \text{height} \end{pmatrix} \tag{14.2}$$

Each value in the vector is called an element. The number of elements in a vector can also be described as the number of dimensions or variables. The vector $\vec{p}$ has two dimensions, which can be described as $\vec{p} \in \mathbb{R}^2$, where $\mathbb{R}$ is the symbol for the real numbers. Vectors with two dimensions can be illustrated in graphs using the x- and y-axis. Say that the vector $\vec{p_1} = (90, 190)$ and we let the x-axis represent weight and the y-axis represent height, then we place $\vec{p_1}$ at the point $(x, y) = (90, 190)$ in a graph:

$$\vec{p} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 90 \\ 190 \end{pmatrix} \tag{14.3}$$

A vector $\vec{x}$ with information about ten variables can be described as $\vec{x} \in \mathbb{R}^{10}$.

14.2 Vector calculations

Vectors have their own rules for addition, subtraction and multiplication. Here follows a brief description based on examples.

14.2.0.1 Addition of vectors.

Let us take two vectors as examples: $\vec{x}_1$ and $\vec{x}_2$, each with n number of elements. Each element contains a number, a value, but here we just call the first element in $\vec{x}_1$ for x_{11}. First element in $\vec{x}_2$ we call x_{21}. The two vectors can be added together and then create a new vector, which we call $\vec{x}_3$. The first element in $\vec{x}_3$ is the sum of element 1 in $\vec{x}_1$ and $\vec{x}_2$, so $x_{31} = x_{11} + x_{21}$. The second element is the sum of element 2 in $\vec{x}_1$ and $\vec{x}_2$, so $x_{32} = x_{12} + x_{22}$. And so on:

$$\begin{aligned}\vec{x}_3 &= \vec{x}_1 + \vec{x}_2 \\ &= (x_{11}, x_{12}, ..., x_{1n}) + (x_{21}, x_{22}, ..., x_{2n}) \\ &= (x_{11} + x_{21}, x_{12} + x_{22}, ..., x_{1n} + x_{2n}) \\ &= (x_{31}, x_{32}, ..., x_{3n})\end{aligned} \tag{14.4}$$

14.2.0.2 Subtraction of vectors.

For subtraction, the corresponding calculation rules as for addition apply. This time too we call our new vector $\vec{x}_3$:

$$\begin{aligned}\vec{x}_3 &= \vec{x}_1 - \vec{x}_2 \\ &= (x_{11}, x_{12}, ..., x_{1n}) - (x_{21}, x_{22}, ..., x_{2n}) \\ &= (x_{11} - x_{21}, x_{12} - x_{22}, ..., x_{1n} - x_{2n}) \\ &= (x_{31}, x_{32}, ..., x_{3n})\end{aligned} \tag{14.5}$$

14.2.0.3 Multiplication of vector by scalar.

Vectors can be multiplied by scalars. This results in a new vector. If we multiply the vector $\vec{x}_1$ by the scalar (the number) s we get:

$$s * \vec{x_1} = s * (x_{11}, ..., x_{1n}) = (sx_{11}, ..., sx_{1n}) \tag{14.6}$$

14.2.0.4 Multiplication of vector with vector.

Multiplication of two vectors does not result in a new vector but in a scalar. This is also called inner product or dot product. This time we take the following two vectors as examples: $\vec{x} = (x_1, x_2)$ and $\vec{y} = (y_1, y_2)$. Each vector has only two elements, where x_1 is the first element in vector $\vec{x}$. We multiply the vectors $\vec{x}$ and $\vec{y}$:

$$\vec{x} * \vec{y} = (x_1, x_2) * (y_1, y_2) = x_1 y_1 + x_2 y_2 = a \tag{14.7}$$

where a is a scalar, a number.

The calculation means that the first element in one vector is multiplied by the corresponding first element in the other vector. The second element in the first vector is multiplied by the second element in the second vector. All these products are summed to a number, which becomes the result, the scalar a. If we multiply three vectors, this becomes in practice like multiplication of a scalar and a vector:

$$\underbrace{(\vec{x}_1 * \vec{x}_2)}_{=\text{scalar}} * \vec{x}_3 = a * \vec{x}_3 \tag{14.8}$$

Regardless of the order in which two vectors are multiplied, the scalar product remains the same. Suppose we have the vectors $\vec{a}$ and $\vec{b}$:

$$\vec{a} * \vec{b} = \vec{b} * \vec{a} = k \tag{14.9}$$

where k is a scalar. Say now that we have the three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$. If we multiply $\vec{a}$ and $\vec{c}$ we get the scalar product $\vec{a} * \vec{c} = s_1$, where the scalar product is a single value. Furthermore, the following also applies:

$$s_3 = \left(\vec{a} + \vec{b}\right) * \vec{c} = \vec{a} * \vec{c} + \vec{b} * \vec{c} = s_1 + s_2 \tag{14.10}$$

where s_1, s_2 and s_3 are scalars. Based on the calculation rules we have already gone through, it can be seen that the following also applies:

$$(k\vec{a}) * \vec{b} = k\left(\vec{a} * \vec{b}\right) \tag{14.11}$$

where k is any scalar and $\vec{a}$ and $\vec{b}$ are vectors. We know that the product of two negative values is a positive value. For vectors, this means among other things that regardless of which elements are in a vector, the scalar product of a vector multiplied by itself can never be negative:

$$\vec{a} * \vec{a} \geq 0 \tag{14.12}$$

This is because if the elements in a vector $\vec{a}$ are negative numbers, the multiplication with the same negative numbers will result in positive values. This in turn also means that the scalar product $\vec{a} * \vec{a} = 0$ if and only if all elements in $\vec{a}$ are equal to 0.

14.2.0.5 Summary of vector calculations.

The above calculation rules can be illustrated with the following two vectors, which each have three elements:

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \tag{14.13}$$

For these two vectors, the following applies for addition:

$$\vec{v}_1 + \vec{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1+4 \\ 2+5 \\ 3+6 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix} \tag{14.14}$$

For subtraction:

$$\vec{v}_1 - \vec{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1-4 \\ 2-5 \\ 3-6 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ -3 \end{pmatrix} \tag{14.15}$$

For multiplication by constant:

$$4 * \vec{v}_1 = 4 * \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4*1 \\ 4*2 \\ 4*3 \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 12 \end{pmatrix} \tag{14.16}$$

For multiplication of the two vectors, the scalar product:

$$\vec{v}_1 * \vec{v}_2 = \begin{pmatrix}1\\2\\3\end{pmatrix} * \begin{pmatrix}4\\5\\6\end{pmatrix} = 1*4+2*5+3*6 = 32 \tag{14.17}$$

14.3 More vectors

Let us calculate some more examples. We have the following three vectors:

$$\vec{a} = \begin{pmatrix}2\\5\\8\end{pmatrix}, \quad \vec{b} = \begin{pmatrix}10\\3\\2\end{pmatrix}, \quad \vec{c} = \begin{pmatrix}22\\14\\17\end{pmatrix} \tag{14.18}$$

We will now calculate:

$$5\vec{a} = 5 * \begin{pmatrix}2\\5\\8\end{pmatrix} \tag{14.19}$$

The scalar 5 is multiplied by each value in the vector:

$$5 * \begin{pmatrix}2\\5\\8\end{pmatrix} = \begin{pmatrix}5*2\\5*5\\5*8\end{pmatrix} = \begin{pmatrix}10\\25\\40\end{pmatrix} \tag{14.20}$$

If we add all three vectors:

$$\vec{a}+\vec{b}+\vec{c} = \begin{pmatrix}2\\5\\8\end{pmatrix} + \begin{pmatrix}10\\3\\2\end{pmatrix} + \begin{pmatrix}22\\14\\17\end{pmatrix} = \begin{pmatrix}42\\22\\27\end{pmatrix} \tag{14.21}$$

If we instead subtract vectors $\vec{a}$ and $\vec{b}$ from $\vec{c}$:

$$\vec{c}-\vec{a}-\vec{b} = \begin{pmatrix}22\\14\\17\end{pmatrix} - \begin{pmatrix}2\\5\\8\end{pmatrix} - \begin{pmatrix}10\\3\\2\end{pmatrix} = \begin{pmatrix}10\\6\\7\end{pmatrix} \tag{14.22}$$

A common way to draw two-dimensional vectors in a two-dimensional graph is as arrows from the origin, the zero point, to the point that the vector's values indicate. Let us take the following two vectors as examples:

$$\vec{a} = \begin{pmatrix}-2\\3\end{pmatrix}, \quad \vec{b} = \begin{pmatrix}1\\2\end{pmatrix} \tag{14.23}$$

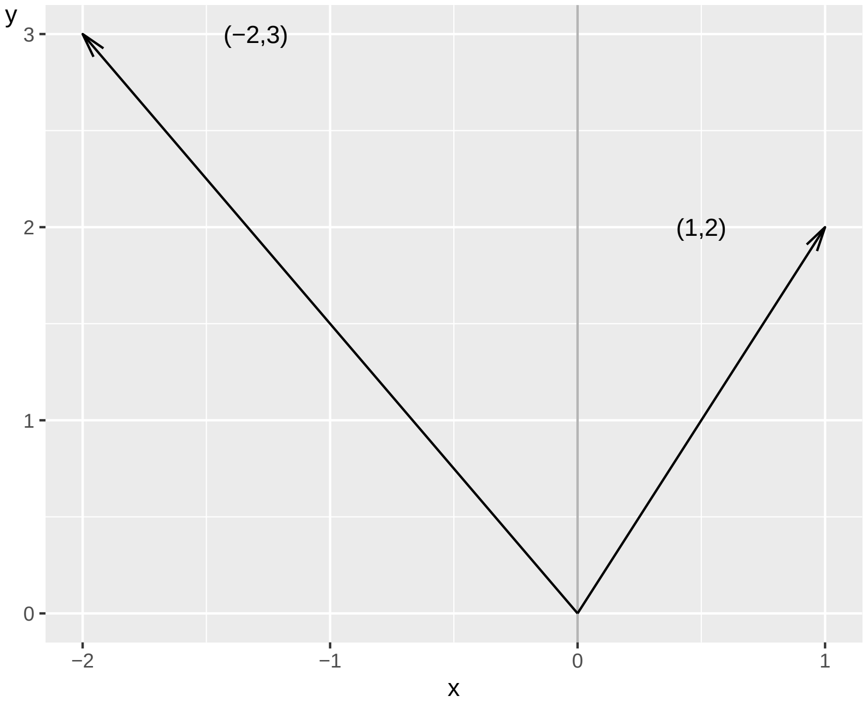


Figure 14.1: Two vectors

Figure 14.1 illustrates the two vectors in the form of two arrows. The arrows go from the zero point $(x, y) = (0, 0)$ to the values that the vectors indicate. The first value in each vector indicates the x-value and the second value indicates the y-value. Now we will calculate the length of each vector's arrow. The vector's length is called norm. For the vector $\vec{a}$ the norm is written $\|\vec{a}\|$ and is calculated as the square root of the sum of the included values:

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2} = (a_1^2 + a_2^2)^{\frac{1}{2}} \tag{14.24}$$

In this case we have:

$$\begin{aligned} \|\vec{a}\| &= \left((-2)^2 + 3^2\right)^{\frac{1}{2}} \approx 3.6 \\ \|\vec{b}\| &= (1^2 + 2^2)^{\frac{1}{2}} \approx 2.24 \end{aligned} \tag{14.25}$$

This should be read as the arrow that goes from the zero point to the point $(x, y) = (1, 2)$ is 2.24 length units long. If the vectors are drawn in a graph where the two axes measure centimeters, then centimeter is the length unit and the arrow is in that case 2.24 cm long.

14.4 Matrices

A matrix is a collection of values organized in rectangle with rows and columns. Matrices are often described with number of rows × number of columns $(r \times k)$ before the name of the matrix. For example 2×3 matrix M:

$$M = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \tag{14.26}$$

The expressions in a matrix are called elements. In matrix M the elements consist of the letters a to f, which represent any real numbers. Matrices are often written with capital letters (M) while individual elements in the matrix are written with lowercase letters and corresponding row and column: m_{11} represents the element in row 1, column 1 in matrix M. Element m_{12} is in row 1 column 2:

$$M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \tag{14.27}$$

If two matrices have the same shape it means that they have the same number of columns and the same number of rows. If we have a matrix A with 5 rows and 7 columns, then it has the same shape as all other matrices that also have 5 rows and 7 columns. There are special calculation rules for how we can use and combine matrices. We begin by going through addition, subtraction, multiplication and division.

14.4.0.1 Addition and subtraction with matrices.

To be able to add or subtract two matrices they must have the same shape. Suppose we have the following two 2×2 matrices A and B:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \; B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \tag{14.28}$$

In addition, each element at position r, k is added to the corresponding element at the corresponding position in the other matrix:

$$A + B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix} \tag{14.29}$$

The result becomes a new matrix. Subtraction is done in the same way:

$$A - B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1-5 & 2-6 \\ 3-7 & 4-8 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -4 & -4 \end{bmatrix} \tag{14.30}$$

14.4.0.2 Multiplication and division with matrices.

Matrices can be multiplied both by individual numbers and other matrices, given certain conditions. We begin by multiplying matrix A by the number 10:

$$10A = 10 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 10*1 & 10*2 \\ 10*3 & 10*4 \end{bmatrix} = \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} \tag{14.31}$$

If we multiply matrix A by the fraction $\frac{1}{10}$ we get:

$$\frac{1}{10}A = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix} \tag{14.32}$$

14.4.0.3 Matrix multiplication.

If we multiply a matrix by another matrix this is called matrix multiplication, and it has its own rules. We illustrate using the following 2×2 matrices A and B:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{14.33}$$

The letters a, b, c and d are any numbers. Matrix multiplication means that the values in column k in matrix A are multiplied by the values in row r in matrix B:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1a + 2c & 1b + 2d \\ 3a + 4c & 3b + 4d \end{bmatrix} \tag{14.34}$$

Matrix AB is a new matrix. Matrix multiplication does not necessarily give the same result if we change the order of the matrices:

$$AB \neq BA \tag{14.35}$$

We see this by calculating BA:

$$BA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a1 + b3 & a2 + b4 \\ c1 + d3 & c2 + d4 \end{bmatrix} \tag{14.36}$$

In matrix multiplication, the number of columns in the left matrix must be equal to the number of rows in the right matrix. The matrix that is created will have the same number of rows as the left matrix and the same number of columns as the right matrix. If we have 3×2 matrix C and 2×3 matrix D:

$$C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad D = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \tag{14.37}$$

and we matrix multiply CD:

$$\begin{aligned} CD &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \\ &= \begin{bmatrix} 1a + 2d & 1b + 2e & 1c + 2f \\ 3a + 4d & 3b + 4e & 3c + 4f \\ 5a + 6d & 5b + 6e & 5c + 6f \end{bmatrix} \\ &= \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \end{aligned} \tag{14.38}$$

where r is the new values in each element in the 3×3 matrix CD.

Not all matrices can be multiplied with all other matrices. If we have the matrices E and F:

$$E = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \tag{14.39}$$

If we try to matrix multiply EF, the value at the first position in the new matrix should be equal to $a * 1 + b * 3 + ... * 5$. We have no element in matrix E that we can multiply by the value 5 in column 1 row 3 in matrix F. Matrix multiplication is not defined for this case.

14.5 More matrices

This section introduces additional examples of useful definitions, operations and calculation rules for matrices in the form of transposition, diagonal matrix, identity matrix and triangular matrix.

14.5.0.1 Transposition.

Transposition of a matrix A means that the elements change places. The transposed version of A, the matrix transpose, is written A^T or A'. If a_{rk} is an element in A at row r column k then the element ends up at position a^T_{kr} in A^T. We illustrate with 2×3 matrix A:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \tag{14.40}$$

Transposed A:

$$A^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} \tag{14.41}$$

If we transpose A^T we get $\left(A^T\right)^T$:

$$\left(A^T\right)^T = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = A \tag{14.42}$$

Transposition does not affect constants. If we multiply matrix A with a constant s and transpose the result we may write:

$$(sA)^T = sA^T \tag{14.43}$$

Transposition of two matrices that are multiplied results in a new matrix where with the transpose of the matrix and where the order of the matrix multiplication is reversed. Thus, if we have two matrices A and B, multiply these as AB and transpose the result $(AB)^T$ we get:

$$(AB)^T = B^T A^T \tag{14.44}$$

14.5.0.2 Diagonal matrix.

A diagonal matrix is a matrix where all elements except the elements in the diagonal are equal to 0. For example:

$$U = \begin{bmatrix} 3 & 0 \\ 0 & \frac{7}{4} \end{bmatrix} \tag{14.45}$$

Usually diagonal matrix refers to square matrices but even rectangular matrices can be diagonal. In that case all elements outside $r = k$ are equal to 0. The expression $r = k$ means row number equal to column number, that is the diagonal in the matrix. For example 2×4 matrix T:

$$T = \begin{bmatrix} 16 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \end{bmatrix} \tag{14.46}$$

Or 3×2 matrix S:

$$S = \begin{bmatrix} a & 0 \\ 0 & z \\ 0 & 0 \end{bmatrix} \tag{14.47}$$

Here we have a column vector V with 2 rows:

$$\vec{v} = \begin{pmatrix} a \\ b \end{pmatrix} \tag{14.48}$$

Now we will create a square diagonal matrix that we call V with the two values from $\vec{v}$ on the diagonal. This can be written as $\text{diag}\left(\vec{v}_r\right)$, where r indicates the number of rows and columns, the dimensions for the new matrix:

$$V = \text{diag}\left(\vec{v}_r\right) = \text{diag}\left(\vec{v}_2\right) = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \tag{14.49}$$

If we want to calculate the sum of the elements in a square diagonal matrix, the result is called trace and can be written as $\text{tr}\left(V\right)$ and is defined:

$$\text{tr}\left(V\right) = \sum_{i=1}^{r} v_{ii} \tag{14.50}$$

14.5.0.3 Identity matrix.

A matrix that is square, has the same number of rows as columns, and whose diagonal values are equal to 1 and where all other values are equal to 0 is called an identity matrix or unit matrix. This type of matrix is often written as I_k where k indicates the number of rows and columns. Here are two examples of identity matrices with dimensions 2×2 and 3×3:

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \ I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{14.51}$$

We have the following 2×3 matrix A:

$$A_{2\times 3} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \tag{14.52}$$

For a matrix A and identity matrices I the following applies:

$$AI_3 = A = I_2 A \tag{14.53}$$

14.5.0.4 Triangular matrix.

A triangular matrix is a square matrix that has zeros on one side of the diagonal. Matrix R is for example triangular:

$$R = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix} \tag{14.54}$$

The letters a to f symbolize any values, not all equal to 0. In this case a, d and f are the values in the diagonal for matrix R.

14.6 Multiplication and transposition

Matrix transposition can among other things be useful in matrix multiplication. Suppose we have the three matrices A, B and C:

$$A = \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 4 \\ 7 & 1 & 8 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 1 & 1 \\ 3 & 0 & 0 \end{bmatrix} \tag{14.55}$$

We may matrix multiply BC, BA and CA:

$$BC = \begin{bmatrix} 2 & 0 & 4 \\ 7 & 1 & 8 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 3 & 1 & 1 \\ 3 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 12 & 4 & 2 \\ 27 & 15 & 8 \end{bmatrix} \tag{14.56}$$

$$BA = \begin{bmatrix} 2 & 0 & 4 \\ 7 & 1 & 8 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 30 \end{bmatrix}$$

$$CA = \begin{bmatrix} 0 & 2 & 1 \\ 3 & 1 & 1 \\ 3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 17 \\ 15 \end{bmatrix}$$

We cannot however multiply the matrices CB, since the number of rows in B is fewer than the number of columns in C(section 14.4). We also cannot multiply AC. We can however transpose A so that we get A^T:

$$A^T = \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix}^T = \begin{bmatrix} 5 & 3 & -1 \end{bmatrix} \tag{14.57}$$

This new matrix A^T we matrix multiply with matrix C:

$$A^T C = \begin{bmatrix} 5 & 3 & -1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 1 \\ 3 & 1 & 1 \\ 3 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 13 & 8 \end{bmatrix} \tag{14.58}$$

Say now that we want to square all values in the column matrix A. Matrix multiplication is not defined for $A^2 = A \times A$, since the number of columns in the left matrix (1) is not equal to the number of rows in the right matrix (3). We calculate $A^T \times A$ but then we get the sum of the squared values:

$$AA^T = \begin{bmatrix} 5 & 3 & -1 \end{bmatrix} \times \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5^2 + 3^2 + (-1)^2 \end{bmatrix} = 3 \tag{14.59}$$

14.7 Determinant

The values in a square matrix can be calculated to a determinant, which is a value and not a matrix. Being able to calculate the determinant of a matrix facilitates many types of calculations later on. The determinant for matrix A can be written $det\,(A)$ or $|A|$, but since the latter is also a way to write absolute value (chapter 2) we stick to $det\,()$ here. In the body text we use italic text $det\,()$ while in the equations we write $\det()$. If we have 2×2 matrix A:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{14.60}$$

Its determinant $det\,(A)$ is then:

$$\det(A) = ad - bc \tag{14.61}$$

If we have the 3×3 matrix B:

$$B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \tag{14.62}$$

Its determinant:

$$\begin{aligned}\det(B) &= a * \det\begin{bmatrix} e & f \\ h & i \end{bmatrix} - b * \det\begin{bmatrix} d & f \\ g & i \end{bmatrix} + c * \det\begin{bmatrix} d & e \\ g & h \end{bmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ &= aei + bfg + cdh - afh - bdi - ceg\end{aligned} \tag{14.63}$$

In section 14.5 we described triangular matrices and in the next section we will go through how to transform a matrix so that it becomes triangular. For a square triangular matrix!triangular matrix, the determinant is the product of its diagonal, that is the values in the matrix's diagonal multiplied with each other. For example if we have 2×2 matrix Z:

$$Z = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \tag{14.64}$$

its determinant $\det(Z)$ can be calculated as:

$$\det(Z) = ac - 0 = ac \tag{14.65}$$

The product ac is the diagonal elements in Z multiplied with each other. Let us now calculate the determinant for the following matrix R:

$$R = \begin{bmatrix} a & b & c \\ 0 & e & f \\ 0 & 0 & i \end{bmatrix} \tag{14.66}$$

Determinant $\det(R)$ is in this case:

$$\begin{aligned}\det(R) &= a * \det\begin{bmatrix} e & f \\ 0 & i \end{bmatrix} - b * \det\begin{bmatrix} 0 & f \\ 0 & i \end{bmatrix} + c * \det\begin{bmatrix} 0 & e \\ 0 & 0 \end{bmatrix} \\ &= a(ei - 0f) - b(0i - 0f) + c(0 - 0e) \\ &= aei - 0 - 0 - 0 \\ &= aei\end{aligned} \tag{14.67}$$

This is the product of the diagonal in matrix R.

14.8 Row reduction in matrix

In section 7.3 we introduced row reduction for equation systems. Row reduction can also be used for matrices. In the previous section we went through how we for a triangular matrix can calculate the determinant as the product of the diagonal. With the help of row reduction it is possible to change a matrix until we get a triangular matrix. In row reduction with matrices we can:

1. Switch places of rows in the matrix, for example switch row 1 with row 2.
2. Multiply a row by a scalar, except 0.
3. Add or subtract all elements in a row with corresponding elements in another row. This can be combined with multiplication by scalar.

Let us take the following matrix H as an example:

$$H = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \tag{14.68}$$

Now we will use row reduction until H becomes triangular. We begin by adding $-4*$ row 1 to row 2:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{matrix} \\ -4r_1 \\ \\ \end{matrix} &= \\ = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix} & \end{aligned} \tag{14.69}$$

We add $-7*$ row 1 to row 3:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix} \begin{matrix} \\ \\ -7r_1 \end{matrix} &= \\ = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} & \end{aligned} \tag{14.70}$$

Then we add $-2*$ row 2 to row 3:

$$\begin{aligned} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \begin{matrix} \\ \\ -2r_2 \end{matrix} &= \\ = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix} & \end{aligned} \tag{14.71}$$

The determinant of matrix H is the product of the diagonal:

$$\det\left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}\right) = 1 \times (-3) \times 0 = 0 \tag{14.72}$$

14.9 Inverse of a matrix

If we divide the number 1 by a real number a we get $1/a = a^{-1}$, which means that $\frac{a}{a} = a^1 a^{-1} = 1$(section 2.6). The number a^{-1} is called the multiplicative inverse of a. There is no definition for division of matrices but there is something called the inverse of a matrix, also called inverse matrix or matrix inverse. A square matrix A is invertible if there exists a matrix B which when matrix multiplied AB results in identity matrix I_r:

$$AB = I_r \tag{14.73}$$

where r is the number of rows and columns in the square matrices A and B respectively. Matrix B is the inverse of A, which is written A^{-1} and is calculated based on the elements in A. We therefore rewrite the equation (14.73) :

$$\begin{aligned} AB &= I_r \\ AA^{-1} &= I_r \end{aligned} \tag{14.74}$$

The inverse matrix has the same dimensions as the current matrix. Another way to describe the condition for whether a matrix A is invertible is that the determinant of A is not equal to 0. If $det\,(A) = 0$ there exists no matrix B that fulfills the condition in equation (14.73) . If we have A^{-1}, the inverse of matrix A, and take the inverse of the matrix inverse, the result becomes matrix A:

$$\left(A^{-1}\right)^{-1} = A \tag{14.75}$$

Let us take the following 2×2 matrix A as example:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{14.76}$$

Inverse matrix A^{-1}:

$$A^{-1} = \frac{1}{\det\left(A\right)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \tag{14.77}$$

The expression $\det\left(A\right)$ is the determinant of matrix A:

$$A^{-1} = \frac{1}{ad - cb} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-cb} & \frac{-b}{ad-cb} \\ \frac{-c}{ad-cb} & \frac{a}{ad-cb} \end{bmatrix} \tag{14.78}$$

If we take the inverse of A^{-1}:

$$\begin{aligned}
\left(A^{-1}\right)^{-1} &= \begin{bmatrix} \frac{d}{ad-cb} & \frac{-b}{ad-cb} \\ \frac{-c}{ad-cb} & \frac{a}{ad-cb} \end{bmatrix}^{-1} \\
&= \frac{1}{\left(\frac{d}{ad-cb}\right)\left(\frac{a}{ad-cb}\right) - \left(\frac{-c}{ad-cb}\right)\left(\frac{-b}{ad-cb}\right)} \begin{bmatrix} \frac{a}{ad-cb} & \frac{b}{ad-cb} \\ \frac{c}{ad-cb} & \frac{d}{ad-cb} \end{bmatrix} \\
&= \frac{1}{\left(\frac{ad-cb}{(ad-cb)^2}\right)} \begin{bmatrix} \frac{a}{ad-cb} & \frac{b}{ad-cb} \\ \frac{c}{ad-cb} & \frac{d}{ad-cb} \end{bmatrix} \\
&= \frac{(ad-cb)^2}{ad-cb} \begin{bmatrix} \frac{a}{ad-cb} & \frac{b}{ad-cb} \\ \frac{c}{ad-cb} & \frac{d}{ad-cb} \end{bmatrix} \\
&= (ad-cb) \begin{bmatrix} \frac{a}{ad-cb} & \frac{b}{ad-cb} \\ \frac{c}{ad-cb} & \frac{d}{ad-cb} \end{bmatrix} \\
&= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\
&= A
\end{aligned} \tag{14.79}$$

For larger matrices the calculation of the inverse often becomes complicated. Already for a 3×3 matrix the calculation becomes relatively extensive. If we for example have the following 3×3 matrix B:

$$B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \tag{14.80}$$

the inverse of matrix B can be defined as:

$$B^{-1} = \frac{1}{\det(B)} \begin{bmatrix} \det\begin{pmatrix} e & f \\ h & i \end{pmatrix} & \det\begin{pmatrix} c & b \\ i & h \end{pmatrix} & \det\begin{pmatrix} b & c \\ e & f \end{pmatrix} \\ \det\begin{pmatrix} f & d \\ i & g \end{pmatrix} & \det\begin{pmatrix} a & c \\ g & i \end{pmatrix} & \det\begin{pmatrix} c & a \\ d & f \end{pmatrix} \\ \det\begin{pmatrix} d & e \\ g & h \end{pmatrix} & \det\begin{pmatrix} b & a \\ h & g \end{pmatrix} & \det\begin{pmatrix} a & b \\ d & e \end{pmatrix} \end{bmatrix} \tag{14.81}$$

where each element is a determinant of four of the elements in B. There are several methods to facilitate the calculation of inverse matrices. One such method is to use row reduction, also called Gauss-Jordan elimination. This works by adding an identity matrix, I_3, next to our 3×3 matrix B, where the identity matrix has the same number of rows and columns as B:

$$BI = \begin{bmatrix} a & b & c & 1 & 0 & 0 \\ d & e & f & 0 & 1 & 0 \\ g & h & i & 0 & 0 & 1 \end{bmatrix} \tag{14.82}$$

Matrix BI has three rows and six columns, 3×6. Now we perform row reduction on this new matrix BI until the left half of the matrix is an identity matrix. That is, columns 1–3 in equation (14.82) should be an identity matrix, instead of columns 4–6. The values in columns 4–6 in matrix BI after the row reduction are the inverse of B. We call this inverse matrix B^{-1}. Let us illustrate with 3×3 matrix D where we use numbers instead of letters:

$$D = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{bmatrix} \tag{14.83}$$

To get D^{-1} we take the matrix DI:

$$DI = \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{bmatrix} \tag{14.84}$$

and do the following calculations:

$$\begin{aligned} DI &= \begin{array}{c} \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \\ -2r_3 \\ -r_1 \end{array} \end{array} \\ &= \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & -2 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{bmatrix} \begin{array}{l} \\ \\ +2r_2 \end{array} \\ &= \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & -2 \\ 0 & 0 & -1 & -1 & 2 & -3 \end{bmatrix} \begin{array}{l} \\ -r_3 \\ \times(-1) \end{array} \\ &= \begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 & -2 & 3 \end{bmatrix} \begin{array}{l} -r_3 \\ \\ \\ \end{array} \\ &= \begin{bmatrix} 1 & 2 & 0 & 0 & 2 & -3 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 & -2 & 3 \end{bmatrix} \begin{array}{l} -2r_2 \\ \\ \\ \end{array} \\ &= \begin{bmatrix} 1 & 0 & 0 & -2 & 4 & -5 \\ 0 & 1 & 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 & -2 & 3 \end{bmatrix} \\ &= ID^{-1} \end{aligned} \tag{14.85}$$

The right part of the matrix in the second to last row is the inverse of D:

$$D^{-1} = \begin{bmatrix} -2 & 4 & -5 \\ 1 & -1 & 1 \\ 1 & -2 & 3 \end{bmatrix} \tag{14.86}$$

14.10 Linear equation systems in matrix

In chapter 7 we introduced linear equations in systems. Linear equation systems can be expressed and calculated using matrices. Let us take the following equation system as an example:

$$\begin{cases} x_1 - 2x_2 & = 3 \\ 4x_1 + 5x_2 & = 6 \end{cases} \tag{14.87}$$

where x_1 and x_2 are variables. We use x_1 and x_2 instead of for example x and y. We will now investigate whether the system has any solutions. We have previously gone through how we may use substitution, for example taking $x_1 = 3 + 2x_2$ from the upper equation and inserting into the lower:

$$\begin{aligned} 4\left(3 + 2x_2\right) + 5x_2 &= 6 \\ x_2^* &= -\frac{6}{13} \end{aligned} \tag{14.88}$$

We insert x_2^* into the first equation and solve for x_1^*:

$$\begin{aligned} x_1 - 2x_2 &= 3 \\ x_1 - 2\left(-\frac{6}{13}\right) &= 3 \\ x_1^* &= \frac{27}{13} \end{aligned} \tag{14.89}$$

Now we will instead use matrices. We begin by setting up matrices in such a way that the result looks the same as the system in equation (14.87) . We place the coefficients in a matrix that we call K:

$$K = \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix} \tag{14.90}$$

The variables x_1 and x_2 we place in a matrix with only one column, a column matrix, that we call X:

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{14.91}$$

When we now matrix multiply KX we get:

$$KX = \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 2x_2 \\ 4x_1 + 5x_2 \end{bmatrix} \tag{14.92}$$

Now we have an expression for the left side in equation (14.87) . We also need a column matrix for the two values that are to the right of the equals sign, the numbers 3 and 6. These we place in the column matrix Y:

$$Y = \begin{bmatrix} 3 \\ 6 \end{bmatrix} \tag{14.93}$$

Now we write the system in equation (14.87) in the form of matrices:

$$\begin{bmatrix}1 & -2\\ 4 & 5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}3\\ 6\end{bmatrix} \tag{14.94}$$
$$KX = Y$$

Now we will find solutions to x_1 and x_2 with the help of row reduction. Row reduction with matrices works in the same way as when we use row reduction for equation systems (section 7.3). We can add and subtract entire rows with each other as well as multiply individual rows with a constant. We begin by setting up matrix K and Y as one matrix and draw a vertical line between their values:

$$[K|Y] = \left[\begin{array}{cc|c} 1 & -2 & 3\\ 4 & 5 & 6\end{array}\right] \tag{14.95}$$

The vertical line separates the right and left sides in the equation (14.87) :

$$\begin{cases} x_1 - 2x_2 & = 3\\ 4x_1 + 5x_2 & = 6\end{cases} \xrightarrow{\text{becomes}} \left[\begin{array}{cc|c} 1 & -2 & 3\\ 4 & 5 & 6\end{array}\right] \tag{14.96}$$

We noted in section 14.5 that $I_2X = X$(equation (14.53)). With the help of row reduction we can rewrite KY so that we get an identity matrix I_2 to the left in matrix KY, whereupon the right column in KY will be the solutions x_1^* and x_2^*. We begin by multiplying row 1 by -4 and adding this to row 2:

$$\begin{aligned} KY &= \left[\begin{array}{cc|c} 1 & -2 & 3\\ 4 & 5 & 6\end{array}\right] {}_{-4r_1} \\ &= \left[\begin{array}{cc|c} 1 & -2 & 3\\ 0 & 13 & -6\end{array}\right]\end{aligned} \tag{14.97}$$

We divide row 2 by 13:

$$\begin{aligned} &\left[\begin{array}{cc|c} 1 & -2 & 3\\ 0 & 13 & -6\end{array}\right] \times \left(\tfrac{1}{13}\right) = \\ = &\left[\begin{array}{cc|c} 1 & -2 & 3\\ 0 & 1 & -\frac{6}{13}\end{array}\right]\end{aligned} \tag{14.98}$$

To row 1 we now add row 2 multiplied by 2:

$$\begin{aligned} &\left[\begin{array}{cc|c} 1 & -2 & 3\\ 0 & 1 & -\frac{6}{13}\end{array}\right]^{+2r_2} = \\ = &\left[\begin{array}{cc|c} 1 & 0 & \frac{27}{13}\\ 0 & 1 & -\frac{6}{13}\end{array}\right]\end{aligned} \tag{14.99}$$

The first two columns to the left are now a 2×2 identity matrix. We have gotten the same result as when we used substitution:

$$\begin{aligned} \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} &= \begin{bmatrix}\frac{27}{13}\\ -\frac{6}{13}\end{bmatrix} \\ \begin{bmatrix}x_1^*\\ x_2^*\end{bmatrix} &= \begin{bmatrix}\frac{27}{13}\\ -\frac{6}{13}\end{bmatrix} \\ X &= Y^*\end{aligned} \tag{14.100}$$

Let us now go through another method to reach the same result. We start from the expression $KX = Y$. If the inverse of K, which we call K^{-1}, exists we have that $KK^{-1} = I_2$(section 14.9). In that case we write:

$$\begin{aligned} KX &= Y \\ K^{-1}KX &= K^{-1}Y \\ X &= Y^* \end{aligned} \tag{14.101}$$

where Y^* contains the solutions x_1^* and x_2^*. Let us illustrate by calculating K^{-1}. In this case we have that the inverse of K is (see equation (14.78)):

$$K^{-1} = \frac{1}{5-(-8)}\begin{bmatrix} 5 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} \frac{5}{13} & \frac{2}{13} \\ -\frac{4}{13} & \frac{1}{13} \end{bmatrix} \tag{14.102}$$

Let us go through the calculation from $K^{-1}KX = K^{-1}Y$:

$$\begin{aligned} K^{-1}KX &= K^{-1}Y \\ \begin{bmatrix} \frac{5}{13} & \frac{2}{13} \\ -\frac{4}{13} & \frac{1}{13} \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} \frac{5}{13} & \frac{2}{13} \\ -\frac{4}{13} & \frac{1}{13} \end{bmatrix}\begin{bmatrix} 3 \\ 6 \end{bmatrix} \\ \begin{bmatrix} \frac{5}{13}*1+\frac{2}{13}*4 & \frac{5}{13}*(-2)+\frac{2}{13}*5 \\ -\frac{4}{13}*1+\frac{1}{13}*4 & -\frac{4}{13}*(-2)+\frac{1}{13}*5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} \frac{5}{13}*3+\frac{2}{13}*6 \\ -\frac{4}{13}*3+\frac{1}{13}*6 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 2+\frac{1}{13} \\ -\frac{6}{13} \end{bmatrix} \\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} \frac{27}{13} \\ -\frac{6}{13} \end{bmatrix} \\ X &= Y^* \end{aligned} \tag{14.103}$$

We can also use an inverse matrix in another way. The expression $KX = Y$ can also be rewritten by transposing both sides:

$$\begin{aligned} (KX)^T &= Y^T \\ X^T K^T &= Y^T \\ \begin{bmatrix} x_1 & x_2 \end{bmatrix}\begin{bmatrix} 1 & 4 \\ -2 & 5 \end{bmatrix} &= \begin{bmatrix} 3 & 6 \end{bmatrix} \end{aligned} \tag{14.104}$$

where X^T, K^T and Y^T are the transposed versions of the matrices. This equation can also be solved using an inverse matrix but the multiplication needs to be done in the following way this time:

$$\begin{aligned} X^T K^T &= Y^T \\ X^T K^T \left(K^T\right)^{-1} &= Y^T \left(K^T\right)^{-1} \\ X^T I_2 &= Y^T \left(K^T\right)^{-1} \\ X^T &= (Y^*)^T \end{aligned} \tag{14.105}$$

$\left(K^T\right)^{-1}$ is the inverse of the transposed version of matrix K, and $(Y^*)^T$ is the transposed version of matrix Y^* which contains the solutions to the equation system. Note also in this case how the order of the matrices determines how we can use matrix multiplication $\left(K^T\right)^{-1}$:

$$\begin{aligned} X^T &= Y^T \left(K^T\right)^{-1} \\ \begin{bmatrix} x_1 & x_2 \end{bmatrix} &= \begin{bmatrix} 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ -2 & 5 \end{bmatrix}^{-1} \\ \begin{bmatrix} x_1^* & x_2^* \end{bmatrix} &= \begin{bmatrix} \frac{27}{13} & -\frac{6}{13} \end{bmatrix} \end{aligned} \tag{14.106}$$

We have once again solved the equation system and gotten the same answer as above.

14.11 Example 1: A bakery

This section and the following two describes simplified examples of how we can use linear algebra to describe real world phenomenons. In later parts we will use vector and matrices for empirical analysis.

Consider a small bakery that makes three products: bread, pastries, and cakes. Each product requires different ingredients:

- Bread needs: 2 kg flour, 0 kg sugar, 1 kg yeast
- Pastries need: 1 kg flour, 3 kg sugar, 0 kg yeast
- Cakes need: 1 kg flour, 4 kg sugar, 1 kg yeast

We organize this information in table 14.1 . The numbers in the table we put as elements in a matrix R(recipes).

Table 14.1: Bakery recipes

	Flour (kg)	Sugar (kg)	Yeast (kg)
Bread	2	0	1
Pastries	1	3	0
Cakes	1	4	1

$$R = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 1 & 4 & 1 \end{bmatrix} \tag{14.107}$$

Now the bakery wants to make 10 breads, 5 pastries, and 3 cakes and we wish to calculate the resources needed. For this we can use matrix multiplication to find total ingredient. First we define the production vector P:

$$P = \begin{bmatrix} 10 & 5 & 3 \end{bmatrix} \tag{14.108}$$

To calculate how much ingredients the bakery need in total for this production we can use matrix multiplication PR. The new matrix will have the same number of rows as the first matrix R(3 rows) and the same number of columns as the second matrix in the multiplication P(1 column), see section 14.4 :

$$PR = \begin{bmatrix} 10 & 5 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 1 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 10*2+5*1+3*1 \\ 10*0+5*3+3*4 \\ 10*1+5*0+3*1 \end{bmatrix} = \begin{bmatrix} 28 \\ 27 \\ 13 \end{bmatrix} \tag{14.109}$$

This is the amounts needed of flour (28 kg), sugar (27 kg) and yeast (13 kg).

14.12 Example 2: Input-output analysis

This section describes another example of how we can use linear algebra in analytical work, by describing how firms in different industries trade with each other. For this we can use what is called input-output analysis (IO). This is based on an input-output table, IO table, where each industry has its own row and its own column. Each row describes the value of everything that companies in an industry sell to other companies in the same industry or other industries. Each column describes the production that is purchased and delivered to each industry respectively. Companies in each industry can trade both with companies in the same industry and companies in other industries.

Table 14.2 shows an example of an IO table with the two industries A and B. Each industry has its own row and its own column. The numbers in the table show the value of the production that is delivered to each industry as well as to final consumption by households and the public sector. In the row for industry A, the numbers show the value of the production that A sells to other companies in the same industry (the value 2) and to companies in industry B (the value 1). The fourth column describes the amount of production for final consumption while the fifth column sums columns 2–4.

Table 14.2: Input-output table

	Industry A	Industry B	Final consumption	Sum
Industry A	2	1	3	6
Industry B	1	3	3	7
Sum	3	4	6	13

The data in table 14.2 we use to calculate how changed production in one part of the economy affects other parts. We begin by placing the table's values in matrices. We collect the flows to and from the industries in a 2×2 matrix Z(the flow matrix), final consumption in column matrix C and the sum in the last column in column matrix S:

$$Z = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad S = \begin{bmatrix} 6 \\ 7 \end{bmatrix} \tag{14.110}$$

The flows to and from industries (Z) we divide by each industry's total production (S):

$$a_{ij} = \frac{y_{ij}}{y_j} \tag{14.111}$$

where y_{ij} is production from industry i that is delivered to industry j, while y_j is the total production that industry j produces. For each value in matrix Z we calculate a value a_{ij} and collect in a 2×2 matrix A.

To do this calculation with matrices we first calculate the multiplicative inverse (section 14.9) of each element in S, that is place the values as denominators in separate fractions with numerator 1. Industry A gets $\frac{1}{6}$ and industry B $\frac{1}{7}$. We collect these values in a column matrix S_m and place in a diagonal matrix, which we describe as $\operatorname{diag}(S_m)$:

$$\operatorname{diag}(S_m) = \begin{bmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{7} \end{bmatrix} \tag{14.112}$$

To calculate A we now matrix multiply Z with $\operatorname{diag}(S_m)$:

$$\begin{aligned} A &= Z * \operatorname{diag}(S_m) \\ &= \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{7} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{6} & \frac{1}{7} \\ \frac{1}{6} & \frac{3}{7} \end{bmatrix} \end{aligned} \tag{14.113}$$

For each industry j we define total production y_j as the sum of input production to all industries plus production to final consumption, which in this case becomes:

$$y_j = a_{j1}y_1 + a_{j2}y_2 + c_j \tag{14.114}$$

where a are the weights we defined in equation (14.111) , y_1 is the total production from industry 1 and y_2 is total production from industry 2, or industry A and B as we called them. We do not know the values for y_j yet but we can calculate it using the matrices. We collect the elements of total production in column matrix Y and can now describe Y as:

$$\begin{aligned} Y &= AY + C \\ Y - AY &= C \\ IY - AY &= C \\ (I - A)\,Y &= C \\ Y &= (I - A)^{-1}\,C \end{aligned} \tag{14.115}$$

Identity matrix I has the same dimensions as A. Column matrix Y describes how much production is required from each industry to companies in the same and other industries as well as to final consumption:

$$Y = (I - A)^{-1}\,S = \begin{bmatrix} \frac{8}{5} & \frac{2}{5} \\ \frac{7}{15} & \frac{28}{15} \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} \approx \begin{bmatrix} 12,4 \\ 15,9 \end{bmatrix} \tag{14.116}$$

This shows how much industry A and B must produce in total for input goods and final consumption. We call the inverted matrix $B = (I - A)^{-1}$. Matrix B has the same dimensions as matrix A. Let us calculate B with the values from table 14.2 :

$$\begin{aligned} B &= (I - A)^{-1} \\ &= \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} \frac{2}{6} & \frac{1}{7} \\ \frac{1}{6} & \frac{3}{7} \end{bmatrix} \right)^{-1} \\ &= \begin{bmatrix} \frac{4}{6} & -\frac{1}{7} \\ -\frac{1}{6} & \frac{4}{7} \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \frac{8}{5} & \frac{2}{5} \\ \frac{7}{15} & \frac{28}{15} \end{bmatrix} \end{aligned} \tag{14.117}$$

If we now in matrix B calculate the sum of a column for an industry j we get the total amount of production for all industries that would be required if the final consumption for industry j were to increase by one unit. This shows how much other industries must deliver in input goods to industry j. This measure is called production multiplier in this context.

If we for matrix B calculate the sum of a row for an industry j in matrix B we get a measure of how much extra production in total in society that is enabled if industry j increases its production of input goods by one unit. This is called input multiplier.

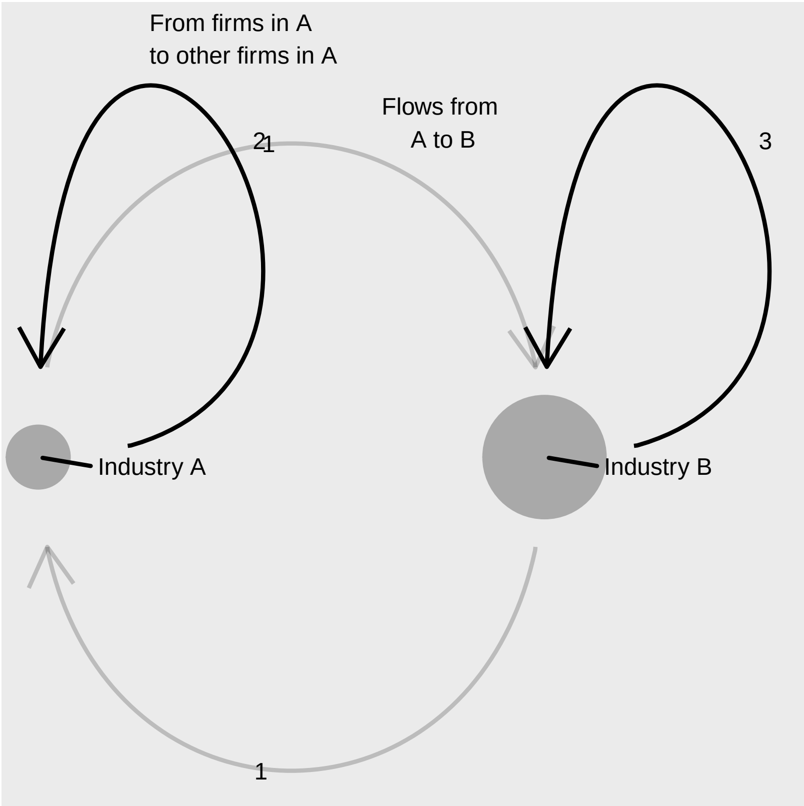


Figure 14.2: A network with the two industries A and B

Figure 14.2 shows an example of how this can be illustrated in a network diagram. The points describe the two industries A and B and the arrows describe the flows to and from the two industries. The values at each arrow represent the values in table 14.2 . The size of the points represents each industry's total production, the rightmost column in the table. In the diagram we have not included any illustration of the production that the industries send to final consumption.

In the same way as the results we calculated above, we can also add other information to our matrices and calculate other mechanisms. Input-output analysis is used among other things to describe how changes in one industry can affect employment in other industries or to describe industries' climate impact. These methods have a long history but began to spread especially within social science during the 1920s, when among others the economist Wassily Leontief published texts about this.

14.13 Example 3: Social networks

Linear algebra can also be used to describe other types of networks and flows between actors, for example a social network of friends. Table 14.3 describes a network of five friends in the form of a matrix with five rows and five columns. The elements in the matrix describe the number of messages that the people send to each other on average per day. Maria for example sends 5 messages to Nushi. Nushi sends 2 messages to Maria.

We can also illustrate this network in a network diagram, which is shown with an example in figure 14.3 . Each arrow represents that communication has occurred between the people. The thickness of the arrows illustrates how many messages have been sent in this direction. The thicker the line, the more messages have been sent.

Table 14.3: Matrix with average number of messages sent per day

	Maria	**Nushi**	**Mohammed**	**Jose**	**Sum**
Maria	0	5	1	3	9
Nushi	2	0	2	6	10
Mohammed	3	1	0	1	4
Jose	1	7	0	0	8
Sum	5	13	3	10	31

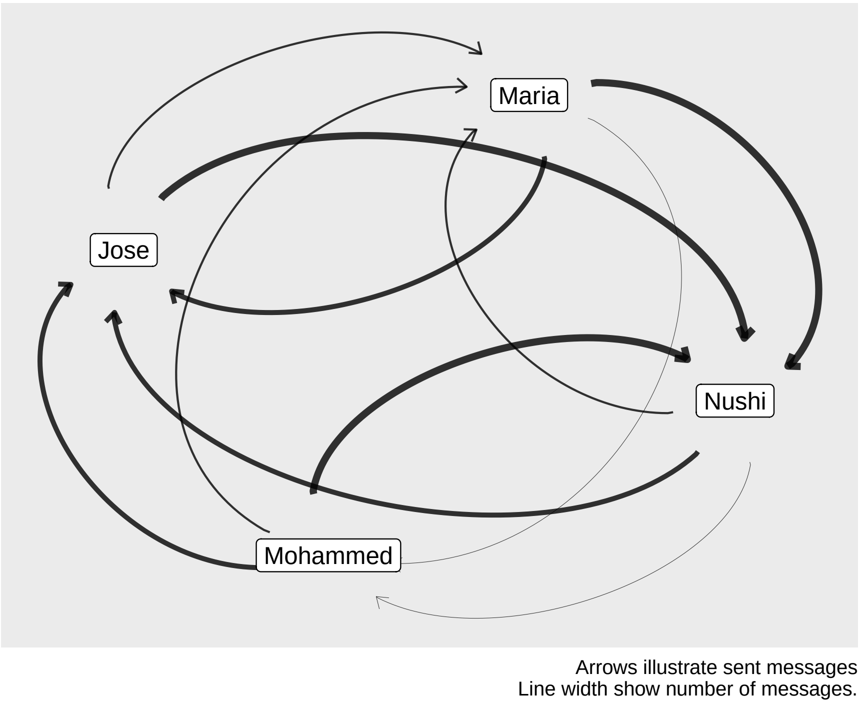


Figure 14.3: Illustration of number of sent messages

In network analyses, one sometimes wants to use measures to calculate how important or influential actors in the network are. A basic measure of this is degree centrality, which in this case is a measure of how many connections each node, person or industry, has to some other node in the network. The concept of centrality also occurs in statistics but then means something else. Degree centrality for a node j is the sum of the number of contacts and can be described with the function:

$$c_j^D(a) = \sum_i^n a_{ji} \tag{14.118}$$

We call degree centrality c^D. In this equation a_{ji} represents whether there is any contact from node j to another node i. Nodes are then for example individual persons, industries or actors, depending on what we study. The letter n symbolizes the amount of nodes in the network. For our social network everyone has sent messages to everyone else, except Jose who has not sent anything to Mohammed. Mohammed thus has $c^D = 3$ while the other three actors have $c^D = 4$.

Often we want to adjust our measures for the maximum number of possible connections. In this case we adjust degree centrality to take into account the maximum number of possible connections in the network by dividing by $n-1$:

$$c_j^D(a) = \frac{\sum_i^n a_{ji}}{n-1} \tag{14.119}$$

Formalized network analysis has existed at least since the 1700s. An early example is a famous text by Leonhard Euler about the seven bridges in Königsberg in Prussia, present-day Kaliningrad in Russia (Euler, 1741). Euler's article laid the foundation for the part of mathematics called graph theory. Leonard Euler is by the way the same Euler as in Euler's number, e.

14.14 Chapter summary

- A mathematical quantity with one value is called scalar. A vector is a quantity with multiple values, for example a person's weight and height. A vector can contain a large amount of values. The values in a vector can be described as variables, dimensions or elements.
- Vectors with the same number of dimensions can be added or subtracted. Example: the vectors $\vec{a} = (a_1, a_2)$ and $\vec{b} = (b_1, b_2)$ give $\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2)$. Multiplication of vector by scalar: $s * \vec{a} = s\,(a_1, a_2) = (sa_1, sa_2)$ where s is any number.
- Multiplication of two vectors results in a scalar, for example $\vec{a}*\vec{b} = (a_1, a_2)*(b_1, b_2) = a_1b_1+a_2b_2 = s$ where s is any number. A vector's norm, length, can be calculated as $\|\vec{a}\| = \sqrt{a_1^2 + a_2^2}$.
- A matrix is a collection of values organized as a rectangle, with columns and rows. Dimensions for a matrix are given with rows $\times$ columns or $r \times k$. Example 2×3 matrix A has 2 rows and 3 columns. The values in a matrix are called elements. Element a_{12} is the value at row 1 column 2. A matrix where $r = k$ is square.
- Matrices with the same shape can be added and subtracted, whereby the value at position rk is added or subtracted with the value in the other matrix at the corresponding position. Example: we have 2×2 matrices A and B where the elements are denoted a_{rk} and b_{rk} respectively. $A + B$ means $a_{11} + b_{11}$ and so on.
- Multiplication of matrix by scalar means that all elements are multiplied by the scalar:

$$sA = \begin{bmatrix} sa_{11} & sa_{12} \\ sa_{21} & sa_{22} \end{bmatrix}$$

.

- Matrix multiplication: the rows in the left matrix are multiplied by the columns in the right matrix. Matrix multiplication of matrices A and B is defined for AB given that number of rows in A = number of columns in B. If we switch places of the matrices we do not necessarily get the same result, that is $AB \neq BA$.

-A^T or A': transposed matrix A, where $a_{rk}^T = a_{kr}$. Generally: $\left(A^T\right)^T = A$ and $(AB)^T = B^T A^T$. Transposition does not affect scalars: $(sA)^T = sA^T$, where s is scalar.

- A matrix where all elements outside the matrix's diagonal are 0 is called diagonal matrix. The elements in the diagonal in a matrix A are those elements a_{rk} where $r = k$, that is $a_{11}, a_{22}, ..., a_{nn}$, where n is any number.
- A diagonal matrix where all values on the diagonal are 1 is called identity matrix or unit matrix. A square matrix where only the elements on one side of the diagonal are equal to 0 is called triangular matrix.
- The determinant of matrix A is denoted $det\,(A)$. Example: matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

. If $c = 0$ or $b = 0$ then $\det(A) = ad$. For a triangular matrix the determinant is equal to the product of the diagonal's elements.

- We can use row reduction in matrices in the same way as for linear equation systems, among other things to edit a matrix until it becomes triangular.

- The inverse of matrix A, the inverse matrix, is written A^{-1}. The inverse matrix can be calculated with Gauss-Jordan elimination: If A has dimensions $r \times k$ we add an $r \times k$ identity matrix I_r, so that we get AI. Edit AI until the right part becomes an identity matrix, which gives IA^{-1}, where the right half is the inverse matrix we seek.
- Vectors and matrices can be used to describe and study networks and flows between actors, for example goods and services between different industries or contacts in a social network.

14.15 Exercises

1. Let $\vec{v}_1 = (2, 3, 1)$ and $\vec{v}_2 = (4, 1, 5)$.

(a) Calculate (vec{v}*{1}+vec{v}*{2}) .

(b) Calculate (vec{v}*{1}-vec{v}*{2}) .

(c) Calculate the scalar product (vec{v}_{1}*vec{v}_{2}) .

Answer: (a) $(6, 4, 6)$ (b) $(-2, 2, -4)$ (c) $2 \cdot 4 + 3 \cdot 1 + 1 \cdot 5 = 16$

2. Let $\vec{a} = (3, 4)$ and $\vec{b} = (1, -2)$.

(a) Calculate (5vec{a}) .

(b) Calculate the norm (leftVert vec{a}rightVert) .

(c) Calculate the scalar product (vec{a}*vec{b}) .

Answer: (a) $(15, 20)$ (b) $\|\vec{a}\| = \sqrt{9+16} = 5$ (c) $3 \cdot 1 + 4 \cdot (-2) = -5$

3. Let

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}$$

.

(a) Calculate (A+B) .

(b) Calculate (A-B) .

(c) Calculate (3A) .

Answer: (a)

$$\begin{bmatrix} 7 & 3 \\ 4 & 4 \end{bmatrix}$$

(b)

$$\begin{bmatrix} -3 & -1 \\ 2 & 4 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 6 & 3 \\ 9 & 12 \end{bmatrix}$$

4. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

.

(a) Calculate (AB) .

(b) Calculate (BA) .

(c) Is (AB=BA) ? What does this illustrate?

Answer: (a)

$$AB = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}$$

(b)

$$BA = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix}$$

(c) $AB \neq BA$

5. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

.

(a) Write out the transpose (A^{T}) .

(b) What are the dimensions of (A) and (A^{T}) ?

(c) Verify that (left(A$^{\{T\}right)}${T}=A) .

Answer: (a)

$$A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$

(b) A (c) Transposing A^T

6. Let

$$D = \begin{bmatrix} 4 & 0 \\ 0 & 7 \end{bmatrix}$$

and

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}$$

.

(a) Write out the (3times3) identity matrix (I_{3}) .

(b) Verify that (AI_{2}=A) by computing the matrix product.

(c) Calculate the trace (text{tr}(D)) .

Answer: (a)

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)

$$AI_2 = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix} = A$$

(c) $\text{tr}(D) = 4 + 7 = 11$

7. Let

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

.

(a) Calculate the determinant (det(A)) .

(b) Find the inverse matrix (A^{-1}) .

(c) Verify that (Acdot A^{-1}=I_{2}) by computing the product.

Answer: (a) $\det(A) = 3 \cdot 4 - 2 \cdot 1 = 10$ (b)

$$A^{-1} = \frac{1}{10}\begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

(c)

$$A \cdot A^{-1} = \frac{1}{10}\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = I_2$$

8. Let

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$$

.

(a) Calculate ((A+B)^{T}) .

(b) Verify that ((A+B)$^{\{T\}=A}$\{T\}+B^{T}) .

(c) Calculate (AB) and verify that ((AB)$^{\{T\}=B}$\{T\}A^{T}) .

Answer: (a)

$$A + B = \begin{bmatrix} 1 & 4 \\ 3 & 3 \end{bmatrix}$$

(b)

$$A^T + B^T = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 4 & 3 \end{bmatrix}$$

(c)

$$AB = \begin{bmatrix} 3 & 7 \\ 1 & 4 \end{bmatrix}$$

9. Solve the system of equations using the inverse matrix: $2x + y = 5$ and $x + 3y = 10$.

(a) Write the system in matrix form (Amathbf{x}=mathbf{b}) .

(b) Calculate (det(A)) .

(c) Find (A^{-1}) and solve for (mathbf{x}=A^{-1}mathbf{b}) .

Answer: (a)

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

(b) $\det(A) = 6 - 1 = 5$ (c)

$$A^{-1} = \frac{1}{5}\begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

10. Let $\vec{u} = (3, 1)$ and $\vec{v} = (2, 4)$.

(a) Calculate the scalar product (vec{u}cdotvec{v}) .

(b) Calculate the projection of (vec{u}) onto (vec{v}) : (text{proj}_{vec{v}}(vec{u})=frac{vec{u}cdotvec{v}}{leftVert vec{v}rightVert ^{2}}vec{v}) .

(c) Interpret: what does this projection represent geometrically?

Answer: (a) $\vec{u} \cdot \vec{v} = 6 + 4 = 10$ (b) $\|\vec{v}\|^2 = 4 + 16 = 20$ (c) The projection is the component of $\vec{u}$

11. A market equilibrium model gives the conditions $4x_1 - x_2 = 10$ and $-2x_1 + 3x_2 = 5$.

(a) Write the system in matrix form (Amathbf{x}=mathbf{b}) .

(b) Calculate (det(A)) .

(c) Find the equilibrium quantities (x_{1}) and (x_{2}) using (A^{-1}) .

Answer: (a)

$$A = \begin{bmatrix} 4 & -1 \\ -2 & 3 \end{bmatrix}$$

(b) $\det(A) = 12 - 2 = 10$ (c)

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$$

12. Let

$$W = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

be a weight matrix and $\vec{v} = (2, 1, 4)$ a score vector.

(a) Calculate the weighted score vector (Wvec{v}) .

(b) Calculate the trace (text{tr}(W)) .

(c) Calculate the norm (leftVert vec{v}rightVert) .

Answer: (a) $W\vec{v} = (6, 5, 8)^T$ (b) $\text{tr}(W) = 3 + 5 + 2 = 10$ (c) $\|\vec{v}\| = \sqrt{4 + 1 + 16} = \sqrt{21} \approx 4.58$

Part II

Data, Statistics and Probability

Part II: Data, Statistics and Probability

This part introduces the foundations of statistical analysis. We begin with how data is collected and organised, and how to describe variation using basic statistical measures. We then cover probability theory — discrete and continuous distributions, the normal distribution, and statistical inference — which provides the mathematical tools needed to evaluate whether patterns in data could have arisen by chance.

Chapter 15

Data and Descriptive Statistics

This chapter introduces fundamental concepts for analytical work: how we collect and organise data, how populations and samples relate to each other, and how to describe variation in data using basic statistical measures.

15.1 Population and Sample

A decisive question in analytical work is how we delimit our problem and how large a part of the world and world history we want to be able to make statements about. These phenomena can be described with the help of the concepts population, sample and superpopulation.

Population refers in statistics to the collection of units, observations, that we study. The population can consist of both a finite or an infinite amount of values. Examples of a finite population could be all cars in a city, all citizens in a country or all heart operations performed at a hospital. The population can also be infinite, for example if we conduct repeated equivalent experiments we can in theory do this an infinite number of times.

Sample is the observations we take from the population. Our hope is that the patterns we observe in the sample are the same patterns that are found in the entire population. The goal is therefore that the sample should be more or less representative of the population. Samples can be created in different ways, which is described a bit more thoroughly in section 15.4 . When we work with samples, which is normally the case within social science, there is always a certain risk and probability that our results are incorrect and could just as well have arisen by chance. In Part III we introduce how we can reason about and calculate probability and chance. Work with data under uncertainty is called statistical inference (statistical reasoning). To study causal relationships (causality) is sometimes called causal inference. The word inference means reasoning, to draw conclusions.

Superpopulation is the theoretical population that we hope to be able to make statements about beyond the study's sample and population. Suppose we perform an analysis on a sample of observations that we take from a population. Often we want to be able to make statements both about the population that exists today and different conceivable populations that may exist in the future. In certain cases we perform analyses to be able to predict what will happen if something changes, that is, to be able to predict different conceivable outcomes. These hypothetical observations belong to what is called the superpopulation.

In section 4.7 we introduced set theory. If we imagine that we have a study where the population consists of a collection of observations that we number from 1 to N and each observation is called x_i where $i = 1, 2, 3, ..., N$. The letter N symbolizes number of observations, regardless of whether it is a small or large value. It is central to remember here that when we study observations from our population there is for each observation x_i from the population, an observable version $(x_i^{\text{observable}})$ that we study, and a non-observable one that we will never see $(x_i^{\text{non-observable}})$. This applies to both the treatment and control groups.

Suppose we conduct a study regarding the effect of a medicine on patients with a certain disease, where the population is all patients with the disease. We collect all N observations from the population and divide these into treatment group (receives medicine) and control group (does not receive medicine). The observations in the treatment group we call $x_{i,B}$ and those in the control group $x_{i,K}$. Regardless of which group each observation belongs to, we work only with what we observe and each observation in each group will have a non-observable, counterfactual, version.

Sometimes it is difficult to exactly define what is a study's exact population and superpopulation. If we study what effect a medicine has on patients' disease symptoms, the population can for example consist of all patients in the entire world with this disease. From these patients we take a sample, for example 100 patients, which we divide into treatment and control groups. By studying this sample, our ambition is to estimate the covariation between medicine and symptoms in the population. We thereby want to be able to make statements about the effect the medicine has in the sample, which is an estimate of the effect the medicine has in the population. We thereby also try to estimate the effects of the medicine in the superpopulation, which in this case is all comparable patients that can be thought to come to exist in the future. Since people are different, it is not always so obvious how general our results are and how far-reaching conclusions are possible.

Suppose we instead are interested in what effect an education has on people's future earnings. The population would in this case be able to be all persons who can be thought to participate in the education, for example all adult people in the ages 18 to 64 years. Or maybe we are only interested in people with certain specific prerequisites. Depending on what is the study's purpose and ambition, it can affect how we define the population.

Even superpopulation can be difficult to define. Suppose we find that the education has a positive impact on the participants' future earnings. If a large number of people studied the education, there would be masses of people with the same education. Then the education might not have the same effect compared to a situation where only a small proportion of the population had studied the education.

15.2 Information and variables

Within analytical work, information about reality is often called data. Data is just another word for information and can consist of anything that we can perceive with our senses, for example numbers, text, images, sound recordings. All analytical work uses information about reality in some form or extent. All data must in turn be interpreted. For example, much analytical work consists of mapping events: What happened where and when? Exactly what type of information is required for an analysis depends on what is the study's purpose. To formulate the study's purpose can sometimes require detailed deliberations.

Social scientific analysis uses information about reality in different ways. When we work with quantitative methods, for example studying how different phenomena vary, we often benefit from categorizing information in data variables. A data variable can contain any type of information, for example street numbers, names of children born in April, the gross national product for different countries or all individual words in a novel organized with one word per row. To use a data variable for calculations, this variable needs to consist of numbers or be rewritten into numbers.

When we define variables, it is important that all information that we collect into a variable has the same properties. If we for example collect information about inhabitants' life expectancy and income, we should place this information in two different variables: one variable for life expectancy and one variable for income. If we mix information in a variable, for example life expectancy and income, it becomes impossible to do calculations on the variable. Within one and the same variable, data should use the same unit of measurement. We do not want to have one observation in centimeters and another in inches in the same column.

A common way to organize data is tables, where each column represents a variable and each row an observation. In the same way that we have clear categories for our variables in the columns, we need to have clear categories for our observations. Exactly what represents an observation depends on what the data describes. It can for example be one observation per individual or per country. Often we also need to delimit what time point an observation refers to, for example an observation can contain information about a person per week. The next row can in that case contain the next observation, for example information about the same person the week after. What the data observations describe is called unit of observation.

Table 15.1: Three variables with four observations

	Variable 1 (y)	Variable 2 (x)	Variable 3 (z)
Observation 1	y_1	x_1	z_1
Observation 2	y_2	x_2	z_2
Observation 3	y_3	x_3	z_3
Observation 4	y_4	x_4	z_4

Table 15.1 shows an example where we have three data variables and call these x, y and z. We number the observations y_1 to y_4, x_1 to x_4 as well as z_1 to z_4. There are no rules for how variables should be named or numbered. It also occurs that variables are called by the same letter and distinguished through numbering, for example the variables x_1, x_2 and so on.

15.3 Different types of data

Data that is used for calculations is usually divided into different data types: ratio data, interval data, ordinal data and nominal data. One way to think about this is that we have a variable, x, with some form of information about reality. The division is then based on the following criteria:

- Ranking: Can the observations in x be ranked?
- Equidistance: Is the distance between two values in x equally large?
- Absolute zero point: Is the variable defined in such a way that there is a value that is the smallest conceivable?

Depending on the answers to these three questions, we may define our variable as one of the four data types. The three criteria and the four data types are summarized in table 15.2 .

Table 15.2: Four types of data

	Ranking	Equidistance	Absolute zero point
Nominal data	No	No	No
Ordinal data	Yes	No	No
Interval data	Yes	Yes	No
Ratio data	Yes	Yes	Yes

Let us go through the four data types and give some examples. Nominal data is not ranked, has no clear distance between each scale step and no absolute zero point. Typical examples of this are categorical variables, for example languages, names of countries, gender. Since the information in this type of data mostly concerns grouping or categorizing, the possibility to use mathematical operations is limited to primarily defining whether an observation belongs or does not belong to one category or another.

Ordinal data has ranking but not equidistance or absolute zero point. A typical example is school grades, or placement in a list or queue, for example place 1, 2, 3 and so on. Another example can be survey responses where respondents are asked to rank alternatives from best to worst. In addition to categorizing, this data type can also use mathematical operations for inequality, for example that position 1 comes before position 2, which comes before position 3. It is thereby possible to calculate the median value for this type of data, but not for example the mean, see section 2.13 .

Interval data allows calculations of difference and addition and subtraction. However not ratios. We therefore cannot in a meaningful way compare the double value of interval data. Typical examples include temperatures in for example degrees Celsius. The difference between –2 and +3 degrees Celsius is the same as +5 and +10 degrees Celsius. But we cannot in the same way calculate these distances as ratios, since 0 degrees Celsius is just a marker for the temperature when water freezes.

Ratio data allows the same mathematical methods as the other three data types, including the calculations of ratios. This includes many common measures within physics and technology, such as for example length, time and mass. Unlike the other data types, it is possible to use division for ratio data and compare relative distances, such as that 10 meters is twice as long as 5 meters.

This division into four types of data is commonly occurring. Other more detailed divisions can also occur. To think about data in these forms can help us to sort information and see what possibilities and limitations that precisely our data offer. This should however primarily be seen as a tool, rather than a rule system.

15.4 Collecting a sample

To understand the idea behind sampling of data, we will in this section briefly go through some overarching examples of how this can happen. First the population must be defined. A popular method is to then take a simple random sample (SRS). As the name suggests, this is based on letting all observations in the population have the same theoretical possibility to be included in the sample. The probability for each individual observation must also be above 0. The observations that are to be included in the sample are chosen randomly, for example through lottery or with the help of a computer's random generator. Say for example that we want to know what all of the adult inhabitants think about Jesus. We may then randomly sample 100 persons and ask them.

Another method is systematic sampling: we give each person a number. Instead of drawing numbers randomly, we select every 1,000th person. This also gives a random sample as long as the numbered list of the population does not contain any patterns.

There are several commonly occurring methods that are not random. One such method is quota sampling, or quota-based sampling. This method is based on that we know of some characteristics in the population that are important for the study. We therefore weight the sample based on these characteristics to create a sample that is more representative of the population. For example, we are interested in what the population of a country thinks about Jesus and think that the population's proximity to the sea is important for explaining variations in faith. We estimate that 40% of the population lives near the sea and therefore control our sample so that 40% of the sample consists of residents in the vicinity of the sea. A problem with this method is that many factors affect people's view of Jesus and if our sampling method is deficient, it risks worsening the result.

To be sure that different parts of the population are represented in our sample, we may use stratified sampling. The population is divided into different groups, strata. From each stratum, observations are chosen randomly. The number of observations does not need to be the same proportion in the sample as in the population, as long as we adjust our calculations for this. Suppose we want to investigate the views on Jesus among the population for a country. In our study we want to make sure that we get enough inhabitants from the northern inland of the country. By doing so we make sure that it is possible to make more precise statements about this specific group. In the northern inland lives about five percent of the population. If we randomly select observations from all people in the country, the risk is that we get a correct proportion of respondents from the northern inland, but these observations will at the same time be too few to make any more detailed observations about the people of the northern inland. For instance, how much does opinions differ among parents and children in the northern inland. To get more reliable results, one may collect extra many observations from a group that actually constitutes a smaller proportion of the full population of the country. When we then calculate on our data material, we must take into account the bias in the sample that we ourselves have created, so we do not risk getting an incorrect picture of the population.

Another method is cluster sampling, also called group sampling. The full population is divided into clusters, groups, and some of these clusters are selected randomly. Thereafter, random samples can be made from each group. For example, we are to ask the students at a school what they think about Jesus. The population is all students at the school. We divide the students into clusters depending on which school class they belong to. The school classes that are to be included in the study are chosen randomly.

Collecting information usually entails several challenges, such as how to best measure a phenomenon, whether the sources are reliable, whether our information can be verified or whether we succeed in interpreting it correctly. These are important aspects of analytical work but there is no room to go through them here.

15.5 Frequency Distribution

To study covariation between variables, observations must have different values; there must be variation within the variables. If we, for example, have ten observations for the variables x and y and all observations have the values $(x, y) = (5, 23)$, then there is no variation within variable x or y. The variables therefore cannot covary either.

One way to study the spread of values is frequency distribution. Suppose we have the following four values: 4, 5, 5 and 6, where the number 5 occurs twice. One way to show the spread in a collection of values is a bar graph, a graph where the frequency of each respective value is reported with bars. Figure 15.1 shows the frequency distribution for the four values.

Now we have a variable x that has the following four values: 3, 4, 6 and 7. Figure 15.2 illustrates this variable in two different bar graphs. In the left graph this is illustrated with a bar of height 1 per value. The right graph shows a so-called histogram. A histogram is a form of bar graph where each bar represents an interval of values. In the graph to the right, the left bar represents the three values 3, 4 and 6. On the vertical y-axis, this bar therefore reaches up to the value 3, because the bar represents three observations. The right bar in the right graph we have one observation, which represents the value 7.

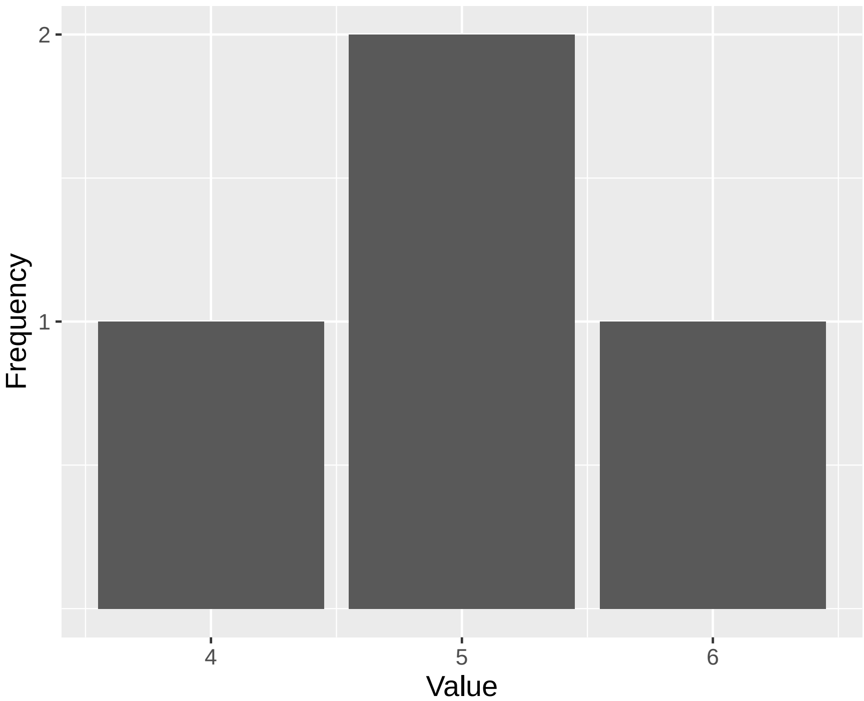


Figure 15.1: Frequency distribution for the values $4, 5, 5, 6$

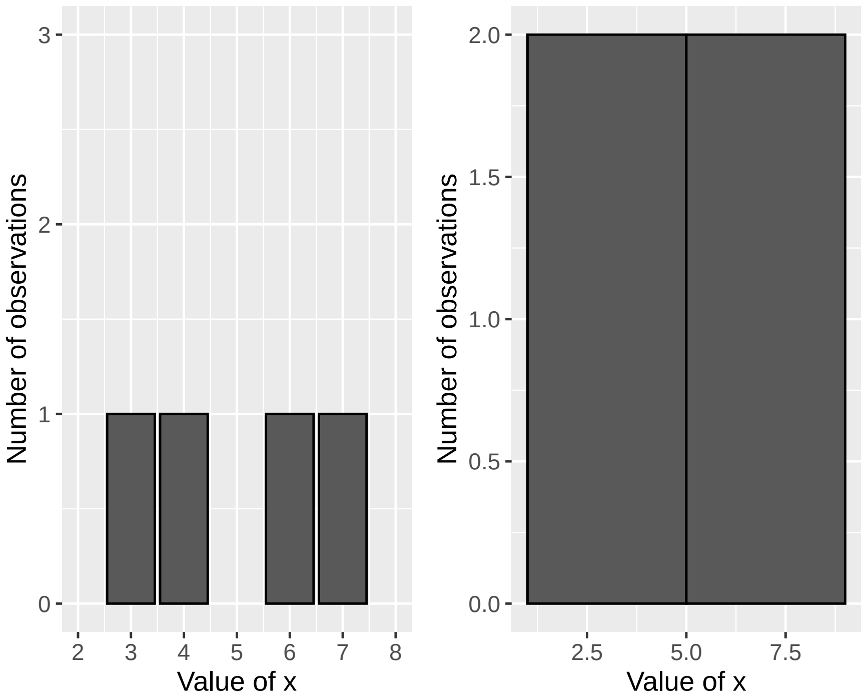


Figure 15.2: Two illustrations of a frequency distribution

Bar graphs and histograms are often used to describe the spread in data. Consider another example with more observations. Figure 15.3 shows a histogram that illustrates the distribution of average life expectancy for men and women in 290 countries and regions in the world, with one average value per gender and country/region. There are 290 observations for men and 290 observations for women. The horizontal x-axis shows the values for expected age, while the number of observations (countries or regions) is shown on the vertical y-axis.

In the graph we see that women on average live longer than men, since the bars for women are located more to the right in the graph. The bar furthest to the left and the right in the graph illustrates that there is a few places in the world where men and women, on average, tend to live shorter as well as longer lives, compared to the other places.

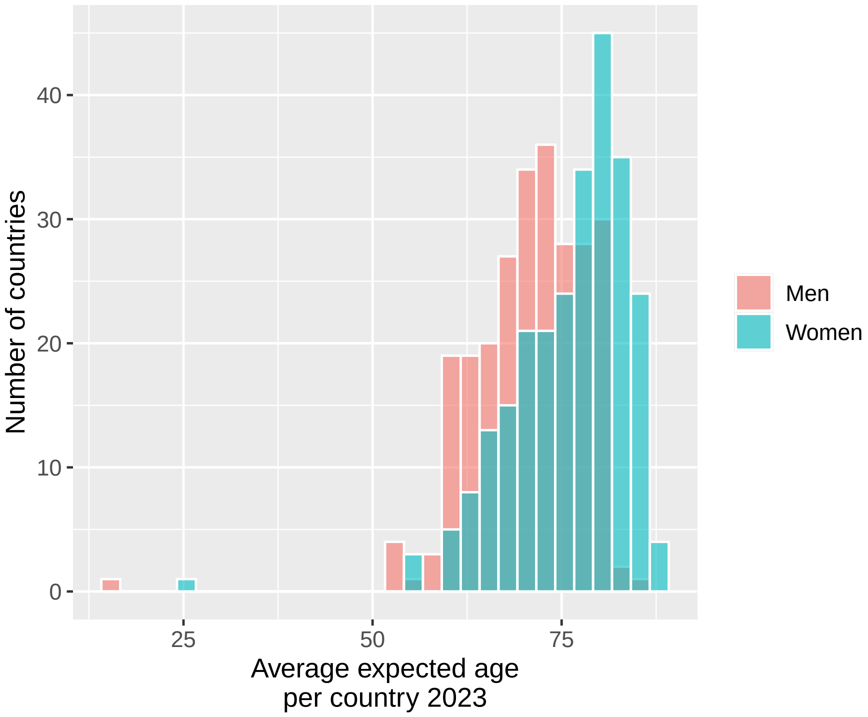


Figure 15.3: Average expected life in countries and regions in the world. Data from Our World In Data.

15.6 Measures of Central Tendency and Measures of Dispersion

For describing what constitutes an average value for a collection of values, we use what are called measures of central tendency. Two examples of measures of central tendency are mean (sum divided by number) and median (the middle value). We introduced both of these measures in section 2.13 . Another useful measure of central tendency is mode, which is the value that occurs most frequently. Consider the following fifteen numbers as an example:

1	1	1	4	5	6	6	6	7	8	8	9	9	9	9

For this collection of values, the mode = 9 since the number 9 occurs most frequently. The median for these numbers is the value 6 since it is the middle value. The mean is the sum divided by the number of values $= 89/15 \approx 5.93$.

To compare how the values are spread around the average value, we use different types of measures of dispersion. One way to get a picture of the spread in a collection of values is to calculate the range, the difference between the highest and lowest value. For the 15 numbers above, the range is $9 - 1 = 8$. A useful measure of dispersion is percentiles, which are the values that divide a collection of values into 100 parts. Similarly, other divisions are also used, such as deciles, quintiles and quartiles. A decile divides a collection of values into ten parts, quintile into five and quartile into four.

Percentiles are indicated with corresponding numbers 1–100, where P_{10} symbolizes the tenth percentile and is the value that is greater than 10% of the values in the collection. Here we are content with a simplified example. Consider all the integers 0 to 1,000. The tenth percentile for this set is $P_{10} = 100$, since the value 100 delimits the bottom 10 percent of the values. The fourteenth percentile is $P_{14} = 140$. The twentieth percentile is $P_{20} = 200$, and so on.

Percentiles P_{10}, P_{20}, P_{30}, P_{40}, P_{50}, P_{60}, P_{70}, P_{80} and P_{90} are the same values as the deciles. Deciles can be written as D_1 for decile 1, which is the same value as P_{10}, D_2 for decile 2, and so on. Percentiles P_{20}, P_{40}, P_{60} and P_{80} are the same values as the quintiles, where $P_{20} = K_1$ for quintile 1. Percentiles P_{25}, P_{50} and P_{75} are the same values as the quartiles, which divide the collection into four parts. Quartiles are often written as Q_1, Q_2 and Q_3. $P_{50} = Q_2$ marks the middle value in the collection, which is the same thing as the median. Sometimes the concepts interquartile range and quartile deviation are used. Interquartile range is the difference between the third and first quartile: $Q_3 - Q_1$. Quartile deviation is interquartile range divided by 2: $(Q_3 - Q_1)/2$.

15.7 Variance and standard deviation

Two other common measures of dispersion are variance and standard deviation. Both of these are very central to all statistical analysis, which is why we go through them a bit more thoroughly here. In section 15.1 we introduced population and sample. When we work with information and data variables, for example data that describes some phenomenon in society, we very rarely have access to reliable information about the population. When we study causal relationships, we lack observations for the counterfactual scenario that we are actually interested in and therefore do not have access to all data. Instead, we are then referred to trying to estimate what the population looks like by studying the sample data we have access to.

Say now that we are interested in variable x. In the population for this variable there are N number of observations and from this population we take n number of observations for our sample. Our goal is to find out what the spread in the population looks like and we use our sample data to study this. Note here that what we are actually constantly seeking are the values that exist in the population. We make calculations on our sample data to estimate the values in the population. The expressions estimate and estimate mean the same thing.

If we have access to population data, the mean is known. To mark this, the population's mean is usually written as μ_x(the Greek letter mu). The mean $\bar{x}$ is an estimate of the population's μ_x, where $\bar{x} = \sum_i x_i/n$.

To study spread, we calculate distance from the mean. In the population, the difference between observation x_i and the mean μ_x is written as $x_i - \mu_x$, where letter i means observation number i. With our sample data we take $x_i - \bar{x}$. The sum of differences from the mean $\sum_i (x_i - \bar{x})$ is always 0, which we see by writing:

$$\begin{aligned}\sum (x_i - \bar{x}) &= \sum x_i - n\bar{x} \\ &= \sum x_i - n\frac{1}{n}\sum x_i \\ &= \sum x_i - \sum x_i \\ &= 0\end{aligned} \tag{15.1}$$

One way to measure deviations from the mean is to estimate mean absolute deviation:

$$\textbf{Mean absolute deviation} = \frac{1}{n}\sum |x_i - \bar{x}| \tag{15.2}$$

We can also replace the mean $\bar{x}$ with some other measure of central tendency such as median or mode. Another way to sum deviations from the mean to a positive value is to square the deviations: $(x_i - \bar{x})^2$. If we divide the sum of the squared differences by the number of observations n, we get the variance for variable x:

$$\textbf{Variance: } \operatorname{var}(x) = \left(\frac{1}{n}\right)\sum_i^n (x_i - \bar{x})^2 \tag{15.3}$$

where the parenthesis $(x_i - \bar{x})^2$ shall be calculated for each observation and summed:

$$\sum_i^n (x_i - \bar{x})^2 = (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + ... + (x_n - \bar{x})^2 \tag{15.4}$$

By squaring each deviation from the mean we get a positive measure. This also means that greater deviations from the mean will have a greater weight for the estimated variance.

In the same way that the mean $\bar{x}$ is an estimate of the population's μ_x, $\mathrm{var}\,(x)$ is an estimate of the variance in the population. The variance in the population is often denoted with the Greek letter small sigma squared, σ_x^2, which indicates that this is the population value for variable x.

Standard deviation for a collection of observations is the positive square root of the variance. For population, standard deviation is often denoted σ_x. Estimated standard deviation for variable x can be written $\hat{\sigma}_x$, sd_x or s_x:

$$\textbf{Standard deviation: } s_x = +\sqrt{\mathrm{var}\,(x)} = \left(\frac{\sum_i^n (x_i - \bar{x})^2}{n} \right)^{1/2} \tag{15.5}$$

When we estimate the mean with sample data ($\bar{x}$), this often deviates from the population's mean (μ_x). This deviation tends to lead to our estimate of the population's variance becoming too small. To correct for this, in the equation for variance (equation (15.3)) we divide by $n-1$ instead of n. This new definition of variance is called corrected variance or sample variance:

$$\textbf{Sample variance: } \mathrm{var}\,(x) = \left(\frac{1}{n-1} \right) \sum_i^n (x_i - \bar{x})^2 \tag{15.6}$$

Division by $n-1$ is called Bessel's correction and often results in the estimated variance being closer to the population's variance. Many computer programs have ready commands for estimating (calculating) variance and then use Bessel's correction as in equation (15.6) . For standard deviation we also use Bessel's correction:

$$\textbf{Sample standard deviation: } s_x = \left(\frac{\sum_i^n (x_i - \bar{x})^2}{n-1} \right)^{1/2} \tag{15.7}$$

If we have a constant a that represents an arbitrary value, the variance for this is $\mathrm{var}\,(a) = 0$. This is because a single value has no spread. If a is multiplied by a variable x, the variance for ax equals $a^2\mathrm{var}\,(x)$. We see this by taking:

$$\begin{aligned} \mathrm{var}\,(ax) &= \left(\frac{1}{n}\right) \sum_i (ax_i - a\bar{x})^2 \\ &= \left(\frac{1}{n}\right) \sum_i (a^2x_i^2 - 2a^2\bar{x} + a^2\bar{x}^2) \end{aligned} \tag{15.8}$$

Since a is not affected by the summation, we factor out all a^2:

$$\begin{aligned} \mathrm{var}\,(ax) &= a^2 \left(\frac{1}{n}\right) \sum (x_i^2 - 2\bar{x} + \bar{x}^2) \\ &= a^2 \left(\frac{1}{n}\right) \sum (x_i - \bar{x})^2 \\ &= a^2 \mathrm{var}\,(x) \end{aligned} \tag{15.9}$$

For standard deviation we get:

$$s_x(ax) = \left(\frac{1}{n}\sum_i^n (ax_i - a\bar{x})^2\right)^{1/2} \tag{15.10}$$
$$= \left(\frac{1}{n}\sum_i^n \left(a^2x_i^2 - 2a^2\bar{x} + a^2\bar{x}^2\right)\right)^{1/2}$$
$$= \left(a^2\right)^{1/2}\left(\frac{1}{n}\sum_i^n (x_i - \bar{x})^2\right)^{1/2}$$
$$= |a|\, s_x$$

Table 15.3: Some calculations with variables x and y

Observation	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
1	3	3	-2	$-0,5$	4	0,25
2	4	2	-1	$-1,5$	1	2,25
3	6	5	1	1,5	1	2,25
4	7	4	2	0,5	4	0,25
Mean	5	3,5				
Sum	20	14			10	5

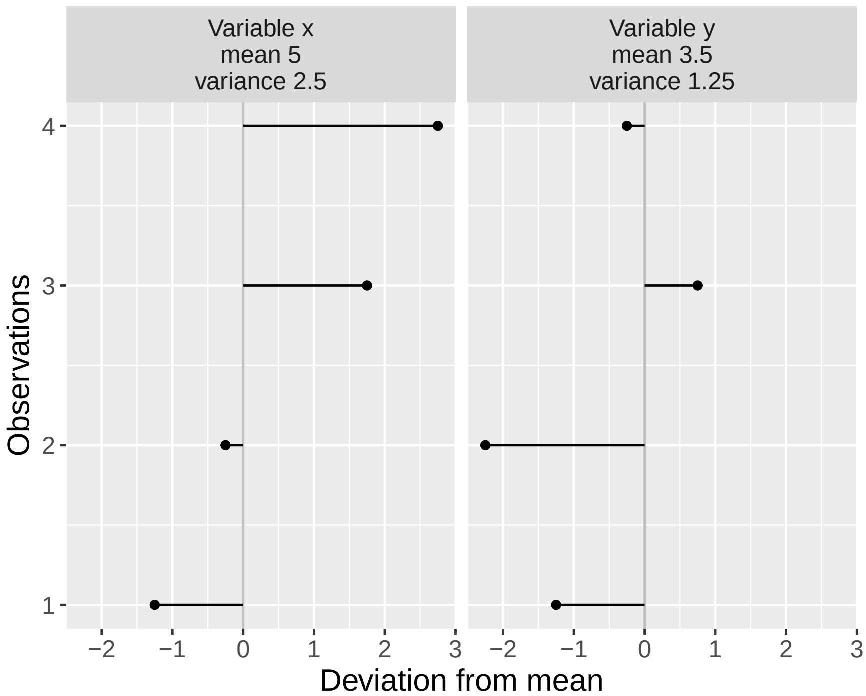


Figure 15.4: Variance for variables x and y

where $|a|$ is the absolute value of the constant a. Now we shall calculate (estimate) the variance with four observations for two variables: x and y. Variable x has the values 3, 4, 6 and 7. Variable y has the values 3, 2, 5 and 4. Table 15.3 summarizes the calculations we need. We use the definition of variance from equation (15.6) :

$$\begin{aligned}\operatorname{var}(x) &= \frac{\sum_i (x_i - \bar{x})^2}{n-1} = \frac{10}{3} \\ \operatorname{var}(y) &= \frac{\sum_i (y_i - \bar{y})^2}{n-1} = \frac{5}{3}\end{aligned} \tag{15.11}$$

The variance for variable x is larger than the variance in variable y. This indicates that the values in x are more spread out from the mean compared to the spread in variable y. Figure 15.4 illustrates deviations from the mean for each respective variable. In the heading for each respective graph, mean and variance are reported. To calculate standard deviation for variables x and y, we take the positive square root of the variance:

$$\begin{aligned}s_x &= \sqrt{\frac{\sum_i^n (x_i - \bar{x})^2}{n-1}} = +\sqrt{\frac{10}{3}} \approx 10.05 \\ s_y &= +\sqrt{\frac{5}{3}} \approx 0.745\end{aligned} \tag{15.12}$$

We summarize some of the measures of central tendency and measures of dispersion with the two tables below. In the upper table, a collection of values is presented. In the lower table 15.4, these are described with measures of central tendency and measures of dispersion.

Table 15.4: Example with measures of central tendency and measures of dispersion.

x_i	12	27	35	38	53	53	55	57	66	69	74	89	98
i	1	2	3	4	5	6	7	8	9	10	11	12	13

Measure	Description	Result
Mean	Sum divided by number	55.8
Median	Middle value	55
Mode	Most frequently occurring	53
Quartiles	Divides the distribution into four parts	$Q_1 = 38$, $Q_3 = 69$
Percentiles	Divides the distribution into parts	$P_{20} = 36.2$, $P_{80} = 72$
Variance	Measures spread among values	539.1
Standard deviation	Positive square root of the variance	23.2
Minimum	Smallest value	12
Maximum	Greatest value	98

15.8 Standardized values

It is difficult to compare variance between two different variables if the variables are measured in different units, for example, such as weight and height. One method to facilitate comparisons between variables is standardization and normalization. Standardized values for a variable x are a measure of how far a value (x_i) in a variable is from the variable's mean $(\bar{x})$ counted in number of standard deviations. This is a very useful measure that is often used in statistics. Standardized values are also called standard scores, z-scores or z-values. Standardized values for a variable x can be calculated in the following way:

$$\textbf{Standardized value: } x_i = z_i = \frac{x_i - \bar{x}}{s_x} \tag{15.13}$$

where x_i is each respective value in variable x and z_i is the standardized value of x_i. $\bar{x}$ is the mean for x and s_x is the standard deviation for variable x. Each observation's distance to the mean is divided by the standard deviation for the current variable. Standardized variables always get the mean 0 and standard deviation 1.

If we take our variables x and y that we used as examples in the previous section and convert these to standardized values, we get the two new variables z_x and z_y. Calculations and results are reported in table 15.5 . Observations 1 and 2 for z_x and z_y become negative since these values lie below their respective means.

Table 15.5: Standardized values

Observation	x_i	y_i	$z_x = (x_i - \bar{x})/s_x$	z_y
1	3	3	−1.1	−0.39
2	4	2	−0.55	−1.16
3	6	5	0.55	1.16
4	7	4	1.1	0.39
Mean, $\bar{x}$ and $\bar{y}$	5	3.5	0	0
Standard deviation, s	1.83	1.29	1	1

Normalization can mean slightly different things in statistics. One meaning is to convert a variable so that all values become between 0 and 1. Normalization of a variable x is done in the following way:

$$\textbf{Normalized value: } x_{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}} \tag{15.14}$$

where x_i is observation i for variable x, x_{min} is the lowest value in variable x and x_{max} is the highest value for the variable. Similar to standardization, this can also be used to compare variables that otherwise differ from each other. Table 15.6 shows normalized values for variables x and y.

Table 15.6: Normalized values

	x_i	y_i	x_{norm}	y_{norm}
1	3	3	0	0.33
2	4	2	0.25	0
3	6	5	0.75	1
4	7	4	1	0.67
Min.	3	2	0	0
Max.	7	5	1	1

15.9 How do we know the things we know?

We described above the concepts superpopulation, population and sample. Since we cannot observe the counterfactual outcome that we are interested in when we study causal relationships, it can be described as us in these situations always working with a sample of data. When we work with samples, there is always more or less uncertainty to what extent we succeed in finding the values we seek for our population and superpopulation.

Even in situations where we actually have access to data for our entire population, there are reasons to regard the information as uncertain, that is, that there is a greater or smaller probability that our results are incorrect. Say for example that we study the economic development in Europe's countries in recent decades. We have gotten hold of a table with data where we have a collection of variables with one observation per year and country. Even if our data contains information about all countries over a long time period, this should be regarded as a sample. Collection of data can entail measurement errors, for example due to technical aids or human mistakes. Or perhaps the data we use does not measure exactly what we want to study, but is an approximate measure of something more complex.

Another reason to treat most information as sample data is that there can be uncertainty around exactly how the population should be defined. We might want to be able to make statements about all patients of a certain age or all adult people in the entire country. Then it is usually a bit uncertain exactly what characteristics our population has. We can also think about these questions in terms of the superpopulation that will exist in the future. Even for these coming observations, there is uncertainty around their exact characteristics.

In the next chapters we will introduce how we can work more carefully and in detail with uncertainty and probability.

Chapter 16

Basic Probability 1: discrete distributions

This chapter introduces probability theory and discrete probability distributions, which have a limited number of possible values, such as two sides of a coin or six sides of a die.

16.1 How we know what we know

Say we design an experiment where we want to study the effects of a medicine and have two patients at our disposal. One patient receives the medicine (treatment) and the other receives no medicine (control). Since we are only comparing two people, there is a high risk that the study's results could just as well be a result of chance. One way to reduce the risk of random results is to use more observations. But even with a large number of observations, we cannot be completely certain that the results are not just as well a result of chance. In this part of the book we will go through how we may work with this type of question using probability theory and mathematical calculations.

Even in our daily lives we use probability in some sense to draw conclusions. Based on previous observations, we often have a feeling for various forms of more or less likely results. In many situations we believe we are more certain of our position precisely because we have observed corresponding phenomena many times before. Our everyday reasoning ability is biased and easily confuses facts and fantasy. Our daily life is also full of unique situations where we cannot repeat exactly the same situation, but must make a subjective assessment from case to case based on previous experience and information.

16.2 Random variables

In previous chapters we have worked with variables defined by a mathematical function over a domain, for example the function $y = x^2$ defined over all real numbers. We have worked with data variables consisting of collected information, observations. Later in this book, when we go through regression analysis, we will work with variables that are predicted using regression models.

When we are now going to work with probability and chance we will use random variables. A random variable is a variable whose results are affected by a random process. By random process we mean any situation whose outcome is uncertain beforehand.

Random variables can be defined by mathematical functions that describe the probability for the values (outcomes) that the variable can assume. Say for example that we have a variable M that can assume two different values: the number 1 or 2. These values are the sample space for variable M: $\{1, 2\}$. The probability for each outcome is 50%. We describe M with the probability function $P(M)$, where the function is called P after the word probability:

$$\begin{aligned} P(M=1) &= \frac{1}{2} = 0.5 = 50\% \\ P(M=2) &= \frac{1}{2} \end{aligned} \tag{16.1}$$

The total probability for the two possible outcomes in M is $0.5 + 0.5 = 1$, that is 100%. Equation (16.1) is an example of a probability distribution. We say that the variable M follows a discrete probability distribution, since the distribution has a limited number of outcomes. An experiment, or trial, is called the process that leads to an outcome. An event refers to one or several outcomes, a subset of the sample space.

A continuous probability distribution has an infinite number of outcomes within one or several intervals. Say for example that a random variable X can assume all possible decimals between 0 and 1 (the sample space) with the same probability. Theoretically, X can assume the value 0.23 with the same probability as 0.9999345. There are infinitely many decimals and therefore infinitely many outcomes. Since there are infinitely many outcomes, the probability, mathematically, for each specific value approaches 0. For a continuously random variable we therefore instead calculate the probability for an interval of outcomes. For example, there is a 50% probability for the event that X assumes a value between 0 and 0.5 and 50% probability that X assumes a value between 0.5 and 1.

There are infinitely many probability distributions. As long as we can come up with a new description of a probability distribution, it exists. Say now that we have a six-sided die T with the numbers 1 to 6 and that each side has an equal probability of coming up, one sixth, $\frac{1}{6}$. The probability for the different outcomes can be described:

$$P(T=t) = \begin{cases} \frac{1}{6} & t = 1,2,3,4,5,6 \\ 0 & \text{otherwise} \end{cases} \tag{16.2}$$

The total probability for all outcomes is $\frac{1}{6} \cdot 6 = 1 = 100\%$. Suppose we instead have a six-sided die X_2 with the integers 1 to 6, which is unbalanced in such a way that the probability for outcome 3 is $\frac{2}{7}$ while the probability for each other outcome is $\frac{1}{7}$. The number 7 in the denominator is not the number of sides on the die but follows from the variable's probability distribution. Variable X_2 can be summarized as:

$$P(X_2=x) = \begin{cases} \frac{1}{7} & x = 1,2,4,5,6 \\ \frac{2}{7} & x = 3 \\ 0 & \text{otherwise} \end{cases} \tag{16.3}$$

Let us now assume that we have two six-sided perfectly balanced dice, one red and one black with the numbers 1 to 6. The probability for each unique result with t number of perfectly balanced six-sided dice can generally be described as $\left(\frac{1}{6}\right)^t = \frac{1}{6^t}$. With one die the probability for each result is $1/6$. With two dice it is $1/6^2 = 1/36$. With two dice there are 36 possible combinations. The probability that we get the point sum 2 is the same as the point sum 12: $1/36$. With three dice $1/6^3 = 1/216$.

Red dice \ Black dice	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

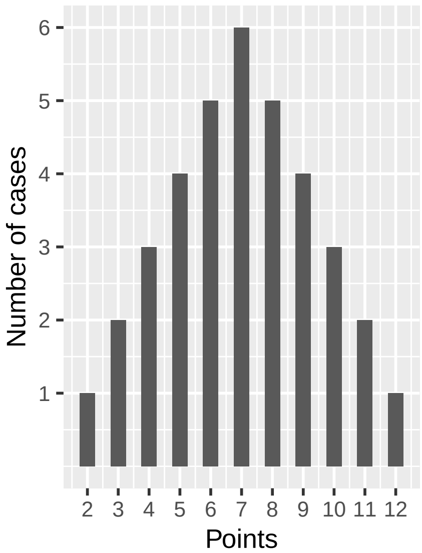


Figure 16.1: Number of cases and result sum for two dice

Now we will calculate the probability for the total point sum for the two dice, which we call variable Z. Figure 16.1 illustrates this with a table of all 36 possible point combinations (left) and a graph showing the number of combinations per point sum from 2 to 12 (right). The probability for outcome $Z = 3$ is 2/36, which we see by adding together the probability for each unique result that gives this point sum, $(\text{red}, \text{black}) = (1, 2)$ and $(2, 1)$. The result that red die = 1 and black die = 2 is considered the same result as red die = 2 and black = 1:

$$P(Z = 3) = \frac{1}{36} + \frac{1}{36} = \frac{2}{36} \tag{16.4}$$

Outcome 7 is given by most combinations: $(1, 6), (6, 1), (2, 5), (5, 2), (3, 4)$ and $(4, 3)$. This outcome therefore has the highest probability:

$$P(Z = 7) = \frac{1 + 1 + 1 + 1 + 1 + 1}{36} = \frac{6}{36} = \frac{1}{6}$$

The probability of getting 3 points or less with the two dice, $Z \leq 3$, can be calculated in the following way. There are three squares out of 36 possible for this result and the probability can therefore be calculated as:

$$P(Z \leq 3) = P(2) + P(3) = \frac{1}{36} + \frac{2}{36} = \frac{3}{36} = \frac{1}{12}$$

16.3 Probability with multiple variables

If we are to calculate the probability for an outcome with regard to multiple random variables, this is called a multivariate probability distribution. If we only have two variables in the probability distribution it is called a bivariate probability distribution. There are several ways that variables can interact in probability. Here are examples of simultaneous and conditional probability.

Simultaneous probability: Suppose we have two random variables X and Y. The outcome in the two variables is determined simultaneously, at the same time, and do not affect each other. Say for example that we roll two dice where the result of one die does not affect the other and we are interested in the dice's total point sum.

Conditional probability: Conditional probability describes the probability that a variable X assumes a value x, given that another random variable Y assumes a value y, which can be written: $P(X = x|Y = y)$.

Table 16.1: Three types of balls

	Filled	Hollow	Sum
Red	40	40	80
Yellow	10	10	20
Sum	50	50	100

Say for example that we are to choose a ball at random from one hundred balls, of which 80 are red and 20 are yellow. Half of the balls of each color are hollow and half are filled, see Table 16.1. The probability of getting a red ball is 80/100=4/5 and a yellow ball 20/100=1/5. The probability for hollow or filled is 1/2 regardless of color. If X = color and Y = hollow/filled then the probability of getting a red ball, given that the ball is hollow:

$$P\left(X=\text{red}|Y=\text{hollow}\right)=\frac{40}{50}=\frac{4}{5}$$

We divide the number of red hollow balls (40) by the number of hollow balls in total (50). The probability that a ball is filled, given that it is yellow is:

$$P\left(Y=\text{filled}|X=\text{yellow}\right)=\frac{10}{20}=\frac{1}{2}$$

16.4 Expected value and variance

For a random variable we cannot estimate (calculate) a mean in the way that we do for a collection of discrete values, such as a collection of numbers. Instead we use what is called expected value, also called expectation value. Expected value is a generalization of the weighted mean. The expected value for a random variable is the sum of each outcome multiplied by its probability:

$$E\left(M\right)=\sum_{i}^{n}m_iP\left(m_i\right) \tag{16.5}$$

where $E\left(\right)$ is called the expected value function, which can also be written $E\left[M\right]$, $\mathbb{E}\left(M\right)$ or EM. In section 15.7 we introduced the population mean μ. The expected value for population X is $E\left(X\right)=\mu_X$. For variable M that has outcomes 1 and 2 each with 0.5 probability, the expected value is:

$$E\left(M\right)=m_1\cdot P\left(m_1\right)+m_2\cdot P\left(m_2\right)=1\cdot 0.5+2\cdot 0.5=1.5$$

Suppose we instead have a random variable H with outcomes 1 and 2 with 75% and 25% probability respectively. The expected value then becomes:

$$E\left(H\right)=1\cdot 0.75+2\cdot 0.25=1.25$$

The expected value function $E\left(\right)$ is a linear function, which can be described as that given that we have the random variables X and Y, the following applies:

$$E\left(X+Y\right)=E\left(X\right)+E\left(Y\right) \tag{16.6}$$

If we have two six-sided perfectly balanced dice X and Y, each with the possible outcomes 1 to 6 where each outcome has 1/6 probability, then the expected value $E\left(X\right)=E\left(Y\right)=3.5$. The expected value for X plus the expected value for Y:

$$E\left(X\right)+E\left(Y\right)=3.5+3.5=7 \tag{16.7}$$

Let us call the point sum for the two dice Z. Its expected value is given by:

$$E(Z) = \frac{2+6+12+20+30+42+40+36+30+22+12}{36} = 7$$

When we added the expected value for X and Y we got result 7, in equation (16.7). Another way to write this is the following:

$$E(Z) = E(X+Y) = E(X) + E(Y) = 7$$

Compare equation (16.6). The linearity of expected value also has significance when we multiply a random variable by a constant. Say for example that we have an arbitrary constant a. If we multiply the expected value $E(X)$ by a this is the same thing as a multiplied by each individual value in variable X:

$$E(aX) = aE(X) \tag{16.8}$$

If we instead add a constant b we may also move this out of the expected value function, as $E(aX+b) = aE(X)+b$. In section 15.7 we defined variance for population and sample. For a random discrete variable X variance can be defined as:

$$\operatorname{var}(X) = \sum_i^n (x_i - \mu_X)^2 P(x_i)$$

where $\mu_X = E(X)$ and $P(x_i)$ is the probability for each value x_i. The expected value we defined in equation (16.5) as $E(X) = \sum_i^n x_i P(x_i)$. For a probability distribution where all values have the probability $\frac{1}{n}$ and n is the number of observations, the variance is:

$$\operatorname{var}(X) = \frac{1}{n}\sum_i^n (x_i - \mu_X)^2$$

This can for example describe the variance for a perfectly balanced six-sided die where all values have the probability $\frac{1}{6}$. We will now see how variance and expected value are connected. We remove the parentheses and move in the summation $\sum_i^n$. Since μ_X is a constant we can move this out of the summation:

$$\operatorname{var}(X) = \sum_i^n x_i^2 P(x_i) - 2\mu_X \sum_i^n x_i P(x_i) + \mu_X^2 \sum_i^n P(x_i)$$

The expression $\sum_i^n x_i P(x_i)$ is the same as the expected value $E(X)$ and μ_X (see equation (16.5)). The expression $\sum_i^n x_i^2 P(x_i)$ can also be written $E(X^2)$. The sum of probabilities for all possible outcomes is by definition $\sum_i^n P(x_i) = 1$, that is 100%. Based on this we now write:

$$\begin{aligned} \operatorname{var}(X) &= E(X^2) - 2\mu_X^2 + \mu_X^2 \\ &= E(X^2) - \mu_X^2 \end{aligned} \tag{16.9}$$

Since $E(X) = \mu_X$ we write:

$$\operatorname{var}(X) = E(X^2) - (E(X))^2 \tag{16.10}$$

Note that $E(X^2)$ is the expected value of X^2, while $(E(X))^2$ is the expected value $E(X)$ squared. Variance can also be expressed as the expected value:

$$\operatorname{var}(X) = E(X - \mu_X)^2 \tag{16.11}$$

One way to think about the meaning of this expression is that variance is a value (a constant) that describes the spread in a variable. The expected value describes this spread in a random variable's population. To see that equation (16.10) and (16.11) are the same thing we write:

$$\begin{aligned}
\operatorname{var}(x) &= E\left(X-\mu_X\right)^2 \\
&= E\left(X-E\left(X\right)\right)^2 \\
&= E\left(X^2-2XE\left(X\right)+E\left(X\right)^2\right) \\
&= E\left(X^2\right)-2E\left(X\right)E\left(X\right)+E\left(X\right)^2 \\
&= E\left(X^2\right)-\left(E\left(X\right)\right)^2
\end{aligned} \tag{16.12}$$

If we now have $\operatorname{var}(aX+b)$, where a and b are constants we get:

$$\operatorname{var}(aX+b)=a^2\operatorname{var}(X)$$

That is, a constant a that is multiplied by the random variable can be moved out of the variance function $\operatorname{var}()$ and multiplied by itself. The second constant b disappears. The positive square root of the variance for X is also in this case the standard deviation and can be written $\sigma_x = s(x) = \sqrt{\operatorname{var}(x)}$, where σ represents the variance in the population (a constant). For standard deviation it applies that:

$$s(aX+b)=|a|\,s(X) \tag{16.13}$$

which means that if we multiply a random variable X by a constant a then its variance and standard deviation are multiplied, but it does not change the form of the spread.

16.5 Conditional expected value

The expected value for a variable can also depend on the outcome of another variable. This is called conditional expected value. If the outcome of variable Y depends on variable X, the conditional expected value for Y, given X can be written:

$$E(Y|X)=\sum_i y_i P(Y|X)$$

Say for example that random variable Y is a perfectly balanced six-sided die, values 1 to 6, probability $\frac{1}{6}$. Random variable X is a perfectly balanced coin with sample space 1 or 2 with probability $\frac{1}{2}$. If $X=1$ the values for Y are 1 to 6. If $X=2$ we add 10 to the value for Y, which is why $Y=11$ to 16. The sample space for the two variables X and Y together: $\{1,2,3,4,5,6,11,12,13,14,15,16\}$. The conditional expected value for variable Y given $X=1$ and $X=2$ can be written:

$$E(Y|X=1)=3.5 \qquad E(Y|X=2)=13.5$$

The conditional expected value Y of X is its own random variable with its own probability distribution for the sample space with integers 1 to 6 and integers 11 to 16. The **law of total expectation** is a useful rule. For the random variables X and Y this can be formulated:

$$E(E(Y|X))=E(Y) \tag{16.14}$$

The expected value of the conditional expected value $E(Y|X)$ is the same as the expected value $E(Y)$. We illustrate using the coin X and the die Y from the previous example. The law of total expectation gives:

$$\begin{aligned} E\left(E\left(Y|X\right)\right) &= E\left(Y|X=1\right)P\left(X=1\right)+E\left(Y|X=2\right)P\left(X=2\right) \\ &= 3.5\cdot\frac{1}{2}+13.5\cdot\frac{1}{2} \\ &= 8.5 \end{aligned} \tag{16.15}$$

The conditional expected value function also has the following property:

$$E\left(XY|X\right)=XE\left(Y|X\right)$$

where X and Y are random variables. The equation means that the expected value of X times Y conditional on X is the same thing as X times the expected value of Y conditional on X. To see this we write:

$$E\left(XY|X\right)=\sum_i xy_iP\left(Y|X\right)=x\sum_i y_iP\left(Y|X\right)=XE\left(Y|X\right)$$

16.6 Law of large numbers

The more experiments that are conducted with a random variable, the greater the probability that the results approach the population value. This also means that the larger sample we take, the greater the probability that an estimated mean corresponds with the population's expected value. This is called the law of large numbers.

We illustrate this with a random variable where the possible outcomes are integers 1 to 6 and where each outcome has the probability 1/6, which is why the expected value is 3.5 (like a six-sided perfectly balanced die). We let the computer choose an integer randomly between 1 and 6. Then choose two values and calculate the mean for these, then three values and so on until we have 1,000 means, where the last mean is calculated from 1,000 randomly created integers between 1 and 6.

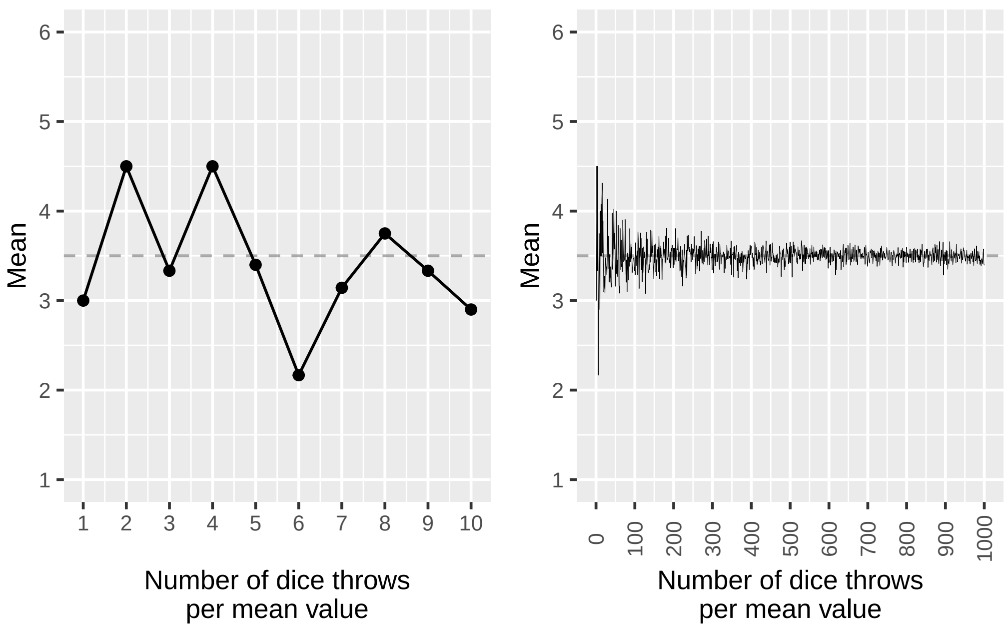


Figure 16.2: Law of large numbers: simulated means from a die (values 1–6) for increasing sample sizes. The dashed line marks the expected value 3.5.

Figure 16.2 illustrates the 1,000 results in two graphs, where the left graph only illustrates the first ten means and the right graph shows all calculated results, including those visible in the left graph. On the horizontal x-axis in both graphs the number of random values used to calculate the mean at that point is described. The point furthest to the right in the right graph is the mean of 1,000 randomly chosen integers between 1 and 6. Exactly as the law of large numbers predicts, these means approach the expected value 3.5.

More formally, the law of large numbers can be formulated in the following way: suppose we have an infinite sequence of random variables $X_1, X_2, ..., X_n$ that have the same expected value μ:

$$E(X_1) = E(X_2) = \cdots = \mu$$

The mean for n of these variables is:

$$\bar{X}_n = \frac{1}{n}\sum_{i}^{n} X_i$$

The law of large numbers can then be expressed as that the following limit value is 1 when n approaches infinity:

$$\lim_{n\to\infty} P\left(\left|\bar{X}_n - \mu\right| < \epsilon\right) = 1 \tag{16.16}$$

where $\left|\bar{X}_n - \mu\right|$ is the absolute value of the mean $\bar{X}_n$ minus the expected value μ. The function $P()$ describes the probability for an outcome. The term ϵ is an arbitrary positive value, for example a very low value close to 0. The entire equation can be read as that the probability that $\left|\bar{X}_n - \mu\right|$ is less than ϵ approaches 100% when the number of random variables X_i grows to infinity, that is $n \to \infty$.

Another way to describe this is that the difference between $\bar{X}_n$ and μ approaches 0 and this difference will be less than the low value ϵ. The definition in equation (16.16) is called the law of large numbers in weak form. The law of large numbers in strong form can be formulated:

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1$$

This should be read as that the mean $\bar{X}_n$ instead becomes equal to the expected value μ when n approaches infinity. The probability for this, $P()$, is equal to 1, that is 100%.

16.7 Uniform probability distributions

We have now in different examples used the probability function $P()$ to describe the probability for specific outcomes. For a random variable M we will now clarify that when we want to define the probability for a specific value we use the function $f(M)$. The two functions f and P describe the same thing so far:

$$f(m) = P(M = m)$$

For discrete probability distributions the probability function f is called probability mass function, PMF. In equation (16.5) we described the expected value for a random discrete variable as possible outcomes multiplied by the probability for these. If we instead of the probability function P use f, the expected value for random variable M can be written:

$$E(M) = \sum_{i}^{n} m_i f(m_i) \tag{16.17}$$

Now we will introduce what is called the cumulative distribution function (CDF), or just the distribution function. The distribution function describes the probability that a random variable assumes a value within a specific interval. Here too we use the probability function $P()$ to for example describe the probability that random variable M assumes a value equal to or less than the value m, which we describe as $P(M \leq m)$. We call the distribution function F:

$$F(m) = P(M \leq m)$$

Say now that we have a random variable X with sample space $\{1, 2, 3, 4, 5, 6\}$, where each outcome has the probability $\frac{1}{6}$:

$$f(x) = P(X = x) = \begin{cases} \frac{1}{6} & x = 1, 2, 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases} \tag{16.18}$$

The first row in the probability function describes the probability 1/6 for integers 1 to 6. The second row describes that for all other outcomes the probability is 0. X follows a uniform discrete probability distribution. The distribution is uniform because each outcome has the same probability. The distribution is discrete because there are a limited number of possible outcomes. That variable X follows a uniform probability distribution can be described with the following expression:

$$X \sim U(a, b)$$

where tilde $\sim$ describes that X follows the probability distribution U, which is an abbreviation for uniform probability distribution. The letters a and b indicate the limits for the interval of the distribution's possible values. In this case $a = 1$ and $b = 6$, which is why $X \sim U(1, 6)$. For a discrete variable that follows a uniform probability distribution, the distribution function can be described generally as:

$$F(x) = P(X \leq x) = \frac{x - a + 1}{b - a + 1}, \quad x = a, a + 1, \dots, b \tag{16.19}$$

where a and b are the lowest and highest integer that X can assume, that is 1 and 6 respectively. The distribution function for X:

$$F(x) = P(X \leq x) = \frac{x}{6}, \quad x = 1, 2, 3, 4, 5, 6 \tag{16.20}$$

For example, this means that $F(2) = 2/6$, which means that the cumulative probability of getting outcome 1 or 2 is equal to 2/6. One may also describe the same distribution function as:

$$F(x) = P(X \leq x) = \begin{cases} 0 & \forall\, x < 1 \\ \frac{x}{6} & 1 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

where we added the top row to describe that $F(x) = 0$ for $x < 1$. The probability that $X \leq x$ then increases stepwise for $x = 1, 2, 3, 4, 5$ and 6 to reach $F(x) = 1$, that is 100%, from $x = 6$ and upward.

The cumulative probability $P(X \leq x)$ must by definition be a value between 0 and 1, between 0 and 100%. What then remains of 100% is the probability that $X > x$:

$$F(x) = P(X \leq x) = 1 - P(X > x) \tag{16.21}$$

The probability $P(X > x)$ we calculate by taking $1 - F(x)$, which we see by rearranging the equation:

$$P(X > x) = 1 - P(X \leq x) = 1 - F(x) \tag{16.22}$$

For variable X the probability of getting 3 to 6 points is:

$$P(X > 2) = 1 - F(2) = 1 - \frac{2}{6} = \frac{4}{6}$$

The expected value for X is then:

$$E(X) = \sum_i x_i f(x_i) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5$$

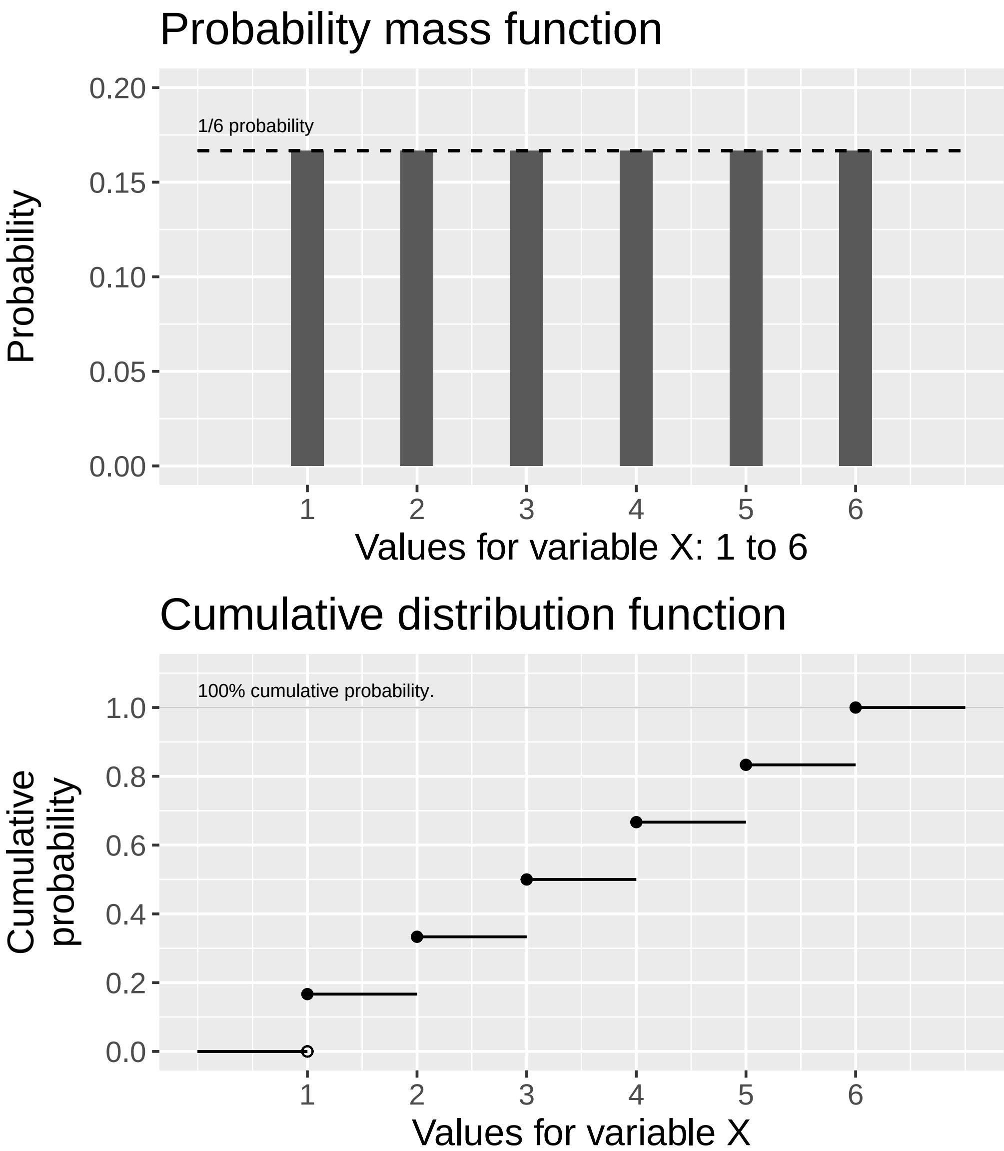


Figure 16.3: Probability mass function (top) and cumulative distribution function (bottom) for a uniform discrete probability distribution that can assume the integers 1 to 6.

Figure 16.3 illustrates the probability function and cumulative distribution function for variable X that can only assume integers 1 to 6 and the probability for each result is $\frac{1}{6}$. The upper graph shows the probability function, equation (16.18).

In the lower graph the cumulative, accumulated, probability that $X \leq x$ is illustrated, which increases stepwise by $\frac{1}{6}$ for each integer on the horizontal x-axis, see equation (16.20). In the lower graph this is illustrated by the horizontal lines between the points, where the cumulative probability is constant between each integer since the variable is discrete and cannot assume any values between these integers. At $X = 6$ the cumulative probability is 100%, since we have then counted all possible outcomes.

16.8 The binomial distribution

Say now that we have a variable M with the two outcomes 1 and 2 respectively, each with 50% probability. One can think of M as a perfectly balanced coin. We now define the probability for outcome 1 as the probability $p = 0.5 = 50\%$. The probability for the second outcome, value 2, must by definition be $1 - p = 1 - 0.5 = 0.5$.

Now we will describe the probability that we with variable M get more of outcome 2. Each individual result of M is independent of the previous result. The probability of getting k number of outcome 2 in n trials can then be described with the following probability mass function:

$$f(k) = P(M = k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{16.23}$$

where

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Equation (16.23) describes the probability function for what is called the binomial distribution. The expression $\binom{n}{k}$ is called the binomial coefficient and $n!$ is n factorial, such as $2! = 2 \cdot 1$. In a binomial distribution the outcome in one trial is independent from other trials, which is why the results from different rounds do not affect each other. It also does not matter in which order we get k outcomes.

Let us now use equation (16.23) to calculate the probability that we with two trials get outcome 2 exactly two times: $n = 2$ and $k = 2$. If we summarize the results as (result trial 1, result trial 2) we get the following results: $\{(1,1),(1,2),(2,1),(2,2)\}$. The probability for the result $(2,2)$ is then:

$$\begin{aligned} f(k) = P(M = k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \binom{2}{2} 0.5^2 (1-0.5)^{2-2} \\ &= \left(\frac{2!}{2!\,(2-2)!}\right) 0.5^2 (0.5)^0 \\ &= 0.25 \end{aligned} \tag{16.24}$$

The probability is $\frac{1}{4} = 0.25$. Each unique value of the four possible results has the probability 25%. There are two possibilities to get outcome 2 once in two trials: result $(1,2)$ or result $(2,1)$. The probability for this is therefore 50%:

$$f(1) = \left(\frac{2!}{1!\,(2-1)!}\right) 0.5^1 (0.5)^{2-1} = \left(\frac{2 \cdot 1}{1 \cdot 1}\right) 0.5 \cdot 0.5 = 0.5$$

Let us now calculate the probability of getting 10 outcomes with 2s in 10 trials:

$$f(10) = \binom{10}{10} 0.5^{10} (0.5)^{10-10} \approx 0.00098 \tag{16.25}$$

For the binomial distribution, the cumulative distribution function F describes the probability that variable M is equal to or less than the value k:

$$F(k) = P(M \leq k) = \sum_{i=0}^{\lfloor k \rfloor} \binom{n}{i} p^i (1-p)^{n-i} \tag{16.26}$$

where $\sum$ describes the sum of probabilities from the value $i = 0$, up to and including the value $\lfloor k \rfloor$. The expression $\lfloor k \rfloor$ is the floor function for the number k. The floor function for a real number gives the largest integer that is equal to or less than this number. Examples: $\lfloor 3.001 \rfloor = 3$, $\lfloor 3.6 \rfloor = 3$ and $\lfloor 3.9 \rfloor = 3$. A similar function is what is called the ceiling function, which is instead written $\lceil k \rceil$ and gives the smallest integer that is equal to or greater than the real number k, for example: $\lceil 3.002 \rceil = 4$ or $\lceil 3.8 \rceil = 4$. Let us calculate the cumulative probability of getting outcome 2 one or zero times in 2 trials. We have $n = 2$ and sum from $i = 0$ up to $k = 1$. From equation (16.26) we get:

$$\begin{aligned}F(1) &= \sum_{i=0}^{\lfloor 1 \rfloor} \binom{2}{i} 0.5^i (1-0.5)^{2-i} \\ &= \binom{2}{0} 0.5^0 (1-0.5)^{2-0} + \binom{2}{1} 0.5^1 (1-0.5)^{2-1} \\ &= 0.75\end{aligned} \tag{16.27}$$

The cumulative probability is 75%. Now we perform 10 trials ($n = 10$) and want to know the probability of getting 3 or fewer of outcome 2:

$$F(3) = \sum_{i=0}^{\lfloor 3 \rfloor} \binom{10}{i} 0.5^i (0.5)^{10-i} \approx 0.001 + 0.01 + 0.044 + 0.117 = 0.172$$

The probability of getting 0, 1, 2 or 3 outcome 2s in 10 trials is approximately 17.2%.

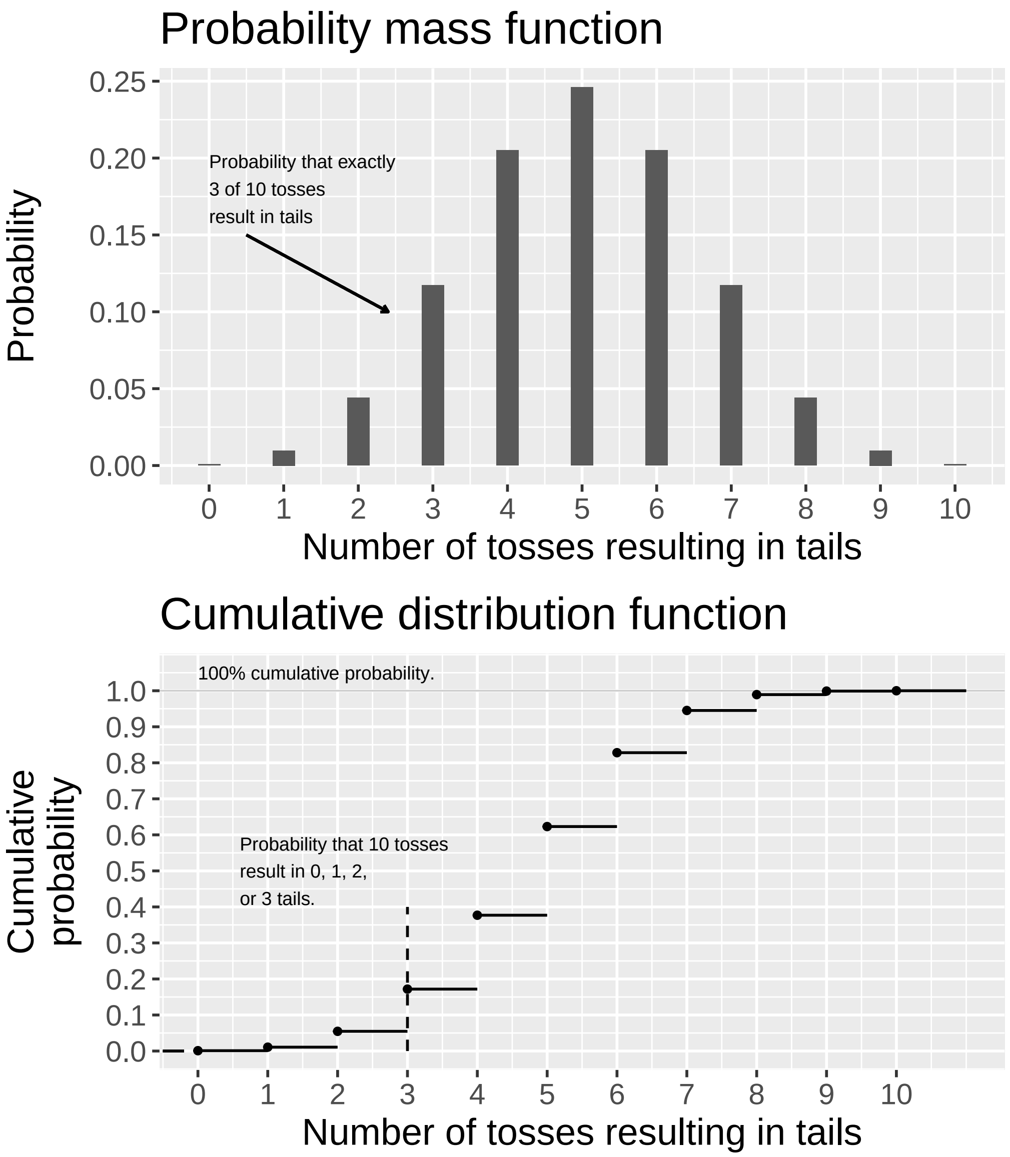


Figure 16.4: Probability mass function (top) and cumulative distribution function (bottom) for a binomial distribution with $n = 10$ and $p = 0.5$.

Figure 16.4 illustrates the binomial distribution's probability mass function (equation (16.23)) and distribution function (equation (16.26)). The graphs describe probabilities for different results, given that we make 10 trials, for example 10 tosses with a coin. The upper graph describes the probability that we with 10 tosses of the coin get different numbers of tails. The almost invisible bar furthest to the left describes the low probability of getting 0 tails in 10 tosses. The almost invisible bar furthest to the right describes the low probability of getting tails on all 10 tosses, see equation (16.25). The middle bar shows the probability of getting 5 tails in 10 trials, approximately 24.6%.

The lower graph describes the cumulative distribution function. Note that the two graphs have different vertical scales, where the lower graph's scale goes from 0 to 100%. In a similar way as before, the distribution function is illustrated by stepwise levels of cumulative probability that the result is equal to or less than the values on the horizontal x-axis. For results under 0 the probability is 0. We cannot get a negative number of outcomes, for example a negative number of tails. This is illustrated by the black horizontal line furthest down to the left in the graph.

At zero the probability increases to just under 0.1%, which is illustrated by the first point with its associated horizontal line, see equation (16.25). Thereafter the cumulative probability increases stepwise since we sum the probability for more results. Furthest to the right in the lower graph the cumulative probability is 100% since the probability of getting 0 to 10 tails in 10 tosses, which covers all possible alternatives, by definition must be 100%. If a random variable M follows the binomial distribution this can be described in the following way:

$$M \sim b\left(n, p\right)$$

where b stands for the binomial distribution, n is the number of trials and p is the probability for one of the two possible values in the current variable. The binomial distribution's expected value is:

$$E\left(M\right) = np \tag{16.28}$$

which is the number of trials n multiplied by the probability p. If $M \sim b\left(2, \frac{1}{2}\right)$ we have the expected value $E\left(M\right) = 1$, which should be read as that we in two trials on average will get one successful result. If we instead have $M \sim b\left(10, \frac{1}{2}\right)$ then $E\left(M\right) = 5$, which we also see in the upper graph in figure 16.4. Variance can also be used as a measure of the spread of the possible values in a probability distribution. The binomial distribution has variance:

$$\operatorname{var}\left(M\right) = np\left(1 - p\right)$$

where we multiply the number of occurrences n with the probability for one value, p, and the probability for the other value, $1 - p$.

16.9 The Poisson distribution

In many situations we want to calculate the probability for an event within an intended time frame. A commonly occurring probability distribution for this type of example is what is called the Poisson distribution, named after the French mathematician Siméon Denis Poisson (1781–1840). Let us illustrate with a classic example. In 1898 the Polish economist and statistician Ladislaus Bortkiewicz described the probability that x number of Prussian soldiers are kicked to death by a horse by accident in a specific year. The calculation was based on data collected between 1875 and 1894 for 14 Prussian cavalry corps.

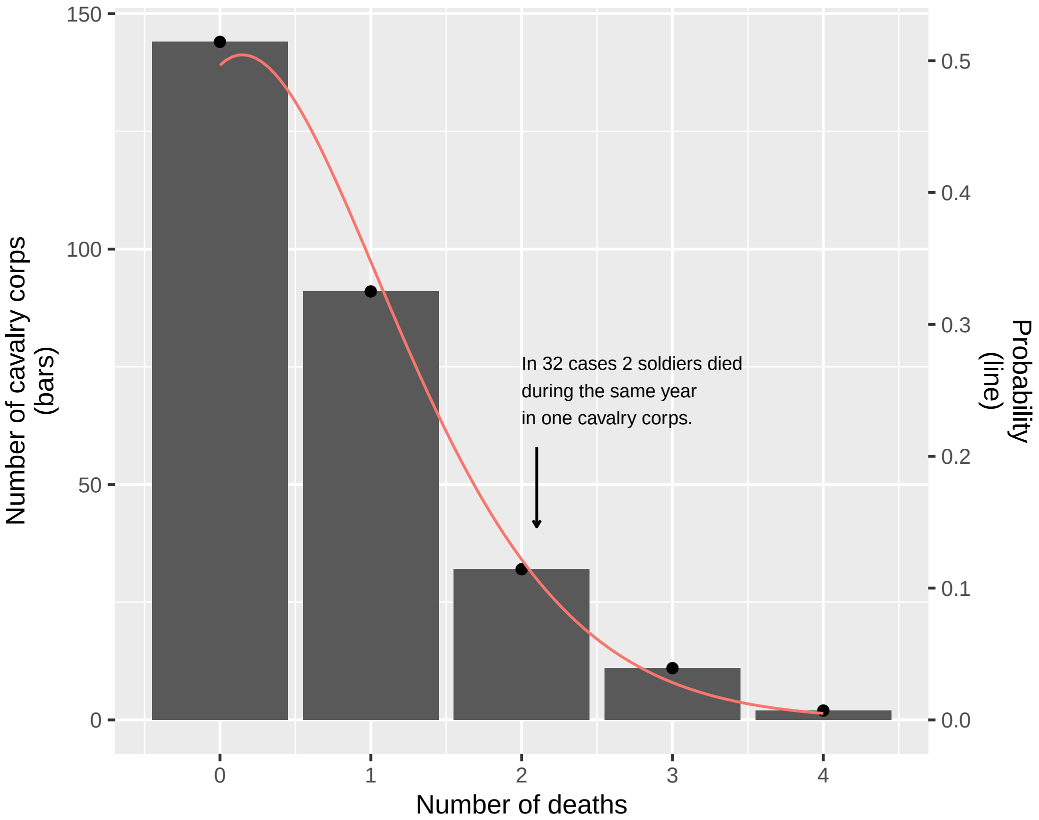


Figure 16.5: Deaths per cavalry corps per year (Bortkiewicz 1898).

Figure 16.5 illustrates the data that Bortkiewicz based his work on, where the bars show the number of deaths per cavalry corps per year. The bar for 0 deaths per year is highest. In 91 cases one death occurred in a cavalry corps during one of the years. In 32 cases 2 soldiers died during one year in one and the same cavalry corps. In two cases 4 soldiers died during one and the same year in one and the same cavalry corps.

When Bortkiewicz studied these numbers he noticed a pattern in the distribution that can be described with the help of the line that follows the bars' highest points. The line is drawn using the following equation, which describes the probability function for a Poisson distribution:

$$f(x, \lambda) = P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!} \tag{16.29}$$

The probability function f for the random variable X with possible outcomes x describes the probability for the number of deaths per cavalry corps per year. The letter e is Euler's number and λ (Greek lambda) is the average number of events, in this case the average number of deaths. Bortkiewicz's data includes 280 observations and sums to 196 deaths, since some of the observations have 0 deaths. The average number of deaths therefore becomes:

$$\lambda = \frac{196}{280} = 0.7$$

Let us estimate the probability that three soldiers are kicked to death in a cavalry corps during one year. We have $x = 3$ and $\lambda = 0.7$. The probability can then be estimated to:

$$f(x = 3, \lambda = 0.7) = e^{-\lambda} \frac{\lambda^x}{x!} = e^{-0.7} \frac{0.7^3}{3!} \approx 0.0284$$

The probability is approximately 2.84%. We see this in figure 16.5 by comparing the height of the bar for 3 deaths with the line. The Poisson distribution's distribution function gives us the probability $P(X \leq x)$:

$$F(x) = P(X \leq x) = e^{-\lambda} \sum_{i=0}^{\lfloor x \rfloor} \frac{\lambda^i}{i!} \tag{16.30}$$

where $\lfloor k \rfloor$ is the floor function of the value k (see equation (16.26)). Function F sums the probability for integers up to x. That variable X follows the Poisson distribution can be written:

$$X \sim \text{Pois}(\lambda), \quad \lambda > 0$$

The mean λ is also the Poisson distribution's expected value and variance:

$$E(X) = \text{var}(X) = \lambda$$

If a variable follows the Poisson distribution this is often called that the variable's values follow a Poisson process. Let us estimate another example with the Poisson distribution. Suppose we run a customer service reception on the telephone that is open a total of 100 hours in a year. On average we receive 50 calls:

$$\lambda = \frac{50}{100} = \frac{1}{2} = 0.5$$

We now want to know the probability that someone calls within the first hour. We call our random variable Y with possible outcomes y. If no one calls we have $y = 0$. The number of calls per hour follows a random Poisson process and we therefore estimate the probability that no one calls during the next hour using equation (16.29):

$$f(y = 0) = e^{-\lambda} \frac{\lambda^x}{x!} = e^{-0.5} \frac{0.5^0}{0!} \approx 0.607$$

The probability for zero calls in the first hour is approximately 61%. The probability for five calls in the first hour is:

$$f(y = 5) = e^{-\lambda} \frac{\lambda^x}{x!} = e^{-0.5} \frac{0.5^5}{5!} \approx 0.00016$$

16.10 Chapter Summary

- A random variable can assume different values. Each value has a probability. An example of a random variable is a perfectly balanced coin that can assume two values with 50% probability each. Or a perfectly balanced six-sided die, where each value has the probability $\frac{1}{6}$. The probability that a random variable X assumes the value x can be written with the function $P(X = x)$, where P is short for probability.
- A random variable's expected value can be described as $E(X) = \sum_x xP(x)$, where $P(x)$ describes the probability for the value x in variable X. If variable X can only assume the values 1 and 2 respectively, each with 50% probability, we get the expected value $E(X) = 1 \cdot 0.5 + 2 \cdot 0.5 = 1.5$.
- Conditional expected value $E(y|x)$ describes the expected value for y given a value for x. The law of total expectation: $E(E(y|x)) = E(y)$ means that the expected value of the conditional expected value y of x is the same as the expected value for y. Another way to describe this is that the conditional expected value is the mean of possible expected values for y conditional on the value for x.
- The variance in a random variable can be described as $\text{var}(x) = E(x^2) - (E(x))^2$ where $\text{var}(x)$ is the function for variance and $E(x)$ is the expected value function.
- Probability distributions can be a result of two or more variables. Suppose we have the two random variables X and Y:
  - Joint probability distribution: the results for X and Y are independent of each other and sum to a coherent probability distribution, for example two dice.

  - Conditional probability distribution: one variable's outcome is dependent on the other variable. Conditional expected value is an example of this.
- A discrete probability distribution has a limited number of possible values. Each value has a probability. The values can have the same or different probability.
- A probability mass function describes the probability for each value in a discrete random variable, for example $f(x) = P(X = x) = \frac{1}{2}$, $x = 1, 2$, where 1 and 2 are the values that variable X can assume. The probability that a random discrete variable X assumes a value that is equal to or less than x is described with the cumulative distribution function, $F(x) = P(X \leq x)$.
- A variable whose values have the same probability can be described as following a uniform probability distribution, which can be written as $X \sim U(a, b)$ where a and b are the lower and upper limits for the interval that the uniform distribution covers. If variable X can assume the values 1 and 2 we get $a = 1$ and $b = 2$.
- The binomial distribution: variable X can assume two values, x_1 and x_2 respectively. The probability for x_1 is equal to $p \in [0, 1]$. The probability of getting k number of results of x_1 in n number of trials is estimated as $f(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. If a random variable X follows the binomial distribution this can be described as $X \sim B(n, p)$ where n is the number of trials and p is the probability for one outcome. Expected value: $E(X) = np$. Variance: $\text{var}(X) = np(1-p)$.
- The Poisson distribution can be used to describe the probability for x number of results within a limited time period or a physical area. The probability function describes the probability for x number of occurrences $f(x) = P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$. Cumulative distribution function: $F(x) = P(X \leq x) = e^{-\lambda} \sum_{i=0}^{\lfloor x \rfloor} \frac{\lambda^i}{i!}$. Expected value $E(X) = \lambda$ and variance $\text{var}(X) = \lambda$.
- The law of large numbers means, in weak form, that given the infinite sequence of variables $X_1, X_2, \ldots, X_n$ with the expected value $E(X_1) = E(X_2) = \cdots = \mu$, and the mean $\bar{X}_n = \frac{1}{n} \sum_i^n X_i$, it holds that $\lim_{n\to\infty} P\left(\left|\bar{X}_n - \mu\right| < \epsilon\right) = 1$. The law of large numbers in strong form: $P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1$.

16.11 Exercises

1. An urn contains 3 red and 2 blue balls. One ball is drawn at random.

(a) What is (P(text{red}))?

(b) What is (P(text{blue}))?

(c) What is (P(text{red or blue}))?

Answer: (a) $P(\text{red}) = 3/5 = 0.6$, (b) $P(\text{blue}) = 2/5 = 0.4$, (c) $P(\text{red or blue}) = 1$.

2. A random variable X takes values 1, 2, 3 with probabilities 0.3, 0.5, 0.2.

(a) Verify that the probabilities sum to 1.

(b) Calculate (E(X)=sum x_{i}P(X=x_{i})).

(c) Calculate (text{var}(X)=E(X$^{\{2\})\text{-}[E(X)]}${2}).

Answer: (a) $0.3+0.5+0.2 = 1$, (b) $E(X) = 1(0.3)+2(0.5)+3(0.2) = 1.9$, (c) $E(X^2) = 1(0.3)+4(0.5)+9(0.2) = 4.1$, $\text{var}(X) = 4.1 - 1.9^2 = 0.49$.

3. Events A and B have $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.1$.

(a) Calculate (P(Acup B)).

(b) Are (A) and (B) independent?

(c) Calculate (P(Amid B)).

Answer: (a) $P(A \cup B) = 0.4 + 0.3 - 0.1 = 0.6$, (b) no: $P(A)P(B) = 0.12 \neq P(A \cap B) = 0.1$, (c) $P(A \mid B) = P(A \cap B)/P(B) = 0.1/0.3 = 1/3$.

4. A random variable X has $E(X) = 5$ and $\text{var}(X) = 4$.

(a) What is (E(2X+3))?

(b) What is (text{var}(2X+3))?

(c) What is the standard deviation (s(2X+3))?

Answer: (a) $E(2X+3) = 2 \cdot 5 + 3 = 13$, (b) $\text{var}(2X+3) = 4 \cdot \text{var}(X) = 16$, (c) $s(2X+3) = \sqrt{16} = 4$.

5. $E(Y \mid X = 1) = 8$ and $E(Y \mid X = 0) = 3$, with $P(X = 1) = 0.4$.

(a) Calculate (E(Y)=E(Ymid X=1)P(X=1)+E(Ymid X=0)P(X=0)).

(b) If (X) represents university education (1=yes, 0=no) and (Y) is income, interpret (E(Ymid X=1)=8).

(c) Does this conditional expectation prove that education causes higher income?

Answer: (a) $E(Y) = 8(0.4) + 3(0.6) = 3.2 + 1.8 = 5.0$, (b) average income among those with university education is 8 units, (c) no — selection bias may explain the difference; education may be correlated with unobserved factors such as family background.

6. A fair die has six equally likely outcomes.

(a) What is (E(X)) for a single roll?

(b) What does the law of large numbers say about the sample mean (bar{X}_{n}) as (nrightarrowinfty)?

(c) Why is this useful for statistical inference?

Answer: (a) $E(X) = 3.5$, (b) $\bar{X}_n \to \mu = 3.5$ as $n \to \infty$, (c) large samples give reliable estimates of population parameters — the sample mean converges to the true mean.

7. X is uniformly distributed on $\{1, 2, 3, 4\}$ (discrete uniform).

(a) What is the probability mass function (f(x))?

(b) What is (E(X))?

(c) What is (P(2leq Xleq3))?

Answer: (a) $f(x) = 1/4$ for $x \in \{1, 2, 3, 4\}$, (b) $E(X) = (1 + 2 + 3 + 4)/4 = 2.5$, (c) $P(2 \leq X \leq 3) = 2/4 = 0.5$.

8. A biased coin has $P(\text{heads}) = 0.3$. It is flipped $n = 5$ times. Let X = number of heads.

(a) Calculate (P(X=0)=(1-p)^{n}).

(b) Calculate (E(X)=np).

(c) Calculate (text{var}(X)=np(1-p)).

Answer: (a) $P(X = 0) = 0.7^5 \approx 0.168$, (b) $E(X) = 5 \cdot 0.3 = 1.5$, (c) $\text{var}(X) = 5 \cdot 0.3 \cdot 0.7 = 1.05$.

9. For a binomial distribution with $n = 10$ and $p = 0.5$:

(a) Calculate (E(X)).

(b) Calculate (text{var}(X)).

(c) Calculate (P(X=0)).

Answer: (a) $E(X) = np = 5$, (b) $\text{var}(X) = np(1-p) = 2.5$, (c) $P(X = 0) = \binom{10}{0} 0.5^0 \cdot 0.5^{10} = (0.5)^{10} \approx 0.001$.

10. Events occur at a rate of $\lambda = 3$ per hour. X follows a Poisson distribution.

(a) What is (E(X))?

(b) What is (text{var}(X))?

(c) Calculate (P(X=0)=e^{-lambda}).

Answer: (a) $E(X) = 3$, (b) $\text{var}(X) = 3$, (c) $P(X = 0) = e^{-3} \approx 0.050$.

11. For a Poisson distribution with $\lambda = 2$:

(a) Calculate (P(X=0)=e^{-2}).

(b) Calculate (P(X=1)=e$^{\text{\{-2\}cdotfrac\{2}}${1}}{1!}).

(c) What is (E(X))?

Answer: (a) $P(X = 0) = e^{-2} \approx 0.135$, (b) $P(X = 1) = 2e^{-2} \approx 0.271$, (c) $E(X) = \lambda = 2$.

12. Two random variables X and Y are independent with $E(X) = 3$, $E(Y) = 4$, $\text{var}(X) = 2$, $\text{var}(Y) = 5$.

(a) Calculate (E(X+Y)).

(b) Calculate (text{var}(X+Y)).

(c) What would (text{var}(X+Y)) be if (X) and (Y) were not independent?

Answer: (a) $E(X + Y) = 3 + 4 = 7$, (b) $\text{var}(X + Y) = 2 + 5 = 7$ (since independent), (c) $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y) + 2\,\text{cov}(X, Y)$; if not independent the covariance term must be added.

Chapter 17

Basic Probability 2: Continuous Distributions

In this chapter we will introduce probability distributions that are continuous and can assume an infinite number of different values. The chapter covers some examples of commonly occurring probability distributions.

17.1 Uniform Continuous Probability Distributions

A continuous random variable can assume an infinite number of different values, for example all decimals between two integers. Let us assume that X follows a uniform continuous probability distribution and has the same probability of assuming all values between 0 and 1. Since the number of possible values is infinite, the probability for a specific value is zero. However, the probability is greater than 0 for an interval of values. The probability for $0 < X \leq 0.5$ is 50% and the probability for $0.5 < X \leq 1$ is also 50%.

For a discrete variable we used a probability function to describe the probability for different outcomes. The corresponding function that describes the probability for possible outcomes in a continuous variable is called a density function.

A continuous probability distribution with constant probability over the possible outcomes is called a uniform probability distribution. The density function for a random continuous variable X can then be written as:

$$f(x; a, b) = P(X = x) = \begin{cases} \frac{1}{b-a} & \forall a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \tag{17.1}$$

where a and b delimit the interval for the possible outcomes for X. If X can assume all values between 0 and 1 we have that $a = 0$ and $b = 1$. To calculate the cumulative probability for a continuous random variable we use a cumulative distribution function, which we call F. In chapter 13 we covered how we with integrals can sum values over continuous intervals. This we also use to derive the cumulative distribution function for a random continuous variable. We take the integral of the density function f in equation (17.1):

$$\int_{-\infty}^{b} f(x)\, dx = \int_{-\infty}^{a} 0\, dx + \int_{a}^{b} \frac{1}{b-a} dx$$

We divide the integral into one part that sums $f(x)$ with respect to x for values from negative infinity, $-\infty$, to a. From equation (17.1) we know that $f(x)$ is 0 for all these values. The integral of 0 therefore also becomes 0. If we take the integral from the value $X = a$ to $X = b$ we get the probability that X will assume any of all possible outcomes. The cumulative probability is then 1, that is 100%:

$$F(x) = P(X \leq x) = \left.\frac{x}{b-a}\right|_a^b = \frac{b-a}{b-a} = 1$$

More generally we describe F as:

$$F(x; a, b) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & b \geq x \end{cases} \tag{17.2}$$

For the values $x < a$, $F(x) = 0$. For values $x \geq b$, $F(x) = 1$, since we cannot have more than 100% probability. Suppose we want to calculate the probability that X takes a value between a and c, where c is a value between a and b:

$$F(x) = \left.\frac{x}{b-a}\right|_a^c = \frac{c}{b-a} - \frac{a}{b-a} = \frac{c-a}{b-a} \tag{17.3}$$

We also calculate the probability that X assumes a value within an interval, for example between 0.3 and 0.5:

$$P(0.3 \leq x \leq 0.5) = \left.\frac{x}{b-a}\right|_{0.3}^{0.5} = \left.\frac{x}{1-0}\right|_{0.3}^{0.5} = \frac{0.5-0.3}{1} = 0.2$$

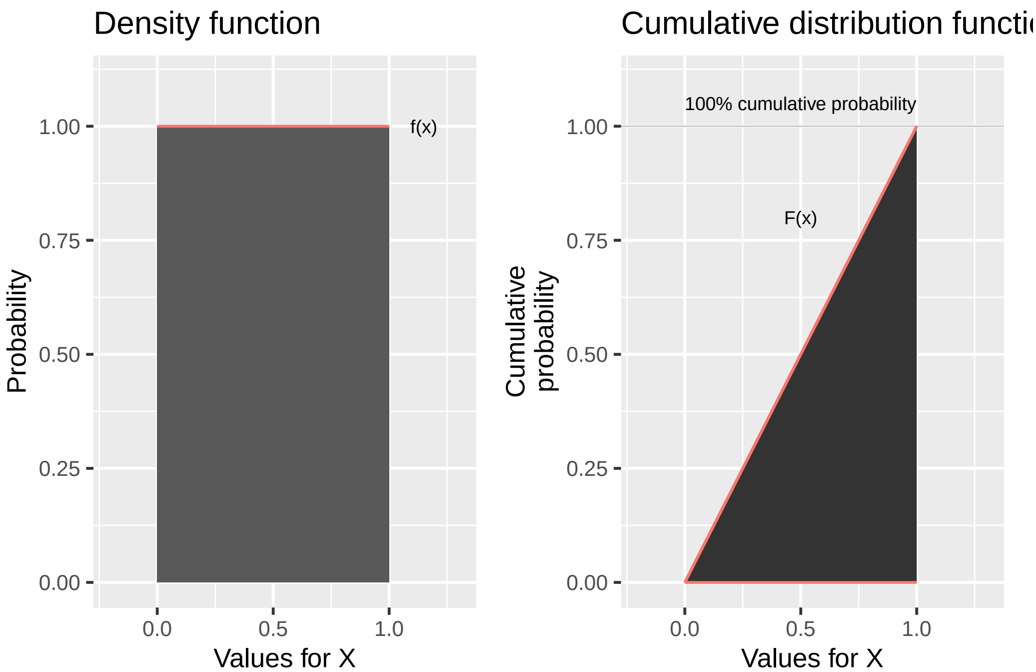


Figure 17.1: Uniform continuous probability: density function (left) and cumulative distribution function (right).

Figure 17.1 illustrates the density function f and cumulative distribution function F for the random continuous variable X, which follows a uniform probability distribution. In the left graph the density function is illustrated in the way it is described in equation (17.1). If we have a continuous random variable, the line that the density function draws in a graph is called a probability curve. The probability on the vertical y-axis is equal to $\frac{1}{b-a}$ for all values between 0 and 1. The density function for a uniform probability distribution is characterized by always looking like a rectangle in a graph. In the right graph the cumulative distribution function is illustrated, equation (17.2). The cumulative probability increases from 0 to 1 over the interval 0 to 1. That variable X follows a uniform continuous probability distribution can in a similar way as for a discrete variable be written:

$$X \sim U(a, b)$$

where a and b are the limits for the interval of possible outcomes.

17.2 Expected Value and Variance for Continuous Random Variables

In section 16.4 we introduced expected value for discrete random variables as the sum of outcomes multiplied by their probability, $E(X) = \sum x f(x)$. The expected value for a continuous random variable is in a similar way the sum of outcomes multiplied by the probabilities. Since we now will sum a continuous interval we use integrals:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx \tag{17.4}$$

which should be read as that we sum from negative infinity to positive infinity all possible outcomes x multiplied by the probability per value, which is given by $f(x)$. The expected value for a random variable X that follows a uniform continuous probability distribution, $X \sim U(a, b)$, can be written:

$$E(X) = \frac{1}{2}(a + b)$$

where a and b delimit possible values for X. If we let $a = 0$ and $b = 1$, X has the expected value:

$$E(X) = \frac{1}{2}(0 + 1) = \frac{1}{2}$$

Variance for a continuous random variable X can be described as the same expected value as variance for discrete random variables:

$$var(X) = E\left((X - \mu_X)^2\right) = \sigma_X^2$$

If we write out the definition of the expected value from equation (17.4):

$$E\left((X - \mu_X)^2\right) = \int_{-\infty}^{\infty} (X - \mu_X)^2 f(x)\, dx$$

The standard deviation is, as for discrete variables, the square root of the variance: $\sigma_X = \sqrt{var(X)} = \sqrt{\sigma_X^2}$. Conditional expected value for continuous random variables X and Y is written:

$$E(Y|X) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y)\, dx\, dy$$

where $f(x, y)$ is the joint probability distribution. The law of total expectation also applies to continuous distributions, $E(E(X|Y)) = E(X)$, as well as the rule that $E(XY|X) = XE(Y|X)$.

17.3 Exponential Distributions

In section 16.9 we introduced the Poisson distribution. Now we will instead of number of outcomes within a time period calculate the probability for a time period until the next outcome. Say for example that your boss usually calls you three times per hour to check how it's going with that report you were supposed to write. This average value we call $\lambda = 3$. The average waiting time for a call is in that case $\theta = \frac{1}{\lambda} = \frac{1}{3}$ hour. The time until the next call is a continuous random variable X that follows what is called the exponential distribution. The probability for the waiting time x can then be described with the following density function:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{17.5}$$

where λ is greater than 0 and e is Euler's number. The exponential distribution can among other things be used to describe the probability for the waiting time until the first event in a Poisson process. Since we have a continuous probability distribution, the probability for each individual outcome is equal to 0. The cumulative distribution function F of the exponential distribution can be written:

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x} \tag{17.6}$$

Function F shows the probability that the waiting time for the next phone call is less than x. If we instead want to calculate the probability that the waiting time is longer than x, that is $X > x$, we take:

$$P(X > x) = 1 - P(X \leq x)$$

Since $P(X \leq x) = F(x)$ we replace the term $P(X \leq x)$ with the definition of the cumulative probability distribution:

$$P(X > x) = 1 - F(x) = 1 - \left(1 - e^{-\lambda x}\right) = e^{-\lambda x} \tag{17.7}$$

With this definition we estimate the probability that the waiting time for the next call from the boss is longer than time x. If the boss usually calls three times per hour then $\lambda = 3$ and $\theta = 1/\lambda = 1/3$. Now we will estimate the probability that the waiting time becomes more than 1 hour, $X > 1$. From equation (17.7) we have that:

$$P(X > 1) = e^{-3 \cdot 1} \approx 0.0496$$

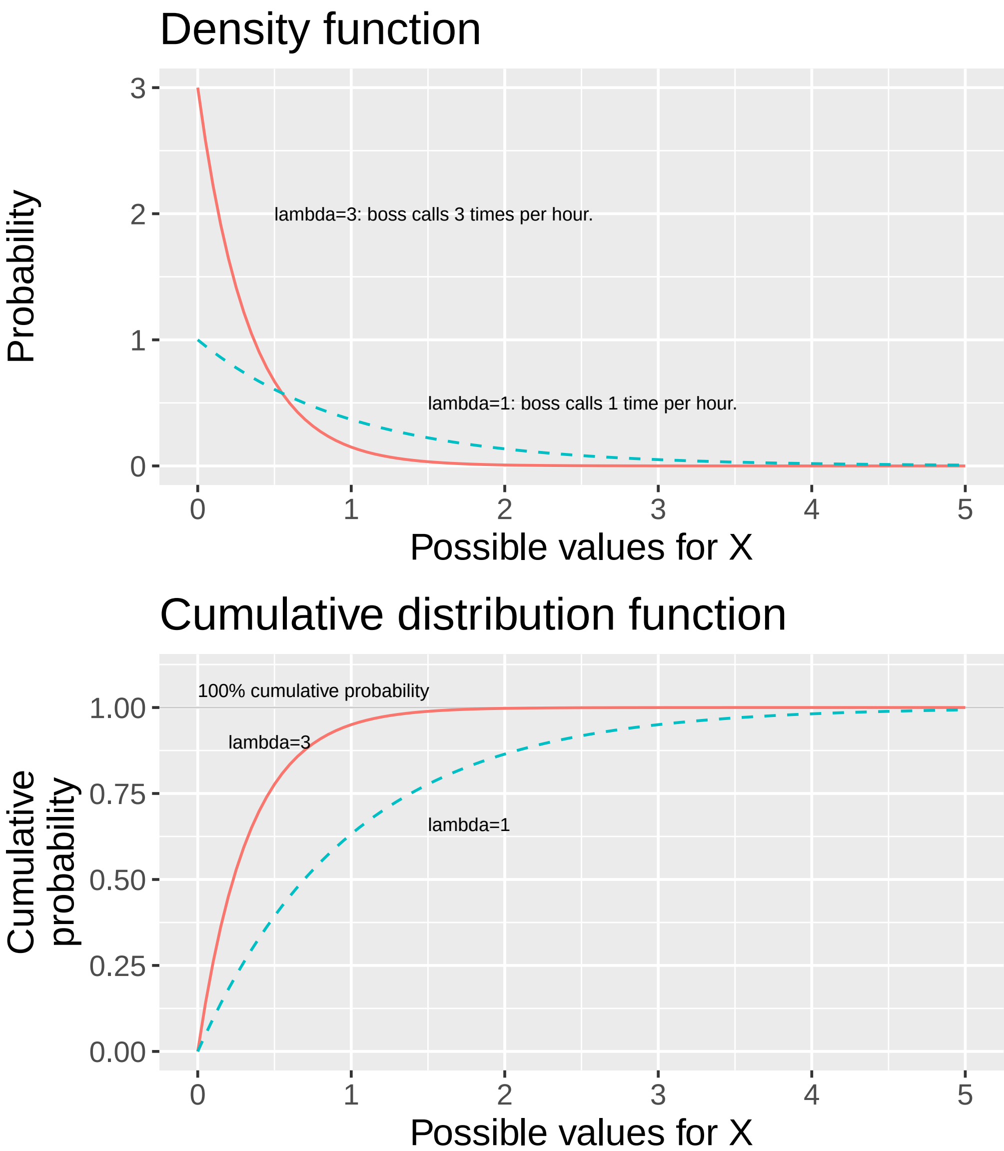


Figure 17.2: The exponential distribution: density function, $f(x)$, and cumulative distribution function, $F(x)$.

The probability that the boss does not call the first hour is approximately 5%. Figure 17.2 illustrates the density function and cumulative distribution function for the exponential distribution with two examples for $\lambda = 1$ and $\lambda = 3$, where λ is the number of events on average over a time interval. The value for λ is what determines the shape of the probability distribution.

The upper graph shows the density function, equation (17.5). Since X is defined for all positive values, the probability never reaches 0 by definition, but only infinitely close. In the lower graph the cumulative distribution function is illustrated. The cumulative distribution function never reaches the value 1, but instead comes infinitely close to 1.

To read the probability of approximately 5% when we calculated $P(X > 1) = 1 - F(x) = e^{-3}$ we look in the lower graph and the solid black line, for which $\lambda = 3$. When $X = 1$ on the x-axis this line is near $y = 1$, that is 100% probability. For $X = 1$ this line is at $y = F(X = 1) \approx 0.95$. This means that $P(X > 1) = 1 - F(x) \approx 0.05$. If a variable X follows the exponential distribution we write:

$$X \sim EXP(\lambda)$$

The expected value for a variable X that follows the exponential distribution:

$$E(X) = \frac{1}{\lambda}$$

$E(X)$ = average waiting time. Variance for the exponential distribution:

$$var(X) = \frac{1}{\lambda^2}$$

For higher values for λ, more events on average per time unit, we get a smaller expected value, shorter waiting time, and smaller variance. In figure 17.2 the line with higher value for λ has steeper slope closer to $x = 0$.

17.4 From Poisson to Gamma

Say now that we have a variable that follows a Poisson process and that we want to calculate the probability that event number k occurs within time interval x, where k is any positive integer. For this we use the Gamma distribution, whose density function f can be written:

$$f(x; k, \theta) = P(X = x) = \frac{x^{k-1} e^{-x/\theta}}{\Gamma(k)\, \theta^k} \tag{17.8}$$

where $\theta = 1/\lambda$ is the average waiting time for an event. θ is the Greek letter lowercase theta and λ is as before the average number of events per time unit. The density function uses three input values: x, the time interval we will calculate the probability for, k, number of events that should have occurred within the course of x, and θ, the average expected waiting time.

The expression $\Gamma(k)$ in the denominator of equation (17.8) is what is called the gamma function, which is a generalization of factorial. There is no room here to go through the gamma function in detail. For our review it suffices to note that for all positive integers $k = 1, 2, 3, ...$ the gamma function can be described as:

$$\Gamma(k) = (k-1)! \tag{17.9}$$

where Γ is the Greek letter capital gamma. If $k = 3$: $\Gamma(3) = (3-1)! = 2 \cdot 1 = 2$. One way to write the gamma function is the following integral:

$$\Gamma(x) = \int_0^\infty y^{x-1} e^{-y}\, dx, \text{ where } y > 0$$

which sums from 0 to positive infinity for all positive real numbers. The cumulative distribution function of the gamma distribution:

$$F(x; k, \theta) = P(X \le x) = 1 - e^{-x/\theta} \sum_{n=0}^{k-1} \frac{(x/\theta)^n}{n!} \tag{17.10}$$

where x is the waiting time, k is the number of the event we will estimate the probability for and θ is the average waiting time for such an event. Let us use function F to estimate the probability that we have to wait for more than two hours before the boss (from the example in the previous section) has managed to call us three times, $k = 3$. The boss usually calls three times per hour, which is why $\lambda = 3$ and $\theta = 1/\lambda = 1/3$ hour. The waiting time two hours we write $x = 2$ and we therefore have $x/\theta = 2/(1/3) = 6$.

We will calculate the probability $P(X > 2)$ but the cumulative distribution function in equation (17.10) is defined so that $F(x) = P(X < x)$. To calculate $P(X > x)$ we write (compare equation (17.7)):

$$P(X > x) = 1 - F(x)$$

We replace $F(x)$ in the right side with the definition from equation (17.10):

$$P(X > x) = 1 - \left(1 - e^{-x/\theta} \sum_{n=0}^{k-1} \frac{(x/\theta)^n}{n!}\right) = e^{-x/\theta} \sum_{n=0}^{k-1} \frac{(x/\theta)^n}{n!}$$

Now we calculate the probability for $X > 2$:

$$P(X > 2) = e^{-6} \sum_{n=0}^{3-1} \frac{6^n}{n!} = e^{-6} \cdot 25 \approx 0.06197 \tag{17.11}$$

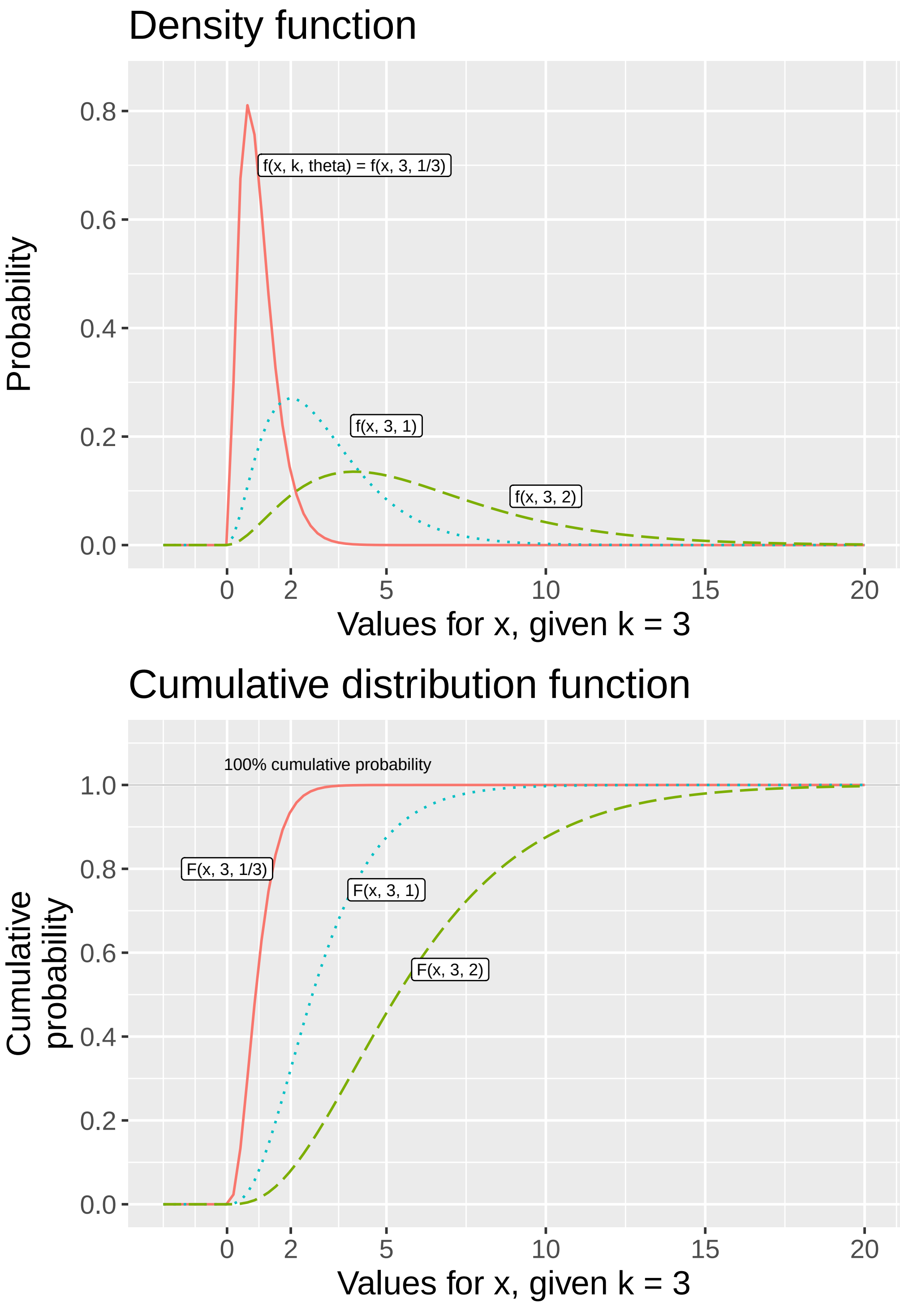


Figure 17.3: The gamma distribution with $k = 3$: density function, $f(x)$, and the cumulative distribution, $F(x)$.

There is approximately 6.2% probability that we have to wait for more than 2 hours for the boss to call us for the third time. Figure 17.3 illustrates the gamma distribution's density function and cumulative distribution function with three different examples. All lines describe the probability that the boss calls three times, $k = 3$, but based on different average waiting times, $\theta = 1/\lambda$, namely $\frac{1}{3}$, 1 or 2. The number $\frac{1}{3}$ corresponds to 20 min and that the boss calls three times per hour. The number 1 means one hour and that the boss calls once per hour. The number 2 corresponds to two hours and means that the boss calls every other hour.

The upper graph illustrates the density function, equation (17.8) and the lower graph illustrates three versions of the cumulative distribution function, equation (17.10). Note in the upper graph how larger values for θ mean greater spread in the probability distribution.

In both graphs $x = 2$ is marked on the horizontal x-axis. The calculation in equation (17.11) can for example be compared with the lower graph and the solid line. At $x = 2$ the line is near $y = 1$, which means that most of the cumulative probability is covered by the values $X < x$. If a random variable X follows the gamma distribution we write:

$$X \sim \Gamma(k, \theta)$$

where k is the number of the event for which we will estimate the probability and θ is the average waiting time, which we described above. Sometimes the symbols $\alpha = k$ and $\beta = \theta = 1/\lambda$ are used instead. The expected value for a variable X that follows the gamma distribution is:

$$E(X) = k\theta$$

If we have $X \sim \Gamma\left(3, \frac{1}{3}\right)$, as the example above with the calling boss, we get the expected value $E(X) = 3 \cdot \frac{1}{3} = 1$. The average waiting time for 3 calls is one hour. The variance for the gamma distribution:

$$var(X) = k\theta^2$$

This means that higher values for θ, longer waiting time, result in greater expected value and variance. This is illustrated in figure 17.3 where the lines in the upper graph with higher values for θ show probability distributions with higher mean values that are also more spread out along the x-axis.

17.5 Chi-squared Distributions

Say now that we have a variable X that follows a gamma distribution with $\theta = 2$ and $k = r/2$ where r is any positive integer. The distribution we then get is called the chi-squared distribution and is central to a large amount of analytical work. The density function in equation (17.8) for X can then be written:

$$f(x; k = r/2) = \frac{x^{\frac{r}{2}-1} e^{-x/2}}{\Gamma\left(\frac{r}{2}\right) 2^{r/2}}, \quad 0 < x < \infty \tag{17.12}$$

where $\Gamma\left(\frac{r}{2}\right)$ is the gamma function. Equation (17.12) describes the density function for the chi-squared distribution. The cumulative distribution function $F(x; r)$ for the chi-squared distribution can be written:

$$F(x; r) = P(X \leq x) = \int_0^x \frac{y^{\frac{r}{2}-1} e^{-y/2}}{\Gamma\left(\frac{r}{2}\right) 2^{r/2}} dy \tag{17.13}$$

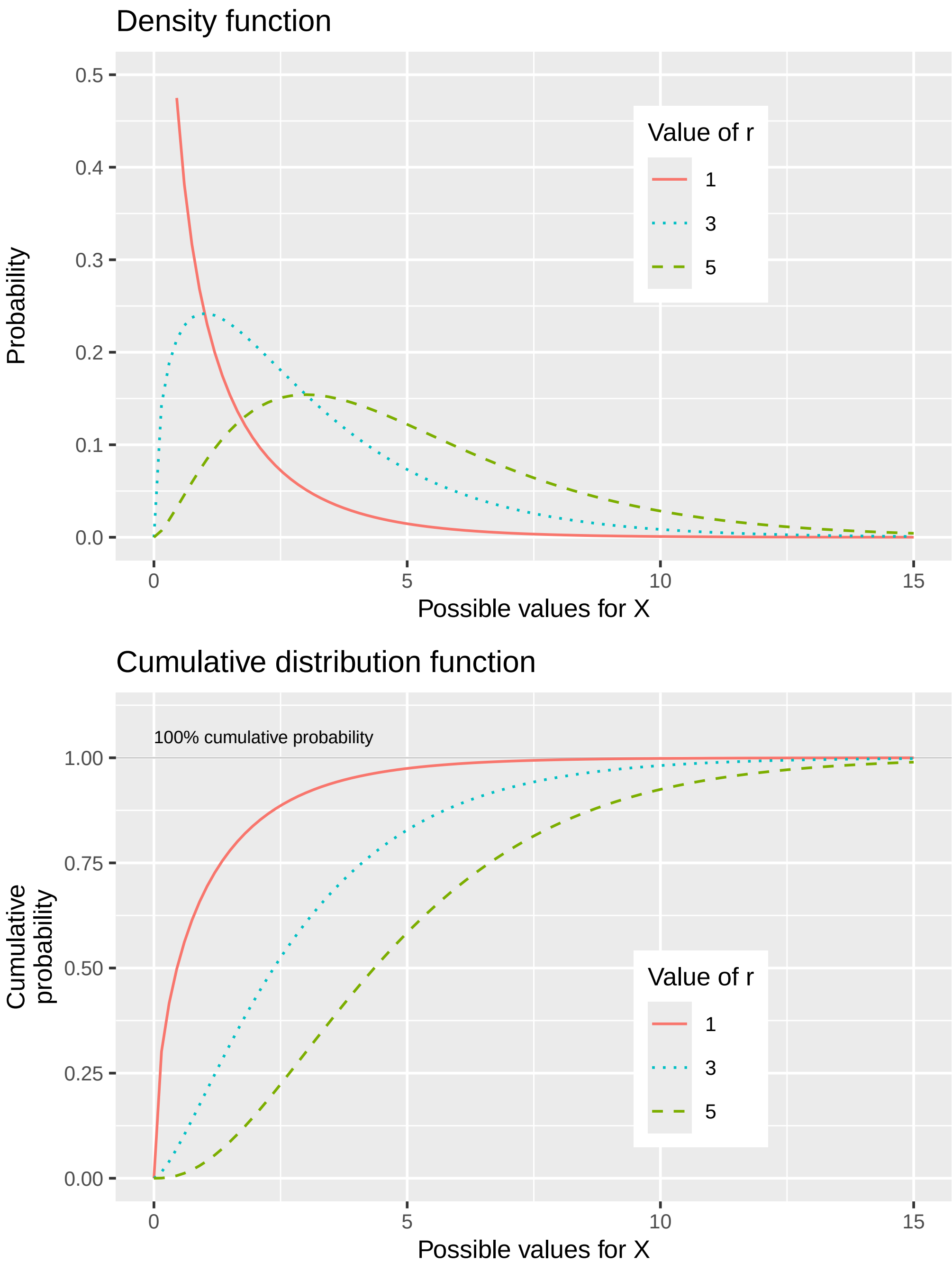


Figure 17.4: Chi-square distribution: density function, $f(x)$, and cumulative distribution function, $F(x)$.

Figure 17.4 describes the density function and cumulative distribution function for the chi-squared distribution. The upper graph describes the density function for three different values of r. The lower graph describes the corresponding cumulative distribution functions with the three values for r. If a variable X follows the chi-squared distribution we describe this as:

$$X \sim \chi^2(r)$$

Alternatively $X \sim \chi^2_r$, where χ is the Greek letter lowercase chi. Generally the expected value for the chi-squared distribution can be described as:

$$E(X) = r$$

Table 17.1: Frequency distribution

Value	5	6	7	8	9	10
Frequency	12	14	15	13	12	11

The expected value for a chi-squared distribution is equal to r. The variance for the chi-squared distribution is equal to $2r$:

$$var\,(X) = 2r$$

This is also illustrated in figure 17.4 in that the lines with higher values for r in the upper graph have higher mean values, further to the right in the graph. The greater variance is illustrated by the line with the lowest value for r also having a steep slope near the vertical y-axis. Since the chi-squared distribution is continuous we use the cumulative distribution function $F\,(x)$ in equation (17.13) to calculate probabilities.

The chi-squared distribution is often used in statistical analysis. Here we give only a simple example. Table 17.1 describes a frequency distribution where the first row indicates the value and the second row indicates observed frequencies of each value.

Say now that we want to know the probability that these observations come from a discrete uniform probability distribution (introduced in section 16.7). We compare the observed frequencies against the value we could have “expected” at uniform probability. If all values $\{5, 6, 7, 8, 9, 10\}$ have the same probability, all values should have the same frequency:

$$\frac{\sum x_i}{n} = \frac{12 + 14 + 15 + 13 + 12 + 11}{6} = 12.83$$

To calculate the probability we use the following statistic:

$$\chi^2 = \frac{\sum_i^n (x_i - \mu)^2}{\mu} \tag{17.14}$$

where in this case we have $\mu = 12.83$, the expected value we want to compare against. The statistic χ^2 follows the chi-squared distribution with $r = n - 1$ degrees of freedom, which in this case becomes $r = n - 1 = 6 - 1 = 5$.

Equation (17.14) calculates the cumulative probability, $F\,(x)$, that our observations deviate from the value we “expect”, based on the hypothetical uniform probability distribution that we want to check if the observations come from. That is, the higher χ^2 we calculate, the lower probability that our observations come from the type of distribution we compare against. We sum the squared deviation from μ for each frequency value. For observation $i = 1$ we get $(x_i - \mu)^2 = (12 - 12.83)^2 = 0.6889$. Our test statistic becomes:

$$\chi^2 = \frac{\sum (x_i - \mu)^2}{\mu} \approx 10.833$$

The chi-squared distribution is so common that we here compare our calculated χ^2 against pre-calculated results for $F\,(x)$. Table 17.2 describes probabilities for calculated results from the chi-squared distribution’s cumulative probability function $F\,(x) = P\,(X \leq x)$. The percentages in the column headers describe how large a proportion of the distribution is equal to or less than the values in this column. In the table we see that our calculated χ^2-value $10.833 > 9.236$, which is in the column for 90%. This indicates that with 90% probability our observations do not follow a uniform probability distribution.

Table 17.2: The chi-squared distribution: critical values with respect to degrees of freedom, r, and cumulative probability

	50%	75%	90%	95%	99%
r=1	0.455	1.323	2.706	3.841	6.635
r=2	1.386	2.773	4.605	5.991	9.210
r=3	2.366	4.108	6.251	7.815	11.345
r=4	3.357	5.385	7.779	9.488	13.277
r=5	4.351	6.626	9.236	11.070	15.086
r=6	5.348	7.841	10.645	12.592	16.812
r=7	6.346	9.037	12.017	14.067	18.475
r=8	7.344	10.219	13.362	15.507	20.090
r=9	8.343	11.389	14.684	16.919	21.666
r=10	9.342	12.549	15.987	18.307	23.209

17.6 F-distributions

Here follows a brief introduction to the F-distribution, which is also commonly occurring in statistical analysis. If we have two independent random variables X and Y that both follow the chi-squared distribution with degrees of freedom r_x and r_y, we calculate a new variable S:

$$S = \frac{X/r_x}{Y/r_y}$$

Variable S follows what is called the F-distribution, which can be written:

$$S \sim F\left(r_x, r_y\right)$$

where the F-distribution's shape is dependent on the values in r_x and r_y. We do not go through the distribution's density function and cumulative distribution function here but are content to just call these $f\left(x\right)$ and $F\left(x\right)$. Figure 17.5 illustrates the density function $f\left(x\right)$ in the upper graph, and the cumulative distribution function $F\left(x\right)$ in the lower graph.

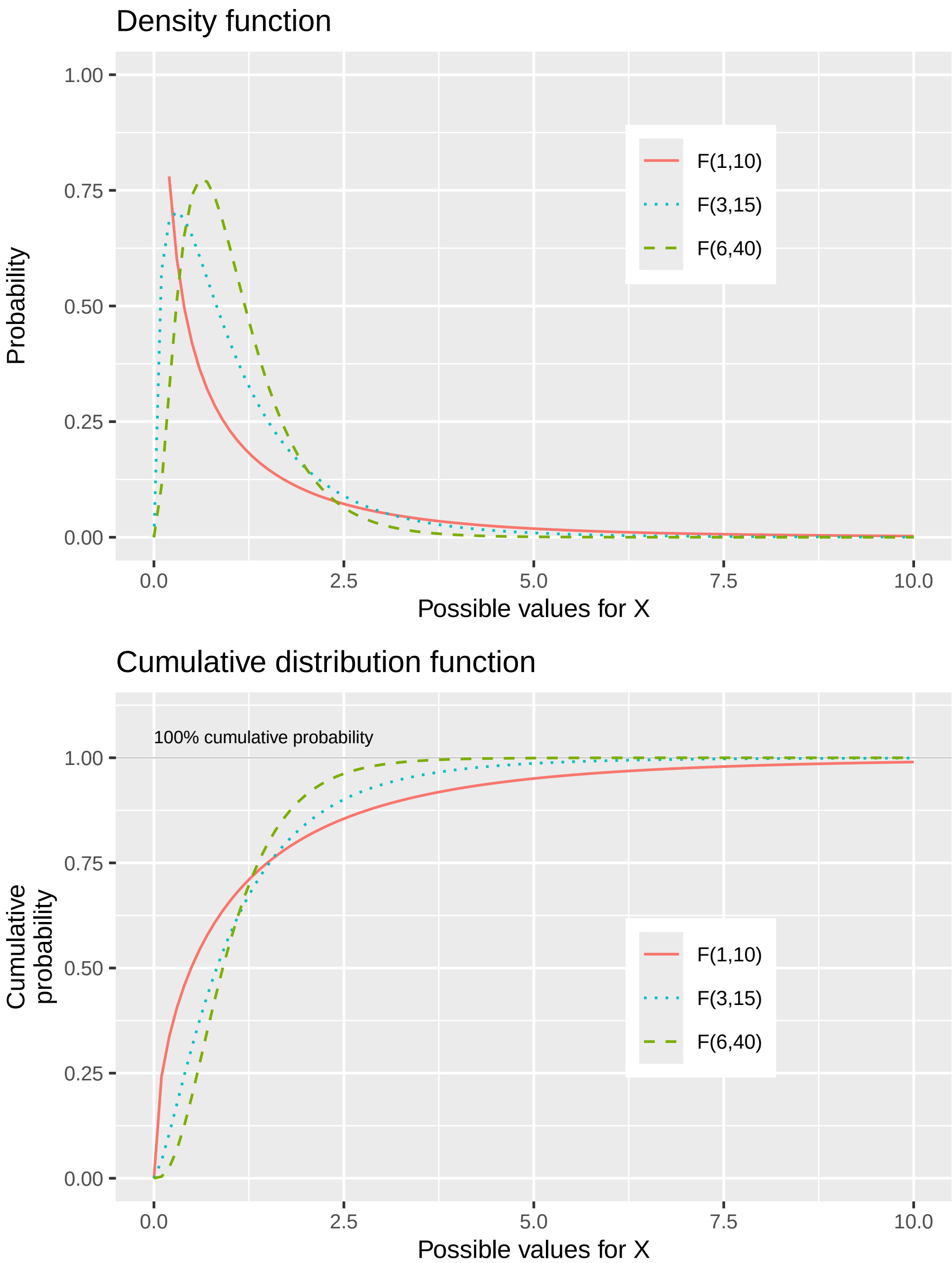


Figure 17.5: The F-distribution: density function, $f(x)$, and cumulative distribution function, $F(x)$.

Table 17.3 describes pre-calculated probabilities for the F-distribution based on its cumulative distribution function, $F(x)$, where the F-distribution can be written as $F(r_x, r_y) = F(\text{df } 1, \text{df } 2)$. The table uses the abbreviation "df" for degrees of freedom. Columns describe different values depending on parameter df_1 (i.e. r_x). The rows are divided for different values on the second parameter df_2 (i.e. r_y). In column 2 with the heading $P(X \leq x)$ the proportion of the distribution that is equal to or less than 0.95 (95%) and 0.99 (99%) of the distribution is described.

When the lines reach the cumulative probability 0.95 or 0.99, near 1 on the vertical y-axis, the values on the x-axis correspond to the value in the table.

Table 17.3: The F-distribution: critical values with respect to the degrees of freedom df 1 and df 2, $F(\text{df }1, \text{df }2)$.

df 2	P(X x)	df1=1	df1=2	df1=3	df1=5	df1=10
5	0.95	6.61	5.79	5.41	5.05	4.74
5	0.99	16.26	13.27	12.06	10.97	10.05
10	0.95	4.96	4.10	3.71	3.33	2.98
10	0.99	10.04	7.56	6.55	5.64	4.85
20	0.95	4.35	3.49	3.10	2.71	2.35
20	0.99	8.10	5.85	4.94	4.10	3.37
30	0.95	4.17	3.32	2.92	2.53	2.16
30	0.99	7.56	5.39	4.51	3.70	2.98
60	0.95	4.00	3.15	2.76	2.37	1.99
60	0.99	7.08	4.98	4.13	3.34	2.63
120	0.95	3.92	3.07	2.68	2.29	1.91
120	0.99	6.85	4.79	3.95	3.17	2.47

17.7 Chapter summary

- A continuous probability distribution has an infinite number of possible values, for example the continuous interval between 0 and 1. The density function, $f(x)$, describes the probability for all these possible values. The cumulative distribution function, $F(x)$, describes the cumulative probability that variable X will assume a value less than or equal to x, $F(x) = P(X \leq x)$.
- We can also calculate $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$. Since the number of possible values in a continuous probability distribution is infinite, the probability for each individual value also approaches 0. Instead we use $F(x)$ and calculate the probability for an interval in the variable: $P(X \leq x)$.
- A uniform probability distribution has the same probability for all values of variable x it is defined for. Density function: $f(x) = 1/(b-a)$, $a \leq x \leq b$, where a and b indicate the interval for which f is defined. Cumulative distribution function: $F(x) = P(X \leq x) = (x-a)/(b-a)$ where $a \leq x < b$. $F(x) = 0$, $x < a$ and $F(x) = 1$, $x \geq b$. Expected value: $E(X) = \frac{1}{2}(a+b)$.
- Exponential distribution's density function: $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$ and $f(x) = 0$ for all other x. Cumulative distribution function: $F(x) = 1 - e^{-\lambda x}$. If variable X follows the exponential distribution this can be written as $X \sim EXP(\lambda)$ where the expected value $E(X) = 1/\lambda$ and variance $var(X) = 1/\lambda^2$.
- Gamma distribution's density function for variable X with values x is written $f(x;k,\theta) = \frac{x^{k-1}e^{-x/\theta}}{\Gamma(k)\theta^k}$, where $\Gamma(k)$ is the gamma function, a generalization of factorial, which for the positive integer k is $\Gamma(k) = (k-1)!$. k is the number of the event for which we will calculate the probability, $\theta = 1/\lambda$ is the average waiting time for such an event, where λ is the average number of events per time unit. x is the time period for which we will calculate the probability. Cumulative distribution function: $F(x) = P(X \leq x) = 1 - e^{-x/\theta}\sum_{n=0}^{k-1}\frac{(x/\theta)^n}{n!}$. Expected value: $E(X) = k\theta$. Variance: $var(X) = k\theta^2$.
- Chi-squared distribution's density function: $f(x;r,\theta) = \frac{x^{\frac{r}{2}-1}e^{-x/2}}{\Gamma(\frac{r}{2})2^{r/2}}$, where $0 < x < \infty$. Cumulative distribution function: $F(x) = P(X \leq x) = \int_0^x \frac{y^{\frac{r-2}{2}}e^{-y/2}}{\Gamma(r/2)2^{r/2}}dy$. If variable X follows the chi-squared distribution it can be written $X \sim \chi_r^2$ or $X \sim \chi^2(r)$, where r is also called degrees of freedom. Expected value: $E(X) = r$. Variance: $var(X) = 2r$.
- F-distribution: If the individual variables X and Y follow the chi-squared distribution with r_x and r_y, the following variable follows the F-distribution: $S = (X/r_x)/(Y/r_y)$. If variable Z follows the F-distribution this can be written $Z \sim F(r_1, r_2)$.

17.8 Exercises

1. X is continuously uniformly distributed on $[2, 8]$.

(a) What is the probability density (f(x))?

(b) Calculate (E(X)=(a+b)/2).

(c) Calculate (operatorname{var}(X)=(b-a)^{2}/12).

Answer: (a) $f(x) = 1/6$ for $x \in [2, 8]$. (b) $E(X) = 5$. (c) $\operatorname{var}(X) = 3$.

2. X is uniformly distributed on $[0, 10]$.

(a) What is (f(x))?

(b) Calculate (P(3leq Xleq7)).

(c) Calculate (E(X)).

Answer: (a) $f(x) = 1/10$ for $x \in [0, 10]$. (b) $P = 0.4$. (c) $E(X) = 5$.

3. A continuous random variable has density $f(x) = 2x$ for $x \in [0, 1]$.

(a) Verify that (int_{0}^{1}f(x),dx=1).

(b) Calculate (E(X)=int_{0}^{1}xf(x),dx).

(c) Calculate (operatorname{var}(X)=E(X$^{\{2\})\text{-}[E(X)]}${2}).

Answer: (a) $\int_0^1 2x\,dx = [x^2]_0^1 = 1$. (b) $E(X) = 2/3$. (c) $\operatorname{var}(X) = 1/18$.

4. A continuous random variable has density $f(x) = 3x^2$ for $x \in [0, 1]$.

(a) Verify that (int_{0}^{1}f(x),dx=1).

(b) Calculate (E(X)).

(c) What is (P(Xleq0.5))?

Answer: (a) $\int_0^1 3x^2\,dx = 1$. (b) $E(X) = 3/4$. (c) $P(X \leq 0.5) = (0.5)^3 = 0.125$.

5. $X \sim \operatorname{Exp}(\lambda = 2)$.

(a) Calculate (E(X)=1/lambda).

(b) Calculate (operatorname{var}(X)=1/lambda^{2}).

(c) Calculate (P(X>1)=e^{-lambdacdot1}).

Answer: (a) $E(X) = 0.5$. (b) $\operatorname{var}(X) = 0.25$. (c) $P(X > 1) = e^{-2} \approx 0.135$.

6. $X \sim \operatorname{Exp}(\lambda = 1)$.

(a) Calculate (E(X)).

(b) Calculate (P(Xleq1)=1-e^{-1}).

(c) Calculate (P(X>2)=e^{-2}).

Answer: (a) $E(X) = 1$. (b) $P(X \leq 1) \approx 0.632$. (c) $P(X > 2) = e^{-2} \approx 0.135$.

7. $X \sim \chi^2(r = 4)$.

(a) Calculate (E(X)=r).

(b) Calculate (operatorname{var}(X)=2r).

(c) What is the chi-squared distribution commonly used for in statistics?

Answer: (a) $E(X) = 4$. (b) $\operatorname{var}(X) = 8$. (c) Testing goodness of fit and independence in contingency tables.

8. $X \sim \chi^2(r = 6)$.

(a) Calculate (E(X)).

(b) Calculate (operatorname{var}(X)).

(c) How is the chi-squared distribution related to the standard normal distribution?

Answer: (a) $E(X) = 6$. (b) $\operatorname{var}(X) = 12$. (c) If $Z \sim N(0,1)$ then $Z^2 \sim \chi^2(1)$; the sum of r squared standard normals follows $\chi^2(r)$.

9. A Gamma distribution has parameters $k = 4$ and $\theta = 2$.

(a) Calculate (E(X)=ktheta).

(b) Calculate (operatorname{var}(X)=ktheta^{2}).

(c) Calculate the standard deviation (s(X)).

Answer: (a) $E(X) = 8$. (b) $\operatorname{var}(X) = 16$. (c) $s(X) = 4$.

10. $X \sim F(r_1 = 2, r_2 = 3)$.

(a) How is the F-distribution constructed from chi-squared variables?

(b) What is the F-distribution commonly used for in regression analysis?

(c) What happens to the F-statistic when the regression model has no explanatory power?

Answer: (a) It is the ratio of two scaled chi-squared variables: $S = (X/r_1)/(Y/r_2)$. (b) Testing whether the regression model explains significant variation (F-test). (c) The F-statistic approaches 1 (the ratio of equal chi-squared variables).

11. Explain the difference between discrete and continuous random variables.

(a) Give an example of a discrete random variable.

(b) Give an example of a continuous random variable.

(c) Why is (P(X=x)=0) for any specific value (x) when (X) is continuous?

Answer: (a) Number of heads in coin flips. (b) Height or waiting time. (c) There are infinitely many values in any interval, so each individual value has zero probability.

12. Two models for waiting times both have $E(X) = 5$: Uniform $[0, 10]$ and $\text{Exp}(\lambda = 0.2)$.

(a) Calculate (operatorname{var}(X)) for the uniform model.

(b) Calculate (operatorname{var}(X)) for the exponential model.

(c) What is an advantage of the exponential model for modeling waiting times?

Answer: (a) $\operatorname{var}(X) = (10 - 0)^2/12 \approx 8.33$. (b) $\operatorname{var}(X) = 1/\lambda^2 = 25$. (c) The exponential distribution is memoryless and allows for long waiting times (heavy right tail).

Chapter 18

Normal and t distributions

This chapter introduces normal and t-distributions, which are central to a large amount of research and analytical work.

18.1 Probability with normal variables

The normal distribution is recognized by its symmetric bell-shaped illustration. It occurs from time to time in social debate and media. Similar to the chi-square and F-distribution, we will also in this case use pre-calculated probabilities and critical values, but before that we shall go through some of the mathematics behind it. If we have a random variable X that follows a normal distribution, the probability density function f of the normal distribution can be written:

$$f(x, \mu, \sigma) = \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}}, \quad -\infty < x < \infty \tag{18.1}$$

The density function f is defined for all values $X = x$ between negative and positive infinity. The letter μ(Greek letter lowercase mu) symbolizes expected value and mean for X, e is Euler's number, approximately 2.718, and π is the number pi, approximately 3.142. The symbol σ(Greek letter sigma) stands for standard deviation for variable X. The cumulative distribution function F of the normal distribution is written:

$$F(x, \mu, \sigma) = P(X \leq x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt \tag{18.2}$$

where we take the integral from negative infinity $-\infty$ to the value x. The normal distribution is a collective name for distributions with certain common characteristics but which can have different mean μ and standard deviation σ. That the random variable X follows a normal distribution can be written $X \sim N(\mu, \sigma)$, where μ and σ represent the current mean and standard deviation for X.

Figure 18.1 gives three different examples of normal distributions where the upper graph illustrates the probability density function of the normal distributions, $f(x)$, and the lower graph shows the cumulative distribution function, $F(x)$. The only difference between the three distributions is that they have different mean (μ) and standard deviation (σ). In the upper graph where the density function is illustrated, all three curves are evenly distributed around their respective means. On both sides of the mean the lines go down, which means that the further from the mean we move, the more unusual the results become. In the tails below and above the mean the probabilities are relatively smaller. A normal distribution with larger variance has thicker tails to the right and left of the mean. In the lower graph the three corresponding cumulative probability functions are illustrated.

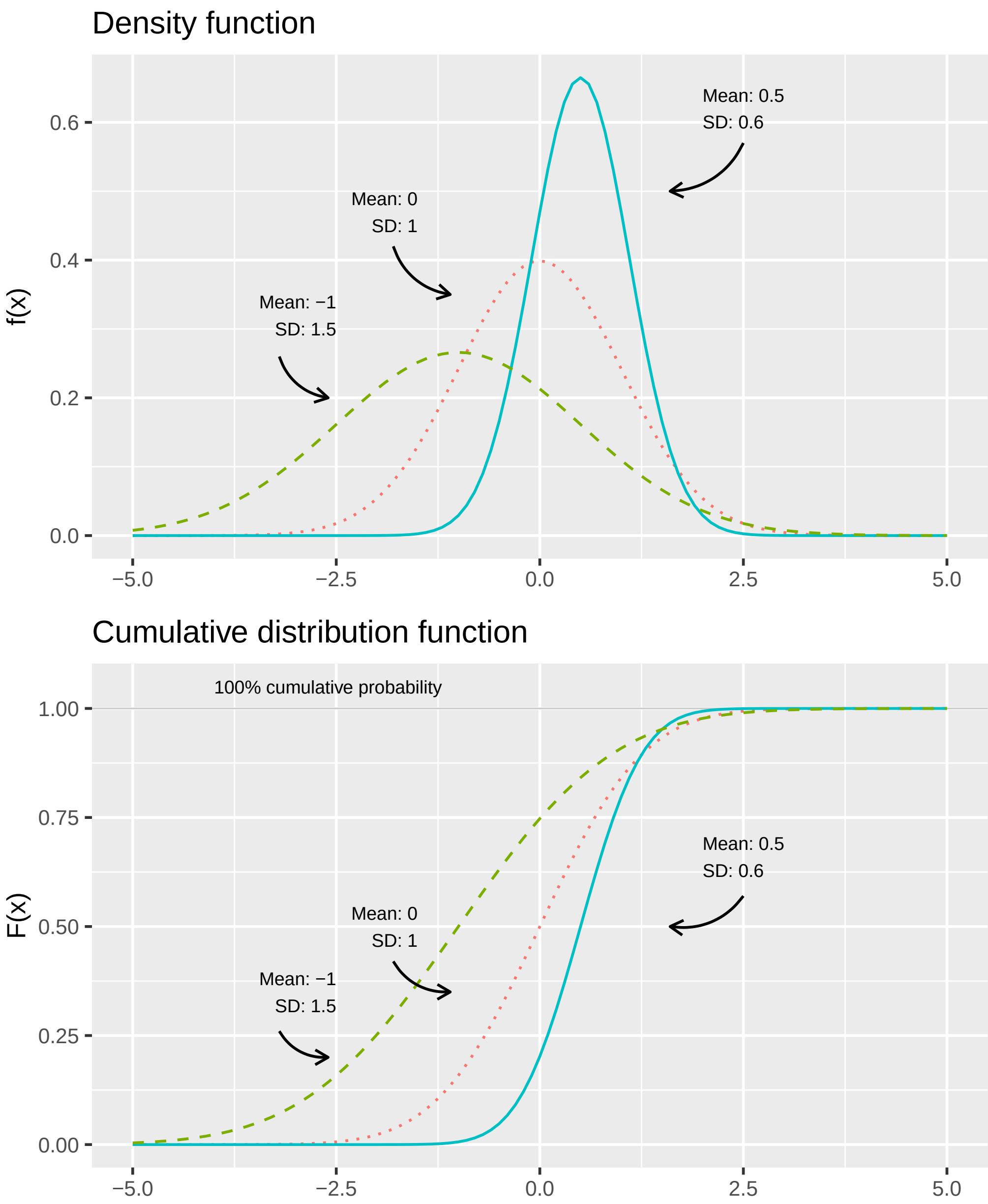


Figure 18.1: The normal distribution: density function $f(x)$ and the cumulative distribution function $F(x)$

18.2 Standardized normal distribution

A normal distribution which have mean $\mu = 0$ and standard deviation $\sigma = 1$ is called the standardized normal distribution or Z distribution. If we put $\mu = 0$ and $\sigma = 1$ into the density function in equation (18.1) we get:

$$f_X\left(x, \mu = 0, \sigma = 1\right) = \frac{e^{-\frac{1}{2}\left(\frac{x-0}{1}\right)^2}}{1 \cdot \sqrt{2\pi}} = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \tag{18.3}$$

The cumulative distribution for the standardized normal distribution is:

$$F_X\left(x, \mu = 0, \sigma = 1\right) = P\left(X \leq x\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}}\, dt \tag{18.4}$$

where we use $F\left(x\right)$ to calculate the probability that variable $X \leq x$, by calculating the integral from $-\infty$

up to and including the value x. The standard normal distribution is so common that its cumulative distribution function (equation (18.4)) is written as $\Phi_X(x) = F_X(x, \mu = 0, \sigma = 1)$, where Φ is the Greek letter capital phi. If X follows a standardized normal distribution we write $X \sim N(0, 1)$.

We rarely encounter any real data that naturally follows a normal distribution with exactly mean 0 and standard deviation 1. In section 15.8 we went through instead how we standardize a collection of values so that these get exactly mean 0 and standard deviation 1:

$$Z_i = \frac{X_i - \bar{X}}{s_X} \tag{18.5}$$

where s_X is estimated standard deviation for X. Standardized normal distributions are so commonly occurring that they are often described as variable Z whose specific values are denoted z. This probability distribution has several known probability measures based on standard deviations, which are often used in statistical analysis. Figure 18.2 illustrates this, where the upper graph describes the probability density function f of the standard normal distribution and the lower graph describes the cumulative distribution function F.

In the upper graph some of the standard deviations below and above the mean 0 are marked. The cumulative distribution function $F(Z \leq z)$ describes, as before, what proportion of all values in variable Z are less than or equal to z. Since the standard normal distribution is also called the Z-distribution, this type of probabilities are often described as precisely Z-values.

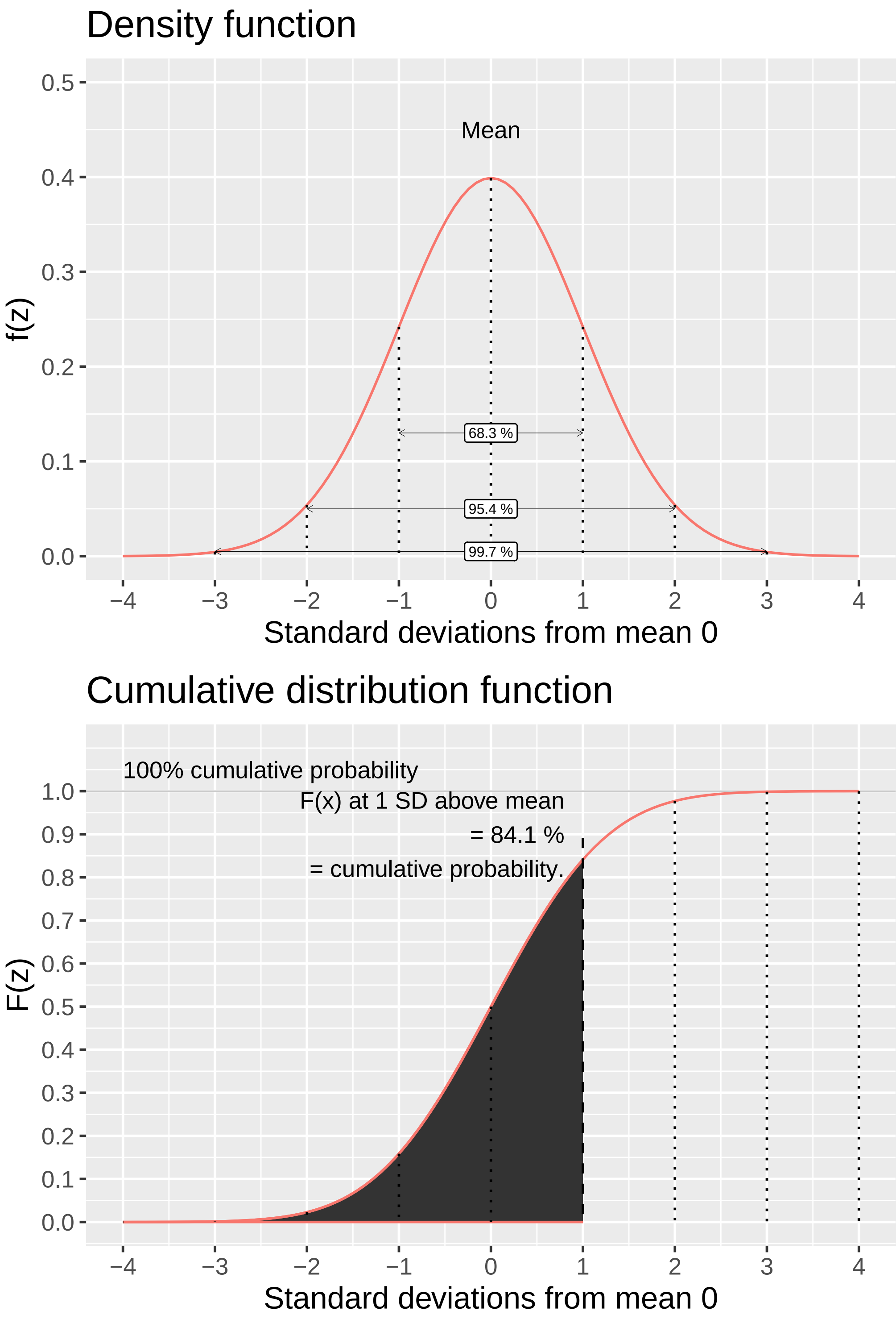


Figure 18.2: Standard normal distribution: density function $f(z)$ and cumulative distribution function $F(Z \leq z)$

Table 18.1 reports pre-calculated probabilities for the standard normal distribution $Z \sim N(0,1)$. The table shows calculated results where the variable Z is less than or equal to a value z, $F(z) = P(Z \leq z)$. Say for example that we want to know the probability that variable Z is equal to or less than the mean plus one standard deviation, that is $Z \leq 1$. We look at the row for $z = 1$ and column 0. This row and column means that $z = 1.0$. The value in the cell shows that $P(Z \leq 1) = 0.8413$. We also see this in figure 18.2, where the lower graph illustrates how the shaded area up to one standard deviation above the mean corresponds to 84.1%.

Table 18.1: Standardized normal distribution, $F(z) = P(Z \leq z)$

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

The normal distribution is symmetric and has equally large outcome space below and above the mean, which means that $F(0) = P(X \leq 0) = 0.5$. That is, 50% of all possible values exist below and above the mean $\mu = 0$ respectively. This means that $F(-x) = 1 - F(x)$. For example:

$$F(-1) = 1 - F(1) = 1 - 0.8413 = 0.1587$$

From the table we see that:

$$F(2) = P(X \leq 2) = 0.97725$$

This also means that:

$$F(-2) = 1 - 0.9772 = 0.02275$$

which illustrates how the area beyond two standard deviations below and above the mean respectively represents approximately 4.55% of the distribution (upper graph in figure 18.2).

We have in various examples compared average expected life expectancy for men and women. Say now that life expectancy for men and life expectancy for women respectively represent two populations and that both of these are approximately normally distributed (see section 18.4 below). Now we want to know if the two populations' means differ, that is if $\mu_{\text{women}} = \mu_{\text{men}}$. To estimate the probability of this we use the standard normal distribution and the following statistic:

$$z = \frac{\bar{X}_{\text{women}} - \bar{X}_{\text{men}}}{\left(\frac{s^2_{\text{women}}}{n_{\text{women}}} + \frac{s^2_{\text{men}}}{n_{\text{men}}}\right)^{1/2}} \tag{18.6}$$

We calculate the z-value with the observations that we used earlier:

$$z \approx \frac{83.95 - 80.38}{\left(\frac{1.032}{290} + \frac{1.663}{290}\right)^{1/2}} \approx 37.0$$

This z-value can be compared against the pre-calculated values for the standard normal distribution's cumulative distribution function $F(Z \leq z)$ in table 18.1. Our calculated z-value deviates so much from the mean 0 that it is not reported in the table, which only extends to $z = 3.09$, where $F(3.09) = 0.999$. In this case our z-value is so far from the mean that $F(z)$ becomes even closer to 0 than the values included in the table. Our z-value thus ends up far out in the standard normal distribution's tail to the right of (above) the mean, which indicates that it is very unlikely that the two populations (men's and women's life expectancy) have the same mean.

18.3 The usefulness of the normal distribution 1

One reason why the normal distribution is so well-known is that many phenomena in reality are distributed approximately like a normal distribution. Figure 18.3 illustrates some examples. In the upper left graph the distribution of height among 3,000 randomly selected prisoners is shown, produced by Scotland Yard and reproduced by Macdonell (1902). This distribution was noted among other things in two famous articles by William Sealy Gosett, under the pseudonym Student (1908a; 1908b).

The solid normal distribution curve uses the same mean and standard deviation as the observations. As can be seen, the distribution of the prisoners' height is close to the normal distribution curve, but not exactly the same. The 3,000 observations for the prisoners are after all discrete values while the normal distribution curve is a continuous variable. Human height, like several other physical measurements, is known to often be approximately normally distributed.

The upper right graph shows total points per game in the North American basketball league National Basketball Association (NBA), for all games from season 1999/2000 to 2018/2019, totaling 24,144 games. These data also follow the normal distribution curve relatively well. The distribution has a small tail to the right that we ignore here, but which arises because Chicago Bulls played against Atlanta Hawks on March 1, 2019, a game that after four overtimes ended with 168-161 after Trae Young scored 49 points. This was the third highest-scoring game in NBA history.

The lower left graph shows 580 observations from Sweden's 290 municipalities on average expected life expectancy for women and men. One observation is a municipality and small municipalities weigh as heavily as large municipalities in the graph. These two distributions also resemble quite well two normal distribution curves.

The lower right graph also shows data from Sweden's municipalities, regarding number of books per municipal resident in municipal public libraries. Number of books is shown on a logarithmic scale, whereupon the observations follow a normal distribution curve relatively well. This is called the variable following the log-normal distribution, which is a logarithmic variant of the normal distribution where the variable's values are in logarithmic form.

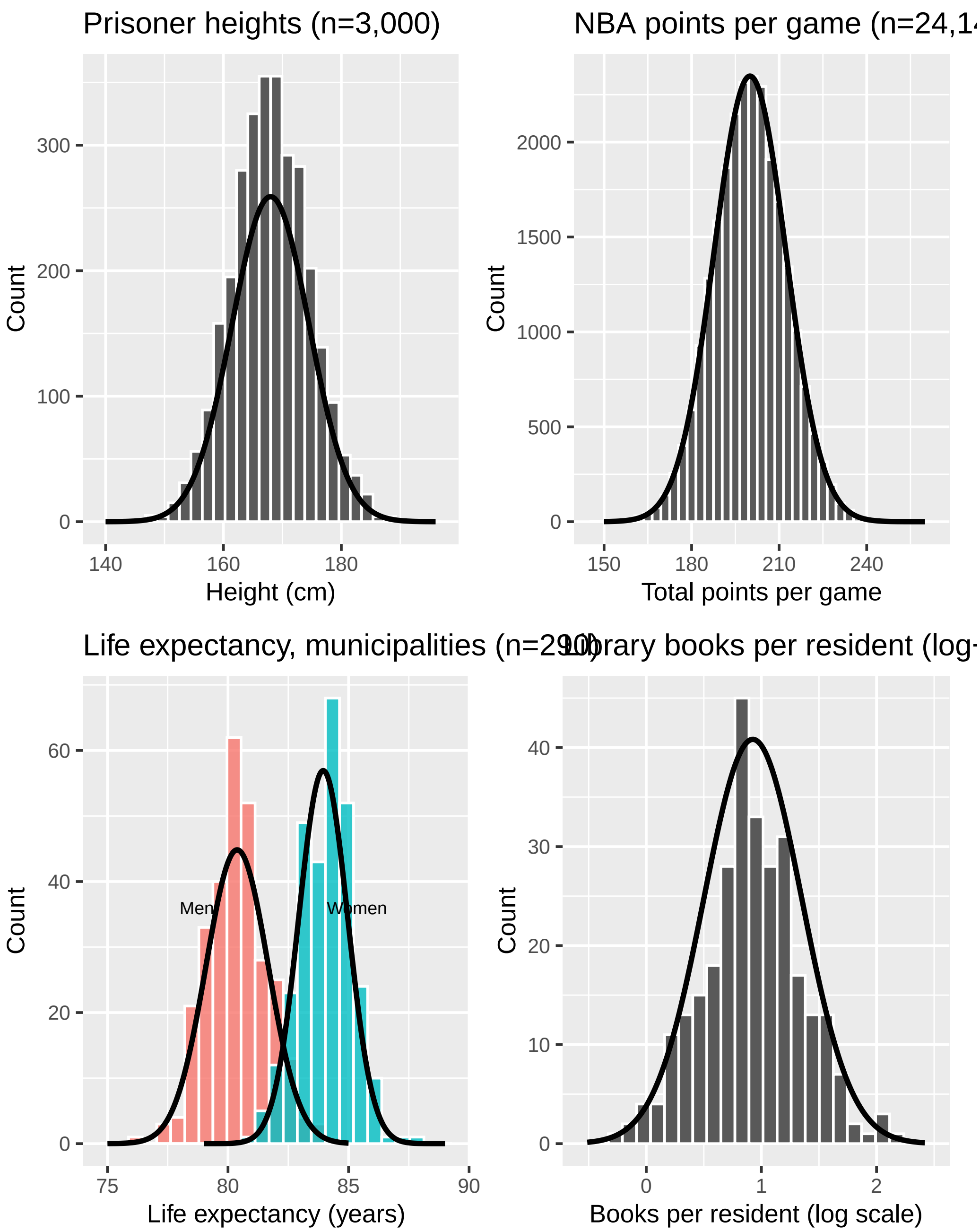


Figure 18.3: Examples from reality of normally distributed phenomena

18.4 The usefulness of the normal distribution 2

Now we shall again look at how probabilities are distributed using random values. We ask the computer to randomly choose 100,000 values, where each value is either 0 or 1. Each time the computer chooses a value both values have the same probability of being chosen: 50%. Just as expected the results are distributed approximately 50% on each of 0 and 1.

Next we let the computer choose 4 values at a time, 0 or 1, and each time with 50% probability each. For the 4 randomly chosen values the computer calculates a mean and then repeats this until we have 100,000 new means. With four random values we have the following five possible results: $\{0,0,0,0\}$, $\{0,0,0,1\}$, $\{0,0,1,1\}$, $\{0,1,1,1\}$ and $\{1,1,1,1\}$, which gives the following possible means: 0, 0.25, 0.5, 0.75 and 1.

The more values we add, the more the distribution of these values begins to resemble a normal distribution. The mean is consistently 0.5, which is because the values that are chosen, and thus the calculated means, are consistently 0 or 1. Figure 18.4 illustrates this with n = 1, 4, 20 and 200.

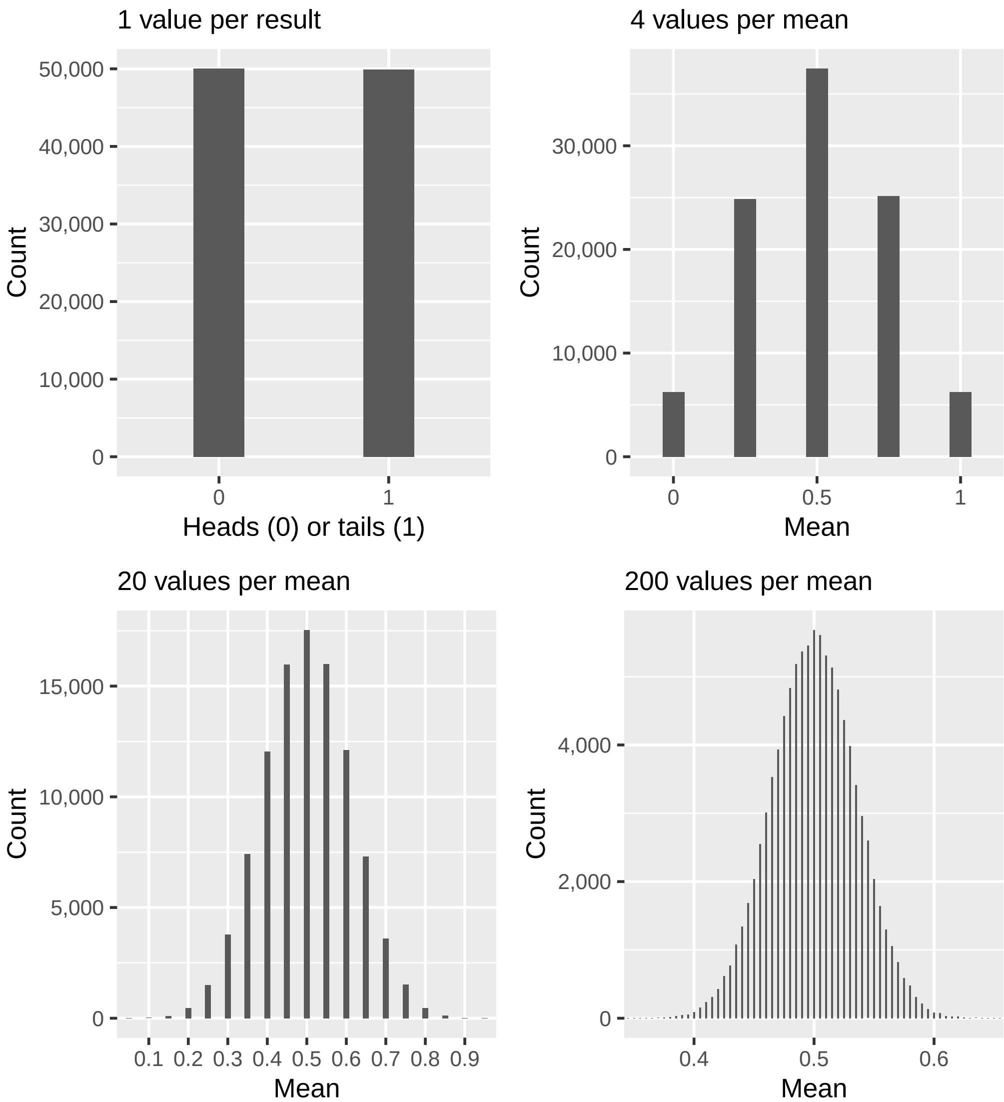


Figure 18.4: Random values: 100,000 means per graph

If we flip a coin and want to know the probability of getting a certain number of outcomes, for example x number of tails on n number of attempts, we use the binomial distribution (see section 16.8). The simulation we let the computer do above is precisely this function with ever higher value for n. As n becomes larger the binomial distribution approaches an approximate normal distribution. We call this an approximation since the binomial distribution by definition is a discrete probability distribution while the normal distribution is continuous. Mathematically this can be described as given that np and $n(1-p)$ are both large numbers, the following applies:

$$b(n,p) \approx N(np,\, np(1-p)) \tag{18.7}$$

The same phenomenon applies to many other probability distributions — given certain conditions, many different probability distributions approach the normal distribution. Another example can be illustrated by simulating perfectly balanced dice, which means we let the computer choose among the integers 1 to 6 randomly where each integer has the same probability: $\frac{1}{6}$.

In figure 18.5 we start with one die and let the computer generate 100,000 values (upper left graph). The bars are almost equally large, which illustrates that each integer has been chosen approximately equally many times. In the next step we let the computer choose two values and sum them (upper right). The lower left graph shows the distribution of point sums with four dice. The lower right graph shows eight dice. The more randomly chosen values that are included in a point sum, the more the distribution of the results begins to resemble a normal distribution.

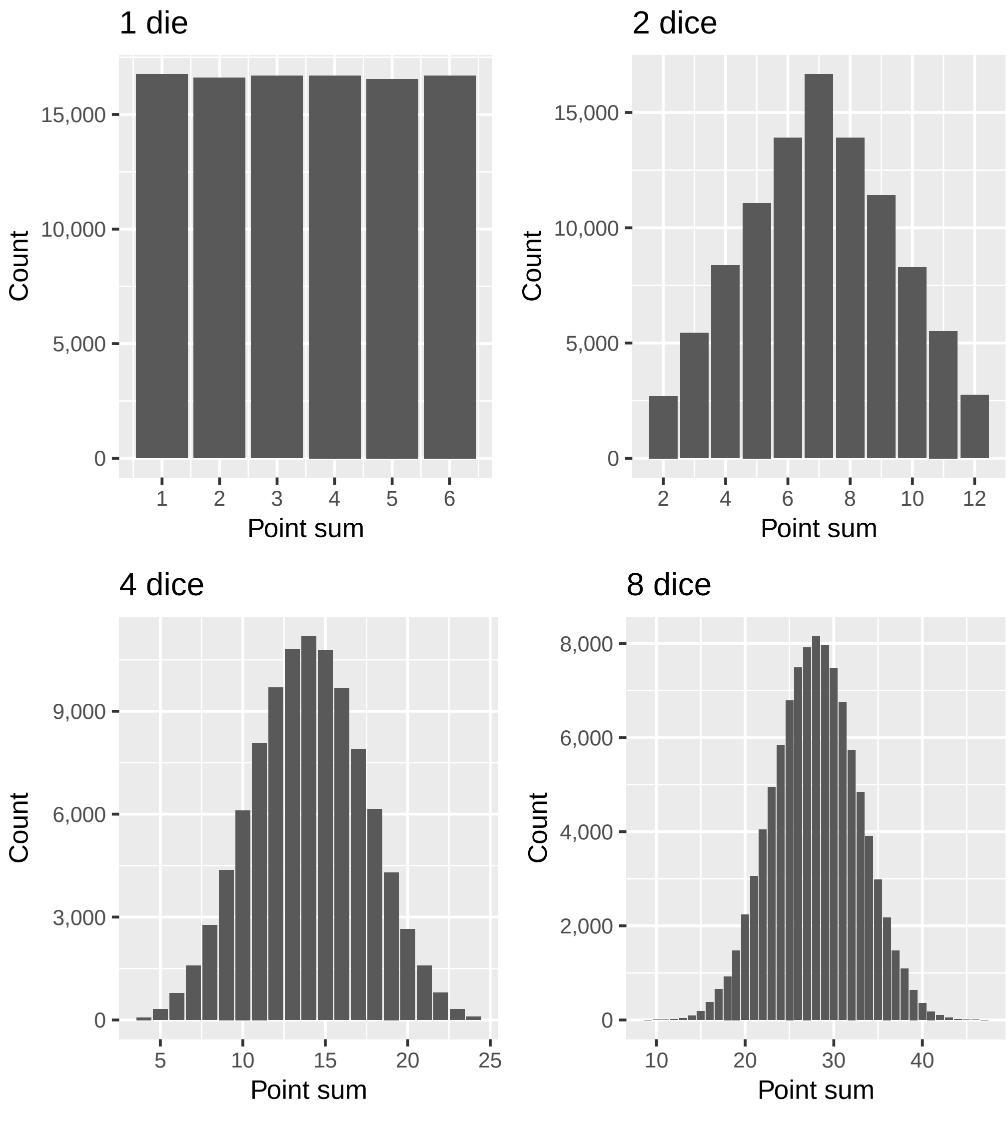


Figure 18.5: Random results: 100,000 sums of points per graph

The dice results can also be described as becoming increasingly approximately normally distributed. The Poisson distribution (see section 16.9) can also approach an approximation of the normal distribution if λ becomes sufficiently large:

$$Pois\left(\lambda\right) \approx N\left(\lambda, \lambda\right)$$

In section 17.4 we introduced the continuous gamma distribution, whose density function is:

$$f\left(x; k, \theta\right) = \frac{x^{k-1}e^{-x/\theta}}{\Gamma\left(k\right)\theta^{k}}$$

where k is the number of the event we shall calculate the probability for and θ is the average waiting time for such an event: $\theta = 1/\lambda$. The gamma distribution approaches an approximation of the normal distribution for large values of k:

$$\Gamma\left(k\right) \approx N\left(\frac{k}{\lambda}, \frac{k}{\lambda^{2}}\right)$$

The chi-square distribution was introduced in section 17.5 and can be described as a gamma distribution with $\theta = 2$ and $k = r/2$ where r is a positive integer. A variable X that follows a chi-square distribution, $X \sim \chi^{2}\left(r\right)$, approaches an approximate normal distribution for large values of r:

$$\chi^{2}\left(r\right) \approx N\left(r, 2r\right)$$

where the approximate normal distribution has mean $= r$ and standard deviation $2r$. We described above how a normal distribution that has mean 0 and standard deviation 1 is called the standardized normal distribution. If the variable $Z \sim N\left(0, 1\right)$ it also applies approximately that:

$$Z^{2} \sim \chi^{2}\left(1\right)$$

Variable Z squared thus follows the chi-square distribution with 1 degree of freedom.

18.5 The usefulness of the normal distribution 3

We have now gone through examples of how the normal distribution is common in real data as well as how other probability distributions resemble the normal distribution in different situations. In this section we shall look at how repeated samples from distributions that themselves are not normally distributed can lead to approximate normal distributions.

We first create four different types of distributions that we call X, Y, Z and W with 10,000 values each. These four made-up collections of values we regard as population data for each variable respectively. Now we shall let the computer take random samples from each of the four collections. The computer takes 300 observations at a time and calculates a mean of these observations. Then the computer repeats the sample again until we have calculated 10,000 means per population variable.

Figure 18.6 illustrates the results of this exercise in eight graphs. The graphs to the left show data over the four different populations X, Y, Z and W, which illustrates that the four populations have different dispersion. Despite the population looking very different in the graphs to the left, repeated samples lead to collections of values that resemble a normal distribution in the graphs to the right. The phenomenon arises when we retrieve sufficiently large amounts of samples from the populations.

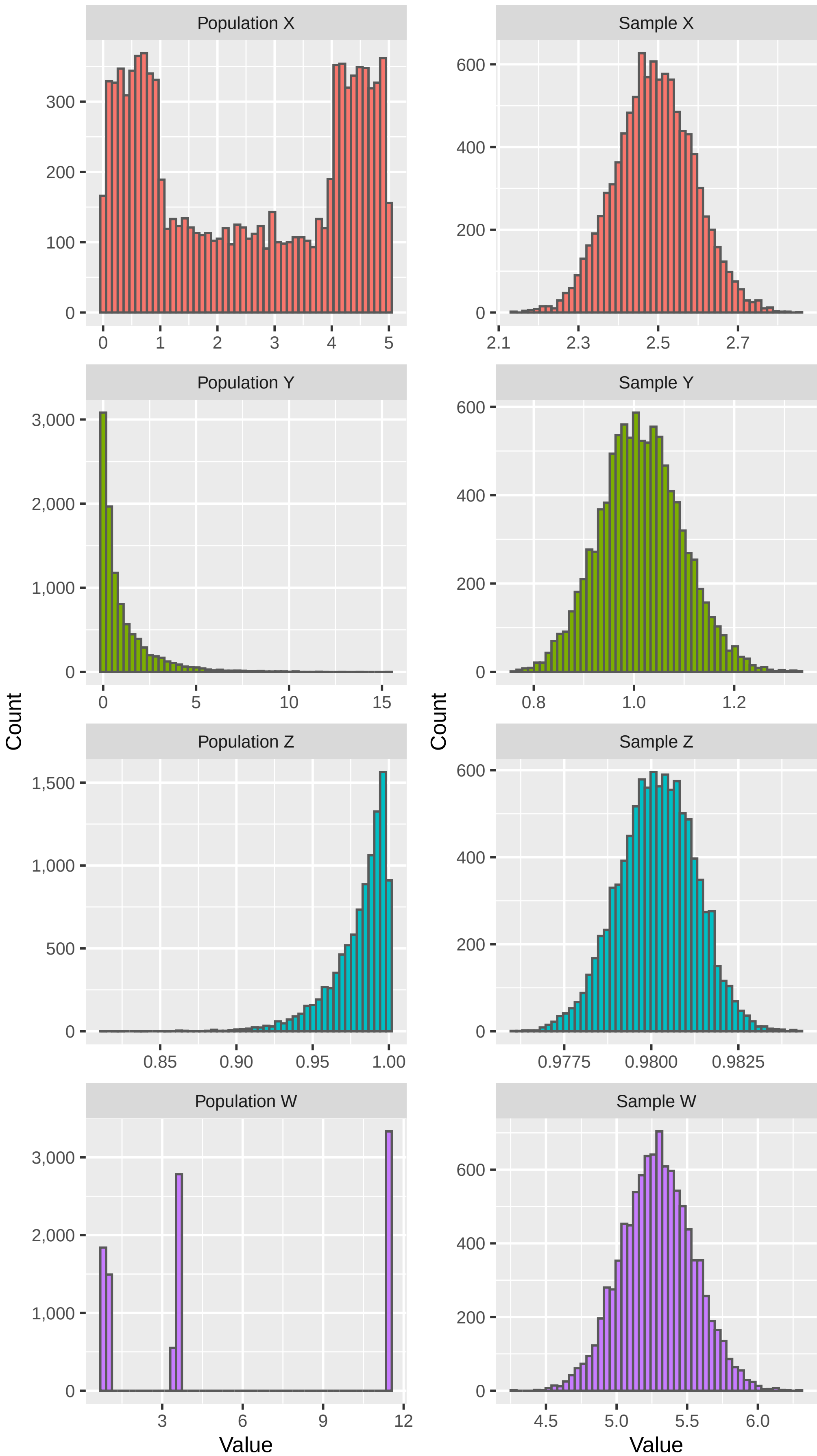


Figure 18.6: Populations (left, 10,000 values each) and 10,000 means of random samples of 300 observations from those populations (right)

This phenomenon is well-known within statistics and is described in what is called the **central limit theorem** (CLT), which is an important theorem within statistics and probability theory. The central limit theorem shows that if we take random samples from a large quantity of variables, the distribution of these samples' means will approach the normal distribution, even if the samples are retrieved from other types of distributions.

The central limit theorem can be described as: we have the random samples $X_1,\ X_2,\ ..., X_n$ from a population with mean μ and standard deviation σ. The mean for the samples is $\bar{X}$. According to the law of large numbers (see section 16.6) the mean of the samples $X_1, ..., X_n$ will converge toward μ when the number of observations in the sample n becomes sufficiently many:

$$\lim_{n\to\infty} \frac{X_1 + \dots + X_n}{n} = \mu$$

When n becomes sufficiently large, the distribution of $\bar{X}_n$ will approach a normal distribution with mean μ and variance σ^2/n, where σ^2 is the population's variance. In addition, the central limit theorem means the following:

$$\lim_{n\to\infty} \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) = N\left(0,1\right) \tag{18.8}$$

where $\sqrt{n}$ is the square root of number of means, $\bar{X}_n$ is n number of means from the random samples, μ is the population's mean and σ is the population's standard deviation. This means that if we are to calculate probability for variables that are not actually normally distributed, we often still start from the normal distribution, as long as the size of the sample is sufficiently large. A common rule of thumb says that if the sample has more than 30 observations, $n > 30$, then the central limit theorem applies.

18.6 T-distribution

Suppose we have a variable X that follows the standardized normal distribution, $X \sim N\left(0,1\right)$, and a variable Y that follows the chi-square distribution with k degrees of freedom. Y is independent from X. In that case we create the new variable T:

$$T = \frac{X}{\sqrt{Y}/\sqrt{k}} \tag{18.9}$$

This variable follows a type of probability distribution that is called the t-distribution with k number of degrees of freedom, where for our examples we assume that $k > 0$. The t-distribution has variance:

$$var\left(T\right) = \frac{k}{k-2} \tag{18.10}$$

A variable X that follows the t-distribution can be described as $X \sim t\left(k\right)$ or $X \sim t_k$ for k degrees of freedom. Like the other probability distributions covered above, the t-distribution has both a density function and a cumulative distribution function. Rather than examining these mathematical functions in detail, we will focus on the distribution's general properties and practical importance.

Figure 18.7 illustrates the t-distribution's density function $f\left(x\right) = P\left(X = x\right)$ in the upper graph and the cumulative distribution function $F\left(x\right) = P\left(X \leq x\right)$ in the lower graph. As can be seen, the t-distribution resembles the standardized normal distribution in shape. The degrees of freedom affect the density function's shape as well as what proportion of the distribution is under the cumulative distribution curve for a given value of x.

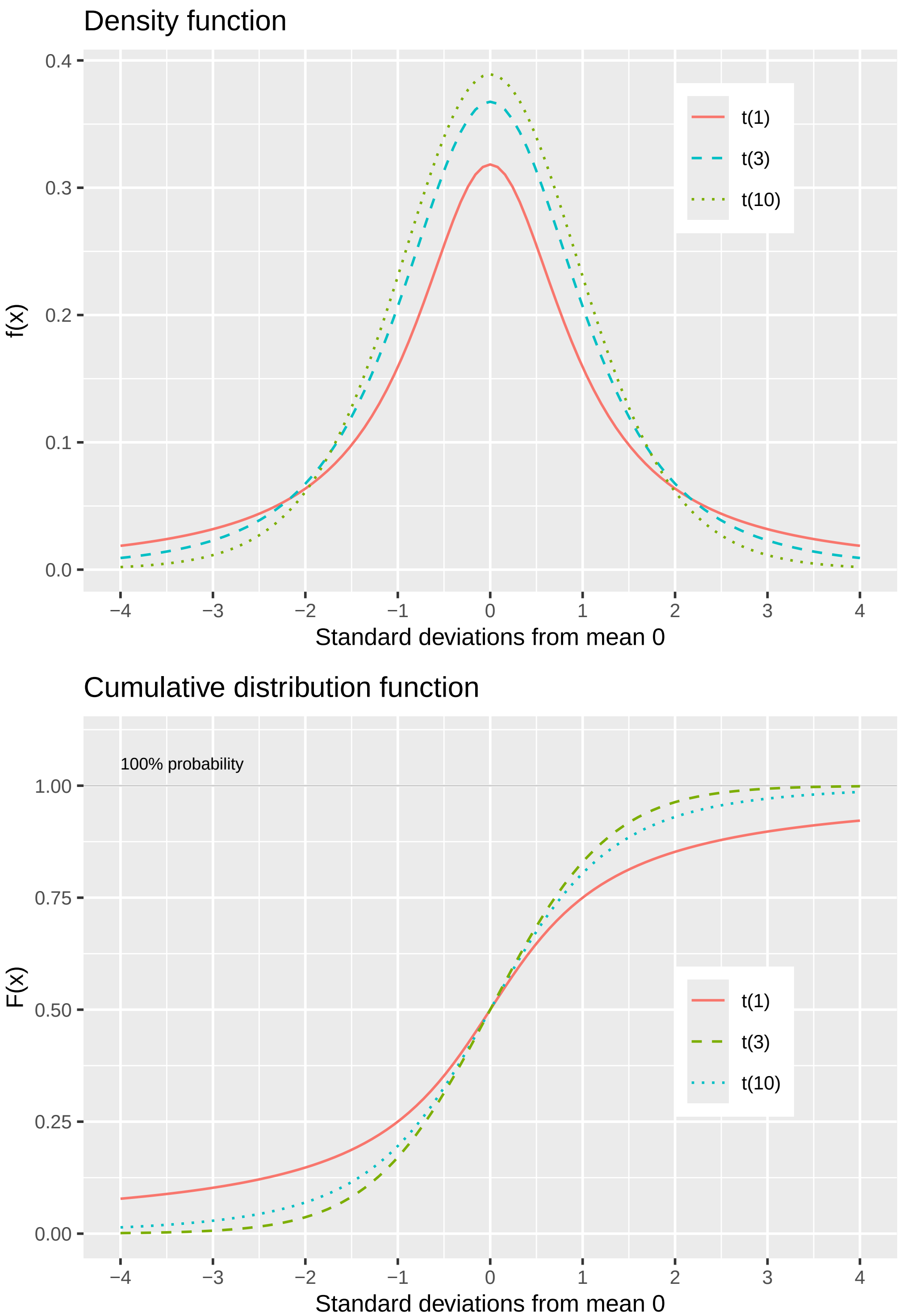


Figure 18.7: T-distribution: density function $f(x)$ and cumulative distribution function $F(x)$

For a variable $X \sim t(k)$ higher values of k mean that the tails become thicker, as in the upper graph in the figure. In the same way as the normal distribution, the t-distribution is also symmetric around the mean. This means that at the same time as we have $F(x) = P(X \leq x)$ we also have:

$$F(-x) = 1 - P(X \leq x) = P(X > x)$$

Table 18.2 describes calculated critical values for the t-distribution based on degrees of freedom k and percentage probability, where the percentages describe significance level. These are values from the

Table 18.2: T-distribution: critical values per degree of freedom. One-tailed test

	90%	95%	97.5%	99%	99.5%	99.9%
1	3.078	6.314	12.706	31.821	63.657	318.309
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.610
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
60	1.296	1.671	2.000	2.390	2.660	3.232
120	1.289	1.658	1.980	2.358	2.617	3.160
∞	1.282	1.645	1.960	2.326	2.576	3.090

cumulative distribution function $F(x)$, depending on number of degrees of freedom. The percentages, the significance level, refer to what proportion of the distribution is equal to or less than this t-value.

If we have a variable $X \sim t(10)$ we look at the row where $k = 10$. The columns with the percentages 90% to 99.9% show what proportion of the distribution's values are equal to or less than $F(x)$ where the value for x is what is shown in the table's cells. On the row for $k = 10$ we see that 90% of the distribution is less than or equal to $x = 1.37$.

If we turn it around and think that for the variable $X \sim t(10)$ we want to know what proportion of the distribution is less than or equal to 2 standard deviations above the mean. We then have $F(x = 2) = P(X \leq 2)$ and look at row $k = 10$. Since we have $x = 1.81$ for 95% and $x = 2.23$ for 97.5%, this means that between 95 and 97.5% of the distribution is equal to or less than $F(x = 2)$. This is illustrated in figure 18.8 where the gray area to the left of the mean represents an equally large proportion of the distribution as the gray area to the right of the mean, which means that $F(x) = F(-x)$.

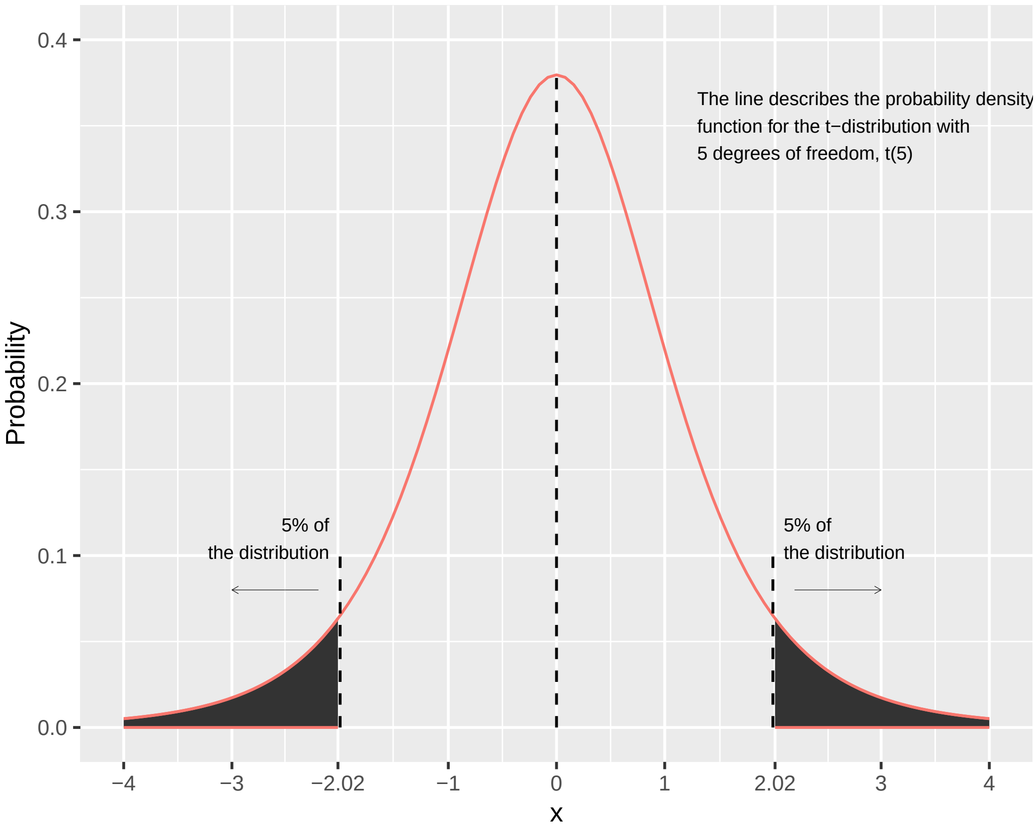


Figure 18.8: T-distribution: density function with shaded significance level

Above we went through how different probability distributions approach approximate normal distributions in different situations. The same thing also applies to the t-distribution. For a t-distribution $t(k)$ it applies that for large values of k:

$$t(k) \approx N(0,1)$$

With many degrees of freedom, high values for k, the t-distribution approaches the standardized normal distribution. A t-distribution with k degrees of freedom where $k > 0$ has mean 0, the same as the standardized normal distribution. As described above, the t-distribution has variance $var(t) = k/(k-2)$. This means that when $k \to \infty$ then it follows that $var(t) \to 1$ which makes the t-distribution approach the standardized normal distribution, $N(0,1)$.

The last row in table 18.2 shows the critical t-values for $k = \infty$, which can be compared against the standardized normal distribution's pre-calculated probabilities. Example: for $k = \infty$, $t = 3.09$ means that $P(X \leq x) = 99.9\%$. If we look in table 18.1 at the bottom in the right corner, row for $x = 3$ and column .09, we see that for this value of x the standardized normal distribution's $F(x) = P(X \leq x) = 0.999$, that is 99.9%.

18.7 Chapter summary

- **The normal distribution**: written $N(\mu, \sigma)$ where μ is mean and σ is standard deviation. Density function: $f(x;\mu,\sigma) = e^{-(x-\mu)^2/(2\sigma^2)} / \left(\sigma\sqrt{2\pi}\right)$ for $-\infty < x < \infty$. Cumulative distribution function: $F(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-(t-\mu)^2/(2\sigma^2)} dt$.
- Since the normal distribution is symmetric we have $F(-x) = 1 - F(x)$. For $F(x=\mu) = P(X \leq \mu) = 0.5$, with 50% of the distribution above and below the mean μ respectively.
- **The standardized normal distribution** has $\mu = 0$ and $\sigma = 1$ and is written $N(0,1)$. Density function: $f(x) = e^{-x^2/2}/\sqrt{2\pi}$. Also called the standard normal distribution or the Z-distribution.

- Many real data within both natural and social sciences follow the normal distribution. Many other probability distributions approach the normal distribution approximately in different situations. Example: the binomial distribution approaches the normal distribution approximately when np and $n(1-p)$ are large numbers: $b(n,p) \approx N(np, np(1-p))$.
- **The central limit theorem (CLT)**: if we take random samples from a variable that follows another probability distribution, the mean of the samples will approach the normal distribution approximately when the number of samples becomes sufficiently large. The mean will according to the law of large numbers approach μ as $n \to \infty$, while the CLT describes how the dispersion around the mean will approach a normal distribution as the number of samples n approaches infinity.
- **The t-distribution**: if $X \sim N(0,1)$ and $Y \sim \chi^2(k)$ the variable $T = X/\left(\sqrt{Y}/\sqrt{k}\right)$ follows the t-distribution, written $T \sim t(k)$ for k degrees of freedom. When k becomes sufficiently large $t(k) \approx N(0,1)$. The t-distribution is symmetric: $F(x=0) = 0.5$ and $F(-x) = 1 - F(x)$.

18.8 Exercises

1. The standard normal distribution is written $Z \sim N(0,1)$.

(a) What are $E(Z)$ and $\mathrm{Var}(Z)$?

(b) By symmetry, what is $P(Z \leq 0)$?

(c) If $P(Z \leq 1.96) = 0.975$, what is $P(Z \geq 1.96)$?

Answer: (a) $E(Z) = 0$, $\mathrm{Var}(Z) = 1$.

(b) $P(Z \leq 0) = 0.5$.

(c) $P(Z \geq 1.96) = 1 - 0.975 = 0.025$.

2. Let $X \sim N(50, 100)$ (mean 50, variance 100, so $\sigma = 10$). Compute the z-score $z = (x - \mu)/\sigma$ for each value.

(a) $x = 70$

(b) $x = 40$

(c) $x = 50$

Answer: (a) $z = (70 - 50)/10 = 2$.

(b) $z = (40 - 50)/10 = -1$.

(c) $z = (50 - 50)/10 = 0$.

3. Suppose $P(Z \leq 1.64) = 0.95$. Use symmetry of the standard normal to answer the following.

(a) What is $P(Z \geq 1.64)$?

(b) What is $P(-1.64 \leq Z \leq 1.64)$?

(c) What is $P(Z \leq -1.64)$?

Answer: (a) $P(Z \geq 1.64) = 1 - 0.95 = 0.05$.

(b) $P(-1.64 \leq Z \leq 1.64) = 0.95 - 0.05 = 0.90$.

(c) $P(Z \leq -1.64) = 0.05$ by symmetry.

4. Let $X \sim N(100, 25)$ (mean 100, $\sigma = 5$).

(a) Convert $x = 110$ to a z-score.

(b) Convert $x = 90$ to a z-score.

(c) Given $P(Z \leq 2) = 0.977$, what is $P(X \leq 110)$?

Answer: (a) $z = (110 - 100)/5 = 2$.

(b) $z = (90 - 100)/5 = -2$.

(c) $P(X \leq 110) = P(Z \leq 2) = 0.977$.

5. According to the central limit theorem (CLT), the sample mean $\bar{X}$ is drawn from a population with mean $\mu = 50$ and variance $\sigma^2 = 100$, based on $n = 100$ observations.

(a) What is $E(\bar{X})$?

(b) What is $\text{Var}(\bar{X})$?

(c) What distribution does $\bar{X}$ approximately follow for large n?

Answer: (a) $E(\bar{X}) = \mu = 50$.

(b) $\text{Var}(\bar{X}) = \sigma^2/n = 100/100 = 1$.

(c) $\bar{X}$ approximately follows $N(50, 1)$ by the CLT.

6. A population has mean $\mu = 200$ and standard deviation $\sigma = 20$. A sample of $n = 400$ is drawn.

(a) What is the standard error $\text{SE}(\bar{X}) = \sigma/\sqrt{n}$?

(b) Compute the z-score for $\bar{X} = 201$.

(c) Is $n = 400$ large enough to apply the CLT? Why?

Answer: (a) $\text{SE} = 20/\sqrt{400} = 20/20 = 1$.

(b) $z = (201 - 200)/1 = 1$.

(c) Yes, $n = 400 \gg 30$, which is the common rule of thumb for the CLT to apply.

7. Let $X \sim \text{Bin}(100, 0.5)$.

(a) Compute $E(X)$ and $\text{Var}(X)$.

(b) Which normal distribution approximates X for large n?

(c) Using $P(Z \geq 1) \approx 0.159$, approximate $P(X \geq 55)$.

Answer: (a) $E(X) = np = 50$, $\text{Var}(X) = np(1 - p) = 25$.

(b) $N(50, 25)$, i.e. $N(\mu = 50, \sigma^2 = 25)$.

(c) $z = (55 - 50)/5 = 1$, so $P(X \geq 55) \approx P(Z \geq 1) \approx 0.159$.

8. Heights are normally distributed with mean 175 cm and standard deviation 8 cm, i.e. $X \sim N(175, 64)$.

(a) Compute the z-score for $x = 183$.

(b) Compute the z-score for $x = 167$.

(c) Given $P(-1 \leq Z \leq 1) \approx 0.683$, find $P(167 \leq X \leq 183)$.

Answer: (a) $z = (183 - 175)/8 = 1$.

(b) $z = (167 - 175)/8 = -1$.

(c) $P(167 \leq X \leq 183) = P(-1 \leq Z \leq 1) \approx 0.683$.

9. The t-distribution $T \sim t(k)$ has k degrees of freedom.

(a) What are $E(T)$ and the symmetry property of $t(k)$?

(b) Compared to $N(0, 1)$, does $t(k)$ have heavier or lighter tails?

(c) As $k \to \infty$, what does $t(k)$ converge to?

Answer: (a) $E(T) = 0$. By symmetry: $F(-x) = 1 - F(x)$ and $F(0) = 0.5$.

(b) Heavier tails — the t-distribution has more probability mass in the tails than the standard normal.

(c) $t(k) \to N(0, 1)$ as $k \to \infty$.

10. Let $T \sim t(10)$.

(a) By symmetry, what is $P(T \leq 0)$?

(b) If $P(T \leq 2.228) = 0.975$, what is $P(T \geq 2.228)$?

(c) Is the critical value 2.228 larger or smaller than 1.96 (the 97.5th percentile of $N(0,1)$)? What does this tell us?

Answer: (a) $P(T \leq 0) = 0.5$ by symmetry.

(b) $P(T \geq 2.228) = 1 - 0.975 = 0.025$.

(c) $2.228 > 1.96$, which tells us that the t-distribution has heavier tails than the standard normal — we need to go further out to capture the same 97.5% of the distribution.

11. When should you use the t-distribution rather than the standard normal for inference about a mean?

(a) Population variance σ^2 is unknown, $n = 25$: use t or z?

(b) Population variance σ^2 is known, $n = 200$: use t or z?

(c) Population variance σ^2 is unknown, $n = 1000$: use t or z? Explain.

Answer: (a) Use t — small sample, unknown variance.

(b) Use z — variance is known, regardless of sample size.

(c) Either works: with $n = 1000$, $t(999) \approx N(0,1)$, so both give virtually the same result. Strictly, t is correct when variance is estimated, but z is a fine approximation.

12. A sample of $n = 100$ gives $\bar{X} = 48$. The population standard deviation is known: $\sigma = 10$.

(a) Compute the standard error $\text{SE} = \sigma/\sqrt{n}$.

(b) Construct a 95% confidence interval using $\bar{X} \pm 1.96 \cdot \text{SE}$.

(c) How would the confidence interval change if n increased to 400?

Answer: (a) $\text{SE} = 10/\sqrt{100} = 1$.

(b) $48 \pm 1.96 \cdot 1 = [46.04,\ 49.96]$.

(c) $\text{SE} = 10/\sqrt{400} = 0.5$, so the interval narrows to $48 \pm 0.98 = [47.02,\ 48.98]$. Larger samples give more precise estimates.

Chapter 19

Statistical analysis and inference

In this chapter, we continue the description of how statistical analysis can be conducted. We go through how we can estimate individual values, points, and intervals. We also introduce the concept of standard error, which gives us a measure of the uncertainty in an estimated value.

19.1 Point estimation

In section 15.1 we introduced the difference between the population, whose values are generally unknown, and the sample data we have access to. Statistical inference is the process by which we attempt to draw conclusions about unknown properties in a population using empirical data. This is what we do when we try to estimate the values in a population by calculating results with sample data.

If we want to know the population mean μ_X, we estimate this with sample data and the usual formula for the mean: $\bar{x} = \sum_i^n x_i/n$. If the mean we calculate aims to make statements about a (larger) population beyond the data we have access to, then this is by definition an estimate of the population mean.

If we want to know the population's unknown variance σ_X^2 we estimate this as $var\,(X) = \left(\frac{1}{n-1}\right)\sum_i^n \left(x_i - \bar{x}\right)^2$.

For the mean and variance, we estimate specific values, which is called estimating points, point estimates, or point estimation. The mean is thus an example of a point estimate. Each estimated mean is a constant, but the equation we use to estimate the mean with a collection of sample observations is called an estimator. The equation we use to estimate the variance is called the variance estimator.

19.2 Standard error

The expected value of the mean estimator is the population mean: $E\left(\bar{x}\right) = E\left(\frac{1}{n}\sum_i^n x_i\right) = \mu_X$. When we estimate the mean $\bar{x}$ we know that this differs more or less from the population value μ_X. Since we do not have access to all information about the population, there are random errors in our estimates. The difference between the estimated $\bar{x}$ and the population value μ_X can be described as the sampling error $\bar{x} - \mu_X$. This sampling error cannot be calculated as long as we do not know the population mean μ_X and is therefore often more to be regarded as a theoretical concept.

To calculate a measure of the uncertainty in a point estimate, we estimate what is called the standard error. The standard error is calculated in different ways depending on which point estimate we study. The standard error of the mean is estimated in the following way:

$$\textbf{SE of the mean} = \frac{s_x}{\sqrt{n}} = \sqrt{\frac{s_x^2}{n}} \tag{19.1}$$

where s_x is the standard deviation, estimated with the sample observations and n is the number of observations in the sample. The equation shows different ways to write the same thing.

The equation can be read as: the smaller the dispersion we have in the observations (the numerator), the smaller the standard error becomes. Since we have the number of observations n in the denominator, this means that the standard error tends to decrease the more observations we include in our sample.

Both of these things are logical. The less dispersed the population's values are, the more likely our calculation is to be correct. The more we know about the population (more observations), the closer our estimate of the population mean will be to the correct mean in the population.

19.3 Confidence interval

Another way to study a population is to estimate intervals, which is also called interval estimation or interval estimate. A particular interval that is often used in statistical analysis is the **confidence interval**, which is an interval between two values within which a specified proportion of repeated samples will contain the population value we seek. We determine ourselves for what proportion of samples we want to estimate the confidence interval, for example 90 or 95%, where the percentages thus indicate for what proportion of samples the interval will include the population value. The percentages are called the confidence level.

We are interested in the population's (unknown) mean μ_X. With our sample data we estimate the mean as $\bar{x} = \sum_i^n x_i/n$. Our estimation depends on what distribution the population has. We start here from a normally distributed variable X with mean μ_X and variance σ_X. The confidence interval can then be defined as:

$$\textbf{Confidence interval:} \quad \bar{x} \pm z\frac{\sigma}{\sqrt{n}} \tag{19.2}$$

where $\bar{x}$ is an estimated mean in a collection of sample observations. The letter z represents a value that we obtain from the standard normal distribution's cumulative distribution function F, which can be read from table 18.1 in chapter 18, depending on which confidence level we have chosen.

The table should be read as: if we choose confidence level 90%, we look up the value 0.95 in the table. We want to exclude 10% of the population's values. By taking 0.95 we exclude 5% in the distribution's upper tail and 5% in the distribution's lower tail. For 0.95 we see in the table that $z \approx 1.65$. If we want to estimate a 95% confidence interval we choose 0.975, which gives $z = 1.96$. For confidence level 99% we choose 0.995, which gives $z = 2.58$.

The symbol $\pm$ describes plus and minus. The confidence interval therefore results in a value below the mean $\bar{x}$ and a value above. The two values are called the confidence interval's lower and upper limits. Since σ(the population's standard deviation) is generally unknown, the estimated confidence interval can be written:

$$\textbf{Estimated confidence interval:} \quad \bar{x} \pm z\frac{s_x}{\sqrt{n}} \tag{19.3}$$

where s_x is the estimated standard deviation. Say for example that we have a data sample of $n = 28$ observations and estimate the mean $\bar{x} = 13$ and standard deviation $s_x = 3.7$. We choose confidence level 90%, therefore $z = 1.65$. This gives the following limits for the confidence interval:

$$\begin{aligned} \bar{x} + z\frac{s_x}{\sqrt{n}} &= 13 + 1.65 \cdot \frac{3.7}{\sqrt{28}} \approx 14.15 \\ \bar{x} - z\frac{s_x}{\sqrt{n}} &= 13 - 1.65 \cdot \frac{3.7}{\sqrt{28}} \approx 11.85 \end{aligned} \tag{19.4}$$

This indicates that if we take repeated samples, the population mean will be in the interval $11.85 < \mu_X < 14.15$ in 90% of cases.

In section 18.2 we calculated the probability that two estimated means came from the same population, by calculating the difference between average life expectancy for men and women respectively in Sweden's municipalities. The confidence interval for the difference between the estimated means for these two populations can be estimated in the following way:

$$(\bar{x}_1 - \bar{x}_2) \pm z \cdot \sqrt{\frac{s^2_{x_1}}{n_1} + \frac{s^2_{x_2}}{n_2}} \tag{19.5}$$

where $\bar{x}_1$ and $\bar{x}_2$ are estimated means for the two populations, z is the value from the standard normal distribution's cumulative distribution function F, s^2 is estimated variance (standard deviation squared) and n is the number of observations in the respective sample. In this case we have the following values:

$$\begin{aligned} \bar{x}_{\text{men}} &= 80.38 \\ \bar{x}_{\text{women}} &= 83.95 \\ s^2_{\text{men}} &= 1.663 \\ s^2_{\text{women}} &= 1.032 \\ n_{\text{men}} = n_{\text{women}} &= 290 \end{aligned} \tag{19.6}$$

This time we choose confidence level 95%, which gives $z = 1.96$. The confidence interval's lower and upper limits then become:

$$\begin{aligned} (80.38 - 83.95) + 1.96\sqrt{\frac{1.032}{290} + \frac{1.663}{290}} &\approx -3.38 \\ (80.38 - 83.95) - 1.96\sqrt{\frac{1.032}{290} + \frac{1.663}{290}} &\approx -3.76 \end{aligned} \tag{19.7}$$

The estimate indicates that with repeated samples the difference between the two populations will in 95% of cases be $3.38 < (\mu_{\text{Women}} - \mu_{\text{Men}}) < 3.76$.

19.4 Hypothesis testing

When we study patterns in data and covariation, we do this by testing hypotheses with statistical tests. A scientific hypothesis must be falsifiable, which means that it must be capable of being disproven with facts. We formulate a hypothesis and use a statistical test to assess the probability that the hypothesis is false. The probability that a hypothesis is false can never be below 0 or above 100%.

Statistical tests use a null hypothesis and an alternative hypothesis. The null hypothesis is often denoted H_0 and the alternative hypothesis is denoted H_1. The two hypotheses are formulated so that they are mutually exclusive. Both hypotheses cannot by definition be true or false simultaneously.

We formulate our hypotheses based on the covariation we want to study. The null hypothesis is generally formulated as there being no covariation — a non-relationship. In simplified terms, we have a theory we want to test with a statistical test. The null hypothesis describes the situation we must accept until we have shown valid reasons to believe otherwise.

For example, suppose we conduct an experiment and study how a medicine covaries with disease symptoms in a group of patients, divided into a treatment group (receives medicine) and a control group (does not receive medicine). Our theory is that the medicine will reduce the patients' disease condition. When translating this to a null hypothesis and an alternative hypothesis, we formulate the null hypothesis as a non-relation between cause and effect. If the medicine has no effect on the disease, the treatment group will be equally sick or sicker than the control group:

$$H_0 : \text{sickness}_{\text{treatment}} \geq \text{sickness}_{\text{control}}$$

The alternative hypothesis H_1 becomes in this case the alternative situation where the disease condition is less in the treatment group than in the control group (which according to our theory is due to the medicine):

$$H_1 : \text{sickness}_{\text{treatment}} < \text{sickness}_{\text{control}}$$

Observe that the hypotheses are not formulated regarding the actual causal relationship of the medicine. The hypotheses are instead formulated regarding observable differences. Our hypothesis test therefore tests a theory regarding the data we study.

Hypothesis tests are often used in connection with regression analysis. Suppose we study whether variations in a phenomenon X cause a certain type of variations in phenomenon Y, which we do using the regression model $Y = a + bX + u$ where Y and X are variables, u is the error term and a and b are the coefficients that we use the least squares method to estimate.

We have reason to believe that X and Y covary and that $b \neq 0$. Based on this we formulate a null hypothesis in the form of a non-relation between the variables. A non-relation between X and Y in our regression model means that $b = 0$, therefore our null hypothesis becomes:

$$H_0 : b = 0$$

The alternative hypothesis is formulated in this case as:

$$H_1 : b \neq 0$$

The null hypothesis and alternative hypothesis thereby cover all possible alternatives: b is either equal to 0 or not equal to 0. This means that if our analysis shows that the covariation between X and Y is either positive or negative, we may have reason to reject our null hypothesis. But we are both interested in estimating the covariation between X and Y and in estimating the probability that the covariation we find could also have arisen by chance.

We test our hypotheses by calculating the probability that we would have gotten the result we get given that H_0 is true. That is, we estimate a slope coefficient b in a regression model and then want to know what is the probability that we would have gotten this estimate, given that H_0 is correct, that is, that we actually have $b = 0$.

19.5 Statistical significance, power and type errors

In classical statistics it is usually described that we should choose in advance with what probability we want to risk rejecting a true H_0 when conducting statistical tests. This is called choosing significance level or alpha value and is denoted α. Another way to describe significance level is that it is the probability that a value deviates from an expected value, given that H_0 is true. For example, what is the probability that we in the population have $b = 0$, given the value we estimate for $\hat{b}$ in our regression model with our sample data? For 10% significance level, $\alpha = 0.1 = 10\%$:

$$1 - \alpha = 1 - 0.1 = 0.9 = 90\%$$

Based on this, one also usually says that a result is statistically significant for $\alpha = 0.1$, or statistically established, given 10% significance level. There is no objective method for choosing significance level. 95 percent is often used, which is common in academic articles within social science, but it is completely arbitrary and there is no mathematical logic behind why we would not use 94%, 96% or some completely different significance level instead.

The probability that H_0 is true is usually denoted by what is called the **p-value**. The p-value gives the probability that we would have gotten an equally extreme value in a statistical test given that H_0 is false. While significance level is generally stated in rounded numbers, for example 90, 95 or 99 percent, the p-value is calculated exactly through the statistical test.

Let us illustrate with an example. In section 18.2 we estimated the difference between the average life expectancy for men and women respectively in Sweden's municipalities by calculating the following statistic:

$$z = \frac{\bar{X}_{\text{women}} - \bar{X}_{\text{men}}}{\left(\frac{s^2_{\text{men}}}{n_{\text{men}}} + \frac{s^2_{\text{women}}}{n_{\text{women}}}\right)^{1/2}} \approx \frac{83.95 - 80.38}{\left(\frac{1.663}{290} + \frac{1.032}{290}\right)^{1/2}} \approx 37.0$$

Let us set this up as a statistical test. We have a theory that life expectancy differs between men and women. We formulate our null hypothesis as the corresponding non-relationship (that there is no difference):

$$\begin{aligned} H_0 &: \mu_{\text{men}} = \mu_{\text{women}} \\ H_1 &: \mu_{\text{men}} \neq \mu_{\text{women}} \end{aligned} \tag{19.8}$$

As the null hypothesis is formulated, this means that we have a two-sided test. Both positive and negative difference can mean that we have reason to reject H_0 as false. If we start from significance level 5%, our calculated z-value is 37.0, which is so high that it is not included in the table and the probability is therefore far above 0.999 (a value very close to 1, thus 100%). The result from our statistical test indicates with good margin that H_0 is false. This is called that the result is statistically significant at the 1% level, which means that it is unlikely that it would have arisen through a random process.

When we use statistical tests and calculate probabilities to reject or not reject the null hypothesis, there is always a risk that we draw the wrong conclusion. **Type 1 error** is what it is called when we in a statistical test reject a null hypothesis H_0 despite H_0 being true. The probability of a type 1 error is the same thing as the significance level α. Based on this we describe $1 - \alpha$ as the probability that we reject H_0 when H_0 is false.

Type 2 error means that we instead accept a null hypothesis H_0 despite this actually being false. The probability of type 2 error is usually denoted by the letter β(Greek letter beta). **Statistical power** is the probability (a value between 0 and 1) that we succeed in rejecting H_0 when H_0 is false:

$$\textbf{Statistical power} = P\left(\text{reject } H_0 \mid H_0 \text{ is false}\right)$$

To summarize:

$$\begin{aligned} \textbf{Type 1 error} &= \alpha = \text{significance level, reject true } H_0 \\ \textbf{Correct inference} &= 1 - \alpha = \text{do not reject true } H_0 \\ \textbf{Type 2 error} &= \beta = \text{do not reject false } H_0 \\ \textbf{Statistical power} &= 1 - \beta = \text{reject false } H_0 \end{aligned} \tag{19.9}$$

The power of a statistical test can be calculated in different ways and is affected by different things. Let us illustrate with an example. Suppose we have 290 observations with sample data for two approximately normally distributed variables with the means $\bar{x}_1 = 80$ and $\bar{x}_2 = 81$ and the same standard deviation $s_1 = s_2 = 1.5$. We want to test whether the means differ:

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2$$

We decide on $\alpha = 0.05$, that is 95% significance level, and estimate the z-value:

$$z = \frac{\bar{x}_2 - \bar{x}_1}{\left(\frac{s_2^2}{n_2} + \frac{s_1^2}{n_1}\right)^{1/2}} = \frac{81 - 80}{\left(\frac{1.5^2}{290} + \frac{1.5^2}{290}\right)^{1/2}} \approx 8.03$$

According to table 18.1, our calculated z-value indicates a statistically significant difference between the two populations at a level above 99%. This means therefore that we reject H_0 as false.

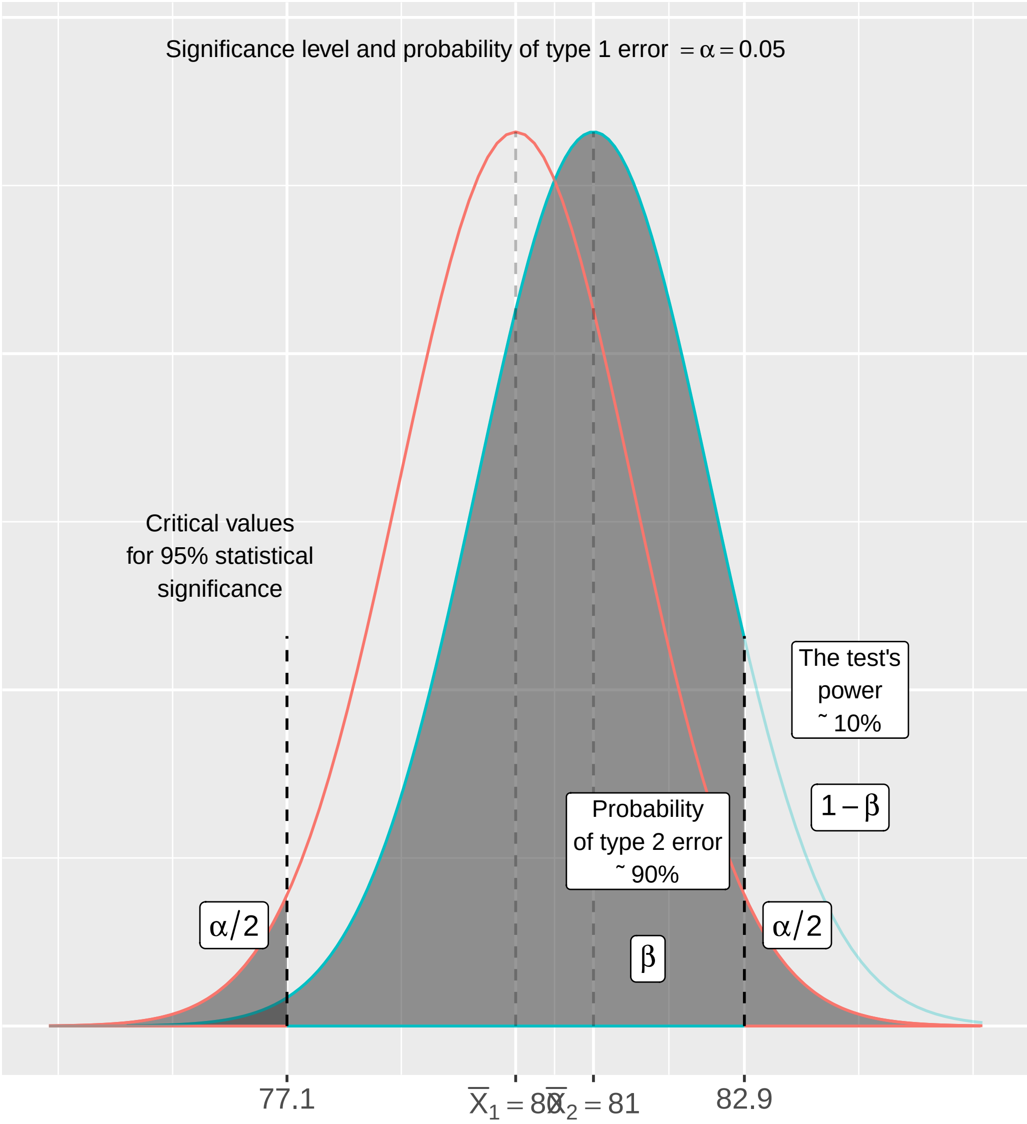


Figure 19.1: Statistical power

To estimate statistical power (the probability of rejecting false H_0) for this test we imagine that there exist two populations X_1 and X_2 that have the population means $\mu_{X_1} = 80$ and $\mu_{X_2} = 81$ as well as standard deviation $\sigma_{X_1} = \sigma_{X_2} = 1.5$. Given our determined significance level $\alpha = 0.05$, a deviation from the mean $\mu_{X_1} = 80$ of 82.93 means that we reject H_0(that the populations have the same mean). But for the other population X_2, approximately 90% of the cumulative probability (proportion of the population) $X_2 < 82.93$. This is in that case the probability that we here commit a type 2 error: $\beta \approx 0.9 = 90\%$. The test's power is therefore $1 - \beta \approx 1 - 0.9 = 0.1 = 10\%$.

Description: Figure 19.1 illustrates this, where the black line shows the normally distributed population X_1 and the gray line shows the population X_2. The dashed lines mark the critical values based on significance level $\alpha = 0.05$. The dark gray areas represent 2.5% of population X_1 in each tail. The light gray area is β, the probability of type 2 error, from which we estimate the test's power $1 - \beta$. If we for example lower the significance level α to $\alpha = 0.1$, then the critical values move closer to the mean for population X_1 and the area that represents β decreases. Conversely, higher α results in less risk for type 2 error (β) but increased risk for type 1 error α.

19.6 Chapter summary

- **Point estimation**: estimates of specific values in a population, for example mean. **Interval estimation**: estimates of intervals in a population.

- **Confidence interval**: interval in a population, within which a specified proportion of repeated samples will contain a population value. For a normally distributed variable: $\bar{x} \pm z \cdot s_x/\sqrt{n}$, where z is taken from the standard normal distribution table for the chosen confidence level.
- Statistical tests are based on a **null hypothesis** H_0, which is generally formulated as a non-relation, and an **alternative hypothesis** H_1, which is formulated as the null hypothesis's antithesis. Not rejecting H_0 does not mean that we have proven that H_1 is true.
- The **p-value** gives the probability of getting an equally extreme test statistic given that H_0 is true.
- In classical statistics, a **significance level** is chosen before the statistical test, denoted α, and is the same thing as the probability of type 1 error: rejecting true H_0.

-β= probability of type 2 error: not rejecting false H_0. **Statistical power** $= 1 - \beta =$ the probability of rejecting false H_0.

19.7 Exercises

1. A sample gives $x = \{4, 6, 8, 10\}$.

(a) Compute the sample mean $\bar{X} = \sum x_i/n$.

(b) Is $\bar{X}$ an unbiased estimator of the population mean μ?

(c) What does "unbiased" mean in this context?

Answer: (a) $\bar{X} = (4 + 6 + 8 + 10)/4 = 7$.

(b) Yes.

(c) Unbiased means $E(\bar{X}) = \mu$ — on average the estimator equals the true population mean.

2. A sample of $n = 100$ gives $\bar{X} = 50$. The known population standard deviation is $\sigma = 10$. Use $P(Z \leq 1.96) = 0.975$.

(a) Compute $\text{SE} = \sigma/\sqrt{n}$.

(b) Construct a 95% confidence interval $\bar{X} \pm 1.96 \cdot \text{SE}$.

(c) Interpret the confidence interval in plain language.

Answer: (a) $\text{SE} = 10/\sqrt{100} = 1$.

(b) $(50 - 1.96,\ 50 + 1.96) = (48.04,\ 51.96)$.

(c) If we repeated the sampling many times, 95% of such confidence intervals would contain the true population mean μ.

3. How does the width of a 95% confidence interval change in each scenario?

(a) The sample size quadruples from n to $4n$.

(b) The confidence level increases from 95% to 99% (critical value 2.576 instead of 1.96).

(c) The population standard deviation σ doubles.

Answer: (a) SE halves ($\sigma/\sqrt{4n} = \text{SE}/2$), so the CI width halves.

(b) The CI widens — a higher confidence level requires a wider interval.

(c) SE doubles, so the CI width doubles.

4. A researcher wants to test whether mean exam score $\mu = 60$.

(a) State H_0 and H_1 (two-sided).

(b) What is a type 1 error in this context?

(c) What is a type 2 error in this context?

Answer: (a) $H_0 : \mu = 60$, $H_1 : \mu \neq 60$.

(b) Rejecting a true H_0 — concluding $\mu \neq 60$ when $\mu = 60$.

(c) Not rejecting a false H_0 — missing a real difference from 60.

5. Test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$. Observed: $\bar{X} = 106$, $n = 36$, $\sigma = 12$.

(a) Compute $\text{SE} = \sigma/\sqrt{n}$.

(b) Compute the test statistic $z = (\bar{X} - \mu_0)/\text{SE}$.

(c) With $\alpha = 0.05$ (reject if $|z| > 1.96$), what is your decision?

Answer: (a) $\text{SE} = 12/\sqrt{36} = 2$.

(b) $z = (106 - 100)/2 = 3$.

(c) $|3| > 1.96$, so reject H_0.

6. Test $H_0 : \mu = 50$ vs $H_1 : \mu > 50$ (one-sided). Observed: $\bar{X} = 53$, $n = 100$, $\sigma = 10$.

(a) Compute SE.

(b) Compute z.

(c) The one-sided critical value at $\alpha = 0.05$ is 1.64. Reject H_0?

Answer: (a) $\text{SE} = 10/\sqrt{100} = 1$.

(b) $z = (53 - 50)/1 = 3$.

(c) $3 > 1.64$, so reject H_0.

7. A two-sided z-test gives $z = 2.5$. Given $P(Z \leq 2.5) = 0.9938$.

(a) Compute the two-sided p-value $p = 2 \cdot (1 - 0.9938)$.

(b) With $\alpha = 0.05$, reject H_0?

(c) With $\alpha = 0.01$, reject H_0?

Answer: (a) $p = 2 \times 0.0062 = 0.0124$.

(b) 0.01240.01, so do not reject H_0.

8. The significance level is $\alpha = 0.05$.

(a) What is the probability of a type 1 error?

(b) If α is lowered to 0.01, what typically happens to β (type 2 error)?

(c) If $\beta = 0.20$, what is the statistical power?

Answer: (a) The probability of type 1 error equals $\alpha = 0.05 = 5\%$.

(b) β typically increases — stricter α makes it harder to reject H_0, raising the type 2 error risk.

(c) Statistical power $= 1 - \beta = 1 - 0.20 = 0.80 = 80\%$.

9. A 95% confidence interval for μ is $(48, 54)$.

(a) $H_0 : \mu = 50$. Does the CI contain 50? What do you conclude?

(b) $H_0 : \mu = 46$. Does the CI contain 46? What do you conclude?

(c) Which significance level α corresponds to a 95% CI?

Answer: (a) Yes, 50 is in $(48, 54)$, so we do not reject $H_0 : \mu = 50$.

(b) No, 46 is outside $(48, 54)$, so we reject $H_0 : \mu = 46$.

(c) $\alpha = 0.05$ (a 95% CI corresponds to the 5% significance level).

10. True mean is $\mu = 55$. We test $H_0 : \mu = 50$.

(a) With $n = 100$, $\sigma = 10$: compute SE and z. Reject at $\alpha = 0.05$?

(b) With $n = 4$, $\sigma = 10$: compute SE and z. Reject at $\alpha = 0.05$?

(c) What does comparing (a) and (b) illustrate about sample size and statistical power?

Answer: (a) SE $= 10/\sqrt{100} = 1$, $z = (55-50)/1 = 5 > 1.96$, reject H_0.

(b) SE $= 10/\sqrt{4} = 5$, $z = (55-50)/5 = 1 < 1.96$, do not reject H_0.

(c) Larger n increases power — with more observations we are more likely to detect a true effect.

11. Consider statistical vs. practical significance.

(a) A study with $n = 1{,}000{,}000$ finds $p = 0.001$. Does this guarantee the effect is large?

(b) A study with $n = 20$ finds $p = 0.08$. Is it possible that a real effect exists?

(c) Name one way to assess practical significance beyond the p-value.

Answer: (a) No — with very large n even trivially small effects can become statistically significant. A tiny p-value does not imply a large or meaningful effect.

(b) Yes — low power with small n may miss real effects (type 2 error). The result is inconclusive rather than evidence of no effect.

(c) Possible answers: effect size (e.g. Cohen's d), confidence interval width, percentage change, or expert judgment on minimum practically relevant difference.

12. Test $H_0 : \mu = 200$ vs $H_1 : \mu \neq 200$. Observed: $\bar{X} = 196$, $n = 64$, $\sigma = 16$.

(a) Compute SE.

(b) Compute z.

(c) With critical value ± 1.96, reject H_0?

Answer: (a) SE $= 16/\sqrt{64} = 2$.

(b) $z = (196 - 200)/2 = -2$.

(c) $|-2| = 2 > 1.96$, so reject H_0.

Part III

Regression and Causality

Part III: Regression and Causality

In Part I we covered the foundations of mathematics and in Part II we covered data, descriptive statistics and probability theory. This part introduces covariation and regression analysis — the central method for studying how variables relate to each other. We begin with the logic of causal analysis and the mathematical measurement of covariation, then move through the least squares method, multiple regression, and statistical testing of regression results. The part closes with methods for studying causal relationships using regression analysis.

Chapter 20

Causality and Covariation

This chapter introduces the logic of causal analysis: how we reason about causes and effects, what role covariation plays as a necessary condition for causality, and how to measure covariation mathematically.

20.1 Cause and effect

To be able to study causal relationships, we need to compare what happened due to a cause with what would have happened if the cause had never occurred. What did not happen is called the counterfactual outcome, also called potential outcome. The causal relationship entails an effect, which is the difference between the actual outcome and the counterfactual outcome. The cause can be anything that can be thought to affect something else, for example human behavior, environmental change, legislation, a process, and so on.

In our daily lives, we constantly assume different types of causal relationships. We drink water to become less thirsty. We pursue education to learn skills and qualify for new jobs. We take medicine to alleviate illness. Our conviction about many such everyday causal relationships is often based on a similar logic to that used in analytical work and research. By collecting observations (experiencing) in our everyday lives and conducting repeated attempts (every day), we have experience that a lot of our everyday actions entail certain effects.

Figure 20.1 illustrates how cause A occurs at a point in time and affects a phenomenon B, whose development follows the solid line diagonally up to the right in the graph. The change in level since the cause occurred is the distance between the horizontal gray line and the solid black line. A common misunderstanding is that it is sufficient to compare this visible change, the difference between the horizontal gray line and the solid black line. But the effect, the result of cause A, is the difference between the solid line and the dashed line. The dashed line describes how the development would have looked if cause A had never happened, the counterfactual outcome.

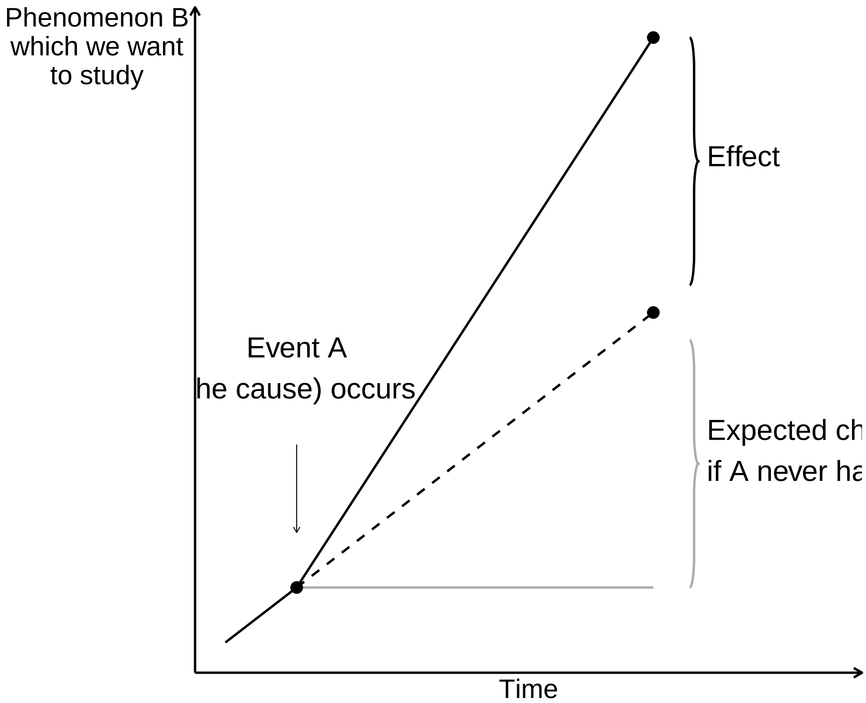


Figure 20.1: Counterfactual outcome

By definition, counterfactual outcomes cannot be observed, since this would require us to have access to a parallel universe where the only difference from our universe is that the cause never happened. To study causal relationships, we instead observe a group of observations that have been treated (exposed to cause A) and a group of observations that have not been treated. These groups are called treatment group and control group. Ideally, the only difference between the two groups should be the treatment itself. We interpret the difference between the groups as an effect.

Say we are to study the effect of how a new medicine affects disease symptoms in patients. We have a group of patients who receive medicine and another group that does not receive medicine. We compare how their disease symptoms develop and interpret the difference between the groups as an effect of the medicine. By studying the control group, we hope, ideally, to be able to observe a result that describes the counterfactual outcome, how the symptoms would have developed for the patients who received medicine if they had never received it. It is important that it is precisely groups with several observations. By comparing several observations, we reduce the risk that the patterns we observe are a result of chance, which we will return to later.

If phenomenon A causes phenomenon B, we call A active treatment. We call the alternative scenario control treatment. The control treatment is usually the passive alternative where we do not perform an action, for example not taking a medicine. Often it is clear from the context what is cause and effect, which is why these are usually called treatment and control. For each unit of analysis, it is only possible to study one result. An observation can only be either treatment or control.

Ideally, we want to study how a patient feels after having taken the medicine, compared to how the same patient would have felt if this person had never taken the medicine. If it is a medicine that is to treat a recurring condition, for example headache, it is tempting to try this by asking the same patient to take the medicine one day and refrain from taking the medicine the next time, and compare the difference. But this is not the counterfactual outcome either. The effect of the medicine might vary for one and the same patient over time. The patient might be in better or worse condition at one time point, which affects both the headache and what effect the medicine has. The patient might get used to the medicine, which also affects both the patient's experience and the effect.

Let us instead imagine that we are to study what effect the government's new policy (cause) has on the amount of job opportunities (outcome). To analyze this, we would preferably compare the outcome, that is, what has actually happened and we may observe, against how the world would have looked if the government's policy had never taken place (the counterfactual outcome). This is of course impossible.

It is possible observe how the world looked before and after the government's policy, but these are two separate observations. The situation before is not the counterfactual outcome. Things happen during the time period that can affect how the world looks and we can never be sure what causes the changes we see. We also cannot be sure that the change we observe is a good measure of the policy's effect, which is what we want to estimate.

Often it is difficult to say exactly what the counterfactual outcome is. If we are interested in the effect of a patient taking a medicine compared to if a patient does not take a medicine, it may sound obvious. But should the patient who does not take the medicine lie completely still during the time or take a walk? Should both patients take a jog or drink and eat something particular?

If we want to study effects of the government's policy, it usually becomes even more difficult to know what the counterfactual outcome is. What would have happened if the government had not implemented its policy? Would the government have implemented some other policy or resigned? If the policy only concerns a smaller part of society, for example only the companies in one industry, we may sometimes treat the affected industry as treatment group and other industries as control group and compare their developments.

When we divide units of analysis, for example patients, into treatment group and control group, there is a risk that we miss some characteristic that will ultimately affect the results in our analysis. The division into groups must therefore occur in such a way that this division itself does not affect the phenomenon we want to study, such as the disease symptoms.

Often there are phenomena that affect both the distribution between treatment and non-treatment as well as the outcome we want to study the effect on. Say for example that we want to study what effect an education has on students' future incomes. If we compare the students' income development with the rest of the population, we risk missing that the students might have some characteristic or prerequisite that would affect their incomes regardless of whether they studied the education or not. This characteristic also affects the students' decision to study the education. Even without the education, the students would therefore have had higher incomes compared to people who did not study the education (compare figure 20.1). This means in that case that a simple comparison of the students' incomes with the rest of the population overestimates the effects of the education.

A method that is often described as the ideal for distributing units of analysis into control and treatment groups is chance (randomization). The idea behind this is that units of analysis have known and unknown characteristics that can affect our analysis. By distributing the units of analysis randomly, these characteristics, if we have sufficiently many observations, will not have any systematic impact on the analysis results. Using this method is called conducting a randomized controlled trial.

Randomized studies often require a large amount of observations and therefore risk becoming costly and complicated. If we for example want to study the effects of an education, one method could be that we select a group of people randomly in the population and let half of these study the education. The part of the group that gets to study the education becomes our treatment group and the rest becomes our control group. Since the participants know that they are participating in an experiment, this can in itself affect the results. The participants might behave in some other way than people who would study the education voluntarily. Letting a large amount of people study an education for free as an experiment with unclear results can become both costly and demanding.

20.2 Variation, Covariation and Association

Suppose we are to study how phenomenon A affects phenomenon B, which we do with the help of a treatment and control group. To be sure that A causes B, we first want to isolate a variation in A that is created externally and is not created by, for example, phenomenon B. This is called observing an exogenous variation in A.

Thereafter we examine whether we find any form of covariation. If phenomenon A on average increases at the same time as phenomenon B on average increases, and decreases at the same time as B decreases, then there is a positive covariation between A and B. If phenomenon A increases at the same time as B decreases and A decreases while B increases, then there is a negative covariation between A and B. Compare positive and negative slope in graphs that we introduced in chapter 4 .

Covariation is a necessary condition for us to be able to claim that there is a causal relationship between two phenomena. Say for example that we study what effect a medicine has on patients' disease symptoms. We control when the patients take the medicine (exogenous variation) and then observe the disease symptoms. We examine whether there is a covariation between use of the medicine and disease symptoms.

Covariation is a condition for us to be able to claim that we have found a causal relationship. It is also a requirement for us to be able to measure an effect (the difference between the treatment and control groups). However, covariation is no proof of a causal relationship. Suppose we observe that neighborhoods with more police patrols tend to have higher crime rates. A simple interpretation might suggest that police cause crime. However, the more likely explanation is that police are deployed to areas with existing crime problems - the causation runs in the opposite direction.

Consider another example. Students who receive tutoring often perform worse on standardized tests than students who don't receive tutoring. Does this mean tutoring hurts academic performance? More likely, students who struggle academically are more likely to seek tutoring in the first place. The tutoring might actually improve their performance compared to what it would have been without help, but not enough to surpass students who never needed extra assistance.

These examples illustrate why establishing causation requires careful analysis beyond simply observing covariation. We must consider alternative explanations: reverse causation (B causes A instead of A causing B), confounding variables (C causes both A and B), or selection effects (certain types of units are more likely to experience the treatment). Even when we find strong covariation, multiple causal stories may be consistent with the same pattern of data.

20.3 Observational Study and Experiment

When we study covariation, it is common to distinguish between observational studies and experiments. In an observational study, we study collected data, for example from a survey or information collected by others. In an experiment, we arrange collection of data to be able to study cause and effect, for example by randomly distributing units of analysis between treatment and control groups. In both observational studies and experiments, we study covariation.

Effects are often mixed up with other phenomena that we also want information about. Say for example that we send out a survey to students who have studied a course and ask if they have had use of the course. Everyone answers yes, which means that the students are satisfied. This is admittedly fun to know but it is not the same thing as the education's effect on the students' lives, since we do not have a counterfactual outcome, or a control group, to compare against.

In section 5.10 we compared GDP and happiness in the world's countries and showed a positive covariation, where inhabitants in countries with higher GDP are more satisfied with their lives. This is an example of an observational study. The covariation does not necessarily show how large an effect an income increase would have on happiness. People in rich countries have on average many advantages compared to inhabitants in poor countries that can be thought to affect both happiness and wealth.

To estimate an effect, we need to control for all other phenomena that can affect the covariation. Full-scale controlled experiments are in many cases unethical and impossible within social science. Say for example that we want to study the relationship between income and happiness. We decide to randomly assign a large group of people into treatment and control groups. The treatment group gets lots of money while the control group gets to live their entire lives in poverty. Even if we were to disregard the ethical aspects, the participants' behavior and experiences will probably be affected by the knowledge that they are participating in an experiment. The results therefore become meaningless regardless of how we go about it.

Due to the difficulties of using experiments, social science is largely instead referred to what is called quasi-experiments, or natural experiments. Quasi-experiments refer to analyses where the division into treatment and control groups is identified by the analyst afterwards. Often this happens by the analyst finding some mechanism that has led to a more or less unintentional random division into treatment and control groups.

Say for example that the government enacts a new law that is to be implemented in all the country's municipalities. Due to some typographical error in a document, the law is delayed by one year for some of the municipalities. We use this to study effects of the new law (the treatment). We have by chance gotten two groups of municipalities: a group where the law takes effect the first year (the treatment group) and another group with municipalities where the law takes effect one year later (the control group).

A well-known historical example of a quasi-experiment was the cholera outbreak that occurred in London in 1854. The physician John Snow noted that mortality in cholera seemed to be connected with the use of dirty water from the River Thames. By studying variations in mortality and water use between different parts of London and over time, Snow argued that the dirty water caused increased mortality among the population. The water use was not designed by any analyst and the inhabitants of London were not randomly divided into treatment and control groups. But since the use of water with different dirtiness was randomly distributed in London, Snow could utilize this to study whether water consumption caused diseases.

We return to these questions in chapter 27 where we introduce some methods for quasi-experimental observational studies.

20.4 Covariation: Mathematical Measures

Having established that covariation is a necessary condition for causality, we now introduce mathematical tools for measuring it.

20.5 Covariation in graphs

In section 20.2 we introduced the concept of covariation and went through how positive covariation means that low values for one variable X on average are associated with low values for another variable Y, and that high values on average are associated with high values. Negative covariation means that high values in one variable X are associated with low values in the other variable Y and low values in X are associated with high values in Y.

One way to study covariation is to illustrate the information we are interested in in graphs. But graphs rarely go particularly far. Here follows an attempt to illustrate these things. Figure 20.2 illustrates three examples that describe covariation between the two variables X and Y. Each graph shows its own collection of observations, where each dot represents an observation with a value for X and Y respectively. In the uppermost graph, no. 1, we see a clear positive covariation. The dots lie almost on a straight line up toward the right in the graph. In the graph we may relatively easily read how much variable Y changes on average when X increases by one unit, in the same way as when we introduced linear functions and graphs in chapter 4 .

In graph no. 2 we can still describe the placement of the dots roughly as along a line up toward the upper right corner. But the dots are more spread out, which makes it a bit harder to say with certainty exactly how much Y changes on average when we go from low to high values on X. It therefore requires a more detailed analysis with the help of calculations if we want to know something more exact about the relationship between the variables.

In the third graph at the bottom it is more uncertain whether we have any meaningful relationship between the variables at all. The dots are spread out as in a large cloud. But even in this graph there is a positive covariation between X and Y, which the author knows since the graph's data is created with the help of the computer.

As a rule, it is required that we calculate covariations more carefully in order for us to be completely sure whether the variables can be assumed to covary or not, and how large this covariation is.

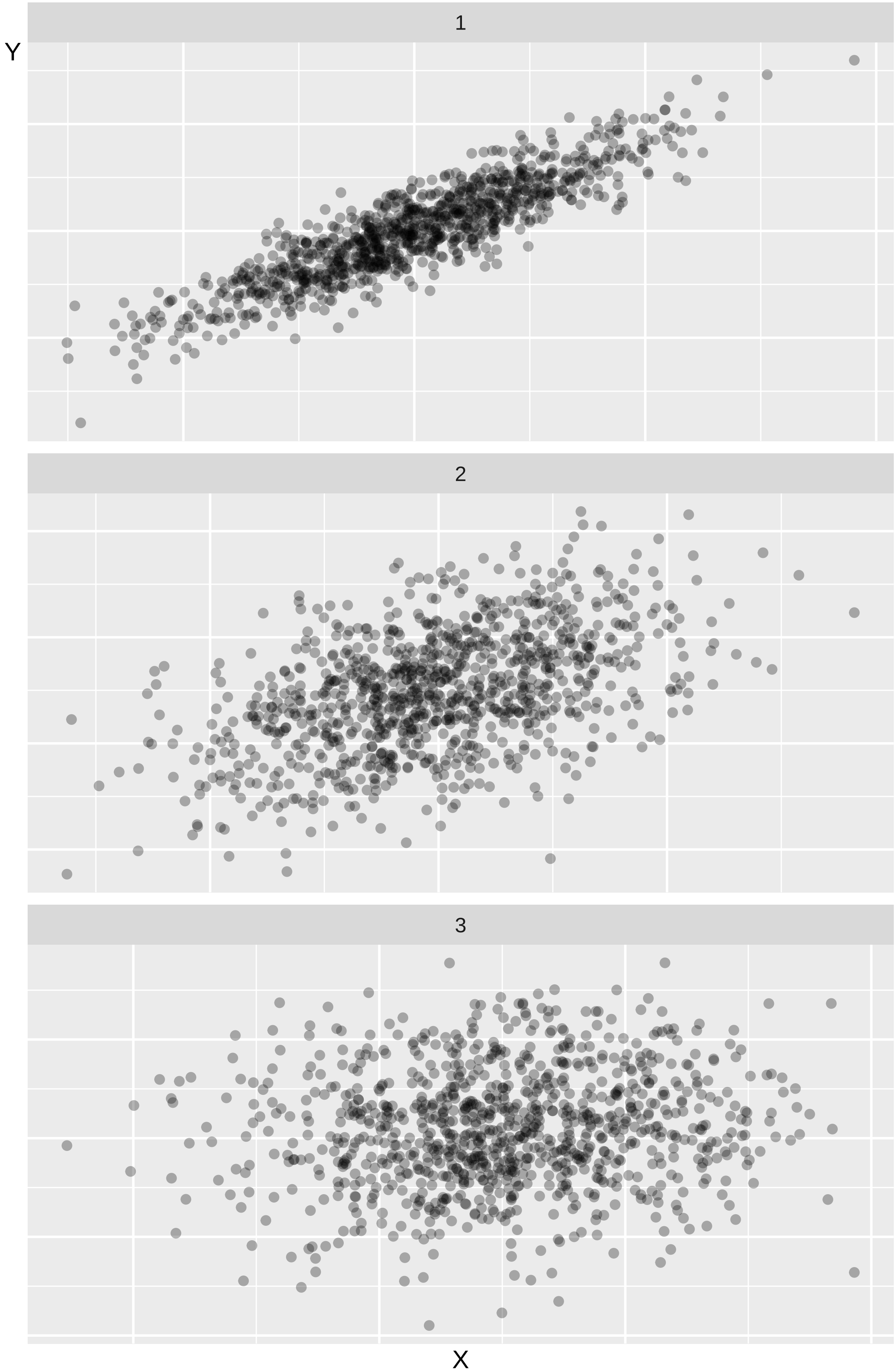


Figure 20.2: Three examples of covariation

20.6 Covariation 1: Covariance

Now we shall introduce our first statistical measure of linear covariation: covariance. Covariance is a measure of covariation between two variables, for example x and y. For population data:

$$cov(x, y) = \frac{1}{n}\sum_{i}^{n}(x_i - \bar{x})(y_i - \bar{y}) \tag{20.1}$$

Our goal is to estimate the covariation in a population and possibly a superpopulation. From our population we take a sample of observations that we use to estimate the covariation.

In previous examples we have worked with populations for one variable. Since we are now interested in covariation, our population consists of two variables. The covariance between x and y in a population is denoted σ_{xy}(compare the notation for the population's variance σ_x). Positive covariance means positive covariation: higher values of x are associated with higher values of y. Negative covariance means negative covariation, that higher values of x are associated with smaller values of y and vice versa. It does not matter in which order we write the variables in the parentheses: $\operatorname{cov}(x, y) = \operatorname{cov}(y, x)$. If we take the covariance for x and the same variable x, we get the variance of x:

$$\begin{aligned} \operatorname{cov}(x, x) &= \left(\frac{1}{n}\right)\sum_{i}^{n}(x_i - \bar{x})(x_i - \bar{x}) \\ &= \left(\frac{1}{n}\right)\sum_{i}^{n}(x_i - \bar{x})^2 \\ &= \operatorname{var}(x) \end{aligned} \tag{20.2}$$

We can also apply Bessel's correction here to reduce the underestimation of the spread in the population that we risk making. This gives us what is called sample covariance:

$$\textbf{Sample covariance: } \operatorname{cov}(x, y) = \left(\frac{1}{n-1}\right)\sum_{i}(x_i - \bar{x})(y_i - \bar{y}) \tag{20.3}$$

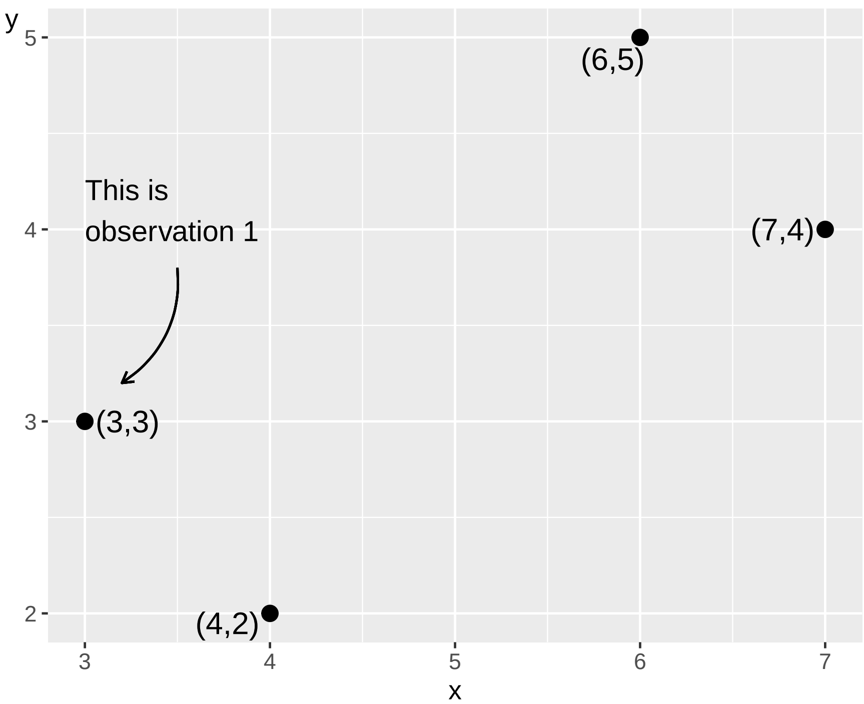


Figure 20.3: Four observations for x and y

Table 20.1: Calculations for covariance between x and y

Observation	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$
1	3	3	−2	−0.5	1
2	4	2	−1	−1.5	1.5
3	6	5	1	1.5	1.5
4	7	4	2	0.5	1
Mean	5	3.5			
Sum					5

Let us estimate the covariance between variables x and y that we used in the previous section. Figure 20.3 describes our four observations in a table to the left and a graph to the right. In the table we have one observation per row and one variable per column. In the graph each observation is represented by a point. The point furthest to the left is observation 1: $(x, y) = (3, 3)$. The point furthest to the right is observation 4: $(x, y) = (7, 4)$. Row 1 in the table consists of the first value for x and y respectively and represents values that belong together in some way. If we work with observed data, collected information, each observation represents an observation unit, for example information about a person or perhaps a country. Our four observations could thus represent four people, four countries or something else. In table 20.1 we have calculated the parts we need to get the covariance between x and y (equation (20.3)):

$$\operatorname{cov}(x, y) = \left(\frac{1}{n-1}\right) \sum_i (x_i - \bar{x})(y_i - \bar{y}) = \left(\frac{1}{3}\right) \times 5 = \frac{5}{3} \tag{20.4}$$

We find that $\operatorname{cov}(x, y) = \frac{5}{3}$. This positive value indicates positive covariation. Covariance is useful for getting an estimate of whether the covariation between two variables is positive or negative, but it is difficult to say much more than that. There are no limitations for which values covariance can take. The value of covariance depends on which unit the variables' values have. For example, we will get different results depending on whether we use a variable that describes income in USD or in thousands of USD.

20.7 Covariation 2: The correlation coefficient

Another measure of linear covariation is Pearson's r, also called Pearson's correlation coefficient or the correlation coefficient. The correlation coefficient for variables x and y is denoted for population ρ_{xy}(Greek rho):

$$\textbf{Correlation coefficient: } \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \tag{20.5}$$

where σ_{xy} is the covariance between x and y in the population, and σ_x and σ_y are standard deviation in the population for each respective variable.

Another way to describe this is that the correlation coefficient is standardized covariance (compare section 15.8). The correlation coefficient can only take values between -1 and 1, where -1 means perfect negative correlation and 1 means perfect positive correlation. If the correlation coefficient equals 0, this indicates that there is no linear covariation between the variables. Since all results for the correlation coefficient are within the interval $[-1, 1]$, different correlation coefficients are comparable.

If we work with sample data, the correlation coefficient is denoted r_{xy} or $corr\,(x, y)$:

$$r_{xy} = \frac{\operatorname{cov}(x, y)}{s_x s_y} \tag{20.6}$$

where we instead have estimated covariance in the numerator and estimated standard deviation for the two variables in the denominator. The different parts in this equation we have defined earlier in this chapter. We write out the equations and add Bessel's correction:

$$
\begin{aligned}
r_{xy} &= \frac{\operatorname{cov}(x,y)}{s_x s_y} \\
&= \frac{\left(\frac{1}{n-1}\right)\sum_i^n (x_i-\bar{x})(y_i-\bar{y})}{\left(\frac{\sum_i^n (x_i-\bar{x})^2}{n-1}\right)^{1/2}\left(\frac{\sum_i^n (y_i-\bar{y})^2}{n-1}\right)^{1/2}} \\
&= \frac{(n-1)^{-1}\sum_i^n (x_i-\bar{x})(y_i-\bar{y})}{(n-1)^{-1}\left(\sum_i^n (x_i-\bar{x})^2\right)^{1/2}\left(\sum_i^n (y_i-\bar{y})^2\right)^{1/2}} \\
&= \frac{\sum_i^n (x_i-\bar{x})(y_i-\bar{y})}{\left(\sum_i^n (x_i-\bar{x})^2\right)^{1/2}\left(\sum_i^n (y_i-\bar{y})^2\right)^{1/2}}
\end{aligned}
\tag{20.7}
$$

Since Bessel's correction is included in all three measures, we cancel $(n-1)^{-1}$ from both numerator and denominator in the last row. Let us again demonstrate by estimating r_{xy} for the two variables x and y and their four observations from figure 20.3 . We reuse the results for covariance (equation (20.4)) and standard deviation for x and y(equation (15.12)):

$$
r_{xy} = \frac{\operatorname{cov}(x,y)}{s_x s_y} \approx \frac{5/3}{\left(+\sqrt{\frac{10}{3}}\right)\left(+\sqrt{\frac{5}{3}}\right)} \approx 0.71 \tag{20.8}
$$

This result indicates a positive correlation. High values of x coincide with high values of y and low values of x coincide with low values of y. Different correlation coefficients can be compared but it is still difficult to interpret this type of result more precisely.

20.8 Chapter summary

- Frequency distribution shows number of observations per value or interval and can be reported in for example a table or a graph. A special type of bar graph that is often used in these cases is histogram, which shows frequencies per interval. In statistics, different measures of dispersion are used to measure distribution of values in a variable.
- Variance in the population for variable x: σ_x. Estimated variance with sample data for the same variable x: $\hat{\sigma_x} = \operatorname{var}(x) = \frac{1}{n}\sum (x_i-\bar{x})^2$ where n is number of observations, x_i is observation i of variable x and $\bar{x}$ is mean.
- Standard deviation for variable x: $+\sqrt{\sigma_x}$. For sample data: $\hat{\sigma}_x = s_x = +(var)^{1/2} = \left(\frac{\sum(x_i-\bar{x})}{n}\right)^{1/2}$.
- Bessel's correction means that we divide by $n-1$ instead of n. This reduces the error that risks arising when we use sample data to estimate measures of dispersion for a population. Bessel's correction can also be used for other measures.
- Standardized value, also called z-value or z-score, indicates a value's difference with the mean divided by standard deviation. For variable x this can be calculated $z_x = \frac{x_i-\bar{x}}{s_x}$. Normalized values for variable x are given by $x_{norm} = \frac{x_i - x_{min}}{x_{max}-x_{min}}$.
- Covariance is a measure of linear covariation between two variables. For the population for x and y: σ_{xy}. Estimated with sample data: $\hat{\sigma}_{xy} = \operatorname{cov}(x,y) = \frac{1}{n}\sum (x_i-\bar{x})(y_i-\bar{y})$.
- The correlation coefficient (Pearson's r) for population: $\rho_{xy} = \frac{\operatorname{cov}(x,y)}{s_x s_y}$. Sample data: $r_{xy} = \sum\left(\frac{x_i-\bar{x}}{s_x}\right)\left(\frac{y_i-\bar{y}}{s_y}\right)$ where s_x is standard deviation for variable x.

20.9 Exercises

1. An experiment shows that the treatment group has average outcome $\bar{Y}_B = 80$ and the control group has $\bar{Y}_K = 60$.

(a) What is the estimated treatment effect (hat{tau}=bar{Y}*{B}-bar{Y}*{K}) ?

(b) If the counterfactual outcome for the treatment group would have been (bar{Y}_{B}^{cf}=65) , what is the true causal effect?

(c) What does the difference between (a) and (b) represent?

Answer: (a) $\hat{\tau} = 80 - 60 = 20$, (b) $\tau = 80 - 65 = 15$, (c) selection bias or confounding.

2. A researcher assigns 50 students randomly to a new reading program (treatment group) and 50 students to the standard curriculum (control group).

(a) What is the treatment in this experiment?

(b) Why is random assignment used instead of letting students choose their group?

(c) Is this an experiment or an observational study?

Answer: (a) the reading program, (b) ensures groups are comparable on average, (c) experiment.

3. A dataset shows that as years of education x increase, average income y also increases. Family wealth z correlates with both variables.

(a) Is the covariation between (x) and (y) positive or negative?

(b) Does this covariation prove that education causes higher income?

(c) What role does (z) play in the analysis?

Answer: (a) positive, (b) no — covariation is not proof of causation, (c) a potential confounder.

4. Classify each variable as nominal, ordinal, interval, or ratio data:

(a) Annual income in dollars.

(b) Education level (primary / secondary / tertiary).

(c) Temperature in Celsius.

Answer: (a) ratio, (b) ordinal, (c) interval.

5. A researcher studies 300 randomly selected Swedish adults to estimate average weekly screen time. The goal is to draw conclusions about all Swedish adults.

(a) What is the sample in this study?

(b) What is the population?

(c) What could the superpopulation be?

Answer: (a) the 300 sampled adults, (b) all Swedish adults, (c) all comparable future populations (e.g. future Swedish adults).

6. A company gives voluntary employees access to a wellness program and later compares their sick-day rates with employees who did not join.

(a) Is this an experiment or an observational study?

(b) What selection problem may arise when comparing the two groups?

(c) How could the study design be improved to reduce this problem?

Answer: (a) observational study, (b) self-selection bias (healthier employees may be more likely to join), (c) randomize enrollment in the program.

7. A new traffic law is introduced in region A, while region B introduces the same law two years later due to an administrative delay.

(a) What is the treatment in this quasi-experiment?

(b) Which region acts as the control group during the first two years?

(c) What assumption must hold for this to be a valid quasi-experiment?

Answer: (a) the new traffic law, (b) region B, (c) the two regions would have had parallel trends without the law.

8. Four treated observations have outcomes 9, 7, 8, 6 and four control observations have outcomes 5, 4, 6, 5.

(a) Calculate the group means (bar{Y}*{B}) and (bar{Y}*{K}) .

(b) What is the estimated treatment effect (hat{tau}=bar{Y}*{B}-bar{Y}*{K}) ?

(c) Under what condition is (hat{tau}) an unbiased estimate of the causal effect?

Answer: (a) $\bar{Y}_B = 7.5, \bar{Y}_K = 5.0$, (b) $\hat{\tau} = 2.5$, (c) random assignment to treatment and control groups.

9. A medical study compares patients who chose to take a new supplement with those who did not, and finds that supplement-takers have better health outcomes.

(a) What is the potential source of bias in this study?

(b) How could randomization solve this problem?

(c) What is this type of bias commonly called?

Answer: (a) selection bias (healthier individuals may self-select into treatment), (b) randomization removes systematic differences between groups, (c) selection bias.

10. A researcher takes a sample of $n = 50$ from a population of $N = 10000$. The sample mean is $\bar{x} = 42$.

(a) What fraction of the population is sampled?

(b) What parameter does (bar{x}=42) estimate if the sampling is random?

(c) Why is random sampling important for the representativeness of the sample?

Answer: (a) $n/N = 50/10000 = 0.005$, (b) the population mean μ , (c) ensures the sample is representative of the population.

11. A study finds that cities with more hospitals have higher mortality rates.

(a) Is this positive or negative covariation between hospitals and mortality?

(b) Does this mean that hospitals cause deaths?

(c) What is the most likely explanation for this pattern?

Answer: (a) positive, (b) no — reverse causation, (c) hospitals are located where there is more illness.

12. A study shows that students who eat breakfast score higher on exams than those who skip it. A researcher concludes that breakfast improves exam performance.

(a) What is the treatment?

(b) Name one possible confounding variable.

(c) How could a randomized controlled trial help establish causality?

Answer: (a) eating breakfast, (b) socioeconomic status, (c) randomly assign students to eat or skip breakfast ensuring groups are comparable.

1. Given the observations $x = \{1, 5, 9\}$:

(a) Calculate the mean (bar{x}=frac{1}{n}sum x_{i}) .

(b) Calculate the sample variance (var(x)=frac{1}{n-1}sum(x_{i}-bar{x})^{2}) .

(c) Calculate the standard deviation (s_{x}=sqrt{var(x)}) .

Answer: (a) $\bar{x} = 5$, (b) $var(x) = 16$, (c) $s_x = 4$.

2. Given the observations $x = \{3, 6, 9, 12\}$:

(a) Calculate the mean (bar{x}) .

(b) Find the median.

(c) What is the mode of the dataset ({3,6,6,9,12}) ?

Answer: (a) $\bar{x} = 7.5$, (b) median $= 7.5$, (c) mode $= 6$.

3. Given $x = \{1, 5, 9\}$ with $\bar{x} = 5$ and $s_x = 4$, calculate the z-score $z_i = \frac{x_i - \bar{x}}{s_x}$ for each observation:

(a) (x_{1}=1) .

(b) (x_{2}=5) .

(c) (x_{3}=9) .

Answer: (a) $z_1 = -1$, (b) $z_2 = 0$, (c) $z_3 = 1$.

4. Given $x = \{2, 4, 6, 10\}$, calculate the normalized value $x_{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}}$:

(a) What are (x_{min}) and (x_{max}) ?

(b) What is the normalized value of (x_{2}=4) ?

(c) What is the normalized value of (x_{3}=6) ?

Answer: (a) $x_{min} = 2, x_{max} = 10$, (b) $x_{norm} = 0.25$, (c) $x_{norm} = 0.5$.

5. Describe the sign of covariance in each case:

(a) Income increases as education increases.

(b) Crime rates are higher in areas with more police (due to reverse causation).

(c) Two variables have (cov(x,y)=0) . What does this indicate?

Answer: (a) positive, (b) positive, (c) no linear covariation between the variables.

6. Given $x = \{1, 3, 5\}$ and $y = \{2, 6, 10\}$:

(a) Calculate (bar{x}) and (bar{y}) .

(b) Calculate (cov(x,y)=frac{1}{n-1}sum(x_{i}-bar{x})(y_{i}-bar{y})) .

(c) What does the sign of (cov(x,y)) indicate?

Answer: (a) $\bar{x} = 3, \bar{y} = 6$, (b) $cov(x, y) = 8$, (c) positive covariation.

7. Given $x = \{1, 3, 5\}$ and $y = \{10, 6, 2\}$:

(a) Calculate (bar{x}) and (bar{y}) .

(b) Calculate (cov(x,y)) .

(c) Interpret the result.

Answer: (a) $\bar{x} = 3, \bar{y} = 6$, (b) $cov(x, y) = -8$, (c) negative covariation (y decreases as x increases).

8. For $x = \{1, 3, 5\}$ and $y = \{2, 6, 10\}$ with $s_x = 2$, $s_y = 4$ and $cov(x, y) = 8$:

(a) Calculate the correlation coefficient (r_{xy}=frac{cov(x,y)}{s_{x}s_{y}}) .

(b) What range can (r_{xy}) take?

(c) What does (r_{xy}=1) mean?

Answer: (a) $r_{xy} = 1$, (b) $-1 \leq r_{xy} \leq 1$, (c) perfect positive linear covariation.

9. For $x = \{1, 3, 5\}$ and $y = \{10, 6, 2\}$ with $s_x = 2$, $s_y = 4$ and $cov(x, y) = -8$:

(a) Calculate (r_{xy}) .

(b) Interpret the value.

(c) Does (r_{xy}=-1) imply causation?

Answer: (a) $r_{xy} = -1$, (b) perfect negative linear covariation, (c) no — correlation does not imply causation.

10. A variable x has $\bar{x} = 10$ and $s_x = 3$.

(a) What z-score does (x_{i}=13) get?

(b) What z-score does (x_{i}=7) get?

(c) What original value has z-score (z=2) ?

Answer: (a) $z = 1$, (b) $z = -1$, (c) $x_i = 16$.

11. For $x = \{2, 4, 6\}$ and $y = \{5, 5, 5\}$:

(a) Calculate (bar{x}) and (bar{y}) .

(b) Calculate (cov(x,y)) .

(c) What does (cov(x,y)=0) tell us about the covariation?

Answer: (a) $\bar{x} = 4, \bar{y} = 5$, (b) $cov(x, y) = 0$, (c) no linear covariation between x and y.

12. Three countries have GDP $x = \{10, 20, 30\}$ (thousands USD) and happiness index $y = \{4, 6, 8\}$.

(a) Calculate (bar{x}) and (bar{y}) .

(b) Calculate (cov(x,y)) .

(c) Calculate (r_{xy}) given (s_{x}=10) and (s_{y}=2) .

Answer: (a) $\bar{x} = 20, \bar{y} = 6$, (b) $cov(x, y) = 20$, (c) $r_{xy} = 1$.

Chapter 21

The Least Squares Method

This chapter introduces the least squares method, which is another method for studying linear covariation. In chapter 20 we introduced different ways to study covariation among two variables. The least squares method can be used to study how one variable covaries with one or several other variables simultaneously. In this chapter, we begin by estimating the covariation between two variables to later introduce several variables.

The least squares method is a central building block within social and natural sciences, which is why we will devote much attention to this in the rest of this book. The least squares method can be traced back to 1805 when the French mathematician Adrien-Marie Legendre (1752–1833) published a book about comets (Legendre, 1806). The earliest example within social science research is Yule (1899), which represents the first known example of using the least squares method with multiple variables.

The chapter will start with some basic logic behind the least square method, and then introduce the mathematical basics on how to calculate covariation with this method. In the next chapter we will introduce some examples from social science.

21.1 Basic logic

In figure 21.1 (same as the graph to the right in figure 5.3) we see a straight black line through the black dots. The line show the average change in happiness (on the vertical axis) as we go from low values of GDP to higher (on the horizontal axis). This line is not any line. It is drawn using a special method called the least squares method. The line is an illustration of a numerical result given by calculations based on the observations (the black dots), where each dot represents a combined value of happiness and GDP.

The straight black line is called the regression line. The least squares method is an example of regression analysis, which is a field within statistics. This book does not cover any other regression methods.

The least squares method is one way to calculate the average change in one variable, which we call the explained (or dependent) variable, as another (the explanatory or independent) variable increase with one unit. In the case of figure 21.1 we have calculated the average change in the variable happiness (the explained variable), as GDP per capita (the explanatory variable) increases with one unit. One unit of GDP per capita is in this case equal to 1,000 USD adjusted for local prices, purchasing power.

Why would we want to calculate this? For instance, this may be interesting if we wish to test some kind of theory of how GDP, or income, affects happiness. The graphical result in figure 21.1 indicate a positive covariation – people in countries with higher GDP is happier on average. But suppose now that we have a theory that says that this happiness is actually caused by temperature. Using least squares we can measure covariation between both temperature and happiness, as well as covariation between GDP and happiness at the same time. This way, we may study to what extent GDP or temperature may explain happiness. This require that we can use least squares with three or more variables. These kind of methods are extremely useful for a lot of analytical work and is introduced in chapter 23 .

Before we move on to the mathematics behind the least squares method it is important to remember that when we study covariation we are usually trying to estimate covariation in a population. But since population values are usually unknown, we use sample data to estimate the population covariation. Estimations of population values are associated with uncertainty. In chapter 24 we will introduce how we can calculate probabilities regarding this uncertainty.

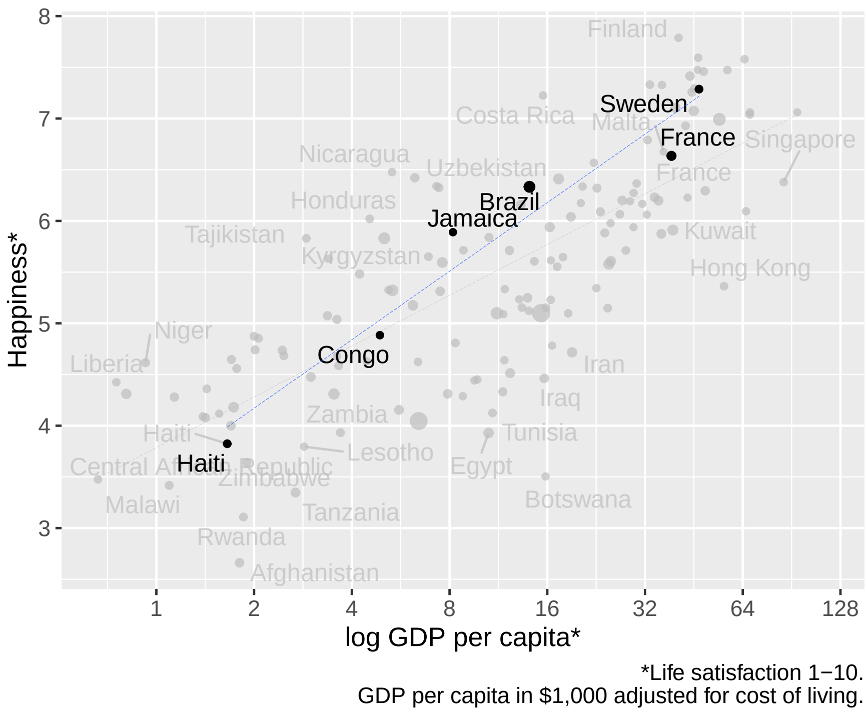


Figure 21.1: GDP and happiness

21.2 Covariation with Straight Line

For our calculations, we will here again use the same made up observations for the variables X and Y as in chapter 20, which are repeated in figure 21.2 . The reason we use these numbers is to make the mathematical calculations easy. In the hypothetical examples here, it is irrelevant exactly what the population represents and calculated examples are used only as illustration of the method.

All straight lines in a graph can be described with an equation of the form $Y = a + bX$, that is, the equation of the straight line (section 4.4). The constant coefficient a is the line's y-intercept or intercept while the constant coefficient b is the line's slope, or slope coefficient. Y and X are in this case our variables. The line we will draw will give us an illustration of the average change in variable Y as we move from relatively lower to relatively higher values on X.

Obs. i	X_i	Y_i
1	3	3
2	4	2
3	6	5
4	7	4

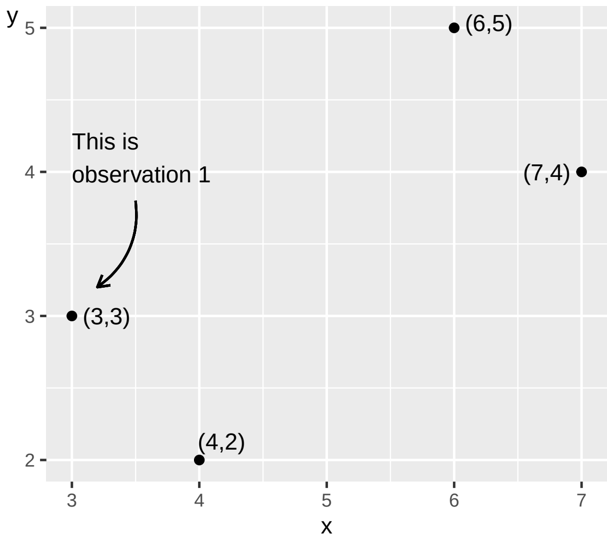


Figure 21.2: Four observations for x and y

Figure 21.3 shows the four points and a straight line, which slopes upward in the graph and illustrates the linear covariation between the points. The least squares method allows us to use information from X and Y to find precisely those values for the coefficients a and b that allow us to draw the straight line through the points where the sum of the vertical squared distance between each point and the straight line is as small as possible.

In the figure's graph, an example of the distance is given in the form of $\hat{v}_3 = 5-4$. Any other straight line would result in a larger summed vertical distance between the points and the line. If the distance between a point and the line equals 5, the squared distance is the same value squared, $5^2 = 25$. Points that lie below the line get a negative distance, for example -5, and the squared value then becomes $(-5)^2 = 25$.

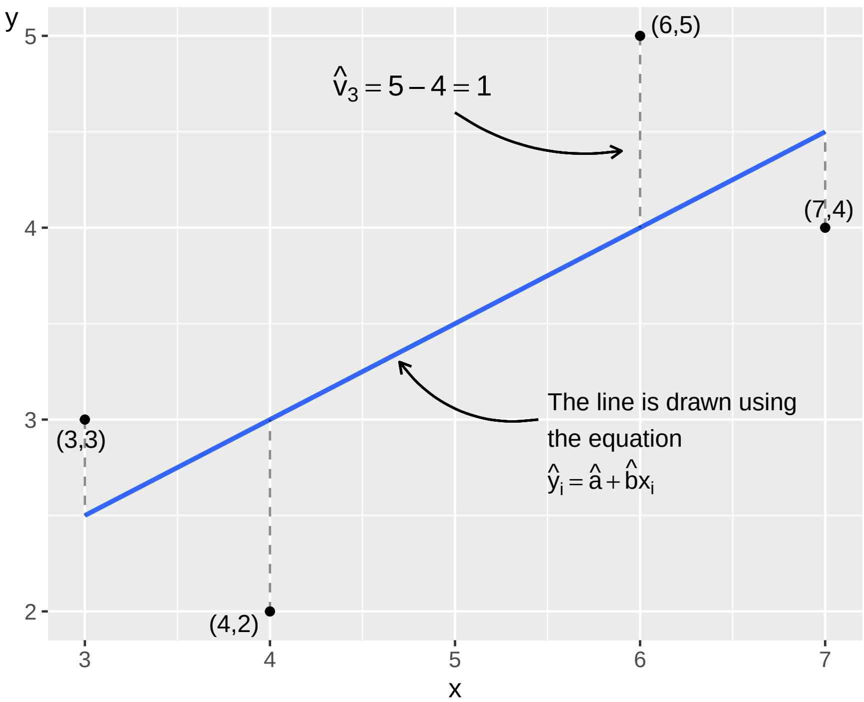


Figure 21.3: A line drawn using the least squares method

21.3 Regression analysis

The least squares method is one of several methods in regression analysis. In regression analysis we determine the variable whose variation we will map based on the variation in one or several other variables. Variation in an explanatory variable is used to explain some or all of the variation in an explained variable.

Regression analysis is a method for mapping patterns in data. We then use this to study causal relationships. But as discussed in earlier sections it is important to remember that patterns (covariation) are no proof of a causal relationship (chapter 20). The term "explanatory variable" therefore does not mean that one variable necessarily is the cause behind the state of the other variable.

We will now, based on the least squares method, estimate the linear covariation between the variables Y and X where we determine that Y is our explained variable and X is our explanatory variable. How we decide this depends on what idea underlies our analysis, for example a theory about GDP and happiness. Based on this, we set up the following regression model:

$$Y = a + bX + V \tag{21.1}$$

Since variable Y stands to the left of the equals sign in the regression model, it is the explained variable. Variable X, to the right of the equals sign, is the explanatory variable. This regression model exists in our population, it is the population model or the population's regression model. With sample data, we can estimate this population model. Depending on the purpose of the study, we may also have an interest in our regression model representing a relation in a superpopulation (section 15.1).

The letters a and b are the model's coefficients or parameters. The slope coefficient b should be read as a marginal increase in variable X being associated with on average b change in variable Y. Coefficient a, the constant, is the line's y-intercept. We do not know what values a and b have in the population model. That is what we want to estimate. The estimated versions of the coefficients' population values a and b we write as $\hat{a}$ and $\hat{b}$. The hat symbol $\hat{()}$ is estimated values, that is, our estimates with sample data of population values (compare figure 21.3). Letters a and b without the hat symbol refer to their population values.

If we have a population with N number of observations, from which we obtain n number of observations for our sample, we use these n observations to estimate the coefficients $\hat{a}$ and $\hat{b}$, and thereafter $\hat{Y}$. The line we will draw is called the regression line (already drawn in figure 21.3). When we have estimated $\hat{a}$ and $\hat{b}$, we can with the help of the observations in variable X also estimate new values for variable Y. The estimated version of Y we call $\hat{Y}$, which is the Y-value on the regression line. The estimated coefficients $\hat{a}$ and $\hat{b}$ that we seek in this case are constants, which means that they have the same values regardless of which values for variable X we compare against.

In the regression model, the letter V represents what is called the error term. Regression analysis aims to measure the extent to which variations in the explanatory variable(s) coincide with variations in the explained variable Y. The error term indicates the difference between each observation's Y-value and the regression line in the population. Estimated error terms are called residuals, written $\hat{V}$, and are the difference between $\hat{Y}$(the Y-values along the regression line) and observed Y:

$$\hat{V}_i = Y_i - \hat{Y}_i \tag{21.2}$$

If we insert the estimated coefficients, the residuals and the variable $\hat{Y}$ into the regression model in equation (21.1) we write:

$$\hat{Y} = \hat{a} + \hat{b}X + \hat{V} \tag{21.3}$$

Our regression model $Y = a + bX + V$ can also be written:

$$Y_i = a + bX_i + V_i \tag{21.4}$$

The letter i is an index for observation number. In this case, we use the four observations in figure 21.2 , which is why $i = \{1, 2, 3, 4\}$. If we take observation $i = 2$, we have $Y_2 = a + bX_2 + V_2$. The two values for a and b are constant for all observations and these therefore do not have any index i.

Before we go through the mathematics around the least squares method, we look at the equation for the regression line in figure 21.3 . If variable X increases by 1, this is associated with an on average 0.5 higher value on Y. This means that the regression line's slope coefficient $\hat{b}$ in this case is:

$$\hat{b} = 0.5 \tag{21.5}$$

The value for the regression model's y-intercept is $\hat{a}$, which we estimate by taking the straight line's equation $Y = a + bX$, but substituting a and b with estimated $\hat{a}$ and $\hat{b}$ and rearranging a bit:

$$\begin{aligned} Y &= \hat{a} + \hat{b}X \\ \hat{a} &= Y - \hat{b}X \end{aligned} \tag{21.6}$$

Now we have a definition for $\hat{a}$. In figure 21.3 , we see that the regression line passes through the point $(X, Y) = (3,\ 2.5)$. If we insert these values for X and Y in the equation for $\hat{a}$, we get:

$$\hat{a} = Y - \hat{b}X = 2,5 - 0.5 * 3 = 1 \tag{21.7}$$

Now we have the results for $\hat{a}$ and $\hat{b}$ as well as the values for variable X. Now we also estimate $\hat{Y}$ and thereby draw the regression line:

$$\hat{Y}_i = \hat{a} + \hat{b}X_i \tag{21.8}$$

Table 21.1 shows our calculations for estimating $\hat{Y}$ with the observed values for $X = \{3, 4, 6, 7\}$. The values in the table are the same as in figure 21.3 . The point on the diagonal line that the vertical dashed lines meet is the values for $\hat{Y}$. If we calculate $\hat{Y}$ for other values for X, we get other points along the regression line.

Table 21.1: Estimation of $\hat{Y}$

1	3	$1 + 0.5 \cdot 3 = 2.5$
2	4	$1 + 0.5 \cdot 4 = 3$
3	6	$1 + 0.5 \cdot 6 = 4$
4	7	$1 + 0.5 \cdot 7 = 4.5$

In figure 21.3 the line slopes upward to the right and $\hat{b}$>0, which means that we have a positive covariation between the variables X and Y. If $\hat{b} < 0$, we have a negative covariation, whereupon the line would have sloped downward to the right in the graphs. Larger values in X then covariate with on average smaller values in Y and smaller values in X coincide with on average higher values in Y. If $\hat{b} = 0$, the line is horizontal and lacks slope, which in that case indicates that we do not find any linear covariation between the variables. Estimated $\hat{Y}$ is also called predicted Y.

21.4 Error term and residual

The error term V represents the variation in our explained variable Y that cannot be explained by the regression model and the explanatory variable X. Our regression model should be designed in such a way that all variation in Y that depends on other factors should be evenly and randomly distributed over the error terms' different values, which we will return to later. In equation (21.8) we defined predicted $\hat{Y}_i = \hat{a} + \hat{b}X_i$. We now insert this definition into our equation for the residual $\hat{V}_i$:

$$\hat{V}_i = Y_i - \hat{Y}_i = Y_i - \left(\hat{a} + \hat{b}X_i\right) = Y_i - \hat{a} - \hat{b}X_i \tag{21.9}$$

where we see that $\hat{V}_i$ is a function of the observations in the variables Y_i and X_i as well as the estimated coefficients $\hat{a}$ and $\hat{b}$. Table 21.2 shows estimated results for $\hat{Y}$ and $\hat{V}$ from the observations in figure 21.2 . In figure 21.3 we read the values for $\hat{V}_i$ as the vertical distance between the diagonal line and each point. For observation 3 in the graph, we have written out the calculation:

$$\hat{V}_3 = Y_i - \hat{Y}_i = 5 - 4 = 1 \tag{21.10}$$

Note that the sum of the residuals $\hat{V}_i$ is 0:

$$\sum_i \hat{V}_i = \sum \left(Y_i - \hat{Y}_i\right) = 0 \tag{21.11}$$

which means that the sum of the vertical distance between the line and the points above the line is equal to the sum of the distance between the line and the points below the line. For the same reason, the mean of the residuals $\sum_i \hat{V}_i/n = 0$. If the sum of the error terms were equal to 0, all observations would lie on the regression line and the regression model would provide a complete explanation for the variation in the explained variable Y. There is no reason why we should strive for such a regression model. There can be many reasons why the sum of the error terms and the residuals are not close to 0, which we will return to later.

Table 21.2: Estimate $\hat{V}$

Observation i	Y_i	$\hat{Y}_i$	$\hat{V}_i = Y_i - \hat{Y}_i$
1	3	2.5	0.5
2	2	3	-1
3	5	4	1
4	4	4.5	-0.5
Sum			0
Mean			0

21.5 Calculating with Least Squares

We estimated $\hat{a}$ and $\hat{b}$ earlier by comparing a regression line in a graph. Now we will use the least squares method to estimate the coefficients $\hat{a}$ and $\hat{b}$ based on the observations in our variables X and Y. In practice, the calculations are usually performed by a computer and many analysis programs have ready-made commands for this. The method does not require advanced mathematics but when we have many variables and observations, the calculations can take a long time if we do them by hand.

In this example, we have only four observations and two variables (figure 21.2), which is why it is relatively simple to do the calculation by hand. Manual calculations help us understand the method better. The least squares method gives us the following definitions for estimating the coefficients $\hat{a}$ and $\hat{b}$:

$$\begin{aligned} \hat{a} &= \bar{Y} - \hat{b}\bar{X} \\ \hat{b} &= \frac{\sum_i \left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sum_i \left(X_i - \bar{X}\right)^2} \end{aligned} \tag{21.12}$$

These equations are called the coefficients' estimators. With the help of sample data, we estimate the coefficients in the population. Let us go through these equations step by step. The definition for $\hat{a}$:

-$\hat{a}$ is the estimated constant coefficient, which indicates the regression line's Y-intercept for $X = 0$.

-$\bar{Y}$ and $\bar{X}$ are means of Y and X, estimated with the sample data we have for each variable.

-$\hat{b}$ is the estimated value of the constant coefficient b, the line's slope. We need $\hat{b}$ to estimate $\hat{a}$.

- The definition of $\hat{b}$:

-$\sum_i$ means the sum of all observations, where the letter i refers to how the observations are indexed: observation 1 to 4. We sum all observations from 1 to 4, which is why we in this case have $\sum_i = \sum_{i=1}^{n=4}$.

- The parenthesis $\left(X_i - \bar{X}\right)$ means that we for each observation i of X subtract the mean $\bar{X}$. The parenthesis $\left(Y_i - \bar{Y}\right)$ means the same thing for each observation i of Y. For each observation we multiply $\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)$.
- The numerator in the definition for $\hat{b}$ means that we for each observation calculate $\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)$ and sum these. This we divide by $\sum_i \left(X_i - \bar{X}\right)^2$, where we for each observation take the parenthesis $\left(X_i - \bar{X}\right)$ squared and sum over all observations.

Table 21.3: Example with the Least Squares Method.

Observation	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$\left(X_i - \bar{X}\right)^2$	$\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)$
1	3	3	-2	-0.5	4	1
2	4	2	-1	-1.5	1	1.5
3	6	5	1	1.5	1	1.5
4	7	4	2	0.5	4	1
Mean $\left(\bar{X}\text{ eller }\bar{Y}\right)$	5	3.5				
Sum $(\sum)$	20	14			10	5

Table 21.3 describes the calculations we need for $\hat{a}$ and $\hat{b}$. Go carefully through each step in the table and make sure you understand what happens in each column. The sum of the observations in variable X is given by $\sum_{i=1}^{n=4} X_i = 3 + 4 + 6 + 7 = 20$. The mean for X is given by $\bar{X} = 20/4 = 5$. We begin with $\hat{b}$ which we need the value for to estimate $\hat{a}$:

$$\hat{b} = \frac{\sum_i \left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sum_i \left(X_i - \bar{X}\right)^2} = \frac{5}{10} = 0.5 \tag{21.13}$$

The result $\hat{b} = 0.5$ means that an increase of X by one unit is on average associated with a 0.5 higher Y. Since we now have $\hat{b}$, we also estimate $\hat{a}$:

$$\hat{a} = \bar{Y} - \hat{b}\bar{X} = 3.5 - 0.5 \times 5 = 1 \tag{21.14}$$

The result $\hat{a} = 1$ means that for $X = 0$, $\hat{Y} = 1$. The results $\hat{a} = 1$ and $\hat{b} = 0.5$ are the same as we saw in section 21.2 . Let us now calculate the points on the straight line by calculating $\hat{Y}_i$. This is given by the regression model, the estimates for $\hat{a}$ and $\hat{b}$ and the observations from the explanatory variable X. We get the following definition for estimated $\hat{Y}$:

$$\hat{Y}_i = \hat{a} + \hat{b}X_i = 1 + 0.5X_i \tag{21.15}$$

where X_i are the observations for X. With $\hat{Y}$ we may also estimate the residual $\hat{V}$. The results are presented in table 21.4 . In the column for $\hat{Y}$ we see the values for the vertical Y-axis along the regression line. We only have values on the regression line that coincide with the values we have for variable X. Try yourself to draw the points in a graph $\left(X_1, \hat{Y}_1\right) = (3, 2.5)$, $(4, 3)$, $(6, 4)$ and $(7, 4.5)$.

Table 21.4: Estimating $\hat{Y}$ and $\hat{V}$

Observation	X_i	Y_i	$\hat{Y}_i = 1 + 0.5X_i$	$\hat{V}_i = Y_i - \hat{Y}_i$	$\left(\hat{V}_i\right)^2$
1	3	3	2.5	0.5	0.25
2	4	2	3	-1	1
3	6	5	4	1	1
4	7	4	4.5	-0.5	0.25
Sum					$\sum_i \hat{V}_i^2 = 2.5$

In section 20.6 we introduced covariance. The estimator for coefficient $\hat{b}$ for the regression model $Y = a + bX + V$ can also be defined as the covariance between X and Y divided by the variance of the explanatory variable X:

$$\begin{aligned}\frac{\operatorname{cov}(X,Y)}{\operatorname{var}(X)} &= \frac{\left(\frac{1}{n}\right)\sum_i^n \left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\left(\frac{1}{n}\right)\sum_i^n \left(X_i - \bar{X}\right)^2} \\ &= \frac{\sum_i^n \left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sum_i^n \left(X_i - \bar{X}\right)^2} = \hat{b}\end{aligned} \tag{21.16}$$

We here see that this is the same definition for $\hat{b}$ as in equation (21.12) .

21.6 Other straight lines

The least squares method helps us find the regression line that minimizes the sum of the vertical distance between the regression line and the observations. All other straight lines result in a larger summed value for the squared distance between these lines and the points.

This we may illustrate with the help of an example where we use the same four observations for x and y in figure 21.2 that we used in the previous section. But instead of the regression line that we estimated based on the least squares method, we now have a horizontal line where $\bar{Y} = 3.5$ for all Y_i, which is the same as the mean for Y in our four observations: $\bar{Y} = (3 + 4 + 5 + 6)/4 = 3.5$. A horizontal line has slope coefficient $b = 0$. Let us calculate the sum of the squared residuals $\left(\sum_i \hat{V}_i^2\right)$ for this line:

$$\begin{aligned}\sum_i \hat{V}_i^2 &= \sum_i \left(Y_i - \hat{Y}_i\right)^2 \\ &= (3 - 3.5)^2 + (4 - 3.5)^2 + (5 - 3.5)^2 + (6 - 3.5)^2 \\ &= 9\end{aligned} \tag{21.17}$$

which is more than 2.5. The regression line we could draw based on our regression model that we estimated with the least squares method is therefore better if we are to minimize the summed vertical distance. Let us exemplify with another line, which passes through observation 1 and 4, that is, the points $(X,Y) = (3,3)$ and $(7,4)$. Based on these values for X and Y, we know that when X increases from 3 to 7, this is associated with an average change in Y from 3 to 4, which is why the line's slope is:

$$\hat{b} = \frac{Y_4 - Y_1}{X_4 - X_1} = \frac{4-3}{7-3} = \frac{1}{4} \tag{21.18}$$

The second coefficient $\hat{a}$ we calculate with the point $(x_1, y_1) = (3,3)$:

$$\hat{a} = Y_i - \hat{b}X_i = 3 - \frac{1}{4} * 3 = \frac{9}{4} \tag{21.19}$$

Since the line passes through observation 1 and 4, we know that the residual for these two is $\hat{V}_i = 0$. For observation 2, we have that $X = 4$, which gives:

$$\hat{Y}_2 = 9/4 + \frac{1}{4} * 4 = 13/4 \tag{21.20}$$

For observation 3, we have $X = 6$ which gives:

$$\hat{Y}_3 = 9/4 + \frac{1}{4} * 6 = 15/4 \tag{21.21}$$

For observation 2 and 3, we have that $Y_2 = 2$ and $Y_3 = 5$:

$$\begin{aligned}\sum_i \hat{V}_i^2 &= \sum_i \left(Y_i - \hat{Y}_i\right)^2 \\ &= \left(Y_1 - \hat{Y}_1\right)^2 + \left(Y_2 - \hat{Y}_2\right)^2 + \left(Y_3 - \hat{Y}_3\right)^2 + \left(Y_4 - \hat{Y}_4\right)^2 \\ &= 0 + (2 - 13/4)^2 + (5 - 15/4)^2 + 0 \\ &= 3.1875 > 2.5\end{aligned} \tag{21.22}$$

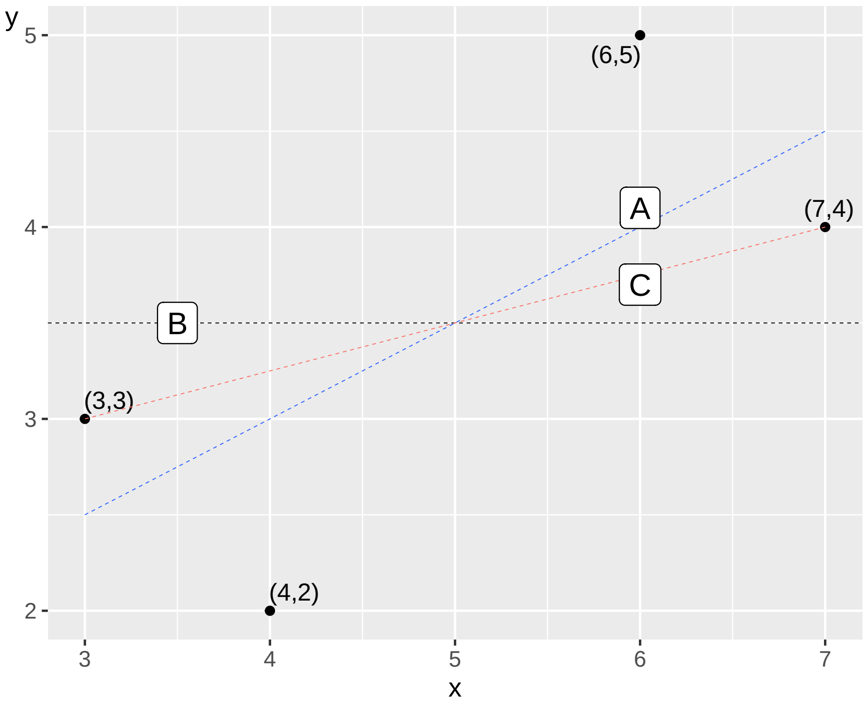


Figure 21.4: Three straight lines

Figure 21.4 illustrates the three lines we have now compared. Line A in the graph is the one we estimated based on the least squares method. Line B is the horizontal line where $\hat{Y} = \bar{Y} = 3.5$. Line C is drawn by the function $Y = 9/4 + \frac{1}{4}X$ and goes through two of the points.

21.7 What the Line Doesn't Tell Us

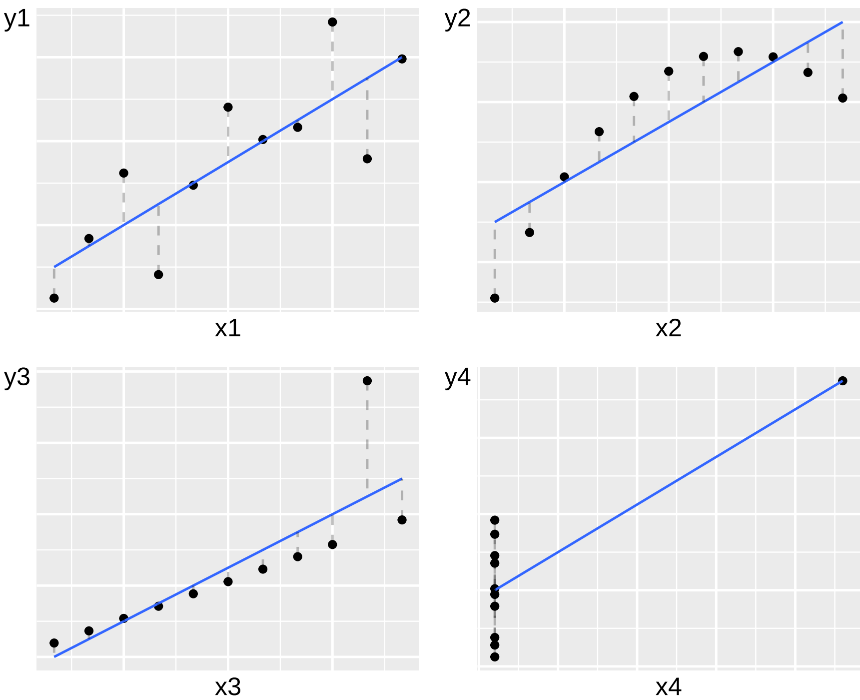


Figure 21.5: Four examples of linear covariation, from [(Anscombe, 1973)](https://www.jstor.org/stable/2682899)

Figure 21.5 shows four examples where combined values for the variables X (horizontal axis) and Y (vertical axis) are plotted. The Y-variables and X-variables in each graph all have the same mean: $\bar{Y} = 7.5$ and $\bar{X} = 9$. The coefficients a and b in the regression model $Y = a + bX + V$ that draw the regression lines in each graph have the same values in all four graphs: $\hat{a} = 3$ and $\hat{b} = 0.5$. The lines slope upward to the right in each graph, which is given by $\hat{b} > 0$. In all four graphs, we thus use the least squares method to find a positive covariation that looks the same in the regression model. The example illustrates that despite the patterns in the four graphs differing drastically, calculated results can easily give the impression that different collections of data have more in common than they actually do. The four graphs were first presented in 1973 by the British statistician Francis John Anscombe (Anscombe, 1973) . The examples illustrate the importance of both calculating covariation and studying data in graphs.

Despite the points in the graphs being placed in different patterns, we get the same positive linear covariation in all four examples. In the upper left graph, we see that the points lie spread along the diagonal line up toward the right corner of the graph. In the upper right graph, the points lie in a concave arc, up toward the graph's right corner. The points' pattern is clearly nonlinear, but we still calculate a linear positive covariation. In the lower left graph, all points except one lie on a straight line. In the lower right graph, all points except one lie on a vertical line.

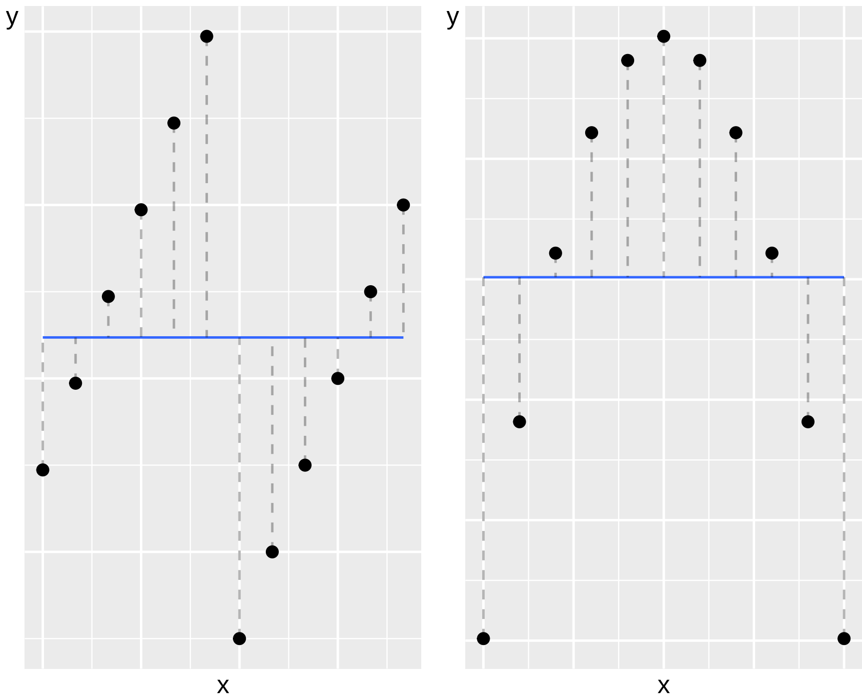


Figure 21.6: Regression lines with zero slope

Consider two more examples on a similar theme. Figure 21.6 describes two graphs where the dots again show combined values of the variables x and y. The straight lines are regression lines calculated based on the least squares method. Now the lines are horizontal, which means that the slope coefficients are equal to 0 in both cases. When x increases, this is not associated with any change in variable y.

But when we look at the graphs in the figure, we clearly see that the dots in the two graphs lie in different patterns, which suggests different types of covariation. In the left graph, the relationship between the variables is clearly positive, but only if we study the observations as two groups. In the right graph, a nonlinear covariation is visible when we look at all observations. If we look only at the left half of the graph, a positive covariation is visible, while the points in the right half of the graph have a negative covariation.

21.8 Another example

Let us now estimate the following regression model:

$$Z = \alpha + \beta K + U \tag{21.23}$$

where Z and K are variables, α and β are the coefficients that we shall estimate based on the least squares method and U is the error term. That we now use other letters does not affect the mathematics. Let us repeat our method:

1. We have an idea that K covariates with Z. This is a statement about covariation, not a theory about a causal relationship.
2. We formulate our statement about covariation in the form of a regression model in equation (21.23) .
3. We use the least squares method to find the estimators (the equations) for $\hat{\alpha}$ and $\hat{\beta}$, where the hat symbol $\hat{}$ indicates that they are estimated versions of α and β.
4. After we have estimated the coefficients, we estimate the regression line's equation $\hat{Z} = \hat{\alpha} + \hat{\beta}K$ and draw the regression line in a graph. The values for $\hat{Z}$ indicate the regression line's values on the vertical axis.

5. The residual $\hat{U} = Z - \hat{Z}$ is the vertical distance (y-axis) between the regression line's $\hat{Z}$ and each observation Z.

Table 21.5: Calculations with Z and K

Observation i	Z_i	K_i	$Z_i - \bar{Z}$	$K_i - \bar{K}$	$(Z_i - \bar{Z})(K_i - \bar{K})$	$(K_i - \bar{K})^2$
1	1	0	−0.5	−2	1	4
2	4	0	2.5	−2	−5	4
3	0	4	−1.5	2	−3	4
4	1	4	−0.5	2	−1	4
Sum	6	8			−8	16
Mean	1.5	2				

The observations for the variables Z and K are described in table 21.5 together with the calculations we need to estimate the regression model in equation (21.23) . For the slope coefficient $\hat{\beta}$ we get:

$$\hat{\beta} = \frac{\sum_i (K_i - \bar{K})(Z_i - \bar{Z})}{\sum_i (K_i - \bar{K})^2} = \frac{-8}{16} = -\frac{1}{2} \tag{21.24}$$

For the other coefficient $\hat{\alpha}$:

$$\hat{\alpha} = \bar{Z} - \hat{\beta}\bar{K} = 1,5 - \left(-\frac{1}{2}\right) 2 = 2.5 \tag{21.25}$$

That $\hat{\beta} < 0$ means that the regression line's slope is negative and that we have a negative covariation between the four observations for Z and K. We now set up an equation for predicted $\hat{Z}$:

$$\hat{Z}_i = \hat{\alpha} + \hat{\beta} K_i = 2.5 - \frac{1}{2} K_i \tag{21.26}$$

With the help of this and the observed values for variable K, we estimate four values for predicted $\hat{Z}$, which are described in table 21.6 . With $\hat{Z}$ we then estimate the four residuals $\hat{U}$, which are described in the same table. The four observations and the regression line that can be drawn with equation (21.26) are illustrated in figure 21.7 .

Table 21.6: Estimated residuals $\hat{U}$

Observation i	Z_i	K_i	$\hat{Z}_i$	$\hat{U}_i = Z_i - \hat{Z}_i$
1	1	0	2.5	–1.5
2	4	0	2.5	1.5
3	0	4	0.5	–0.5
4	1	4	0.5	0.5

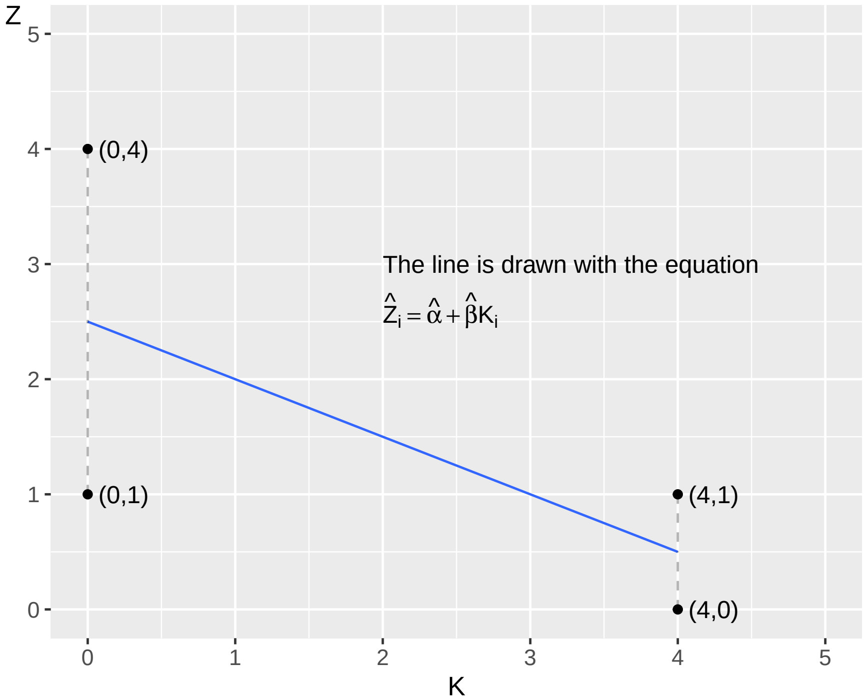


Figure 21.7: Four observations and a regression line

21.9 Which model fits the data best?

To compare how well different statistical models fit the data they aim to describe the pattern for, there are several commonly used methods. An important part in the evaluation of the regression model is to compare the residuals, that is, the distance between the data we used in our explained variable Y and the predicted values for the same variable $\hat{Y}$.

One way to compare how well a regression model $Y = a + bX + V$ fits the data that the model is estimated against is to compare the Sum of Squared Errors (SSE):

$$SSE = \sum_{i}^{n} \hat{V}_i^2 = \sum_{i}^{n} \left(Y_i - \hat{Y}_i\right)^2 \tag{21.27}$$

SSE is the amount of variation in the explained variable Y that cannot be explained by the regression model. Another useful measure is the Sum of Squares due to Regression (SSR):

$$SSR = \sum_{i}^{n} \left(\hat{Y}_i - \bar{Y}\right)^2 \tag{21.28}$$

SSR can be described as the variation in Y that can be explained by the regression model. As mentioned earlier, the concept "explained" in this context does not necessarily mean causal explanation. The total sum of squares is the sum of SSE and SSR. TSS can be interpreted as the amount of variation that occurs in Y before the regression model is estimated and written as the sum of the squared difference between observed Y_i and the mean $\bar{Y}$:

$$\begin{aligned} TSS &= SSR + SSE \qquad (21.29) \\ \sum_{i}^{n} \left(Y_i - \bar{Y}\right)^2 &= \sum_{i}^{n} \left(\hat{Y}_i - \bar{Y}\right)^2 + \sum_{i}^{n} \left(Y_i - \hat{Y}_i\right)^2 \end{aligned}$$

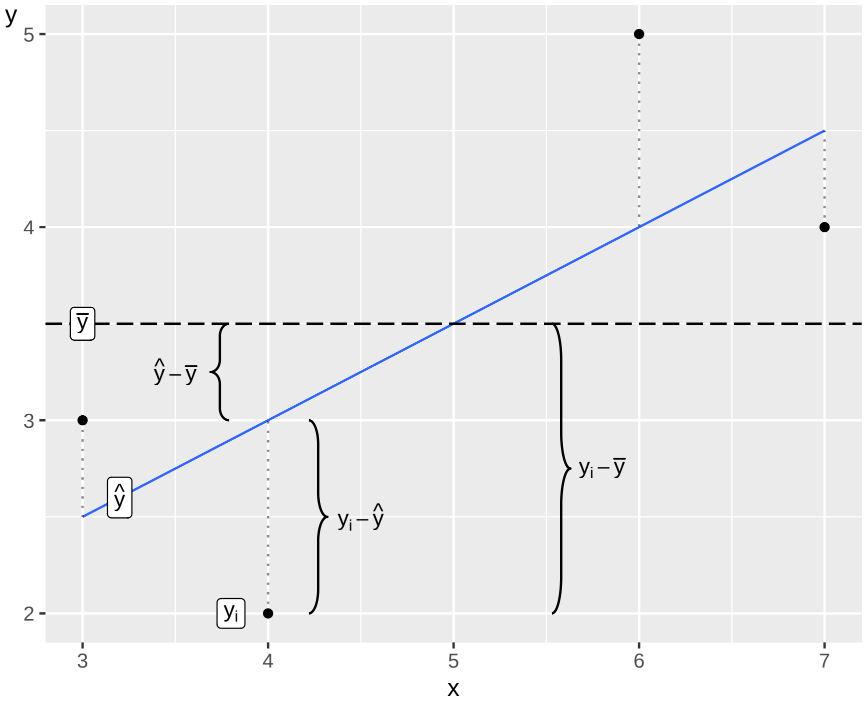


Figure 21.8: Difference between $Y_i, \bar{Y}_i$ and $\hat{Y}_i$

One way to illustrate the differences between SSE, SSR and TSS is given in figure 21.8 . We use here again the four observations for $X = \{3, 4, 6, 7\}$ and $Y = \{2, 3, 4, 5\}$ that we used in our first example in this chapter. In the figure the residuals are highlighted with dashed vertical lines between the values of Y_i and the regression line $\hat{Y}_i$.

The abbreviations TSS, SSR and SSE are common to use. Unfortunately, other abbreviations also occur to describe the same thing. The essential thing is of course what each designation or abbreviation represents in each context, thus the mathematics itself.

Based on the sums of squares, we can estimate the coefficient of determination, which is usually denoted R^2:

$$R^2 = \frac{SSR}{TSS} = \frac{TSS - SSE}{TSS} = 1 - \frac{SSE}{TSS} = 1 - \frac{\sum \left(Y_i - \hat{Y}_i\right)^2}{\sum \left(Y_i - \bar{Y}\right)^2} \tag{21.30}$$

R^2 measures what proportion of the variation in the explained variable, for example Y, that can be explained by the regression model and the variation in the explanatory variable, for example X. R^2 can take values between 0 and 1. $R^2 = 0$ means that nothing of the variation in Y can be explained by the regression model and the variation in X. $R^2 = 1$ means that all variation in variable Y is explained by the regression model and the variation in variable X.

Put simply, R^2 is a measure of how well we manage to explain the variation in Y. In social science we study human behavior, which often is the result of many phenomenons and complex relationships. It is therefore only natural that R^2 is usually relatively low for any regression model which try to explain human life.

For instance, say we wish to explain variations in happiness between countries, as discussed in the introduction to this chapter. We can easily come up with a long list of variables which might affect how the population of different countries feel about their lives. Most regression models will therefore probably only explain a limited part of the variation in happiness.

To study how well a regression model fits against data, we compare the mean sum of squares, also called mean squares. To get the mean squares, we divide each sum of squares by the number of degrees of freedom. In statistics, degrees of freedom refers to the number of parameters that can vary in a calculation, which we will return to below. The Mean Squares of Errors (MSE):

$$MSE = \frac{SSE}{n-p} = \frac{\sum_i \left(Y_i - \hat{Y}_i\right)^2}{n-p} \tag{21.31}$$

The denominator $n-p$ shows the number of degrees of freedom, where n is the number of observations and p is the number of estimated coefficients in the regression model. In figure 21.8 we have $n = 4$. If we have the regression model $Y = a + bX + V$, we have $p = 2$ for the coefficients a and b. Simplified, a low MSE can mean that the regression model's predicted $\widehat{Y}$ ends up relatively close to the observations Y. The Mean Squares of the Regression (MSR) is calculated by dividing SSR by the number of explanatory variables:

$$MSR = \frac{SSR}{p-1} = \frac{\sum_i^n \left(\hat{Y}_i - \bar{Y}\right)^2}{p-1} \tag{21.32}$$

In the numerator we here take p(number of estimated coefficients) minus 1, since coefficient a is not multiplied by any explanatory variable.

Table 21.7: Variables X and Y and some calculations

Observation i	X_i	Y_i	$\hat{Y}_i$	$Y_i - \hat{Y}_i$	$Y_i - \bar{Y}$	$\left(Y_i - \hat{Y}_i\right)^2$	$\left(Y_i - \bar{Y}\right)^2$
1	3	3	2.5	0.5	-0.5	0.25	0.25
2	4	2	3	-1	-1.5	1	2.25
3	6	5	4	1	1.5	1	2.25
4	7	4	4.5	-0.5	0.5	0.25	0.25
Mean	5	3.5					
Sum						2.5	5

Table 21.8: Variables Z and K and some calculations

Obs. i	Z_i	K_i	$\hat{Z}_i$	$\hat{\epsilon}_i =$ $Z_i - \hat{Z}_i$	$Z_i - \bar{Z}$	$\left(Z_i - \hat{Z}_i\right)^2$	$\left(Z_i - \bar{Z}\right)^2$
1	1	0	2.5	-1.5	-0.5	2.25	0.25
2	4	0	2.5	1.5	2.5	2.25	6.25
3	0	4	0.5	-0.5	-1.5	0.25	2.25
4	1	4	0.5	0.5	-0.5	0.25	0.25
Mean	1.5	2					
Sum						5	9

Earlier in this chapter we have used two different collections of observations (figure 21.2 and table 21.5) to estimate the two regression models in equation (21.1) and (21.23) based on the least squares method. One way to compare which regression model best fits a dataset is to compare MSE for each regression model. We begin with the regression model in equation (21.1) : $Y = a + bX + V$. Table 21.7 repeats the observations for the variables x and y as well as some calculations that we need, which gives MSR:

$$MSE_{\text{model 1}} = \frac{\sum_i \left(Y_i - \hat{Y}_i\right)^2}{n} = \frac{2,5}{4} = 0.625 \tag{21.33}$$

Let us calculate MSE also for the second regression model, from equation (21.23) : $Z = \alpha + \beta K + U$, with the variables Z and K. Table 21.8 repeats the variables' values and summarizes some calculations, which gives:

$$MSE_{\text{model 2}} = \frac{5}{4} = 1.25 \tag{21.34}$$

Since $MSE_{\text{model 2}} > MSE_{\text{model 1}}$, this suggests that the first regression model's predicted $\hat{Y}$ lies closer to observed Y, compared to how well the second regression model succeeds in predicting the explained variable Z in that model.

MSR is calculated using the observations we use and the result can therefore change if we add or subtract observations, alternatively if we specify our regression models in some other way. Let us also estimate R^2. For the first regression model $Y = a + bX + V$ we get (see table 21.7):

$$R^2_{\text{model 1}} = 1 - \frac{\sum_i \left(Y_i - \hat{Y}_i\right)^2}{\sum_i \left(Y_i - \bar{Y}\right)^2} = 1 - \frac{2.5}{5} = \frac{1}{2} \tag{21.35}$$

The result $R^2 = 1/2$ indicates that half of the variation in Y can be explained by the regression model and the variations in X. Let us also calculate R^2 for the second regression model $Z = \alpha + \beta K + \epsilon$(see table 21.8):

$$R^2_{\text{model 2}} = 1 - \frac{\sum_i \left(Z_i - \hat{Z}_i\right)^2}{\sum_i \left(Z_i - \bar{Z}_i\right)^2} = 1 - \frac{5}{9} = \frac{4}{9} \tag{21.36}$$

Since $\frac{4}{9} < \frac{1}{2}$, this indicates that the first regression model has a higher degree of explanation compared to the second model.

21.10 Derivation of the Estimators

For the regression model:

$$Y_i = a + bX_i + V_i \tag{21.37}$$

where Y_i and X_i are variables and V_i is the error term, we introduced estimators for the coefficients $\hat{a}$ and $\hat{b}$ in equation (21.1) . Now we shall go through how we derive these. The least squares method allows us to define the estimators $\hat{a}$ and $\hat{b}$ so that we minimize the sum of the squared residuals $\sum_i \hat{V}_i^2 = \sum \left(Y_i - \hat{Y}_i\right)^2$. The estimators can therefore be derived from a minimization problem (see chapter 11). The sum of the squared residuals becomes:

$$\begin{aligned}
\sum_{i=1}^{n} \hat{V}_i^2 &= \hat{V}_1^2 + \hat{V}_2^2 + ... + \hat{V}_n^2 \\
&= \left(Y_1 - \hat{Y}_1\right)^2 + \left(Y_2 - \hat{Y}_2\right)^2 + ... + \left(Y_n - \hat{Y}_n\right)^2 \\
&= \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2
\end{aligned} \tag{21.38}$$

where n is the number of observations, $\hat{Y}$ is predicted Y. To be able to estimate $\hat{Y} = \hat{a} + \hat{b}X$, we need to estimate $\hat{a}$ and $\hat{b}$. From equation (21.37) we take the definition of $\hat{Y}_i$ and insert into equation (21.38) :

$$\begin{aligned}
\sum_{i=1}^{n} \hat{V}_i^2 &= \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 \\
&= \sum_{i=1}^{n} \left(Y_i - \left(\hat{a} + \hat{b}X_i\right)\right)^2 \\
&= \sum_{i=1}^{n} \left(Y_i - \hat{a} - \hat{b}X_i\right)^2
\end{aligned} \tag{21.39}$$

Now we have an equation where the sum of squared residuals $\sum_i \hat{V}_i^2$ is a function of estimated $\hat{a}$ and $\hat{b}$. It is this function that we shall minimize with respect to $\hat{a}$ and $\hat{b}$, which we write as a minimization problem:

$$\min_{\text{w.r.t. } \hat{a},\hat{b}} \sum_{i=1}^{n} \hat{V}_i^2 = \sum_{i=1}^{n} \left(Y_i - \hat{a} - \hat{b}X_i\right)^2 \tag{21.40}$$

We differentiate the right-hand side with respect to $\hat{a}$ and $\hat{b}$. By setting the results equal to 0, we get two first-order conditions:

$$\begin{aligned}
\left[\sum_{i=1}^{n} \hat{V}_i^2\right]'_{\hat{a}} &= -2\sum_{i=1}^{n} \left(Y_i - \hat{a} - \hat{b}X_i\right) = 0 \\
\left[\sum_{i=1}^{n} \hat{V}_i^2\right]'_{\hat{b}} &= -2\sum_{i=1}^{n} X_i \left(Y_i - \hat{a} - \hat{b}X_i\right) = 0
\end{aligned} \tag{21.41}$$

where both results arise by differentiating the parenthesis (moving down exponent 2) and differentiating the contents of each parenthesis. When we differentiate with respect to $\hat{a}$, we get -1. When we differentiate with respect to $\hat{b}$, we get $-X_i$. We put -2 to the left of the summation sign since these things are not affected by the summation. From the two first-order conditions, we solve for $\hat{a}$ and $\hat{b}$. We begin with the first condition:

$$\begin{aligned}
-2\sum_{i=1}^{n} \left(Y_i - \hat{a} - \hat{b}X_i\right) &= 0 \\
(-2)\sum_{i=1}^{n} Y_i - (-2)\sum_{i=1}^{n} \hat{a} - (-2)\sum_{i=1}^{n} \hat{b}X_i &= 0
\end{aligned} \tag{21.42}$$

Since all terms are multiplied by -2, we cancel this factor. Since $\hat{a}$ is constant over all observations, the term $\sum_i^n \hat{a}$ is the same thing as the number of observations times $\hat{a}$: $\sum_i^n \hat{a} = n\hat{a}$. We now write:

$$\begin{aligned}
0 &= (-2)\sum_{i=1}^{n} Y_i - (-2)\sum_{i=1}^{n} \hat{a} - (-2)\sum_{i=1}^{n} \hat{b}X_i \\
0 &= \sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} \hat{a} - \sum_{i=1}^{n} \hat{b}X_i \\
0 &= \sum_{i=1}^{n} Y_i - n\hat{a} - \sum_{i=1}^{n} \hat{b}X_i \\
n\hat{a} &= \sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} \hat{b}X_i
\end{aligned} \tag{21.43}$$

We divide by n so that the term farthest to the right becomes $\frac{1}{n}\sum \hat{b}X_i$. The mean for X can be written $\bar{X} = \frac{1}{n}\sum_i^n X_i$ which gives:

$$\begin{aligned}\hat{a} &= \frac{1}{n}\sum_{i=1}^{n} Y_i - \frac{1}{n}\sum_{i=1}^{n}\hat{b}X_i \\ \hat{a} &= \bar{Y} - \hat{b}\bar{X}\end{aligned} \tag{21.44}$$

Now we have an expression for $\hat{a}$. To derive $\hat{b}$, we begin with the second first-order condition in equation (21.41) . We take the opportunity here to simplify the writing of the summation signs $\sum_i^n = \sum$ but the meaning is still the same. Summation is done over the expressions that have subscript i:

$$\begin{aligned}0 &= \sum_i -2x_i\left(Y_i - \hat{a} - \hat{b}X_i\right) \\ &= \sum X_iY_i - \sum X_i\hat{a} - \sum X_i\hat{b}X_i\end{aligned} \tag{21.45}$$

Since the constants are not summed, we move these to the left of the summation signs. In the term farthest to the right we have $x_i * x_i = x_i^2$. We substitute $\hat{a}$ with the definition from equation (21.44) :

$$\begin{aligned}0 &= \sum X_iY_i - \hat{a}\sum X_i - \hat{b}\sum X_iX_i \\ &= \sum X_iY_i - \hat{a}\sum X_i - \hat{b}\sum x_i^2 \\ &= \sum X_iY_i - \left(\bar{Y} - \hat{b}\bar{X}\right)\sum X_i - \hat{b}\sum X_i^2 \\ &= \sum X_iY_i - \bar{Y}\sum X_i + \hat{b}\bar{X}\sum X_i - \hat{b}\sum X_i^2 \\ &= \sum X_iY_i - \bar{Y}\sum X_i - \hat{b}\left(\sum X_i^2 - \bar{X}\sum X_i\right)\end{aligned} \tag{21.46}$$

We move $\hat{b}$ to the left side of the equals sign:

$$\begin{aligned}\hat{b}\left(\sum X_i^2 - \bar{X}\sum X_i\right) &= \sum X_iY_i - \bar{Y}\sum X_i \\ \hat{b} &= \frac{\sum X_iY_i - \bar{Y}\sum X_i}{\sum X_i^2 - \bar{X}\sum X_i}\end{aligned} \tag{21.47}$$

Now we have an expression for $\hat{b}$. For it to look the same as in section 21.2 , we rewrite a bit. The final result remains the same:

$$\begin{aligned}\hat{b} &= \frac{\sum X_iY_i - \bar{Y}\sum Y_i}{\sum X_i^2 - \bar{X}\sum X_i} \\ &= \frac{\sum X_iY_i - \bar{Y}\sum_i X_i - \bar{X}\sum y_i + \bar{X}\sum Y_i}{\sum X_i^2 - \bar{X}\sum X_i - \bar{X}^2\sum 1 + \bar{X}^2\sum 1} \\ &= \frac{\sum\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sum\left(X_i - \bar{X}\right)^2}\end{aligned} \tag{21.48}$$

Now we have the same thing as equation (21.47) . Note that the first condition in equation (21.41) means:

$$\sum_i \hat{V}_i = 0 \tag{21.49}$$

and the second condition can be described as:

$$\sum X_i \hat{V}_i = 0 \tag{21.50}$$

This we will have use for later on.

21.11 Chapter summary

- The least squares method (often referred to as *Ordinary Least Squares*, OLS) is a method for studying linear covariation between two or more variables and thereby describing patterns in data. The least squares method is one of several methods within regression analysis.
- Based on an idea about covariation, we formulate a regression model $Y_i = a + bX_i + V_i$ where Y_i is the explained variable and X_i is the explanatory variable, both noted for observation i. a and b are the coefficients that we shall estimate. V is the error term, which measures the variation in the explained variable Y that the regression model and the explained variable X cannot explain. The estimated error term is called residual $\hat{V}$ and indicates the difference between estimated $\hat{Y}_i$ and observed Y_i. Symbols with a hat over them are estimated versions.
- The least squares method allows us to define how a and b shall be estimated so that the sum of the squared residuals $\sum_i^n \hat{V}_i^2 = \sum_i^n \left(Y_i - \hat{Y}_i\right)^2$ is minimized. We estimate the coefficients $\hat{a}$ and $\hat{b}$ and thereafter $\hat{Y}$ and $\hat{V}$.
- The coefficients are estimated as $\hat{a} = \bar{Y} - \hat{b}\bar{X}$ and $\hat{b} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$ where $\bar{Y}$ and $\bar{X}$ are means.

  The results for $\hat{a}$ and $\hat{b}$ are constants.

-b indicates the line's slope. Result $\hat{b} > 0$ indicates positive covariation. $\hat{b} < 0$ indicates negative covariation. Result $\hat{b} = 0$ indicates that there is no linear covariation between X and Y.

-a indicates the y-intercept. If coefficient $\hat{a} = 0$, we have the model $\hat{Y}_i = \hat{b}X_i + \hat{V}_i$, for which the straight line passes through the origin, $(0, 0)$. If coefficient $\hat{b} = 0$, we have the model $\hat{Y}_i = \hat{a} = \bar{Y}$, where $\hat{Y}_i$ is equal to the mean for Y, meaning $\hat{a} = \bar{Y}$ for all observations i.

21.12 Exercises

1. Given $X = \{0, 1, 2, 3\}$ and $Y = \{1, 3, 4, 4\}$:

(a) Calculate (bar{X}) and (bar{Y}) .

(b) Calculate (sum_{i}(X_{i}-bar{X})^{2}) .

(c) Calculate (sum_{i}(X_{i}-bar{X})(Y_{i}-bar{Y})) .

Answer: (a) $\bar{X} = 1.5, \bar{Y} = 3$, (b) $\sum(X_i - \bar{X})^2 = 5$, (c) $\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 5$.

2. Continued from exercise 1, using $\hat{b} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$ and $\hat{a} = \bar{Y} - \hat{b}\bar{X}$:

(a) Calculate the slope (hat{b}) .

(b) Calculate the intercept (hat{a}) .

(c) Write the estimated regression equation (hat{Y}*{i}=hat{a}+hat{b}X*{i}) .

Answer: (a) $\hat{b} = 1$, (b) $\hat{a} = 1.5$, (c) $\hat{Y} = 1.5 + X$.

3. Given the regression equation $\hat{Y} = 1.5 + X$:

(a) What is the fitted value (hat{Y}) when (X=2) ?

(b) If (Y=4) when (X=2) , what is the residual (hat{V}=Y-hat{Y}) ?

(c) What property do OLS residuals always satisfy regarding their sum?

Answer: (a) $\hat{Y} = 3.5$, (b) $\hat{V} = 0.5$, (c) OLS residuals always sum to zero.

4. Using $X = \{0, 1, 2, 3\}$, $Y = \{1, 3, 4, 4\}$ and $\hat{Y} = 1.5 + X$:

(a) Calculate (SSE=sum(Y_{i}-hat{Y}_{i})^{2}) .

(b) Calculate (TSS=sum(Y_{i}-bar{Y})^{2}) .

(c) Calculate (R^{2}=1-frac{SSE}{TSS}) .

Answer: (a) $SSE = 1$, (b) $TSS = 6$, (c) $R^2 = 5/6 \approx 0.83$.

5. Given $cov(X, Y) = 12$ and $var(X) = 4$:

(a) Calculate (hat{b}=frac{cov(X,Y)}{var(X)}) .

(b) If (bar{X}=3) and (bar{Y}=10) , calculate (hat{a}) .

(c) Write the regression equation.

Answer: (a) $\hat{b} = 3$, (b) $\hat{a} = 1$, (c) $\hat{Y} = 1 + 3X$.

6. Interpret the regression result $\hat{Y} = 500 + 50X$, where X is years of education and Y is monthly income in USD:

(a) Interpret the intercept (hat{a}=500) .

(b) Interpret the slope (hat{b}=50) .

(c) What does (R^{2}=0.6) tell us?

Answer: (a) predicted income with 0 years of education is 500 USD, (b) each additional year of education is associated with 50 USD higher income on average, (c) 60% of the variation in income is explained by the model.

7. A regression of happiness Y on GDP per capita X gives $\hat{Y} = 3 + 0.5X$.

(a) What is the predicted happiness when (X=4) ?

(b) If observed happiness is 6 when (X=4) , what is the residual?

(c) Does the positive slope mean that GDP causes happiness?

Answer: (a) $\hat{Y} = 5$, (b) $\hat{V} = 1$, (c) no — correlation does not imply causation.

8. For the regression $\hat{Y} = \hat{a} + \hat{b}X$, describe what happens when:

(a) (hat{b}=0) .

(b) (hat{b}>0) .

(c) (hat{b}<0) .

Answer: (a) no linear covariation between X and Y , (b) positive covariation, (c) negative covariation.

9. For two observations $X = \{1, 3\}$ and $Y = \{2, 8\}$:

(a) Calculate (bar{X}) and (bar{Y}) .

(b) Calculate (hat{b}) .

(c) Calculate (hat{a}) .

Answer: (a) $\bar{X} = 2, \bar{Y} = 5$, (b) $\hat{b} = 3$, (c) $\hat{a} = -1$.

10. Given $TSS = 20$ and $SSE = 5$:

(a) Calculate (R^{2}) .

(b) Calculate (SSR=TSS-SSE) .

(c) Interpret the value of (R^{2}) .

Answer: (a) $R^2 = 0.75$, (b) $SSR = 15$, (c) 75% of the variation in Y is explained by the model.

11. Given the regression $\hat{Y} = 2 + 0.5X$, predict Y for:

(a) (X=10) .

(b) (X=0) .

(c) (X=6) .

Answer: (a) $\hat{Y} = 7$, (b) $\hat{Y} = 2$, (c) $\hat{Y} = 5$.

12. A set of residuals from an OLS regression is $\hat{V} = \{-1, 2, -1, 0\}$.

(a) Do the residuals sum to zero?

(b) Calculate (SSE=sumhat{V}_{i}^{2}) .

(c) If (TSS=12) , calculate (R^{2}) .

Answer: (a) yes, sum $= 0$, (b) $SSE = 6$, (c) $R^2 = 0.5$.

Chapter 22

Covariation in Social Science

This chapter reviews some simple examples of variation and covariation with social science data.

22.1 GDP and Life Expectancy in different countries

We have in previous examples discussed the relation between income and life expectancy. Now we shall compare average life expectancy with average income for inhabitants in a number of countries. The data used in this section are taken from the database www.ourworldindata.com, which has a lot of free data you can download and work with. We begin by comparing only a small number of countries. Figure 22.1 describes the data we shall use. In the table to the left are found the values for average life expectancy (abbreviated L) and GDP per capita, (G) for six selected countries in the year 2022.

To the right in the figure the variables' frequency distribution, dispersion, is illustrated in two histograms. The upper graph shows income and the lower graph shows life expectancy. On the horizontal x-axis the values for each respective variable are shown. On the vertical y-axis the number of observations is shown. For each of the two variables, life expectancy and income, the observations are grouped in up to four bars.

Country	L	G
Brazil	74.87	14640.08
Congo	65	4187.51
France	82.48	39065.71
Haiti	63.95	1546.42
Jamaica	71.48	7481.67
Sweden	83.05	47126.13

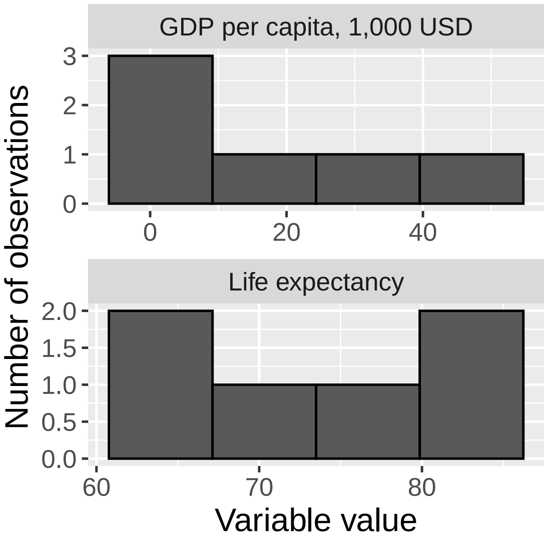


Figure 22.1: GDP per capita and average life expectancy

Let us estimate variance (section 15.7) for each variable. Variance for the variable life expectancy (L):

$$\operatorname{var}(L) = \frac{\sum_i \left(L_i - \bar{L}\right)^2}{n-1} = \frac{341.1156}{5} \approx 68.22 \tag{22.1}$$

where $L_i - \bar{L}$ indicates the deviation from the mean for the variable life expectancy. Variance for GDP per capita, variable G, is:

$$\mathrm{var}\,(G) = \frac{\sum_i \left(G_i - \bar{G}\right)^2}{n-1} = 373.886 \tag{22.2}$$

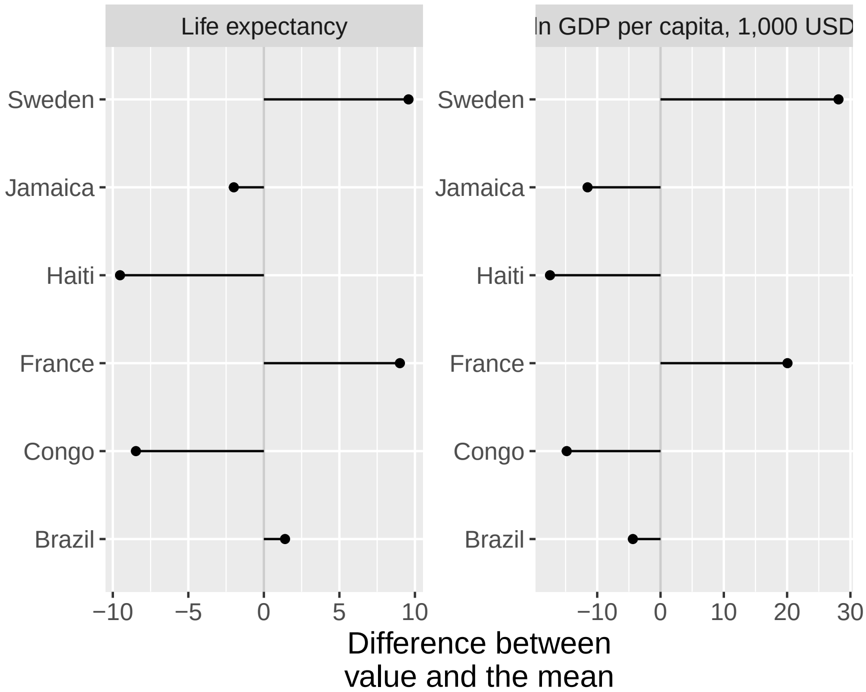


Figure 22.2: Life expectancy and GDP per capita: difference to the mean

The results are illustrated in figure 22.2 with deviations from the mean, with life expectancy in the graph to the left and GDP per capita in the graph to the right.

Standard deviation for life expectancy L is:

$$s_L = \sqrt{\mathrm{var}\,(L)} = \sqrt{\frac{\sum_i^n \left(L_i - \bar{L}\right)^2}{n-1}} \approx 8.26 \tag{22.3}$$

Standard deviation for income M:

$$s_G = \sqrt{\mathrm{var}\,(G)} = \sqrt{\frac{\sum_i^n \left(G_i - \bar{G}\right)^2}{n-1}} \approx 19.336 \tag{22.4}$$

The results show that life expectancy has a standard deviation of 0.98 years and average income has a standard deviation of 21.5. We have an idea that life expectancy and income can be expected to covary linearly, even though we do not have a theory about a causal relationship. We therefore estimate the covariance between the variables. Large parts of the calculation are found in table 22.2 :

$$\mathrm{cov}\,(L, G) = \left(\frac{1}{n-1}\right) \sum_i \left(L_i - \bar{L}\right)\left(G_i - \bar{G}\right) \approx 151.696 \tag{22.5}$$

The result indicates a positive covariance, greater GDP is associated with on average higher average life expectancy. As was described in section 20.6 this measure is often difficult to interpret. We cannot assess whether the covariation we find should be regarded as high or low (strong or weak).

Table 22.1: Life expectancy L and GDP G, and standardized values

Country	L_i	G_i	$L_i - \bar{L}$	$G_i - \bar{G}$	z_L	z_G
Brazil	74.87	14.64	1.4	-4.37	0.17	-0.23
Congo	65	4.19	-8.47	-14.82	-1.03	-0.77
France	82.48	39.07	9	20.06	1.09	1.04
Haiti	63.95	1.55	-9.52	-17.46	-1.15	-0.9
Jamaica	71.48	7.48	-1.99	-11.53	-0.24	-0.6
Sweden	83.05	47.13	9.58	28.12	1.16	1.45
Mean	73.47	19.01				
Standard deviation	8.26	19.34				

Note: Data from Our World in Data. GDP per capita adjusted for inflation and cost of living between countries. GDP shown in \$1,000. All values rounded to second decimal.

Let us now calculate standardized values for life expectancy L and GDP G, by dividing the difference to the mean by standard deviation. We call the standardized values of each respective variable z_L and z_G. Table 22.1 describes the results.

The standardized values are easier to compare and already in the table we see that the variables display a tendency toward positive covariation: several observations whose values deviate negatively in variable z_L also deviate negatively in variable z_G, and vice versa for several positive values. We also see that the standardized size of the deviations is of similar magnitude.

To estimate standardized covariance we use the measure correlation coefficient (Pearson's r) which we defined in equation (20.7) as $r_{XY} = \frac{\text{cov}(X,Y)}{s_X s_Y}$ where $\text{cov}(X,Y)$ is estimated covariance between X and Y, and s_X respectively s_Y are estimated standard deviation for each respective variable. The correlation coefficient $r_{L,G}$ then becomes:

$$r_{L,G} = \frac{\text{cov}(L,G)}{s_L s_G} \approx 0.9498 \tag{22.6}$$

This result also indicates positive covariation, where higher life expectancy covaries with higher average income. As noted in section 20.7 the correlation coefficient r is also a difficult to interpret.

Now we shall estimate the covariation between L and G by using regression analysis and the least squares method. When we use regression analysis we must also determine which variable should be the explanatory variable, that is the variable that should be on the left side of the equals sign in our regression model. We decide to map to what extent variations in life expectancy L can be "explained" by the regression model and the explanatory variable average income G. We cannot draw conclusions about causation, since we cannot be certain that there is not some other phenomenon that affects one or both variables in the model. We have the following regression model:

$$L = \alpha_1 + \alpha_2 G + e \tag{22.7}$$

where L is life expectancy, G is GDP, α_1 is the coefficient for y-intercept, α_2 is the slope coefficient and e is the error term. Based on the least squares method we shall now estimate $\hat{\alpha_1}$ and $\hat{\alpha_2}$. That we use Greek letters for the coefficients has no mathematical significance. In table 22.2 we have the calculations required to give us the slope coefficient $\hat{\alpha}_2$:

$$\hat{\alpha_2} = \frac{\sum\left(G_i - \bar{G}\right)\left(L_i - \bar{L}\right)}{\sum\left(G_i - \bar{G}\right)^2} \approx \frac{758.48}{1869.41} \approx 0.4057 \tag{22.8}$$

This we use to estimate the y-intercept $\hat{\alpha}_1$:

$$\hat{\alpha_1} = \bar{L} - \hat{\alpha_2}\bar{G} \approx 73.47 - 0.4057 * 19.01 \approx 65.758 \tag{22.9}$$

Table 22.2: Calculations with L and G

L_i	G_i	$\tilde{L}$	$\tilde{L}^2$	$\tilde{G}$	$\tilde{G}^2$	$\tilde{L}\tilde{G}$	$\hat{L}_i$	$\hat{e}_i$	$\hat{e_i}^2$	$L_i - \hat{L}_i$	$\left(L_i - \hat{L}_i\right)^2$
74.87	14.64	1.4	1.96	-4.37	19.08	-6.12	71.7	3.17	10.07	3.17	10.07
65	4.19	-8.47	71.74	-14.82	219.64	125.53	67.46	-2.46	6.04	-2.46	6.04
82.48	39.07	9	81.09	20.06	402.31	180.62	81.61	0.87	0.75	0.87	0.75
63.95	1.55	-9.52	90.67	-17.46	304.9	166.27	66.39	-2.44	5.94	-2.44	5.94
71.48	7.48	-1.99	3.96	-11.53	132.85	22.93	68.79	2.69	7.22	2.69	7.22
83.05	47.13	9.58	91.69	28.12	790.63	269.25	84.88	-1.83	3.36	-1.83	3.36

Note: $\tilde{L} = L_i - \bar{L}$, $\tilde{G} = G_i - \bar{G}$

The result $\hat{\alpha}_2 = 0.4057$ means that 1,000 USD higher GDP per capita on average coincides with approximately 0.4057 years longer average life expectancy (almost five months). Our y-intercept $\hat{\alpha_1}$ equals 65.758, which can be interpreted as that if there had existed a country with zero GDP per capita ($G_i = 0$) the average life expectancy there would have been 65.758 years. Let us now estimate predicted average life expectancy $\hat{L_i}$ and the residuals $\hat{e_i}$:

$$\begin{aligned} \hat{L_i} &= \hat{\alpha_1} + \hat{\alpha_2}G_i \\ \hat{e_i} &= L_i - \hat{L_i} \end{aligned} \tag{22.10}$$

The results are described in table 22.2 and illustrated in figure 22.3 . Please check that the table is correct. Next we estimate R^2 for the regression model in equation (22.7) :

$$R^2 = 1 - \frac{\sum_i\left(L_i - \hat{L}_i\right)^2}{\sum_i\left(L_i - \bar{L}\right)^2} \approx 1 - \frac{33.4}{341} \approx 0.902 \tag{22.11}$$

This indicates that approximately 90% of the variation in average life expectancy in these countries could be explained by the regression model and the variation in our explanatory variable, GDP per capita.

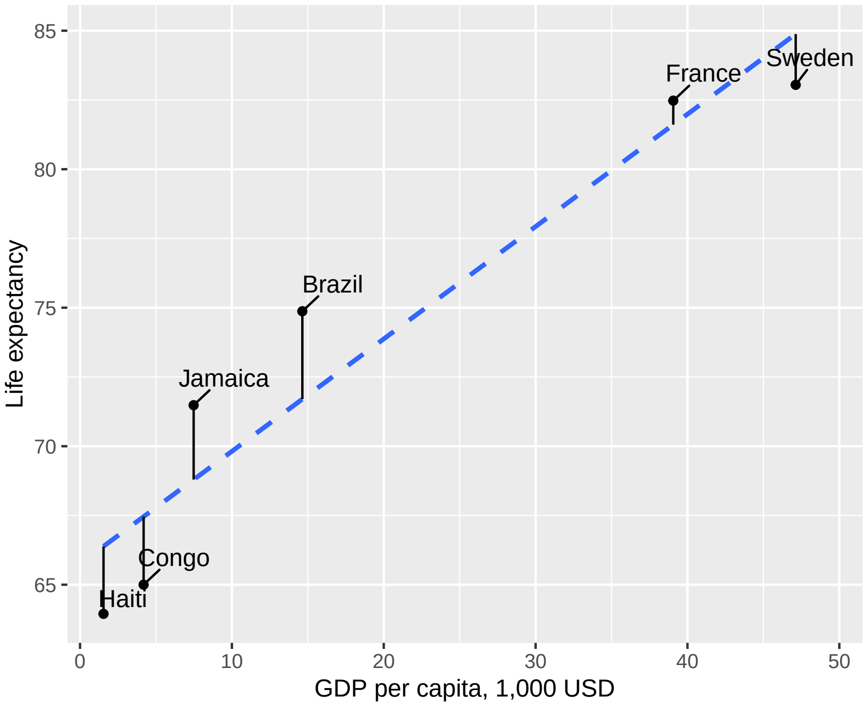


Figure 22.3: Linear covariation between life expectancy and GDP per capita for six countries

22.2 Covariation with a lot of countries

In the previous section we used a few observations for a smaller sample of countries. Now we shall do the corresponding calculation but use data on life expectancy and GDP per capita for 2022 for 164 countries. Table 22.3 lists some measures of central tendency and dispersion for the data we shall use now.

Table 22.3: Measures of Central Tendency and Dispersion for GDP per capita and Life Expectancy

	GDP per capita, G	Life expectancy, L
Minimum	0.72	54.08
Maximum	88.37	84.05
Mean	19.11	73.01
Median	13.03	74.18
Variance	337.51	47.84
Standard deviation	18.37	6.92
N = number of observations	164	164

Note: Data from Our World in Data

We have previously used histograms to illustrate variables. Another type of graph that is often used to illustrate distribution of values is box plots, which are also called box-and-whisker plots. Now we shall compare the distribution of our two variables in histograms and box plots simultaneously.

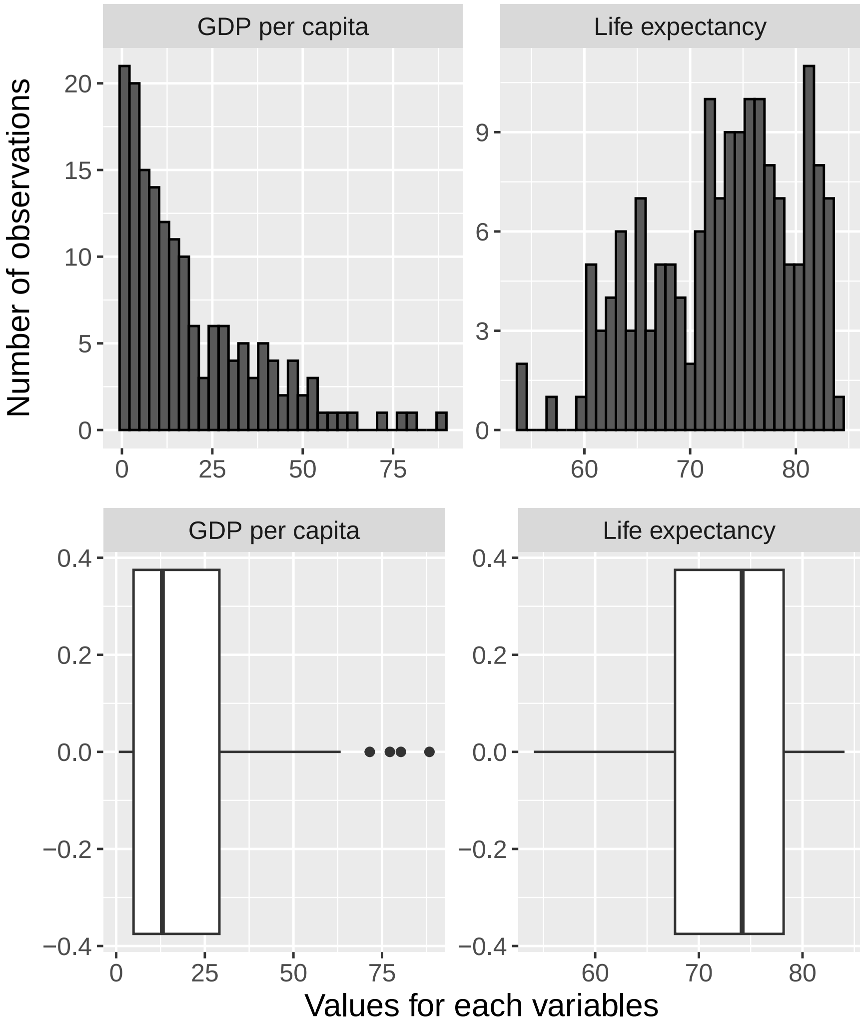


Figure 22.4: Histograms and box plots for life expectancy and GDP per capita

Figure 22.4 illustrates the distribution of the two variables income and life expectancy for all countries. The upper graphs show histograms while the lower graphs are box plots. In the histogram for GDP, the upper graph to the left, we see how the distribution has a slightly longer "tail" upward, which arises because a few countries have unusually high GDP per capita, while the great majority are clustered around the distribution's median and mean, to the left in the graph. In the histogram for life expectancy, upper graph to the right, we see that the values are somewhat more evenly distributed below and above the mean, which in turn means that the median and mean lie close to each other.

Also in the two box plots at the bottom of the figure we can read the mean at the line in the middle of the white boxes. The white rectangles' left and right edges respectively are the first and third quartile, Q_1 and Q_3. The lines outside the rectangle mark 1.5 * the interquartile range, that is $1.5 * (Q_3 - Q_1)$. The covariance between L and G is:

$$\operatorname{cov}(L, G) \approx 94.84 \tag{22.12}$$

Now we shall estimate the following regression model with data for all countries in the dataset:

$$L = \beta_1 + \beta_2 G + v \tag{22.13}$$

where v is the error term and β_1 and β_2 are the coefficients we estimate using the least squares method. Life expectancy L is a linear function of GDP per capita G. The model thus has the same form as the one in equation (22.7) . The reason we now define the new coefficients β_1 and β_2 for the regression model is that we want to clarify that this is a different regression model than the previous one. Now we use more observations, which therefore by definition is a different collection of data than last time.

Although the mathematics with this many observations becomes relatively extensive to perform manually, by hand, this type of calculation takes less than a second for a computer. This time we get the following result for our regression model:

$$L = \hat{\beta}_1 + \hat{\beta}_2 G + v \approx 67.645 + 0.281 * G_i + v_i \tag{22.14}$$

This also indicate a pretty strong positive correlation between GDP and life expectancy. The result is not that far from the results we found in the previous section when we only used data for a few countries. The regression result is illustrated in figure 22.5 where the regression line (the straight dashed line) between the points is calculated using the least squares method. In the graph the vertical distances between the points and the regression line are also drawn, which corresponds to the residuals. Note how we in the graph on an overview level can see how the observations for the two variables are distributed. A few points lie a bit further to the right in the picture, which corresponds to the countries with unusually high GDP.

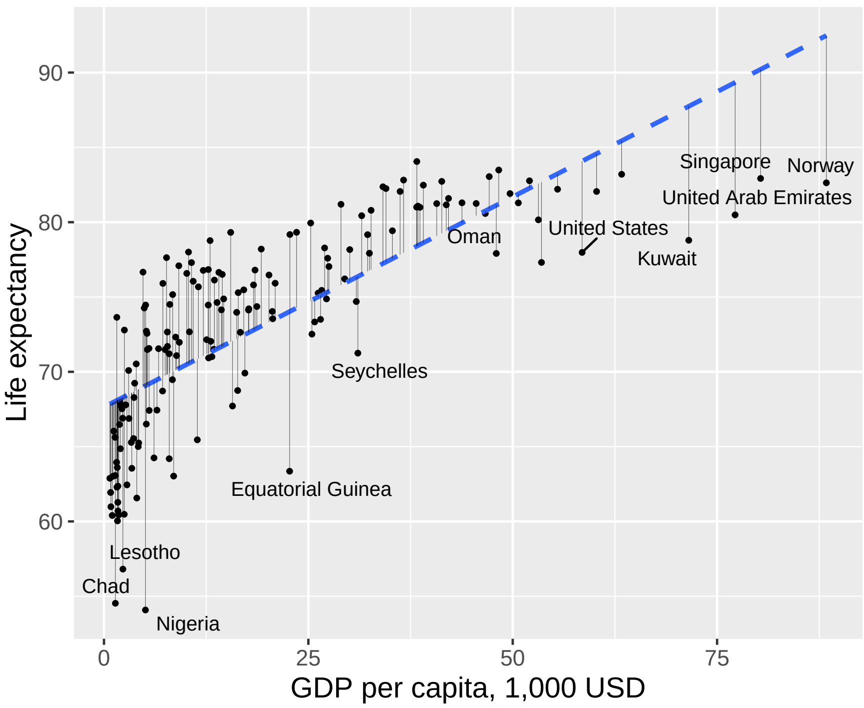


Figure 22.5: Covariation between life expectancy and GDP per capita

We have now produced results for all countries in our dataset. This is large amounts of information and can seem like a powerful result. This is an example of an observational study where we observe the covariation that existed at a historical point in time between two phenomena. We could admittedly argue theoretically for why income could be thought to have a positive impact on life expectancy. Wealthy people have greater opportunity to live relaxed lives and enjoy various advantages that money brings. But few people would probably claim that it is the money itself that makes people with higher income live longer.

Regardless of the strength of this type of argument, our investigation says very little about the specific causal relationship, exactly how it works and how large an effect income has on life expectancy. Sometimes this type of objection matters less but sometimes they have great significance. It depends on what purpose our analysis has and what type of conclusions we hope to be able to draw. If we for example want to find an exact method for how people should live longer, we need to establish a more exact causal relationship and be able to measure its size. If we then want to implement a political measure, we also need to understand what other consequences our proposal will have on the world.

22.3 Linear trend over time

In section 5.9 we went through how we through logarithmic transformation of GDP can get a more linear picture of the development of the economy. Table 22.4 shows (once again) GDP for the US in billions USD (nominal values), and the same values in natural log, $\ln(\text{GDP})$(compare table 3.7). In the table we use the abbreviations G for GDP and T for years. Values between 0 and 1 is negative in logarithmic form since $\frac{1}{a^2} = a^{-2}$.

Table 22.4: GDP and the natural log of GDP

Year T	GDP (G_t)	ln GDP $(\ln G_t)$
1800	0.486	-0.72
1850	2.656	0.9768
1900	21.197	3.05
1950	299.827	5.703
2000	10,250.952	9.235

Table 22.5: Calculations with the least squares method

Row	G_t	T_t	$\tilde{G}$	$\tilde{T}$	$\tilde{T}^2$	$\tilde{T}\tilde{G}$	$\ln G_t$	$\widetilde{\ln G}$	$\tilde{T}\widetilde{\ln G}$
1	0.486	1800	-2,114.54	-100	10,000	211,453.76	-0.72	-4.37	437.1
2	2.656	1850	-2,112.37	-50	2,500	105,618.38	0.9768	-2.67	133.63
3	21.197	1900	-2,093.83	0	0	0	3.05	-0.596	0
4	299.83	1950	-1,815.2	50	2,500	-90,759.83	5.703	2.05	102.69
5	10,250.95	2000	8,135.93	100	10,000	813,592.84	9.235	5.586	558.56
Mean	2,115.02	1900							
Sum					25,000	1,039,905.15			1,231.99

Note: $\tilde{G} = G_t - \bar{G}$, $\tilde{T} = T_t - \bar{T}$, $\widetilde{\ln G} = \ln G_t - \overline{\ln G}$, $\tilde{T}\widetilde{\ln G} = \left(T_t - \bar{T}\right)\left(\ln G_t - \overline{\ln G}\right)$

Now we shall estimate the linear trend over time for GDP using regression analysis. We shall estimate two regression models and compare the results. One regression model where we use the variable $G_t =$ GDP in billions USD in year t and one model with $\ln G_t =$ natural logarithm of GDP. The explanatory variable in both models is time (years). In the first regression model we use the variable GDP as a function of the variable year:

$$G_t = \alpha_1 + \alpha_2 \text{year}_t + e_t \tag{22.15}$$

where α_1 and α_2 are coefficients we shall estimate using the least squares method and e_t is the error term. Note now that we use a linear model to describe a variable whose data is not particularly linear. In the table above the values for the variable G_t increase exponentially.

In the second regression model we use the natural log of GDP:

$$\ln G_t = \beta_1 + \beta_2 \text{year}_t + \epsilon_t \tag{22.16}$$

where β_1 and β_2 are coefficients, ϵ_t is the error term. Note that even though we now use $\ln b$ we still estimate the covariation against the variable year without logarithmic transformation of this variable. We use different letters to describe the coefficients in this model to clarify that these are two different regression models. We begin by estimating the first model in equation (22.15) . In table 22.5 the calculations we need to estimate $\hat{\alpha}_2$ are described:

$$\hat{\alpha_2} = \frac{\sum (\text{year}_t - \bar{\text{year}}) (G_t - \bar{G})}{\sum (\text{year}_t - \bar{\text{year}})^2} = \frac{1,039,905.15}{25,000} = 41.6 \tag{22.17}$$

For $\hat{\alpha}_1$ we get:

$$\hat{\alpha_1} = \bar{G} - \hat{\alpha_2}\bar{\text{year}} \approx 2,115.0236 - 41.6 \times 1,900 = -76,917.8 \tag{22.18}$$

The slope coefficient indicates the average annual linear GDP growth between year 1800 and 2000 in the US. Our result thus indicate that GDP on average has increased by approximately $41.6 billions per year. The result for the y-intercept $\hat{\alpha}_1$ indicates that in year zero GDP was minus $76,917.8 billion, which is impossible. This result comes from the use of a linear model to describe a development that is not linear.

For the second regression model, equation (22.16) with the natural log of GDP, we instead get the following result for $\hat{\beta}_2$:

$$\hat{\beta}_2 = \frac{\sum (\ln G_t - \bar{\ln G}) (\text{year}_t - \bar{\text{year}})}{\sum (\ln G_t - \bar{\ln G})^2} \approx \frac{1,231.986}{25,000} \approx 0.04928 \tag{22.19}$$

Estimation of $\hat{\beta}_1$:

$$\hat{\beta}_1 = \bar{\ln G} - \hat{\beta}_2\bar{\text{year}} \approx 2,115.0236 - 0.04928 \times 1900 \approx -89.98 \tag{22.20}$$

Since we now use the natural log of GDP the estimated coefficient $\hat{\beta}_2$ is an approximate measure of how the model estimates the annual percentage growth rate: $0.04928 \approx 4.9\%$. The other coefficient, y-intercept $\hat{\beta}_1$, gives us the level of ln GDP at year zero. To get the level in billions USD we now need to set -89.98 as exponent on the number e, which we calculate with the function exp (), the anti-logarithm of the natural log:

$$\exp\left(\hat{\ln G_t}\right) = e^{\hat{\ln b_t}} = \hat{b}_t \tag{22.21}$$

For year zero we then get:

$$\begin{aligned} \ln\left(\hat{G}_t\right) &= \hat{\beta}_1 + \hat{\beta}_2\text{year}_0 \\ \ln\left(\hat{G}_0\right) &= -89.98 + 0.04928 \times 0 \\ \ln\left(\hat{G}_0\right) &= -89.98 \end{aligned} \tag{22.22}$$

From this we see that our model estimates GDP year zero to:

$$\text{GDP year zero } = \exp(-89.98) = e^{-89.98} = \frac{1}{e^{89.98}} \text{ bn USD.} \tag{22.23}$$

The number $e^{89.98}$ is a high value, a bit over 1 duodecillion, a 1 followed by 39 zeros. This means that $1/e^{89.98}$ is a very low value, close to 0. This is logically a more credible result than any negative number, which we estimated GDP to be in year 0 in the first regression model. What this exercise primarily illustrates is that the natural log of GDP has a more linear correlation with time, compared to GDP. But none of our estimates of GDP in year 0 should be taken particularly seriously. We use simple models and just a few observations to compare complex processes.

The slope coefficient for time in each respective regression model gives us the average growth rate for GDP, as well as the growth rate for the natural log of GDP. Results are shown in table 22.6 with predicted values (the regression lines) for each regression model, the variables $\hat{G_t}$ and $\widehat{\ln G_t}$ as well as the residuals $\hat{e_t}$ and $\hat{\epsilon_t}$.

In the column furthest to the right we have $\exp\left(\widehat{\ln G_t}\right)$, the level of GDP that is predicted with the regression model that uses the natural log of GDP. Based on this we then calculate the sum of squared residuals. For the regression model $G_t = \alpha_1 + \alpha_2 \text{year}_t + e_t$ we get:

$$\sum_t \hat{e_t}^2 = 39,549,636.76 \tag{22.24}$$

For the regression model $\ln G_t = \beta_1 + \beta_2 \text{year}_t + \epsilon_t$:

$$\sum_t \hat{\epsilon_t}^2 \approx 1.31 \tag{22.25}$$

To convert the results for the natural log variable to billions of dollars we use the exponential function:

$$\exp\left(\sum_t \hat{\epsilon_t}^2\right) \approx \exp(1.31) \approx 3.71 \tag{22.26}$$

As can be seen in table 22.6 the predicted results for $\hat{G_t}$ from the regression model in equation (22.15) are further from observed GDP, G_t, compared, to the difference between predicted values for $\widehat{\ln G_t}$(from the regression model in equation (22.16)). This is particularly clear for the lower values of GDP, during the 1800s.

Table 22.6: Predicted GDP

G_t	year_t	$\hat{G_t}$	$\hat{e_t}$	$\ln G_t$	$\widehat{\ln G_t}$	$\hat{\epsilon_t}$	$\exp\left(\widehat{\ln G_t}\right)$
0.49	1800	$-2,044.6$	2,045.08	-0.72	-1.28	0.56	0.28
2.66	1850	35.21	-32.56	0.98	1.19	-0.21	3.27
21.2	1900	2,115.02	$-2,093.83$	3.05	3.65	-0.6	38.46
299.83	1950	4,194.83	$-3,895.01$	5.7	6.11	-0.41	451.9
10,250.95	2000	6,274.64	3,976.31	9.24	8.58	0.66	5,310.49

This we also see in the sums of the squared residuals. The higher value for $\sum \hat{e_t}^2$ compared to $\sum \hat{\epsilon}_t^2$ indicates that the total distance between the observations and the predicted values is larger in the first model, without logarithmic transformation, compared to in the second. The same thing is also seen in figure 22.6 where two graphs show GDP and the natural log of GDP, as well as the linear trend lines drawn using the models we estimated above. Just as our results indicate, the logarithmic values allow themselves to be better described with a linear regression model, compared to the non-logarithmic values.

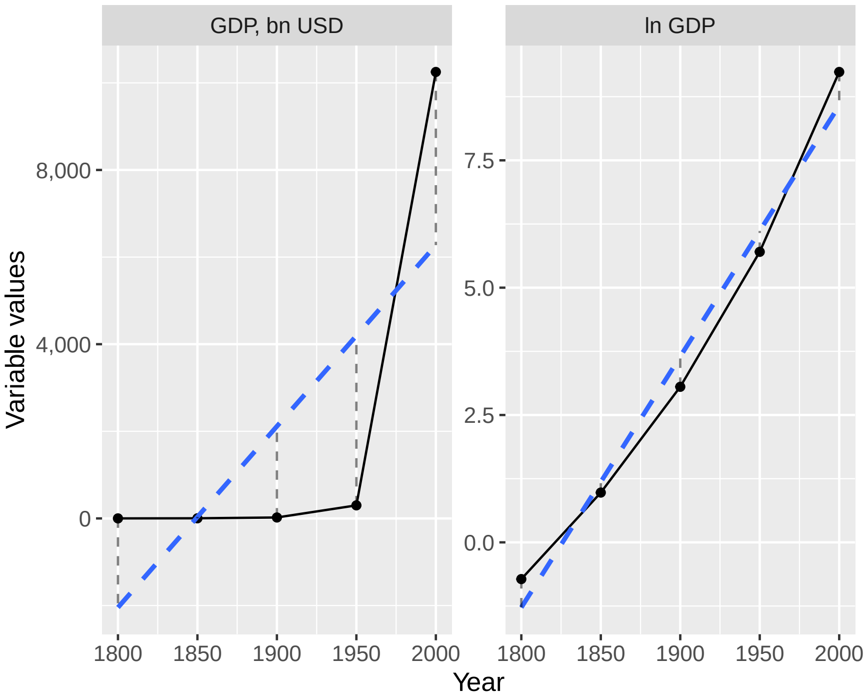


Figure 22.6: GDP and ln GDP with their respective regression lines

22.4 Cycles and waves

In this section we will also compare GDP in the US over time. But instead of the explained variable or the coefficients, we are now interested in the residuals of the following regression model with the natural log of GDP as the explained variable:

$$\ln G_t = \beta_1 + \beta_2 \text{year}_t + \epsilon_t \tag{22.27}$$

For this regression model the residual $\hat{\epsilon}_t$ indicates the difference between observed $\ln G_t$ and estimated $\widehat{\ln G_t}$. In the previous section we estimated the coefficients for this regression model to $\hat{\beta}_1 = -89.98$ and $\hat{\beta}_2 = 0.04928$. When we study economic development we are often interested in both the long-term trend and deviations from this trend. The deviation from the trend is in this case a measure of more short-term variation in GDP growth. This type of short-term variations has been the subject of great discussion within social science. In this case we measure deviations from the trend by observing the residuals.

Now we shall compare different deviations from estimated measures of long-term trend. This provides additional illustrations of the least squares method and how certain data lie closer to the predictions we estimate using a linear regression model. We have already estimated the residual $\hat{\epsilon}_t = \ln G_t - \widehat{\ln G_t}$. Now we want to calculate the deviation from trend in billions of USD, which we for year t call d_t:

$$\begin{aligned} d_t &= G_t - \exp\left(\widehat{\ln G_t}\right) \\ &= G_t - e^{\widehat{\ln G_t}} \\ &= G_t - \hat{G}_t^{\text{from ln}} \end{aligned} \tag{22.28}$$

The term G_t is GDP in billions USD. In the first row we use the exponential function $\exp()$ to set the logarithmic expression $\widehat{\ln G_t}$ as exponent with base e, which gives us $\hat{G}_t$. In the last row we mark this term with the text "from ln", as a reminder that this predicted G_t is from the regression model with $\ln G_t$. This gives us our first result.

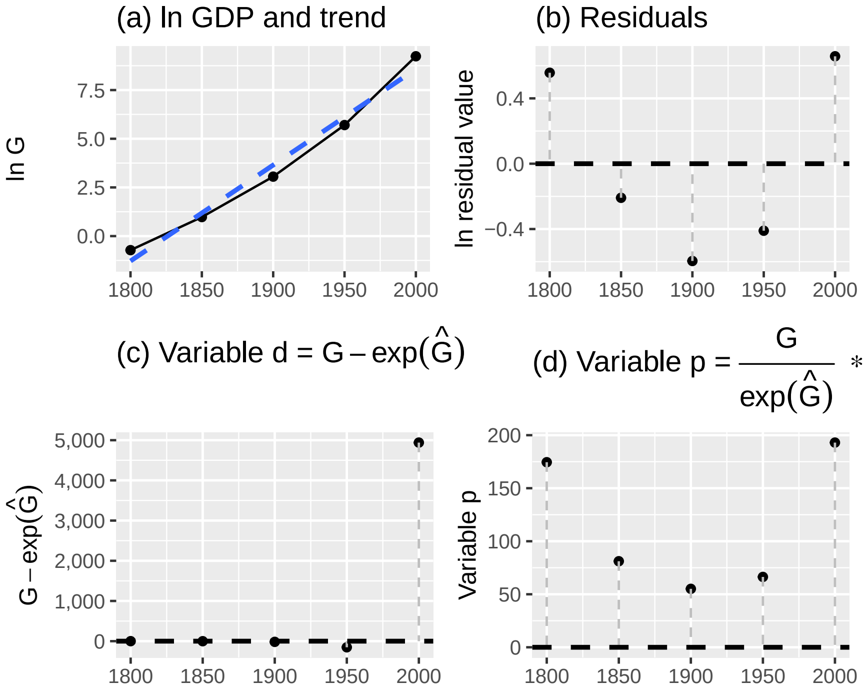


Figure 22.7: GDP and deviations from trend

Table 22.7: Trend and Deviations in GDP Growth

year_t	G_t	$\ln G_t$	$\widehat{\ln G_t}$	$\exp\left(\widehat{\ln G_t}\right)$	$\hat{\epsilon}_t$	d_t	p_t
1800	0.49	−0.72	−1.28	0.28	0.56	0.21	174.53
1850	2.66	0.98	1.19	3.27	−0.21	−0.62	81.16
1900	21.2	3.05	3.65	38.46	−0.6	−17.26	55.12
1950	299.83	5.7	6.11	451.9	−0.41	−152.08	66.35
2000	10,250.95	9.24	8.58	5,310.49	0.66	4,940.46	193.03

Next we shall calculate the deviation from trend as percent of GDP for each respective year. This gives us a relative measure of how deviations each respective year compared to trend. We call this variable p_t:

$$p_t = \frac{G_t}{\exp\left(\widehat{\ln G_t}\right)} \times 100 \tag{22.29}$$

In most situations, all of these measures are not that interesting. This is mostly an exercise to illustrate different ways to look at data and thereby hopefully better understand the math behind our calculations. The results from our calculations are reported in table 22.7 and illustrated in four graphs in figure 22.7 . The upper left graph (a) shows the long-term trend in GDP growth estimated using the least squares method, which is the same as in figure 22.6 . Deviations from the trend are in graph (a) the residual $\hat{\epsilon}_t$ in the regression model in equation (22.15) . The upper right graph (b) shows the deviations from this trend line where the vertical y-axis in the graph measures the value of the residuals $\hat{\epsilon}_t$(in logarithmic form). Put simply, the right graph is a rotated version of the left graph, so that the trend line is now horizontal. In graph (c), lower left, the deviation from trend is illustrated by variable d_t, which measures differences between observations and the regression line in USD. In the lower right graph (d) we see deviation from trend as percent of GDP for the current year, variable p_t.

22.5 Wealth and happiness

In section 5.10 we compared the covariation between self-assessed life satisfaction and GDP per capita. Table 22.8 repeats this data and now we will use this to estimate the covariation between these phenomena using two regression models. The first model:

$$H_i = \gamma_1 + \gamma_2 Y_i + e_i \tag{22.30}$$

where H_i is happiness in country i, Y_i is GDP per capita, e_i is the error term and we shall estimate the two coefficients γ_1 and γ_2 using the least squares method. The second regression model:

$$H_i = \delta_1 + \delta_2 \log_{10} Y_i + \epsilon_i \tag{22.31}$$

where $\ln Y_i$ is $\log_{10}$ (GDP per capita), ϵ_t is the error term and δ_1 and δ_2 are the coefficients we shall estimate. The results for the first model (equation (22.30)):

$$\begin{aligned} \hat{\gamma_2} &= \frac{\sum \left(H_i - \bar{H}\right)\left(Y_i - \bar{Y}\right)}{\sum \left(Y_i - \bar{Y}\right)^2} = 0.05622 \\ \hat{\gamma_1} &= \bar{H} - \hat{\gamma_2}\bar{Y} = 4.7311 \end{aligned} \tag{22.32}$$

For the other model (equation (22.31)) the results are:

$$\begin{aligned} \hat{\delta_2} &= \frac{\sum \left(H_i - \bar{H}\right)\left(\log_{10} Y_i - \bar{\log_{10}} Y\right)}{\sum \left(\log_{10} Y_i - \bar{\log_{10}} Y\right)} = 2.222 \\ \hat{\delta_1} &= \bar{H} - \hat{\delta_2}\bar{\log_{10}} Y = 3.503 \end{aligned} \tag{22.33}$$

This time we leave the details of the calculation as an exercise for the reader. If we focus on the slope coefficients see that according to the first model an increase in GDP per capita of 1,000 PPP USD is associated with on average around 0.056 higher self-assessed happiness. According to the second model one unit higher $\log_{10}$ GDP per capita coincides with around 2.22 units higher life satisfaction. As can be seen in table 22.8 an increase of $\log_{10}$ GDP per capita with one corresponds to almost a tenfold increase of GDP per capita, for example the approximate difference between Congo and Sweden or the difference between Haiti and Brazil.

Table 22.8: Happiness and GDP in six countries

Country	Happiness, H	GDP per capita, Y	$\log_{10} Y$
Haiti	3.824	1.653	0.218
Congo	4.884	4.881	0.689
Jamaica	5.89	8.194	0.913
Brazil	6.333	14.103	1.149
France	6.635	38.606	1.587
Sweden	7.287	46.949	1.672

Note: Data from Our World in Data, www.ourworldindata.org.

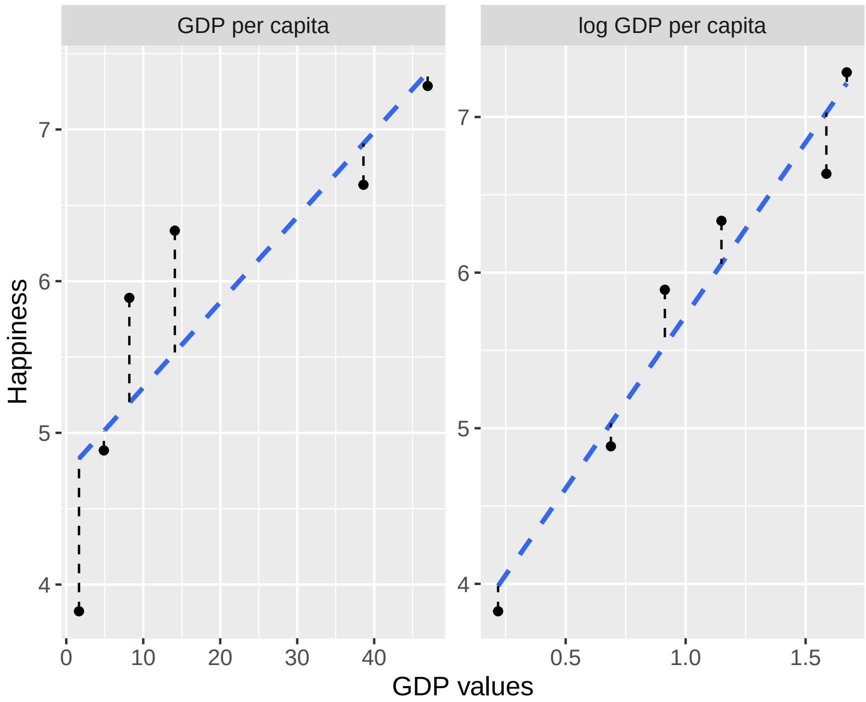


Figure 22.8: Happiness and GDP per capita

Let us also compare R^2, how large a proportion of the variation in happiness can be explained by the variation of the explanatory variables Y respectively $\log_{10} Y$. For the first model we get:

$$R^2_{\text{model 1}} = 1 - \frac{\sum \left(H_i - \hat{H}_i\right)^2}{\sum \left(H_i - \bar{H}\right)^2} \approx 0.719 \tag{22.34}$$

For the other model:

$$R^2_{\text{model 2}} = 1 - \frac{\sum \left(H_i - \hat{H}_i\right)^2}{\sum \left(H_i - \bar{H}\right)^2} \approx 0.948 \tag{22.35}$$

According to this measure the first model explains about 72% of the variation in average happiness between countries. The second model explains about 95% of the variation in happiness between countries.

Figure 22.8 illustrates the data we used and the two regression models' prediction lines. The result indicates that higher and relatively higher GDP covary with higher self-assessed life satisfaction, where we called the latter "happiness". When we compare the covariation with relative income differences (log GDP) almost all variation in happiness can be explained by the variation in the explanatory variable log GDP per capita. A relatively higher measure of R^2 means that the sum of the residuals is relatively smaller. This can also be seen in the graphs if we compare the distance between the points and the regression lines.

This is an observational study. For the same reasons as we discussed earlier we cannot draw any more detailed conclusions about causal effects from these results.

22.6 Star Wars

Statistical analysis is fundamentally a quantitative method. We use numbers to calculate new numbers. If we instead want to study other types of data, this needs to be quantified in some form and written with numbers. There are many situations where we may benefit from statistical analyses of data material that at first glance is not associated with statistics. Here follows a simple example as illustration.

Table 22.9: Movies and characters in Star Wars

Film	Obi-Wan Kenobi	Anakin Sky-walker/ Darth Vader	Luke Skywalker	Leia Organa	Chewbacca	Han Solo (alive)	R2-D2
Star Wars 1 The Phantom Menace	1	1	0	0	0	0	1
Star Wars 2 Attack of the Clones	1	1	0	0	0	0	1
Star Wars 3 Revenge of the Sith	1	1	1	1	1	0	1
Star Wars 4 A New Hope	1	1	1	1	1	1	1
Star Wars 5 The Empire Strikes Back	1	1	1	1	1	1	1
Star Wars 6 Return of the Jedi	1	1	1	1	1	1	1
Star Wars 7 The Force Awakens	0	0	1	1	1	1	1
Star Wars 8 The Last Jedi	0	0	1	1	1	0	1
Star Wars 9 The Rise of Skywalker	0	0	1	1	1	0	1

We shall now calculate to what extent some of the characters in Star Wars appear in the same films. Table 22.9 lists seven of the characters in their respective columns and nine of the films (the three trilogies, episodes 1–9) on their respective rows. In each cell the value 0 or 1 is shown depending on whether the character appeared in the film: 0 = does not appear, and 1 = appears. Of these characters the robot R2-D2 has been in the most films, all nine. We shall now estimate pairwise correlations, the covariation between two characters at a time. We have here 7 characters and can therefore estimate a total of 21 combinations, which we find using equation (2.155) from section 2.16 :

$$\begin{aligned}
\binom{n}{k} &= \frac{n!}{(n-k)! * k!} \\
&= \frac{7!}{(7-2)!2!} \\
&= \frac{7 * 6 * \cancel{5 * 4 * 3 * 2 * 1}}{\cancel{5 * 4 * 3 * 2 * 1} * 2 * 1} \\
&= 21
\end{aligned} \tag{22.36}$$

We estimate pairwise correlation with the correlation coefficient (equation (20.7)):

$$r_{xy} = \frac{\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\left(\sum_i^n (x_i - \bar{x})^2\right)^{1/2} \left(\sum_i^n (y_i - \bar{y})^2\right)^{1/2}} \tag{22.37}$$

For this example we replace the variables x and y with two of the columns with zeros and ones in table 22.9 . We calculate a value for r_{xy} for each respective pair of characters, such as Obi-Wan Kenobi and Anakin Skywalker/Darth Vader and vice versa. If we call Obi-Wan Kenobi variable x and Anakin Skywalker/Darth Vader variable y the correlation coefficient becomes:

$$\rho_{xy} = \frac{\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\left(\sum_i^n (x_i - \bar{x})^2\right)^{1/2} \left(\sum_i (y_i - \bar{y})^2\right)^{1/2}} = \frac{2}{2^{\frac{1}{2}+\frac{1}{2}}} = 1 \tag{22.38}$$

A correlation coefficient equal to 1 means maximum possible covariation. This we also see in the table. All films where Obi-Wan Kenobi appears also have Anakin Skywalker/Darth Vader. Let us calculate the correlation between Leia Organa, x, and Han Solo, y:

$$\begin{aligned}
\rho_{xy} &= \frac{\sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\left(\sum_i^n (x_i - \bar{x})^2\right)^{1/2} \left(\sum_i (y_i - \bar{y})^2\right)^{1/2}} \\
&\approx \frac{0.889}{(1.247)^{1/2} (1.491)^{1/2}} \\
&\approx 0.478
\end{aligned} \tag{22.39}$$

The correlation becomes in this case weaker, approximately 0.478. Leia appears in several films that Han is not in. As was described in section 20.7 it is difficult to interpret different results for the correlation coefficient more precisely. Despite this, this type of vague indication can be useful if we for example want a rough overview of how the different characters relate to each other. With more detailed information we could also do more detailed analysis, such as for example number of scenes where different characters appear simultaneously.

Hopefully this is an illustrative example of how statistical methods can be used even in situations that at first glance perhaps are not associated with statistics. In a similar way these methods and other types of calculations can be used to map patterns even in other types of data that are not quantified, such as for example texts. Pairwise correlations can for example give an overview of which words are used in different documents or in different sections in one and the same document. If one uses computers it is fast to do this type of estimations for large amounts of information, even if this requires millions of calculations.

22.7 Chapter summary

- By estimating covariation between different phenomena we can discover important aspects of data. The least squares method can among other things be used to study to what extent life expectancy and income covary. A positive covariation indicates that longer life expectancy coincides with higher incomes.

- The least squares method can also be used to estimate a linear trend over time, for example the average rate of change in the economy's incomes.
- Sometimes we may benefit from mapping patterns of covariation even in data that we perhaps do not spontaneously associate with statistical analysis, for example covariation between different words in texts or between the characters in the Star Wars films.

22.8 Exercises

1. A regression of life expectancy Y on GDP per capita X (thousands USD) gives $\hat{Y} = 50 + 0.4X$.

(a) What is the predicted life expectancy for a country with (X=30) ?

(b) Interpret the slope coefficient (hat{b}=0.4) .

(c) Does this regression prove that higher GDP causes longer life expectancy?

Answer: (a) $\hat{Y} = 50 + 0.4 \times 30 = 62$, (b) each additional 1000 USD in GDP per capita is associated with 0.4 years higher life expectancy on average, (c) no — correlation does not imply causation.

2. Three countries have GDP $X = \{10, 20, 30\}$ (thousands USD) and life expectancy $Y = \{60, 65, 70\}$.

(a) Calculate (bar{X}) and (bar{Y}) .

(b) Calculate (hat{b}) .

(c) Calculate (hat{a}) and write the regression equation.

Answer: (a) $\bar{X} = 20, \bar{Y} = 65$, (b) $\hat{b} = 0.5$, (c) $\hat{a} = 55, \hat{Y} = 55 + 0.5X$.

3. A regression of employment Y on year X gives $\hat{Y} = -3000 + 2X$.

(a) What is the predicted employment in year 2010?

(b) What is the predicted employment in year 2020?

(c) Interpret the slope (hat{b}=2) .

Answer: (a) $\hat{Y} = 20$, (b) $\hat{Y} = 40$, (c) employment increases by 2 units per year on average.

4. A regression of happiness on GDP gives $R^2 = 0.85$.

(a) What does (R^{2}=0.85) tell us?

(b) What does the remaining 15% represent?

(c) Does a high (R^{2}) prove causation?

Answer: (a) 85% of the variation in happiness is explained by GDP, (b) variation in happiness not explained by GDP, (c) no.

5. Describe the expected sign of $\hat{b}$ for each relationship:

(a) Income (X) and years of education (Y) .

(b) Temperature (X) and hot chocolate sales (Y) .

(c) Number of doctors per capita (X) and hospital mortality (Y) .

Answer: (a) positive, (b) negative, (c) positive (reverse causation: doctors are placed where there is more illness).

6. For 4 countries with $X = \{5, 10, 15, 20\}$ and $Y = \{40, 50, 60, 70\}$:

(a) Calculate (bar{X}) and (bar{Y}) .

(b) Calculate (hat{b}) .

(c) Calculate (hat{a}) and write the regression equation.

Answer: (a) $\bar{X} = 12.5, \bar{Y} = 55$, (b) $\hat{b} = 2$, (c) $\hat{a} = 30, \hat{Y} = 30 + 2X$.

7. A regression of CO $_2$ emissions Y on year X gives $\hat{Y} = -18000 + 10X$.

(a) What is the predicted emission in year 2000?

(b) Interpret the slope (hat{b}=10) .

(c) Is the estimated trend necessarily a causal relationship?

Answer: (a) $\hat{Y} = 2000$, (b) emissions increase by 10 units per year on average, (c) no — it is a descriptive trend, not necessarily causal.

8. Two OLS regression models for the same data have $R^2_{\text{model 1}} = 0.9$ and $R^2_{\text{model 2}} = 0.3$ with $TSS = 100$.

(a) Which model fits the data better?

(b) What is (SSE) for model 1?

(c) What is (SSE) for model 2?

Answer: (a) model 1, (b) $SSE_1 = 10$, (c) $SSE_2 = 70$.

9. A study finds that countries with more dentists per capita also have higher rates of heart disease.

(a) Is the covariance between dentists and heart disease positive or negative?

(b) Does this mean dentists cause heart disease?

(c) What is a possible confounding variable?

Answer: (a) positive, (b) no — confounding variable, (c) wealth (richer countries have more dentists and also more heart disease risk factors).

10. A regression of ice cream sales Y (dollars) on temperature X (Celsius) gives $\hat{Y} = 100 + 20X$.

(a) What are predicted sales when (X=25) ?

(b) What does the intercept (hat{a}=100) represent?

(c) If actual sales are 700 when (X=25) , what is the residual?

Answer: (a) $\hat{Y} = 600$, (b) predicted sales at 0°C are 100 dollars, (c) $\hat{V} = 100$.

11. A pairwise correlation analysis of film characters shows: characters A and B appear together in $r = 0.9$ of scenes, characters A and C in $r = 0.1$ of scenes, and characters B and C in $r = -0.2$.

(a) Which pair of characters most often appears together?

(b) What does (r=-0.2) between B and C indicate?

(c) What type of data is used in this kind of analysis?

Answer: (a) A and B, (b) B and C tend not to appear together, (c) binary co-occurrence data.

12. A linear time trend gives $\hat{Y} = 10 + 3X$ where X is years since 2000.

(a) What is the predicted value in year 2000 (when (X=0))?

(b) What is the predicted value in year 2010 (when (X=10))?

(c) Interpret the slope (hat{b}=3) .

Answer: (a) $\hat{Y} = 10$, (b) $\hat{Y} = 40$, (c) Y increases by 3 units per year on average.

Chapter 23

Least Squares with Multiple Variables

In this chapter, we will introduce regression models with multiple explanatory variables. Suppose we are studying the effect of X on an outcome Y among people. We also want to know whether the participants' gender and age affect Y. We could certainly divide all participants and estimate separate regression models for all age groups separately for men and women. But this risks becoming confusing and leading to misleading results.

Instead, we will use a regression model where, in addition to variable X, we add the variables gender and age. This allows us to adjust our estimate of the effect that X has on Y, taking these variables into account. In many situations, we must perform this type of analysis to obtain correct estimates of the covariation we seek.

An important result in this context is that observations from all variables in a regression model can affect the results for all coefficients in the model. That is, when we adjust for age and gender, perhaps the estimated covariation between X and Y also changes.

Large parts of this review are abstract and may seem removed from reality. Later on, examples follow of how we can use this within social science. The method is completely central to an enormous amount of analytical work and research.

23.1 Regression analysis with three variables

Let us now start from a regression model where the explained (dependent) variable Y is explained by the two explanatory (independent) variables X and Z:

$$Y_i = a + bX_i + cZ_i + V_i \tag{23.1}$$

The letters a, b and c are constant coefficients and V_i is the error term for observation i. We will now estimate the coefficients $\hat{a}$, $\hat{b}$ and $\hat{c}$ and thereafter $\hat{Y}$ as well as $\hat{V}$. The coefficient $\hat{b}$ will give us the average change in Y that is associated with an increase of X by one unit, given the values in Z. Coefficient $\hat{c}$ shows the average change in Y that is associated with an increase of Z by one unit, given the values in X.

That is, by using both explanatory variables X and Z in the same regression model, the calculation will show the covariation between Y and X, taking into account variations in Z. The covariation between Y and Z will be estimated taking into account variations in X. This is a central aspect of this type of regression analysis and we return below to what this means.

Just as when we only had one explanatory variable in the regression model, we shall now, when we have two explanatory variables, find those values for the constant coefficients a, b and c that minimize the squared residuals $\sum \hat{V}^2$, compare equation (21.40) . We therefore again describe our calculation as a minimization problem:

$$\min_{\hat{a},\hat{b},\hat{c}} \sum_{i=1}^{n} \hat{V}_i^2 = \min_{\hat{a},\hat{b},\hat{c}} \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 \tag{23.2}$$

Predicted $\hat{Y}$ can be defined from our regression model as:

$$\hat{y}_i = \hat{a} + \hat{b}x_i + \hat{c}z_i \tag{23.3}$$

This definition of $\hat{Y}$ can be inserted into our minimization problem:

$$\min_{\hat{a},\hat{b},\hat{c}} \sum_{i=1}^{n} \hat{V}_i^2 = \min_{\hat{a},\hat{b},\hat{c}} \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2 = \min_{\hat{a},\hat{b},\hat{c}} \sum_{i=1}^{n} \left(Y_i - \hat{a} - \hat{b}X_i - \hat{c}Z_i\right)^2 \tag{23.4}$$

In this equation we know Y_i, X_i and Z_i, since these are our observed data. We have three factors to take into account in the form of the three constants $\hat{a}$, $\hat{b}$ and $\hat{c}$. We calculate the first-order conditions by differentiating the expression in equation (23.4) with respect to a, b and c respectively and setting each result equal to 0. Since we have three factors we get the following three results:

$$\begin{aligned}
\frac{\partial}{\partial \hat{a}}\left(\sum_{i=1}^{n} \hat{V}_i^2\right) &= \sum -2\left(Y_i - \hat{a} - \hat{b}X_i - \hat{c}Z_i\right) \\
\frac{\partial}{\partial \hat{b}}\left(\sum_{i=1}^{n} \hat{V}_i^2\right) &= \sum -2X_i\left(Y_i - \hat{a} - \hat{b}X_i - \hat{c}Z_i\right) \\
\frac{\partial}{\partial \hat{c}}\left(\sum_{i=1}^{n} \hat{V}_i^2\right) &= \sum -2Z_i\left(Y_i - \hat{a} - \hat{b}X_i - \hat{c}Z_i\right)
\end{aligned} \tag{23.5}$$

In a similar way as we did in section 21.10 we set the first condition equal to 0 and solve for a definition of $\hat{a}$:

$$\begin{aligned}
0 &= -2\sum y + \sum \hat{a} + \sum \hat{b}x_i + \sum \hat{c}z_i \\
n\hat{a} &= ny - \hat{b}nx_i - \hat{c}nz_i \\
\hat{a} &= \bar{y} - \hat{b}\bar{x} - \hat{c}\bar{z}
\end{aligned} \tag{23.6}$$

Compare the definition for the y-intercept when we only have two variables in equation (21.12) . Coefficient $\hat{a}$ is a function of the observations in all three variables included in the regression model: Y, X and Z. From here we continue to solve for the slope coefficients $\hat{b}$ and $\hat{c}$. Here, however, we only look at the final results.

The equations below describe the estimators for the two slope coefficients $\hat{b}$ and $\hat{c}$. To compress the algebra we write $\tilde{X}_i = X_i - \bar{X}$ and correspondingly for $\tilde{Y}_i$ and $\tilde{Z}_i$. This means that the coefficients' estimators are described based on the observations' deviations from their respective means:

$$\begin{aligned}
\hat{b} &= \frac{\left(\sum \tilde{y}_i\tilde{x}_i\right)\left(\sum \tilde{z}_i^2\right) - \left(\sum \tilde{y}_i\tilde{z}_i\right)\left(\sum \tilde{x}_i\tilde{z}_i\right)}{\left(\sum \tilde{x}_i^2\right)\left(\sum \tilde{z}_i^2\right) - \left(\sum \tilde{x}_i\tilde{z}_i\right)^2} \\
\hat{c} &= \frac{\left(\sum \tilde{y}_i\tilde{z}_i\right)\left(\sum \tilde{x}_i^2\right) - \left(\sum \tilde{y}_i\tilde{x}_i\right)\left(\sum \tilde{x}_i\tilde{z}_i\right)}{\left(\sum \tilde{x}_i^2\right)\left(\sum \tilde{z}_i^2\right) - \left(\sum \tilde{x}_i\tilde{z}_i\right)^2}
\end{aligned} \tag{23.7}$$

All three coefficients $\hat{a}$, $\hat{b}$ and $\hat{c}$ depend on the observed values in the three variables in the regression model: Y, X and Z. We see this because values for all three variables are included in each respective equation. This means that even though $\hat{b}$ measures the covariation between Y and X, $\hat{b}$ is a function of Y, X and Z. And despite the fact that the slope coefficient $\hat{c}$ measures the covariation between the variables Y and Z, $\hat{c}$ is also a function of observations in all three variables Y, X and Z.

Had we had more variables and coefficients in our regression model, the equations would have become even more extensive. At the end of this chapter we will go through another way to derive equations for the coefficients in a regression model regardless of the number of variables and coefficients.

If we add a variable to our analysis, this can affect the results for all coefficients included in the model. Say that the variable Z should be included in the model but that this for some reason is not included in the analysis. In that case the result for variable X will become misleading. We do not get the correct result until we include Z. This is central to understanding this type of method and analytical work in general.

Table 23.1: Variables Y, X and Z

i	y_i	x_i	z_i	$\tilde{y}_i$	$\tilde{x}_i$	$\tilde{z}_i$	$\tilde{y}_i\tilde{x}_i$	$\tilde{y}_i\tilde{z}_i$	$\tilde{x}_i\tilde{z}_i$	$\tilde{x}_i^{\,2}$	$\tilde{z}_i^{\,2}$
1	3	3	1	–0.5	–2	–0.5	1	0.25	1	4	0.25
2	2	4	4	–1.5	-1	2.5	1.5	–3.75	–2.5	1	6.25
3	5	6	0	1.5	1	–1.5	1.5	–2.25	–1.5	1	2.25
4	4	7	1	0.5	2	0.5	1	–0.25	–1	4	0.25
Mean	3.5	5	1.5								
Sum							5	−6	−4	10	9

Note: $\tilde{y}_i = y_i - \bar{y}$*, as well as for* $\tilde{x}_i$ *and* $\tilde{z}_i$

Now we will estimate a regression model based on some observations. For this we reuse the fictitious variables Y, X and Z with four observations each that we used in chapter 21 when we introduced the least squares method. All three variables are reported in table 23.1 with some calculations that we need. In section 21.2 we used the variables Y and X to estimate the regression model $Y_i = a + bX_i + V_i$ and found then that $\hat{a} = 1$ and $\hat{b} = 0.5$. Now we will estimate the coefficients for the following regression model:

$$y_i = a + bx_i + cz_i + v_i \tag{23.8}$$

where Y, X and Z are the variables, a, b and c are the coefficients we will estimate and V is the error term. In equation (23.7) we have the definitions for how we estimate $\hat{b}$ and $\hat{c}$. In table 23.1 we have some of the results we need:

$$\begin{aligned} \hat{b} &= \frac{(5)(9) - (-6)(-4)}{(10)(9) - (-4)^2} \approx 0.28 \\ \hat{c} &= \frac{(-6)(10) - (5)(-4)}{(10)(9) - (-4)^2} \approx -0.54 \end{aligned} \tag{23.9}$$

Table 23.2: Calculations of $\hat{Y}$ and $\hat{V}$

y_i	x_i	z_i	$\hat{y}_i$	$\hat{v}_i$
3	3	1	3.2	−0.2
2	4	4	1.86	0.14
5	6	0	4.59	0.41
4	7	1	4.34	−0.34

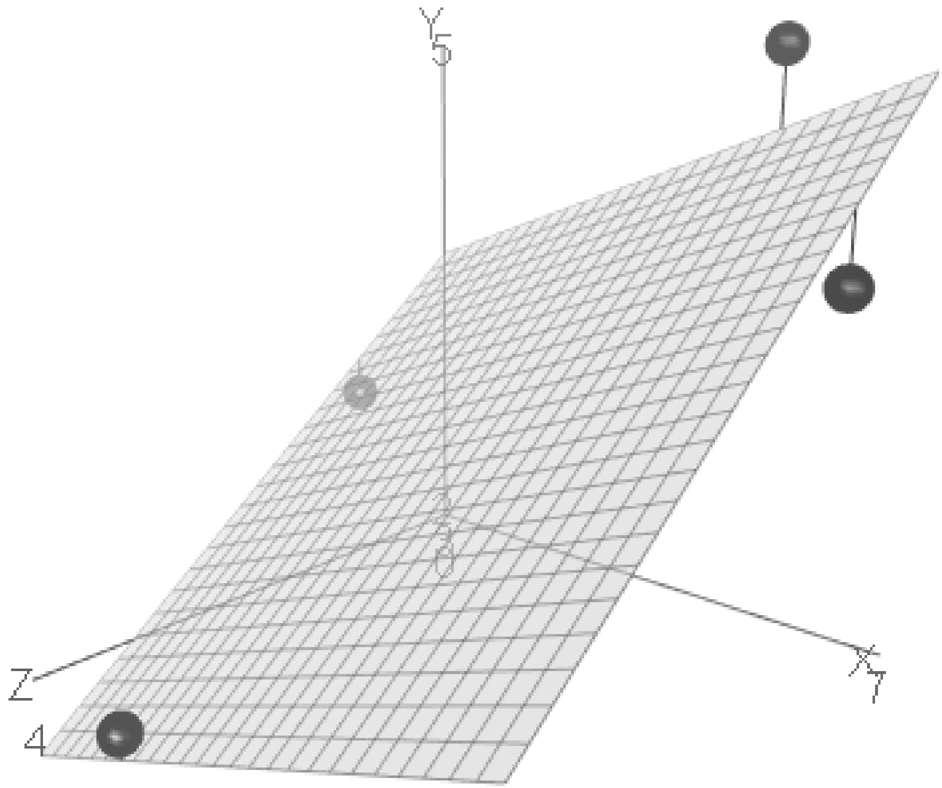


Figure 23.1: Results from the regression model $y_i = a + bx_i + cz_i + v_i$

When we estimated the regression model $Y = a + bX + V$ we found that $\hat{b} = 0.5$. When we now added the variable Z to the regression model we see how the result for the slope coefficient $\hat{b}$ goes from 0.5 to 0.3. We use the results for $\hat{b}$ and $\hat{c}$ to estimate $\hat{a}$:

$$\begin{aligned} \hat{a} &= 3.5 - \hat{b} * 5 - \hat{c} * 1.5 \\ &= 3.5 - 0.28 * 5 - (-0.54) * 1.5 \\ &\approx 2.89 \end{aligned} \tag{23.10}$$

We summarize our estimated coefficients by inserting the results into our regression model:

$$y_i = \hat{a} + \hat{b}x_i + \hat{c}z_i + v_i = 2.89 + 0.28x_i - 0.54z_i + v_i \tag{23.11}$$

Now we may also estimate predicted $\hat{Y}_i$ and the residual $\hat{V}_i$, which is summarized in table 23.2 with rounded results. When we have three variables it is more difficult to illustrate covariation in a graph. Despite this, an attempt is made in figure 23.1 where the four observations are placed in the graph based on their values for Y, X and Z. The vertical axis is the Y-axis while the variables X and Z each have their own horizontal axis. The black dot at the top of the graph is observation 3 whose values are $(Y, X, Z) = (5, 6, 0)$.

Since the regression model has three variables, the regression line $\left(\hat{Y}\right)$ now becomes a plane surface with three dimensions, which is illustrated by the grid. This plane surface is angled with regard to the two variables X and Z, depending on their respective slope coefficient. Since the slope coefficient $\hat{c} < 0$ the grid slopes downward along the Z-axis seen from the Y-axis. Since $\hat{b} > 0$ the grid slopes upward along the X-axis seen from the Y-axis.

23.2 Factor variables

Regression analysis is a quantitative method in the sense that we use quantitative data to create quantitative results. We can also use qualitative data in our analysis, information that is not originally expressed in numbers, as long as we assign numbers to this. One such example is factor variables, also called categorical variables. Factor variables typically contain nominal data, see section 15.3 .

Let us start with factor variables that only assume two different values, for example if we send out a survey and ask recipients to answer "Yes" or "No". A common method is then to replace one response alternative with the number 0 and the other with the number 1. Another example is a variable that indicates whether a row in a table contains information about a man or a woman, whereupon the variable can assume the value 0 for one gender and 1 for the other gender. It does not matter which variable value gets which number, as long as we keep track of what represents what when we calculate.

An explanatory variable in a regression model that only assumes the values 0 or 1 is called a dummy variable or indicator variable. Data that can only assume two values is called binary. Often 0 or 1 is used precisely to describe the information in a binary variable, even though 0 and 1 symbolize something else (compare section 22.6).

Table 23.3: Average earnings for men and women

Country	Men	Women
France	19.9	17.6
United Kingdom	18.2	15.3
Sweden	24	21.6

Note: Data from ILOSTAT. Earnings in 2021 US dollars (USD), adjusted for inflation and local prices (PPP = Purchasing Power Parity).

Table 23.3 shows average hourly earnings for men and women respectively in three countries 2020, calculated in US dollars, adjusted for local prices (Purchasing Power Parity). Now we will study the differences in average earnings between male and females. We begin by formulating a regression model:

$$W_i = a + bG_i + e_i \tag{23.12}$$

Table 23.4: Data for W and G

Country	W_i	G_i	$\tilde{W}$	$\tilde{G}$	$\tilde{G}^2$	$\tilde{W}\tilde{G}$
France	19.9	0	0.452	-0.5	0.25	-0.226
United Kingdom	18.2	0	-1.19	-0.5	0.25	0.597
Sweden	24	0	4.54	-0.5	0.25	-2.27
France	17.6	1	-1.85	0.5	0.25	-0.924
United Kingdom	15.3	1	-4.14	0.5	0.25	-2.07
Sweden	21.6	1	2.2	0.5	0.25	1.1
Mean	19.4	0.5				
Sum					1.5	-3.79

Note: Data from ILOSTAT. Earnings in 2021 PPP USD. $\tilde{W} = W_i - \bar{W}_i$, $\tilde{G} = G_i - \bar{G}_i$.

where W_i is average earnings in country i and G_i is gender which has the value 0 for men and 1 for women. We then estimate the regression model in the same way as before, where the coefficients are estimated by the least squares method. Table 23.3 gives us parts of the calculation:

$$\hat{b} = \frac{\sum \left(G_i - \bar{G}\right)\left(W_i - \bar{W}\right)}{\sum \left(G_i - \bar{G}\right)^2} \approx \frac{-3.7945}{1.5} \approx -2.53 \tag{23.13}$$
$$\hat{a} = \bar{W} - \hat{b}\bar{G} \approx 19.445 - (-2.53)\,0.5 \approx 20.71$$

The coefficient $\hat{b}$ is negative, which means that the variable G has a negative covariation with W. Our dummy variable is defined $G = 1$ for women. Variable W is associated with an 2.53 units lower value for women compared to men, which means that women on average in 2020 had 2.53 lower earnings per hour than men in these countries. The result is illustrated in figure 23.2 .

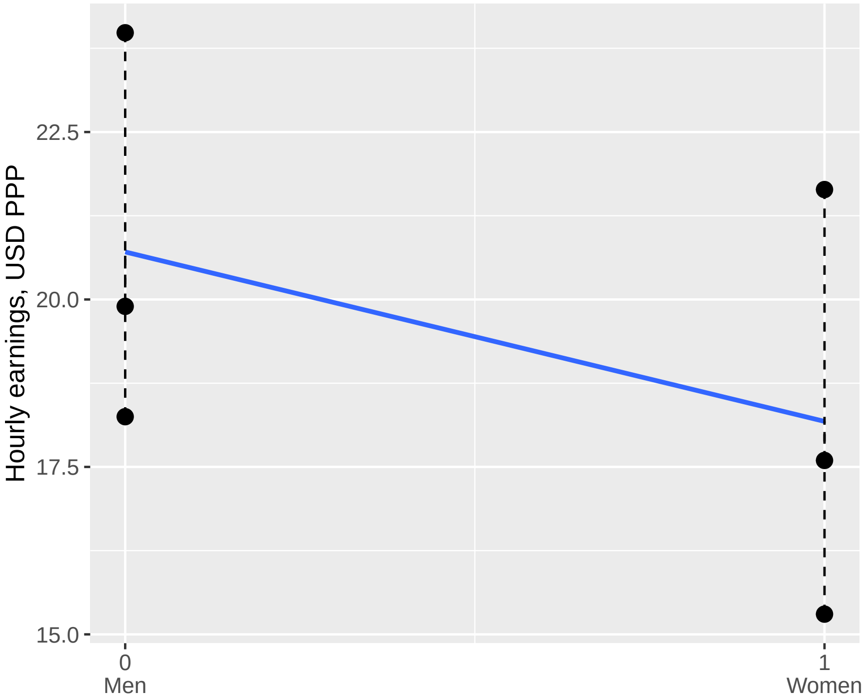


Figure 23.2: Average earnings for men and women in three countries

Dummy variables can also be useful for categorizing factor variables with more than two values. Let us again compare differences in earnings between the three countries in the previous example, but instead of the difference between women and men we will now calculate the difference in average earnings between the countries. Table 23.5 presents the variables, where Y_i now indicates average earnings for all residents in the countries. The variables K_1 and K_2 are two dummy variables for the countries in the following regression model:

$$Y_i = a + bK_{\text{United Kingdom}} + cK_{\text{Sweden}} + e_i \tag{23.14}$$

Table 23.5: Three countries and two dummy variables

Country	Y_i	$K_{\text{United Kingdom}}$	K_{Sweden}
France	18.8	0	0
United Kingdom	16.8	1	0
Sweden	22.8	0	1

Note: Average earnings 2020. Data from ILOSTAT. Earnings in 2021 PPP USD.

where e is the error term. Our dummy variables represent the countries United Kingdom and Sweden, one dummy variable fewer than the number of countries. When both dummy variables in the model equal 0 we get the estimates for the third country, France. Regression analysis with dummy variables is sometimes called fixed effects. Our regression model calculates the same thing as we already see in the data material. The purpose of this exercise is to understand the meaning of dummy variables in a regression model. We estimate the regression model in equation (23.14) with the least squares method:

$$\begin{aligned}\hat{Y} &= \hat{a} + \hat{b}K_{\text{United Kingdom}} + \hat{c}K_{\text{Sweden}} \\ &= 18.763 - 1.97K_{\text{United Kingdom}} + 4.086K_{\text{Sweden}}\end{aligned} \tag{23.15}$$

The result $\hat{a} = 18.763$ is approximately 18.8, which match our data for France. This is the result for earnings $\hat{Y}$ if we set $K_{\text{United Kingdom}} = K_{\text{Sweden}} = 0$.

We get $\hat{b} = -1.97$, which means that the average earnings in United Kingdom equals $18.763 - 1.97 = 16.793 \approx 16.8$, same as the table. Coefficient $\hat{c} = 4.086$ means that the average earnings in Sweden equals $18.763 + 4.086 = 22.849 \approx 22.8$.

When you have a regression model where a categorical variable is included, for example region, you can often see formulations of regression models with abbreviations for dummy variables. Say for example that we would like to estimate the regression model in equation (23.14) with data on 150 countries. In that case we would want to use 149 dummy variables (number of countries minus 1). The last country's result is given by setting all dummy variables equal to zero.

23.3 Ceteris paribus

As we went through in the introduction to this chapter, regression analysis with multiple explanatory variables is often useful for getting a correct picture of the covariation between multiple phenomena. By controlling for other relevant variables in our model, the slope coefficient for another variable in the model can change.

This is what we saw in section 23.1 , where the coefficients a, b, c in a regression model of the type $Y_i = a + bX_i + cZ_i + V_i$ do not only depend on how each individual variable covaries with Y, but also on the relationships between all variables included in the regression model. This is called estimating the covariation for X and Y while holding constant the other variables (which is included in the model). In our particular model we estimate the covariation between Y and X, while holding Z constant. And we estimate the covariation between Y and Z, while holding X constant.

Table 23.6: Data on earnings and life expectancy per country and gender

Country and gender	L	I	G
France, Women	85.137	17.597	1
France, Men	79.179	19.897	0
Sweden, Women	84.284	21.641	1
Sweden, Men	80.597	23.982	0
United Kingdom, Women	82.400	15.302	1
United Kingdom, Men	78.362	18.250	0
Mean	81.6598	19.4448	0.5

Note: Data from ILOSTAT (earnings, I), Our World in Data (life expectancy, L). Earnings in 2021 PPP USD.

We illustrate this with the help of data on average life expectancy and earnings for women and men respectively in three countries, described in table 23.6 . The column furthest to the right describes a dummy variable for gender where women = 0 and men = 1. Let us begin by comparing the relationship between life expectancy and earnings with the following regression model:

$$L = a_1 + a_2 I + \epsilon \tag{23.16}$$

where L is life expectancy, I is earnings, a_1 and a_2 are the coefficients we will estimate with the least squares method and ϵ is the error term. The slope coefficient a_2 we estimate to:

$$\hat{a_2} = \frac{\sum_i \left(I_i - \bar{I}\right)\left(L_i - \bar{L}\right)}{\sum_i \left(I_i - \bar{I}\right)^2} \approx \frac{-5.7321}{47.6187} \approx -0.1204 \tag{23.17}$$

Coefficient a_1 we estimate to:

$$\hat{a_1} = \bar{L} - \hat{b}\bar{I} \approx 81.6598 - (-0.1204)\,19.4448 \approx 84 \tag{23.18}$$

The calculations are not described here.

According to our calculations there seems to be a negative relationship between life expectancy L and earnings I, which means that people with higher earnings on average live shorter lives. Let us now add the variable gender G to our regression model. In table 23.6 this is a dummy variable that only takes the value 0 or 1. We will now estimate the following regression model:

$$L = b_1 + b_2 I + b_3 G + V \tag{23.19}$$

where L, I and G are our variables, b_1, b_2 and b_3 are the coefficients we will estimate using least squares, and V is the error term. To estimate the slope coefficients we now use the definitions in equation (23.7) . We begin with the slope coefficient b_2:

$$\begin{aligned}\hat{b}_2 &= \frac{\left(\sum \tilde{L}_i\tilde{I}_i\right)\left(\sum \tilde{G}_i^2\right) - \left(\sum \tilde{L}_i\tilde{G}_i\right)\left(\sum \tilde{I}_i\tilde{G}_i\right)}{\left(\sum \tilde{I}_i^2\right)\left(\sum \tilde{G}_i^2\right) - \left(\sum \tilde{I}_i\tilde{G}_i\right)^2} \\ &\approx \frac{(-5.73)\,(1.5) - (6.84)\,(-3.79)}{(-3.79)\,(1.5) - (14.4)^2} \\ &\approx 0.3044\end{aligned} \tag{23.20}$$

Tilde over a letter describes deviation from the mean, for example: $\tilde{G} = G_i - \bar{G}$. The slope coefficient b_3:

$$\begin{aligned}\hat{b}_3 &= \frac{\left(\sum \tilde{L}_i\tilde{G}_i\right)\left(\sum \tilde{I}_i^2\right) - \left(\sum \tilde{L}_i\tilde{I}_i\right)\left(\sum \tilde{I}_i\tilde{G}_i\right)}{\left(\sum \tilde{I}_i^2\right)\left(\sum \tilde{G}_i^2\right) - \left(\sum \tilde{I}_i\tilde{G}_i\right)^2} \\ &\approx \frac{(6.84)\,(47.6) - (-5.73)\,(-3.79)}{(47.6)\,(1.5) - 14.4} \\ &\approx 5.3311\end{aligned} \tag{23.21}$$

Finally we estimate the coefficient b_1:

$$\begin{aligned}\hat{b_1} &= \bar{L} - \hat{b_2}\bar{I} - \hat{b_3}\bar{G} \\ &\approx 81.66 - 0.3 * 19.44 - 5.33 * 0.5 \\ &\approx 73.0746\end{aligned} \tag{23.22}$$

Let's put our estimates into the regression model in equation (23.19) :

$$L = \hat{b}_1 + \hat{b}_2 I + \hat{b}_3 G = 73.0746 + 0.3044I + 5.3311G \tag{23.23}$$

The result now indicates a positive relationship between our explained variable life expectancy (L) and the explanatory variable earnings (I) taking into account gender (G). People with higher earnings live on average longer lives. When we estimated the regression model in equation (23.16) we found that earnings and life expectancy related negatively.

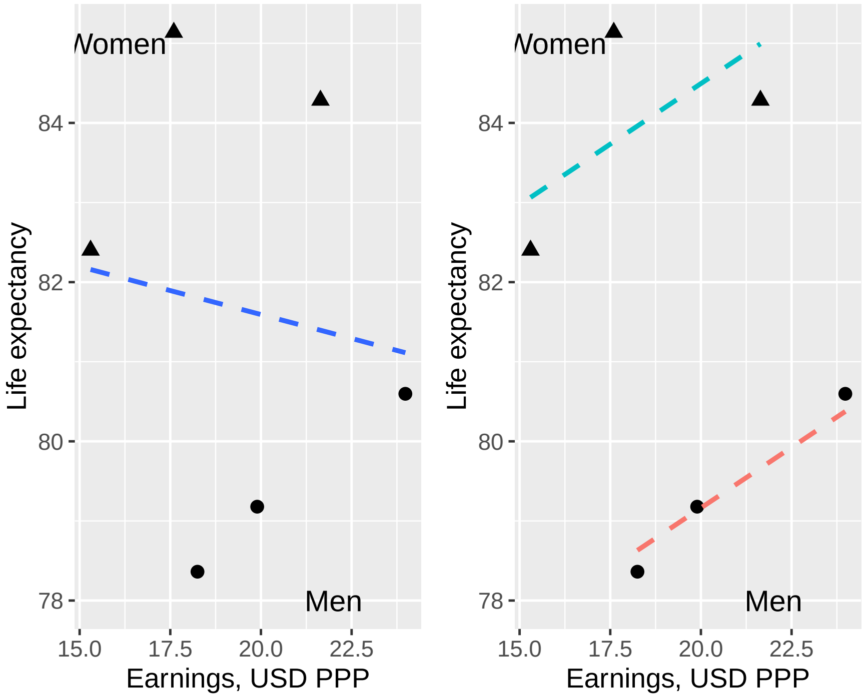


Figure 23.3: Life expectancy, earnings and gender

Figure 23.3 illustrates the relationship between life expectancy and earnings in two diagrams, one each for the regression models in equation (23.16) and (23.19) . In both graphs we have marked which observations belong to men and women respectively. We have three dots for men and three for women, since we have three countries with two observations per country. The left graph illustrates the result in the first regression model in equation (23.19) . Above we estimated a negative relationship between life expectancy and earnings for this model. In the graph the regression line is illustrated with a dashed line.

The right graph illustrates the result in the second regression model in equation (23.19) . In the graph we have drawn two regression lines to illustrate the significance of this dummy variable. The two regression lines are estimated with the same regression model, but illustrate the difference between $G = 0$ and $G = 1$. If $G = 0$ we get the upper regression line, which runs through the dots that illustrate the observations for women. If the dummy variable $G = 1$ we instead have the lower regression line, which runs through the three dots for men.

23.4 Least Squares with Matrices

In section 21.10 we derived the coefficient estimators for the least squares method for a regression model with two variables. In this section we will derive a more general expression for the estimators based on a regression model with any number of coefficients and variables. For this we will use matrices. We begin with the assumptions:

- Suppose we have a regression model with many coefficients, variables and observations. The model can be summarized as $y_i = b_1 + b_2x_1 + b_3x_2 + ... + b_kx_{k-1} + v_i$ where y_i is the explained variable for observation i. b_1 to b_k are the coefficients we will estimate using the least squares method. x_1 to x_{k-1} are the explanatory variables. v_i is the error term.

- The first coefficient b_1 is the constant term and is not multiplied by any explanatory variable (similar to a in the model $y = a + bx$). In matrix notation, this constant term is represented by multiplying b_1 with a column of ones. Therefore, the first column in X consists of the number 1 in each cell, followed by one column for each explanatory variable x_1 to x_{k-1}. We collect all coefficients b in a column matrix B, with as many rows as coefficients. Since we have k coefficients, B has the dimensions $k \times 1$.
- We collect our explanatory variables x_1 to x_{k-1} in matrix X. Each column in matrix X contains the values for one explanatory variable. The first column in X consists of the number 1 in each cell, which we will multiply with b_1. The next column in X contains observed values for the first explanatory variable, x_1, followed by one column per explanatory variable, x_2 to x_{k-1}. The number of rows in X is the same as the number of observations, n. Matrix X has the dimensions $n \times k$.
- We collect the error term, $v_1, v_2, ..., v_n$, in matrix V. We have as many error terms as observations, n.

Our regression model can now be written with matrices:

$$\begin{gathered}\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1} = \begin{bmatrix} 1 & x_{11} & x_{21} & ... & x_{k1} \\ 1 & x_{12} & x_{22} & ... & x_{k2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & x_{2n} & ... & x_{k-1,n} \end{bmatrix}_{n\times k} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \end{bmatrix}_{k\times 1} + \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}_{n\times 1} \\ Y = XB + V \end{gathered} \tag{23.24}$$

We get a column matrix Y, a matrix X with a column with ones and the variables, a column matrix B with the coefficients and a column matrix V with the error terms.

Based on our regression model we now formulate our minimization problem using matrix notation. We seek the estimated coefficients $\hat{B}$ that minimize the sum of squared residuals. We begin by inserting $\hat{B}$ and $\hat{V}$ into our equation and rearrange a bit so that we get an equation where the residuals $\hat{V}$ are a result of the other terms:

$$\hat{V} = Y - X\hat{B} \tag{23.25}$$

$\hat{V}$ is a column matrix and we cannot get squared residuals by multiplying $\hat{V}\hat{V}$, since this is not defined for matrices (the number of columns in the left matrix must equal the number of rows in the right matrix). To get the sum of squared residuals in matrix $\hat{V}$ we instead multiply the transposed $\hat{V}^T$ with $\hat{V}$(see section 14.6):

$$\hat{V}^T\hat{V} = [v_1 + v_2 + ... + v_n] \times \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = [v_1^2 + v_2^2 + ... + v_n^2] \tag{23.26}$$

This equation can be rewritten by using the definition of $\hat{V}$ from equation (23.25) :

$$\begin{aligned} \hat{V}^T\hat{V} &= \left(Y - X\hat{B}\right)^T \left(Y - X\hat{B}\right) \\ &= Y^TY - Y^TX\hat{B} - \hat{B}^TX^TY + \hat{B}^TX^TX\hat{B} \end{aligned} \tag{23.27}$$

Since $Y^T\hat{B}X = \left(\hat{B}^TX^TY\right)^T$ we write:

$$\hat{V}^T\hat{V} = Y^TY - 2\hat{B}^TX^TY + \hat{B}^TX^TX\hat{B} \tag{23.28}$$

Now we have an expression for the squared residuals in $\hat{V}^T\hat{V}$ and it is this equation we will minimize with respect to the coefficients in matrix $\hat{B}$: $\min_{\hat{B}}\left(\hat{V}^T\hat{V}\right)$. We differentiate equation (23.28) with respect to $\hat{B}$ and set the result equal to 0, which gives us the first-order conditions:

$$\frac{\partial \hat{V}^T\hat{V}}{\partial \hat{B}} = -2X^TY + 2X^TX\hat{B} = 0 \tag{23.29}$$

From this we solve for $\hat{B}$. The first step is to move the negative term $-2X^TY$ to the right-hand side:

$$\begin{aligned} -2X^TY + 2X^TX\hat{B} &= 0 \\ \left(X^TX\right)\hat{B} &= X^TY \end{aligned} \tag{23.30}$$

We know that the matrix $\left(X^TX\right)$ is square and symmetric with dimensions $k \times k$, where k is the number of explanatory variables plus the constant. Given that the inverse of this matrix exists, $\left(X^TX\right)^{-1}$, we multiply both sides by this and thereby get a definition of $\hat{B}$. The multiplication of the inverted matrix $\left(X^TX\right)^{-1}$ and $\left(X^TX\right)$ equals the identity matrix I, a matrix with the value 1 on the diagonal. Since $I\hat{B} = \hat{B}$ we get:

$$\begin{aligned} \left(X^TX\right)^{-1}\left(X^TX\right)\hat{B} &= \left(X^TX\right)^{-1}X^TY \\ I\hat{B} &= \left(X^TX\right)^{-1}X^TY \\ \hat{B} &= \left(X^TX\right)^{-1}X^TY \end{aligned} \tag{23.31}$$

We now have an expression for the estimated coefficients $\hat{B}$ as a function of the explanatory variables in X and the explained variable in Y. The matrices Y and X include all observations for each respective variable. This means that all coefficients in $\hat{B}$ are a result of all variables in Y and X, regardless of the number of variables. All variables in the regression model can therefore affect the results for all coefficients in the regression model.

23.5 Estimation with matrices

Let us now reuse the first regression model we went through in chapter 21 , $Y = a+bX+V$, and estimate this with matrices and the observations in table 23.7 . We begin by describing the regression model with matrices:

$$\begin{aligned} Y =& XB + V \\ \begin{bmatrix}3\\4\\6\\7\end{bmatrix} =& \begin{bmatrix}1 & 3\\1 & 2\\1 & 5\\1 & 4\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} + \begin{bmatrix}v_1\\v_2\\v_3\\v_4\end{bmatrix} \end{aligned} \tag{23.32}$$

Table 23.7: The variables x and y

	y_i	x_i
1	3	3
2	4	2
3	6	5
4	7	4

Our estimator is:

$$\hat{B} = \left(X^T X\right)^{-1} X^T Y \tag{23.33}$$

If we calculate these matrices we get the estimated coefficients:

$$\begin{aligned}
\hat{B} &= \left(X^T X\right)^{-1} X^T Y \\
\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} &= \left(\begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 6 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 4 \\ 1 & 6 \\ 1 & 7 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 3 & 4 & 6 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 5 \\ 4 \end{bmatrix} \\
\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} &= \begin{bmatrix} 4 & 20 \\ 20 & 110 \end{bmatrix}^{-1} \begin{bmatrix} 14 \\ 75 \end{bmatrix}
\end{aligned} \tag{23.34}$$

We calculate the inverse matrix, in the way we described in section 14.9 :

$$\begin{aligned}
\begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} &= \begin{bmatrix} 4 & 20 \\ 20 & 110 \end{bmatrix}^{-1} \begin{bmatrix} 14 \\ 75 \end{bmatrix} \\
&= \left(\frac{1}{4 * 110 - (-20) * (-20)} \right) \begin{bmatrix} 110 & -20 \\ -20 & 4 \end{bmatrix} \begin{bmatrix} 14 \\ 75 \end{bmatrix} \\
&= \left(\frac{1}{440 - 400} \right) \begin{bmatrix} 110 & -20 \\ -20 & 4 \end{bmatrix} \begin{bmatrix} 14 \\ 75 \end{bmatrix} \\
&= \begin{bmatrix} \frac{110}{40} & \frac{-20}{40} \\ \frac{-20}{40} & \frac{4}{40} \end{bmatrix} \begin{bmatrix} 14 \\ 75 \end{bmatrix} \\
&= \begin{bmatrix} \frac{11}{4} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{10} \end{bmatrix} \begin{bmatrix} 14 \\ 75 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}
\end{aligned} \tag{23.35}$$

Table 23.8: Variables y, x and z

	y_i	x_i	z_i
1	3	3	1
2	2	4	4
3	5	6	0
4	4	7	1

Our result shows that $\hat{a} = 1$ and $\hat{b} = 0.5$, which is the same thing we arrived at earlier. Let us also do the same thing for the regression model from the introduction of this chapter:

$$y_i = a + bx_i + cz_i + v_i \tag{23.36}$$

which can be written with matrices as $Y = XB + V$, where matrix X contains the variables x and z and matrix B contains the coefficients a, b and c. V is the error terms and Y is the explained variable. The observations we will use are repeated in table 23.8 . To estimate the coefficients we now take:

$$\hat{B} = \left(X^T X\right)^{-1} X^T Y \tag{23.37}$$

$$\begin{bmatrix}\hat{a}\\ \hat{b}\\ \hat{c}\end{bmatrix} = \left(\begin{bmatrix}1 & 1 & 1 & 1\\ 3 & 4 & 6 & 7\\ 1 & 4 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 3 & 1\\ 1 & 4 & 4\\ 1 & 6 & 0\\ 1 & 7 & 1\end{bmatrix}\right)^{-1}\begin{bmatrix}1 & 1 & 1 & 1\\ 3 & 4 & 6 & 7\\ 1 & 4 & 0 & 1\end{bmatrix}\begin{bmatrix}3\\ 2\\ 5\\ 4\end{bmatrix}$$

$$= \begin{bmatrix}4 & 20 & 6\\ 20 & 110 & 26\\ 6 & 26 & 18\end{bmatrix}^{-1}\begin{bmatrix}14\\ 75\\ 15\end{bmatrix}$$

$$\approx \begin{bmatrix}2.9\\ 0.3\\ -0.5\end{bmatrix}$$

Table 23.9: Variables Z and K

	Z_i	K_i
1	1	0
2	4	0
3	0	4
4	1	4

The last row contains the rounded results $\hat{a} = 2.9$, $\hat{b} = 0.3$ and $\hat{c} = -0.5$. This is the same thing we arrived at in section 23.1 . In section 21.8 we used four observations for the variables Z and K, which are repeated in table 23.9 . With these observations we estimated the regression model $Z = \alpha + \beta K + \epsilon$, where ϵ is the error term and α and β are the model's coefficients, which we found were $\hat{\alpha} = 2.5$ and $\hat{\beta} = -0.5$.

With matrices we write this regression model as $Z = KB + E$, where Z is the explained variable, B is a matrix with the coefficients α and β, K is a column matrix with the explanatory variable and E is the error terms. The estimator can in this case be written:

$$\hat{B} = \left(K^T K\right)^{-1} K^T Z \tag{23.38}$$

This gives us the following estimates:

$$\begin{bmatrix}\hat{\alpha}\\ \hat{\beta}\end{bmatrix} = \left(\begin{bmatrix}1 & 1 & 1 & 1\\ 0 & 0 & 4 & 4\end{bmatrix}\begin{bmatrix}1 & 0\\ 1 & 0\\ 1 & 4\\ 1 & 4\end{bmatrix}\right)^{-1}\begin{bmatrix}1 & 1 & 1 & 1\\ 0 & 0 & 4 & 4\end{bmatrix}\begin{bmatrix}1\\ 4\\ 0\\ 1\end{bmatrix} = \begin{bmatrix}2.5\\ -0.5\end{bmatrix} \tag{23.39}$$

which is the same result as in section 21.8 .

23.6 Interaction

Interaction effect means that we combine two or more of the explanatory variables into a new variable. Thereby we check whether an explanatory variable's relationship with the explained variable varies depending on the values in another explanatory variable. Consider the following regression model as an example:

$$y = a + bx + cz + \epsilon \tag{23.40}$$

where ϵ is the error term, a, b and c are coefficients, x and z are explanatory variables and y is the explained variable. Now we have reason to believe that the relationship between y and x may also depend on what value the variable z has. We also believe that z relates to y in its own way, which in turn may also vary with the values in variable x.

Say for example that we believe the relationship between y and x is slightly positive for low values of z, but that the slope is different for higher values of z. This interaction effect we estimate by creating a new variable $k = x * z$, where we for each observation multiply the value in x with the value in z. We add the new variable k to our regression model with its own slope coefficient:

$$\begin{aligned} y_i &= a + bx + cz + dxz + \epsilon \\ &= a + bx + cz + dk + \epsilon \end{aligned} \tag{23.41}$$

Coefficient a, the model's intercept, is just as before the value that y takes when all other variables equal zero ($x = z = k = 0$). Coefficient d which is multiplied by $k = xz$ measures the average change in y that is associated with an increase of variable x by one unit and an increase of variable z by one unit. Coefficient b indicates how large a change in y is associated with one unit higher x. To estimate the total relationship between x and y we also need to add $d * z$, since variable z is included as a multiplier in variable k. The total relationship between x and y can therefore be summarized as:

$$\frac{\Delta y}{\Delta x} = b + d * z \tag{23.42}$$

where Δy means change in variable y and Δx is change in variable x. The letters b, d and z are the same as in the regression model in equation (23.41) . For those values when $z = 0$, b measures the change in y that coincides with an increase of x by one unit. In a corresponding way we get the total relationship between variable z and y through the following expression:

$$\frac{\Delta y}{\Delta z} = c + d * x \tag{23.43}$$

For the values $x = 0$, the slope coefficient c indicates how much y increases when z increases by 1. Let us illustrate the interaction effect with the help of data. In section 23.3 we estimated the relationship between life expectancy and earnings for men and women in three countries. Now we will use the same data again but this time add an interaction effect.

Table 23.10: Data on earnings and life expectancy per country and gender

Country and gender	L	I	G	$I * G$
France, Women	85.137	17.597	1	17.597
France, Men	79.179	19.897	0	0
Sweden, Women	84.284	21.641	1	21.641
Sweden, Men	80.597	23.982	0	0
United Kingdom, Women	82.400	15.302	1	15.302
United Kingdom, Men	78.362	18.250	0	0
Mean	81.6598	19.4448	0.5	

Note: Data from ILOSTAT (earnings, I), Our World in Data (life expectancy, L). Earnings in 2021 PPP USD.

Table 23.10 describes the same data that we used in section 23.3 but with the addition of a new variable that is created by multiplying the two variables earnings I and gender G. The new variable has the heading "Interaction, $I * G$". Since G is a dummy where three of six observations have the value 0, the new variable also gets the value 0 in these cases. In section 23.3 we estimated two regression models. First a model for the relationship between life expectancy and earnings, with the following result:

$$L = \hat{a_1} + \hat{a}_2 I = 92 - 4.2 * I \tag{23.44}$$

The second model also included the dummy variable gender G:

$$L = \hat{b}_1 + \hat{b}_2 I + \hat{b}_3 G = 69.6 + 6.64I - 6.66G \tag{23.45}$$

Now we will estimate the following regression model:

$$L = c_1 + c_2 I + c_3 G + c_4 GI + v \tag{23.46}$$

where c_1, c_2, c_3 and c_4 are coefficients we will estimate with the least squares method and v is the error term. We recognize the variables in the model from table 23.10 : life expectancy L, earnings I, gender G and the interaction term GI. With matrices we may describe our regression model in the following way:

$$L = CX + V \tag{23.47}$$

L is a column matrix with the values for the explained variable life expectancy L. C is a column matrix with all coefficients in the model, c_1 to c_4. X is a matrix with all explanatory variables in the model and the observations for each variable in its own column. V is a column matrix with the error terms $v_1, v_2, ..., v_n$ where n is the number of observations. To estimate the coefficients we have the estimator, with corresponding setup as in equation (23.31) but with other letters:

$$\hat{C} = \left(X^T X\right)^{-1} X^T L \tag{23.48}$$

Matrix X is in this case:

$$X = \begin{bmatrix} 1 & 17.597 & 1 & 17.597 \\ 1 & 19.897 & 0 & 0 \\ 1 & 21.641 & 1 & 21.641 \\ 1 & 23.982 & 0 & 0 \\ 1 & 15.302 & 1 & 15.302 \\ 1 & 18.25 & 0 & 0 \end{bmatrix} \tag{23.49}$$

All elements in the first column in matrix X have the value 1 since the first coefficient in the regression model is not multiplied by any variable. The other values in the matrix are taken from the observations for the three explanatory variables in table 23.10 . The transposed version of X is written X^T. The values in X^T:

$$X' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 17.597 & 19.897 & 21.641 & 23.982 & 15.302 & 18.25 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 17.597 & 0 & 21.641 & 0 & 15.302 & 0 \end{bmatrix} \tag{23.50}$$

The values in matrix L are also taken from table 23.10 :

$$L = \begin{bmatrix} 85.137 \\ 79.179 \\ 84.284 \\ 80.597 \\ 82.4 \\ 78.362 \end{bmatrix} \tag{23.51}$$

As in previous examples we estimate the regression model's coefficients with the equation $\hat{C} = \left(X^T X\right)^{-1} X^T L$. Our result is now:

$$\hat{C} = \left(X^T X\right)^{-1} X^T L \tag{23.52}$$
$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} \approx \begin{bmatrix} 71.4732 \\ 0.3818 \\ 8.121 \\ -0.1427 \end{bmatrix}$$

This we may put in our regression model:

$$\begin{aligned} L &= \hat{c}_1 + \hat{c}_2 I + \hat{c}_3 G + \hat{c}_4 GI \\ &= 71.4732 + 0.3818I + 8.1211G - 0.1427 * GI \end{aligned} \tag{23.53}$$

where the capital letters now indicate individual variables, not matrices. Now we shall estimate how the interaction term GI contributes to the covariation between the explanatory I and G and the explained variable L. In equation (23.42) we saw that the total effect for a variable x must include both the coefficient for that variable as well as the interaction effect: $\frac{\Delta y}{\Delta x} = b + d * z$. To estimate the total covariation between earnings I and life expectancy L we therefore take:

$$\frac{\Delta L}{\Delta I} = \hat{c}_2 + \hat{c}_4 G = 0.3818 - 0.1427 * G \tag{23.54}$$

The total relationship $\Delta L / \Delta I$ can be read as that for men, $G = 0$, on average one extra PPP USD in annual earnings on average coincides with 0.3818 years longer life expectancy. For women, $G = 1$, one extra PPP USD in annual earnings on average coincides to $0.3818 - 0.1427 = 0.2391$ years (almost three months) longer average life expectancy.

Men have lower life expectancy than women, but the relationship between life expectancy and earnings is more positive for men compared to women. The total relationship between gender G and average life expectancy L we can estimate from our regression model to:

$$\frac{\Delta L}{\Delta G} = \hat{c}_3 + \hat{c}_4 * I = 8.1211 + 0.1427 * I \tag{23.55}$$

The coefficient c_1 in our regression model we estimated to 71.4732 (equation (23.53)). This is the average life expectancy that our regression model indicates that people have, where the variables I and G equal zero, that is, a woman $G = 0$ without earnings $I = 0$.

If the person without earnings is a woman, $G = 1$, $\hat{c}_3 = 8.1211$ in equation (23.55) indicates that we expect that person to on average live 8.121 years longer. This relationship changes however at higher earnings.

Figure 23.4 illustrates parts of the results in a diagram where we see the relationship between life expectancy and earnings. Two regression lines are drawn in the diagram, one for women and one for men. The regression line for women is drawn with the equation:

$$L = \hat{a} + \hat{b}\,(I|G = 1) = 79.5943 + 0.2391\,(I|G = 1) \tag{23.56}$$

where $(I|G = 1)$ means that we only use the observations for life expectancy (L) in table 23.10 that represent women. The coefficients $\hat{a}$ and $\hat{b}$ are estimated with the least squares method . The regression line for men is drawn in the diagram with the equation:

$$L = \hat{c} + \hat{d}\,(I|G=0) = 71.4732 + 0.3818\,(I|G=0) \tag{23.57}$$

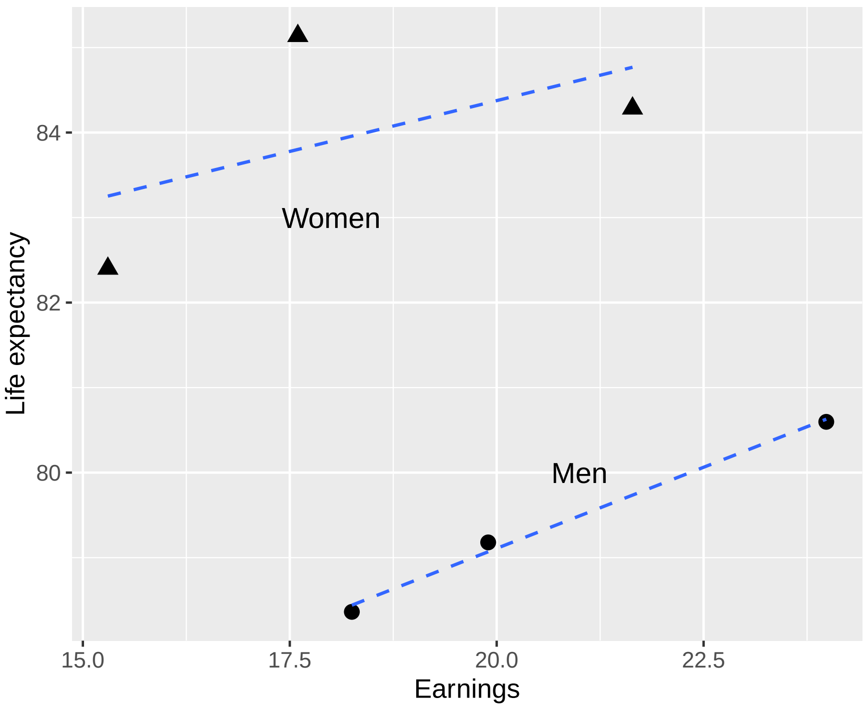


Figure 23.4: Life expectancy and earnings for women and men

where $(I|G=1)$ means that we only use the observations in table 23.10 that represent men. By estimating the relationship between life expectancy and earnings for men and women separately we reach the same conclusion as when we estimate the regression model with interaction effects. The regression results for only women in equation (23.56) are the same as the results from the regression model in equation (23.53) with $G=0$, since the first two coefficients in both regression models are the same: $\hat{c}_1 = \hat{a}$ and $\hat{c}_2 = \hat{b}$.

The results in the regression model for men in equation (23.57) can simply be described as us adding together the coefficients in equation (23.53) . The coefficient $\hat{c} = \hat{c}_1 + \hat{c}_3 = 71.4732 + 8.1211 = 79.5943$. And coefficient $\hat{d} = \hat{c}_2 + \hat{c}_4 = 0.3818 - 0.1427 = 0.2391$. The slope coefficient in the regression model for only men, $\hat{d}$, is the same value as $\Delta L/\Delta I$ for $G=1$ in equation (23.54) . This also means that the difference between the two slope coefficients in the regression models for women and men respectively is $\hat{b} - \hat{d} = 0.2391 - 0.3818 = -0.1427$, which is the same value that we found for $\hat{c}_4$ in equation (23.53) .

23.7 Adjusted R^2

In section 21.9 we introduced the measure R^2, which measures what proportion of the variation in the explained variable, for example y, that can be explained by the variation in the explanatory variable, for example x:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \tag{23.58}$$

where $y_i - \hat{y}$ indicates the difference between observed y_i and predicted $\hat{y}_i$. The expression $y_i - \bar{y}$ indicates the difference between observed y_i and the mean $\bar{y}$. Say now that we have a regression model with k number of explanatory variables:

$$y = b_0 + b_1 x_1 + b_2 x_2 + ... + b_k x_k + e \tag{23.59}$$

where all b are constant coefficients and all x are explanatory variables. Coefficients and variables are numbered to mark that they differ from each other. The term e is the error term. Let us rewrite the calculation of R^2 by reminding ourselves of how $\hat{y}$ is defined from the regression model:

$$\begin{aligned} R^2 &= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \\ &= 1 - \frac{\sum \left(y_i - \left(\hat{b}_0 + \hat{b}_1 x_1 + ... + \hat{b}_k x_k + \hat{e}_i\right)\right)^2}{\sum (y_i - \bar{y})^2} \end{aligned} \tag{23.60}$$

where i marks the number of the observation. Now we will add to our regression model in equation (23.59) another explanatory variable: x_{k+1} and its slope coefficient b_{k+1}. The new regression model will have $k+1$ number of explanatory variables. R^2 for this new model becomes:

$$R^2 = 1 - \frac{\sum \left(y_i - \left(\hat{b}_0 + \hat{b}_1 x_1 + ... + \hat{b}_k x_k + \hat{b}_{k+1} x_{k+1} + \hat{e}_i\right)\right)^2}{\sum (y_i - \bar{y})^2} \tag{23.61}$$

The squared parenthesis in the numerator increases, or alternatively remains unchanged if the new term $b_{k+1}x_{k+1} = 0$. R^2 for this new model is equal to or greater than R^2 for the model with fewer variables. If we add more variables, R^2 risks increasing even though the new regression model we thereby create cannot explain variations in the explained variable more than the previous regression model.

This does not mean that R^2 is a worthless measure but we must be aware of these characteristics when we use it. Another common measure is adjusted R^2, which is denoted R^2_{adj} or $\bar{R}^2$:

$$\bar{R}^2 = 1 - (1 - R^2)\,\frac{n-1}{n-k-1} \tag{23.62}$$

Table 23.11: Calculations with y and $\hat{y}$

i	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$y_i - \bar{y}$	$(y_i - \hat{y}_i)^2$	$(y_i - \bar{y})^2$
1	3	3.2	−0.2	−0.5	0.041	0.25
2	2	1.86	0.14	−1.5	0.018	2.25
3	5	4.59	0.41	1.5	0.164	2.25
4	4	4.34	−0.34	0.5	0.114	0.25
Mean	3.5					
Sum					0.3378	5

where k is the number of explanatory variables and n is the number of observations. For the regression model $y = a + bx + v$ we have one explanatory variable x with an associated coefficient, therefore $k = 1$. In the regression model $y = a + bx + cz + v$ we have the two explanatory variables x and z, therefore $k = 2$.

Let us now estimate R^2 and $\bar{R}^2$ for the regression model $y = a + bx + cz + e$ from the chapter's introduction. Table 23.11 summarizes with rounded results the calculations we need, which gives R^2:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \approx 1 - \frac{0.34}{5} \approx 0.9324 \tag{23.63}$$

We estimate $\bar{R}^2$ (adjusted R^2):

$$\begin{aligned}\bar{R^2} &= 1 - (1 - R^2)\,\frac{n-1}{n-k-1} \\ &= 1 - 0.0676 * \left(\frac{4-1}{4-2-1}\right) \\ &\approx 0.7973\end{aligned} \tag{23.64}$$

The result shows that our estimated $\bar{R^2} < R^2$. The difference arises due to the adjustment for the number of explanatory variables.

23.8 If we have too few variables

With the least squares method we can estimate the covariation between two variables and hold constant other important variables by including these in the regression model. We may describe it as the goal of the method being that the covariation we estimate between the explained variable y and one of our explanatory variables, for example x, is the covariation that applies all else being equal. Sometimes the Latin phrase ceteris paribus is used.

But to be able to estimate the covariation between two variables all else being equal, the method requires that we in our regression analysis have in one way or another taken into account everything else. What is not included in the regression model must covary with our explanatory variable y in such a way that it is only random and can be captured by the regression model's error term. If we for example conduct a randomized experiment, create treatment and control groups through random assignment and compare the outcome for the groups with regression analysis, then our ambition with the random assignment is to ensure that nothing disturbs the covariation between the variables in the regression analysis.

If we work with a model of the type $y = a + bx$ the result can become seriously misleading if the relationship between x and y actually depends on variable z. If some other variable, for example z, can be expected to affect y in a way that is not captured by the variable x that already is included in the model, we may need to include z to get a correct result. If one or more variables that should be included in the regression model are missing, this is called the regression model being underspecified.

Table 23.12: Variables y, x and z

	y_i	x_i	z_i
1	3	3	1
2	2	4	4
3	5	6	0
4	4	7	1
Mean	3.5	5	1.5

The impact a missing variable in a regression model has on the other variables is called omitted variable bias (often abbreviated OVB). We illustrate OVB by comparing the two regression models with the fictitious data we used in section 23.1 . Table 23.12 repeats the observations we will use. We reuse the regression model:

$$y_i = \hat{a}_1 + \hat{b}_1 x_i + v_i = 1 + 0,5x_i + v_i \tag{23.65}$$

where we number the coefficients $\hat{a}_1$ and $\hat{b}_1$ to indicate that this is model 1. We also have the results from the second regression model, which we have also estimated previously:

$$y_i = \hat{a}_2 + \hat{b}_2 x_i + \hat{c}_2 z_i + e_i = 2.9 + 0.3x_i - 0.5z_i + e_i \tag{23.66}$$

where we number the coefficients $\hat{a}_2$ and $\hat{b}_2$ with 2. We see that $\hat{b}_1 = 0.5$ when we have one explanatory variable (x) and that $\hat{b}_2 = 0.3$ when we have two explanatory variables (x, z). Given that variable z should be included in our model, we note that there is a bias in the result for $\hat{b}$. In this simple fictitious example it is possible to estimate the size of the bias, that is, how incorrect our estimated result for $\hat{b}$ becomes due to our omitting z. For this we use the observations in table 23.1 to estimate the covariation between the explanatory variables x and z. With the least squares method we now estimate a regression model where x is a function of z:

$$x_i = \hat{\beta}_1 + \hat{\beta}_2 z_i + u_i = 5.7 - 0.4 z_i + u_i \tag{23.67}$$

where u_i is the error term. The calculation of the coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ is not described here but is left as an exercise for the reader. The results for these are rounded. The bias in the result for $\hat{b}$ that arises in the first regression model (equation (23.65)) is what is called omitted variable bias (OVB):

$$\begin{aligned} \textbf{Omitted Variable Bias:} \quad &= \hat{b}_2 + \hat{c}_2 \hat{\beta}_2 \\ &= 0.3 + (-0.5)(-0.4) \\ &= 0.5 \end{aligned} \tag{23.68}$$

which is the same result as what we found for $\hat{b}_1$ in the first regression model where we only have one explanatory variable x. This example is fictitious and here we easily get to know that z should be included in the regression model and moreover we have observations for all involved variables. The purpose of this exercise is however to illustrate a more complex problem. Whether an additional variable, such as z, should be included in a regression model is often a more complicated question, many times without any simple answer. In many situations this is primarily a qualitative question that is best answered by the theory behind the analysis. Unfortunately there is rarely or never any simple clear answer to exactly which variables should be included in a regression model. It depends on what is being studied and in what way, that is, what the purpose and method of the investigation is.

The calculation we have performed in this section of bias as a result of a missing variable illustrates the importance of not missing any variable. In practice we may sometimes reason our way to conclude that a variable, a phenomenon, is missing in our regression model but simultaneously face the problem that no data exists for this phenomenon. That we lack data is however no reason to ignore a problem. It simply means that our results become misleading, regardless of whether we have all the data we wish for or not.

23.9 If we have too many variables

In the previous section we described how the result in a regression analysis can become incorrect if one or more variables are missing in the regression model. It is easy in such a situation to be tempted to insert more variables into the model for safety's sake. But since all variables included in a regression model can affect also the results concerning all other variables, this can lead to other problems. As an example, our estimates can become incorrect if we insert wrong variables into the model. Having too much information in a regression model leads to the regression model becoming overspecified.

Regression analysis can among other things be used to predict future outcomes. A regression model with many variables may possibly explain variations in the explained variable's historical observations. But if the result is too dependent on a large and complex regression model, it risks reducing our possibilities to make statements about future observations.

If we suspect that our regression model may be overspecified, there are several methods for testing and handling this. A simple method is to remove one or a few observations at a time and estimate the regression model without this or these. A regression model that is overspecified will have difficulty predicting the observations that are removed. But just as with underspecification, there is unfortunately no simple method that can guarantee identification of exactly which variables should be included in a regression model and often it is simply something we must reason and work our way towards in each individual analysis and situation.

23.10 The error term does not need to approach zero

In regression analysis it is easy to believe that the error term V always should approach 0, that is, that the regression model should create a line that lies very close to the points and thereby be able to explain all variation in the explained variable Y. That might be the case for some models. But not in general.

If we use regression analysis to study a causal relationship and estimate what effect a phenomenon X has on Y. Given that we have controlled for all important factors that affect the effect, the error terms capture variation in Y that is not explained by our analysis. The error terms instead capture other things that do not affect the effect we are interested in. As long as the error terms are random noise, there is no reason why the sum of error terms should approach 0. X can still affect Y even if many other things can also affect Y.

To understand this we may think of an example. Suppose we want to study whether medicine X affects headache Y in a collection of patients, divided into treatment and control groups. We find that medicine X affects headache Y. But medicine X is of course not the only thing that affects a headache. Other potential factors include a patient's age, food, drink, general health and so on. Our analysis is correctly designed (there is a true effect between medicine and headache) but cannot explain all variation in Y. The variation we cannot explain is captured by the error term.

The error terms can also represent uncertainty in measurements, such as our variables being data about people that contain small random measurement errors. Many social science phenomena are studied based on scarce information, partly because people are complex and a lot of parts of life is not easy to measure in any simple way. We often work with approximate measures and more or less subjective estimates.

Many times our regression model also describes a simplification of the relationship between the variables. In reality the relation we study is not exactly linear. But even so, linear covariation often works surprisingly well. In these cases the error terms also do not become 0. In certain situations it would require very much work to construct a non-linear regression model that represents the relationship between the variables. But a completely correct model might still not help for the specific purpose with our analysis. Linear models instead give a rough and often sufficiently good picture of the relationship between variables, even if the error terms do not become 0.

23.11 Chapter summary

- Regression models can be described with matrices: $Y = XB + V$ where Y is a column matrix with n observations for the explained variable, X is an $n \times k$ matrix where all rows in the first column have the value 1 and the other columns consists of the observations for the explanatory variables $x_1, ..., x_{k-1}$. B is a column matrix with all coefficients $b_0, ..., b_k$, where b_0 is the constant. V is a column matrix with n error terms. The least squares estimator: $\hat{B} = (X'X)^{-1} X'Y$ where all coefficients in the model are a function of all explanatory variables in the model.
- The purpose of a regression model with multiple explanatory variables is to estimate the covariation between one explanatory variable and the explained variable, taking into account the covariation between all the other explanatory variables and the explained variable. This is called estimating the covariation for one explanatory variable while holding constant the variation in all other explanatory variables.
- The goal of regression analysis is that we should take into account or control for all phenomena that are significant for the covariation we are studying. The phenomena that are not included in the model are represented by the error term. Underspecified: one or more variables are missing in the regression model. Overspecified: variables should be removed from the regression model. Both over- and underspecification can lead to incorrect results.

-R^2 is a rough measure of what proportion of the variation in the explained variable Y that can be "explained" by the regression model and the explanatory variables. R^2 risks increasing if we add more variables. Adjusted R^2 does not necessarily increase because more explanatory variables are included in the model.

23.12 Exercises

1. A multiple regression gives $\hat{Y} = 2 + 3X_1 + 1.5X_2$, where X_1 is years of education and X_2 is years of experience:

(a) What is the predicted (Y) when (X_{1}=4) and (X_{2}=2) ?

(b) Interpret the coefficient (hat{b}*{1}=3) holding (X*{2}) constant.

(c) Interpret the coefficient (hat{b}*{2}=1.5) holding (X*{1}) constant.

Answer: (a)\begin_inset Formula $\hat{Y} = 2+12+3 = 17$, (b) one additional year of education is associated with 3 units higher Y on average, holding experience constant, (c) one additional year of experience is associated with 1.5 units higher Y on average, holding education constant.

2. A regression model includes a gender dummy variable D (1 = female, 0 = male): $\hat{Y} = 500+50X+100D$.

(a) What is the predicted income for a male with (X=10) ?

(b) What is the predicted income for a female with (X=10) ?

(c) What does the coefficient 100 represent?

Answer: (a)\begin_inset Formula $\hat{Y} = 1000$, (b)\begin_inset Formula $\hat{Y} = 1100$, (c) the average income difference between females and males, holding X constant.

3. A multiple regression gives $\hat{Y} = 10 + 2X_1 + 3X_2$.

(a) What is the ceteris paribus effect on (hat{Y}) of a one-unit increase in (X_{1}) ?

(b) If (X_{1}) increases by 2 and (X_{2}) stays constant, what is the change in (hat{Y}) ?

(c) Why is the ceteris paribus condition important in multiple regression?

Answer: (a) 2, (b) $\Delta\hat{Y} = 4$, (c) it allows us to isolate the effect of one variable while controlling for others.

4. Given $\hat{Y} = 1 + 2X_1 + 3X_2$, calculate the predicted value for:

(a) (X_{1}=1,X_{2}=2) .

(b) (X_{1}=3,X_{2}=1) .

(c) (X_{1}=0,X_{2}=4) .

Answer: (a)\begin_inset Formula $\hat{Y} = 9$, (b)\begin_inset Formula $\hat{Y} = 14$, (c)\begin_inset Formula $\hat{Y} = 13$.

5. A regression with a country dummy D_S (1 = Sweden, 0 = other) gives $\hat{Y} = 100 + 5X + 20D_S$.

(a) What is the predicted (Y) for a non-Swedish country with (X=10) ?

(b) What is the predicted (Y) for Sweden with (X=10) ?

(c) What does the coefficient 20 represent?

Answer: (a)\begin_inset Formula $\hat{Y} = 150$, (b)\begin_inset Formula $\hat{Y} = 170$, (c) the average difference in Y for Sweden compared to other countries, holding X constant.

6. Discuss model specification:

(a) What is an underspecified model?

(b) What is an overspecified model?

(c) What problem can underspecification cause?

Answer: (a) a model missing one or more relevant explanatory variables, (b) a model with irrelevant variables included, (c) omitted variable bias — the coefficients may be incorrect.

7. Model A has $R^2 = 0.65$ with 1 explanatory variable. Model B has $R^2 = 0.70$ with 4 variables and adjusted $R^2 = 0.66$.

(a) Which model has higher (R^{2}) ?

(b) Why use adjusted (R^{2}) when comparing models?

(c) Which model is preferred according to adjusted (R^{2}) ?

Answer: (a) model B, (b) adjusted R^2 penalizes for adding variables that do not improve fit, (c) model B (adjusted R^2 0.66 > 0.65).

8. In the matrix form of OLS, $Y = XB + V$:

(a) What does the matrix (X) contain?

(b) What does the vector (B) contain?

(c) Write the OLS estimator (hat{B}) .

Answer: (a) a column of ones and the explanatory variables, (b) the regression coefficients $\hat{a}, \hat{b}_1, \hat{b}_2, \ldots$, (c)\begin_inset Formula $\hat{B} = (X'X)^{-1}X'Y$.

9. A regression includes an interaction term: $\hat{Y} = 5 + 2X_1 + 3X_2 + 1.5X_1X_2$.

(a) What is the effect of (X_{1}) on (hat{Y}) when (X_{2}=0) ?

(b) What is the effect of (X_{1}) on (hat{Y}) when (X_{2}=2) ?

(c) What does the interaction term (1.5X_{1}X_{2}) capture?

Answer: (a) 2, (b)\begin_inset Formula $2 + 1.5 \times 2 = 5$, (c) that the effect of X_1 on Y depends on the value of X_2 .

10. Given $n = 100$, $k = 4$ explanatory variables, $TSS = 500$, $SSE = 100$:

(a) Calculate (R^{2}) .

(b) Calculate adjusted (bar{R}^{2}=1-frac{SSE/(n-k-1)}{TSS/(n-1)}) .

(c) Is (bar{R}^{2}) higher or lower than (R^{2}) ?

Answer: (a)\begin_inset Formula $R^2 = 0.8$, (b)\begin_inset Formula $\bar{R}^2 = 1 - \frac{100/95}{500/99} \approx 0.791$, (c) $\bar{R}^2$ is lower than R^2 .

11. A regression of income Y on education X_1 and age X_2 gives: $\hat{Y} = 100 + 80X_1 + 10X_2$.

(a) What is the predicted income for a person with (X_{1}=10) years of education and (X_{2}=30) years of age?

(b) By how much does income increase per additional year of education, holding age constant?

(c) By how much does income increase per additional year of age, holding education constant?

Answer: (a)\begin_inset Formula $\hat{Y} = 100 + 800 + 300 = 1200$, (b) 80, (c) 10.

12. A researcher adds a variable X_3 to a model and observes that R^2 increases from 0.72 to 0.74 but adjusted R^2 decreases from 0.70 to 0.69.

(a) Why does (R^{2}) always increase (or stay the same) when a variable is added?

(b) What does the decrease in adjusted (R^{2}) suggest about (X_{3}) ?

(c) Should the researcher keep (X_{3}) in the model?

Answer: (a) adding variables always reduces SSE or leaves it unchanged, which mechanically increases R^2 , (b) X_3 does not contribute enough to justify its inclusion, (c) no, based on adjusted R^2 .

Chapter 24

Regression Analysis with Statistical Tests

This chapter introduces how we estimate the probability that the results from our regression model could have occurred by chance.

24.1 The Population Model and Statistical Tests

In the previous chapters we introduced the concepts of population and sample, and we described how to use regression analysis and the least squares method to estimate a regression model. Our goal is to estimate the relationship that exists in the *population*, which we call the population's regression model:

$$Y = a + bX + V \tag{24.1}$$

where Y and X are variables, a and b are coefficients, and V is the error term. With sample data we want to estimate the population values for the constant coefficients a and b and find values as close as possible to the population values. In this process there is always some uncertainty, regardless of how many observations we have or how high quality our data is.

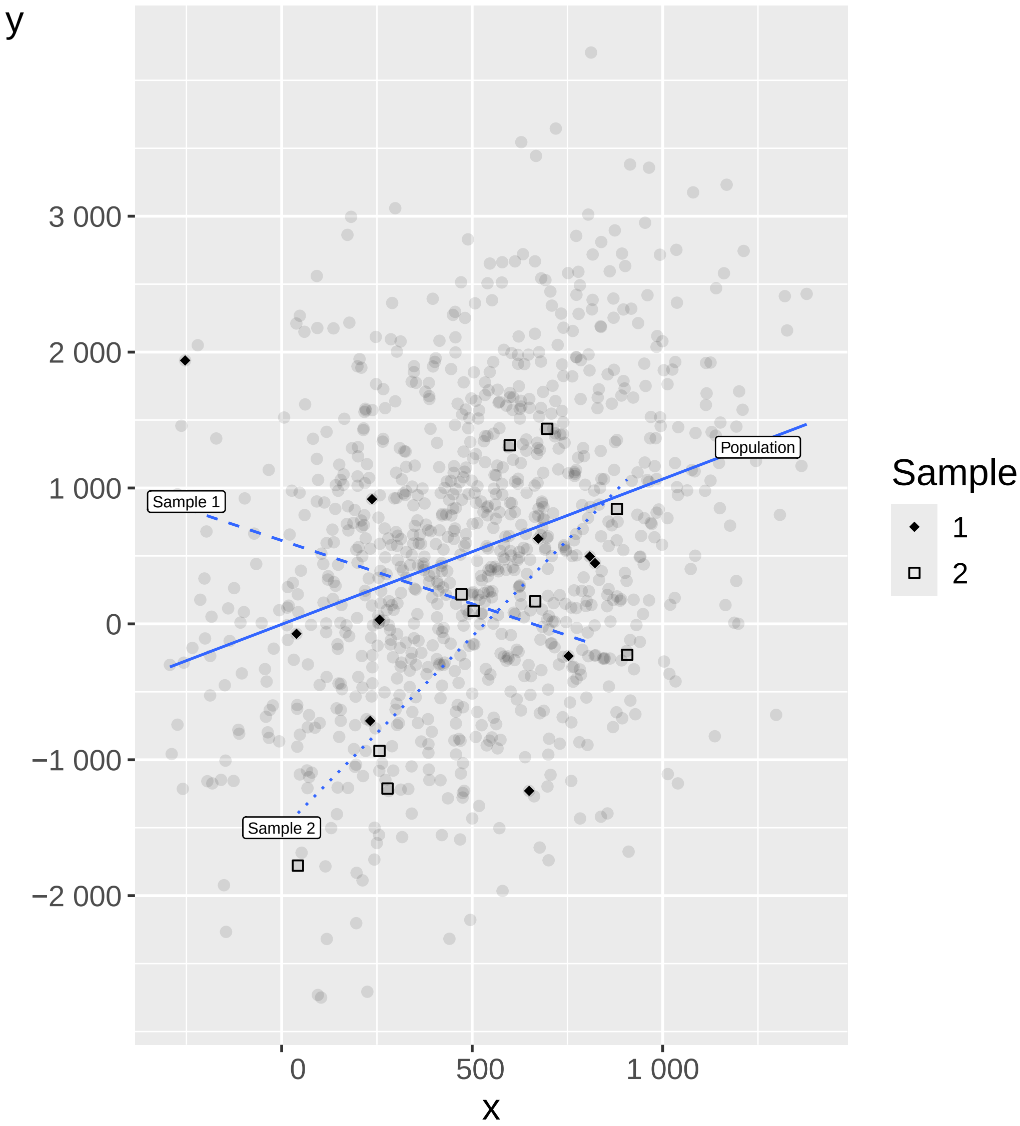


Figure 24.1: Population regression line and two sample regression lines

Description: Figure 24.1 illustrates this with fictitious data. The gray dots are the population in which there exists a positive covariation between the two normally distributed variables Y and X. The positive covariation is illustrated by the solid line, which is the regression line for the population. From the population of points we have taken two random samples of a few observations and estimated one regression line per sample. Neither of the two samples gives a correct picture of the population's covariation.

In chapter 19 we introduced statistical tests with null hypothesis and alternative hypothesis. Now we apply these tools to regression coefficients. We start from the population model $Y = a + bX + V$. For both coefficients a and b we set up separate statistical tests and formulate separate null and alternative hypotheses. We are primarily interested in the slope coefficient b because it describes whether there is a linear covariation between X and Y. A common way to formulate the null hypothesis for b is:

$$H_0 : b = 0, \qquad H_1 : b \neq 0$$

We can also formulate corresponding hypotheses for the intercept: $H_0 : a = 0$ and $H_1 : a \neq 0$. If we estimate $\hat{b} > 0$ but cannot reject $H_0 : b = 0$ at the chosen significance level α, this indicates that the estimate $\hat{b} > 0$ could just as well be the result of a random process and we therefore have no reason to assume that $b \neq 0$.

24.2 Starting Points and Assumptions

Statistical tests for regression coefficients require that certain assumptions hold. These may at first glance seem abstract but will appear more logical as the chapter progresses. They are also discussed more thoroughly in a later chapter. The assumptions are:

1. **Linear model.** Our regression model is linear in its coefficients (the coefficients have exponent equal to one) and specified in exactly the way that the relationship between the variables in the population looks.
2. **Random variables.** Y is a random variable from a population with unknown properties. The explanatory variable X is either from a random sample (observational study) or from a designed process (for example, a controlled experiment). Both Y and X are assumed to be random, with each observation (Y_i, X_i) independent and identically distributed (i.i.d.).
3. **Zero conditional mean.** The conditional expected value for the error terms is zero: $E(V \mid X) = 0$. This means that the error terms v_i do not covary with the explanatory variable X, so $cov(v, X) = 0$.
4. **Constant variance (homoscedasticity).** The error terms have constant variance σ_V^2 regardless of which value of X we compare against: $var(V \mid X) = \sigma_V^2$, where $\sigma_V^2 > 0$. This means that along the population regression line, the error terms have the same variance for all values of X.
5. **Uncorrelated errors.** The error terms for different observations do not correlate with each other.
6. **Normal errors.** The errors of the population model follow a normal distribution, regardless of which values of X we compare against: $V \mid X \sim N(0, \sigma_V^2)$.

In this chapter we use these assumptions as theoretical starting points. When we work with real data it is common to encounter situations where one or more assumptions are not fulfilled.

24.3 The Variance of the Regression Model

We described in earlier chapters how to estimate $\hat{b}$ as:

$$\hat{b} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

The estimated coefficient $\hat{b}$ is an estimate from a particular sample. The estimator $\hat{b}$ is a function of the random variables x and y, so its expected value equals the population value: $E(\hat{b}) = b$.

Given assumption 6 (that the error terms follow a normal distribution) we assume that the coefficients' estimators $\hat{a}$ and $\hat{b}$ follow a t-distribution. To test $H_0 : b = 0$ vs $H_1 : b \neq 0$ we use a two-sided t-test where the t-value is estimated with:

$$t = \frac{\hat{b} - b_0}{se(\hat{b})} = \frac{\hat{b}}{se(\hat{b})} \tag{24.2}$$

where $b_0 = 0$ is the null hypothesis value, and $se(\hat{b})$ is the standard error of $\hat{b}$.

To estimate the standard error we take the square root of the variance of the estimator $\hat{b}$, which is defined as:

$$var(\hat{b}) = \frac{\sigma_V^2}{\sum_i (x_i - \bar{x})^2} \tag{24.3}$$

The variance of $\hat{b}$ depends on σ_V^2— the unknown variance of the error terms in the population. Since we cannot observe σ_V^2 directly, we estimate it using the **mean squared residual** (MSR):

$$\hat{s}_V^2 = \frac{\sum_i (y_i - \hat{y}_i)^2}{n - p} \tag{24.4}$$

where p is the number of estimated parameters ($p = 2$ in simple regression: $\hat{a}$ and $\hat{b}$), so the degrees of freedom are $k = n - p$. Replacing σ_V^2 with $\hat{s}_V^2$ gives the estimated standard error of $\hat{b}$:

$$se(\hat{b}) = \sqrt{\frac{\hat{s}_V^2}{\sum_i (x_i - \bar{x})^2}} = \sqrt{\frac{\sum_i (y_i - \hat{y}_i)^2}{(n-p)\sum_i (x_i - \bar{x})^2}} \tag{24.5}$$

For the intercept $\hat{a}$:

$$var(\hat{a}) = var(\hat{b}) \cdot \frac{\sum_i x_i^2}{n}, \qquad se(\hat{a}) = se(\hat{b}) \cdot \sqrt{\frac{\sum_i x_i^2}{n}} \tag{24.6}$$

24.4 Regression analysis with t-test

Numerical example. Consider a dataset with $n = 4$ observations: $x = \{3, 4, 6, 7\}$ and $y = \{3, 2, 5, 4\}$. The OLS estimates are $\hat{b} = 0.5$ and $\hat{a} = 1$, so $\hat{y} = 1 + 0.5x$.

i	x_i	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$
1	3	3	2.5	0.5	0.25
2	4	2	3.0	−1.0	1.00
3	6	5	4.0	1.0	1.00
4	7	4	4.5	−0.5	0.25
Sum					**2.50**

With $\bar{x} = 5$ we have $\sum(x_i - \bar{x})^2 = 4 + 1 + 1 + 4 = 10$. With $p = 2$ and $k = n - p = 2$ degrees of freedom:

$$\hat{s}_V^2 = \frac{2.5}{2} = 1.25, \qquad se(\hat{b}) = \sqrt{\frac{1.25}{10}} = \sqrt{0.125} \approx 0.354$$

$$t = \frac{\hat{b}}{se(\hat{b})} = \frac{0.5}{0.354} \approx 1.414$$

The critical value from the t-table with $k = 2$ degrees of freedom and $\alpha = 0.05$(two-sided) is $t^*_{2,\,0.025} \approx 4.3$. Since $|t| = 1.414 < 4.3$ we do not reject H_0: the slope is not statistically significantly different from zero at the 5% level.

For the intercept $\hat{a} = 1$, with $\sum x_i^2 = 9 + 16 + 36 + 49 = 110$:

$$se(\hat{a}) = 0.354 \cdot \sqrt{\frac{110}{4}} = 0.354 \cdot 5.24 \approx 1.854$$

$$t(\hat{a}) = \frac{1}{1.854} \approx 0.54$$

Again $|t| < 4.3$, so we do not reject $H_0 : a = 0$ either.

24.5 Regression analysis with confidence interval

A confidence interval for a regression coefficient gives a range of plausible values for the population parameter. For the slope b:

$$\hat{b} \pm t^*_{k,\,\alpha/2} \cdot se(\hat{b}) \tag{24.7}$$

where $t^*_{k,\alpha/2}$ is the critical value from the t-table with $k = n - p$ degrees of freedom and significance level $\alpha/2$ in each tail.

Using the numerical example with $t^*_{2,0.025} = 4.303$:

- **CI for $\hat{b}$:** $0.5 \pm 4.303 \times 0.354 \approx 0.5 \pm 1.52$, giving $(-1.02,\ 2.02)$
- **CI for $\hat{a}$:** $1 \pm 4.303 \times 1.854 \approx 1 \pm 7.97$, giving $(-6.97,\ 8.97)$

Both confidence intervals contain zero, which is consistent with not rejecting H_0 at the 5% level. With only $n = 4$ observations the critical value is large and the intervals are wide. Confidence intervals and hypothesis tests are equivalent: if the CI does not contain the hypothesized value b_0, the test rejects H_0.

24.6 Confidence Interval for the Regression Line

We can also construct a confidence interval for the *predicted mean value* $\hat{Y}$ at a specific value x_p. The predicted value is $\hat{Y}_{x_p} = \hat{a} + \hat{b}\,x_p$, and its standard error is:

$$se(\hat{Y}_{x_p}) = \sqrt{\hat{s}^2_V \left(\frac{1}{n} + \frac{(x_p - \bar{x})^2}{\sum_i (x_i - \bar{x})^2} \right)} \tag{24.8}$$

The confidence interval for the mean response at x_p is $\hat{Y}_{x_p} \pm t^*_{k,\alpha/2} \cdot se(\hat{Y}_{x_p})$.

The standard error in equation (24.8) increases as x_p moves away from $\bar{x}$. This means that the CI is narrowest at the mean of X and widens toward the extremes, producing the characteristic funnel-shaped confidence band shown in figure 24.2.

For the numerical example at $x_p = 3$:

$$\hat{Y}_3 = 1 + 0.5 \times 3 = 2.5$$

$$se(\hat{Y}_3) = \sqrt{1.25 \left(\frac{1}{4} + \frac{(3-5)^2}{10} \right)} = \sqrt{1.25 \times 0.65} \approx 0.901$$

$$\text{CI: } 2.5 \pm 4.303 \times 0.901 \approx 2.5 \pm 3.88 = (-1.38,\ 6.38)$$

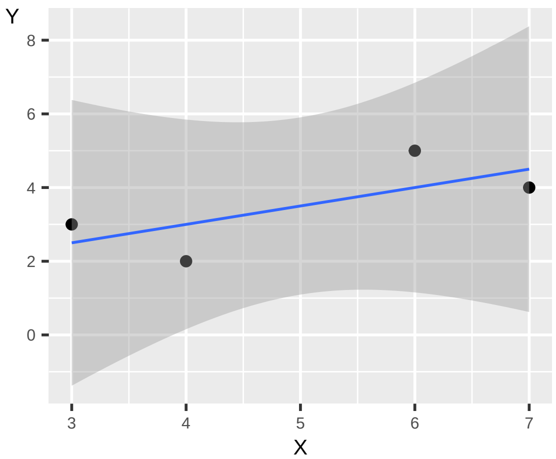


Figure 24.2: Regression line with 95% confidence interval

24.7 Statistical Power in Regression

The precision of our estimates depends on sample size. With a small sample, $se(\hat{b})$ is large, requiring a large observed effect to reach statistical significance. This leads to low statistical **power** — the probability of detecting a true effect when it exists.

To illustrate, suppose the true population model is $Y = 100+5X+V$ where $V \sim N(0, 80)$ and $X \sim U(0, 1)$. We draw 2,000 samples and estimate the regression model for each, testing $H_0 : b = 0$ at $\alpha = 0.05$. Figure 24.3 shows the distribution of the 2,000 estimates $\hat{b}$ for two sample sizes: $n = 500$ and $n = 50{,}000$. Black bars show statistically significant results; gray bars show results that are not statistically significantly different from zero.

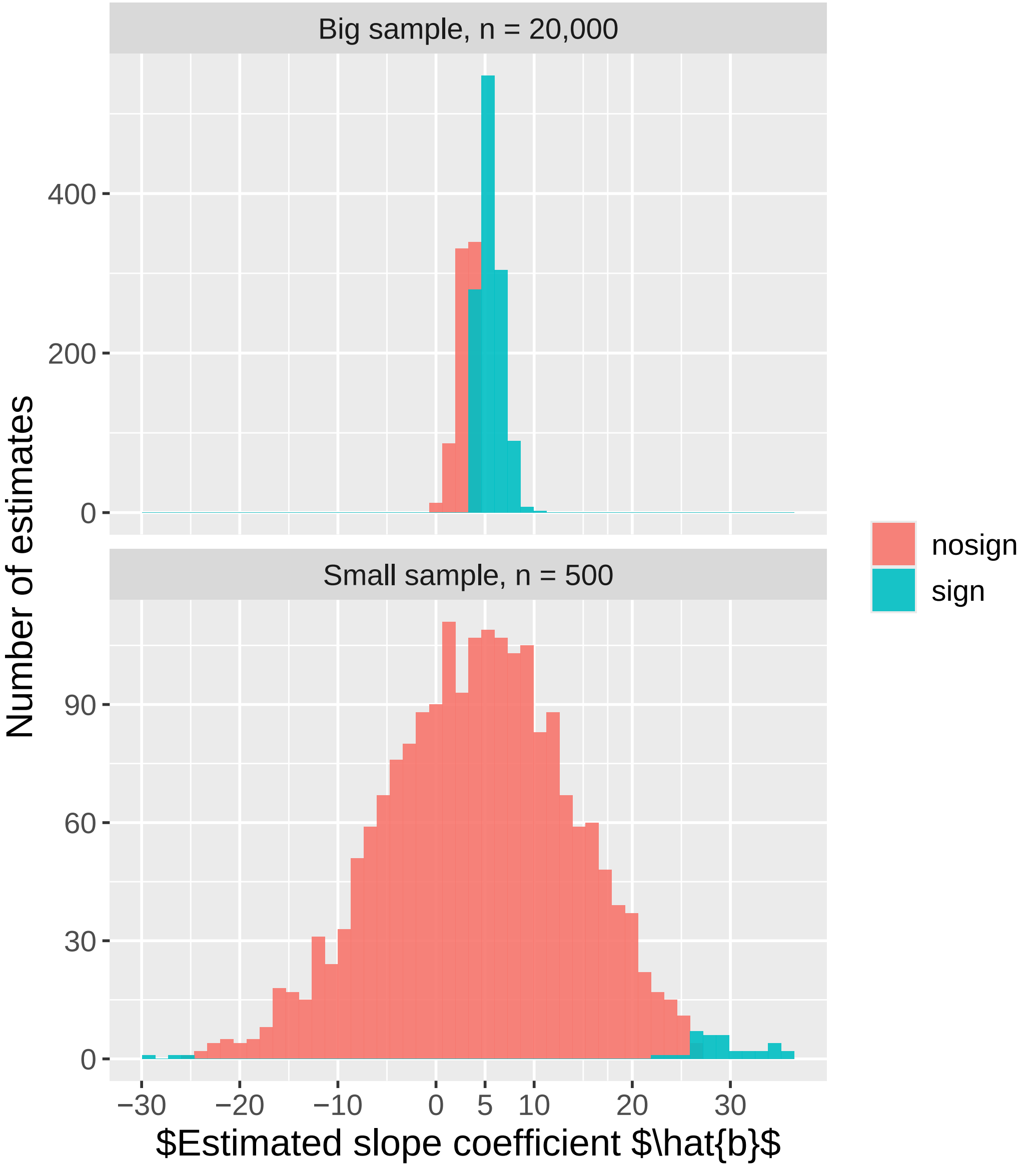


Figure 24.3: Statistical power in regression

In the upper panel ($n = 500$) the estimates have a large spread around the true value $b = 5$ and are mostly not statistically significant. The statistically significant results are concentrated far from the population value — distributed around -30 or 30. This is the **winner's curse**: when statistical power is low, only extreme estimates can cross the significance threshold, giving a misleading picture of the true effect size.

In the lower panel ($n = 50{,}000$) most estimates are statistically significant and all are clearly centered around the population value $b = 5$.

The more observations we include in our sample, the more accurate the results become, all else equal. By calculating the power of a test we can adjust our analysis to minimize the risk of Type I and Type II errors.

24.8 Statistical and Economic Significance

A statistically significant estimate does not necessarily mean that a result has great importance for the real phenomenon we are studying. Statistical analysis often refers separately to **economic significance** (sometimes called practical significance) — how much importance our results have in reality. This question generally has no straightforward answer. It depends on the purpose of the study, the type of data, and how the results should be interpreted.

Many people experience, for example, that weather affects their mood. But few would compare normal variations in weather with the effect that earnings or work environment has on their lives. If we succeeded in measuring the causal effect of these phenomena with a regression analysis, the slope coefficient for the weather variable would be relatively small. The slope coefficient for earnings would be larger. Both coefficients could be statistically significantly different from zero. But only one of the variables would have an economically significant effect — that is, have great importance for people's lives.

Statistical significance depends on the ratio $\hat{b}/se(\hat{b})$. With a very large sample, even a tiny effect can be statistically significant. A coefficient $\hat{b} = 0.001$ might be statistically significant at the 1% level while being completely unimportant in practice. Interpreting regression results requires both statistical rigor and subject-matter judgment about what magnitudes matter.

24.9 Chapter Summary

- Our goal with regression analysis is to estimate the regression model in a population, the true regression model. With sample data we estimate the model $Y = a + bX + V$ where a and b are coefficients, Y and X are variables, and V is the error term.
- Assumptions for regression analysis with the least squares method (do not necessarily need to match real data): (i) the regression model is linear; (ii) Y is a random variable, X is either random or from a designed process, observations are i.i.d.; (iii) $E(V \mid X) = 0$ and V and X do not covary; (iv) error terms have constant variance $var(V \mid X) = \sigma_V^2 > 0$; (v) error terms do not correlate; (vi) $V \mid X \sim N(0, \sigma_V^2)$.
- We estimate $\hat{a}$ and $\hat{b}$ and can then perform statistical tests for each coefficient, for example a two-sided t-test with $H_0 : b = 0$, $H_1 : b \neq 0$. For coefficient b we estimate $t = (\hat{b} - b_0)/se(\hat{b})$ where b_0 is the null hypothesis value and $se(\hat{b})$ is the standard error. If the calculated $|t| > t^*$ given the chosen significance level α, this indicates that H_0 can be rejected as false.
- A 95% confidence interval for b is $\hat{b} \pm t^*_{k,\,0.025} \cdot se(\hat{b})$. If the CI does not contain zero, the slope is statistically significant.
- Note the difference between statistical significance (can we reject H_0 at the chosen significance level?) and economic significance (what importance does the result have for reality?).

24.10 Exercises

1. Consider the population regression model $Y = \alpha + \beta X + \varepsilon$.

(a) What does β represent?

(b) What does ε represent?

(c) What is the difference between β (population) and $\hat{b}$ (sample estimate)?

Answer: (a) β is the effect of a one-unit increase in X on Y in the population — the true slope.

(b) ε is the random error term capturing unobserved factors that affect Y beyond X.

(c) β is a fixed but unknown population parameter; $\hat{b}$ is a random estimate computed from sample data that varies from sample to sample.

2. OLS gives $\hat{b} = 3$ with $se(\hat{b}) = 1.5$. Test $H_0 : b = 0$.

(a) Compute $t = \hat{b}/se(\hat{b})$.

(b) With critical value $t^* = 1.96$, reject H_0?

(c) Construct a 95% confidence interval $\hat{b} \pm 1.96 \cdot se(\hat{b})$.

Answer: (a) $t = 3/1.5 = 2$.

(b) $2 > 1.96$, so yes, reject H_0.

(c) $3 \pm 1.96 \times 1.5 = (0.06,\ 5.94)$. The CI does not contain zero, consistent with rejecting H_0.

3. Simple regression with $n = 20$ observations and $k = 1$ regressor. The degrees of freedom are $df = n - k - 1$.

(a) What are the degrees of freedom?

(b) OLS gives $\hat{b} = 2$, $se(\hat{b}) = 0.8$. Compute t.

(c) The critical value $t^*(df = 18,\ \alpha = 0.05) \approx 2.10$. Reject $H_0 : b = 0$?

Answer: (a) $df = 20 - 1 - 1 = 18$.

(b) $t = 2/0.8 = 2.5$.

(c) $2.5 > 2.10$, so yes, reject H_0.

4. Estimated model: $\hat{y} = 5 + 2x$, $t = 4$, $p < 0.001$.

(a) Interpret the slope $\hat{b} = 2$.

(b) What does $t = 4$ tell us?

(c) Is the slope statistically significant at $\alpha = 0.05$?

Answer: (a) A one-unit increase in X is associated with a 2-unit increase in Y.

(b) $\hat{b}$ is 4 standard errors from zero — strong evidence against $H_0 : b = 0$.

(c) Yes. With $t = 4$ and $p < 0.001$, the slope is statistically significant at any conventional level.

5. How does the precision of $\hat{b}$ change in each case?

(a) The sample size n increases.

(b) The residual variance s_e^2 increases.

(c) The variance of X increases.

Answer: (a) $se(\hat{b})$ decreases — more observations yield more precise estimates.

(b) $se(\hat{b})$ increases — greater unexplained variation makes the slope less precise.

(c) $se(\hat{b})$ decreases — more variation in X means more information about the slope, improving precision.

6. Test $H_0 : b = 0$ vs $H_1 : b > 0$ (one-tailed). $\hat{b} = 1.5$, $se(\hat{b}) = 0.8$.

(a) Compute t.

(b) One-tailed critical value at $\alpha = 0.05$ is 1.64. Reject H_0?

(c) Two-tailed critical value at $\alpha = 0.05$ is 1.96. Reject H_0 in a two-sided test?

Answer: (a) $t = 1.5/0.8 = 1.875$.

(b) $1.875 > 1.64$, so yes, reject H_0 in the one-tailed test.

(c) $1.875 < 1.96$, so no, do not reject H_0 in the two-sided test. The result is significant in one direction but not two-sided.

7. OLS gives $\hat{b} = 4$, $se(\hat{b}) = 1$, $t^* \approx 2$ (large df, $\alpha = 0.05$).

(a) Construct the 95% CI: $\hat{b} \pm 2 \cdot se(\hat{b})$.

(b) Does the CI contain 0? What does this tell us about significance?

(c) Does the CI contain 3?

Answer: (a) $4 \pm 2 \times 1 = (2,\ 6)$.

(b) 0 is not in $(2, 6)$ — $\hat{b}$ is statistically significant at $\alpha = 0.05$.

(c) Yes, 3 is in $(2, 6)$, so we cannot reject $H_0 : b = 3$ at this significance level.

8. $\hat{b} = 0.5$, $se(\hat{b}) = 0.3$.

(a) Compute t.

(b) With two-tailed critical value 1.96 ($\alpha = 0.05$), reject $H_0 : b = 0$?

(c) With critical value 1.64 ($\alpha = 0.10$), reject H_0?

Answer: (a) $t = 0.5/0.3 \approx 1.67$.

(b) 1.671.64, reject at $\alpha = 0.10$. The result is sensitive to the chosen significance level.

9. Model: $\hat{y} = 1 + 3x_1 + 2x_2$, $se(\hat{b}_1) = 1$, $se(\hat{b}_2) = 2.5$.

(a) Compute t_1 and t_2.

(b) Which coefficient is statistically significant at $\alpha = 0.05$ (critical value 1.96)?

(c) Interpret $\hat{b}_1 = 3$ ceteris paribus.

Answer: (a) $t_1 = 3/1 = 3$, $t_2 = 2/2.5 = 0.8$.

(b) Only $\hat{b}_1$ is significant ($t_1 = 3 > 1.96$); $\hat{b}_2$ is not ($t_2 = 0.8 < 1.96$).

(c) A one-unit increase in X_1 increases Y by 3 units, holding X_2 constant.

10. $n = 50$, $k = 1$, $R^2 = 0.30$. The F-statistic is $F = \frac{R^2 \cdot (n-k-1)}{(1-R^2) \cdot k}$.

(a) Compute F.

(b) With critical value $F^*(1, 48) \approx 4.04$, reject H_0 (all slopes = 0)?

(c) What does the F-test test in simple regression?

Answer: (a) $F = 0.30 \times 48/(0.70 \times 1) \approx 20.6$.

(b) $20.6 > 4.04$, so yes, reject H_0.

(c) In simple regression the F-test tests whether the slope equals zero — the same hypothesis as the t-test (and $F = t^2$).

11. A coefficient $\hat{b} = 0.001$, $se = 0.0003$, $t = 3.33$, $p = 0.001$.

(a) Is the coefficient statistically significant at $\alpha = 0.05$?

(b) A 1-unit increase in X increases Y by 0.001 units. Is this economically significant?

(c) How can statistical significance coexist with negligible economic significance?

Answer: (a) Yes — $t = 3.33 > 1.96$, so statistically significant at $\alpha = 0.05$.

(b) Debatable — it depends on the scale of X and Y. A 0.001-unit effect may be negligible in practice.

(c) Very large n makes even tiny effects statistically detectable. Statistical significance tells us the effect is real; it does not tell us the effect is important.

12. A regression is run with $n = 100$, $\hat{b} = 5$, $se(\hat{b}) = 2$.

(a) Compute t.

(b) Construct a 95% CI (use $t^* \approx 1.98$ for $df = 98$).

(c) Does the CI contain 0? Conclusion?

Answer: (a) $t = 5/2 = 2.5$.

(b) $5 \pm 1.98 \times 2 \approx (1.04,\ 8.96)$.

(c) 0 is not in the CI, so $\hat{b}$ is statistically significant at $\alpha = 0.05$.

Chapter 25

Statistical Tests in Regression Analysis with Multiple Variables

This chapter describes how we can use statistical tests with regression analysis when we have two or more explanatory variables. For this we use matrices and linear algebra, introduced in chapter 14.

25.1 Starting Points and Assumptions

We start from the regression model written with matrices for a population with unknown characteristics — the population model:

$$Y = XB + V \tag{25.1}$$

where Y is an $n \times 1$ column matrix with the explained variable, B is a $k \times 1$ column matrix for all coefficients, and V is an $n \times 1$ column matrix with n error terms. X is an $n \times k$ matrix with the explanatory variables in separate columns and the observations in separate rows. The regression model includes the intercept b_0 as the first coefficient, which is why all elements in the first column of matrix X have the value 1.

Theoretical assumptions:

1. The population's regression model describes a linear relationship between X and Y that exists in the population.
2. X and Y are random variables from a population whose characteristics are unknown. Explanatory variables can also be the result of a controlled process. Observation $(Y_i, X_{1,i}, \dots, X_{k,i})$ is independent and identically distributed (i.i.d.).
3. The explanatory variables in X have no strong correlation among themselves. None of the variables X_j can be described as a linear combination of any of the other variables X_k.
4. The error terms are not correlated with each other: $cov(v_i v_j \mid X) = 0$ for $i \neq j$.
5. The conditional expected value of the error terms is zero: $E(V \mid X) = 0$, and V and X do not covary: $cov(V, X) = 0$.
6. The error terms have constant variance σ_V^2 across all values of all explanatory variables X, described as:

$$E(VV^T \mid X) = \sigma_V^2 I = \begin{bmatrix} \sigma_V^2 & 0 & \cdots & 0 \\ 0 & \sigma_V^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \sigma_V^2 \end{bmatrix}$$

where V^T is the transposed column matrix and $\sigma_V^2 I$ is an $n \times n$ diagonal matrix.

7. The error terms follow the normal distribution: $V \mid X \sim N(0, \sigma_V^2 I)$.

25.2 Estimators of the Coefficients

Based on the least squares method we estimate the coefficients as:

$$\hat{B} = (X^T X)^{-1} X^T Y \tag{25.2}$$

The expected value of the estimator $\hat{B}$ is the population model's coefficients: $E(\hat{B}) = B$, making the estimator **unbiased**. We can show this by substituting $Y = XB + V$:

$$\begin{aligned} \hat{B} &= (X^T X)^{-1} X^T Y \\ &= (X^T X)^{-1} X^T (XB + V) \\ &= (X^T X)^{-1} (X^T X) B + (X^T X)^{-1} X^T V \\ &= B + (X^T X)^{-1} X^T V \end{aligned} \tag{25.3}$$

where in the fourth line we use $AA^{-1} = I$ and $IA = A$. Taking the expected value conditional on X and using $E(V \mid X) = 0$:

$$E(\hat{B} \mid X) = B + (X^T X)^{-1} X^T E(V \mid X) = B$$

Given that the assumptions hold, the least squares method gives unbiased estimates: if we were to take an infinite number of samples from the same population, the mean of the estimated coefficients would equal the population values.

To test whether each coefficient in B is statistically significantly different from zero we set up null and alternative hypotheses $H_0 : b_j = 0$ and $H_1 : b_j \neq 0$, where b_j is one of the coefficients in B. Given assumption 7 (normal errors), the estimator $\hat{B}$ follows the t-distribution, and the t-statistic for each coefficient is:

$$t = \frac{\hat{b}_j}{se(\hat{b}_j)}$$

25.3 The Variance-Covariance Matrix

We describe the variance and standard errors for all coefficient estimators using the **variance-covariance matrix**:

$$var(\hat{B} \mid X) = E\left[(\hat{B} - B)(\hat{B} - B)^T \mid X\right]$$

From equation (25.3) we have $\hat{B} - B = (X^T X)^{-1} X^T V$. Substituting:

$$\begin{aligned} var(\hat{B} \mid X) &= E\left[(X^T X)^{-1} X^T V V^T X (X^T X)^{-1} \mid X\right] \\ &= (X^T X)^{-1} X^T E(V V^T \mid X)\, X (X^T X)^{-1} \end{aligned} \tag{25.4}$$

where we factor out X since the expectation is conditional on X. By assumption 6, $E(VV^T \mid X) = \sigma_V^2 I$, giving:

$$var(\hat{B} \mid X) = \sigma_V^2 (X^T X)^{-1} \tag{25.5}$$

Since σ_V^2 is unknown we estimate it with $\hat{s}_V^2 = \sum (y_i - \hat{y}_i)^2 / (n - p)$, yielding the **estimated variance-covariance matrix**:

$$var(\hat{B} \mid X) = \hat{s}_V^2 (X^T X)^{-1} = \frac{\sum (y_i - \hat{y}_i)^2}{n-p} (X^T X)^{-1} \tag{25.6}$$

Structure for simple regression. For the model $Y = a + bX + V$ with two coefficients, the estimated variance-covariance matrix is a 2×2 matrix:

$$var(\hat{B} \mid X) = \hat{s}_V^2 (X^T X)^{-1} = \begin{bmatrix} var(\hat{a}) & cov(\hat{a}, \hat{b}) \\ cov(\hat{b}, \hat{a}) & var(\hat{b}) \end{bmatrix}$$

The diagonal elements are the variances of the estimators. Taking their square roots gives the standard errors needed for the t-tests. The off-diagonal elements are the covariances between estimators. For simple regression, the explicit formulas reduce to those derived in chapter 24:

$$var(\hat{b}) = \frac{\sum (y_i - \hat{y}_i)^2}{(n-p) \sum (x_i - \bar{x})^2}, \qquad var(\hat{a}) = var(\hat{b}) \cdot \frac{\sum x_i^2}{n}$$

The variance-covariance matrix illustrates how each explanatory variable affects the results for the statistical tests for all coefficients in the model. Adding or removing a variable from the regression model can change the estimated coefficients and their standard errors.

25.4 T-tests for Multiple Regression

We now have all the tools needed to use statistical tests when we have more than two coefficients. We illustrate with the regression model from chapter 14:

$$y_i = a + bx_i + cz_i + v_i \tag{25.7}$$

with the four observations in the table below.

i	y_i	x_i	z_i
1	3	3	1
2	2	4	4
3	5	6	0
4	4	7	1

The OLS estimates are $\hat{a} \approx 2.89$, $\hat{b} \approx 0.28$, $\hat{c} \approx -0.54$.

We test three null hypotheses: $H_0 : a = 0$, $H_0 : b = 0$, $H_0 : c = 0$(each against the corresponding two-sided alternative). The t-statistics are:

$$t_{\hat{a}} = \frac{\hat{a}}{se(\hat{a})}, \qquad t_{\hat{b}} = \frac{\hat{b}}{se(\hat{b})}, \qquad t_{\hat{c}} = \frac{\hat{c}}{se(\hat{c})}$$

From the calculations above: $\sum (y_i - \hat{y}_i)^2 \approx 0.338$, $p = 3$ parameters, $n - p = 1$ degree of freedom, so $\hat{s}_V^2 \approx 0.338$. With $\bar{x} = 5$, $\bar{z} = 1.5$:

$$\sum (x_i - \bar{x})^2 = 10, \qquad \sum (z_i - \bar{z})^2 = 9, \qquad \sum (x_i - \bar{x})(z_i - \bar{z}) = -4$$

The standard errors are:

$$se(\hat{b}) = \left(0.338 \cdot \frac{9}{10 \cdot 9 - (-4)^2}\right)^{1/2} = \left(0.338 \cdot \frac{9}{74}\right)^{1/2} \approx 0.203$$

$$se(\hat{c}) = \left(0.338 \cdot \frac{10}{74}\right)^{1/2} \approx 0.214, \qquad se(\hat{a}) \approx 1.22$$

Giving t-statistics:

$$t_{\hat{a}} = \frac{2.89}{1.22} \approx 2.37, \qquad t_{\hat{b}} = \frac{0.28}{0.203} \approx 1.38, \qquad t_{\hat{c}} = \frac{-0.54}{0.214} \approx -2.52$$

With only $k = n - p = 1$ degree of freedom, the critical value at $\alpha = 0.05$(two-sided) is $t^* \approx 12.7$. None of the t-statistics exceed this critical value, so we reject none of the three null hypotheses. The estimated coefficients could have arisen through a random process.

The R output below verifies these calculations:

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.892      1.220    2.37    0.254
## x              0.284      0.203    1.40    0.395
## z             -0.541      0.214   -2.53    0.240
```

```
##             (Intercept)       x       z
## (Intercept)      1.4883 -0.2328 -0.1598
## x               -0.2328  0.0411  0.0183
## z               -0.1598  0.0183  0.0457
```

The variance-covariance matrix returned by `vcov()` shows the variances (diagonal) and covariances (off-diagonal) for the three coefficient estimators. The square root of the diagonal gives the standard errors shown in the coefficient table.

25.5 F-test for the Regression Model

Another commonly used statistical test in connection with regression analysis is the **F-test**, which tests whether a regression model as a whole can explain variation in Y better than a simpler model. Consider two models:

$$\text{Model 1 (unrestricted):} \quad y = b_0 + b_1 x_1 + \cdots + b_k x_k + u \tag{25.8}$$
$$\text{Model 2 (restricted):} \quad y = a + v \tag{25.9}$$

We formulate the null and alternative hypotheses:

$$H_0 : b_1 = b_2 = \cdots = b_k = 0, \qquad H_1 : \text{at least one } b_i \neq 0$$

Let $q = df_{m2} - df_{m1} = k$ be the number of restrictions (the number of slopes set to zero). The F-test statistic is:

$$F = \frac{SSR_{m2} - SSR_{m1}}{SSR_{m1}} \cdot \frac{n-k-1}{q} \tag{25.10}$$

where $SSR = \sum(y_i - \hat{y}_i)^2$ is the sum of squared residuals. An equivalent formula using R^2 is:

$$F = \frac{R^2}{1-R^2} \cdot \frac{n-k-1}{k} \tag{25.11}$$

The F-statistic follows an $F(q,\ n-k-1)$ distribution under H_0. We reject H_0 if $F > F^*_{q,\, n-k-1}$ at the chosen significance level α.

Partial F-test. We can also compare two nested models where the restricted model is not the intercept-only model. If we drop q variables from the unrestricted model (k variables, R_U^2) to get the restricted model (R_R^2):

$$F = \frac{(R_U^2 - R_R^2)/q}{(1 - R_U^2)/(n-k-1)} \tag{25.12}$$

Numerical example. Using the same four observations and the model $y_i = a + bx_i + cz_i + v_i$:

- Unrestricted m1: $SSR_{m1} \approx 0.338$
- Restricted m2 ($y = a + v$, i.e., $\bar{y} = 3.5$): $SSR_{m2} = \sum(y_i - \bar{y})^2 = 5$

With $n = 4$, $k_{m1} = 2$, $q = 2$:

$$F = \frac{5 - 0.338}{0.338} \cdot \frac{4-2-1}{2} = \frac{4.662}{0.338} \cdot \frac{1}{2} \approx 6.9$$

The critical value $F^*(2,1)$ at $\alpha = 0.05$ is approximately 199. Our $F = 6.9 < 199$, so we do not reject H_0. With only 1 degree of freedom in the denominator, the critical value is extremely large and almost nothing is significant — a consequence of having so few observations.

25.6 Chapter Summary

- The population model written with matrices $Y = XB + V$ describes a linear relationship between the explained Y and the explanatory variables in X. Given that the theoretical assumptions hold, the OLS estimator $\hat{B} = (X^T X)^{-1} X^T Y$ is unbiased: $E(\hat{B}) = B$.
- For each coefficient in B we can perform a two-sided t-test: $H_0 : b_j = 0$, $H_1 : b_j \neq 0$. The t-statistic is $t = \hat{b}_j / se(\hat{b}_j)$ where the standard errors come from the diagonal of the variance-covariance matrix.
- The estimated variance-covariance matrix is $var(\hat{B} \mid X) = \hat{s}_V^2 (X^T X)^{-1}$. The diagonal elements are the variances of the individual coefficient estimators; the off-diagonal elements are covariances between estimators. Including or excluding variables from the model affects both the coefficient estimates and the statistical tests for all coefficients.
- The F-test tests whether the regression model as a whole — all slope coefficients jointly — is statistically significantly different from zero. Under $H_0 : b_1 = \cdots = b_k = 0$ the F-statistic follows an $F(k, n-k-1)$ distribution.
- Statistical power applies to multiple regression as well. With a small sample, a larger estimated deviation is required to reject the null hypothesis.

25.7 Exercises

1. Multiple regression with $n = 100$, $k = 3$ regressors, $R^2 = 0.45$. F-statistic: $F = R^2(n-k-1)/[(1-R^2)k]$.

(a) Compute F.

(b) Critical value $F^*(3,96) \approx 2.70$. Reject H_0?

(c) What is H_0 in the F-test for overall significance?

Answer: (a) $F = 0.45 \times 96/(0.55 \times 3) \approx 26.2$.

(b) $26.2 > 2.70$, so yes, reject H_0.

(c) $H_0 : b_1 = b_2 = b_3 = 0$ — all slope coefficients are simultaneously equal to zero.

2. Estimated model: $\hat{y} = 2 + 3x_1 + x_2 - 2x_3$, $se(\hat{b}_1) = 1$, $se(\hat{b}_2) = 0.5$, $se(\hat{b}_3) = 0.8$.

(a) Compute t_1, t_2, t_3.

(b) Which coefficients are statistically significant at $\alpha = 0.05$ (critical value $|t| > 1.96$)?

(c) Interpret the significance of $\hat{b}_3 = -2$.

Answer: (a) $t_1 = 3/1 = 3$, $t_2 = 1/0.5 = 2$, $t_3 = -2/0.8 = -2.5$.

(b) All three: $|t_1| = 3 > 1.96$, $|t_2| = 2 > 1.96$, $|t_3| = 2.5 > 1.96$.

(c) x_3 has a statistically significant negative effect on Y: a one-unit increase in x_3 is associated with a 2-unit decrease in y, holding other variables constant.

3. A multiple regression has $n = 50$ observations and $k = 4$ regressors.

(a) How many degrees of freedom does the t-test have?

(b) If we add 3 more regressors (total $k = 7$), what are the new degrees of freedom?

(c) Why does losing degrees of freedom raise the critical t-value?

Answer: (a) $df = 50 - 4 - 1 = 45$.

(b) $df = 50 - 7 - 1 = 42$.

(c) Fewer observations relative to parameters means more uncertainty about each coefficient. The t-distribution has heavier tails with fewer degrees of freedom, requiring a larger t-statistic to reject H_0.

4. OLS gives $\hat{b}_1 = 5$, $se(\hat{b}_1) = 2$, $t^*(\alpha = 0.05) \approx 2.01$.

(a) Construct the 95% CI for b_1.

(b) Does the CI contain 0? Is $\hat{b}_1$ statistically significant?

(c) Does the CI contain 3? Can we reject $H_0 : b_1 = 3$?

Answer: (a) $5 \pm 2.01 \times 2 = (0.98,\ 9.02)$.

(b) 0 is outside $(0.98, 9.02)$ — $\hat{b}_1$ is statistically significant.

(c) 3 is inside $(0.98, 9.02)$ — we cannot reject $H_0 : b_1 = 3$ at $\alpha = 0.05$.

5. X_1 and X_2 are highly correlated ($r = 0.95$).

(a) What effect does high multicollinearity have on $se(\hat{b})$?

(b) How does this affect t-statistics?

(c) Can the F-test still be significant even if individual t-tests are not?

Answer: (a) High multicollinearity increases $se(\hat{b})$ — the estimates become imprecise because the data cannot distinguish the separate effects of X_1 and X_2.

(b) Larger se means smaller t-statistics, making it harder to reject H_0 for individual coefficients.

(c) Yes — the F-test evaluates joint significance, and the two variables together may still explain variation in Y even if their individual contributions cannot be separated.

6. Unrestricted model: $k = 3$, $R^2_U = 0.50$, $n = 100$. Restricted model (drop x_2 and x_3): $R^2_R = 0.40$. Partial F: $F = [(R^2_U - R^2_R)/q]/[(1 - R^2_U)/(n - k - 1)]$ with $q = 2$.

(a) Compute F.

(b) Critical value $F^*(2, 96) \approx 3.09$. Reject H_0?

(c) What is H_0 in this partial F-test?

Answer: (a) $F = (0.10/2)/(0.50/96) = 0.05/0.00521 \approx 9.6$.

(b) $9.6 > 3.09$, reject H_0.

(c) $H_0 : b_2 = b_3 = 0$ — the two dropped variables have no joint effect on Y.

7. Consider the variance-covariance matrix of $\hat{B}$ in a model with 2 slope coefficients (plus intercept).

(a) What are the dimensions of this matrix?

(b) What do the diagonal elements represent?

(c) What do the off-diagonal elements represent?

Answer: (a) 3×3 (including the intercept) or 2×2 if only the slopes are considered.

(b) The diagonal elements are the variances $se(\hat{b}_j)^2$ — the squared standard errors of individual coefficient estimates.

(c) The off-diagonal elements are covariances between pairs of coefficient estimates, showing how the estimation error in one coefficient is related to that of another.

8. In simple regression ($k = 1$), the F-statistic equals t^2.

(a) If $t = 2.5$, what is F?

(b) Why is the F-test always two-sided?

(c) Why is the F-test especially useful with multiple regressors?

Answer: (a) $F = 2.5^2 = 6.25$.

(b) Squaring t removes the sign, so F only detects the magnitude of deviation from zero — it cannot distinguish positive from negative effects.

(c) The F-test simultaneously tests whether all slopes are jointly zero, which cannot be done by multiple individual t-tests (due to the multiple testing problem and inter-correlations among regressors).

9. Model 1: $\hat{y} = a + \hat{b}_1 x_1$, $R^2 = 0.30$, $n = 50$. Model 2: $\hat{y} = a + \hat{b}_1 x_1 + \hat{b}_2 x_2$, $R^2 = 0.31$.

(a) Does Model 2 always fit better according to R^2?

(b) Adjusted $\bar{R}^2$ penalizes for extra parameters. Should x_2 be included?

(c) Use the partial F-test: if $t_2 = 0.8$ and critical value is 1.96, should x_2 be retained?

Answer: (a) Yes — adding a variable never lowers R^2. This is why R^2 alone cannot tell us whether to include an extra variable.

(b) If $\bar{R}^2$ increases when x_2 is added, include it; if it decreases, exclude it. With only a 1-percentage-point gain in R^2 and one extra parameter, $\bar{R}^2$ likely decreases.

(c) $t_2 = 0.8 < 1.96$ — $\hat{b}_2$ is not statistically significant. Drop x_2.

10. A researcher estimates $\hat{b}_2 = 0.5$, $se(\hat{b}_2) = 0.25$, $n = 200$, $k = 5$.

(a) Compute t_2.

(b) Construct a 95% CI (use $t^* \approx 1.97$ for $df = 194$).

(c) Reject $H_0 : b_2 = 0$ at $\alpha = 0.05$?

Answer: (a) $t_2 = 0.5/0.25 = 2$.

(b) $0.5 \pm 1.97 \times 0.25 \approx (0.01,\ 0.99)$.

(c) $t = 2 > 1.97$, so yes, reject H_0. The CI does not contain 0, confirming significance.

Chapter 26

Conditions of the Least Squares Method

In the previous chapters we mentioned different starting points and assumptions for regression analysis with the least squares method. This chapter goes through these assumptions in somewhat more detail and discusses their implications.

26.1 Gauss-Markov

The theoretical assumptions for the least squares method are described in the **Gauss-Markov theorem**, named after Carl Friedrich Gauss and Andrey Markov. The theorem describes the conditions for the least squares method estimator to have the smallest sampling variance among all linear unbiased estimators. The word "unbiased" refers to the model estimating the population value correctly. The conditions can be presented in slightly different ways and the description depends partly on what type of data we are studying.

If one or more of these conditions are not fulfilled, the results we estimate using the least squares method risk becoming misleading. When we analyze real data, it is common that one or more conditions are not fulfilled. This may sound discouraging, but fortunately there are also several established methods for handling these challenges. This chapter provides occasional simple examples of solutions, but for those interested there is a large amount of literature to delve deeper into.

26.2 Assumption 1: Linearity

We start from the following regression model for the population, described with matrices:

$$Y = XB + V \tag{26.1}$$

where we have n number of observations and k number of explanatory variables $x_1, \dots, x_k$ and $k+1$ number of coefficients $b_0, b_1, \dots, b_k$. The letters in the regression model symbolize the same matrices as described in previous examples.

The regression model describes a linear relationship that is found in the population. Linearity means that all coefficients in a regression model have exponent $= 1$. The following models are not linear:

$$Y = a + \frac{1}{b}X + e$$
$$\log Z = \alpha + \beta D + \epsilon$$

In the first regression model we have $\frac{1}{b} = b^{-1}$. In the second model we have that $\log(Z_1 + Z_2) \neq \log(Z_1) + \log(Z_2)$, where Z_1 and Z_2 are two arbitrary values in variable Z.

Table 26.1: US GDP 1800–2000. Data from Measuring Worth (www.measuringworth.org/usgdp/).

Year	GDP (billions USD)	ln GDP
1800	0.486	-0.7
1810	0.714	-0.3
1820	0.718	-0.3
1830	1.032	0.0
1840	1.590	0.5
1850	2.656	1.0
1860	4.410	1.5
1870	7.899	2.1
1880	10.592	2.4
1890	15.607	2.7
1900	21.197	3.1
1910	33.746	3.5
1920	89.246	4.5
1930	92.160	4.5
1940	102.899	4.6
1950	299.827	5.7
1960	542.382	6.3
1970	1073.303	7.0
1980	2857.307	8.0
1990	5963.144	8.7
2000	10250.952	9.2

It can be difficult to test whether a regression model is a good linear approximation of the data we are studying. We may for instance inspect the data in a scatter graph or histogram, and compare different measures of dispersion. We can both compare how the explained variable covaries with our explanatory variables, and compare how the residuals are distributed around a regression line.

Let us illustrate this by comparing US GDP every tenth year during the period 1800–2000 using data from Measuring Worth,[1] see Table 26.1, with years, GDP and logarithmic GDP. In Figure 26.1 the linear trend over time for GDP and ln(GDP) respectively is illustrated in separate graphs. In the left graph the trend is estimated based on the regression model:

$$Y_t = a + b\,\mathrm{YEAR}_t + V_t$$

where Y_t is GDP in year t and YEAR_t is the year variable. In the right graph we have estimated the regression model:

$$\ln Y_t = c + d\,\mathrm{YEAR}_t + U_t \tag{26.2}$$

where $\ln Y_t$ is logarithmic GDP in year t.

A linear model fits the data relatively poorly when we use GDP, which we see in the left graph. The data points in the graph are placed approximately in the shape of a lying L. The development is exponential and not linear. The regression line does capture the positive development but does not give a representative picture of the long-term trend in GDP. During the years 1800–1850, to the left in the graph, the regression line is below 0. This means that the regression model predicts that GDP was negative all years before 1850. Since GDP is a measure of everything that is produced, bought and sold in a society, this is not possible. The linear model does however capture the long-term trend in logarithmic GDP relatively well, which is seen in the right graph where the points follow the regression line for all the years 1800–2000.

[1] Data available at www.measuringworth.org/usgdp/.

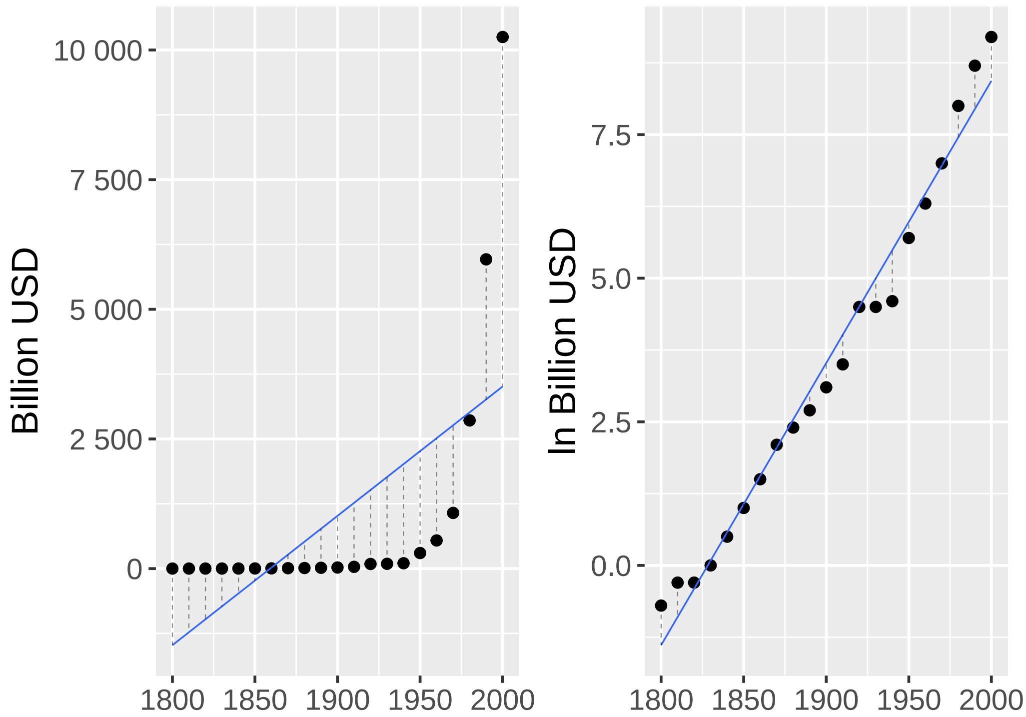


Figure 26.1: Linear trend for GDP and ln GDP.

Another way to compare how well the linear regression model captures the covariation in the data is to compare the residuals. In the two graphs vertical lines connect each point and the regression line. In the left graph the distance between the points and the regression line varies considerably — in a way that they do not in the right graph.

Logarithms can often be used to create a more linear relationship between variables. This often works well for exponential series. Unfortunately there is no general solution that works for all regression models. Note however how the data points in the right graph do not describe an exact linear relationship.

In practice, we often encounter populations with variables with nonlinear relationships. But even then we benefit from a linear regression model as a linear approximation, rather than an exact description of the population. When this is more or less suitable depends on what the purpose of the analysis is.

26.3 Assumption 2: Independent and Identically Distributed Observations

In the regression model the explained variable Y is a random variable drawn from the population. The explanatory variables in matrix X can either be random (as in an observational study) or created through a controlled process, for example in a controlled experiment where we ourselves design the treatment. Regardless, none of the variables are correlated with the error term.

In the description of the least squares method we assume that each observation $(y_i, x_{1,i}, \dots, x_{k,i})$, $i = 1, \dots, n$, is **independent and identically distributed** (i.i.d.). That is, each observation is drawn in a random sample from a population with the same distribution. That the observations are independent means that the observations do not covary with each other in a systematic way.

Say for example that we want to study the covariation between some characteristics that people have. The characteristics are our variables and each person has their own observation. We draw a random sample from the population, which fulfills the assumption of i.i.d.

But say now that we want to follow people over time, so that each observation is information about a person at a point in time. One of the variables is earnings. Even though people's earnings vary over time, we will be able to see patterns where different observations can be followed over time. The observations for a person's earnings are therefore not independent of each other and therefore also do not fulfill the assumption of i.i.d.

26.4 Assumption 3: Expected Value and Variance of the Error Terms

The expected value of the error terms with respect to the explanatory variables X is $E(V|X) = 0$. This also means that the error terms V do not covary with any of the explanatory variables X. This can be described as us having **exogenous** explanatory variables X. If any explanatory variable $x_1, \dots, x_k$ is correlated with the error term v, this is called the variable being **endogenous**. In addition we assume that the error terms V have the same variance, regardless of observation and variable:

$$var(V|X) = \sigma^2 I_n$$

where the symbol σ^2 symbolizes the population's constant variance and I_n is an $n \times n$ identity matrix. That the error term exhibits constant variance is usually called **homoscedasticity**. Its opposite is called **heteroscedasticity**, which means that the variance in the error term varies across the observations in one or more explanatory variables X.

Figure 26.2 illustrates heteroscedasticity by comparing the residuals' covariation against the explanatory variable X. The points' distance to the regression line varies across X. Further to the right in the graph the distance between the line and the points is larger compared to the points further to the left. The residuals' variance is thus not constant (homoscedasticity) across different observations but varies (heteroscedasticity).

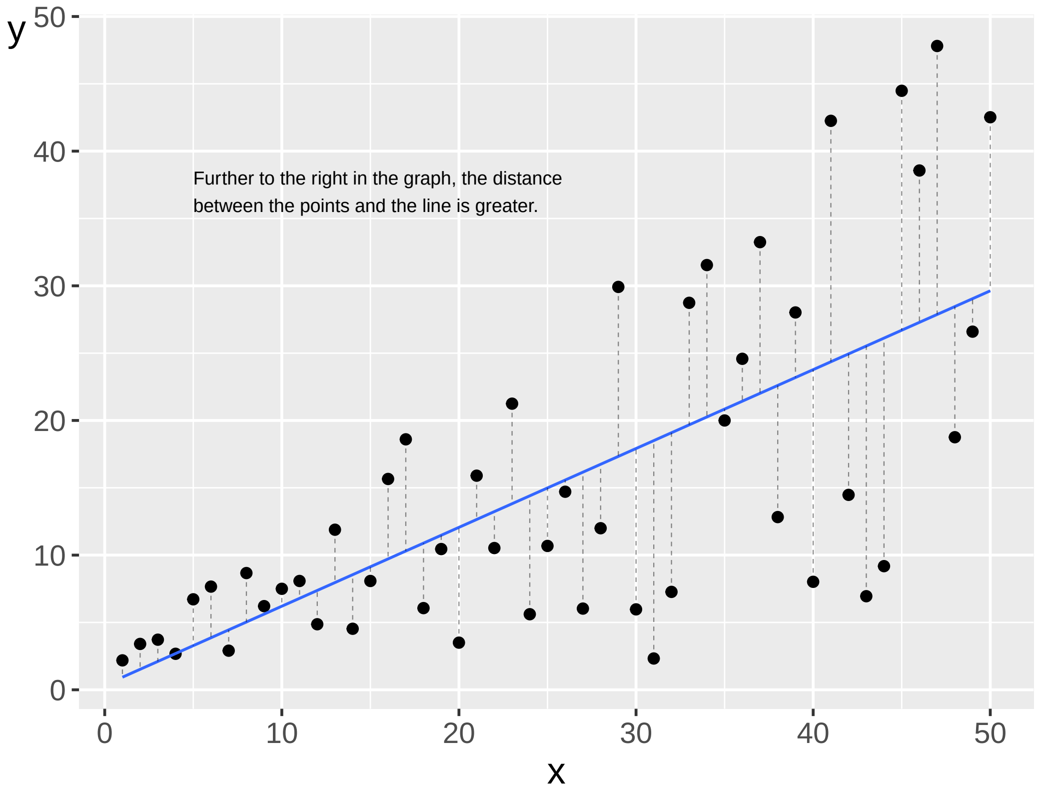


Figure 26.2: Heteroscedasticity: the distance between points and the regression line increases with X.

A common assumption for the error terms is that these follow a normal distribution (see Section 26.7 below). Combined with the assumption of homoscedasticity this means that the error terms' dispersion around the regression line follows a normal distribution with the same shape, regardless of which values for the explanatory variables X we compare against. With heteroscedasticity we may for example have normally distributed error terms but with different variance. Figure 26.3 illustrates an example where we have three normal distributions with different variance along a regression line.

Figure 26.3: Normal distributions with different variance along a regression line.

If the error term's variance is not constant across different observations (heteroscedasticity), this can lead to the statistical tests (for example t-test and F-test) giving incorrect results. It is therefore often important to check for heteroscedasticity, which can be done in different ways. One way is to compare the dispersion of the residuals in graphs.

Above we described how we in many situations can use a linear regression model to describe a linear approximation for a relationship between two variables that is actually nonlinear. In such cases it is also common that the error terms do not have constant variance, that is, we find heteroscedasticity.

26.5 Assumption 4: Multicollinearity

The explanatory variables in the regression model must not correlate too strongly with each other. If a variable is a linear function of another variable in the regression model this is called them exhibiting **collinearity**. If several variables exhibit high correlation this is called **multicollinearity**. Suppose we have the explanatory variable x_1= earnings in USD, and x_2= earnings in thousands of USD. Variable x_1 can be described as a linear function of x_2: $x_1 = 1{,}000 \cdot x_2$. The relationship between x_1 and x_2 is called *perfect collinearity.* If x_1 is included in the regression model then it does not add any information to our analysis to also add x_2.

Multicollinearity can affect the results in the regression analysis and disturb the calculation of the slope coefficients. In addition it can cause the variance in slope coefficients to become larger than what would otherwise have been the case. This can in turn reduce the probability that we in a t-test discover a statistically significant covariation between the variables concerned and the explained variable. Instead we risk accepting a false null hypothesis, a type 2 error.

The assumption about the absence of multicollinearity does not mean that two or more variables in a regression model may not covary at all. In regression analysis it is common that many variables have some form of relationship and that they therefore often covary more or less. The problem arises when they exhibit too high correlation. Perfect collinearity is usually easy to remedy by removing one of the variables from the regression model. Sometimes however it happens that two variables happen to have high correlation. In such cases it is difficult to give general recommendations.

To detect collinearity we can study covariation between variables in graphs. A common relatively simple check is to estimate the correlation coefficient r_{xy} for all pairs of explanatory variables. There is no exact value for r_{xy} that guarantees high collinearity, but a common recommendation is that if two explanatory variables exhibit $r_{xy} > 0.8$, this should be regarded as an indication of problematically high collinearity.

Another common method for checking the occurrence of multicollinearity is to estimate the **Variance Inflation Factor (VIF)**. Suppose we have the following regression model:

$$Y = a + bX + cZ + V \tag{26.3}$$

To estimate VIF we take the auxiliary regression:

$$X = \beta_1 + \beta_2 Z + U$$

We estimate R^2 for this regression and thereafter the VIF factor for $\hat{b}$:

$$VIF(\hat{b}) = \frac{1}{1 - R^2} \tag{26.4}$$

A rule of thumb is that if the VIF factor for an explanatory variable is over 5 or 10, this indicates high multicollinearity.

26.6 Assumption 5: Correlation Between Residuals

The error terms in the regression model are assumed to be uncorrelated, which can be described as the covariance $cov(v_i, v_j) = 0$ for observation i and j(where i and j refer to any observations). Another way to describe this is that the error term should not covary with itself. This phenomenon is called **autocorrelation**, or serial correlation. We usually do not know the error terms in the population but we may examine the occurrence of autocorrelation among our estimated residuals.

Autocorrelation can for example arise if our regression model estimates the covariation between variables over time. We illustrate this by again comparing the example with the linear trend for the natural log of GDP (Section 26.2):

$$\ln Y_t = c + d\,\mathrm{YEAR}_t + U_t$$

Figure 26.4 illustrates the covariation between the estimated residuals $\hat{U}_t$ from this model. In the left graph only the residuals are shown, where the marked horizontal line at $Y = 0$ is the regression line. The points in the left graph (the residuals) form the shape of a U. Residuals that are positive, above the 0-line, are often followed by another positive residual. And the residuals that are negative, below the 0-line, are often followed by another negative residual.

The pattern in the left graph is confirmed in the right graph where we instead see the covariation between residual $\hat{U}_t$ and the residual for the previous year $\hat{U}_{t-1}$. The straight diagonal line in the graph is the regression line from the regression model $\hat{U}_t = \beta\hat{U}_{t-1} + \epsilon$. The correlation is positive, which indicates autocorrelation.

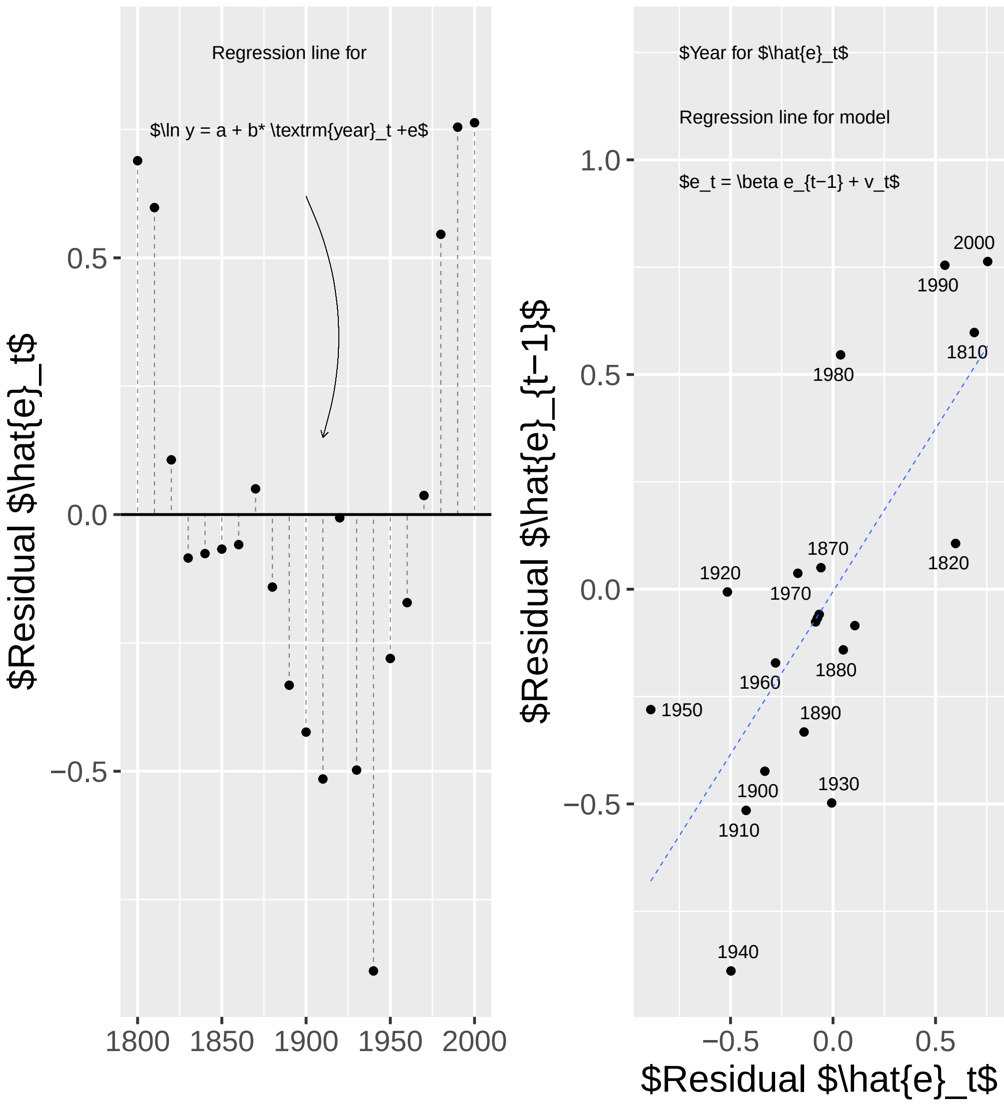


Figure 26.4: Residuals from the ln GDP regression model.

Autocorrelation does not affect the coefficients in our original regression model. But autocorrelation can lead to the standard error becoming smaller than it actually is, which in turn can cause the t-tests for the slope coefficients to give a higher t-value than otherwise and we may reject a true null hypothesis (type 1 error).

That is, in the t-test for slope coefficient b we estimate the t-value $t = \frac{\hat{b}-b_0}{se(\hat{b})}$, where $se(\hat{b})$ is the standard error for estimated $\hat{b}$. If we test the hypothesis that $b = 0$ then $b_0 = 0$ and $t = \frac{\hat{b}}{se(\hat{b})}$. If the standard error in the denominator of the t-value decreases then our estimated t-value increases and we will then get a value further out in the tails of the t-distribution.

26.7 Assumption 6: Normally Distributed Error Terms

In addition to the above conditions for the least squares method, an additional condition is often mentioned in this context: that the error term is normally distributed across all observations and explanatory variables in X. We describe this as $V|X \sim N(0, \sigma^2 I_n)$ where σ^2 is the variance and I_n is an $n \times n$ identity matrix. Given that the error terms and the residuals are approximately normally distributed we use t-tests to calculate the probability that our estimated coefficients are a result of a random process.

Even if we have good reasons to expect that the error term follows a normal distribution, in practice error terms can follow any distribution. This condition too we therefore check by studying the residuals.

A first step can be to compare how the residuals are distributed in graphs. Let us illustrate this using the same data for logarithmic GDP from Figure 26.1. The regression model is $\ln Y_t = c + d\,\text{YEAR}_t + U_t$. If we take the estimated residuals from that model and compare how they are distributed around their own mean, we get the distribution shown in Figure 26.5.

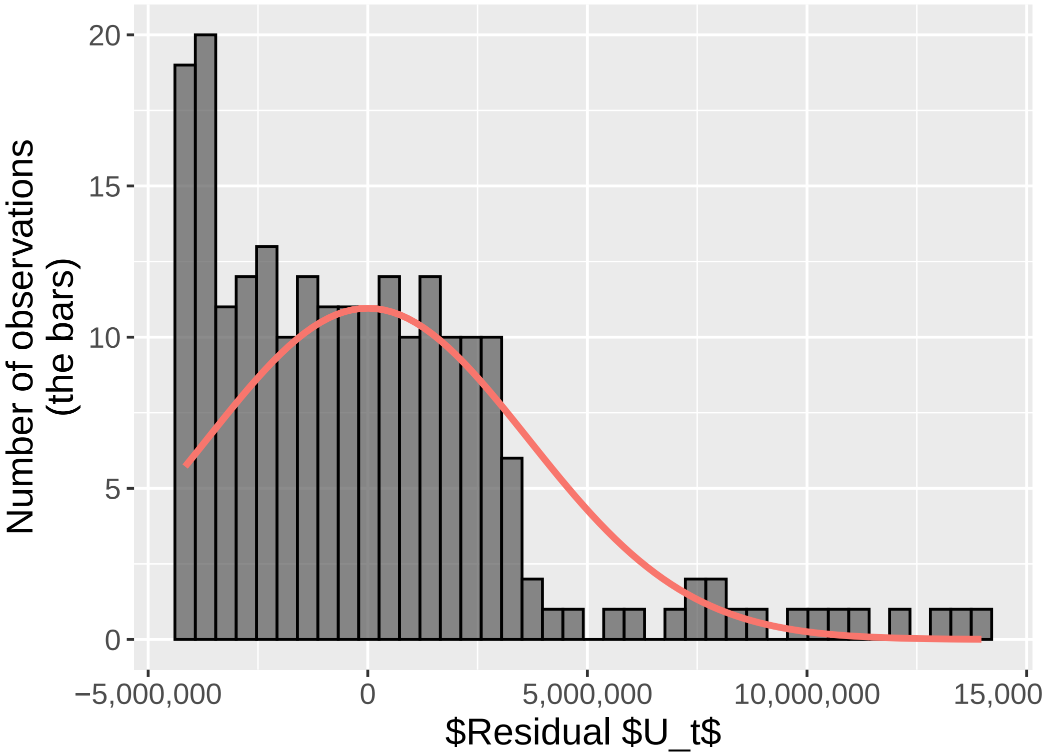


Figure 26.5: Residuals from the ln GDP regression model.

The bars show the number of residuals. In the graph we have also drawn a curve for a normal distribution with the same mean and standard deviation as the estimated residuals. The bars do not really follow the normal distribution curve. The residuals show a U-shaped pattern over time (as seen in Figure 26.4), which means their distribution is also non-normal — with fewer observations near zero than a normal distribution would imply.

Let us look at another example. Consider a regression model:

$$L_i = a + bM_i + e_i \tag{26.5}$$

where L_i is mean life expectancy for the inhabitants in country i and M_i is GDP per capita. Figure 26.6 illustrates estimated results from this regression model in two graphs. In the graph to the left we see a positive covariation between life expectancy and earnings. In the graph to the right the residuals are shown, where the bars show the proportional distribution of the residuals and a normal distribution density curve with the same mean and variance as the residuals.

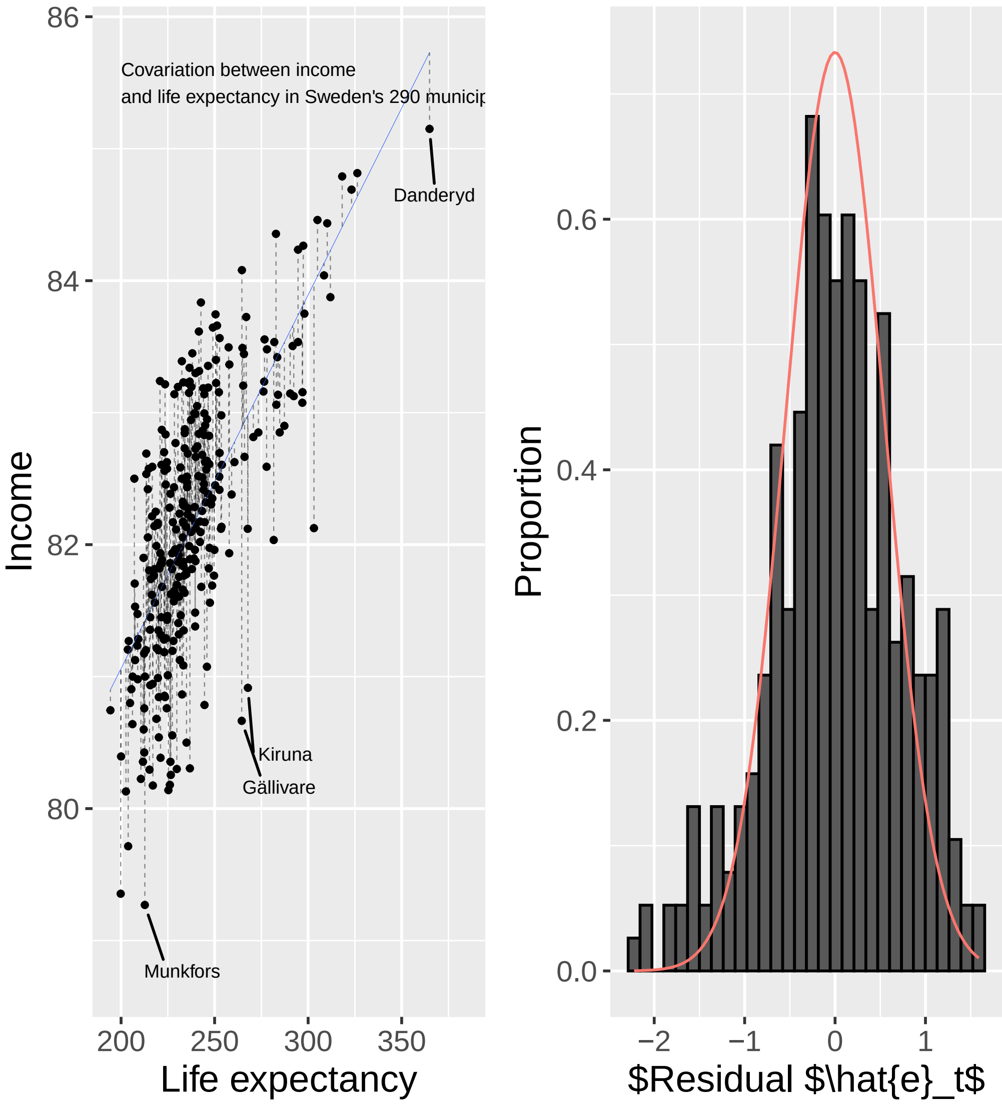


Figure 26.6: Life expectancy and income in Sweden's 290 municipalities.

We compare the bars to the density curve to judge if the residuals' distribution resembles a normal distribution. Whether we accept the residuals' distribution as an indication that the population error terms follow a normal distribution depends, at least partly, on how big deviations from a normal distribution we wish to tolerate.

Sometimes a transformation of the variables in the regression model, for example taking logarithms, can lead to the residuals following a normal distribution. There are also specific methods for how to perform a regression analysis and instead assume that the error terms follow other types of probability distribution, which there is no room to go through here.

26.8 Adjusted Standard Errors

In this chapter we have gone through the conditions of the least squares method as well as examples of phenomena that violate these. Some of these deviating phenomena can be remedied by thinking through the model's design. Some deviating phenomena can be handled by adjusting the data we study.

There are also many well-known alternative methods for estimating regression models and using statistical tests. A well-known such example is what is called **robust standard errors**, also called heteroscedasticity-robust standard errors or heteroscedasticity-consistent standard errors. Sometimes the same thing is called White's standard errors after an article by Halbert White (1980) that contributed to the spread of this method within economics. Here follows a brief description, largely based on Hansen

Table 26.2: Variables y, x and z.

y	x	z
3	3	1
2	4	4
5	6	0
4	7	1

(2022) chapter 4. Many computer programs for statistical analysis have simple shortcut commands for all such calculations.

Robust standard errors is a summary term for a type of adjustment that we can perform on the residuals so that these can be used for regression analysis based on the least squares method, even in the presence of heteroscedasticity. Suppose we have the following regression model for the population:

$$Y = XB + V$$

where Y and X are matrices with explained and explanatory variables respectively, B is the coefficients and V is the error terms. We have previously defined the estimator $\hat{B} = (X^T X)^{-1} X^T Y$ and the variance-covariance matrix:

$$var(\hat{B}|X) = (X^T X)^{-1} X^T E[VV^T|X] X (X^T X)^{-1} \tag{26.6}$$

Given that the error terms exhibit constant variance σ^2(homoscedasticity), $E(VV^T|X) = \sigma^2 I$ and $var(\hat{B}|X) = \sigma^2 (X^T X)^{-1}$. If we work with sample data it is common that the residuals exhibit irregular variance (heteroscedasticity). What we now seek is an alternative version of the variance-covariance matrix that takes into account whether we have heteroscedasticity in our data.

It is possible to estimate the residuals' variance as $\hat{s}^2 = \frac{\sum(y_i - \hat{y}_i)^2}{(n-p)}$. The letters V in the expression $E(VV^T|X)$ are the error terms in the population, which we do not know. Instead we use the estimated residuals $\hat{V}$ and get $E(\hat{V}\hat{V}^T|X) = \text{diag}(\hat{v}_1^2, \hat{v}_2^2, \ldots, \hat{v}_n^2)$, which is a diagonal matrix where the diagonal elements consist of the squared estimated residuals. We insert this into the variance-covariance matrix in equation (26.6) and rewrite:

$$var(\hat{B}|X)^{\text{HC0}} = (X^T X)^{-1} \left(\sum_{i=1}^{n} X_i X_i^T \hat{v}_i^2 \right) (X^T X)^{-1} \tag{26.7}$$

This is a variant of the variance-covariance matrix, robust for heteroscedasticity. We mark this version of the variance-covariance matrix with "HC0": $var(\hat{B}|X)^{\text{HC0}}$. Estimated results in $var(\hat{B}|X)^{\text{HC0}}$ we use to for example estimate t-tests for the coefficients in regression models where the data exhibit heteroscedasticity.

In the chapter on multiple regression we saw how the variance-covariance matrix $var(\hat{B}|X)$ is symmetric, the elements in the diagonal in $var(\hat{B}|X)$ are the variance for each coefficient's estimator, while the other elements are covariance for the coefficients' estimators. The difference with $var(\hat{B}|X)^{\text{HC0}}$ consists in how the estimates are estimated.

There are also additional variants of $var(\hat{B}|X)$. The following version is often denoted $var(\hat{B})^{\text{HC1}}$ and aims to adjust for $\hat{v}_i^2$ risking becoming too low:

$$var(\hat{B})^{\text{HC1}} = \frac{n}{n-k} (X^T X)^{-1} \left(\sum_{i=1}^{n} X_i X_i^T \hat{v}_i^2 \right) (X^T X)^{-1} \tag{26.8}$$

where n is the number of observations and k is the number of coefficients in the regression model. The factor $\frac{n}{n-k}$ means that HC1 is often recommended over HC0. We illustrate the differences between the three versions of the variance-covariance matrix $var(\hat{B})$, $var(\hat{B})^{\text{HC0}}$ and $var(\hat{B})^{\text{HC1}}$ with the three variables y, x and z, the four observations in Table 26.2 and the following regression model:

$$Y = XB + V, \qquad \begin{bmatrix} 3 \\ 2 \\ 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 4 & 4 \\ 1 & 6 & 0 \\ 1 & 7 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

where V is the error terms. We estimated this regression model in a previous chapter, found the residuals $\hat{V} \approx \{-0.2, 0.14, 0.41, -0.34\}$ and the variance $var(\hat{V}) = \hat{s}^2 \approx 0.338$. We do not go through the calculations in detail here but primarily describe the results:

$$\begin{aligned} var(\hat{B}|X) &= \hat{s}^2 (X^T X)^{-1} \\ &\approx \begin{bmatrix} 1.49 & -0.23 & -0.16 \\ -0.23 & 0.04 & 0.02 \\ -0.16 & 0.02 & 0.05 \end{bmatrix} \end{aligned} \tag{26.9}$$

For $var(\hat{B}|X)^{\text{HC0}}$ we get:

$$\begin{aligned} var(\hat{B}|X)^{\text{HC0}} &= (X^T X)^{-1} \left(\sum_i^n X_i X_i^T \hat{e}_i^2 \right) (X^T X)^{-1} \\ &\approx \begin{bmatrix} 0.247 & -0.041 & -0.025 \\ -0.041 & 0.0086 & 0.002 \\ -0.025 & 0.002 & 0.0066 \end{bmatrix} \end{aligned} \tag{26.10}$$

For $var(\hat{B}|X)^{\text{HC1}}$:

$$\begin{aligned} var(\hat{B}|X)^{\text{HC1}} &= \left(\frac{n}{n-k} \right) (X^T X)^{-1} \left(\sum_i^n X_i X_i^T \hat{e}_i^2 \right) (X^T X)^{-1} \\ &\approx \begin{bmatrix} 0.987 & -0.164 & -0.102 \\ -0.164 & 0.034 & 0.008 \\ -0.102 & 0.008 & 0.026 \end{bmatrix} \end{aligned} \tag{26.11}$$

where $\frac{n}{n-k} = \frac{4}{4-3} = 4$. Note how the elements in the diagonal for each variance-covariance matrix represent the variance for the coefficients' estimators. Of these three matrices we get the highest values in $var(\hat{B}|X)$. Second highest values are given by $var(\hat{B}|X)^{\text{HC1}}$, while the lowest variance for the coefficients' estimators is found in $var(\hat{B}|X)^{\text{HC0}}$. These are three variants of the variance-covariance matrix among many conceivable ones.

26.9 Chapter Summary

- The Gauss-Markov theorem describes the theoretical assumptions that must hold for the least squares method's estimator to have the smallest sampling variance among all linear unbiased estimators. A common, but not necessary, assumption in regression analysis with the least squares method is that the error terms follow a normal distribution.
- It is common that one or several of the assumptions in the Gauss-Markov theorem are not fulfilled. There are different methods and tests to check if the conditions are fulfilled as well as different methods to handle deviations from the conditions.
- As an example, in the presence of heteroscedasticity one can use adjusted standard errors (robust standard errors).

26.10 Exercises

1. The Gauss-Markov theorem states that OLS is BLUE when its assumptions hold. (a) What does BLUE stand for? (b) Name three assumptions of the Gauss-Markov theorem. (c) What is the key consequence if OLS is BLUE?

Answer: (a) Best Linear Unbiased Estimator. (b) Any three of: $E(\varepsilon|X) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$ (homoscedasticity), no perfect multicollinearity, no autocorrelation. (c) OLS has the smallest variance among all linear unbiased estimators.

2. The true relationship is $Y = aX^2 + \varepsilon$ but we estimate $\hat{y} = \hat{a} + \hat{b}X$. (a) Which Gauss-Markov assumption is violated? (b) How can non-linearity be detected visually? (c) How can we address non-linearity while still using OLS?

Answer: (a) The linearity assumption. (b) A residuals-vs-fitted plot will show a curved pattern. (c) Include X^2 as an additional regressor.

3. Homoscedasticity assumes $\text{Var}(\varepsilon_i) = \sigma^2$ for all i. (a) What is heteroscedasticity? (b) A residuals-vs-fitted plot shows variance growing with $\hat{y}$. What does this indicate? (c) Does heteroscedasticity bias $\hat{b}$?

Answer: (a) $\text{Var}(\varepsilon_i)$ varies across observations. (b) Heteroscedasticity — the variance of the error term grows with the fitted values. (c) No — $\hat{b}$ remains unbiased; only the standard errors are affected.

4. X_1 and X_2 have correlation $r = 0.99$. (a) What is this problem called? (b) What happens to $\text{se}(\hat{b})$ when regressors are nearly collinear? (c) Name one practical remedy.

Answer: (a) Multicollinearity. (b) $\text{se}(\hat{b})$ inflates — estimates become less precise. (c) Drop one of the collinear variables, or collect more data.

5. The Gauss-Markov theorem requires $E(\varepsilon|X) = 0$. (a) What does this assumption mean in words? (b) What is violated if an omitted variable Z affects Y and is correlated with X? (c) Is $\hat{b}$ biased in this case?

Answer: (a) Errors are uncorrelated with X on average — the model is correctly specified. (b) The assumption $E(\varepsilon|X) = 0$ is violated. (c) Yes — omitted variable bias.

6. Autocorrelation in error terms: $\text{Cov}(\varepsilon_t, \varepsilon_{t-1}) \neq 0$. (a) In which type of data is autocorrelation most common? (b) A plot of residuals vs. time shows a clear wave pattern. What does this suggest? (c) What is one consequence for OLS standard errors?

Answer: (a) Time series data. (b) Autocorrelation in the residuals. (c) OLS standard errors are unreliable — typically underestimated, inflating t-statistics.

7. Three OLS violations and their consequences. (a) Heteroscedasticity: is $\hat{b}$ unbiased? (b) Omitted variable correlated with X: is $\hat{b}$ unbiased? (c) Measurement error in X: is $\hat{b}$ biased?

Answer: (a) Yes — heteroscedasticity does not bias $\hat{b}$. (b) No — an omitted variable correlated with X causes omitted variable bias. (c) Yes — attenuation bias (the coefficient is biased toward zero).

8. The Gauss-Markov theorem does not require normally distributed errors. (a) Is OLS unbiasedness affected by non-normal errors? (b) When is normality of ε important for exact inference? (c) With $n = 1000$, does non-normality of ε strongly affect t-tests? Why?

Answer: (a) No — unbiasedness holds regardless of the error distribution. (b) Normality is needed for exact small-sample t-tests and F-tests. (c) No — by the Central Limit Theorem, $\hat{b}$ is approximately normally distributed for large n.

9. Robust standard errors. OLS: $\hat{b} = 3$, $\text{se}_{\text{OLS}} = 0.5$. Robust: $\text{se}_{\text{robust}} = 0.8$. (a) Which standard error is more reliable under heteroscedasticity? (b) Does using robust SE change $\hat{b}$? (c) If $\text{se}_{\text{robust}} = 2$ instead, compute the robust t and decide whether to reject $H_0 : b = 0$ at $\alpha = 0.05$.

Answer: (a) The robust standard error is more reliable under heteroscedasticity. (b) No — only the standard error changes, not the coefficient estimate. (c) $t = 3/2 = 1.5 < 1.96$, so we do not reject H_0.

10. Diagnosis — identify the most likely problem in each scenario. (a) Residuals vs. fitted plot shows a funnel/fan shape. (b) Residuals vs. time plot shows a regular wave pattern. (c) Correlation matrix shows

$r(X_1, X_2) = 0.97$.

Answer: (a) Heteroscedasticity. (b) Autocorrelation. (c) Multicollinearity.

11. Model specification. (a) Adding a relevant omitted variable Z (correlated with X): how does $\hat{b}$ change? (b) Adding an irrelevant variable: what happens to $\hat{b}$ and efficiency? (c) Name the fundamental trade-off in choosing covariates.

Answer: (a) $\hat{b}$ changes — the original estimate suffered from omitted variable bias. (b) $\hat{b}$ is unaffected on average but the standard error increases (less efficiency). (c) Omitted variable bias vs. efficiency loss from over-controlling.

12. Heteroscedasticity in a wage regression: $\hat{y} = 10000 + 2000x$ where x is years of education. (a) Why might errors have larger variance for high-education workers? (b) What is the effect on OLS standard errors if heteroscedasticity is ignored? (c) Name the HC0 and HC1 variants of robust standard errors in terms of what they estimate.

Answer: (a) High earners have more variability in wages — bonuses, equity, and career diversity grow with education. (b) Standard errors are biased and inference is unreliable. (c) HC0 and HC1 are heteroscedasticity-consistent estimators of the variance-covariance matrix; HC1 applies a small-sample degrees-of-freedom correction by multiplying by $n/(n-k)$.

Chapter 27

Regression Analysis for Causal Relationships

This chapter introduces some examples of how we can use regression analysis to study causal relationships. The chapter provides a brief overview of some of the most well-known methods. Within each method there are many variants and also combinations of different methods, which cannot be described here.

27.1 We need special methods to study cause and effect

In chapter 20 we introduced counterfactual analysis for studying causal relationships. In the previous chapter we went through how we can include several explanatory variables in our regression model and thereby study the covariation between two variables with regard to the variation in other variables.

Say now, greatly simplified, that we have a theory that a medicine (X) reduces the amount of disease symptoms (Y) in patients. We have found a group of patients who all have the same amount of disease symptoms, which we have divided into a treatment group (that receives medicine) and a control group (that does not receive medicine). To examine the medicine's effect on the disease symptoms we set up the following regression model:

$$Y = \alpha + \beta X + \epsilon \tag{27.1}$$

By observing the covariation between medicine and disease symptoms the estimated $\hat{\beta}$ will give us a measure of the medicine's X effect on the disease symptoms Y. If the medicine works we expect that β will be negative, that is, that the medicine X reduces symptoms Y. One problem, among several possible ones, is that for us to be able to interpret $\hat{\beta}$ as a correct measure of the medicine we must be certain that no other phenomena affect the medicine and the disease symptoms.

Suppose that we know from previous research that there are several other things that both affect the medicine's effect and the disease symptom itself. This can involve things like gender, age and so forth that we can include in our study.

But in addition there are some additional phenomena that we have strong reasons to believe are important for both the medicine and the symptoms, but for which we lack data. And if we lack data, we cannot use it in our regression model.

27.2 Illustration with a DAG

Figure 27.1 illustrates an example of how this sort of problem can look. The arrow from X to Y describes the causal relationship that we are interested in. For us to be able to estimate what effect X has on Y we must partly establish that when a variation in X occurs, caused without connection to Y, then a variation in Y also occurs. In this case we decide that the treatment group takes the medicine, which is the induced variation in X. This is called exogenous variation, that is, a change in X that comes "from outside". The variation does not come from Y.

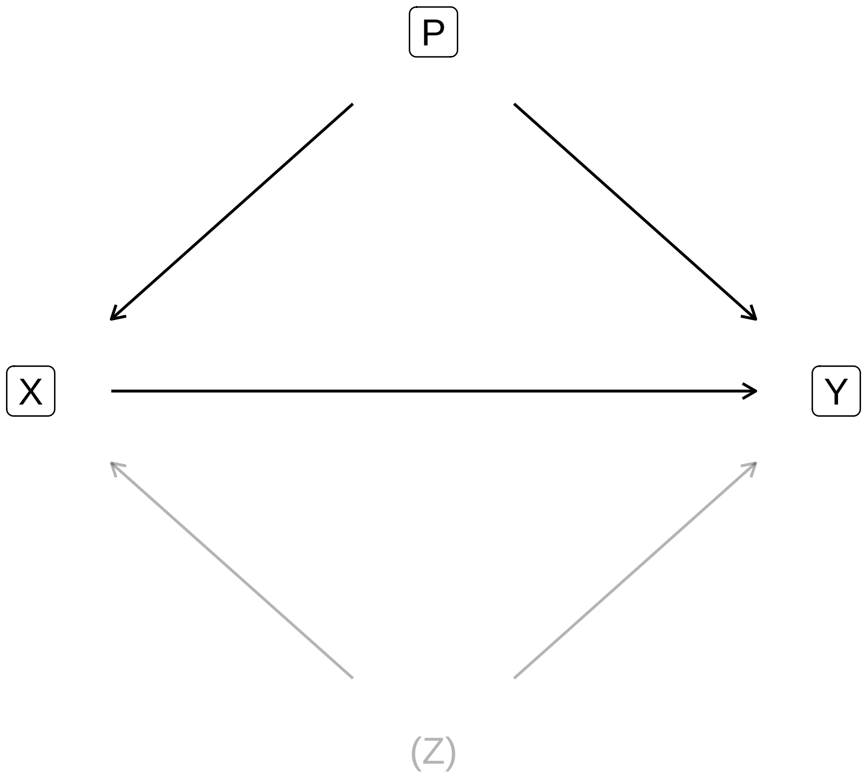


Figure 27.1: To estimate the effect of X on Y, we must control for P and Z

But to estimate the effect of X on Y we must also take into account the two phenomena P and Z that affect both X and Y. Phenomenon P is observable but phenomenon Z refers to something for which data is lacking and for which there may never be data. If we do not adjust our analysis for P and Z, the variations we observe in X and Y, as well as the covariation between them, will be wholly or partly caused by P and Z.

The graph in figure 27.1 is called a Directed Acyclic Graph (DAG). The graph describes an example and additional challenges of a similar kind are also conceivable, but are not discussed here.

In the previous chapter we saw how the absence or presence of other variables can affect all results in the regression model. It is therefore important that we think carefully about how we design our study. If we perform a controlled experiment, for example randomly divide the patients into 2 groups and give one group medicine, our possibilities to control for all important phenomena increase. But not even a randomized experiment is any guarantee that our analysis will be correct. We must still think carefully about the conditions of our study and ensure, as best we can, that no other phenomenon affects our analysis.

To design our study and regression model we can produce different types of materials, study other covariations, try different solutions and compare results. This work is something that analysts must decide from case to case. Regardless of how advanced method or how many observations and variables we have, there is no method for proving exactly which variables should be included in an analysis.

Sometimes it is tempting to believe that if we find some incredibly smart theoretical model that describes how the entire world works, that is in itself proof of how a regression model should be designed. Ideas and theories are the foundation for all analysis but that is not in itself proof for including or excluding precisely the variables we have at our disposal. This type of difficulty has been well-known for a long time and discussed extensively within social science. Two recommended texts on the topic are Hendry (1980) and Leamer (1983).

27.3 Both the presence and absence of variables can affect our results

As we went through in section 20.3 it is often impossible to conduct controlled experiments within social science. It is easy to believe that this means we might as well make statements about causal relationships since no one knows anyway. This is simply not true. The difficulty of making statements about causal relationships does not lower requirements.

Instead of experiments, social scientists are often referred to quasi-experimental observational studies where treatment and control groups for example are created by analysts managing to identify a process that, without intention, distributes a sufficiently large amount of observations randomly into treatment and control. Even in these situations we must examine whether our assumptions in the analysis are reasonable, for example that the distribution really occurred randomly. One may for instance do this by comparing the treatment and control groups' characteristics and thereby check that the only important difference between the groups is the treatment itself. If the observable characteristics are equivalent it is sometimes save to assume that the non-observable characteristics are too.

Many such methods are based precisely on no phenomenon affecting the distribution (selection) to the treatment and control group. Say for example that we are to study what effect an education (X) will have on participants' future earnings (Y). If we compare the participants' earnings with the rest of the population we risk other non-observable characteristics (Z) of the participants both explaining why they participated in the education (X) and their earnings (Y). We therefore cannot estimate the effect in this way. Characteristic Z can for example be that the students on average have better conditions or higher ambitions than the population at large.

There are several different well-known methods for how to identify or create treatment and control groups in quasi-experimental studies. In this chapter some well-known such methods are introduced in overview and it is described how to use regression analysis to study cause and effect.

27.4 Regression discontinuity

Regression discontinuity is a method for identifying and comparing treatment and control groups in an observational study. The method is based on us being able to observe variations around a threshold with significance for both treatment and outcome. Small variations around the threshold can be assumed to be random, which is why this can be used to divide observations into treatment and control.

Let us again imagine that we are to study what effect an education has on participants' future earnings. Admission to the education occurs with an entrance exam where one must get at least 92 points to be admitted. We now think that those who got just over and under 92 points have equivalent skills. Some had a bad day and some had an unusually good day. The students who ended up just over 92 points and were admitted become our treatment group. The students who ended up just under 92 points and were not admitted become our control group. To check whether the treatment and control groups have equivalent skills we may compare the relevant characteristics, such as age, previous performance, previous education and so forth.

Consider another example. Many countries have age limits for when one may buy stronger alcoholic beverages. That alcohol is directly harmful and also indirectly increases the risk of various negative effects is well-known. At the same time it can be difficult to estimate in detail the harmful effects (phenomenon Y) of increased access to alcohol (phenomenon X) at the societal level since people who fare badly may have other destructive habits (phenomena P and Z).

One way to estimate the effect is to use the age limit for purchasing alcoholic beverages as a threshold to estimate variations in consumption and harm around the time when people get increased access to alcohol. Carpenter and Dobkin (2009) present such an analysis for the USA, where the age limit is 21 years. Their work shows that drinking increases relatively sharply shortly after the 21st birthday while traffic accidents, suicides and various alcohol-related deaths increase.

Exactly how we should design our regression model when we use regression discontinuity depends on what type of effects we want to study. We illustrate here two examples based on the data that Carpenter and Dobkin (2009) present. We begin with the following regression model:

$$Y_i = a_1 + a_2 X_i + a_3 T_i + u_i \tag{27.2}$$

where a_1, a_2 and a_3 are coefficients. Y_i is the number of alcohol-related deaths per 100,000 inhabitants in respective age group i, where each age group is divided into months: 19 years and 1 month, 19 years and 2 months, and so on. Variable X is age calculated in months. T is a dummy variable that is $T = 0$ for the age groups that are under 21 and $T = 1$ for the age groups that have turned 21. u_i is the error term for age group i. The coefficient a_3 that is multiplied by T will in this regression model describe the effect that the 21st birthday has on alcohol-related mortality.

The coefficient a_2 in the regression model in equation (27.2) estimates the covariation between mortality (Y) and age (X). But this covariation can also be thought to change when a person turns 21. To control for this we add an interaction term for X*T:

$$Y_i = b_1 + b_2 X_i + b_3 T_i + b_4 (X_i * T_i) + V_i \tag{27.3}$$

where b_1, b_2, b_3 and b_4 are coefficients, Y, X and T are the same variables as in the regression model in equation (27.2) and V is the error term. The interaction term $b_4 (X_i * T_i)$ estimates changes in the covariation between alcohol consumption and mortality.

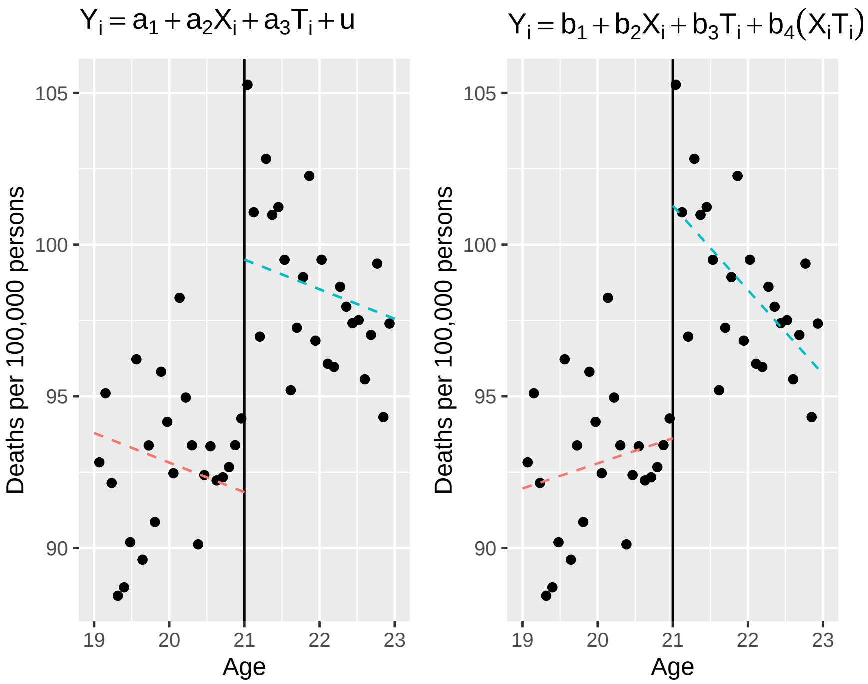


Figure 27.2: Regression discontinuity. Data from Carpenter and Dobkin [(2009)](https://doi.org/10.1257/app.1.1.164)

The two regression models are illustrated in figure 27.2 where both graphs use the same data, obtained from Carpenter and Dobkin (2009) . The vertical line in the middle of each graph marks the time when the people in the study turn 21. To the left of the line the average number of deaths is shown for people just under 21 and to the right deaths for those who have just turned 21.

In the left graph we see two regression lines, where both are estimated with the regression model in equation (27.2) . The regression line to the left, for the age groups under 21, is drawn with $T = 0$ and slopes downward. For the age groups over 21, to the right in the left graph, the regression line is drawn with $T = 1$. The right regression line in the left graph shifts up at 21 years of age, which indicates that mortality rises sharply after the 21st birthday as people have the opportunity to buy alcohol. The two regression lines in the left graph have the same slope, since we only estimate one slope coefficient for age in the regression model in equation (27.2) .

In the right graph we also see two regression lines that are estimated based on the regression model in equation (27.3) . For the age groups over 21, to the right in the graph, the line shifts up. The slope also changes. For the age groups under 21, to the left in the right graph, a positive correlation between alcohol-related mortality and age is visible. For the age groups over 21 a negative correlation between alcohol-related mortality and age is instead visible.

27.5 Instrumental variable

Another way to handle the problems that observable and non-observable phenomena like P and Z entail is to use instrumental variable analysis. The method is based on us finding another phenomenon S that is not affected by Z, does not affect Y directly but affects X, and thereby Y. A random variation in S causes a variation in X which in turn affects Y. The effect S has on X is possible to isolate and thereby also estimate the variation in X (caused by S) that affects Y. Figure 27.3 illustrates the basic logic in this. Phenomenon S affects only Y via its impact on X.

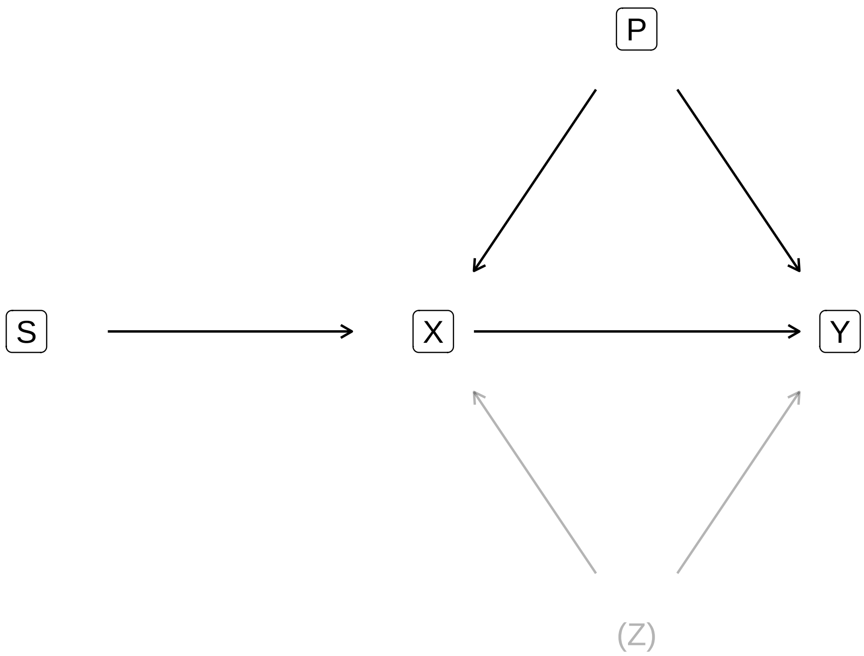


Figure 27.3: Instrumental variable S

Instrumental variable analysis is used for different things within social science. The oldest known example was presented in a book by Philip G. Wright (1928) and was based on work together with his son Sewal. Wright wanted to estimate the price elasticity in supply and demand for linseed oil, that is, percentage changes in supply or demand at percentage price changes (see section 10.10 and 8.4).

If we directly compare the covariation between price and quantity in society we cannot distinguish whether we observe the shape of the supply and demand curves (elasticity), or observe different points where supply and demand meet (price and quantity in equilibrium). To handle this Wright instead observed different phenomena that can be thought to shift the supply or demand curve, without affecting the other curve. For example Wright argued that the yield per cultivation hectare affects supply without affecting demand, which is why short-term changes in yield can be used as an instrumental variable to study how the supply curve moves while the demand curve is constant.

Instrumental variables are also commonly occurring within studies concerning cause and effect. Angrist (1990) compared in a well-known study economic effects for American soldiers who participated in the Vietnam War. One way to study this is to compare former military personnel's earnings with the rest of the male population. The selection of soldiers risks however distorting the results, if the men who did military service perhaps had other conditions than the rest of the population. For example, perhaps many men knew that they had poor conditions in the labor market and therefore chose voluntarily to enlist.

In the early 1970s the American state arranged a lottery that gave young adult men a number randomly, which determined whether they would be forced to go to Vietnam. The lottery (instrumental variable S) explains variation in war participation (variable X) but has no direct impact on participants' earnings (variable Y). However S can affect Y via the effect S has on X and thereby on Y. Angrist's analysis shows that the men who through the lottery were assigned military service had clearly worse earnings ten years after the war compared to corresponding men who did not participate in the war. This broke at least partly with other previous studies that instead found that war veterans on average had better economy.

In practical terms regression analysis with instrumental variables can proceed in slightly different ways. Here we content ourselves with a simpler description of a popular method, called Two-Stage Least Squares (abbreviated 2SLS), which in Swedish means roughly the least squares method in two stages. We have the regression model:

$$Y = a + bX + u \tag{27.4}$$

where a and b are coefficients, u is error term and Y and X are the variables. We have good reasons to believe that X causes changes in Y but that another factor (phenomenon Z) explains variations in both X and Y. Mathematically this means that X covaries with the error term u, that is, the covariance (cov) between X and u is not equal to 0: $cov\,(X, u) \neq 0$(see chapter 26).

To handle this we use the instrument S, which correlates with X but not with Y. Another way to describe this is that S does not covary with the error term u: $cov\,(S, u) = 0$. The variable S explains variations in X and thereby indirectly also affects Y. What we are now seeking is a measure of what influence S has on X and thereby on Y, which we can describe with the following regression model:

$$X = c + dS + v \tag{27.5}$$

This model captures the variation in X that is caused by S. The predicted variable $\hat{X}$ from this regression model we then use to create the following regression model:

$$Y = e + f\hat{X} + \epsilon \tag{27.6}$$

where e and f are coefficients, ϵ is error term and Y is still the explained variable that we are primarily interested in. Given that we have done right and not missed any other important phenomenon, the slope coefficient f can now be interpreted as the effect that $\hat{X}$ has on Y. For more comprehensive descriptions and discussions about instrumental variables see for example Angrist & Krueger (2001) and Bollen (2012) .

27.6 Difference-in-differences

Another way to distribute observations between treatment and control groups and study causal relationships and effects is the method called difference-in-differences (diff-in-diff or DiD). The method is based on us identifying a treatment group whose development we follow over time. We also manage to identify a group that is untreated but that we have good reasons to believe illustrates the parallel development that our treatment group would have had if cause X had never taken place. By comparing the difference between the groups before and after a treatment the difference in the difference can be interpreted as a measure of effect size.

Figure 27.4 illustrates the method where we compare the development for a control and treatment group. The time before X occurs the outcome variable Y for the two groups develops in parallel. When X occurs the upper solid line shows the development for the treatment group. The dashed line for the treatment group shows the counterfactual development that we theoretically assume that the treatment group would have had if no treatment X had taken place. The dashed line follows the control group in parallel and is based on the theoretical assumption that the two groups would have followed each other in precisely this way.

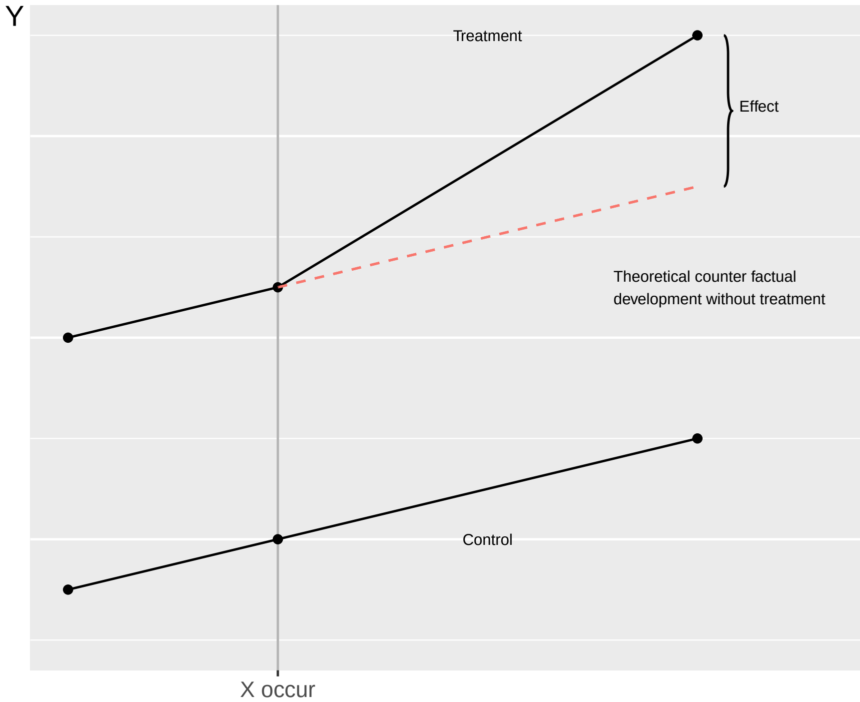


Figure 27.4: Difference-in-differences

A well-known example of this method was presented in an academic article by Card and Krueger (1994) . During the spring of 1992 minimum wages were raised in New Jersey, a state in the US, from $4.25 to$ 5.05. Corresponding wages were unchanged in the nearby state of Pennsylvania. Card and Krueger used this difference to study how minimum wages and employment within the fast food industry in the two states changed between February and November 1992.

The authors' results indicate that despite minimum wages being raised in New Jersey but not in Pennsylvania, no increased differences in employment between the states are noticeable among the groups affected by the minimum wages. These results, together with several others on similar themes, contributed to a renewed discussion about how both minimum wages and wages in general function in a social economy.

Let us now describe a regression model to estimate difference-in-differences. Suppose we have data over individuals distributed across treatment and control groups and want to compare these before and after a treatment (as in figure 27.4). We set up the regression model:

$$Y_{it} = a + bT_t + cG_i + d\left(T_t * G_i\right) + u_{it} \tag{27.7}$$

where a, b, c and d are coefficients. Y_{it} is the explained variable we are interested in for individual i at time point t. u_{it} is the error term per individual and time point. We have only two time points, before and after, which means that $t = \{1, 2\}$. The number of individuals is all participants in the treatment and control groups. T_t is a dummy variable for time point 1 respectively 2, that is $T = 0$(before) respectively $T = 1$(after). G_i is a dummy variable for the control group $(G = 0)$ respectively the treatment group $(G = 1)$. The term bT_t measures any difference in the y-intercept a before and after treatment. The expression $T_t * G_i$ is a combined dummy variable that equals 1 for the treatment group at time point $t = 2$(after treatment). The slope coefficient d measures the difference-in-differences between the two groups, which we interpret as an effect of cause X.

27.7 Pros and cons with these methods

This section briefly goes through some known challenges with the methods described in the chapter. Challenges do not mean that a method is bad. Understanding a method's limitations is important for us to understand methods in depth. We began this chapter by noting that there is no objective method for determining which variables should be included in a regression model. Unfortunately there is also no objective method for testing whether our theoretical assumptions in an effect analysis are correct. The design of an analysis is a largely qualitative aspect of analytical work. We need to familiarize ourselves with a specific situation and have good knowledge of the processes behind the emergence of the experimental situation. We must in each study examine and think about whether our treatment and control groups are comparable in the way that our analysis requires.

We need to think about and study carefully what is a suitable instrumental variable and ensure that its impact on the outcome variable goes precisely via our explanatory variable. If we are to use the difference-in-differences method we need to be sure that the groups are comparable in precisely the way that our regression model describes. In the same way we must be sure that the threshold we use for our regression discontinuity really is useful for dividing the observations into control and treatment.

Sometimes quasi-experiments can be conducted based on variations in nature, such as weather or natural disasters. Sometimes precisely this type of quasi-experiment is described as natural experiments. But even then we must ensure that our treatment and control groups are comparable and that no phenomenon P or Z affects our analysis.

Often we want to be able to generalize our results to large groups of observations that lie outside the data we have access to for our study. By using quasi-experiments we may capture a causal effect. But to be able to measure causal effects we must often delimit our analysis carefully and then the results also risk becoming more limited. For example the results in Angrist (1990) indicate that military service in Vietnam during the 1970s as a result of state coercion had a negative impact on soldiers' future earnings. But most American soldiers in the Vietnam War participated voluntarily and their earnings were perhaps affected in other ways compared to those who were forced (Angrist, 1998) .

In section 27.6 we described how Card and Krueger (1994) concluded that raised minimum wages in New Jersey did not lead to decreased employment compared to the development in the nearby state of Pennsylvania. This does not mean that we can draw the conclusion that a raise of all wages simultaneously in the entire labor market would not affect employment.

An individual study should as a rule be interpreted carefully. There is today an enormous amount of research around masses of phenomena. It is first when we get a more coherent picture of the research literature that it is possible to form a more general opinion about the world. On the other hand there are unfortunately many questions that are important for society but that probably will never be possible to study with either controlled experiments or quasi-experiments.

27.8 Causality without covariation

If we design our analysis to study causal relationships it is required that we find covariation for us to be able to claim that a causal relationship exists. In practice it is however easy to imagine different situations where a causal relationship exists between two variables, but where a regression analysis still will not find any covariation. This is yet another reason to be humble before estimated results. This section gives some simple examples of why we could fail to find covariation between two phenomena despite there being a causal relationship between them.

27.8.0.1 Wrong sample.

Suppose we investigate why so few students want to take our course in mathematics and statistics. To find this out we conduct a survey among the students who are already taking our course. Regardless of how we design this type of survey the study will not give us particularly much useful information. The students who are already taking the course are probably not representative of the entire population of potential students, which is the group we are really interested in. That is, there may be a result, a causal relationship, in a population. But our study examines a sample where we will not find any results for the population we are really interested in (compare section 24.1 below).

27.8.0.2 Wrong variables.

In the previous chapter 23 we went through how both the presence and absence of variables in a regression model can affect all results from the model. Suppose we investigate the covariation between disease X and its impact on mortality and people's lives in general. This particular disease has however been known for a long time and in recent years several effective medicines have been developed that slow its effects. If we miss taking into account the counteracting treatments our results will also become misleading.

27.8.0.3 Wrong model.

In section 21.7 we saw examples of covariation that we do not succeed in capturing with a linear regression model (figure 21.6). Say for example that the government introduces an education program for the unemployed and we are to study what effect this has on participants' opportunities to get jobs. But the effect of the program varies between time periods, depending on the economy's ups and downs. In bad times the education program shortens participants' time in unemployment. In good times people are encouraged to participate in an education they do not need. When we estimate the covariation between education and participants' opportunities for work we miss controlling for these circumstances and find no relationship, despite such a one existing, sometimes.

27.9 Chapter summary

- Within social science it is difficult to conduct controlled experiments, which is why we instead often use quasi-experimental observational studies. Often we want to identify a process that means that the distribution to treatment and control groups has occurred randomly. We must also adjust for all observed and non-observed phenomena that can be thought to affect cause and outcome.
- Regression discontinuity is a way to compare treatment and control groups through random variation around a threshold. Differences between the groups can be interpreted as an effect.
- Instrumental variable analysis can be used to study how X causes an effect in Y. An instrumental variable Z can predict variation in variable X and thereby Y, but has no direct impact on Y.
- The difference-in-differences method means that we assume that the counterfactual outcome can be described as a parallel trend of a control group. Difference in the differences between treatment and control before and after a treatment is interpreted as effect.
- Causal relationships require some form of covariation. But causal relationships can also exist when we do not find covariation, if we for example have the wrong sample, wrong variables or wrong regression model.

27.10 Exercises

1. A researcher wants to study the effect of a job training program on employment.

(a) Why is it difficult to conduct a randomized experiment?

(b) Describe a quasi-experimental approach.

(c) What assumption is needed for the quasi-experiment to be valid?

Answer: (a) ethical and practical constraints prevent random assignment, (b) exploit an unintended or administrative mechanism that randomizes treatment, (c) the mechanism must be independent of potential outcomes (as-good-as-random assignment).

2. A regression of wages on education gives $\hat{b} = 100$. Adding family background as a control reduces $\hat{b}$ to 60.

(a) What does the decrease in (hat{b}) suggest?

(b) What is the name of this problem?

(c) Is the original estimate biased upward or downward?

Answer: (a) family background is a confounder that was omitted from the original model, (b) omitted variable bias, (c) upward (the original estimate confounded education with family background).

3. A scholarship is awarded to students scoring ≥ 70 on a test. Students just above 70 earn more later than students just below 70.

(a) What is the threshold in this regression discontinuity design?

(b) Why is this a quasi-experiment?

(c) What is the key identifying assumption?

Answer: (a) score of 70, (b) assignment near the threshold is as good as random, (c) no other factor changes discontinuously at the threshold.

4. Geographic proximity to a university is used as an instrumental variable Z for education X in a wage regression.

(a) What condition must the instrument (Z) satisfy (relevance)?

(b) What is the exclusion restriction?

(c) Why would proximity to a university affect wages only through education?

Answer: (a) Z must be correlated with X (proximity predicts education), (b) Z affects wages only through education (not directly), (c) proximity to a university does not directly affect wages apart from the education it encourages.

5. Region A introduces a minimum wage policy; region B does not. Employment before: A = 100 , B = 120 . Employment after: A = 105 , B = 125 .

(a) What is the change in employment in region A?

(b) What is the change in employment in region B?

(c) What is the difference-in-differences estimate of the policy effect?

Answer: (a) $\Delta A = 5$, (b) $\Delta B = 5$, (c) $\hat{\tau} = 5 - 5 = 0$ (no estimated effect).

6. The DiD estimator is $\hat{\tau} = (\bar{Y}_{B,\text{after}} - \bar{Y}_{B,\text{before}}) - (\bar{Y}_{K,\text{after}} - \bar{Y}_{K,\text{before}})$. Given: $\bar{Y}_{B,\text{after}} = 80$, $\bar{Y}_{B,\text{before}} = 60$, $\bar{Y}_{K,\text{after}} = 70$, $\bar{Y}_{K,\text{before}} = 55$:

(a) Calculate the change in the treatment group.

(b) Calculate the change in the control group.

(c) Calculate (hat{tau}) .

Answer: (a) $80 - 60 = 20$, (b) $70 - 55 = 15$, (c) $\hat{\tau} = 20 - 15 = 5$.

7. A DiD study uses cities A (treated) and B (control).

(a) What is the parallel trends assumption?

(b) Why is this assumption untestable for the post-treatment period?

(c) How can we check it using pre-treatment data?

Answer: (a) without treatment, both groups would have followed the same trend over time, (b) we cannot observe the counterfactual post-treatment trend for the treated group, (c) verify that the two groups had similar trends before the treatment.

8. A tax benefit is given to firms with fewer than 50 employees. A researcher compares firms with 48–49 employees to firms with 50–51 employees.

(a) What is the treatment in this regression discontinuity design?

(b) Why compare firms just around the threshold?

(c) What assumption makes the two groups comparable?

Answer: (a) the tax benefit, (b) firms just above and below the threshold are otherwise similar, (c) firms are comparable on either side of the threshold (continuity assumption).

9. Treated group outcomes: before = 50 , after = 70 . Control group outcomes: before = 40 , after = 50 .

(a) What is the change in the treated group?

(b) What is the change in the control group?

(c) What is the DiD estimate?

Answer: (a) $70 - 50 = 20$, (b) $50 - 40 = 10$, (c) $\hat{\tau} = 20 - 10 = 10$.

10. A regression shows that education has an estimated effect of $\hat{b} = 100$ on wages.

(a) Why can we not necessarily interpret this as a causal effect?

(b) Name one source of endogeneity.

(c) What quasi-experimental method could address this?

Answer: (a) education may be correlated with unobserved ability or family background, (b) omitted variable bias / self-selection, (c) instrumental variable, regression discontinuity, or difference-in-differences.

11. Explain the concepts:

(a) What is omitted variable bias?

(b) What is endogeneity?

(c) What does the error term in a regression represent?

Answer: (a) bias from omitting a variable correlated with both X and Y , (b) X is correlated with the error term, (c) all unobserved factors that affect Y and are not included in the model.

12. A new school policy is introduced in some municipalities but not others. The introduction was determined by a random administrative lottery.

(a) What type of quasi-experimental design is this?

(b) What makes the lottery useful for causal identification?

(c) What assumption must hold for the non-lottery municipalities to serve as a valid control group?

Answer: (a) natural experiment / randomized lottery, (b) the lottery randomly assigns treatment, removing selection bias, (c) that municipalities without the lottery would have had parallel trends to lottery municipalities absent the policy.

Part IV

Computer Programs for Social Scientists

Part IV: Computer Programs for Social Scientists

This part of the book introduces computer programs that can be used for analysis. In separate chapters, a brief introduction is given to three analysis tools: the computer program Microsoft Excel, the program Stata, and the programming language R.

Chapter 28

Why Use Computers?

Analysis tools like Excel, Stata and R offer an enormous range of features and commands. When working with these programs, there are typically many ways to solve the same problem. There is rarely a single right answer about which method is best — it depends on the task, the context, and the user.

The aim here is to get you started quickly and equip you to learn more on your own. This text is deliberately brief and does not aim for depth. The instructions are best understood as examples of how the programs can be used. As you gain experience, you will soon discover other approaches to the problems covered here.

No advanced prior knowledge of computers, analysis programs, mathematics, or statistics is required. Basic computer literacy is assumed, including familiarity with concepts such as hard drive, memory, and files, as well as how to download and save files from the internet.

Keyboard shortcuts are used throughout. On Windows, the Ctrl key is commonly used; the equivalent on Mac is the CMD key.

28.1 Three Analytical Tools

Each tool is introduced through similar examples covering:

1. Saving and editing information, primarily by working with data organized in tables.
2. Importing and exporting information from and to files on the hard drive.
3. Mathematical operations and commands for working with matrices.
4. Describing quantitative data using measures of central tendency and dispersion.
5. Commands for calculating probabilities with common probability distributions.
6. Creating charts and graphics to illustrate data and mathematical functions.
7. Commands for basic regression analysis.

Microsoft Excel is part of the Microsoft Office suite. First released in 1985, Excel is today a versatile spreadsheet and data analysis program. It is one of the most widely installed applications in the world, used across organizations of all kinds for analytical work and many other purposes.

Stata is widely used for statistical analysis at universities, government agencies, and private companies. First released in 1985, it is today one of the world's best-known analysis programs. Stata is developed by StataCorp (www.stata.com).

R is a programming language designed for statistical analysis and other purposes. First introduced in the 1990s, R has since become one of the world's most popular programming languages. Unlike Excel and Stata, R is open source and available as a free download.

All three tools offer far more functionality than we can cover here. Each also supports add-ons and extensions that can be downloaded free of charge. For Stata and R in particular, users routinely create and share their own functions, which rapidly expands what the programs can do.

Many people have strong opinions about which program, tool, or method is best. Such debates are often unproductive. The right choice depends on the task, the objectives, available resources, and other factors

— and often the best program is simply the one you are most comfortable with.

That said, there are good reasons to become proficient with more than one tool. Different programs have distinct strengths and weaknesses. Sometimes a particular tool offers an unusually elegant solution to a specific problem. Familiarity with different tools can spark new ideas and suggest better solutions. And sometimes you simply have to use a program because everyone else in your organisation does.

Once comfortable with a tool, it is easy to stop there. But computers and software evolve constantly, and it is worth keeping an open mind and periodically investing time in learning something new.

28.2 Programming: Why Bother?

Much of this book is devoted to written commands and programming code for Stata and R. For readers unfamiliar with programming, this may seem unnecessarily complex. That said, most programs require only a few hours to get started. Once you grasp the underlying logic, learning new solutions becomes straightforward. In data analysis, knowing around 20 core commands is usually enough to handle most tasks, and additional commands can be looked up as needed. Programming languages for data analysis also share many similarities, so once you know one, picking up another becomes much easier.

Computers are especially useful for at least two types of analytical work:

1. **Complex tasks**: intricate calculations that would be impractical to carry out by hand.
2. **Repetitive tasks**: performing the same operation many times, or finding every instance of a word in a large text.

Most work involves more repetition than it first appears — similar tasks arise again and again. Identifying which tasks repeat is itself an important part of the work. The more skilled you become with analysis tools, the better you get at recognising where a computer can help.

Beyond the technical, programming is an excellent way to document your work, which matters for several reasons. Data projects often involve substantial preparation before any results are produced: correcting errors, reformatting data, adding columns, merging tables. Every step carries the risk of a mistake, and with large datasets those mistakes are easy to miss. When all steps are written as code, errors are far easier to find and fix.

28.3 You Have to Practice

Learning to use a computer program cannot be done by reading alone. You need to spend time working independently, encountering problems, and making mistakes. To get the most from this book, follow the examples and work hands-on with the programs.

Most programs include built-in help resources. The internet offers a wealth of additional material: exercises, tutorials, videos, free textbooks, and more. AI tools can also help generate code and suggest which functions to use for specific problems.

Many programs are designed to let users create and share their own functions freely online. This means the range of available commands is constantly growing. All programs are also regularly updated by their developers.

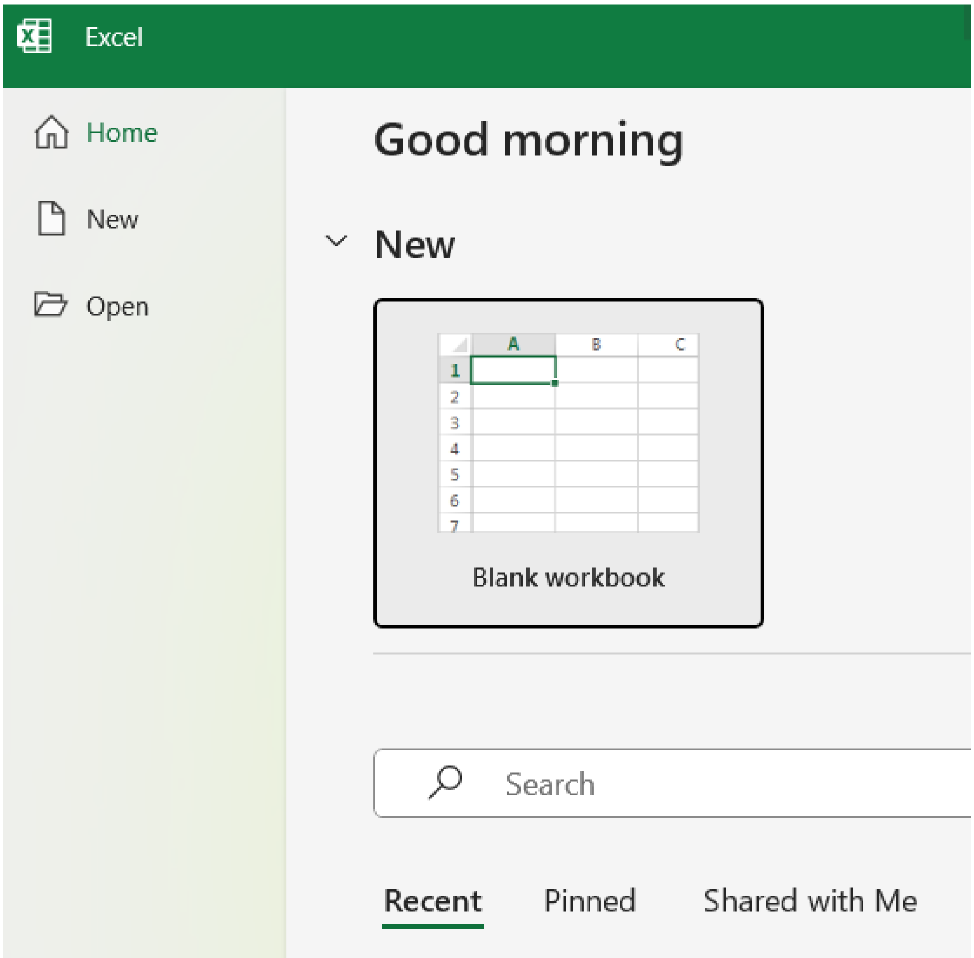
Chapter 29

Microsoft Excel

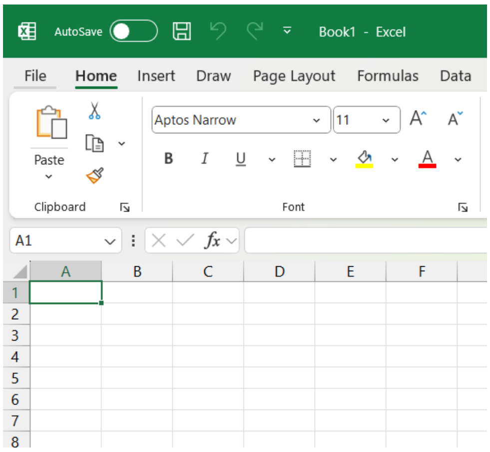
This chapter introduces the computer program Microsoft Excel, hereafter simply Excel. The examples are written using the version of the program called Excel for Microsoft 365 with Windows 11 and English language settings. Excel may look and function slightly differently depending on your computer's language settings. Name of functions and commands in Excel depend on language settings.

29.1 General about Excel

When we open Excel, we first see a screen where it is possible to choose what type of document we want to create, see the figure below. Select the option “Blank workbook”. You will now get a blank workbook that contains an empty worksheet, see the figure below. At the bottom left we see the tab “Sheet1”. Sheet1 is the empty worksheet. If we click on the circled plus sign next to the tab name Sheet1, a new tab is created: Sheet2.



Excel start screen



Empty workbook

Excel looks different depending on what language the program or computer is set to. The examples in this chapter are primarily in English. If your program is set to Swedish language, it will show "Blad1" instead of "Sheet1" at the bottom left of the workbook. To change the user language in Excel, go to File menu > More... > Options > Language > Office Display Language.

There are several different ways to work in Excel. Excel supports several different types of programming languages, for example. Here we primarily describe how to work in the worksheets and using the menus. The boxes in Excel's worksheets are called cells. The cells are numbered according to column and row. The top cell in the upper left is called A1. The cell to the right of this is B1. The cell below A1 is called A2, and so on. In the cells we can write numbers, letters, or mathematical calculations. Click on a cell to place the cursor in that cell. Write an equals sign and a mathematical expression. When we have finished writing the expression, we press the Enter key. A mathematical expression in a cell can look like this, for example: `=1+2`. After we press Enter, the cell changes to show the result of the mathematical expression: 3. Division where we divide 1 by 2 is written: `1/2`. Multiplication is written with an asterisk, for example: `1*2`. To calculate power expressions, such as 2^3, we write: `=2^3`.

Computer programs distinguish between numbers and text strings. A text string can contain numbers, which are then perceived as text characters. We indicate that something is a text string by using quotation marks. Example: If we write `=13`, the program treats it as a number, a numeric value. If we write `="13"`, the program treats this as a text string.

Computer programs generally understand information in a less flexible way than we humans do. For a computer, the words “human” and “Human” are different because in the latter we use a capital letter for H. This is called case-sensitive input.

If we want to be sure about how the program handles information in a cell, we can change the cell’s data format by selecting one or more cells and right-clicking. In the menu that appears, we select “Format Cells…”. A new dialog box appears where we choose how Excel should handle information in the cell. For example, whether the information should be handled as text, a number with decimals, as a currency, a date, and so on.

Changing the data format can also affect how the information in the cell is displayed, for example the number of decimals. Changing the number of decimals does not affect the information stored in the cell. If we change the format from, for example, a number to text, however, the actual content is also affected.

29.2 Import and edit data

The workbook can be saved as a file on your computer’s hard drive. Excel has several proprietary file formats where .xlsx is the standard format for the type of exercises we will do in this chapter. In Excel’s worksheets, we see the data we are working with and we edit directly in the actual data material. This creates a significant risk that we might accidentally overwrite important information or change something that shouldn’t be changed. Therefore, be careful to continuously save your files in such a way that you can easily undo your work steps.

Another common file format that Excel can read is .csv. The abbreviation csv stands for Comma-Separated Values. Csv files contain data tables where different observations are separated by a comma character.

Data in csv files can also be separated by other types of characters, for example semicolons or tabs (a number of spaces). If we try to open a csv file in Excel, it is important that the program knows which character separates the data. Csv files are not unique to Excel but can be read by many other programs, including the two other analysis tools described in this book. There are also several other ways to import and export data to and from Excel from various types of sources, which there isn’t space to describe here. You can find more examples under the Data menu > Get & Transform Data.

In Excel, we edit the information in each cell by placing the cursor on a cell or in the cell. We also select multiple cells and press Delete to erase the information in all these cells at once. If we press Ctrl + -, the entire cell or cells that are selected are deleted. When we delete one or more cells, the cells to the right or below the deleted cells move left or up to replace the deleted cells.

We select entire rows by clicking on a row’s row number at the far left of the worksheet, and entire columns by clicking on the column’s letter at the top of the worksheet. If we delete an entire row, all rows below move up and replace the deleted row. If we delete a column, all columns to the right of this column move left and replace the deleted column. To select all cells in an area or a worksheet, we press Ctrl + A.

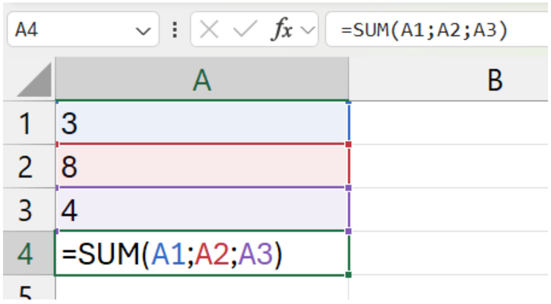
29.3 Worksheet functions

In Excel’s cells, we can use Excel’s own worksheet functions. These functions are ready-made commands that perform different types of operations. Excel currently has 484 functions of this type. In this book, we only go through a small selection. Microsoft provides lists and descriptions of all functions on their website: https://support.microsoft.com/sv-se/excel

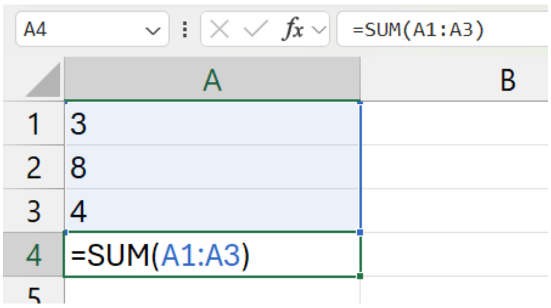
All functions are written with the function's name followed by parentheses. Example: `SUM()`. The function is called SUM. In the parentheses after the function's name, we write the arguments that the function should use. For example `SUM(3,8)`. In this case, we have two arguments in the form of the numbers 3 and 8. Each argument is separated with a comma (,). If we place the cursor in a cell in our Excel sheet and write `=SUM(3,8)` and press Enter, the function returns the sum of 3+8, that is, the value 11. The cell then displays the value 11, but the function remains in the cell. If we click on the cell again, or select the cell and press F2, we may edit the function we used.

Which arguments a function can use varies between different functions. The arguments can consist of references to other cells in Excel, in the same or another worksheet. The figure below shows an example with `SUM(A1,A2,A3)`. A1, A2, and A3 are interpreted by Excel as references to the corresponding cells. The function retrieves the value in each cell and calculates the sum. The SUM function is used in this example in cell A4, where the function's result is returned.

We can also refer to several adjacent cells as one and the same argument. The figure below shows an example where we use the command `=SUM(A1:A3)`. The colon indicates that we are referring to all cells from cell A1 through cell A3. Between cell A1 and A3, there is only cell A2, which is also included. When we write `SUM(A1:A3)`, the cells A1:A3 are used as one argument to the `SUM()` function. We do not use a comma to separate the reference to the cells. If we want to refer to the entire column A, we can instead write A:A, for example `SUM(A:A)`. If we refer to the entire column, we must write the command `=SUM(A:A)` in a cell that is not located in the same column A. If we write the command in a cell in column A, a circular reference is created. Circular reference refers to a cell that refers, among other things, to itself. Circular reference can also occur if a cell refers to a cell that indirectly refers to itself, for example through a third cell.



Each argument is a cell reference



Multiple cells as one argument

All worksheet functions in Excel have their own help pages. These are often worth reading and contain practical tips on how to use the function in different ways. To find a help section about a function, we can do one of the following:

- Go via Help menu > Help. Search for the function's name.
- Place the cursor in a cell and write the function's name, don't press Enter. Press F1.
- Place the cursor in a cell and write the function's name. The program now shows a box just below the cell with some instructions for what type of arguments the function needs. Click with the cursor in this box on the function's name.

Excel has several functions for different types of mathematical operations. Here are some examples:

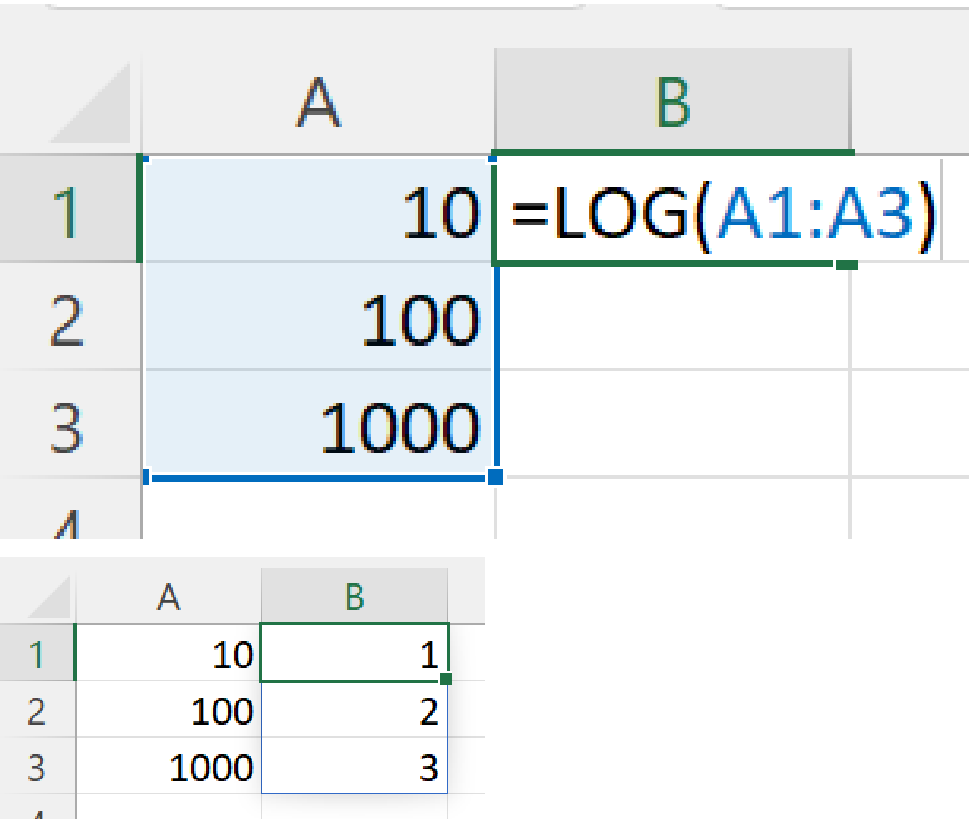
- `ABS()`: Returns the absolute value of a number. Example: `ABS(-4)` returns result 4, since $|-4| = 4$.

- `AVERAGE()`: Returns the average of a collection of numbers. Example: `AVERAGE(4,5,12)` returns the value 7.
- `MEDIAN()`: Returns the median of a collection of numbers. Example: `MEDIAN(4,5,12)` returns the value 5.
- `LOG()`: Returns the base-10 logarithm of a number. Example: `LOG(100)` returns result 2, since $\log_{10} 100 = 2$ and $10^2 = 100$. If we want, we can specify a different base than 10 as the second argument in the function. Example: `LOG(100,2)` returns the logarithm for the number 100 with base 2, which is approximately 6.644.
- `LN()`: Returns the logarithm of a number with base e, the natural logarithm. Example: `LN(10)` returns result approximately 2.3, since $\ln 10 \approx 2.3$.
- `SUMPRODUCT()`: Returns the sum of products of two or more collections of values. See example in the section on combining functions below.
- `FACT(a)` where $a > 0$. Returns a factorial. Example: `FACT(3)` $= 3! = 3 * 2 * 1$.
- `PERMUT(a,b)` where a and b are arbitrary numbers and $a > b$. Returns the number of permutations: $\frac{a!}{(a-b)!}$.

29.4 Quick ways to fill multiple cells

Often we want to calculate several similar results next to each other. Several sums or several logarithmic values. This section goes through some examples of how we can do this. All Excel functions we use in this book can use information from another cell as arguments.

Some functions are designed to use a value, for example from a cell, to return a result. Above we used, for example, the `LOG()` function which returns the logarithm of a number. If we nevertheless specify multiple values as arguments, the function can still return multiple results, but then in separate cells. The figure below shows an example where we have three numbers in column A. In cell B1 we write `LOG(A1:A3)` which retrieves the information in cells A1 to A3 as an argument to the `LOG()` function. When we press Enter, the program returns the results from the function in cells B1, B2, and B3. In cell B1 we get the result $\log_{10} 10 = 1$. In cell B2 we get $\log_{10} 100 = 2$, in cell B3 we get $\log_{10} 1000 = 3$.

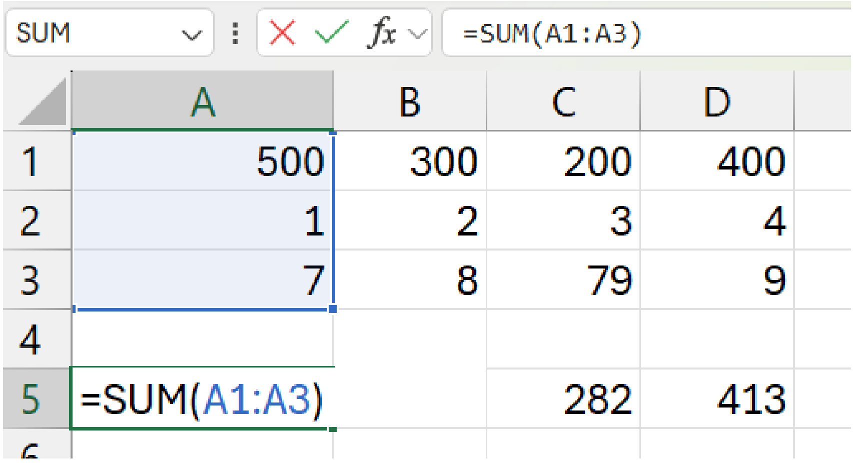


Reference multiple cells at the same time

Some functions are designed to use multiple values as one argument, for example as we described that the

`SUM()` function calculates the sum of a collection of numbers. If we refer to three cells in the parentheses of `SUM()`, we get the sum of the values in those cells. Sometimes, however, we want to calculate the sum of different combined values next to each other.

The figure below shows an example where we want to calculate the sum of the cells in rows 1 to 3 in each respective column. One way to get one sum per column is to select the cells in row 5 in the columns we want to calculate the sum for, in this case columns A to D. Without clicking anywhere, we then begin typing the content we want in cell A5: `=SUM(A1:A3)`. Press Ctrl + Enter. Now cells B5, C5, and D5 are also filled with the `SUM()` function and references to the cells in rows 1 to 3 in each respective column.



One sum per column

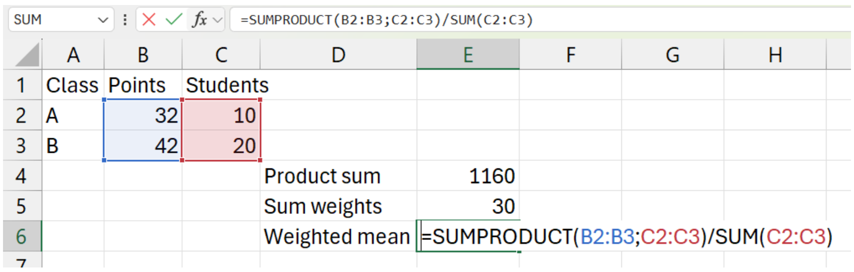
29.5 Combining functions within calculations

A weighted average for variable X can be calculated as $\bar{X}_{\text{weighted}} = \frac{\sum_i^n x_i v_i}{\sum_i^n v_i}$ where x_i is the value for observation i and v_i is the weight for this observation. Say that the two school classes A and B have 10 and 20 students respectively. On the latest math test, the students in the two classes scored 32 and 42 points respectively. The weighted average of their test results:

$$\bar{X}_{\text{weighted}} = \frac{\sum_i^n x_i v_i}{\sum_i^n v_i} = \frac{10 * 32 + 20 * 42}{10 + 20} = \frac{1160}{30} \approx 38.7$$

Now we will use Excel to calculate a weighted average. Excel has no ready-made function for this, but we can combine the `SUMPRODUCT()` and `SUM()` functions. In the numerator of the equation above we have the sum of products $\sum_i x_i v_i$ and in the denominator we have the sum $\sum_i v_i$.

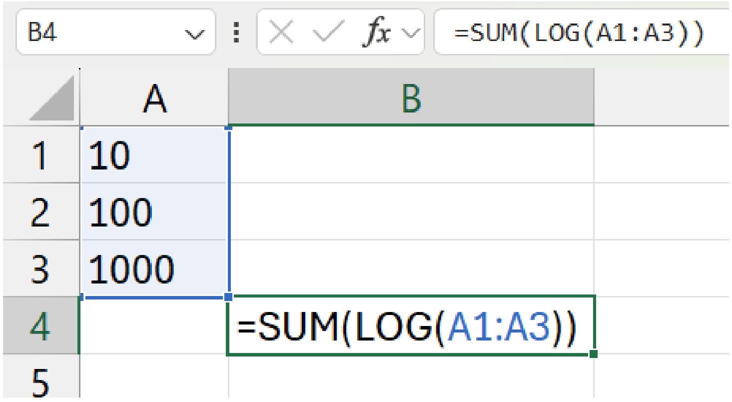
The figure below shows an example of how it can look in Excel. Column B contains the test scores for classes A and B. Column C contains the number of students. In cell E4 we calculate the sum of products in the numerator. In cell E5 we calculate the sum in the denominator. In cell E6 the weighted average is calculated by combining the two functions. In the parentheses of `SUMPRODUCT()` we refer as the first argument to cells B2:B3. As the second argument we refer to cells C2 and C3. The `SUMPRODUCT()` function then calculates the following expression: 32 * 10 + 42 * 20 = 1160. If we divide this by the sum of the weights 10 + 20, the result is approximately 38.7. This is the result we calculate in cell E6.



Calculate a weighted mean

We can also combine functions by writing one or more functions in the parentheses of another function. The figure below shows an example. In cells A1, A2, and A3 we have the values 10, 100, and 1000. We want to use the base-10 logarithm of each value and calculate the sum of the logarithmic values. In cell B4 we write `=SUM(LOG(A1:A3))` and get the result 6:

$$\log 10 + \log 100 + \log 1,000 = 1 + 2 + 3 = 6$$

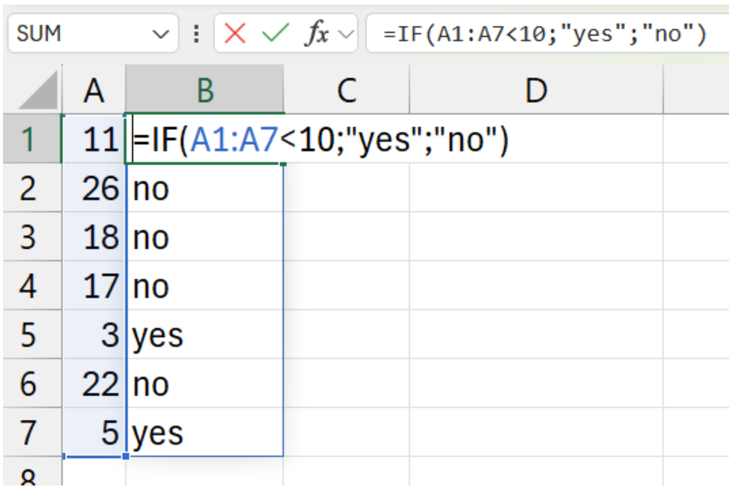


Example of a function within a function

29.6 Logic conditions

In data analysis, we often use logical conditions. A logical condition is true or false. With numbers, we can describe it as the condition being either 1 (true) or 0 (false). Excel's `IF()` function checks whether a logical condition is met and returns a value based on that. We can define which value the function should return if the condition is true or false.

The figure below shows an example where we have a collection of numbers in column A. In column B we use the `IF()` function to test the logical condition of whether the value in column A is under 10. We refer to all cells A1 to A7 simultaneously, see the section on filling multiple cells above. Depending on whether the logical condition is true or false, the function returns the values "yes" or "no" in cells B1 to B7.

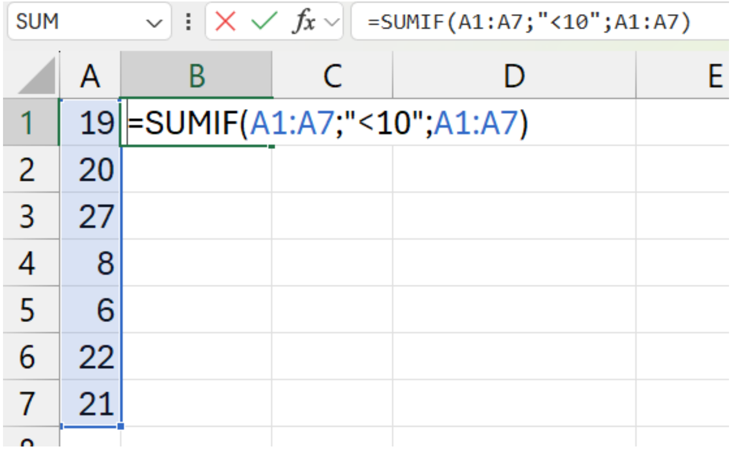


Example with function IF()

The `SUMIF()` function calculates the sum of a collection of values if these values meet a logical condition. The figure below shows an example where we sum the values in column A that are less than 10. The `SUMIF()` function needs three arguments:

1. The cells for which the condition should be tested. In this case: A1 to A7.

2. The condition. This is formulated with quotation marks. In this case `"<10"`. If we want to check whether the cells are equal to the value 10, we write only the value itself: `"10"`.
3. Cells that should be summed. In this case: A1 to A7.


SUM | =SUMIF(A1:A7;"<10";A1:A7)

	A	B	C	D	E
1	19	=SUMIF(A1:A7;"<10";A1:A7)			
2	20				
3	27				
4	8				
5	6				
6	22				
7	21				

Example with function SUM.IF()

The figure above shows an example where we calculate the sum of the values in column A that are less than 10. The answer is returned in cell B1. The sum in this case becomes $8 + 6 = 14$.

If we want to calculate a sum with multiple conditions, we can use the `SUMIFS()` function. The first argument is the range to be summed, after which we can add any number of conditions and condition ranges. This is useful, for example, if we have values that are grouped at multiple levels. For example, a list of employees (column A) at a company who work in different departments (column B) with different job tasks (column C) and we want to calculate the sum of salaries (column D) per department and job task.

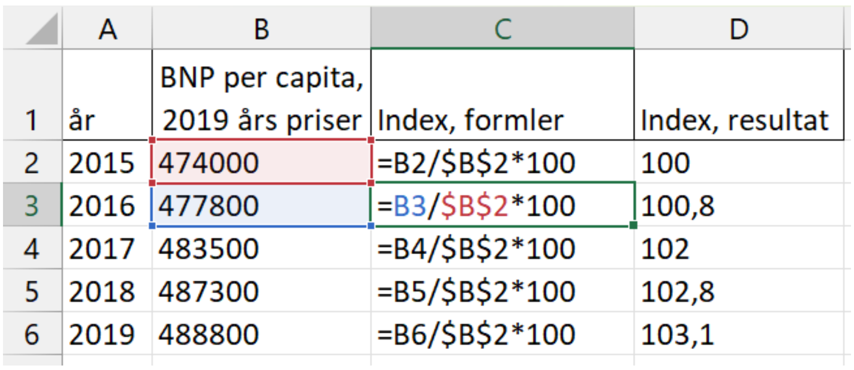
Above we described the `AVERAGE()` function for calculating averages. Averages can also be calculated with conditions, for which we can use the `AVERAGEIF()` and `AVERAGEIFS()` functions, which work in the same way as `SUMIF()` and `SUMIFS()`.

29.7 Calculate index with Excel

To compare relative differences in, for example, a time series with values, we can use an index with a base year. The base year's value can, for example, be 100. Index for year t is calculated as:

$$\text{Index}_t = \frac{\text{Value}_t}{\text{Value}_{\text{base year}}} * 100$$

In Excel, we can calculate this by referring to cells with our time series and using division and multiplication. The figure below shows an example where we have years in column A and data on Sweden's GDP in column B. In column C we write the formulas for our index. Column D shows the result with two decimal places.



	A	B	C	D
1	år	BNP per capita, 2019 års priser	Index, formler	Index, resultat
2	2015	474000	=B2/$B$2*100	100
3	2016	477800	=B3/$B$2*100	100,8
4	2017	483500	=B4/$B$2*100	102
5	2018	487300	=B5/$B$2*100	102,8
6	2019	488800	=B6/$B$2*100	103,1

Calculate an index

There are several ways to make this calculation faster and more convenient. We start by writing the formula in cell C2: `=B2/ $B$ 2*100`. This gives the result 100, which is shown in cell D2. The reason we use dollar signs $here is to be able to fill in the rest of the cells in column C faster. To fill in the other cells in column C, we start by selecting cells C3 to C6 in one of the following ways:

- Use the mouse cursor to select cell C3, hold down the mouse button and drag to cell C6 so C3:C6 is selected.
- Use the mouse cursor to select cell C3. Release the mouse. Hold down the Shift key and use the mouse cursor to click on C6. Now all three cells C3:C6 are selected.
- Select cell C3. Release the mouse. Hold down the Shift key and press the Down arrow key until we reach cell C6.

When cells C3 to C6 are selected, press Ctrl + D. The keyboard shortcut Ctrl + D copies the formula in cell C3 to the other cells and changes the references to the corresponding cells in column B. The numerator in the fraction, which we wrote as B2, changes for each row to B3 on row 3, B4 on row 4, and so on. The dollar sign means that the cell reference `$B$ 2` in the denominator is locked and remains the same even in the other cells, which we want because this is the base year. If we want to do the same type of copying in cells to the right of the leftmost cell in a selection, we can use the keyboard shortcut Ctrl + R.

Another way to achieve the same thing is to first select cells C3:C6. Then begin, without selecting any other cell, typing the equation `=B2/ $B$ 2*100`. Press Ctrl + Enter, whereupon all cells are filled with the corresponding equation where the cell reference B2 is updated in the other cells but `$B$ 2` is locked and remains the same.

29.8 Matrix calculations

A matrix is a collection of values organized in a row, a column, or in multiple rows and columns. In accordance with the calculation rules for matrices, there are special functions in Excel for calculating with matrices. To add or subtract two matrices, we can use regular mathematical operations in a similar way to what we have gone through above. The figure below shows an example with matrix A in cell A1:B2 and matrix B in cell D1:E2. Note that we now use a colon to describe ranges between two cells in different columns and on different rows.

To subtract matrix A-B, we select cell D4:E5. It is in this 2×2 field that we will get the result of the calculation. Without moving the cursor, we then write `=A1-D1` and press Ctrl + Enter. Although what we wrote appears in cell D4, cells D5, E4, and E5 are now automatically filled with corresponding references to matrix A and B. The figure below shows only the formulas. The result becomes a 2×2 matrix with the value -4 in each cell:

$$A - B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

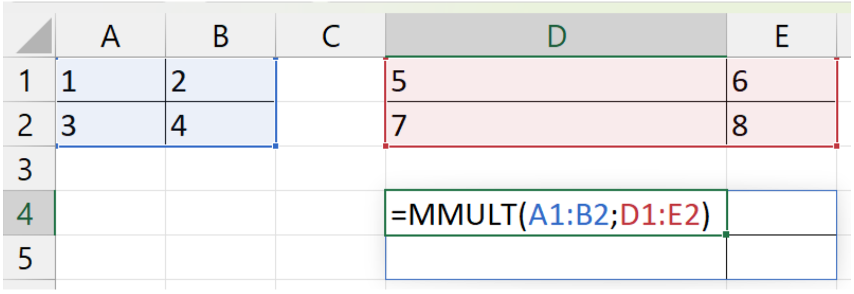
	A	B	C	D	E
1	1	2		5	6
2	3	4		7	8
3					
4				=A1-D1	=B1-E1
5				=A2-D2	=B2-E2

Subtract matrix $A - B$

Given that the number of rows in A equals the number of columns in B, we can calculate the matrix product AB:

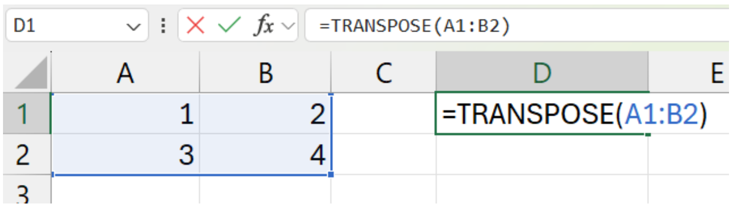
$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1*5+2*7 & 1*6+2*8 \\ 3*5+4*7 & 3*6+4*8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Matrix multiplication can be performed with Excel's `MMULT()` function. In the parentheses of `MMULT()` we specify two arguments with references to the cells that contain the values for the two matrices we want to multiply. The figure below shows an example where we matrix multiply the values in the four cells A1:B2 (matrix A) with the four cells D1:E2 (matrix B). The `MMULT()` function is used in cell D4, where the first argument in the parentheses is A1:B2 and the second argument is D1:E2. The arguments are separated as usual with a comma. The result becomes a new 2×2 matrix, whose values will be shown in the four cells D4:E5.

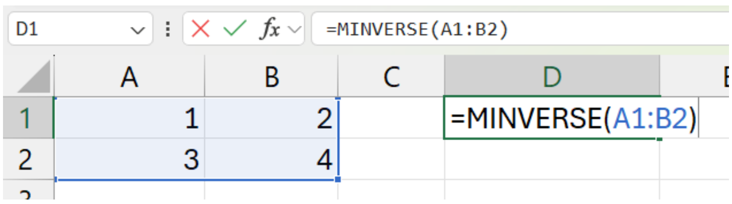


Matrix multiplication with MMULT()

We can transpose a matrix with the `TRANSPOSE()` function by specifying in the function's parentheses the matrix we want to transpose. We can calculate the inverse of a matrix with the `MINVERSE()` function, specifying in the parentheses the matrix for which we want to calculate the inverse matrix. The figure below shows an example where we transpose the 2×2 matrix in cells A1:B2. The result will be shown in D1:E2. The figure after that shows an example where we use the `MINVERSE()` function to calculate the inverse matrix of the 2×2 matrix in cells A1:B2. The result will be shown in cells D1:E2.



Transpose matrix with TRANSPOSE()



Calculate inverse matrix with MINVERT()

29.8.1 Solving a system of equations

Say now that we have the following system of equations:

$$\begin{cases} x_1 - 2x_2 & = 3 \\ 4x_1 + 5x_2 & = 6 \end{cases}$$

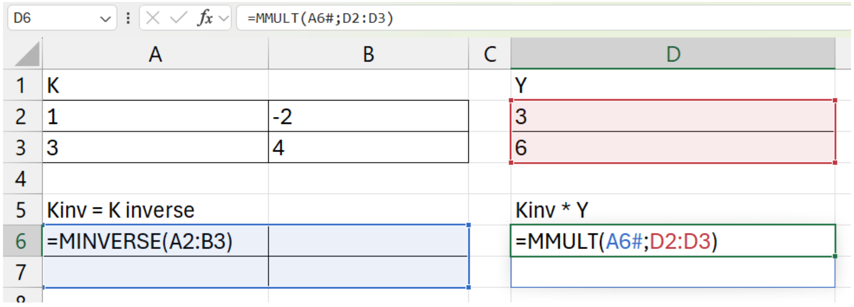
This system has the unique solution $x_1^* = \frac{27}{13}$, $x_2^* = -\frac{6}{13}$. There are several ways we can use Excel to help find these solutions. One way is to use matrix calculations. Let us write the system of equations with matrices:

$$KX = Y \quad \begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

The solution can now be given by matrix multiplying both sides with the inverse matrix K^{-1}:

$$K^{-1}KX = K^{-1}Y \quad \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{27}{13} \\ -\frac{6}{13} \end{bmatrix}$$

The figure below shows how we can calculate these matrices in Excel. Cells A2:B3 contain matrix K. Cells D2:D3 contain matrix Y. Cell A6:B7 contains the inverse matrix K^{-1} and cell D6:D7 contains $Y^* = K^{-1}Y$.

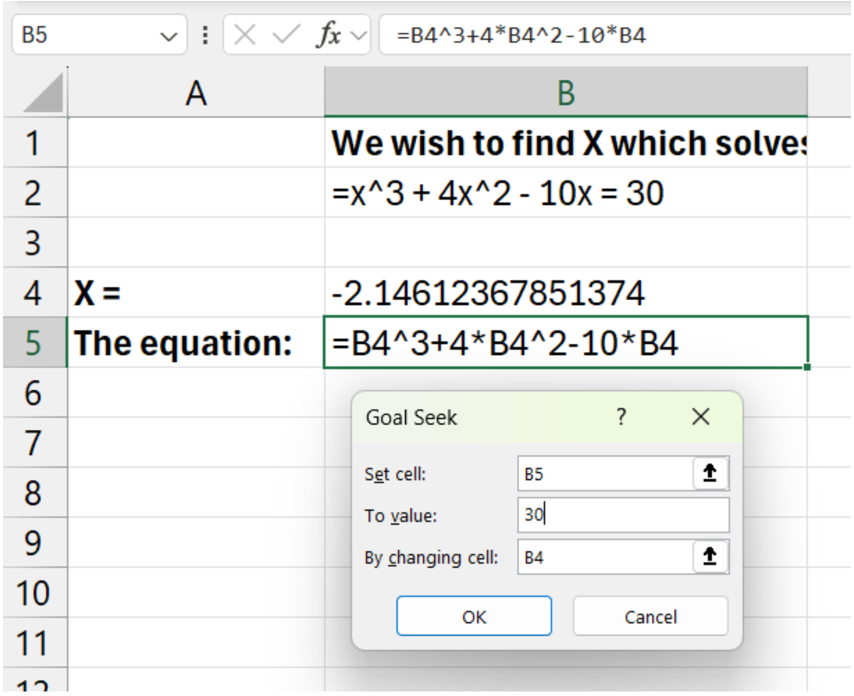


Solving systems of equations with matrix algebra

Excel also has functions in the menus that we can use to solve equations with unknown variables. Say as an example that we want to find the value for x in the following polynomial:

$$x^3 + 4x^2 - 10x = 30$$

For this we can use Excel's Goal Seek function, which we access by clicking on the Data menu > What-If Analysis > Goal Seek... The Goal Seek dialog box opens, where we must fill in three values. Set cell: the equation we want to solve. To value: the value that the equation should equal, in this case 30. By changing cell: the cell that we use as a reference for the variable x. The figure below shows an example where we have entered the left side of the equation above, that is, without =30. Instead of x we refer to the empty cell B4. If we click OK, Excel searches for the solution.



Goal seek

29.9 Input-output analysis

Matrices can be used for input-output analysis and network analysis, among other things. Input-output analysis often starts from flow tables that describe how companies in different industries in an economy produce input goods for their own and other industries in the economy, as well as goods and services for final consumption. We call the flow matrix Z. We take the multiplicative inverse of total production for industry j that goes to input goods and final consumption per industry, as $\frac{1}{s_j}$. These inverses for all industries we collect in matrix S_m. From this we can define matrix A:

$$A = Z * \operatorname{diag}\left(S_m\right)$$

where $\operatorname{diag}\left(S_m\right)$ is a diagonal matrix with the multiplicative inverse of total production per industry along the diagonal. We define matrix C as a column matrix with the production per industry that goes to final consumption. The total production required from industry j for all input goods and final consumption we call y_j. We collect the values for this for all industries in matrix Y and write:

$$\begin{aligned} Y &= AY + SY - AY && (29.1)\\ &= S\left(I - A\right)Y && (29.2)\\ &= SY && (29.3)\\ &= \left(I - A\right)^{-1} S && (29.4) \end{aligned}$$

where I is an identity matrix with the same dimensions as A. The values in Y are the total production required for this economy taking into account both input goods and final consumption. We call $B = \left(I - A\right)^{-1}$ where B has the same dimensions as A. The sum of column j in B is the output multiplier for industry j. The sum of row j in B is the input multiplier for industry j. These things we can calculate in Excel using the functions we used above.

The figure below shows an example with the matrices:

$$Z = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, S = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$$

	A	B	C	D	E
1	Z		C	S	
2	2	1	3	6	
3	1	3	3	7	
4					
5	Sm		diag(Sm)		
6	=1/D2		=A6	0	
7	=1/D3		0	=A7	
8					
9	A			I	
10	=MMULT(A2:B3;C6:D			1	0
11				0	1

Input-output analysis with Excel, step 1

We calculate:

$$A = Z * \operatorname{diag}(S_m) = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} * \begin{bmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{7} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{7} \\ \frac{1}{6} & \frac{3}{7} \end{bmatrix}$$

Then we can calculate:

$$B = (I - A)^{-1} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{7} \\ -\frac{1}{6} & \frac{4}{7} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{8}{5} & \frac{2}{5} \\ \frac{7}{15} & \frac{28}{15} \end{bmatrix}$$

From this matrix we can then calculate the column and row sums to get the output and input multipliers for each industry. Total production required from each industry is given by column matrix Y:

$$Y = (I - A)^{-1} S = \begin{bmatrix} \frac{8}{5} & \frac{2}{5} \\ \frac{7}{15} & \frac{28}{15} \end{bmatrix} * \begin{bmatrix} 6 \\ 7 \end{bmatrix} = \begin{bmatrix} 12.4 \\ 15.9 \end{bmatrix}$$

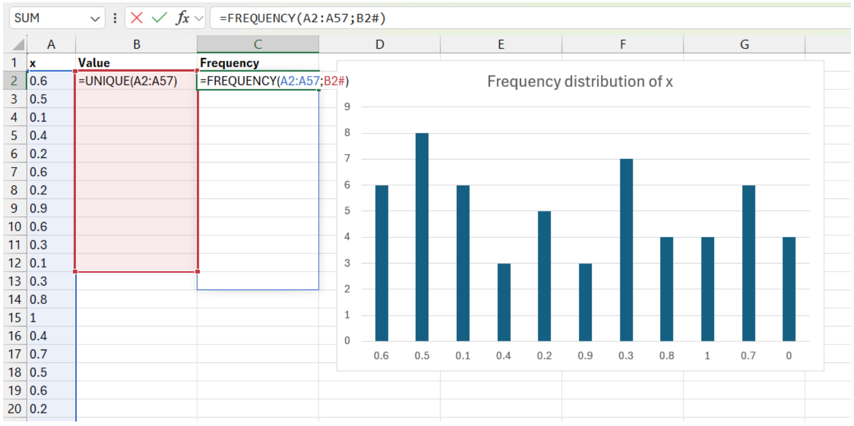
The figure below shows the calculations for matrix B and Y, where we use the same functions as before.

9	A			I	
10	=MMULT(A2:B3;C6:D7)			1	0
11				0	1
12					
13	**I-A**				
14	=D10-A10	=E10-B10			
15	=D11-A11	=E11-B11			
16					
17	B=(I-A)^1			S	
18	=MINVERSE(A14:B15)			6	
19				7	
20					
21	Y=B*S				
22	=MMULT(A18#;D18:D19				
23					

Input-output analysis with Excel, step 2

29.10 Variance and covariance

If we want to calculate how a collection of values is distributed, we can calculate the frequency distribution, among other things — the number of observations per unique value or the number of observations within different intervals, for example using the `FREQUENCY()` function. The figure below shows an example. In column A we have variable x which contains a collection of values, all between 0 and 1. In column B we use the `UNIQUE()` function, which returns the unique values from the numbers in column A. The unique values in this case are 0.1, 0.2, 0.3, ..., 0.9 and 1. The results will be shown in cells B2:B11. In column C we use the `FREQUENCY()` function to count the number of occurrences in column A of each respective value in column B. The result is shown in the chart to the right in the image.



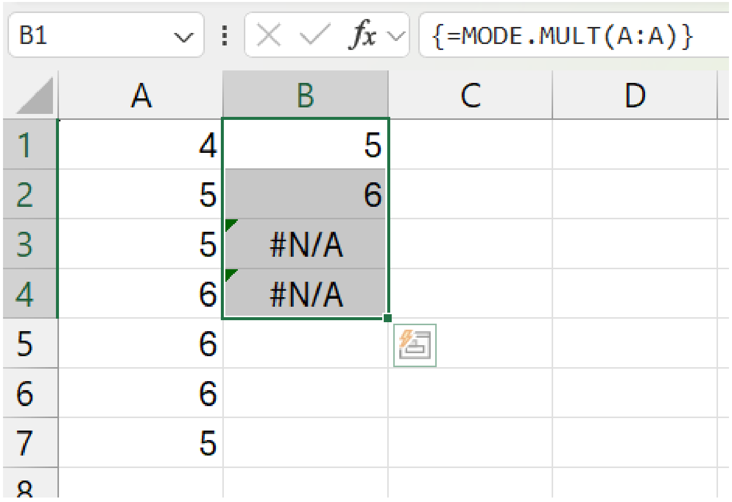
Frequency distribution

We have described above the `AVERAGE()` function for mean, and the `MEDIAN()` function. Mode describes the value in a collection that occurs most frequently. For example, among the values 3, 3, 15, 27, the mode is 3, because it occurs twice and the other values only occur once each. In Excel, mode can be calculated with the `MODE.SNGL()` function. In the parentheses we are required to specify cells or values on which the mode should be calculated. Example: `MODE.SNGL(A:A)` returns the most common value in column A.

If there are multiple values that occur most frequently in a collection, we can use the `MODE.MULT()` function. If we use this function as usual in a cell, it returns the mode for a collection of numbers. The figure below shows an example with `MODE.MULT()`. In column A we have the values 4, 5, 5, 6, 6, 6, 5. The values 5 and 6 occur three times each.

To show multiple modes, we select cells B1 to B4. Without clicking anywhere else, we now write `=MODE.MULT(A:A)`. Press Ctrl + Shift + Enter. This command creates what is called an array formula. When `MODE.MULT()` is used as an array formula, it returns multiple modes, if there are multiple. If only one value in the collection is most common, the function returns the same value in all cells.

The result is shown in cells B1 to B4. Since the values 5 and 6 are most common with three occurrences each, these are shown in cells B1 and B2. In the collection of values, only one other value occurs, 4, once. Since no other values occur as frequently as the numbers 5 and 6 in the collection, the value #N/A! is returned in cells B3 and B4.



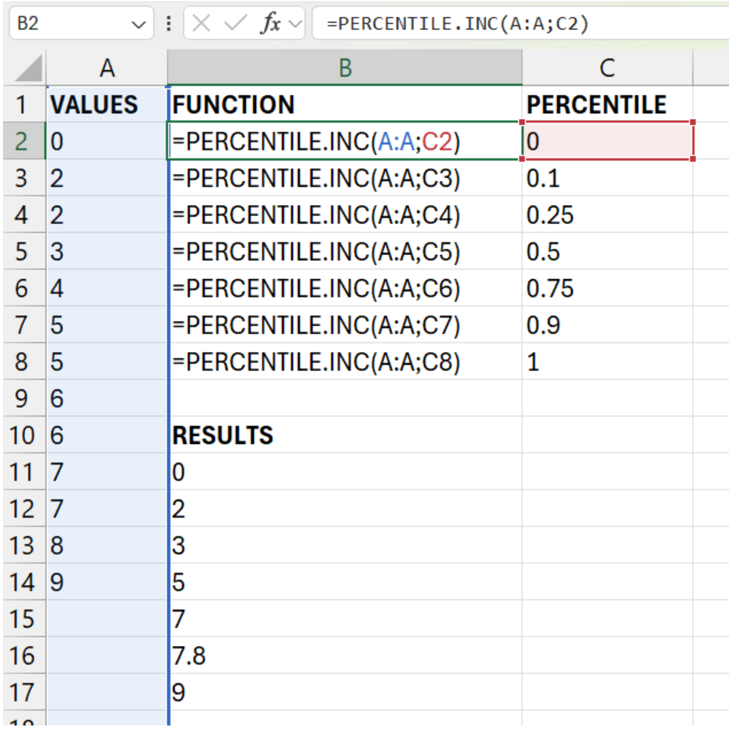
Example with MODE.MULT()

A common type of measure of dispersion is percentiles, deciles, and quartiles. A percentile divides a collection of values into 100 equal parts. The 10th percentile is greater than 10 percent of the values in the collection and less than 90 percent. Deciles instead divide the collection into ten equal parts. The 10th percentile is the same as the 1st decile. The 9th decile is the same as the 90th percentile. Quartiles divide the collection of values into four equal parts. The 25th percentile is the same as the 1st quartile. The 50th percentile is the same as the 2nd quartile, 5th decile, and the median.

In Excel, we can calculate percentiles, deciles, and quartiles with the `PERCENTILE.INC()` and `PERCENTILE.EXC()` functions. Both of these functions calculate percentiles. In the parentheses of the functions, we must specify two arguments: First, the values on which the percentiles should be calculated, or a reference to the cells that contain the values. The second argument should be a value that defines which percentile we want to return, by specifying a value between 0 and 1.

The difference between the functions is that for `PERCENTILE.INC()` we can specify the values 0 and 1 and all decimals in between. If we specify 0, we get the lowest value in the collection, and if we specify 1, we get the highest value in the collection. For `PERCENTILE.EXC()`, the boundary values 0 and 1 are excluded. We must instead specify a decimal value that at least gives us the next value above or below the boundary values.

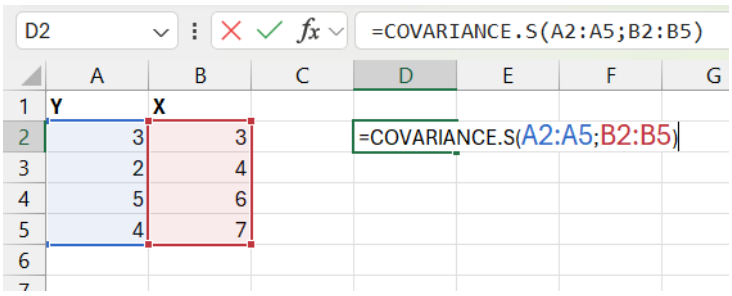
The figure below shows an example where we have a collection of values in column A. In column B we calculate using the `PERCENTILE.INC()` function the 0th, 10th, 25th, 50th, 75th, and 90th percentiles in cells B2 to B8. The results that the functions return are also pasted in cells B11 to B17.



Calculate percentiles with PERCENTILE.INC()

If we want to estimate the variance for a sample from a population, we can use the `VAR.S()` function. If the observations we have access to are population data, we can instead use the `VAR.P()` function. The corresponding functions for estimating standard deviation are `STDEV.S()` for sample data and `STDEV.P()` for population data. In the parentheses for each respective function, we specify the values, or reference to cells with the values, that we want to perform the analysis on. For example: `VAR.S(1,2,3)` or `VAR.S(A:A)`.

Excel also has several functions for estimating covariation. To estimate the covariance between two variables, we can use the `COVARIANCE.P()` function if we have population data and the `COVARIANCE.S()` function if we have sample data. The figure below shows an example where we estimate the covariance between variables X and Y. Data for the variables can be found in cells A2:B5. We enter `COVARIANCE.S()` in cell D2 and specify in the parentheses references to the variables' data in cells A2:A5 and B2:B5 respectively.



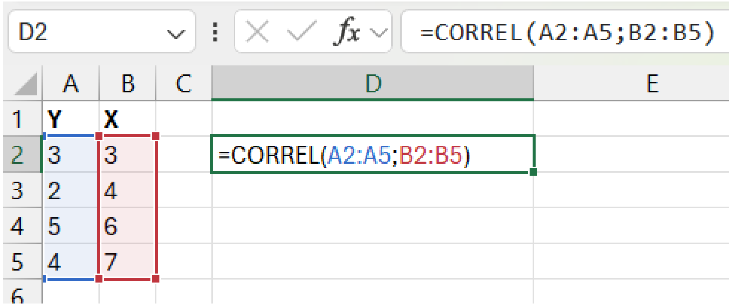
Covariance with COVARIANCE.S()

To estimate the correlation coefficient, Pearson's r, we can use the `CORREL()` function. In the parentheses we specify references to the values for the two variables we want to estimate the covariation between. The figure below shows an example where we estimate the correlation coefficient between Y and X. We enter `CORREL()` in cell D2 and refer to data in cells A1:A5 and B1:B5 respectively.

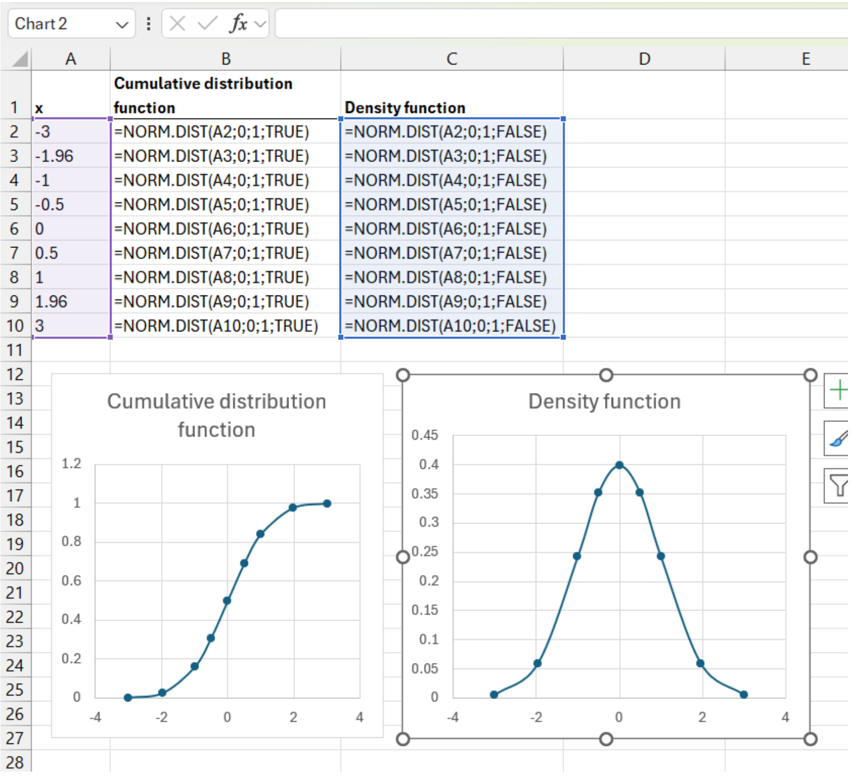
29.11 Probability

Excel has several ready-made functions for calculating probabilities based on some of the most well-known probability distributions. To calculate probabilities for a random variable that follows a normal distribution, we can for example use the `NORM.DIST()` function. The function has four arguments: the input value x, the distribution's mean, the distribution's standard deviation, and a logical argument regarding whether we want to use the normal distribution's cumulative distribution function $F(x)$ or density function $f(x)$.

The figure below shows an example where we use `NORM.DIST()` in two columns, one for the cumulative distribution function and one for the density function. In all cells we specify mean 0 and standard deviation 1, that is, the standard normal distribution. In column A we find the input values for x: -3 to 3. The figure shows the formulas in all cells in columns B and C. In the charts below these cells we can see the returned values illustrated in a chart for the cumulative distribution function (left) and one for the density function (right).



Correlation with CORREL()



Calculate probabilities for the normal distribution

If we want to know which value x corresponds to a specified probability in the normal distribution's cumulative distribution function $F(x)$, we can use Excel's `NORM.INV()` function. The function has three arguments: a value for the probability, the distribution's mean, and the distribution's standard deviation. Say as an example that we have a standard normal distribution, like the example in the figure above. Since the mean marks half of the probability distribution, $x = 0$ gives that the cumulative distribution function $F(0) = 0.5$. If we want to know which value for x gives this cumulative probability, we write `=NORM.INV(0.5, 0, 1)` in a cell in our worksheet and press Enter.

There are several functions that correspond to `NORM.DIST()` but for other probability distributions. If we want to work with the standard normal distribution and avoid having to specify mean and standard deviation, we can use the `NORM.S.DIST()` function.

The `T.DIST()` function returns the probability for a left-tailed t-distribution based on three arguments: variable value x, degrees of freedom, and whether the function should return values for the distribution function (TRUE) or density function (FALSE). The corresponding function for getting probabilities for a right-tailed t-distribution is `T.DIST.RT()`, which only has two arguments: variable value x and degrees of freedom.

The `T.DIST.2T()` function returns the probability for a two-tailed t-distribution based on two arguments: variable value x and degrees of freedom. To instead obtain the variable value x that corresponds to a specified probability for a two-tailed t-distribution, we can use the `T.INV.2T()` function. This function has two arguments: probability and degrees of freedom. If we for example write `T.INV.2T(0.05, 300)`, the function returns approximately 1.9679. If we then specify `T.DIST.2T(1.9679, 300)` based on this, this function in turn returns the value 0.05. That is, 5 percent.

Table 29.1: Probability distributions in Excel

Probability Distribution	**Example Functions in Excel**
Binomial distribution	BINOM.DIST()
Poisson distribution	POISSON.DIST()
Exponential distribution	EXPON.DIST()
Gamma distribution	GAMMA.DIST()
Chi-square distribution	CHISQ.DIST(), CHISQ.DIST.RT(), CHISQ.INV(), CHISQ.INV.RT()
F-distribution	F.DIST(), F.DIST.RT(), F.INV(), F.INV.RT()
Normal distribution	NORM.DIST(), NORM.INV()
T-distribution	T.DIST(), T.DIST.RT(), T.DIST.2T(), T.INV.2T()

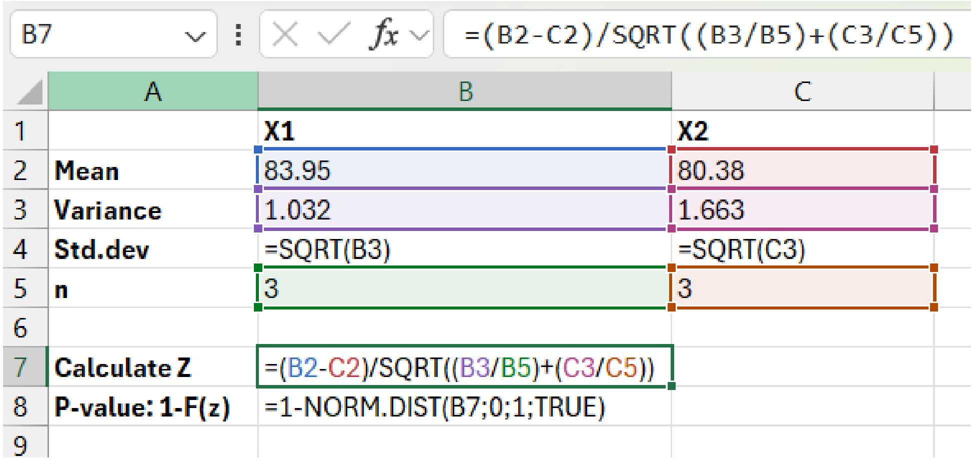
The table above describes examples of other functions in Excel for various common probability distributions. See each function's help section for more information about how that particular function works. Excel has no ready-made functions for uniform probability distributions.

Excel also has ready-made functions for statistical tests based on several common probability distributions. The normal distribution is symmetric with half of the probability distribution below and above the mean respectively. This means that the distribution function $F(-x) = 1 - F(x)$. For a one-tailed statistical test with the normal distribution and confidence level $\alpha = 0.05$, we set the critical value in one tail so that the calculated statistic must be greater than 95% of the distribution. For a two-tailed test with the same α, the critical value in each respective tail corresponds to $\alpha/2 = 0.025$, which means that the critical region in each respective tail corresponds to 2.5% of the distribution.

Say as an example that we have two normally distributed variables with means $\bar{X}_1 = 83.95$ and $\bar{X}_2 = 80.38$ with variance (standard deviation squared) $s_1^2 = 1.663$ and $s_2^2 = 1.032$ and $n_1 = n_2 = 3$. We use a z-test (based on the standardized normal distribution) to calculate the probability that these two variables come from the same population with the same mean:

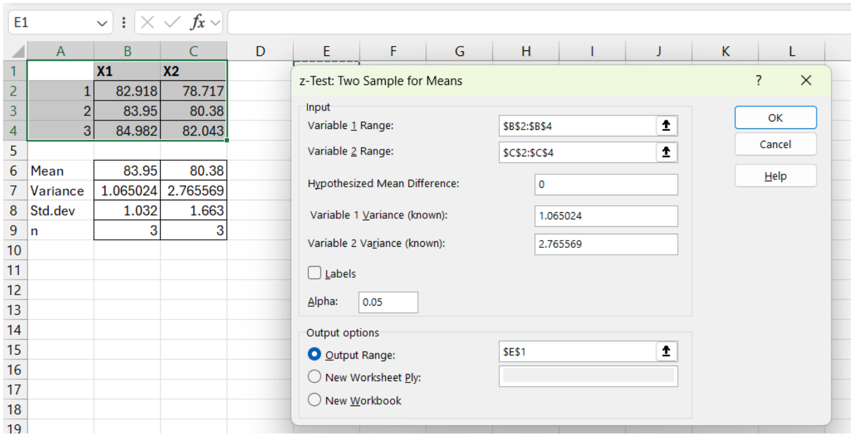
$$z = \frac{\bar{X}_1 - \bar{X}_2}{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^{\frac{1}{2}}} \approx \frac{83.95 - 80.38}{\left(\frac{1.032}{3} + \frac{1.663}{3}\right)^{\frac{1}{2}}} \approx 3.767$$

which we can compare against the standard normal distribution's distribution function. Our calculated z-value is so far from the mean that the probability that the two samples would come from a population with the same mean is under one hundredth of a percent. The figure below shows how we can calculate this with Excel. Rows 2 to 5 contain information from samples for variables X_1 and X_2. In cell B7 we calculate the z-value above. In cell B8 we calculate the p-value by taking $1 - F(z)$ where $z = 3.767$, which gives $p \approx 0.000083$. That is, the probability that the two populations have the same mean is very low.



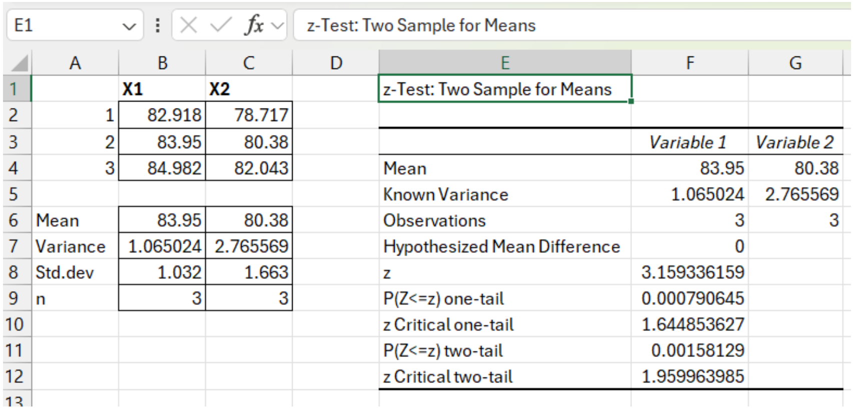
Z test

Excel also has statistical functions that we can access through the menus. For the next step we first need to install the Analysis ToolPak add-in. Go to the File menu > Options. The Excel Options dialog box opens. Select the Add-ins tab. Under the Manage menu, select Excel Add-ins. Click Go... and check the box for Analysis ToolPak. Click OK. Now we have more tools for statistical analysis, including the Data Analysis dialog box. Go to the Data menu > Data Analysis. A dialog box opens. Select "z-Test: Two Sample for Means". Click OK. A new dialog window opens, see the figure below.



Z-test via dialog interface

In the worksheet we have three observations for X_1 and X_2 respectively in cells B2:B4 and C2:C4. In the dialog window for the z-test, we enter in the Variable 1 Range box a reference to the cells that contain the observations for X_1: B2:B4. In the Variable 2 Range box we enter a reference to the observations for X_2: C2:C4. In the Hypothesized Mean Difference box we enter the difference we want to test whether it exists between the two variables. In this case we enter 0 since we want to compare whether the two samples come from the same population and therefore have the same mean. In the variance boxes we enter the variance for each respective variable, which in this example is specified in cells B7 and C7 respectively. Under Output options we can choose where the results should be pasted. If we specify cell E2, the results are displayed in the same worksheet, see the figure below.



E1 | fx | z-Test: Two Sample for Means

	A	B	C	D	E	F	G
1		X1	X2		z-Test: Two Sample for Means		
2	1	82.918	78.717				
3	2	83.95	80.38			*Variable 1*	*Variable 2*
4	3	84.982	82.043		Mean	83.95	80.38
5					Known Variance	1.065024	2.765569
6	Mean	83.95	80.38		Observations	3	3
7	Variance	1.065024	2.765569		Hypothesized Mean Difference	0	
8	Std.dev	1.032	1.663		z	3.159336159	
9	n	3	3		P(Z<=z) one-tail	0.000790645	
10					z Critical one-tail	1.644853627	
11					P(Z<=z) two-tail	0.00158129	
12					z Critical two-tail	1.959963985	
13							

Results from Z-test

29.12 Regression analysis

This section describes different methods for performing regression analysis with Excel. First we go through how we can estimate the coefficients in the regression model using matrix calculations, and then how we can use various shortcuts in Excel.

29.12.1 Regression analysis using matrices

Now we will estimate the regression model:

$$y = a + bx + u$$

where y and x are variables, a and b are coefficients and u is the error term. A regression model can be written with matrices as

$$Y = BX + U$$

where Y is a column matrix with the dependent variable and its observations. Y has as many rows as observations. Matrix B is the coefficients. Matrix X contains the explanatory variables. The first column in matrix X has the value 1, because in the regression model above we have coefficient a, which is not multiplied by any variable (the model's y-intercept). U is a column matrix with the error terms (number of rows = number of observations). The least squares estimator $\hat{B}$ can be described with the following equation:

$$\hat{B} = \left(X^T X\right)^{-1} X^T Y$$

where X^T is the transposed matrix X and $\left(X^T X\right)^{-1}$ is the inverse of matrix $X^T X$. We will now estimate the regression model with an example where we use four observations for the variables $y = \{3, 4, 6, 7\}$ and $x = \{3, 2, 5, 4\}$. The first observation is $(y, x) = (3, 3)$ and the second $(y, x) = (4, 2)$. Matrices Y and X are:

$$Y = \begin{bmatrix} 3 \\ 4 \\ 6 \\ 7 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 3 \\ 1 & 2 \\ 1 & 5 \\ 1 & 4 \end{bmatrix}$$

The figure below shows how we can estimate the regression model using Excel's matrix functions that we introduced above. In cells A2:A5 we find the values for variable y, which are also the elements in matrix Y. In cells C2:D5 we find the values for matrix X. In cell A8 we use the `TRANSPOSE()` function to create the transposed matrix X^T. In cell A12 we use `MINVERSE()` to create the inverse matrix $\left(X^T X\right)^{-1}$. In cell

A16 we use the `MMULT()` function to matrix multiply X^T with Y. In cell C16 we use `MMULT()` to matrix multiply $(X^T X)^{-1}$ with $X^T Y$, which gives $\hat{B}$. In the two cells C16 and C17 we obtain the estimated coefficients $\hat{a} = 1$ and $\hat{b} = 0.5$ respectively. From here we can continue and for example calculate t-tests for the coefficients, but for this we will instead use shortcuts in the next section.

C16 =MMULT(A12#;A16#)

	A	B	C	D
1	Y		X	
2	3		1	3
3	2		1	4
4	5		1	6
5	4		1	7
6				
7	X^T			
8	=TRANSPOSE(C2:D5)			
9				
10				
11	(X^T X)^-1			
12	=MINVERSE(MMULT(A8#;C2:D5))			
13				
14				
15	X^T Y		B^	
16	=MMULT(A8#;A2:A5)		=MMULT(A12#;A16#)	=a
17				=b
18				

Regression analysis with matrices

29.12.2 Regression analysis with LINEST()

In Excel, we can estimate regression models using the `LINEST()` function. We estimate again the regression model $y = a + bx + u$. In the parentheses of `LINEST()` we first need to specify the observations for our dependent variable, which in this case is y. After a comma we specify a reference to the cells where we have observations for our explanatory variables, which in this case is only variable x.

The figure below shows an example where column A contains the values for variable y and column B contains variable x. In cell C6 we use the `LINEST()` function. In the parentheses of the function we refer to the values in columns A and B. The function returns the estimated coefficients $\hat{b}$ and $\hat{a}$ in cells C6 and D6. Note that the function first reports the slope coefficient $\hat{b}$ and then $\hat{a}$. The result is exactly as before: $\hat{a} = 1$ and $\hat{b} = 0.5$.

	A	B	C	D
1	Y	X		
2	3	3		
3	2	4		
4	5	6		
5	4	7		
6			=b	=a
7			=LINEST(A2:A5;B2:B5)	
8				

Regression analysis with LINEST()

Let us take another example. We will now use `LINEST()` to instead estimate the regression model:

$$Y = c + dX + eZ + \epsilon$$

where Y, X and Z are variables and c, d and e are the coefficients. ϵ is the error term. The figure below shows an example where we use `LINEST()` and in the parentheses refer to cells A2:A5 (variable Y) and B2:C5 (variables X and Z). In cells D6, E6 and F6 the three coefficients $\hat{e} \approx 2.89$, $\hat{d} \approx 0.28$ and finally $\hat{c} \approx -0.54$ are returned, which is thus in reverse order compared to how we set up our regression model.

	A	B	C	D	E	F
1	Y	X	Z			
2	3	3	1			
3	2	4	4			
4	5	6	0			
5	4	7	1			
6				=c	=b	=a
7				=LINEST(A2:A5;B2:C5)		
8						

LINEST() with two explanatory variables

The `LINEST()` function has two optional arguments, const and stats, which we can choose to leave unchanged. The const and stats options are controlled by the logical arguments TRUE and FALSE. The default setting for `LINEST()` is that const=TRUE and stats=FALSE, which is used if we don't specify anything else in the parentheses of `LINEST()`. The results illustrated in the figure above are created with these default settings.

If we instead specify const=FALSE, then the regression model $Y = dX + eZ + \epsilon$ is estimated, without the y-intercept c. The second optional argument in `LINEST()` is stats, which controls whether `LINEST()` should also report results for t-tests for the coefficients, F-test for the entire model, and estimate R^2.

The figure below shows an example where we set the option stats=TRUE, with the same four observations for variables y, x and z in columns A, B and C. The results are reported in cells D6:F10. When stats=TRUE, `LINEST()` reports results for a regression model with k coefficients a_1 to a_k with corresponding standard errors se_1 to se_k. Estimated coefficients are shown in the first row and standard errors in the second row. The regression statistics include R^2, estimated standard error for the residuals $\hat{s}_{\hat{\epsilon}}$, F-statistic, degrees of freedom for the F-test df_F, the sum of squared explained variation SSE, and the residual sum of squares SSR.

D7 | `=LINEST(A2:A5;B2:C5;;TRUE)`

	A	B	C	D	E	F	G
1	Y	X	Z				
2	3	3	1				
3	2	4	4				
4	5	6	0				
5	4	7	1				
6				=c	=b	=a	
7				-0.54054	0.283784	2.891892	
8				0.213667	0.202703	1.219964	
9				0.932432	0.581238	#N/A	
10				6.9	1	#N/A	
11				4.662162	0.337838	#N/A	
12							

Regression analysis using LINEST() and stats = TRUE

The top row is in the example row 6 in the worksheet with the estimated coefficients $\hat{c} \approx -0.54$, $\hat{d} \approx 0.28$ and $\hat{e} \approx 2.89$. The second row, row 7, is the standard error for each respective coefficient: $\hat{s}_{\hat{c}} \approx 0.21$, $\hat{s}_{\hat{d}} \approx 0.2$ and $\hat{s}_{\hat{e}} \approx 1.22$.

Cells D8 to D10 report $R^2 \approx 0.93$, F-statistic for the F-test for the entire regression model $F = 6.9$, and the sum of squares of the explained variation $SSE = \sum_i^n \left(\hat{Y}_i - \bar{Y}\right)^2 \approx 0.34$.

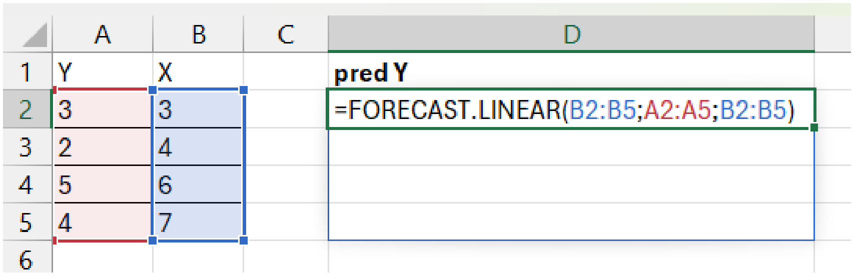
Cells E8 to E10 report estimated standard error for the residuals $\hat{s}_{\hat{\epsilon}} \approx 0.58$, degrees of freedom for the F-test $df_F = 1$, and the residual sum of squares $SSR = \sum_i^n \epsilon_i^2 \approx 4.66$.

In cells F9:F11, #N/A! is returned, which just means that the function does not return any value for these cells. This is not an error. See Excel's help section for more information about the `LINEST()` function and the results.

If we have a regression model with one explanatory variable, we can estimate predicted $\hat{Y}$ with the `FORECAST.LINEAR()` function. From older versions of Excel there are also the similar functions `FORECAST()` and `TREND()`. The figure below shows an example where we predict the regression model:

$$\hat{Y} = a + bX + U$$

where a and b are coefficients and U is the error term. In cells A2:B5 we find the observations for variables Y and X. In cell B2 we write the `FORECAST.LINEAR()` function. In the parentheses we specify three arguments: (1) the X-values that should be used to calculate the predicted values for $\hat{Y}$(which can be any X-values we want), (2) the known Y-values, and (3) the known X-values. The result is shown in cells C2:C5.



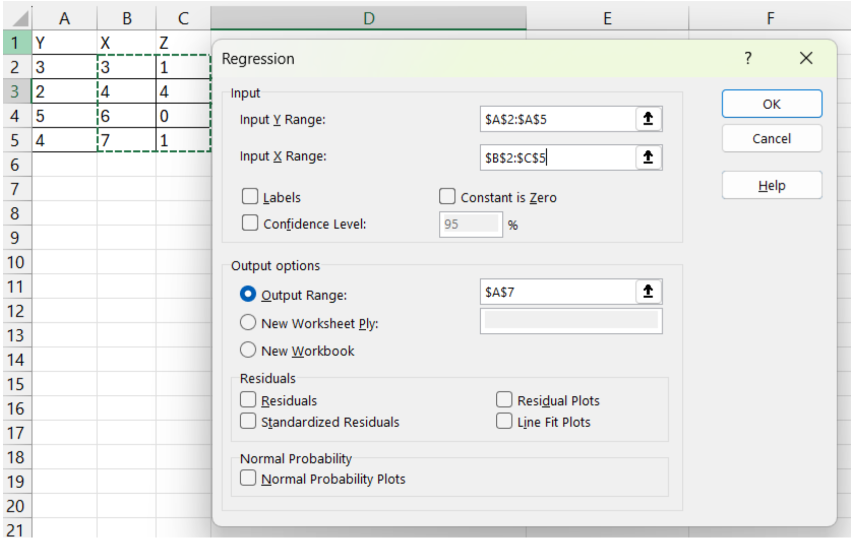
Prediction of $\hat{Y}$ with FORECAST.LINEAR()

29.12.3 Regression analysis with Analysis ToolPak

We can also perform regression analysis in Excel through the Data Analysis dialog box, which we installed using Analysis ToolPak, see the section on probability above. Click on Data menu > Data Analysis. A new dialog window opens. Select "Regression" and click OK. The dialog window for regression opens, see the figure below. We will now again estimate the regression model:

$$Y = c + dX + eZ + \epsilon$$

In the Y Input Range box we specify a reference to the cells that contain the dependent variable's data, including the variable header with the name of the variable. In this example this is cells A1:A5. Check the Labels box, which indicates that we include the variable name in the top cell in the input range. In the X Input Range box we specify a reference to all cells that contain the explanatory variables, which in this case are X and Z. The cells are B1:C5. Under Output options we can choose the settings for where our results should be displayed. In the figure below we have chosen that the results should be displayed in cell A7.



Dialog window for regression analysis

If we click OK, we get the results shown in the figure below, in the form of three different tables with results from the regression analysis. The top table, rows 10–14, first shows “Multiple R”, which stands for the correlation coefficient, also called Pearson’s r, which here indicates the correlation between the dependent variable Y and predicted $\hat{Y}$. The correlation coefficient takes a value between -1 (perfect negative correlation) and 1 (perfect positive correlation). A result of 0 means no correlation.

R Square (R^2) and Adjusted R Square measure what proportion of the variation in the dependent variable can be explained by the regression model and the variation in the explanatory variables. Standard Error here refers to the square root of the mean square sum of the residuals:

$$(MSR)^{\frac{1}{2}} = \left(\frac{\sum (y_i - \hat{y})^2}{p-1}\right)^{\frac{1}{2}}$$

where p is the number of estimated coefficients in the regression model. The last row in the top results table is the number of observations, which in this example is 4.

The second table, rows 16–20, is called the ANOVA (Analysis Of VAriance) table where the headings stand for df = degrees of freedom, SS = sum of squares, MS = mean squares, F = F-statistic, and the p-value for the F-test. The F-statistic is calculated:

$$F = \left(\frac{SSR_{m2} - SSR_{m1}}{SSR_{m1}}\right) / \left(\frac{q}{n - k_{m1} - 1}\right)$$

where SSR stands for the residual sum of squares, $q = df_{m1} - df_{m2}$ and df stands for degrees of freedom and k is the number of explanatory variables. The F-test tests the null hypothesis that the population coefficients for the two explanatory variables are zero:

$$H_0 : d = e = 0$$

against the alternative hypothesis that at least d or e is statistically significantly different from zero.

The third and bottom table, rows 22–25, contains the estimates for the coefficients and associated statistical results. In the first column A, the name of each respective coefficient is given, where “Intercept” refers to the first coefficient c, the y-intercept, which is not multiplied by any variable. The other two, X and Z, refer to coefficients d and e which are multiplied by their respective variables.

The columns in the third and bottom table to the right of the coefficients’ names are Coefficients (the estimates), estimated standard error, t Stat, and P-value for a t-test for the null hypothesis:

$$H_0 : c = 0,\ H_1 : c \neq 0$$

That is, the p-value indicates the probability that the coefficient is statistically significantly different from zero. The columns thereafter show estimated bounds for the 95% confidence interval for each respective coefficient.

7	SUMMARY OUTPUT								
8									
9	*Regression Statistics*								
10	Multiple R	0.965625409997289							
11	R Square	0.932432432432432							
12	Adjusted R Square	0.797297297297297							
13	Standard Error	0.581238193719097							
14	Observations	4							
15									
16	ANOVA								
17		*df*	*SS*	*MS*	*F*	*Significance F*			
18	Regression	2	4.66216216216216	2.33108108108108	6.9	0.259937622455018			
19	Residual	1	0.337837837837838	0.337837837837838					
20	Total	3	5						
21									
22		*Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
23	Intercept	2.89189189189189	1.21996419495053	2.37047275966092	0.254141987430942	-12.6092229399521	18.3930067237358	-12.6092229399521	18.3930067237358
24	X Variable 1	0.283783783783784	0.202702702702703	1.4	0.394863086577493	-2.29179825733271	2.85936582490028	-2.29179825733271	2.85936582490028
25	X Variable 2	-0.540540540540541	0.213667409470837	-2.5298221281347	0.23964588261831	-3.25544239072506	2.17436130964398	-3.25544239072506	2.17436130964398

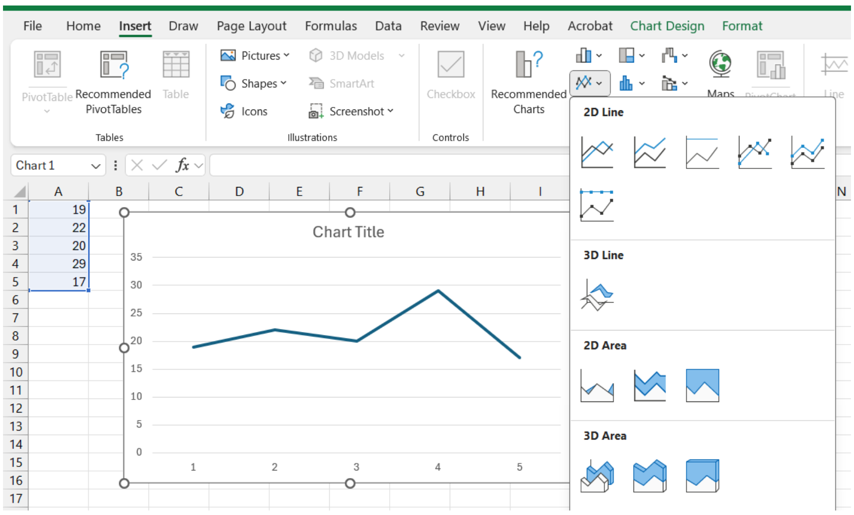
Regression results from the Data Analysis menu

There are no pre-installed functions in Excel for estimating regressions with robust standard errors. We can do this with our own calculations, for example with matrix calculations. If you search online, you can also find functions that private users have created themselves that you can download. In the chapters below we go through how we can easily perform this type of regression with commands in Stata and R.

29.13 Creating plots

Excel has shortcuts for creating several common charts, such as line charts, scatter plots, bar charts, and more. In the menus we find the functions for creating charts under the Insert heading. A simple way to create a chart quickly is to first select the cells with data that we want to illustrate in a chart. Then we click on the Insert menu and choose what type of chart we want to create.

The figure below shows an example where we have created a line chart using the menu with the five values in column A. To create the line chart we select the Insert menu, click on the symbol for line charts next to the larger icon for "Recommended Charts" and choose the first option under 2-D Line.



Create a line plot

When Excel creates a chart this way, the program makes some guesses about how we want our selected chart to look. Sometimes the program guesses correctly and sometimes it guesses incorrectly. In the chart in the figure above, the values in column A have been placed on the vertical y-axis. If we wanted the values on the horizontal x-axis instead, we would need to instruct the program about this. We can also select multiple things simultaneously and create a chart based on this.

To edit a chart that we have created, we can click on the chart. When we select the chart, the two new menu headings Chart Design and Format open, at the far right among the menus at the top. Under these menus we can then choose various ready-made options to change the appearance of the chart. For example, add points along the line, add or remove titles for the y- and x-axes, and more.

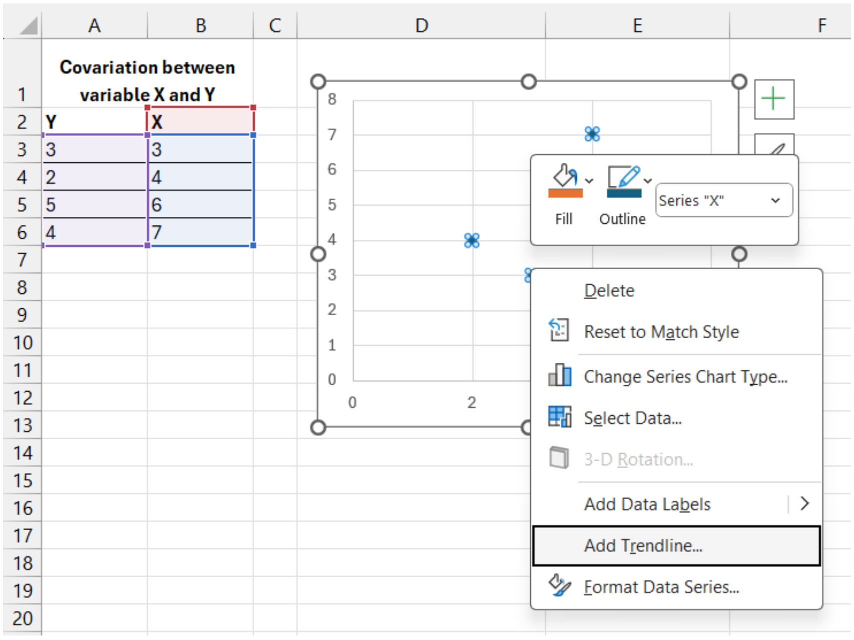
We can also right-click on different parts of the chart and get different options, depending on which parts of the chart we click on. If we want to keep the chart but change which values are shown in the chart, we can right-click somewhere in the chart and choose the Select Data... option. This way we can add or remove variables and observations. By clicking on a chart's different parts, we can change appearance, color and shape, and even what type of chart, if we would prefer a bar chart or scatter plot, for example. There isn't space to go through all possible options for how we can or should edit our charts here. There are several help sections regarding these things in Excel. Many things are also relatively simple to understand by just experimenting by selecting different parts of the chart, trying the options in the menus, and the different options that appear when we right-click in the chart.

Scatter plots are often used to illustrate the covariation between two variables. One way to calculate a linear covariation between two variables is the least squares method. Say that we have the regression model:

$$Y = a + bX + u$$

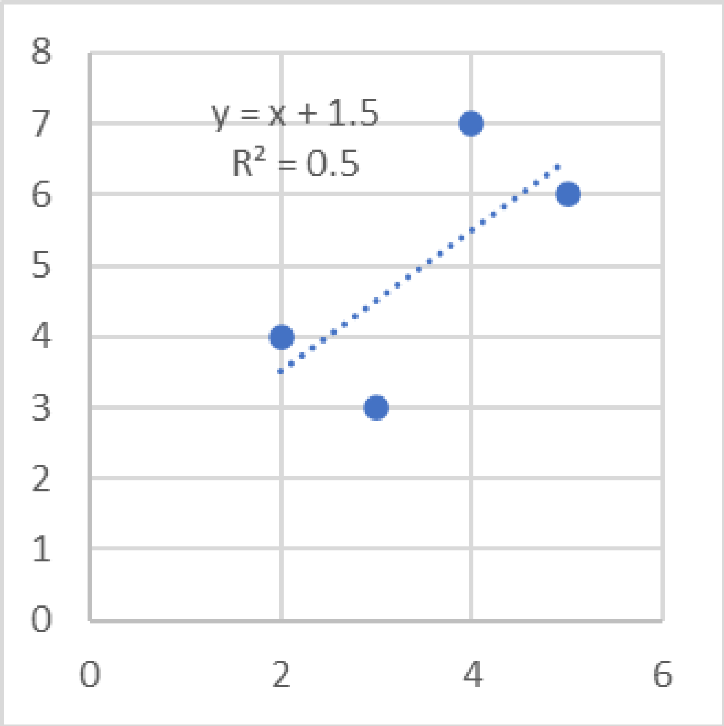
We have four observations with values for variables Y and X that we will use to calculate the constants a and b. The letter u is the error term. With Excel we can create a scatter plot that shows the covariation between Y and X. The figure below shows an example with a scatter plot, which is created by selecting Insert menu > Scatter. In the chart we can also add the regression line that we calculate using the least squares method.

After we have created the chart, we right-click on one of the points in the chart. This opens a menu where we select the option "Add Trendline...". A new dialog box opens where we can specify what type of trendline we want to create. To create a linear regression line, calculated using the least squares method, we select the "Linear" option.



Add the regression line to a scatter plot

In the same menu that we use to define the trendline, we can also choose the option "Display Equation on chart". If we click this option, the equation that draws the regression line is written out in the chart. We can also choose to show the calculated R^2 by clicking the option "Display R-squared value on chart". The figure below shows the chart with trendline and equation after we have clicked these options.



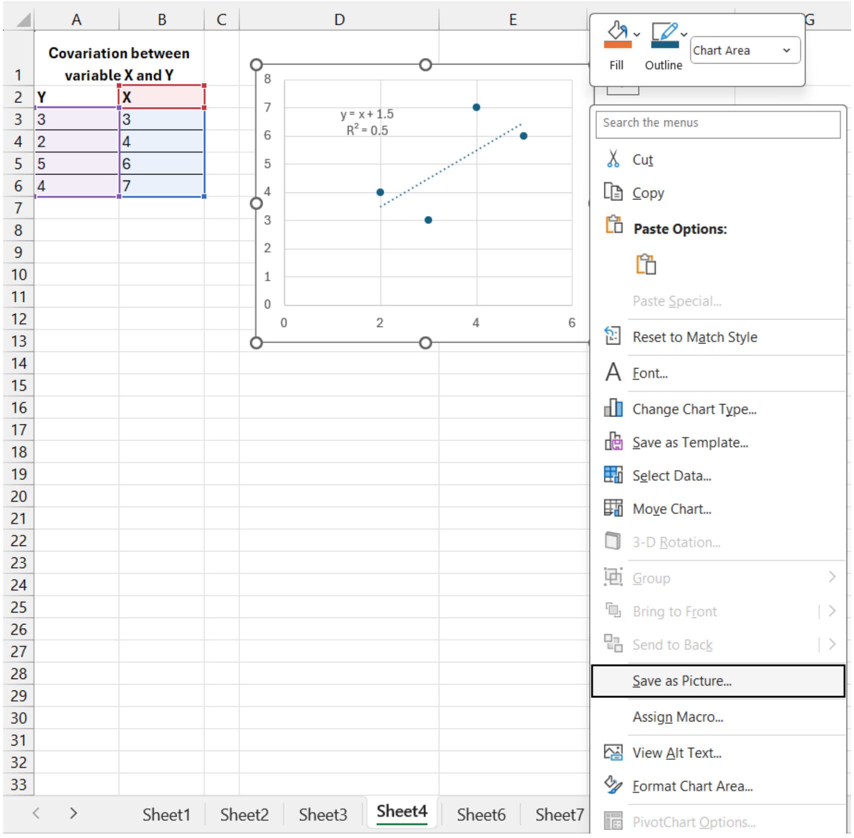
Finished chart with regression line

29.13.1 Presenting results in other programs

When we have calculated a result in an analysis program, we often want to present our results, for example tables and charts, in another program, such as in a text that we write in Microsoft Word or in a presentation that we make in Microsoft PowerPoint. There are several ways to do this and the methods can vary depending on which analysis program we are working in, as well as which program we want to export the results to.

If we work in Microsoft's programs Excel, Word and PowerPoint, it is usually easy to copy and paste results from one program to another. The programs are designed to be able to work together. To export a table in Excel, for example, we can select the cells we want to show, copy (Ctrl + C), place the cursor in the other program at the location where we want to paste the table, and paste (Ctrl + V). The same method also works with charts we have created in Excel.

We can also paste tables and charts from Excel into other programs as images. This method often works in both Microsoft's programs Word and PowerPoint, as well as in programs from other manufacturers. Charts can also be exported as images directly from Excel to an image file on the computer's hard drive. This can be done by right-clicking on the outer surface of a chart, the outer frame, and selecting Save as Picture.... This opens a new dialog box where we can choose which image file format we want to save the chart as and in which directory on the computer's hard drive we want the file to be saved, see the figure below.



Save chart as image from Excel

29.14 Example with GDP per working hour

Now we will go through an example where we use the data shown in the table below. The figures show millions of gross domestic product (GDP) per working hour calculated in US dollars, adjusted for price changes over time (inflation) and price differences between countries (purchasing power adjusted). Purchasing power adjusted values are called Purchasing power parities in English.

The data covers Sweden, Denmark, Finland, Norway, USA, and the United Kingdom for the years 1990, 2000, 2010, and 2020 and is taken from the database that the think tank The Conference Board has online. The database contains, in addition to this data, other variables and many more countries and years, which can be downloaded for free at the following address: <www.conference-board.org/data/>

Table 29.2: GDP per working hour

Country	1990	2000	2010	2020
Denmark	62.6	76.6	84.2	95.9
Finland	47	64.3	73.5	77
Norway	68.1	89.3	95.9	101.1
Sweden	54.6	68.4	81.6	89.3
United Kingdom	50	63.1	70.4	73.4
United States	53.2	63.6	79.5	87.4

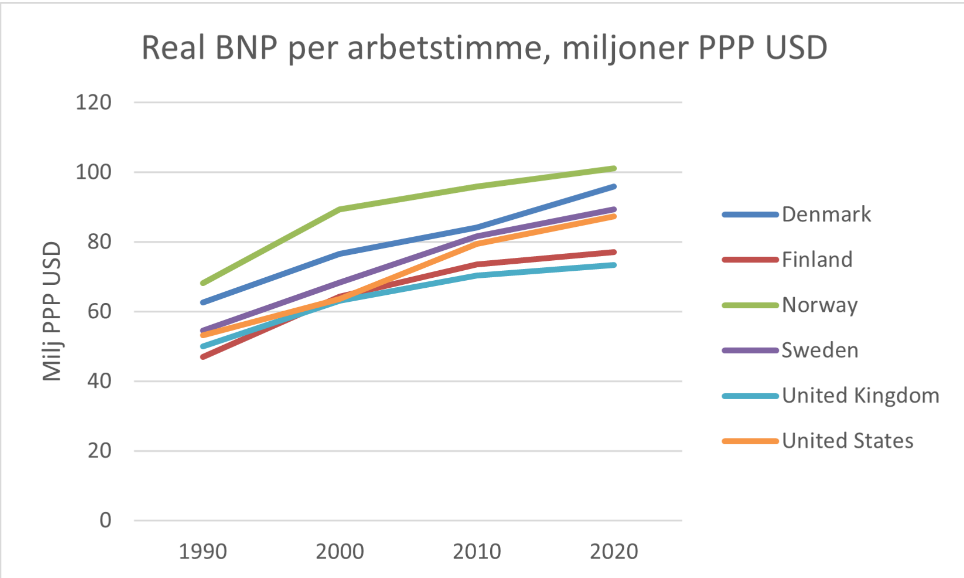
Source: The Conference Board. Real GDP purchasing power adjusted millions USD, divided by hours worked per year.

GDP indicates the value of all goods and services produced during the relevant year, divided by all hours worked the same year for each respective country. GDP per working hour is therefore a measure of how productive all companies are in a country. Since we measure the value of everything produced, this is also a measure of income level. We adjust for inflation to thereby get an estimate of the value of incomes, how much we can buy for the money. We adjust for purchasing power in different countries to adjust for the fact that goods and services cost different amounts in different countries.

There is no method for measuring all these things exactly, which is why the figures should be regarded as approximate. Working time is calculated partly by asking people how much they work, which for natural reasons is difficult to estimate exactly. It is also difficult to adjust for price changes over time and between countries, which is why this too must be interpreted with some caution. But the fact that things are difficult to measure does not mean that the figures are unusable. The measurement methods have been refined over decades and can be used to make several interesting observations. Authorities, organizations, and experts have also synchronized their methods worldwide to make the figures comparable over time and between countries. This means that even though we should be careful about drawing strong conclusions about small variations, we can still get an idea of overall differences between countries and years.

To work with these figures, we need to copy and paste them into an Excel sheet, or write them down. You can also download the material from The Conference Board yourself and make your own calculations. If we want to know the average productivity for the countries in the table, we can use Excel's `AVERAGE()` function. We can, for example, calculate average productivity per year by referring in the parentheses of `AVERAGE()` to the six boxes per year.

A good way to get a bit more overview of the data is to create charts. The figure below shows an example of how it can look in a line chart created in Excel. In the chart we can clearly see that Sweden and the USA are mostly in the middle of the group during these years. At the top are Norway and Denmark. At the bottom are Finland and the United Kingdom.



GDP per working hour, illustrated in chart in Excel

The chart is created with one of the common presets available in Excel. It is possible to change the appearance and colors in many ways by, for example, selecting all or parts of the chart and clicking in the Chart Design or Format menus, and for example choosing among the options Quick Layout, Add Chart Element, or Change Colors.

GDP per working hour increases over the period in the long term. One way to compare the development is to create an index per country. We choose 1990 as the base year and give this year the index value 100. We calculate index x_{it} for country i in year t with the following formula:

$$x_{it} = \frac{GDP_{it}}{GDP_{i,1990}} * 100$$

The figure below shows how this calculation can look in Excel. The data on GDP per working hour is collected in cells B2:E7. Index per country is calculated in cells B11:E16, compare the section on calculating index above. To fill in all cells faster, we can use one of the tricks we went through in the example in that section. We can, for example, enter the formula in cell B11: `=B2/ $B$ 2`, select cells

B11:E16 and press Ctrl + D and Ctrl + R. This makes the formula in B11 copy to all other cells we selected.

The cell reference in our index calculation's numerator, B2, is updated. The cell reference in the denominator `$B$ 2` is locked on column B but not on row 2. The cell reference will therefore update the row but not the column, which means that each country will use the year 1990 as the base year, which is what we want.

	A	B	C	D	E
1	**country**	**1990**	**2000**	**2010**	**2020**
2	Denmark	62,6	76,6	84,2	95,9
3	Finland	47	64,3	73,5	77
4	Norway	68,1	89,3	95,9	101,1
5	Sweden	54,6	68,4	81,6	89,3
6	United Kingdom	50	63,1	70,4	73,4
7	United States	53,2	63,6	79,5	87,4
8					
9	Index				
10		**1990**	**2000**	**2010**	**2020**
11	**Denmark**	=B2/$B2 *100	=C2/$B2 *100	=D2/$B2 *100	=E2/$B2 *100
12	**Finland**	=B3/$B3 *100	=C3/$B3 *100	=D3/$B3 *100	=E3/$B3 *100
13	**Norway**	=B4/$B4 *100	=C4/$B4 *100	=D4/$B4 *100	=E4/$B4 *100
14	**Sweden**	=B5/$B5 *100	=C5/$B5 *100	=D5/$B5 *100	=E5/$B5 *100
15	**United Kingdom**	=B6/$B6 *100	=C6/$B6 *100	=D6/$B6 *100	=E6/$B6 *100
16	**United States**	=B7/$B7 *100	=C7/$B7 *100	=D7/$B7 *100	=E7/$B7 *100

Calculate index per country

Another way to compare the long-term development is to estimate the following regression model per country:

$$G_{it} = a_i + b_i T_{it} + \epsilon_{it}$$

where G_{it} is the natural logarithm of GDP per working hour in country i in year t, a_i is a coefficient for country i(the y-intercept, the constant), b_i is the slope coefficient for country i with respect to the explanatory variable T_{it} which is the year for country i in year t, and ϵ_{it} is the error term for country i in year t.

We take the natural logarithm of GDP per working hour, because we can then interpret slope coefficient b as an approximate measure of average percentage increase in GDP per working hour per year (when variable T increases by one).

To estimate this regression model in Excel, we can for example use `LINEST()`. As the first argument we use the natural logarithm of the values in the GDP table above. As the second argument we use the years 1990, 2000, 2010, and 2020. The figure below shows how this can look in an Excel sheet. In cells B11:E16 we use the `LN()` function to take the natural logarithm of the values in cells B2:E7. In cells G11:G16 we use the `LINEST()` function to estimate a regression model per row 11–16, which gives us an estimate per country.

	A	B	C	D	E	F	G	H
1		**1990**	**2000**	**2010**	**2020**			
2	Denmark	62,6	76,6	84,2	95,9			
3	Finland	47	64,3	73,5	77			
4	Norway	68,1	89,3	95,9	101,1			
5	Sweden	54,6	68,4	81,6	89,3			
6	United Kingdom	50	63,1	70,4	73,4			
7	United States	53,2	63,6	79,5	87,4			
8								
9	logaritmerade värden						Koefficient	
10		**1990**	**2000**	**2010**	**2020**		**b**	**a**
11	Denmark	=LN(B2)	=LN(C2)	=LN(D2)	=LN(E2)		=REGR(B11:E11;$B$10:$E$10)	
12	Finland	=LN(B3)	=LN(C3)	=LN(D3)	=LN(E3)		=REGR(B12:E12;$B$10:$E$10)	
13	Norway	=LN(B4)	=LN(C4)	=LN(D4)	=LN(E4)		=REGR(B13:E13;$B$10:$E$10)	
14	Sweden	=LN(B5)	=LN(C5)	=LN(D5)	=LN(E5)		=REGR(B14:E14;$B$10:$E$10)	
15	United Kingdom	=LN(B6)	=LN(C6)	=LN(D6)	=LN(E6)		=REGR(B15:E15;$B$10:$E$10)	
16	United States	=LN(B7)	=LN(C7)	=LN(D7)	=LN(E7)		=REGR(B16:E16;$B$10:$E$10)	

One regression model per country

29.15 Some other useful functions

Excel has many more functions and options than can be covered here. Here are some brief tips on additional functions that are good to know about. You can easily find more information about each respective function in the program's help sections and online.

- `ROUND()`, `ROUNDDOWN()`, `ROUNDUP()`, `TRUNC()`: Functions for rounding numbers.
- `COUNT()`, `COUNTA()`, `COUNTIF()`, `COUNTIFS()`, `COUNTBLANK()`: Functions that return the number of cells that are not empty or that contain values, with or without consideration of conditions.
- `FIND()` and `SEARCH()`: Return the starting position for text strings (that we want to search for) in other text strings.
- `LEFT()`, `LEFTB()`, `RIGHT()`, `RIGHTB()`: Return one or more characters in a text string counted from the left or right.
- `MID()` and `MIDB()`: Return a number of characters from a text starting at the position we specify.
- `DATE()`, `YEAR()`, `WEEKNUM()`, `DAYS()`: Examples of functions for working with dates and time. Computer programs often handle dates and time specifications in special ways. As an example, we can use the `DATE()` function to create a date specification from three separate values, for example if we have the values for year, month, and day specified in separate columns. The `DAYS()` function returns the number of days between two specified dates.
- `INDEX()`, `MATCH()`, `LOOKUP()`: Functions for finding positions or unique values in a search range based on a specified search term.

29.16 Chapter Summary

- Excel's worksheet functions are written with the function's name followed by parentheses, for example `SUM()`. In the parentheses we specify the arguments the function requires. The `SUM()` function calculates the sum of the numbers that we specify in its parentheses. In the parentheses of functions we can also refer to cells with values. Other examples of functions: `AVERAGE()`, `MIN()`, `MAX()`, `LOG()`, `LN()`, `FACT()`, and `PERMUT()`.

- The worksheet functions can be used for many different types of work tasks, such as mathematical operations, calculating probability, and estimating regression models. Many functions use logical conditions, which are true or false. A logical condition in a function can often be answered by specifying TRUE or FALSE. To calculate with matrices we can use functions such as `MMULT()`, `TRANSPOSE()`, `MINVERSE()`.

- To create charts we can go to the Insert menu and choose chart type, for example 2-D Line. After we have created the chart we can also edit its appearance by clicking on its different parts, such as the axes' titles and scales.

- We can estimate regression models with the `LINEST()` function or using Analysis ToolPak and the Data Analysis dialog box. Excel has several functions for working with common probability distributions, for example `BINOM.DIST()` and `NORM.DIST()`.

Chapter 30

Stata

This chapter introduces the statistical analysis program Stata, whose latest version at the time of writing is called Stata 19. To follow along with the examples, you need to have access to the program on your computer. For instance, you can purchase a license at www.stata.com. If you are a student at a university, you may have access to a free license or special discounts through your university. This introduction to Stata is brief and aims to get the reader started with their own work as quickly as possible, thereby learning more through practice. There is abundant material offering more comprehensive introductions to Stata, see for example

www.stata.com/features/documentation

On this webpage, we find among other things the Stata User Guide:

www.stata.com/manuals/u.pdf

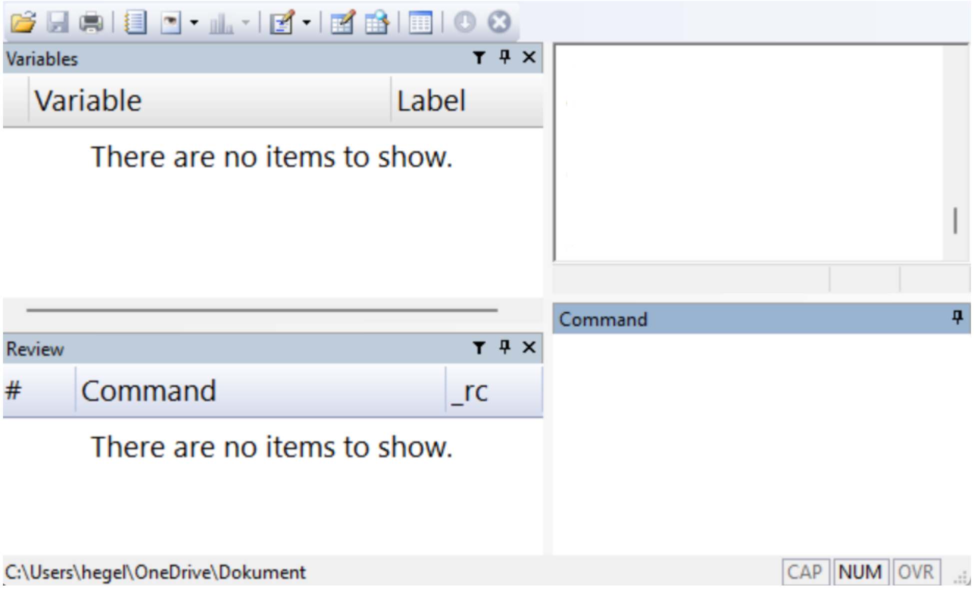
Stata also has support for a Swedish interface, meaning that menus and other elements are displayed in Swedish. See the link for how to change your settings:

www.stata.com/features/overview/swedish-interface/

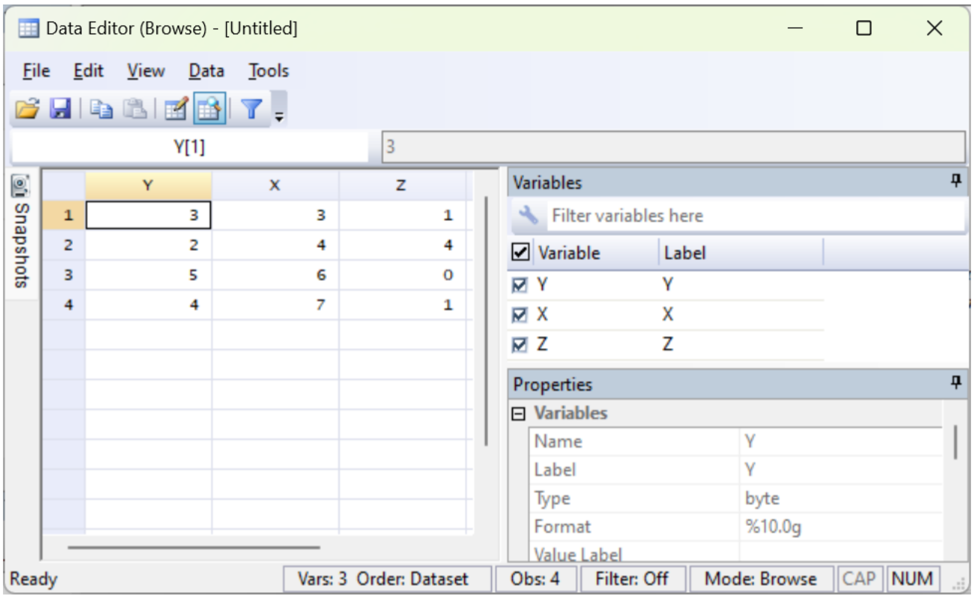
30.1 General info about Stata

The figure below shows an example of what Stata can look like. In the image we see four boxes, windows. In the upper left we see Variable, which shows a list of the variables (columns) we currently have access to. In the lower left, Command, shows a list of the most recently entered commands. In the lower left is the Command window. In the upper right we see the results window, Results.

It is also possible that additional windows other than those shown in the image are displayed. If any of the windows in the image are not shown when we open the program, we can open this and other windows by going to the Window menu and choosing among the options there: Command, Results, Review, Variables and more.



When we open Stata for the first time, we see no table. To look at the data that is loaded in the program, we can open the Data Editor window. This can be done through the menu Window > Data Editor, or with the keyboard shortcut Ctrl + 8. It is also possible to access the Data Editor via the icons at the top below the menus in Stata's window. When we open the Data Editor window for the first time, it is empty.



The figure above shows an example of what the data window can look like. In the image we have a table with the three variables Y, X and Z with four observations, where $Y = \{3, 2, 5, 4\}$, $X = \{3, 4, 6, 7\}$ and $Z = \{1, 4, 0, 1\}$. Note how the variable names appear as column headers. The variable names are thus not values among the observations, as they would be in an Excel sheet, for example.

In the boxes to the right in the image we can see a bit more information about the variables, such as what type of data each variable contains and what data format the variables have. This information about our data is sometimes called metadata. In the image, the first observation for variable Y is selected. The cell contains the value 3.

In the Properties window we see that the entire column with variable Y consists of data of the type float. Float is a data format for real numbers. To the right in the image we also see that variable Y has the format %10.0g, which describes how the real numbers in the column are displayed, for example number of decimals, see below. When we read a file with a data table from the hard drive into Stata, the table opens by default in the program's data window.

Somewhat simplified, we can describe it as Stata working with one data table at a time, or at least one main table. All commands we give the program are interpreted by the program as wanting to perform on this main table, unless otherwise specified. This is a difference from, for example, Microsoft Excel where we can have several tables with different types of information in the same worksheet, together with charts and more, for example.

This means, among other things, that a column in Stata is intended to contain only one type of information. All data in the column is classified as one data type, for example real values or text strings. As a comparison, in Excel we have the possibility to have different types of information saved in the same column or on the same row.

The Data Editor window can be opened in two modes: Browse, with data write-protected, and Edit, without write protection. If we open the Data Editor in Edit mode, we must be careful not to accidentally edit data and lose information.

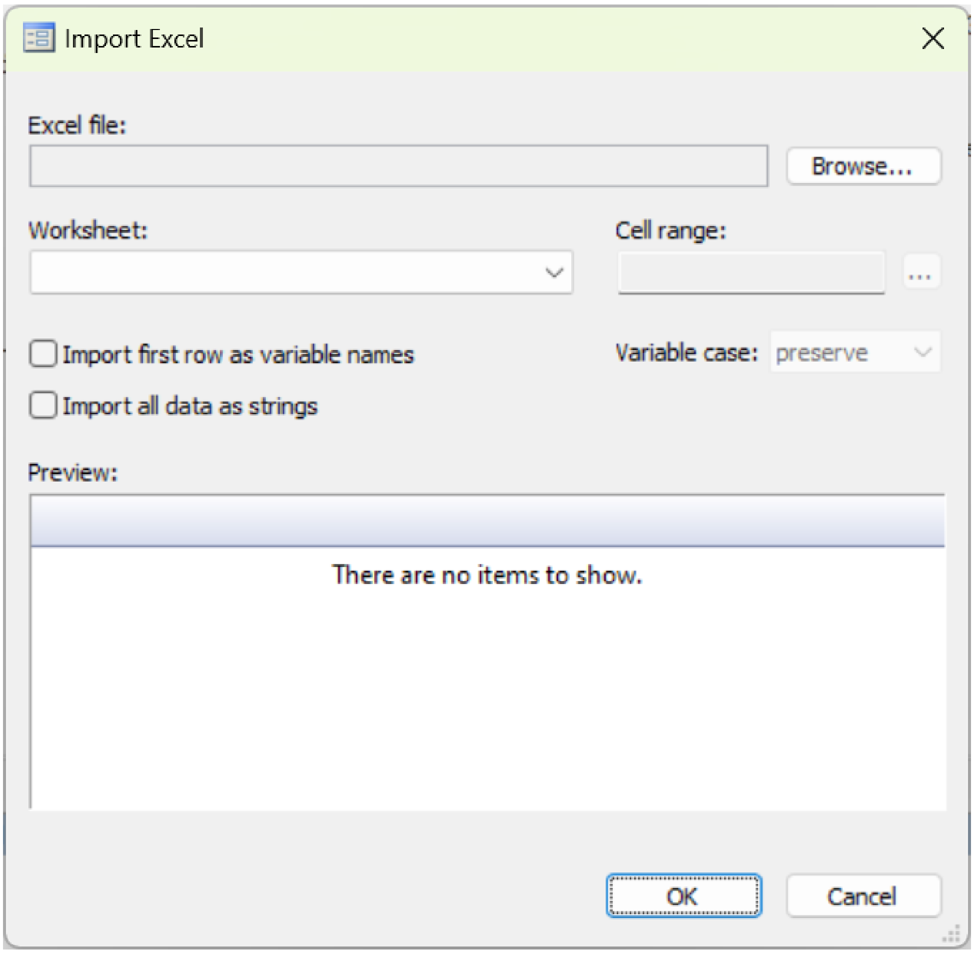
30.2 Commands in Stata

In Stata we can control the program using the menus or by writing commands, also called writing program code or scripts. Stata has its own language, its own syntax, which is specially designed for data analysis. The chapter focuses primarily on how we can write code. We begin with an example of how we can use the menus.

30.2.1 Commands with menus

In Stata we often work with data retrieved from existing data files, for example data that is saved in an Excel sheet or some other program and data format. We can also enter data manually, see the section on entering data manually below. Say we want to import an Excel file to Stata from the computer's hard drive (see also the section on importing and exporting data below). We can then choose menu File > Import… > Excel spreadsheet (*.xls,* .xlsx). A new dialog box opens, where we can choose different options for how the file should be imported to Stata. Under Import… we can also choose to import other file types.

Stata can retrieve the information that is saved in the Excel sheet. If we then work with this data in Stata, the Excel sheet is not affected. Note that this is a difference compared to if we open the same file in Excel. If we edit an Excel sheet in Excel and save the file again, old information is overwritten. If we want to overwrite the Excel file in Stata, we must export the data we are working with to the hard drive and overwrite the old file. This is practical because it reduces the risk of us overwriting useful information.



The figure above shows what the dialog box for importing Excel files can look like. In the top box in the dialog box we choose which Excel file we want to import data from. By clicking on the Browse... button we get the opportunity to click our way through the directories to the file. Excel sheets can contain several tabs (worksheets) with data. In the box for Worksheet in the dialog box we specify the name of the tab in the Excel sheet from which we want to retrieve data. Under Cell range we can specify which cells in the Excel sheet's tab the data should be retrieved from. To retrieve data in cells A1 to B4 we can write A1:B4.

Stata generally uses a period as the decimal separator, which is standard in many countries in the world and therefore also in many computer programs. In Sweden, a comma is generally used as the decimal separator, which can sometimes cause problems when we are going to use computer programs. If the data we want to import is saved with commas, Stata can generally read our data but then interprets the information as text data. We can easily remedy this, which we will return to below.

30.2.2 Commands using do-files

All commands we can perform using the menus can also be performed with written commands. Sometimes we also benefit from combining the methods. When we specify a command through the menus, Stata shows how the same command would have looked as a written command. Stata has a large collection of functions pre-installed, which can be accessed both through menus and Stata's own programming language.

Written commands can be entered either in the lower right window Command in Stata or in the window called the Do-File editor. If we write commands in the Command window, the command is executed but the text we wrote is not saved. What we write in the Do-file editor can be saved as a text file, which is why we focus on that method here. To open the Do-file editor we can click on the menu Window > Do-file editor > New Do-file editor. We can also use the keyboard shortcut Ctrl + 9.

A practical way to work with Do-files is to begin the file by specifying the working directory for this particular work project. By specifying a working directory, we instruct Stata where on the hard drive the program should retrieve files from and where the program should save files, if we give instructions to export information to a file.

The first time we want to define a working directory, it may be easiest to do this using the menu File > Change Working Directory…. This opens a dialog box where we can click our way to the directory on our computer that we want to use as the working directory. Select the directory and click OK. If we choose, for example, a directory called Documents that is located on the computer's hard drive C, and everything is correct, the following text will now appear in the results window:

```
. cd "C:\Documents"
```

Written commands in Stata are often written with the name of a function followed by the arguments the function should use. In many functions we can then, if we want, write a comma and specify various options. To choose a working directory we use the function `cd`, which is an abbreviation for current directory.

Now we can copy the text `cd "C:\Documents"` from the results window and paste it into the do-file editor. The text we write in the do-file editor can be sent to the program, which then performs various actions. Sending written commands, program code, to the program is called executing the program or running the code.

To run a line of program code in the do-file editor, we select one or more characters on the line we want to run and click Execute selection (do) or press Ctrl + D. To run all code in the current do-file we can use the keyboard shortcut Ctrl + D. The code that is run is reported in the results window along with any results from the execution. If we are unsure about which working directory we have chosen, we can use the function `pwd` (abbreviation for present working directory).

The text in the do-file editor can be saved as a file on your computer's hard drive, and then gets the file extension .do. A do-file is a plain text file (in Windows these can be saved as .txt files). Thanks to the file extension .do, however, the program recognizes this file as a do-file for Stata. That is, if we save the text in the do-file editor, only the actual text is saved, which does not take up much space on the hard drive. Any results that the code creates when it is run are not saved, such as data, tables and charts.

Say now that we want to import an Excel file called my_file.xlsx. The file is saved in the directory we specified as the working directory. For this we can use the function `import excel` and specify filename and other options. If we have run the `cd` command above and specified our working directory, we can place our Excel file in this directory and write the following command:

```
import excel my_file.xlsx
```

If we want, we can also write a comma and define which tab in the Excel file data should be retrieved from by defining the option `sheet()`. If we do not specify a tab, data is retrieved from the first tab in the file. If the tab we want to retrieve data from is called sheet1, we can write:

```
import excel my_file.xlsx, sheet(sheet1)
```

We can also specify which cells we want to retrieve data from in this tab by defining the option `cellrange()`:

```
import excel my_file.xlsx, sheet(sheet1) cellrange(A1:B5)
```

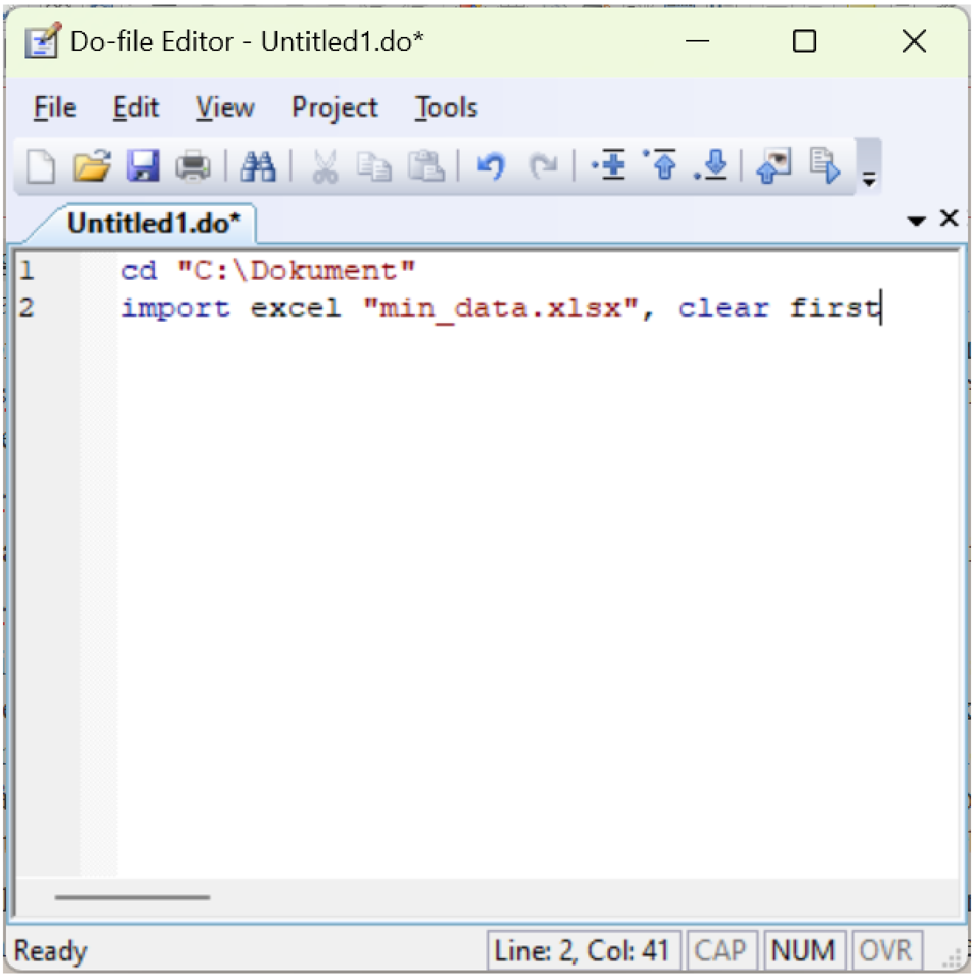
By specifying `cellrange(A1:B5)`, only the information stored in cells A1 to B5 in the current tab and file is imported. If we already have data open in Stata's main table (the data editing window, Data Editor), the program will abort the command with a warning that data loaded in Stata will be lost: no; data in memory would be lost. This is a helpful warning.

Always be careful to save the information we are working with if we want to be sure not to lose our work. After we have saved our work, we can either use the command `clear all` to delete all data that is open in Stata. Then we can import data and the program will not give any error message. A faster method is to, after we have saved our data, use the option `clear`, which also deletes all data in Stata's main table:

```
import excel my_data.xlsx, clear
```

In Stata's data window, we noted how the variable names are specified as column names above the observations. If the first, top row in the data we import is the name of the variables, we can specify the option `first`. The program will then use the information in this top row to name the variables. For example, we can write like this:

```
import excel my_data.xlsx, clear first
```



The figure above shows what it can look like in the do-file editor in Stata. To run the code we select some of the characters on both lines and click Run or use the keyboard shortcut Ctrl + R. Written commands in Stata are generally written one per line. We can also let a command continue on the next line by adding the symbols `///` at the far right of a line. For example like this:

```
import excel ///
    min_data.xlsx, ///
    clear first
```

In the example above we use indentation, the spaces at the beginning of line two and three. This is often practical to make the code more readable, but it is only a matter of taste and does not affect the program. If we are unsure about how something should be written in the code, we can use the menus. If we click through a command using the menus, Stata prints out how the same command can be written as code. We can then reuse this text for our do-file, in the way we went through above for choosing the current working directory.

Another way to learn more about the options for a function and also find new functions is to look up the help pages for a function, which we find via the functions `help` or `search`, followed by the name of the function we want to read about. If we want to read about `import excel` we can for example write:

```
help import excel
```

The result is displayed in the window called Viewer. Often a list of search results is displayed that we can click through.

Many times we also benefit from documenting our work to simply remember exactly what we have done.

It can be surprising how quickly one can forget things one worked with just a few days earlier. For the same reason, programming facilitates collaboration with other analysts. If someone has written program code that clearly describes what happens in an analysis, it facilitates the possibility for others to continue the work.

A central part of writing readable code is to write good comments in the code that explain what the code does. We do this so that we ourselves, or other users we want to show the code to, can more easily follow along with the code. Many people have an amazing ability to surprisingly quickly forget what we have done and thought. To write comments in Stata's do-files we use the asterisk `*`. A line in the do-file that begins with `*` is ignored by the program. We can therefore use `*` to add comments that describe what the code does. Here is an example:

```
* Line that begins with asterisk is ignored by the program.
* Next line defines working directory.
cd "C:\Documents"
* Next line imports data from an Excel file.
import excel my_data.xlsx, clear first
```

Stata has a large number of ready-made functions pre-installed when we install the program. There are also additional functions that we can download for free from the internet. On this webpage there are a large number of links with more information about many different types of functions and tools: www.stata.com/links/resources-for-adding-features/

One of these is the Statistical Software Components (SSC) database, which is run by Boston College in the USA. This database has a large number of functions that are largely created by other users on a voluntary basis. SSC is so well-known that there is a special function for installing new functions from there: `ssc install`. One such function that we can download to Stata from SSC is called outreg2. Right now we don't need to worry about what outreg2 can be used for, we will return to that below. To install outreg2 we can write and run the following code:

```
ssc install outreg2
```

When we run this command, Stata will connect to the internet and download a package with the function `outreg2` as well as instructions for how the function works to our computer and save it on the hard drive. The function is now installed in Stata and we can then use one or more new functions that came with the package. After installation we can also read the function's help section by running the command `help outreg2`. An installation generally only needs to be performed once. The next time we restart Stata, outreg2 is already installed in our program.

30.3 Importing and exporting data

Stata can read data from different types of files. Above we went through how we can import an Excel file of type xls or xlsx to Stata. Stata's own file format for data files is called .dta. Dta-files generally contain a table with columns and rows, where each column represents a variable, as well as information about the different variables. To import the information in a dta-file named my_stata_data.dta, saved in the current working directory on the computer's hard drive (see above), to Stata we can use the function `use`. For example like this:

```
use my_stata_data.dta
```

Before we import data to Stata we need to remove the data we have open in the program's data window, just as we went through above. We can either do this with the command `clear all`, or use the option `clear` (same as when we used the function `import excel` above):

```
use my_stata_data.dta, clear
```

To export the data that is open in Stata's main table (Data Editor) to a dta-file we can use the function `save`. For example like this:

```
save my_stata_data.dta
```

We will go through more about how this works later, when we give examples with data. As mentioned in the previous section, we can import Excel files to Stata with the function `import excel`. We can also export data from Stata to Excel files such as .xlsx using the function `export excel`.

Stata also has ready-made functions for importing and exporting data from and to file formats from other analysis programs. As an example, we can import data files from the program SAS with the function `import sas`. To import data files from the program SPSS we can use the function `import spss`.

A particularly useful file format is .csv (Comma-Separated Values). Csv-files are an example of a type of data file that many analysis programs can both import data from and export data to. Csv-files often consist of, as the name suggests, data where each value is separated by a comma. However, it is possible to save information in this file format and separate the values with some other symbol, for example semicolon ; which often occurs.

To import a csv-file to Stata we can use the function `import delimited`, which is a function for importing data files with comma- or tab-separated values. After the function's name we write the filename, for example my_datafile.csv. We can also specify the option `delimiters()` and in the parentheses for this option describe what type of character the data values in the file are separated with.

```
import delimited my_datafile.csv, delimiters(";")
```

The function also has several other options, which you can read more about on the function's help pages. To export data from Stata to a csv-file we can use the function `export delimited`. Read more about this function in its help section.

30.4 Enter data manually

So far we have described how we can import data from existing data files on the computer's hard drive. We can also enter data manually if we prefer. As we went through above, Stata generally works with one main table, that is the rows and columns shown in the Data Editor window. When we specify commands and run code in Stata, the program assumes that all commands refer to this data table, unless we specify otherwise.

If we want to start with an empty table and enter data ourselves, we begin with the command `clear all`, to remove the variables we have open. To enter new variables we must start by specifying how many observations our table should have. We do this with the command `set obs`, followed by a number that specifies the number of observations. For example:

```
set obs 4
```

This defines that we want to have 4 observations. To create a new variable in our main table we can use the function `generate`, followed by the name of the new variable, an equals sign and the information we want the variable to contain. Here is an example where we create a new variable named `new_variable` where all observations have the value 1:

```
generate new_variable = 1
```

Say now that we instead want to create a variable that we call `n_obs` where each row contains the current row number for this observation. We can do this by writing:

```
generate n_obs = _n
```

The code `_n` instructs the program to retrieve the number of this row for each row of our new variable `n_obs`. In this case, this will create a new column with the numbers 1, 2, 3 and 4, since we defined above that we want to have 4 rows. If we instead prefer to enter each observation manually, we can also open the Data editor window in editing mode, edit mode. Then we can click on the box we want to edit and change the value there.

To enter observations manually we can also use the function `input`, followed by the name of the new variable we want to create. For example `input Y`. On the next line we then specify the value for observation 1. The line below that we specify the value for observation 2, and so on. Below follows an example of code with comments where we use the commands `clear all`, `set obs`, `generate` and `input`. Feel free to try using the same code in Stata and check the result in the program's data window.

```
* Remove all variables that are open.
clear all
```

```
* Next line defines number of observations to 4.
set obs 4

* Create a variable that specifies row number.
generate n_obs = _n

* Create variables Y and X
input Y X
3 3
2 4
5 6
4 7
```

Now that we have written some code in our do-file and hopefully also added some comments that explain what the code does, it is high time to save our script as a file on the computer's hard drive. Press Ctrl + S or choose in the menu File > Save as.... What is now saved is, as mentioned, only the actual text in the do-file. The data that the file can create, for example, is not saved. This is generally just good. The text takes up very little space on the computer's hard drive and it often goes quickly to run through even relatively long do-files.

Often we also want to save the actual data table that we are working with in Stata. Partly because we may have done something special that we are afraid of losing. Or because we want to show our data to someone else. To save the information in Stata's data table in a file on the hard drive, we must use one of the functions we went through above for exporting data (see the section on importing and exporting data). For example:

```
save my_data.dta
```

to save the table as a Stata file. Or if we want to save our data in an Excel file that can more easily be opened in Excel, we can use the following command:

```
export excel "my_data_from_stata.xlsx", replace
```

30.5 Edit Columns and Rows

A large part of data analysis generally consists of editing columns and rows. In Stata we can, as described above, create new columns with for example the function `generate`. If we want to remove two variables `variable_X` and `variable_Y` from the main table we can use the function `drop`, for example:

```
drop variable_X variable_Y
```

When we specify the name of the variables after the function `drop`, these variable names must match the name of the variables in the data table. The program distinguishes between small and capital letters so our spelling of the variables must be exactly the same as the spelling in the data table. If the program cannot find one or more of the variables we name in the data table, we get an error message.

If we want to remove all variables and observations from the data table we can write `drop _all` or `clear all`, as we did above. If we instead want to choose which variables we want to keep and delete the rest, we can use the function `keep`. For example like this:

```
keep variable_X variable_Y
```

which lets us keep these two columns and deletes all other columns in the main table. If we have a variable and want to change the name of it, we can use the function `rename` followed by existing variable and new variable name. For example:

```
rename variable_old_name variable_new_name
```

To sort observations in numerical or alphabetical order we can use the functions `sort` or `gsort`. The function `sort` sorts in ascending order. The function `gsort` sorts in either ascending or descending order. For example:

```
sort var_X
```

sorts all observations in the entire data table by arranging the values in variable var_X in ascending order. We get the same result if we instead use the command `gsort var_X`. To sort in descending order we can write:

```
gsort -var_X
```

We can also add several variables to sort the table in several steps. For example:

```
gsort country year
```

sorts the table by placing the values in variable country in ascending order. Within each unique value in variable country, variable year is sorted in ascending order. Each variable in Stata has a data format, which partly controls how the values in a variable are displayed in the data table and partly controls how the program handles the values. To change format on one or more variables we can use the function `format`. For example like this:

```
format %9.0g x y z
```

This command sets the three variables x y and z to have the format right-aligned general, with a maximum of 9 characters visible in the variable. If we write 0 as decimal for the g-format, the program determines how many decimals are shown, depending on the maximum number of allowed characters (in this case 9). If we want to use commas instead of periods we can replace the period in the format %9.0g with a comma %9,0g. The command to change format on variables x, y and z then reads:

```
format %9,0g x y z
```

Read more about the function `format` and its various options in the function's help pages, which we as usual find by calling the command `help format`.

Stata, like many other computer programs, handles numbers and text strings in different ways. A number can be saved as a text string and is then regarded as text. Stata cannot perform mathematical operations on text strings, even if a string consists only of something that we humans regard as a number. There are special display formats for both numbers and text. If we want to convert a variable from numbers to text strings we can use the function `tostring` and either add the option `replace` (which overwrites existing variable) or `gen(my_new_variable)`, which creates a new variable that we can name:

```
* Convert the numerical variable to text:
tostring X, replace
* Keep numerical variable and create new text variable:
tostring X, gen(X_as_string)
```

If we instead have a variable that is saved as text but contains numbers, and we want to recode so that the program regards the numbers in the variable as numbers, we can use the function `destring`. This function also has the options `replace` and `gen()`:

```
* Convert the text variable to numerical:
destring Z, replace
* Keep text variable and create new numerical variable:
destring Z, gen(Z_as_numerical)
```

The function `destring` can be used if we have data with numbers where commas are used as decimal separators and Stata interprets these numbers as text strings. With the option `dpcomma`, `destring` can convert numbers saved with decimal commas to numbers with periods instead. For example:

```
destring Z, replace
```

If we want to overwrite all or parts of an existing variable with new values we can use the function `replace` (not the same thing as the option replace that we used above). Say we have a variable called income that contains values for income. If we want to take the natural logarithm of the observations in the variable we can use the following command:

```
replace income = log(income)
```

Table 30.1: Mathematical Functions in Stata

Function	Description
ln()	Natural logarithm.
log()	Natural logarithm, same as ln().
log10()	Base-10 logarithm.
exp()	Exponential function, antilogarithm.
sqrt()	Square root.
int()	Round to integer.
round()	Round. round(x) rounds x to integer. round(x,y) rounds x to y number of decimals.
sum()	Summation of a collection of values, for example a variable.
max()	Returns maximum of a collection of values.
min()	Returns minimum.

The function `replace` assumes that the variable we name after `replace` (in this case income) is found in the data table. We must spell the name of this variable exactly the same as the variable in the data table. It is not enough that the variable is mentioned in the script. It is only after we have created the variable income that we can run the function `replace`. The function `log()` here returns the values in the variable income in logarithmic form. Since we use the function `replace`, the existing variable income is deleted and replaced with the new variable with the same name with logarithmic values. If the variable income contains any negative values, this command will result in these being created as missing values in the new variable, since the logarithm is not defined for negative values. Missing values in Stata are displayed as a period in the data table.

There are also other functions that can be used to perform mathematical operations. Note that these functions in Stata's normal mode can only be used if we save the result of the function somewhere, for example in a variable. We cannot, for example, use the following as a separate command:

```
log(5)
```

We can, however, create a variable with the same function:

```
generate x = log(5)
```

The function `replace` can also be used to change the value of a specific observation. For this we add one or more conditions. Conditions can in this case be added using the command `if`. Here is an example where we replace the value on row 2 in variable X with 10, regardless of what value was there previously.

```
replace X = 10 if _n==2
```

The expression `_n==2` means that the row number must be equal to 2. When we describe equal to as a condition, this is written with two equals signs. The same applies in several other programming languages. The table below gives examples of inequality signs we can use to describe conditions.

Table 30.2: Inequality Signs for Conditions

Sign	Meaning
>	greater than
>=	greater than or equal to
<	less than
<=	less than or equal to
!=	not equal to

The last expression `!=` can be used to, for example, specify which observations should not be affected by a change. Say, for example, that we want to change all observations for variable X except row 2 which should remain unchanged. We can then write like this:

```
replace X = 10 if _n!=2
```

We can also combine several conditions using the following expressions:

- `&` and
- `|` or

Here is an example where we set all values for variable X to 10 for observations on row 3 or above and where variable Y is above 5:

```
replace X = 10 if _n>=3 & Y>5
```

By combining conditions with the functions `drop` and `keep` we can select which rows in the data table we want to keep or delete. The following command deletes rows 1 to 9:

```
keep if _n>=10
```

The following command deletes rows 1 to 5:

```
drop if _n<=5
```

Or:

```
drop if _n==1 | _n==2 | _n==3 | _n==4 | _n==5
```

This is thus an unnecessarily cumbersome solution and the first example with the condition `if _n<=5` is preferable in this case. In Stata, missing values are written as a period `.` in the data table. When we perform operations on variables where missing values occur, these are treated somewhat differently, depending on which functions we use. It is therefore important to check this carefully in each case. When we use conditions we can also address missing values by writing a period. Say we have the data shown in the table below (variables Z and K):

Row	Z	K
1	10	.
2	20	1
3	30	1
4	40	2

and specify the following command:

```
generate L = Z + K if K!=.
```

This will create a new variable L with the values ., 21, 31, 42. Row 1 gets missing value because variable K=. on this row (missing value). We get the same result if we instead specify the command:

```
generate L = Z + K
```

This means that the missing value is interpreted as unknown. Since the program does not know what value Z=10 should be added to on row 1, the result in the new variable L is also left as a missing value. Let us now look at the following command:

```
generate Zsum = sum(Z)
```

This creates a new variable `Zsum` using the function `sum()`. The function `sum()` gives us in this case the cumulative sum of the values in variable Z, that is: $Zsum = \{10, 30, 60, 100\}$. We can compare this with if we instead sum variable K with the following command:

```
generate Ksum = sum(K)
```

In this case, the missing value K in the first observation is treated as 0, not as a missing value. To calculate a constant in the form of the total sum we can instead use the function `total()`, for example:

```
generate K_total_sum = total(K)
```

When we work with data analysis we often benefit from being able to transpose our data, that is change whether our data is organized in width or height. In Stata's data table we have some type of information for two people, A and B, for the years 2020, 2021 and 2022. The development per person is described in width where each year has its own column. Variable names in Stata cannot begin with a number so the three columns with years are named x2020, x2021 and x2022.

Person[1] A

	Person	x2020	x2021	x2022
1	A	2	4	3
2	B	11	8	10

If we want to display this information in a chart in Stata it becomes easily complicated, since Stata's functions for displaying charts are primarily designed based on the idea that a variable is organized in one column, see the section on creating plots below. In this case we have one and the same variable spread across three columns, depending on the year.

Now we are going to transpose this table so that we get all the numbers in columns x2020, x2021 and x2022 in one and the same column that we call x. We also want to have the years from the column headers gathered in a column next to x. To transpose the table to long format we can use the function `reshape long`. We must then describe to the function which columns we want to retrieve data from and gather in a new column x. This can in this case be done like this:

```
reshape long x, i(Person) j(year)
```

The first information we specify is x, which is the prefix of the three columns with years. It is from these three columns that the function `reshape` now retrieves data that should be organized lengthwise. The program finds these three columns precisely because all three are named x... something. Then we describe in the parentheses for `i()` that we want to keep the column Person. In the parentheses for `j()` we finally specify the name of the new variable where the information that comes after "x" in the name of the three columns from which we retrieve data should be placed. In this case it is three years, which is why we decide to name this variable year.

Person[1] A

	Person	year	x
1	A	2020	2
2	A	2021	4
3	A	2022	3
4	B	2020	11
5	B	2021	8
6	B	2022	10

30.6 The function egen

Alongside `generate` there is also the function `egen` which can be used to create new variables, both in similar ways as `generate` and with more complex options and functions. If we start with the following example:

```
egen Xs = sum(X)
```

This creates a new variable Xs with a constant total sum for variable X. This is thus the same result as when we combine `generate` and `total()`. However, `egen` has several useful functions. One example among many is the function `std()` which returns standardized values of a variable. We can for example write:

```
egen standardized_X = std(X)
```

Say now that we have ten variables that we have named var_1, var_2, ..., var_10. Now we want to create a new column where we get the sum per row in these 10 variables. We can then use `egen` together with the function `rowtotal()` with the following command:

```
egen row_sum = rowtotal(var_1:var_10)
```

We use the colon symbol to indicate that we want to include all columns (variables) in the data table from var_1 to var_10. The function `egen` also has other functions for doing calculations per row, such as `rowmean()` which returns the mean per row, `rowmax()` which returns the maximum value per row and `rowmiss()` which returns the number of missing values per row. To read more about different possibilities and options with `egen`, see the function's help section which we find with the command `help egen`.

We can also do calculations on entire or parts of columns. Above we mentioned for example how we can combine `generate` and `sum()` to calculate cumulative sum, as well as `egen` and `sum()` to calculate sum per column. Say now that we have a data table where variables id, X and Y are numerical and variable Z contains text strings. We can think of variable id as a variable for identifying analysis units, where each unique value 1 and 2 respectively represents for example a person, a household or an organization. Variable nr is also categorical with the two unique values a and b respectively.

If we now for example want to calculate the sum of the numerical variable X per unique value in variable id (per person/household/organization) we can use the following commands:

```
bysort id: egen sumX = sum(X)
```

The prefix `bysort id:` sorts the data table based on variable id in ascending values. In addition, this groups the table based on the unique values in variable id. Each unique value in id, values 1 and 2 respectively, forms its own group within which the following command is executed. Then we use the function `egen` in combination with `sum()`, just as we did earlier. But now we get the sum of the values in variable X per group, which in this case means per unique value in id. For group 1 the sum in the new variable becomes 50. For group 2 the sum becomes 149. The new variable sumX thus gets the values 50, 50, 149, 149. If we prefer we can also divide this into two steps and write the following commands that give the same result:

```
sort id
by id: egen sumX = sum(X)
```

We can also group on the unique combinations of several variables. For example:

```
bysort id nr: egen X_sum = sum(X)
```

This creates 3 unique groups: id=1 with nr=a, id=2 with nr=a, and id=2 with nr=b. The sum of variable X for these three groups is: 50, 72, 77.

With the help of functions like `generate`, `replace` and `egen` we can create different types of results in tables. These functions can for example be used to compile an overview of extensive data material, for example by calculating number of observations, summing values, calculating mean and standard deviation.

If we then want to present our results in another program, for example Excel, Word or PowerPoint, it may again be time to export our data to a file on the hard drive, for example using the function `export` excel. There are also special functions for exporting tables to various specific formats. To export results from Stata to MS Word it is recommended to check out the function asdoc, which can be installed with the following command:

```
ssc install asdoc
```

Stata has, as far as the author knows, no pre-installed function for calculating weighted means. Based on the functions we have gone through above we can relatively easily create our own calculation. There are also functions developed by users that can be combined with egen to calculate weighted means. See the following link: https://ideas.repec.org/c/boc/bocode/s418804.html

30.7 Join tables

Often when we work with data analysis we want to be able to merge two or more tables with each other into one table. This can for example happen by having a table to which we add rows, where the new rows have the same variables as our table. Sometimes we also want to add columns, where all or some observations in our table match observations in the data that is added.

One way to do this in Stata is by using the function merge. The function is based on having a collection of variables open in the program (in the data table) and another collection saved in a dta-file on the hard drive. Both tables have at least one common column with one or more observations with the same values. Say for example that we are working with a table with information about height for person 1, 2 and 3. The two variables are called person and height. In our working directory we have saved the file weight.dta, which contains information about weight for person 1, 2 and 3. The two variables in the file weight.dta are called person and weight. Now we want to add the variable weight to our data table in Stata using the function merge. For this to work, both the data file on the hard drive and our data table must be sorted on the same variable. Then we can use the following command:

```
merge person using weight.dta
```

If you have done correctly and run the command, the variable weight is added to the data table and each value is matched against the values in the variable person. The program also creates the new variable _merge with information about which values found a matching observation in the data tables. See the help section for the function merge for more information. Below follows a somewhat longer code that creates all data directly in the script, in case you want to try yourself. You must first specify the working directory where the file should be saved.

```
clear all
input person weight
1 55
2 66
3 77
end
sort person
save weight.dta, replace
drop _all
input person height
1 160
2 170
3 180
end
sort person
merge person using weight.dta
```

All versions of Stata before Stata 16 could only have one table open at a time. If we work in Stata 16 or later we can also have other tables saved in the program's memory. To save a data table temporarily in the program's memory we can use the function frame create. We can then edit this with frame change. We can in turn use this among other things to check whether a variable in one table matches a variable in the other table, using the function frlink. Read more in these functions' help sections.

30.8 Calculate index using Stata

Index for year t can be calculated in the following way:

$$\text{Index}_t = \frac{\text{Value}_t}{\text{Value}_{\text{base year}}} * 100$$

The base year's value is here 100. Say now that we have GDP per capita for the years 2015–2019 in the way described in the table below (see the chapter on Excel for a similar example). If we want to enter this data manually into Stata's data table we can use the following command where we call the variables year and gdp:

Year	GDP per capita, 2019 prices
2015	474,000
2016	477,800
2017	483,500
2018	487,300
2019	488,800

```
input year gdp
2015 474000
2016 477800
2017 483500
2018 487300
2019 488800
end
```

We will now index gdp and specify year 2015 as the base year. We can do this in several ways. For example, we can use square brackets to refer the base year's value to the first observation in the data table in the calculation of the index. To be sure that the first observation really is year 2015, we start by sorting the table with the function `gsort`:

```
gsort year
generate index_gdp_1 = gdp/ gdp[1] *100
```

We can also save the base year as a scalar:

```
scalar gdp2015 = gdp[1]
```

and then calculate the index variable:

```
generate index_gdp_2 = gdp/ gdp2015 *100
```

30.9 Matrix calculations and MATA

Alongside Stata's regular functions, the program also has a special programming language, a special syntax, for working with matrices and linear algebra called Mata. Commands in Mata work somewhat differently compared to Stata's regular functions and commands that we went through above. Stata's regular functions generally assume that we want to use or edit data in the program's data table. We also cannot write regular mathematical operations, 1+1, in Stata's command window, unless we simultaneously specify where this information should be saved, such as in a variable. With Mata we can work in a different way with both mathematical operations and with information outside the program's main table.

To use Mata we begin the script with the command `mata`, write in the commands for Mata we want to use and end with the command `end`. We can then for example write a mathematical calculation without defining that the answer should be saved somewhere. Here is an example where we ask the program to calculate the number 2+2:

```
mata
2+2
end
```

If we run this command, the answer is displayed directly in Stata's results window. Say now that we want to define the following matrix A:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

This can be done with the following command:

```
mata
A = (1,2\3,4)
end
```

We name the matrix A and inside the parentheses we specify which elements the matrix should contain. Commas `,` separate the values on the same row. Backslashes `\` separate rows. Since we have two values, a line break and two more values, the program understands that we want to create a 2×2 matrix. To create a row vector R we can use the following command in Mata:

```
R = (3,2,5,4)
```

To create a column vector K we can write:

```
K = (3\2\5\4)
```

In Stata's do-files we can write comments that the program ignores by starting a line with asterisk `*`. In Mata we cannot use this notation but instead write comments by starting a line with `//` or writing `/* comment */`. For example:

```
mata
// next line defines matrix A
A = (1,2\3,4)
/* next line ends Mata input */
end
```

Matrix A is now saved in the program's memory but is not displayed in Stata's data table. To create a matrix we can, besides Mata, also use Stata's function `matrix define`. We then do not need to specify `mata` first but instead write:

```
matrix define A = (1,2 \ 3,4)
```

This gives the same result as the example with Mata above, and creates a 2×2 matrix A that is saved in the program's memory. Another method is to first define the matrix's dimensions using the function `J()` and then add the matrix's elements. For example like this:

```
matrix define A = J(2,2,.)
```

The expression `J(2,2,.)` creates a 2×2 matrix and the period is interpreted as all elements should contain missing values. Another example:

```
matrix define B = J(4,7,1)
```

The command creates a matrix B with 4 rows, 7 columns and where all elements contain the number 1. To display the content of matrices in Stata's memory we can use the function `matrix list` and specify the name of a matrix. For example:

```
matrix list A
```

The result, the elements in matrix A, is then described in Stata's results window. To check which matrices are currently saved in the program's memory we can use the following command:

```
matrix dir
```

To remove a matrix A that we have saved in Stata's memory we can use the command:

```
matrix drop A
```

To delete all matrices in Stata's memory we can write:

```
matrix drop _all
```

If we want to name the rows in a matrix we can use the function `matrix rownames`, for example like this:

```
matrix rownames A = "row 1" "row 2"
```

To name the columns in a matrix we can instead write:

```
matrix colnames A = "column 1" "column 2"
```

If we want to rename an existing matrix we can write:

```
matrix rename old_name new_name
```

Say now that we are going to transpose matrix A and create the following matrix

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

For this we write `A'`, for example in the following way:

```
matrix define A_transpose = A'
```

To display the content of this new matrix we can use `matrix list A_transpose`. The diagonal in matrix A above contains the elements 1 and 4. To extract the diagonal from this matrix we can use the function `diagonal()`, which returns a vector with these elements. For example:

```
matrix define d_A = diagonal(A)
```

If we want to create a diagonal matrix from these values directly we can instead use the function `diag()`, like this:

```
matrix define diag_A = diag(A)
```

The function `diag()` can also be used to create a diagonal matrix from a vector. Here is an example where we first define the vector $\vec{a} = \begin{bmatrix} 1 & 3 & 2 & 4 & 5 \end{bmatrix}$:

```
* Define vector a
a = (1,3,2,4,5)
* Create diagonal matrix from this vector
diag(a)
```

To create an identity matrix we can use the function `I()`. The command `I(n)` creates an $n \times n$ identity matrix, where n is any integer. The command `I(m,n)` creates an $m \times n$ matrix with the value 1 on the diagonal and 0 in all other elements. Here is an example where we create a 2×2 identity matrix:

```
* Create identity matrix
matrix define identity_2 = I(2)
* Display the matrix
matrix list identity_2
```

To calculate the inverse of matrix A, which is real and square, we can use the function `luinv()`. For example like this:

```
matrix A_inverse = luinv(A)
```

To calculate the determinant of a matrix A, which is a scalar, we can instead of `matrix define` use the function `scalar define`, as well as the function `det()`. To display the content of the saved scalar determinant_A we then use the function `scalar list`, instead of `matrix list`.

```
scalar define determinant_A = det(A)
scalar list determinant_A
```

Scalars are a type of object that Stata uses to save information temporarily in memory. Another type of object Stata can use to save information is macros. There are local and global macros. To refer to a named saved local macro we write the name of a local macro with accent ` before and apostrophe ' after. To refer to a global macro we write `$macro_name`, that is we use a dollar sign in front of the name. To display a general list of which macros Stata has saved in memory right now we can also use the function `macro dir`.

Scalars and macros can be used for different things when we program, for example if we want to refer to and report special results or reuse these in new calculations in our code. We can also save text strings in this way by using quotation marks. Here is an example where we create two macros and save numbers in them. A local macro that we call alpha and a global macro that we call beta:

```
* Save local macro
local alpha = 12
display `alpha'
* Save global macro
global beta = 23
display $beta
```

We can now refer to all of these directly in the same way as we refer to macros.

To perform mathematical operations with scalars and matrices we can use regular mathematical symbols, such as for example:

```
matrix define AA = A * A
```

which means that we matrix multiply A by itself. Here is another example where we take the determinant of A, determinant_A, and multiply this by matrix A:

```
matrix define dA = determinant_A * A
```

Say now that we have two 2×2 matrices A and B and want to perform the following calculation:

$$A - B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

For this we can use the following commands:

```
matrix define A = (1,2 \ 3,4)
matrix define B = (5,6 \ 7,8)
matrix define A_minus_B = A - B
```

The commands for matrices and scalars we have described here do not affect variables and values in Stata's data table, but if we want we can retrieve information from the data table and save as matrices in memory, and we can also export information from the matrices to Stata's data table. Say that we have four observations in Stata's data table for the two variables $Y = \{3, 2, 5, 4\}$ and $X = \{3, 4, 6, 7\}$. To save variable Y in Stata's data table as a matrix in memory, we can use the function `mkmat`, for example like this:

```
mkmat Y, matrix(Y_matrix)
```

where the option `matrix()` lets us specify the name of the matrix we create. In memory, a column matrix with four elements and the same values as in variable Y is now saved. To export the values in a matrix to Stata's data table we can use the function `svmat`, for example like this:

```
svmat Y, names(Y_from_matrix)
```

where with the help of the option `names()` we can specify how the columns we create in Stata's data table should be named. All commands for matrices described above can also be written using Mata, but the difference is that we then do not need to write out `matrix define` and similar parts of the commands. Instead we can just write out the operations directly. If we want to retrieve data from variables and save in a matrix we can use `mkmat` but if we want to be able to use this matrix in mata we can instead use the function `putmata`. To retrieve information from mata and save as variables in Stata's data table we can use the function `getmata`. Here follows an example of some operations in Mata:

```
mata
// Create matrix A
A = (1,2 \ 3,4)
/* Write out the name of A to display the content */
A
// Matrix multiply A^2
A * A
// Transpose A
A_t = A'
A_t
// Return the diagonal from A
diagonal(A)
// Create diagonal matrix from the diagonal of A
diag(A)
// Create 2x2 identity matrix
I(2)
// Define vector a and create a diagonal matrix
a = (1,3,2,4,5)
diag(a)
// Calculate trace for A
B = trace(A)
B
// Calculate inverse matrix of A
A_inverse = luinv(A)
// Determinant of A
det(A)
det(A) * A
end
```

Note that with Mata we can also use Stata's mathematical functions and return results directly without saving the results anywhere new. For example:

```
mata
// Create a matrix
x = J(3,4,5)
x
// create a vector
z = (1,2,3,4,5)
z
// Calculate the sum of the elements
sum(x)
sum(z)
end
```

30.10 Input-output analysis

Say that we want to use Stata to do an input-output analysis. In the chapter on Excel we saw how we can define matrix A:

$$A = Z * \mathrm{diag}\left(S_m\right)$$

where Z is the flow matrix and $\mathrm{diag}\left(S_m\right)$ is a diagonal matrix with the multiplicative inverse of total production per sector along the diagonal. We define matrix C as a column matrix with production for final consumption per sector. The total production from sector j required for all inputs and final consumption we call y_j, which we collect in matrix Y, which can be written as:

$$Y = (I - A)^{-1} S$$

where I is an identity matrix with the same dimensions as A. Each element in Y is the total production required with regard to inputs and final consumption. Matrix $(I - A)^{-1}$ we call B, which has the same dimensions as A. In B, column sum j is the production multiplier and row sum j is the input multiplier for sector j.

We can calculate these matrices using Mata in Stata. In the example below we define the matrices manually. As we described above, it is also possible to retrieve the matrices' elements from the data table using the function `mkmat`. We start by defining matrices Z, C and S and diag (S_m). Then we can calculate matrix A, B and finally Y. Feel free to try it yourself in your own program and check the results:

```
mata
// Define matrix Z, C and S
Z = (2,1 \ 1,3)
Z
C = (3\3)
S = (6\7)
Sm = (1/6 \ 1/7)
// Define A
A = Z * diag(Sm)
A
B = luinv( I(2) - A )
B
Y = B* S
Y
end
```

30.11 Variation and covariance

Stata has several functions that can create descriptive statistical measures and present these in tables. One example is the function `summarize`. As an argument to the function we write the name of the variable or variables we want information about, for example:

```
summarize Y
```

The result of the command is displayed in Stata's main window, the results window. If we write several variables, such as `summarize Y X`, one variable per row is listed in the results table. We can also add the option `detail` to get more detailed descriptive information about each variable.

Here follows an example where we produce detailed statistics about variables Y and X. The result includes percentiles, number of observations, mean, standard deviation (Std.dev), variance, skewness and kurtosis. The last two concepts are different measures of how skewed the frequency distribution for the variable is.

```
summarize Y X, detail
```

To get information about what type of data a variable has, we can use the function `codebook`, for example:

```
codebook Y
```

The result shows that variable Y is numeric, has values between 2 and 5, of which all 4 are unique. At the bottom we also get a frequency table that shows the number of observations for each unique value the variable has: 2, 3, 4 and 5. The frequency distribution can be presented in slightly different ways depending on the variable being described. As an example, for a continuous variable with many observations, `codebook` may instead return percentiles. If we want `codebook` to return a more compact table with overview information, we can instead use the option `compact`. If we do not write out any name of a variable, the entire data table is described:

```
codebook, compact
```

There are several ways to export this type of result to other programs or save the results as a file on the hard drive. We mentioned above the function `asdoc`. To export results from commands with `codebook` specifically, there is also `codebookout`, which also requires installation with `ssc install codebookout`.

Another function that can be used to describe the values in a quantitative column (variable) is `tabstat`.

With the help of the option `statistics()` we can define what type of measures we want to display. Here is an example:

```
tabstat X Y, ///
    statistics(min, max, mean, variance, sd)
```

Stata has several pre-installed functions for analyzing covariation between two or more variables. If we for example want to estimate the correlation or covariance between two variables in Stata's data table we can use the function `correlate`. To estimate the correlation coefficient (Pearson's r) we can use the following command:

```
correlate Y X
```

To instead estimate the covariance we can write:

```
correlate Y X, covariance
```

Both commands above return a matrix with pairwise estimates of correlation and covariance respectively. We can also use the function `pwcorr`. These functions also have several options that we can use to control what type of results are returned and how these are presented, see the functions' help sections.

30.12 Probability

Stata also has several pre-installed functions for working with common probability distributions. As an example, if we want to know the cumulative probability for the standard normal distribution, given a specified z-value, we can for example use the function `normal()`. In the parentheses we specify a z-value and the function returns the proportion of the distribution that is equal to or less than this z. As long as we do not use Mata we cannot use this type of function without saving the results somewhere. The following command therefore results in an error message:

```
normal(0)
```

One way to use the function outside Mata is to save the result in a local macro, for example like this:

```
local zd = normal(0)
display `zd'
```

With Mata, however, the function returns results without us saving these anywhere new. For example:

```
mata
normal(-1)
normal(0)
normal(1)
end
```

If we instead want to calculate the probability for exact values on the standard normal distribution, based on its density function, we can use the function `normalden()` and specify a z-value in the parentheses. If we specify two arguments in the parentheses for `normalden()` we can change the mean μ for the normally distributed density function we want to use. If we specify three arguments we can also change the standard deviation σ. Here follow some examples:

```
mata
// Probability for the value 0
normalden(0)
// Probability for 1 if mean = 1
normalden(1,1)
// Probability for 2 if mean = 2, standard deviation = 1
normalden(2,2,1)
end
```

The table below describes some examples of functions in Stata that can be used to calculate with different known probability distributions. See each function's help section for more information about how these work and available options, for example:

```
help runiform
```

In the help sections there is also information about additional functions that can be used to calculate probability in Stata. Stata also has several pre-installed functions for statistical tests. See for example the function `ttest` and its help section.

Table 30.3: Probability distributions in Stata

Probability Distribution	Functions in Stata
Uniform distribution	`runiform()`, `runiformint()`
Binomial distribution	`binomial()`, `binomialp()`, `binomialtail()`, `rbinomial()`, `rnbinomial()`
Poisson distribution	`poisson()`, `poissonp()`, `invpoisson()`, `rpoisson()`
Chi-squared distribution	`chi2()`, `chi2den()`, `chi2tail()`, `rchi2()`
Gamma distribution	`gammap()`, `gammaden()`, `rgamma()`
F-distribution	`F()`, `Fden()`, `Ftail()`
Normal distribution	`normal()`, `normalden()`, `invnormal()`, `rnormal()`
T-distribution	`t()`, `tden()`, `ttail()`, `rt()`

30.13 Regression analysis

This section introduces how we can work with regression analysis in Stata. First by using matrix calculations (compare the chapter on Excel). Then with the help of ready-made functions.

30.13.1 Regression analysis with matrices

Say that we have four observations for the two variables Y and X saved in Stata's data table. Now we are going to estimate the following regression model:

$$Y = a + bX + u$$

where a and b are the coefficients and u is the error term. We can write this regression model with matrices as:

$$Y = XB + U$$

where Y, B and U are column matrices and X is a matrix where the first column has the value 1 in each element and column 2 is the explanatory variable X. As we described in the chapter on Excel, the coefficient estimator for the least squares method can be written with matrices in the following way:

$$\hat{B} = \left(X^T X\right)^{-1} X^T Y$$

To estimate $\hat{B}$ we start by saving the variables from the data table as matrices using the function `mkmat`. For example like this:

```
* Create two matrices
mkmat Y, matrix(Y_mat)
mkmat X, matrix(X_mat)
* Add column 1 to matrix X_mat
matrix define X_mat = ((1\1\1\1), X_mat)
```

Now we have what we need to estimate $\hat{B}$, which we can do with the following command:

```
matrix define B_hat = inv(X_mat' * X_mat) * X_mat' * Y_mat
```

If we instead want to use mata to estimate $\hat{B}$ we can first use `putmata` to save the two matrices that we need to work with them in mata, for example like this:

```
putmata Y X=(1 X), replace
```

This command first creates the column matrix Y, from variable Y in Stata's data table. Then matrix X is created by giving all elements in the first column the value 1 and the remaining columns are defined by variable X in Stata's data table. The option `replace` can be added to replace any previous matrices saved in mata with the same name as these. Then we can perform the same operations on the matrices as we did above. Below follows an example of how we can write these commands. All results are printed in Stata's results window:

```
mata
* Display the content of matrix Y and X
Y
X
* Estimate the coefficients
Bhat = luinv(X'*X)*X'*Y
* Display the result
Bhat
end
```

30.13.2 Regression analysis using `regress`

Stata has a large number of pre-installed functions for statistical analysis and regression analysis. There are also additionally a large number to download for free on the internet. Say now that we again have four observations for the two variables $Y = \{3, 2, 5, 4\}$ and $X = \{3, 4, 6, 7\}$. We will now again estimate the regression model $Y = a + bX + U$ where a and b are the coefficients and U is the error term. To estimate this model with the least squares method in Stata we can use the function `regress` in the following way:

```
regress Y X
```

The result of this command is shown below. In the image three different collections of results are visible. At the top left an ANOVA table is shown with estimated sum of squares, mean sum of squares and degrees of freedom for SSE (sum of squares explained), which in Stata's results appears on the row Model. The row labeled Residual shows SSR and the bottom row, Total, shows SST.

At the top right a list with mixed results is shown: At the top the number of observations is shown and then the calculated F-value as well as the calculated probability based on this F-value that we would have gotten this type of result if this test's null hypothesis were true. This F-test is automatically formulated as all coefficients for the regression model's explanatory variables being equal to zero, which in this case means $H_0 : b = 0$. The result in this example indicates that we do not have reason to reject the F-test's null hypothesis with high statistical probability.

Below the two results that concern the F-test, R^2 and adjusted R^2, that is $\bar{R}^2$, are reported. At the bottom among these results is Root MSE, which is the square root of the mean sum of squares for the residuals, that is: Root MSE $= \sqrt{MSR}$.

The estimated coefficients $\hat{a}$ and $\hat{b}$ are in the lower table under the heading Coef. From top to bottom in this table the estimated coefficients for the explanatory variable(s) are shown. In this case it is thus only $\hat{b} = 0.5$. At the far left in the bottom row is _cons, which refers to this being an estimate concerning the regression model's constant, the y-intercept, which here is estimated to $\hat{a} = 1$.

The next column to the right of the coefficients' estimates is the coefficients' standard errors (Std. Err.). Then the calculated t-value is given for a t-test where the null hypothesis, unless otherwise specified, is formulated as the coefficient in question being zero. For the top row the t-test's null hypothesis is thus $H_0 : b = 0$. The next column has the heading $P > |t|$ and gives the probability that the absolute value of the calculated t exceeds the critical t-value for the current t-distribution. The two columns at the far right show the lower and upper bounds for estimated confidence intervals for each coefficient with confidence level $\alpha = 0.05$.

The function **regress** assumes that the first variable we write out is the explained variable in a linear regression model and that if we write out more variables after this, these are set as explanatory variables. This means that if we write the following command:

```
regress Y X Z
```

then Stata interprets this as us wanting to estimate the following regression model:

$$Y = a + bX + cZ + u$$

where a, b and c are coefficients. If we have a dataset with a continuous variable Y and a factor variable Z we can for example manually create dummy variables in our data table using the functions and commands we went through in the sections on editing columns/rows and on the function `egen`. The disadvantage of this solution is that it is time-consuming and increases the risk of mistakes, especially if we need to create many new variables. If our dummy variables are only to be used to estimate the regression model, we can fortunately use a faster method. By writing `i.` before the variables that we want to be treated as indicator variables (factor variables), the program will automatically create dummy variables for all unique values in the current variable, excluding one value that becomes the base level. For example like this:

```
regress Y i.Z
```

If our factor variable Z has three unique values, this means that Stata in this case will estimate the following regression model:

$$Y = \alpha_1 + \alpha_2 D_1 + \alpha_3 D_2 + \epsilon$$

where D_1 and D_2 are dummy variables for two of the three unique values in variable Z. The three α_i are the coefficients that we estimate with the least squares method. ϵ is the error term. For **regress** there are also options if we want to create interaction terms without having to edit our data table. Say for example that we have variables Y, X and Z and want to estimate the following regression model:

$$Y = \beta_1 + \beta_2 X + \beta_3 Z + \beta_4 XZ + v$$

where β_i are the coefficients and v is the error term. To estimate this model with the least squares method in Stata we can use the following commands:

```
regress Y X##Z
```

This causes the program to add both the interaction term for XZ and the coefficients X and Z separately. The command therefore gives the same result as the following command:

```
regress Y X Z X#Z
```

If we use one hash symbol, only the actual interaction term, XZ, is included. If we use two hash symbols `##`, both the interaction term and each variable included in this term separately are included.

Say now that we would like to estimate the regression model Y=a+bX+cZ+u, with robust standard errors, that is with an adjusted variance-covariance matrix, we can use the option `vce()`. In the parentheses for `vce()` we write which method we want to use to adjust the standard errors and the variance-covariance matrix. For adjustment with HC1 we can use the following command:

```
* Regression analysis with HC1
regress Y X Z, vce(robust)
```

See more options in the function's help section. The regression results report a large number of figures. If we want to use any of these figures specifically we can retrieve these by using one of the functions **return list**, **ereturn list** or **sreturn list** directly after **regress**. These three functions return information that is saved by Stata in slightly different ways. If we run the command **ereturn list** directly after the command with **regress** we get a list of information that was created in memory from the latest run of **regress** and from which we can now retrieve one or more values.

Information is divided into scalars, macros, matrices and functions. From the list returned by `regress` and `ereturn list` we can refer to named scalars and macros as local macros, for example like this:

```
regress Y X
ereturn list
display `e(r2)'
```

To display a matrix we can write:

```
matrix list e(b)
```

If we want to save any matrix from the results from `regress` we can for example use the function `matrix define`. From the list of results we can among other things retrieve matrix `e(b)` which contains estimated coefficients and matrix `e(V)` which contains the variance-covariance matrix. For example like this:

```
regress Y X
* Save the variance-covariance matrix
matrix define var_covar = e(V)
* Save matrix B
matrix define B = e(b)
```

30.14 Export results to disk

When we estimate regression models we often want to present all or parts of the results in some other form than Stata's results window, such as in Microsoft Word. Stata has several ready-made functions for exporting results to different file formats on the hard drive, for example the functions `putexcel`, `putdocx` or `putpdf`. There are also "homemade" functions by other users that we can download for free from the internet (for an introduction to some of these, see www.stata.com/support/faqs/data-management/copying-tables/). One of these functions is `outreg2`, which requires that we download and install it, which we can do by running the following command:

```
ssc install outreg2
```

If we estimate a regression model with the function `regress`, as above, we can directly after this run a command with the function `outreg2` to export our regression results to a file on the hard drive. The function `outreg2` has several options that we can use to influence exactly which results are saved, to what type of file the results are saved and how the result presentation looks in more detail. `outreg2` can among other things save the results in an Excel file or a Word document, but also other file formats, which we can read more about in the function's help section. Here is an example:

```
* Estimate the regression model Y = a + b*X + u
regress Y X
* Export the results to a Word and an Excel file
outreg2 using my_results, word excel replace
```

This command first runs the function `regress` to estimate the regression model, exactly as above. Then we run `outreg2`, which then assumes that we want to save precisely the latest results we produced. The expression `using my_results,` means that we instruct the function `outreg2` to save data in a file that we name my_results. After the comma we specify three options: `word`, `excel` and `replace`. The option `word` instructs the function to save the results in an rtf-file (Rich Text Format, a format for Word documents). The option `excel` instructs the function to save the results in an xml-document, which is a file format for Excel files. The option `replace` means that in case there is already a file called either my_results.rtf or my_results.xml, these old files are overwritten without warning.

Another useful function for exporting results in a similar way is the functions `eststo` and `estout`. These work in such a way that we use `eststo` to save regression results in memory and can then retrieve these results with `estout` and either display them in Stata's results window in edited form, or export them to a file on the hard drive, such as an Excel sheet or a Word document. Here is an example:

```
* Use eststo to save two regression results:
eststo: regress Y X
eststo: regress Y X Z
```

```
estout
```

We can also add different options to influence how the results are presented. Here follows another example with several options. We start with the command `eststo clear`, which clears the memory from results saved with `eststo`. If we do not clear the memory, `eststo` continues to save new tables together with old results.

This time we have added to `estout` among other things the option `cells()`, in whose parentheses we can specify which results are presented and in what format the numbers should be displayed. The next option is `varlabels()` which we can use to define which labels should be used for the explanatory variables, which is written in the form `varlabel(variable 1 label 1 variable 2 label 2)`. We use the option `mlabels()` to set headings on the columns, which names the regression models we estimated. With the option `stats()` we can instruct whether we want to present some other type of result, such as R^2 and number of observations. The result for the code below is shown in the figure:

```
eststo clear
eststo: regress Y X
eststo: regress Y X Z
estout, cells(b(star fmt(2)) se(par fmt(2)) p) ///
    varlabels(X variable_x Z variable_Z _cons Constant) ///
    mlabels("Model_1" "Model_2") ///
    stats(r2 N, labels("R squared" "Obs"))
```

30.15 Creating plots

Stata has a large number of functions and options for creating different types of charts. If we want to create a histogram we can use the function `histogram`. This function has different options that can affect the appearance of the chart, which we can read more about on the function's help page.

Before we create the histogram we will create a random variable, whose values are drawn from a normal distribution, which we can do using the function `rnormal()`. Each observation is drawn from a known population but exactly which value each observation gets is drawn randomly, using the computer's random generator.

The computer's random generator is however not more random than that we can control its conditions. If we want to be sure that the random draw that creates the variable gives exactly the same result every time we run the script, we can define a random seed with the function `set seed`. We then write `set seed` and any number, for example `set seed 1`. Read more about random seeds in this function's help section.

Here follows an example where we first define the random seed to the value 1, create 300 observations for a variable based on a normal distribution and then create a histogram with the function `histogram`:

```
clear all
* Set number for random seed
set seed 1
set obs 300
* Create variable x from the standard normal distribution
generate x = rnormal()
* Display a histogram over x
histogram x
```

The command `histogram` opens a new window in Stata, Graph, which displays the chart. Note that the letter x is interpreted by the program as the name of a variable in the data table. If no variable is loaded with that name (note small and capital letters) then the program will return an error message.

Even though the image is displayed on the screen, it has not been saved on the computer's hard drive yet. To save this as a file in our working directory we must use the function `graph export`, and name the file to be saved. By specifying a file extension we also choose file format, for example .png or .pdf. To read more about which file formats the function `graph export` supports, see the function's help section. If we run the following command, the latest chart we created is saved as an image file of type png:

```
graph export my_histogram.png
```

Above we performed commands per group in the data table. We can also create charts per group. Here is an example where we create two groups, identified with variable id and values 1 and 2 respectively. The first 200 observations we create from a standard normal distribution. The last 200 observations we create from a normal distribution with mean 1 and standard deviation 2. When we create the histogram we now use the option `by()` and specify which variable we want to use for grouping, which in this case is id.

```
clear all
set seed 1
set obs 400
* Create the first 200 observations of id
generate id = 1 if _n<201
* Create the remaining 200 observations of id
replace id = 2 if _n>200
* Create values for x, observation 1-200
generate x = rnormal() if id==1
* Create values for x, observation 201-400
replace x = rnormal(1,2) if id==2
* Create one histogram per group in the same chart
histogram x, by(id)
graph export two_histograms.png
```

The result shows two histograms. In the left chart we see the distribution of variable x for the observations where id=1. In the right chart the values for x for id=2 are shown.

Say now that we have the two variables Y and X with four observations. To create a scatter plot with these four observations, we can use the function `scatter` followed by the name of the variables. The variable we write first ends up on the y-axis and the second on the x-axis. For example like this:

```
scatter Y X
```

The function `scatter` can also use the option `by()` to divide into several charts, depending on the unique values in a variable in our data table. If we want to combine different types of data presentations in the same chart we can add different chart types in the same command. Say for example that we want to create a scatter plot with a regression line in it. We then begin the entire command with the function `twoway`, which is a general function for creating many different types of charts. Then we put the scatter plot in parentheses: `(scatter Y X)`. Then we add parentheses to create the regression line `(lfit Y X)`. The following command creates a scatter plot with regression line:

```
twoway (scatter Y X) (lfit Y X)
```

The function `twoway` can be combined with different types of functions and commands to create different types of charts and influence their appearance. If we instead of the regression line want to draw a chart where we have both points and a line that connects the points, we can combine the functions `twoway`, `scatter` and `line`. Here follows an example where we also use the options `color()` and `legend()` in the parentheses for the function `line`. By placing `color()` and `legend()` in respective parentheses we can influence how `scatter` and `line` display the data in the chart:

```
twoway ///
    (scatter Y X, color(black)) ///
    (line Y X, color(black) ///
        legend(position(3) cols(1)))
```

We write `color(black)` to define the color of the points created by `scatter`, and the line created by `line`, respectively, as black. In the parentheses for `line` we also add the option `legend()` where we define the legend's position using `position()`. The position is specified in this case as placement according to the dial on a clock with 12 numbers. Position 3 = 3 o'clock = to the right of the chart.

The option `cols()` in the parentheses for `legend()` in the parentheses for `line`, specifies the number of columns in the legend, which is why `cols(1)` means that everything written in the legend is placed in one column.

We can also name the charts we create by specifying the option `name()` and writing any name without spaces in the parentheses. Then we can combine our saved charts, for example to save them in one and the same image file. To check which charts are saved in memory we can use the function `graph dir`. Here is an example:

```
* Create a scatter plot and call it graf_1
twoway scatter Y X, title("Scatter Plot") name(graf_1)
* Create a line chart and call it graf_2
twoway line Y X, title("Line Chart") name(graf_2)
* Check which charts are saved in memory
graph dir
```

To combine the two charts we can then use the function `graph combine`, for example like this:

```
graph combine graf_1 graf_2
```

This creates a common image for the two charts. In the help section for the function `graph combine` we can read about different options for influencing the layout of the image and the charts. The image we now create with `graph combine` can then be exported as an image file to the computer's hard drive, in the same way as we did above with `graph export`. For example like this:

```
* Create an image of the two charts in memory
graph combine graf_1 graf_2
* Export the image as an image file to the hard drive
graph export "Combo graf.pdf"
```

In the help section for `twoway` we can read about several other functions and commands for creating different types of charts. Above we introduced the function `histogram` for drawing histograms, which is a form of bar chart. If we instead want to draw a regular bar chart based on two or more variables we can use the functions `twoway bar`. For example like this:

```
twoway bar Y X
```

The above command creates a bar chart where the bars are at the height of variable Y on the y-axis and variable X on the x-axis. If we prefer to draw the corresponding bars horizontally instead we can add the option `horizontal`:

```
twoway bar Y X, horizontal
```

To draw lines with mathematical functions we can use the functions `twoway function`. Below follows an example where we draw a chart for the two mathematical functions $y = x^2$ and $z = x - x^2$. We use the option `range()` to control which x-values are drawn out, the option `color()` to control the lines' color and `lpattern()` to control the lines' form. We also use `legend()` to define and place the legend:

```
twoway ///
(function y=x^2 , ///
    range(-1 1) color(red) ///
    lpattern(dash)) ///
(function z=x-x^2, ///
    range(-1 1) color(black) ///
    lpattern(dot) ///
    legend(position(3) cols(1)))
```

There are also different options for influencing a chart's appearance in more detail. As an example, if we want to add a title to the chart we can use the function `title()`, for example:

```
twoway scatter Y X, title("Here is a scatter plot")
```

To specify titles for the y- and x-axes we can for example use the functions `xtitle()` and `ytitle()`. In the parentheses for these we first write the text we want to specify as the titles' headings. In the parentheses for `xtitle()` and `ytitle()` we can also specify different options to influence the axis titles' appearance, for example the function `orientation()`, if we want to rotate the text, and `placement()`, if we want to specify a specific placement. To change the color of the chart's background we can use the function `graphregion()`, which in this case must be placed in one of the parentheses for `scatter` or `line`. In the parentheses for `scatter` and `line` we can also specify options to define how we want the points and line to look, for example with the functions `color()`, to define color, or `lpattern()` to define the line's form.

Here is an example where we use line breaks and indentation to make the entire command a bit more readable. Note that most of the command is placed in the parentheses for `scatter`:

```
twoway ///
(scatter Y X, ///
    color(black) ///
    legend(off) ///
    xtitle("X-variabeln") ///
    ytitle("Y-variabeln", ///
        orientation(horizontal) placement(12)) ///
    graphregion(color(white)) ) ///
(line Y X, ///
    color(black) ///
    lpattern(dash))
```

Say now that we want to display all data in one and the same chart but separate the observations using colors or patterns. To achieve this we can for example use the function `scatter` twice in separate parentheses and define color manually in each individual case. Here is an example with code:

```
twoway ///
(scatter x y if id==1, ///
    color(black) ///
    legend(label(1 "1"))) ///
(scatter x y if id==2, ///
    color(red) ///
    legend(label(2 "2")))
```

We can also let the program choose colors based on the unique values in a variable, using the option `colorvar()`. For example like this:

```
scatter x y, colorvar(id)
```

Sometimes we want to draw scatter plots with many different combinations for two variables at a time in our data table. We can do this with the function `graph matrix` followed by two or more variables. The program then creates one chart per pairwise combination of the variables we specified. Say for example that we have four observations of the three variables $X = \{3, 2, 5, 4\}$, $Y = \{3, 4, 6, 7\}$ and $Z = \{-1, 2, 3, 7\}$. To draw scatter plots with the combinations (X, Y), (X, Z) and (Y, Z) we can use the following command:

```
graph matrix Y X Z
```

The result shows all pairwise combinations. In the diagonal the variable names are described: Y, X and Z. The three charts below and above the diagonal respectively are different combinations of the three variables. The middle chart in the top row has Y on the y-axis and X on the x-axis. The middle chart in the left column has Y on the x-axis and X on the y-axis. The data points in these two charts are however exactly the same. If we only want to show the charts below the diagonal we can use the option `half`:

```
graph matrix Y X Z, half
```

30.16 Iterations and loops

An incredibly useful part of programming concerns repetitions and how we can instruct the program to perform repeated work tasks with small changes in each round. Within programming this is often referred to as iterations or loops. We will here primarily use the term loops to describe some examples of functions. Already in the examples above we have here and there used repetitions. For example when we instruct the program to create a variable and on each row write in the row number using the call `_n`.

Stata has several functions that are specifically designed for working with repetitions where we can more freely design what the repetition should lead to and what should be performed at each repetition, each round. One such example is the function `forvalues`, which goes through a loop based on a series of numbers that we choose. We start by writing out an example and explain below what happens in this example step by step:

```
forvalues local_macro = 1/3 {
    display `local_macro'
}
```

The function `forvalues` defines a local macro (see above) which we here call `local_macro`. The expression `local_macro` is thus chosen by the user and if we prefer we can write just a single letter or some other expression that begins with letters without spaces. The expression `= 1/3` means that the function goes through the integers 1 to 3 and for each number performs the commands described between the curly brackets `{ }`. The first curly bracket `{` must be written on the same line as `forvalues` and no code may come after it, except for comments.

In this simple example we instruct the function to report for each loop round what value the local macro `local_macro` currently has, using the function `display` that we introduced above. The result is shown in Stata's results window.

Instead of `1/3`, which instructs `forvalues` to go through the integers 1 to 3, we can write `1(1)3`. This means that the function starts at 1. For each loop round it goes one integer forward, until it reaches the number 3, which becomes the value for the last loop round. The expression `1(10)100` would in this context mean that the function starts at 1, jumps 10 numbers at a time, until it reaches 100. The integers 1 to 3 can also be written `1:3`. If we want to count downward from 3 to 1 we can instead write `3(-1)1`.

Loops with `forvalues` can be used to do many different things. Say as an example that we have a data table with the variable year, which covers the years 2000–2020, and variables Y, X and Z. For each year we have many observations with information in variables Y, X, Z concerning many people's lifestyle habits. Say now that we would like to estimate one or more regression models but only with data from one year at a time. One way to do this is the following command where we create a loop with `forvalues`:

```
forvalues y = 2000(1)2020 {
    regress Y X Z if year==`y'
}
```

Within the curly brackets `{}` we now estimate a regression model Y=a+bX+cZ+u, per year. The condition `if year==`\y'‘ means that the regression analysis only uses observations where the variable year = the current year for this loop round.

Another function in Stata for creating loops is `foreach`. Unlike `forvalues`, which creates loops over numbers, `foreach` can create loops with both numbers and other types of objects, such as text strings, variables and macros. Depending on what type of values the loop should go through, the command is written in slightly different ways. For example:

```
foreach a in x y z {
    display "`a'"
}
```

The loop goes through the letters x y z. In each round a text string is displayed, marked with quotation marks and the content of the local macro a. The result is that the three letters are shown in Stata's results window. Let us now compare two similar commands with `foreach` that give different results. We start with this:

```
* foreach in
foreach a in 1/3 {
    display `a'
}
```

If we run the above command, only one number is shown as a result: the number 0.33333... (that is 1/3). This is because the function does not interpret `1/3` as a list of numbers but as precisely a mathematical expression for a number, which is why the loop is only run 1 round. If we write `foreach a in 1/3 1/4` then the loop is run two rounds and the local macro a equals 1/3 the first round and 1/4 the second round (one third and one fourth respectively). Let us now look at the next example:

```
* foreach of numlist
foreach a of numlist 1/3 {
    display `a'
}
```

This time we have replaced the word `in` with the expression `of numlist`. By doing this we instruct the program that the expression `1/3` should be interpreted precisely as a list with numbers, that is the integers 1/3. This time the loop is run three rounds and the local macro a gets the value 1, 2 and 3 respectively. There are several other ways to set up loops with `foreach`, which we can read more about in the function's help section. We reach this as usual with the command `help foreach`.

Above we showed an example of how we could create a loop that goes through all the years and performs a regression analysis per year. Another way to create such a loop is to use the function `levelsof`, which creates a list of the unique values of a variable. Say as an example that we have a variable year with years and we want to create a loop that goes through each unique year. We can accomplish this with the following command:

```
levelsof year, local(the_years)
display "`the_years'"
```

The first line in the example above saves the unique years in the local macro `the_years`. The second line returns the values in `the_years` in the form of a text string, and adds a space between each unique value, like "2000 2001 2002" …and so on. We can in turn combine this with `foreach`. Here is an example where we estimate a regression model with variables Y, X and Z for each year:

```
foreach y of local the_years {
    regress Y X Z if year == `y'
}
```

In the first line we create the loop using `foreach` and add the command `local`, which describes to the function `foreach` that the next concept in the command is a list of local macros, that is the values we have collected in the macro `the_years`. Note that in the first line we refer to `the_years` without the characters accent ` and apostrophe '.

30.17 Example with GDP per hour

Now we will again use data on real purchasing power adjusted GDP per working hour, calculated in millions of US dollars, which we introduced in the chapter on Excel. The data is repeated in the table below, where we see observations for the countries Sweden, Denmark, Finland, Norway, USA and United Kingdom for the years 1990, 2000, 2010 and 2020.

Table 30.4: GDP per Working Hour

country	gh1990	gh2000	gh2010	gh2020
Denmark	62.6	76.6	84.2	95.9
Finland	47	64.3	73.5	77
Norway	68.1	89.3	95.9	101.1
Sweden	54.6	68.4	81.6	89.3
United Kingdom	50	63.1	70.4	73.4
United States	53.2	63.6	79.5	87.4

Source: The Conference Board. Real GDP purchasing power adjusted millions USD, divided by working hours per year.

To work with this data in Stata we need to import it to the program. We can do this for example by opening the data table in Stata in editing mode (edit mode) and then copying the table and pasting it into Stata. Be careful to replace the commas with periods, for example using the function `destring` (see the section on editing columns and rows).

Many of Stata's functions are designed for data organized with one variable per column, that is long format. The table above is organized in width, where the values per country should be read from left to right. To transpose the table to long format we can use the function `reshape`, which we also introduced earlier:

```
reshape long gh, i(country) j(year)
```

Since we specify `gh` after `reshape long`, the command searches for variables whose names begin with these letters. In the table the four columns with the years begin with these letters. It is these columns that will now be organized more in height. In the parentheses for `i()` we specify the variable country that already exists in the table. In the parentheses for `j()` we specify a new variable where we want to save the information from the names of the variables we transpose. In this case it is the years we want to save in a new variable, which is why in this example we choose to call this new variable year. This organizes all data from the table in a long format with the variables country, year and gh.

	country	year	gh
1	Denmark	1990	62.6
2	Denmark	2000	76.6
3	Denmark	2010	84.2
4	Denmark	2020	95.9
5	Finland	1990	47
6	Finland	2000	64.3
7	Finland	2010	73.5
8	Finland	2020	77
9	Norway	1990	68.1
10	Norway	2000	89.3
11	Norway	2010	95.9

The figure above shows the first rows after transposition. Say now that we are going to calculate a mean per year. Since our data is now organized in long format we can do this with the following command:

```
bysort year: egen year_mean = mean(gh)
```

The function `bysort` sorts the entire table based on the values in the variable year (smallest to largest). With the function `egen` we create a new variable that we choose to call `year_mean`, to which we calculate means of the variable gh (GDP per working hour). The function `bysort` creates a grouping per unique value in the variable year, and since we use the prefix `bysort year`, a mean is calculated per year. Now we will calculate an index over GDP per working hour per country. We can do this with the following command:

```
* Sort the table by country and year
sort country year
* Group the table per country
by country: generate gh_index = gh/gh[1] * 100
```

When we have data in long format it is also easier to create charts that give us an overview of the development of GDP per working hour in recent decades in these countries. The code below gives an example of how we can create the chart:

```
twoway ///
(line gh year if country=="Denmark") ///
(line gh year if country=="Sweden") ///
(line gh year if country=="Finland") ///
(line gh year if country=="Norway") ///
(line gh year if country=="United States") ///
(line gh year if country=="United Kingdom", ///
legend(order(1 "DK" 2 "SWE" 3 "FIN" 4 "NOR" 5 "USA" 6 "UK") ///
position(3) cols(1)))
```

Another way to create a similar chart is the following command:

```
twoway line gh year, colorvar(country)
```

With the same functions we can also create a chart over the indices we created per country:

```
twoway line gh_index year, colorvar(country)
```

Let us now estimate the following regression model per country:

$$y_{it} = a_i + b_i t_{it} + u_{it}$$

where y_{it} is the natural logarithm of GDP per working hour in country i year t, a_i is a coefficient (y-intercept, the constant) for country i, t_{it} is year t for country i year t and u_{it} is the error term. Estimated $\hat{b}_i$ will give us an approximate measure of average percentage change of y_{it} per year and country. To estimate one model per country we can for example use `foreach`:

```
foreach a in Denmark Sweden Finland ///
Norway `United States' `United Kingdom' {
    regress gh year if country=="`a'"
}
```

We can also combine `levelsof` with `foreach`. Below follows an example where we also add a command that saves the slope coefficient from each regression result:

```
* Retrieve unique values for the variable country
levelsof country, local(tc)
* Create local macro
local vlist
* Create loop over the unique values in country
foreach c of local tc {
    regress gdphour year if country=="`c'"
    * Round and save the slope coefficient in a local macro
    local temp = round(_b[year], .01)
    * Add the result to the local macro coef
    local coef `coef' `temp'
}
```

To display the macro coef we can use the function `display`:

```
display "`coef'"
1.08 .99 1.06 1.17 .78 1.19
```

30.18 Chapter Summary

- Stata is a program designed for data analysis where we generally work with one main table at a time, which is displayed in the program's data window. Information can also be saved temporarily in the program's memory in various ways. Stata can be controlled through the menus and through written commands. The written commands can be given in the program's Command window or through do-files. The do-files can be saved as plain text documents.

- Data to the program can be entered manually via the data window, via commands in the Command window or the do-files. We can also import data files in different formats from the hard drive to the program, whereupon these are opened in the program's main data table, the data window Data editor. Stata generally works with one main table with data. Variables and observations for this data table are what we can see in the data window.
- Stata has a large number of functions pre-installed. For example, we can create and edit variables in Stata's data table using the functions `generate` and `replace`. To change specific values or observations we can combine `replace` with conditions, such as `replace x=1 if y==2`. Another useful function for creating and editing variables is `egen`, which has many options and additions. To merge the columns in two tables with matching observations we can use the function `merge`.
- In Stata we can work with matrix calculations by for example using the function `matrix define`. We can also use Mata, which is a special programming language that we can work with in Stata to for example perform matrix calculations. To save data from Stata's data table in matrices we can use the functions `mkmat` and `putmata`. To save matrices as data we can use `svmat` or `getmata`.
- Stata has several pre-installed functions for calculating variation and covariation. For example `correlate` and `pwcorr`. Stata also has several functions for working with common probability distributions, such as `normal()`, `runiform()` and `binomial()`.
- To estimate the regression model $y = a + bx + u$ we can use the command `regress y x`. Using the functions `return list`, `ereturn list` and `sreturn list` we can work with the estimates that `regress` returns.
- To create charts we can use the function `twoway` in combination with for example `scatter`, `line` or `lfit`. By specifying options to these we can define how the charts should look and how data should be illustrated. For example `twoway (line x y, color(red))`.
- To create loops with repetitions in Stata we can use the functions `forvalues` and `foreach`.

Chapter 31

R

This chapter introduces the analysis tool R. R is the name of a programming language and the name of a computer program in which we can use the programming language R. There are several other programs that also have support for the programming language R. One such program is RStudio, which we use in the examples in this chapter. To be able to use R it is recommended that you install two software programs on your computer: R and the development environment RStudio. Both are open source software and can be downloaded for free at the following links:

- R: https://cran.r-project.org/
- RStudio: <www.rstudio.com/products/rstudio/#Desktop>

31.1 General Overview of R

Figure 31.1 shows the R program. This program needs to be installed on your hard drive for you to be able to use R in RStudio, but we never need to open the R program. We can instead work in RStudio. Figure 31.2 shows how RStudio generally looks when you open it for the first time. In the bottom left you see the console. In the top right the Global Environment is displayed. In the bottom right help sections and charts are shown, among other things.

In the top left of RStudio you see the script window (Script). If you don't see it, you can open a new script window by pressing Ctrl + Shift + N (Windows) or Cmd + Shift + N (Mac). You can also open the script window by choosing in the menu `File > New File... > R Script`.

We can control RStudio through the menus or through written instructions, that is writing R code. In RStudio you can enter commands ("code") in the console. If you place the cursor in the console and write `1 + 2` and press Enter, the result is output directly, that is the number 3. Neither the command nor the result is saved in the program's memory.

We can also write R code in the script window in the top left. To run a command in your script you can place the text cursor at the command, on the same line, and press `Ctrl` + `Enter` (Windows) or `Cmd` + `Enter` (Mac). You can for example try writing `1 + 2` on a line and running this. The result is output in the console. But unlike when we write commands in the console, the command remains in the script when we have run the command. We can also select exactly the code we want to run and send this to the program with the same commands. If we want to run the entire script and *all* commands found in the script, we instead press `Ctrl` + `Shift` + `Enter` (Windows) or `Cmd` + `Shift` + `Enter` (Mac). We can also use the "Run" button that you find in the script window's upper right corner.

To write comments in R scripts, which are ignored by the program, we can use the hash symbol `#`. For example like this:

```r
# Lines that begin with hash symbol are comments.
# Next line is a mathematical calculation.
1+2
```

If we write the above in an R script, select all these lines or place the cursor on the first line, and press Ctrl + Enter, this is run as a command in R. The first and second lines are ignored by the program and are only reported in the console as text without action. The third line is run as a calculation and the result is returned in the console.

The commands you write in a script can be saved as a regular text file on the computer. To save your script you press Ctrl + S (Windows) or Cmd + S (Mac). You can also choose menu `File > Save`. Your script is saved as a text file on the computer's hard drive with the file extension `.R`. If you have saved your script in an `.R` file on the computer's hard drive, you can, the next time you start RStudio, open the file and continue working with your code.

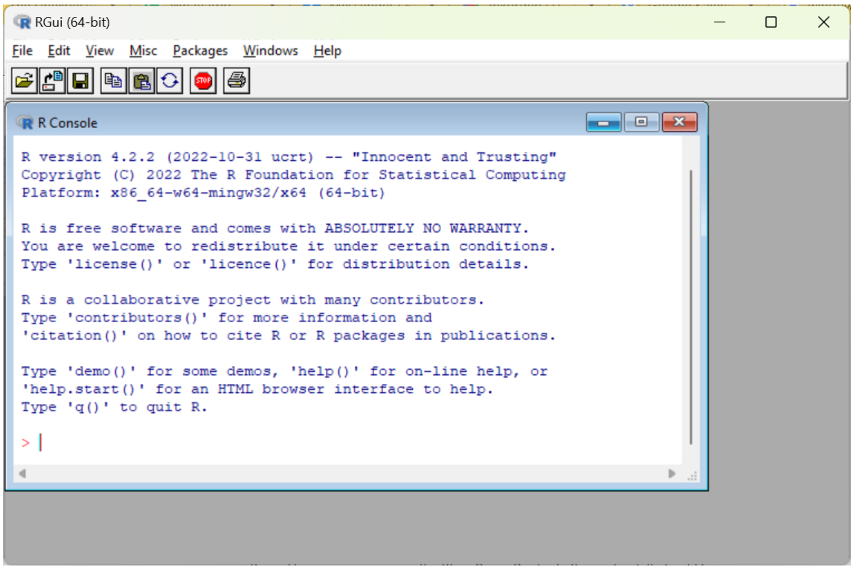


Figure 31.1: The program R

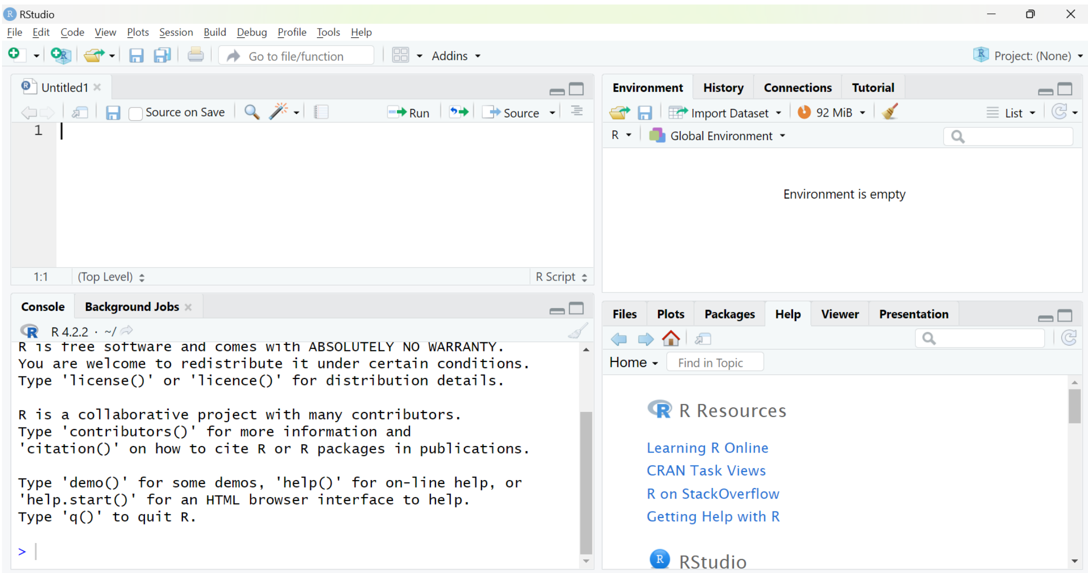


Figure 31.2: The program RStudio

31.2 R Objects

Even though we save a script with for example the code `1+2`, only this text is saved, not the result of the code, in this case the value 3. R "remembers" things by creating R objects. The program can have many objects saved simultaneously. An R object can contain many different types of information, like a number, a table or a list with several tables. This means that we can have access to many different types of information simultaneously among the objects in the program's memory. It also means that data in R can be organized in many different ways and some of these ways may at first glance seem a bit more abstract compared to for example Excel and Stata. We will here only go through a few examples of this.

To save some form of information as an R object we use the function `<-` and write the name we want to use for our new R object to the left of the arrow. To the right of the arrow we write the information that should be saved in the object. Here is an example where we save the number 3 in an object, which we name `my_r_object`. To run the below as code, write the text in an R script and press Ctrl + Enter:

```
my_r_object <- 3
```

The keyboard shortcut for `<-` is Alt + –, which also inserts spaces before and after `<-`. If we prefer we can use the equals sign `=` instead of `<-`. For example:

```
my_r_object = 3
```

When an object is saved in memory in R, the name of the object appears in the Global Environment, in the top right of RStudio. To see what an object contains we can write the name of the object in the console or in a script and run this object name as a command. For example:

```
# Create an object
my_r_object <- 3
# Display the object's content
my_r_object
```

To indicate that something is text, a text string, we use quotation marks. For example:

```
a <- "Hello world!"
```

When we write R scripts we can often use line breaks when we want, as long as the program understands from the code that the command is not finished. As an example we can write `a <-` and then use a line break, since the program will then continue searching for some type of information that should be saved in object `a`. The code simply cannot end after `<-` without some type of information.

All characters between quotation marks are treated as a text string, even numbers. We therefore cannot use text strings for mathematical calculations. Each object must have a unique name for this particular work session. If you use the same name when you create two different objects, the first object disappears and is replaced with the new object with the same name. The program reports no warning for this. You can use the same name on different objects in different scripts or work projects. Each R object has a class that describes the object. To check what class an object has we can for example use the functions `class()` or `str()`.

31.3 Functions in R

R has many pre-installed functions that we can use for different types of operations. Functions in R are generally written with the name of the function followed by parentheses, for example `mean()`, `sum()`, `log()`. An R function can perform both simple and complex work tasks. In the parentheses for each function we have the possibility to specify different types of arguments and options. For example, we can specify as an argument in the parentheses for the function `sum()` the numbers we want to calculate the sum for: `sum(1,2,3)`. Some functions return results even when called entirely without values in their parentheses, but generally at least some type of input data is required.

We can use objects as arguments in functions, whereupon the information in the object is sent to the function. Results from functions can in turn generally be saved in objects. For example:

```
number_1 <- 3
number_2 <- 4
number_3 <- 5
sum_of_numbers <- sum(number_1, number_2, number_3)
```

The R function `c()` can be used to send several values as a coherent argument. If we write `c(1,2,3)` the function returns these three numbers as a collection. To calculate the mean of the three numbers we can then write `mean(c(1,2,3))`.

All functions have specific names for the arguments that the function uses. Often the first argument in a function is called `x`. For the function `mean()`, `x` refers to the input values we want to calculate the mean for, that is: `mean(x=c(1,2,3))`. Or for example:

```
three_numbers <- c(1,2,3)
mean(x=three_numbers)
```

Usually we can omit the name of the first argument and only write for example `mean(c(1,2,3))`. In each function's help section there is a description of the function's arguments and options. The help sections often contain practical examples and links to other useful functions. To read a help section we can write a question mark and the function's name, `?mean`, alternatively use the function `help()`, like `help(mean)`.

When starting a new R script or work project, it is good practice to clear the R objects saved in the global environment, to avoid accidentally using stale objects. To delete all previously saved objects in the global environment we can use the following command:

```
rm(list=ls())
```

If we only want to delete a specific object from the environment we can use the function `rm()` and specify the name of the object in the parentheses, for example: `rm(a)` or `rm(my_r_object)`.

If we want we can enter line breaks in the parentheses for functions, which can sometimes make the text more readable. Here is an example where we create an object with a collection of text strings:

```
names <-
c("Erik",
"Maria",
"Leif")
```

On the first line in the above example the program understands that the command is not finished because we must specify what information we want to save in object `names`. On lines 2 and 3 the program understands that the command is not finished since a closing parenthesis for the function `c()` is missing.

There are several ways that missing values can be written in R. A common way is `NA`, abbreviation for *not available*. Other common notations are `NaN` (Not a number), `NA_real` (missing value real number) and `NA_character` (missing value text string). Functions in R handle missing values differently but often we must specify in our code how we want each function to handle missing values. As an example, the following command will return a missing value, `NA`:

```
mean( c(1,2,NA) )
```

That is, if we ask the function `mean()` to calculate the mean of a collection of values, of which one of these values is `NA`, then we will get back a missing value. Many times we want to calculate the mean of the numbers that are not missing, even if there are one or more missing values in a collection. In that case there is an option in the function `mean()` called `na.rm`. We can define this to `TRUE` or `FALSE`. The default setting for the function `mean()` is that `na.rm=FALSE`. If we instead define it to `na.rm=TRUE` the function calculates the mean of the values that are not `NA`. The following command therefore instead returns the value 1.5:

```
mean(c(1,2,NA), na.rm=TRUE)
```

Table 31.1 describes a list of R functions for some common mathematical operations. See each function's help section for instructions regarding how these work as well as available options. The help sections can for example be accessed with the function `help()`, for example: `help(mean)` or with the command `?mean`.

Table 31.1: Mathematical functions in R.

R function	**Description**
`log()`	Returns the logarithm. `log(x=c(1,2,3))` returns the logarithm of the three numbers 1, 2 and 3.
`sum()`	Summation of a collection of values.
`mean()`	Mean of a collection of values.
`weighted.mean()`	Weighted mean of a collection of values.

R function	Description
`median()`	Median of a collection of values.
`sqrt()`	Square root of one or more values.
`factorial()`	Returns the factorial of one or more positive integers. `factorial(3,2)` returns the results 6 and 2 as separate values.
`round()`	Round. `round(x,y)` rounds x to y number of decimals.
`min()`	Minimum value of a collection of values.
`max()`	Maximum value of a collection of values.

31.4 R Packages

Users constantly develop new functions for R and upload these in the form of R packages on the internet. R packages contain one or more new functions. Many packages we can easily download and install in R using the function `install.packages()`. A particularly useful such package is `tidyverse`, which contains several other packages with many functions. To download and install the package we run the following code:

```
install.packages("tidyverse")
```

When you run the above code, a longer download and installation starts and RStudio generally then shows different types of information and warnings. You don't need to do anything but let the program finish downloading. For this to work you need to have access to the internet and have the ability to save files on the computer's hard drive (administrator rights).

After you have installed a package it is saved on your computer. The next time you open R you don't need to run `install.packages()` again. But after each time we restart the R program we must activate the packages we have downloaded and installed, which we do with the function `library()`. This function thus activates packages that we have already installed. Often it can be practical to begin a script with for example:

```
library("tidyverse")
```

To summarize, we generally need to run the command `install.packages()` only once. We need to run the command `library("tidyverse")` every time we restart R or RStudio if we want to use the functions in tidyverse.

Sometimes packages that we have installed are updated. To check and update an already installed package we can in RStudio click on the menu `Tools > Check For Package Updates...`.

In the tidyverse package there is the magrittr package and there is the special pipe function `%>%`. The pipe function sends the result of a command on the left forward to the next function on the right. The pipe function places the sent result as the first argument in the next function. In this way we can create an easily comprehensible chain of operations. Example:

```
# The following two lines give the same result:
my_result <- sum( log( c(3,4,5) ) )
my_result <- c(3,4,5) %>% log() %>% sum()
```

The last line we can read as "take the numbers in `c()`" AND THEN "use the function `log()`" AND THEN "use the function `sum()`". The result is saved in the object `my_result`. The keyboard shortcut for `%>%` is Ctrl + Shift + M.

Now that we have written a few lines in our R script and hopefully also added some comments that explain what the code does, it is high time to save our script as a file on the hard drive. Press Ctrl + S or choose the menu `File > Save as...`. What is now saved is, as mentioned, only the actual text in our R script. The data that the script creates or edits, as well as the results that are created, are not saved with this command. We will return below to how we can export results of various kinds.

31.5 Importing Data

To register data in R we can, as mentioned, create objects and save information in these, as individual values or with the function `c()`. R can save and organize information in many different forms. As an example, the function `c()` creates collections. Tables are another, of several, forms that R can save information in. R can in turn save several different types of tables, each with their special properties. One of the standard formats for tables in R is `data.frame`. Another type of table is `tibble`, which is defined in the tibble package that is included in the tidyverse package, which we installed above. To create an object with a table of type tibble we can use the function `tibble()`. For example like this:

```
# Create a table of type tibble with variables y, x and z.
my_data_table <- tibble(
    y = c(3,2,5,4),
    x = c(3,4,6,7),
    z = c(1,4,0,1))
```

To instead create a table of type `data.frame` we can use the same code as above but replace the word `tibble` with the expression `data.frame`. When we run many of the functions introduced here, these return precisely a table of type tibble. As described above, each R object has an object class, depending on what type of information is saved in the object. Tibble is a type of object class. To display the content of the object with the table we can use the name of the object as a command:

```
my_data_table
```

The result is displayed in RStudio's Command window. If the object contains a large table, the first rows and columns are displayed. In addition to entering data manually as commands, we can also import data from files saved on the hard drive. When we are going to do this we often benefit from defining our working directory. The working directory is the place where RStudio will look for files that we instruct the program to retrieve information from, if we don't specify another location. It is also the directory that the program will try to save files in, if we instruct the program to export information to the hard drive and don't specify another specific location.

We can specify working directory with the function `setwd()`. In the parentheses for `setwd()` we write a text string with the directory address we want the script to work against. When we work with R and RStudio in Windows, the directory address may not contain any backslashes \ (reverse slash). If necessary they need to be changed to slashes `/`. The following will not work:

```
setwd("C:\Documents")
```

Write instead:

```
setwd("C:/Documents")
```

To define working directory we can also in RStudio use menu `Session > Set Working Directory > Choose Directory...`. This can also be useful if we have difficulty finding the address to our working directory.

R has many functions that we can use to import data from files saved on the hard drive. We have already mentioned the file format `.R` which is the file format that R scripts are saved as. R also has several own file formats for saving data: `.rds`, `.rdata` and `.rda`. To export a single R object we can use the function `saveRDS()`. Here are two examples that give exactly the same result:

```
saveRDS(my_data_table, file = "my_file.rds")
my_data_table %>% saveRDS("my_file.rds")
```

To import data from an rds file on the hard drive we can use the function `readRDS("my_file.rds")`.

We can also export the information in several R objects to one and the same file on the hard drive if we instead use the file format `.rdata` or `.rda`. These two file formats are different names for the same file type, where `.rda` is an abbreviation for `.rdata`. To export information to an rda file we can use the function `save()`. Say as an example that we have two objects that we want to export to the same file:

```
save(my_object_1, my_object_2, file="my_file.rda")
```

To import the same information we can use the function `load()`:

```
load("my_file.rda")
```

R can also read many different types of file formats, for example from Excel and Stata. If we want to import a data file of a particular type to the program we need to find a function that can read precisely this file format.

To import data from or export data to files with Excel's file format `.xls` or `.xlsx` we can for example install the R packages `readxl` and `writexl` and from these use the functions `read_xlsx()` and `write_xlsx()` respectively. Here is an example:

```
# Run install.packages() once
install.packages("readxl")
# Run library() after each restart of R
library("readxl")
# Retrieve data from an Excel file
my_data <- read_xlsx(path="my_excel_file.xlsx")
# Export the same data to an Excel file on the hard drive
install.packages("writexl")
library("writexl")
my_data %>% write_xlsx(path="my_R_data.xlsx")
```

Functions that aim to import or export data to and from files can also in many cases have many useful options. For example, we can with the help of these functions choose which tabs in an Excel sheet we want to read data from, or write data to. The packages `readxl` and `writexl` also contain other functions that can be useful, which we find for example with the help of the command:

```
help(package="readxl")
```

Another package with similar functions is `xlsx`, which we can also install and load using the functions `install.packages()` and `library()`. If some type of option or function is missing in a package, it can sometimes be valuable to try another similar package. We mentioned earlier the file format `.csv`, comma-separated files. Like many other programs, including Excel and Stata, R can also read csv files. If we want to import data from a csv file saved on the hard drive to RStudio we can for example use the function `read.csv()`, which is pre-installed in R when we install the program on the computer. For example like this:

```
# Import csv file
my_data <- read.csv(file="my file on the hard drive.csv")
```

For the command to work, the file `my file on the hard drive.csv` must be saved in the working directory we specified. The command imports the information in the file and saves it in the object `my_data`. The information is saved as a table of type `data.frame`, which is determined by the function `read.csv`. If we want the information to be saved as a table of type tibble we can for example write:

```
my_data <- read.csv("my_file.csv") %>% as_tibble()
```

If we instead want to export data from RStudio to a csv file on the hard drive we can use the function `write.csv()`, for example like this:

```
my_data_object %>% write.csv(file="my_file.csv")
```

31.6 Working with Data Tables

R can, as mentioned, handle information in many different ways and in many different forms, where two variants are in data tables of type tibble and data.frame, which were introduced above. This section gives some examples of how we can edit tables of this type and is based on the following object with table, which we described above:

```
my_data <- tibble(
    y = c(3,2,5,4),
    x = c(3,4,6,7),
    z = c(1,4,0,1))
```

To display the information in an object we can, as mentioned, call the object name, which in this case is `my_data`. If we want to look at the table in a similar way as when we for example work in Excel and Stata, we can use the function `view()`:

```
view(my_data)
```

The function `view()` opens a new tab in RStudio that displays the table. To get more comprehensive information about a table we can use the functions `str(my_data)` and `summary(my_data)`. For continuous variables, `summary()` returns percentiles, which can give us a comprehensive picture of the frequency distribution of a variable.

If we want to know names of columns or rows in a table we can use the functions `colnames(my_data)` and `rownames(my_data)`. Note that all these functions return information that we, if needed, can save in new R objects. If we for example want to create a list of the variables in a table we can for example use the following command:

```
my_variables <- colnames(my_data)
```

The object `my_variables` now contains only the name of the variables. No other information about the variables is saved in `my_variables`, such as what type of data is found in the table `my_data`. To display the first rows in a table we can use the function `head()`, which returns any number of rows counted from the top in a table. To display the last rows in the table we can use `tail()`, which returns rows counted from the bottom. For example like this:

```
# Display the 22 top rows
head(x = my_data, n = 22)
# Next line gives the same result
my_data %>% head(22)
# The 3 rows at the bottom of the table
my_data %>% tail(3)
```

The functions `head()` and `tail()` can also be used if we for example want to reuse a smaller part of a table. Say for example that we have a table in an object called `my_table` and we want to export the first 100 rows in this table to a csv file on the hard drive. We can do this with the following code:

```
my_table %>% head(100) %>% write.csv("the_data_file.csv")
```

To edit columns and rows in a table we can use many functions from the dplyr package. This package is included in the tidyverse package, which we installed and activated above. To read more about this package we can use the following command:

```
help(package="dplyr")
```

The functions below work, unless otherwise stated, in a similar way:

1. The first argument in the function is an object with a table. For example the object `my_data`.
2. Then we describe what we want to do with the table.
3. The result of the function also becomes a table.

As an example, if we want to change the name of a variable we can use the function `rename(new_name = old_name)`:

```
my_data %>% rename( var_y = y )
```

We use pipe `%>%` to specify the object `my_data` as the first argument in the function `rename()`. The function `rename()` describes what we want to do with the table (change the name of one or more variables). The result of the function `rename()` is in turn also a table of type tibble. For the variable to be changed in the object `my_data` we must also overwrite the old object by reusing the name of the object, for example like this:

```
my_data <- my_data %>% rename( var_y = y )
```

If for some reason we want to leave the first object untouched we can create a new object by simply naming the new object something else, for example:

```
my_data_2 <- my_data %>% rename( var_y = y )
```

If we have a table and want to remove columns we can use the function `select()`. For example like this:

```
# Keep columns y and x
my_data %>% select(y,x)
# Keep all columns except y
my_data %>% select(-y)
# Keep columns y and x but rename them
my_data %>% select( var_x = x, var_y = y )
```

If we want to remove specific rows from a table we can use the function `filter()`. For example like this:

```
# Keep all rows where variable y>5
my_data %>% filter(y>5)
# Keep all rows where variable x equals 3
my_data %>% filter(x==3)
```

The double equals signs are used to formulate conditions. We can also use `<`, `>`, `<=`, `>=` and `!=` where the last means "not equal to". That is, `filter(x!=3)` would here mean that we want to keep all rows in the table where variable `x` is not equal to 3. We can combine several conditions by separating these with commas in the function `filter()`. The following means that we want to keep only those observations where both conditions are met:

```
my_data %>% filter(x>2, y>3)
```

To describe how several conditions must be met simultaneously we can use the symbol `&` (and). To describe how one of several conditions must be met we can use the function `|` (or). For example:

```
# Keep the rows where x>2 OR y>2
my_data %>% filter(x>2 | y>2)
```

Just as before, we must also here use the function `<-` if we want the object `my_data` to be changed in memory. The functions `select()` and `filter()` return a table but if we don't overwrite the old object `my_data`, R will not note that any changes have been made.

To sort a table by observations, from smallest to largest value, we can use `arrange()`. To sort from largest to smallest we can combine this with the function `desc()`. Here are two examples:

```
# From smallest to largest
my_data %>% arrange(x)
# From largest to smallest
my_data %>% arrange(desc(x))
```

We can change the order of columns with the function `relocate()`. Example:

```
my_table %>% relocate(z,x,y)
```

To create new columns and edit old ones in a table we can use the function `mutate()`. For example:

```
# Create new variable from existing variable x
my_data %>% mutate(ln_x = log(x))
# Create new variable with value 1 on all rows
my_data %>% mutate(new_var = 1)
```

The function `mutate()` can also be combined with functions such as `as.numeric()` and `as.character()` to change data format on a variable in a table. If we for example have a variable `x` saved as text and want to read this as numbers instead we can as part of a command use the following:

```
my_data %>% mutate(x = as.numeric(x))
```

If we want to instruct R that a variable consists of text strings we can instead use the following:

```
my_data %>% mutate(x = as.character(x))
```

By combining `mutate()` with conditions we can edit specific rows, observations, for example with regard to existing values in one or more variables. The function `row_number()` returns row numbers in a table. We can use this to edit specific observations in a variable. For example:

```
# Change the value of variable x, row 4
my_data %>% mutate(x = 3 if row_number()==4)
```

To edit specific observations we can also benefit from combining the function `mutate()` with the functions `if_else()` and `case_when()`. In the function `if_else()` we first specify a condition, then comma and the value we want `if_else()` to return if the condition is met, and finally comma and the value that should be returned if the condition is not met. We can for example use this to edit all rows that meet a condition. For example:

```
my_data %>% mutate(new_variable =
if_else( row_number() <=2, 1, 0))
```

This command creates the new variable `new_variable` and gives it the value 1 for rows 1 and 2 and the value 0 for all other rows. With the function `case_when()` we can specify any number of conditions with associated results. Here follows an example where we add the new variable `k`:

```
my_data <- my_data %>%
mutate(k = case_when(
    row_number()<=2 ~ 1,
    row_number()==3 ~ 2,
    TRUE ~ 3) )
```

The last argument in the above example, `TRUE ~ 3`, means that the rest of the observations in the variable, those that don't meet any of the other conditions, get the value 3. Just as in previous examples we must use `<-` to overwrite the old object if we want the table saved in the object `my_data` to be changed.

If we use `mutate()` and `case_when()` to edit an existing variable `x`, we can specify `TRUE ~ x`, to let the remaining observations keep previous values. For example:

```
my_data %>%
mutate(x = case_when(
    row_number()==1 ~ 1,
    TRUE ~ x))
```

To group the rows in a table based on the values in a variable in the table we can use the function `group_by()`. In an example above we created the variable `k`. Let us group the table based on the unique values in `k`. Here follows an example. All commands that follow after the grouping will take this into account.

```
my_data %>%
group_by(k) %>%
mutate(sum_per_k_value = sum(x, na.rm=TRUE))
```

If we save the table after the grouping, the grouping will be saved too. To remove the grouping we can use the function `ungroup()`. For example:

```
my_data %>% group_by(k) %>%
mutate(sum(x)) %>% ungroup()
```

Another good package for working with tables is `tidyr`, which is also included in the tidyverse package. Since we activated tidyverse earlier with the command `library("tidyverse")` we have already loaded the tidyr package. When we restart R or RStudio we must run the command `library("tidyverse")` again.

Two good functions in tidyr are `pivot_longer()` and `pivot_wider()`. With the function `pivot_longer()` we can transpose or pivot a table from wide to long format. With `pivot_wider()` we can pivot a table to wide format. Here follows an example where we move around the values in a small table. By working with a small table with few rows and columns it becomes easier to follow what happens in each step.

```
# Create a table and add a new variable with row numbers
my_table <- tibble( a=c(32,14), b=c(5,66) ) %>%
mutate(row=row_number())
# Pivot the table to long format
my_table <- my_table %>% pivot_longer(cols=a:b)
# Pivot the table to wide format
my_table %>% pivot_wider(names_from=name, values_from=value)
```

Both `pivot_longer()` and `pivot_wider()` have different useful options, which we can read more about in their respective help sections. See also the help sections for each package:

```
help(package="tidyr")
help(package="dplyr")
```

If we want to retrieve a variable from a table and convert this to a vector or a collection with values, in the way we can create collections of values with the function `c()`, we can for example use the function `pull()`. For example:

```
my_table %>% pull(x)
```

where `x` in the parentheses for `pull()` is the name of a variable in the table `my_table`. We can also use the dollar sign, the `$`-operator. Say we have an object `my_table` that contains a table with variables `x`, `person` and `gdp2`. If we want to save a variable from the table in its own new object we can for example use the following commands:

```
var_x <- my_table$x
var_person <- my_table$person
gdp2_var <- my_table$gdp2
```

If we want to retrieve a specific value we can use square brackets and specify a position, such as a column, a row or a specific element in a table. The following command retrieves the value in variable `gdp2` on row 7:

```
gdp2_row7 <- my_table$gdp2[7]
```

31.7 Join Tables

To merge columns in two tables we can use the functions `full_join()`, `inner_join()`, `left_join()` and `right_join()`. These functions are built in a similar way but serve slightly different functions. The two R objects with tables that should be merged are defined in each function as `x` and `y`, for example `full_join(x= table_1, y= table_2)`. If the tables contain columns (variables) that have the same name, the function guesses that we want to match observations in these columns against each other in the two tables. The function `full_join()` merges all observations from two tables. As an example:

```
# table_1 contains persons 1, 2 and 3
table_1 <- tibble(person=c(1,2,3), age=c(10,20,30))
# table_2 contains persons 1, 2 and 4
table_2 <- tibble(person=c(1,2,4), height=c(120,190,180))
# The following two lines give the same result
full_join(x=table_1, y=table_2, by=c("person"="person"))
table_1 %>% full_join(table_2)
```

In the last line we don't define the option `by`, which is why `full_join()` instead assumes that we want to match the observations in the variables named `person` in the two tables. If we want to match two columns that don't have the same name in the two tables we can specify that in the option `by`. In the parentheses for `by=c()` we can also specify that we want to match on several variables.

The result is displayed in RStudio's Command window. Note the difference when we instead use `inner_join()`, which only merges the observations that match an observation in both tables:

```
table_1 %>% inner_join(table_2)
```

The function `left_join()` keeps all columns in the first (the left) table. The function `right_join()` keeps all columns in the second table. Note the difference in the results from the following 2 commands:

```
table_1 %>% left_join(table_2)
table_1 %>% right_join(table_2)
```

The results are shown in figure 31.3. With `full_join()` we keep all information we have about persons 1 to 4. With `inner_join()` we only keep information about the persons for whom we have values in all columns. With `left_join()` and `right_join()` we keep all persons from the left and right table respectively and lose the person in each case who is not found in the other table.

full_join()

person	age	height
1	10	120
2	20	190
3	30	NA
4	NA	180

Figure 31.3: Merging of table_1 and table_2

inner_join		
person	age	height
1	10	120
2	20	190

Figure 31.4: Merging of table_1 and table_2

left_join		
person	age	height
1	10	120
2	20	190
3	30	NA

Figure 31.5: Merging of table_1 and table_2

right_join()

person	age	height
1	10	120
2	20	190
4	NA	180

Figure 31.6: Merging of table_1 and table_2

31.8 Calculate Index and Export Tables

Index for a variable year t is calculated in the following way:

$$\text{Index}_t = \frac{\text{Value}_t}{\text{Value}_{\text{base year}}} \times 100$$

where value refers to the value for the variable we want to create an index for. We will now calculate an index for GDP per capita. Table 31.2 describes our data with GDP per capita for the years 2015–2019. To create a table with these observations and variables in R we can use the following command:

```
bnp_data <- tibble(
    year = 2015:2019,
    gdppercap = c(474000, 477800, 483500, 487300, 488800))
```

Table 31.2: Variables year and GDP.

year	GDP per capita, 2019 prices
2015	474,000
2016	477,800
2017	483,500
2018	487,300
2019	488,800

To then calculate an index for GDP we can use the following command where we use the function `mutate()` to add a variable to the table with the index calculation:

```
bnp_data <- bnp_data %>%
arrange(year) %>%
mutate(g_index = gdppercap / gdppercap[1] * 100)
```

Since we use `<-` the old object `bnp_data` is overwritten with the new object. The only difference between the new and old object is that we have added a column with the variable `g_index`. Say now that we want to export our table with our index to a file on the hard drive or display the result in another program, for example in a document we are writing in Microsoft Word. We can do this in several ways. We can for example export our table to an Excel file using the function `write_xlsx()`, which we introduced above. For example like this:

```
bnp_data %>%
write_xlsx(path="my_gdp_index_table.xlsx")
```

This saves all data from the object `bnp_data` to the new Excel file `my_gdp_index_table.xlsx`. If we only want to save parts of the table we can for example select rows and columns using the functions `filter()` and `select()`. Saving tables in Excel can for example be useful if we want to present results in one of Microsoft's programs Word or PowerPoint. We can also create tables directly in RStudio with a specific layout that we choose ourselves. A particular package that can be interesting in this context is `gt`, which we can install and load with `install.packages("gt")` and `library("gt")`. This package contains a large number of functions for precisely this purpose. You can read more about the package at: https://gt.rstudio.com/articles/intro-creating-gt-tables.html

31.9 Matrix Calculations

In R we can, as mentioned, write mathematical expressions directly in our script or in the Command window. The same applies to calculations with matrices, given that we instruct the program that we want to perform matrix calculations. We start by creating the following matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

To define a matrix we can use the function `matrix()`, for example as in the following command:

```
A <- matrix(c(1,2,3,4), nrow=2, byrow=TRUE)
```

The first argument in `matrix()` is the data that should fill the elements in the matrix, by using the function `c()` and specifying the numbers 1, 2, 3 and 4. The option `nrow=2` defines that the matrix should have 2 rows. The option `byrow=TRUE` specifies that the data that is entered should be read in row-wise. The values 1 and 2 therefore become the elements in the top row of the matrix. The following command results in exactly the same matrix:

```
A <- matrix(c(1,3,2,4), nrow=2)
```

To multiply two numbers in R we can, just as in many other programs, use asterisk, for example `2*3`. To matrix multiply two matrices we can instead use the command `%*%`. Here is an example where we define a new matrix `B` and multiply this with matrix `A`:

```r
B <- matrix(c(4,2,3,5), nrow=2)
A %*% B
```

To add or subtract with matrices we can use `+` and `-` as usual. For example `A - B` or `A + B`. To transpose a matrix we can use the function `t()`, for example `t(A)` to transpose matrix `A`. The inverse of matrix A can in mathematics be described as A^{-1}. In R we can calculate the inverse of `A` with the function `solve()`:

```r
# Calculate the inverse of matrix A:
solve(A)
```

The function `diag()` can be used both to return the elements in the diagonal from a matrix, give the diagonal in an existing matrix new values or create a new diagonal matrix from for example a vector. For example like this:

```r
# We start from matrix A:
A <- matrix(c(1,3,2,4), nrow=2)
# Return the diagonal in A
diag(A)
# Create a diagonal matrix from other values
c(1,5,4) %>% diag()
# Give the elements in the diagonal in A new values
diag(A) <- c(5,6)
# Last command changed the content in A
A
```

If we only specify a number in the parentheses for `diag()` the function returns a square identity matrix with the same number of rows and columns as the number we specify, where all values on the diagonal equal 1. For example: `diag(2)` returns a 2×2 identity matrix. To calculate the sum of the diagonal, the trace, in matrix `A` we can use function `trace()`, for example:

```r
# Calculate trace for A
trace(A)
```

To calculate the determinant of matrix `A` we can use the function `det()`, for example `det(A)`. To save the value that is returned in a new object we must as usual use `<-`. In the example above we also combined `diag(A)` with `<-` to assign new values to the diagonal elements in `A`. In a similar way we can also give the columns in matrix `A` new names. For example like this:

```r
matrix_A <- matrix(c(1,3,2,4), nrow=2)
# Display the content
matrix_A
colnames(matrix_A) <- c("col1","col2")
# Note the difference
matrix_A
```

R objects that contain matrices and vectors are not the same thing as an R object that contains a table of type tibble. If we want to use our matrix as a table we can convert the matrix's content to a tibble table with the function `as_tibble()`. For example like this:

```r
matrix_A <- matrix(c(1,3,2,4), nrow=2)
colnames(matrix_A) <- c("col1","col2")
table_A <- matrix_A %>% as_tibble()
```

31.10 Input–Output Analysis

This section goes through the corresponding example with input-output analysis that we used in the Excel and Stata chapters. We defined matrix A:

$$A = Z \cdot \text{diag}\,(S_m)$$

where Z is the flow matrix and $\text{diag}(S_m)$ is a diagonal matrix with the multiplicative inverse of total production per sector along the diagonal. Matrix C is a column matrix with production for final consumption per sector. The total production from sector j required for all inputs and final consumption is y_j, collected in matrix Y, which can be defined as:

$$Y = (I - A)^{-1} S$$

where I is an identity matrix with the same dimensions as A. We call $B = (I - A)^{-1}$, which has the same dimensions as A, where column sum j is the production multiplier and row sum j is the input multiplier for sector j. To perform this calculation in R we start by defining the matrices Z, C, S, S_m and $\text{diag}(S_m)$:

```
Z <- matrix(c(2,1,1,3), nrow=2)
C <- matrix(c(3,3), ncol=1)
S <- c(6,7)
mat_S <- matrix(c(6,7), nrow=2)
diag_Sm <- diag(1/S)
```

Then we can calculate A, B and Y:

```
A <- Z %*% diag(Sm)
B <- solve(diag(2) - A)
Y <- B %*% mat_S
```

Note how we, if we wanted to, can write the entire calculation of Y in one command without creating any objects:

```
solve(diag(2) -
        (matrix(c(2,1,1,3), nrow=2) %*%
         diag(1/c(6,7))
        )
     ) %*% c(6,7)
```

31.11 Variation and Covariance

Say that we have an R object `my_data` that contains the variables `x`, `y` and `z` (same as above):

```
my_data <- tibble(
    y = c(3,2,5,4),
    x = c(3,4,6,7),
    z = c(1,4,0,1))
```

Now we will estimate and calculate mean, variance, standard deviation as well as some percentiles and quartiles for the variables in this table. Parts of this information we can get through the function `summary()`, for example:

```
my_data %>% summary()
```

We can also use one of the many functions that are pre-installed in R, such as `mean()`, `var()`, `sd()` and `quantile()`. For example like this:

```
# Calculate mean for variable x
my_data$x %>% mean()
# Calculate mean using summarize
my_data %>%
summarize(mean(x),
mean(z))
```

With the function `quantile()` we can calculate percentiles and thereby also deciles and quartiles. In the parentheses for the function we need to specify a variable and which percentiles we want to calculate. If we have a collection of values saved in the object `x` we can use the following command:

```
quantile(x, .1)
```

The function then returns percentile 10. The notation `.1` is in R the same value as `0.1`. To calculate several percentiles we can for example use the function `c()`. Here is an example for calculating the 25th, 50th and 75th percentile (first and third quartiles as well as the median):

```
quantile(x, c(.25, .5, .75))
```

Here follow examples of how we can use `quantile()` with the table in the object `my_data`:

```
# Calculate 30th percentile for x
my_data$x %>% quantile(.3)
# Calculate 3 different percentiles for x
my_data %>%
summarize(Three_p = quantile(x, c(.1,.2,.3)))
```

To estimate variance and standard deviation for a variable we can, as mentioned, use the functions `var()` and `sd()`. In the parentheses for `sd()` we can only specify one variable at a time. If we only specify one variable in the parentheses for `var()` the function returns the sample variance. For example:

```
my_data %>% select(y) %>% var()
```

If we specify several variables in the parentheses for `var()` the function instead returns the covariance. R also has a specific function for calculating covariance, called `cov()`. We continue to use the object `my_data` with the table with four observations for variables `y`, `x` and `z`. In this case the following two commands give exactly the same result in the form of a 3×3 matrix with estimated variance and covariance between the three variables `x`, `y` and `z` in our table:

```
# The function var()
my_data %>% var()
# The function cov()
my_data %>% cov()
```

In the above commands we specify the entire object `my_data` as input value to the functions `var()` and `cov()`. This works because all three variables in the table are quantitative and data is saved so that R understands that they are numerical values. If any variable in the table is not numerical, for example a variable with text strings, the functions instead return an error message.

If we instead want to calculate the correlation coefficient between two variables we can use the function `cor()`. If we specify several variables as input value for `cor()`, for example in the form of a table with only quantitative variables, the function returns the pairwise correlation between all variables. Below follows an example where we send the table in `my_data`, which contains three variables, to the function `cor()`:

```
my_data %>% cor()
```

The result is shown in figure 31.7.

```
           y          x          z
y  1.0000000  0.7071068 -0.8944272
x  0.7071068  1.0000000 -0.4216370
z -0.8944272 -0.4216370  1.0000000
```

Figure 31.7: Results of cor()

31.12 Probability

R has several pre-installed functions for calculating probabilities based on common probability distributions. Several of these are built in the same way. If we want to create 100 values drawn from a continuous uniform probability distribution we can use the function `runif()` in the following way:

```
runif(100)
```

The default setting is that the minimum value for this probability distribution is 0 and the maximum value is 1. We can also specify new values for minimum and maximum through the options `min` and `max`. For example:

```
runif(100, min=0, max=100)
```

As usual we must use the function `<-` if we want to save these values in an R object. If we instead want to know probabilities based on a density function for this type of probability distribution we can use the function `dunif()`. The function `punif()` returns values for the cumulative probability function. The function `qunif()` returns the inverse of the cumulative distribution function, the quantile function. The corresponding R functions for the normal distribution are called `rnorm()`, `dnorm()`, `pnorm()` and `qnorm()`. If we only specify one argument in the parentheses for these functions, all of these are based on the standard normal distribution. The following code gives the cumulative probability for $z = 0$:

```
pnorm(0)
```

The functions have different options for defining the distribution, for example:

```
pnorm(2, mean=2, sd=1)
```

Table 31.3 describes some examples of functions that we can use to work with some common probability distributions in R. See each function's help section for instructions on how the functions can be used and different options. R also has several pre-installed functions for several common statistical tests, for example `t.test()`.

Table 31.3: Probability Distributions in R.

Probability Distribution	**Examples of functions in R**
Uniform	`dunif(), punif(), qunif(), runif()`
Poisson distribution	`dpois(), ppois(), qpois(), rpois()`
Exponential distribution	`dexp(), pexp(), qexp(), rexp()`
Gamma distribution	`dgamma(), pgamma(), qgamma(), rgamma()`
Chi-squared distribution	`dchisq(), pchisq(), qchisq(), rchisq()`
F-distribution	`df(), pf(), qf(), rf()`
Normal distribution	`dnorm(), pnorm(), qnorm(), rnorm()`
T-distribution	`dt(), pt(), qt(), rt()`

31.13 Regression Analysis

This section introduces how we can work with regression analysis in R. First by estimating results ourselves using matrix calculations. Then how we can use ready-made R functions to achieve the same results.

31.13.1 Regression Analysis with Matrices

Now we will estimate the following regression model:

$$Y = XB + U$$

where Y, B and U are column matrices with the explained variable Y, the coefficients B and the error terms U. X is a matrix with the explanatory variables, where the first column has the value 1 in each element, for the constant (y-intercept). The estimator for the coefficients can be written with matrices as:

$$\hat{B} = \left(X^T X\right)^{-1} X^T Y$$

We may perform matrix calculations directly in our R script and therefore start by defining the matrices Y and X:

```r
Y <- matrix(c(3,2,5,4), nrow = 4)
X <- matrix(c(1,1,1,1, 3,4,6,7), nrow=4)
```

To estimate $\hat{B}$ we can then use the functions `t()`, `solve()` and matrix multiplication with `%*%`.

```r
Bhat <- solve(t(X) %*% X) %*% t(X) %*% Y
# Check the result
Bhat
```

31.13.2 Regression Analysis with `lm()`

R also has many ready-made functions for regression analysis and even more can be downloaded in the form of R packages. To estimate a linear regression model based on the least squares method we can use the function `lm()`.

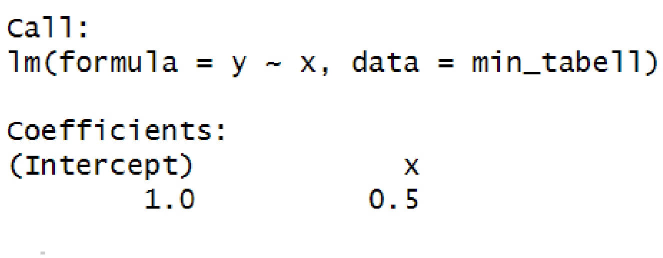
```r
# Create a table
my_table <- tibble(
    y = c(3,2,5,4),
    x = c(3,4,6,7),
    z = c(1,4,0,1))
# Estimate the regression model
lm(formula = y~x, data= my_table)
```

If we run the code above, the estimated coefficients a and b from the regression model $y = a + bx + u$ are displayed in the Console window, where y and x are variables and u is the error term. The result is shown in figure 31.8. If we want to add more variables to the regression model we can do so with the plus sign. As an example, say we instead want to estimate the following regression model:

$$y = \beta_1 + \beta_2 x + \beta_3 z + v$$

where β are the coefficients and v is the error term. To estimate this model with `lm()` we can use the following command:

```r
lm(y ~ x + z, data = my_table)
```



```
Call:
lm(formula = y ~ x, data = min_tabell)

Coefficients:
(Intercept)            x
        1.0          0.5
```

Figure 31.8: Results of lm()

Say now that we want to estimate the following regression model with the interaction term xz:

$$y = \alpha_1 + \alpha_2 x + \alpha_3 z + \alpha_4 xz + \epsilon$$

where y, x and z are the variables, α are the coefficients and ϵ is the error term. The variable xz is variable x multiplied by variable z. To estimate this model with the function `lm()` we can do this with the following command:

```r
lm(y ~ x + z + x:z, data = my_table)
```

The term `x:z` is the interaction term. Another way to get the same result is the following notation:

```
lm(y ~ x*z, data=my_table)
```

The notation `x*z` means that the program adds both `x` and `z` as separate explanatory variables as well as an interaction term for `xz`.

If we want to estimate the regression model $y = \beta_1 + \beta_2 x + \beta_3 z + v$ with robust standard errors, that is an adjusted variant of the variance-covariance matrix, we can do this for example with the function `lm_robust()`. In the parentheses for `lm_robust()` we fill in as usual the regression model and data. As an option we can then specify which method we want to use to adjust the standard errors and the variance-covariance matrix by defining the argument `se_type`. For example:

```
# Use adjustment HC0
lm_robust(y ~x, data=my_table, se_type="HC0")
# Use adjustment HC1
lm_robust(y ~x, data=my_table, se_type="HC1")
```

See the function's help section for more options and more information. When we run a command with `lm()`, several other results are estimated and calculated. To work with these we can for example save the lm result in a new object and use the function `summary()`. Here follows an example of commands. The result from `lm()` combined with `summary()` is shown in figure 31.9:

```
lm_results <- lm(y~x, data = my_table)
lm_results %>% summary()
```

```
> lm_results %>% summary()

Call:
lm(formula = y ~ x, data = min_tabell)

Residuals:
   1    2    3    4 
 0.5 -1.0  1.0 -0.5 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.0000     1.8540   0.539    0.644
x             0.5000     0.3536   1.414    0.293

Residual standard error: 1.118 on 2 degrees of freedom
Multiple R-squared:    0.5,	Adjusted R-squared:   0.25 
F-statistic:     2 on 1 and 2 DF,  p-value: 0.2929
```

Figure 31.9: Results from lm() and summary()

We can also study the results returned from `lm()` using the functions `str()` and `head()`. For example like this: `lm_results %>% str()`, or `head(lm_results)`. We can also use the `$`-operator. Here is an example where we estimate the regression model $y = a + bx + u$ and then retrieve the predicted values for $\hat{y}$ as well as the residuals $\hat{u}$. We save both of these in two new R objects:

```
lm_results <- lm(y~x, data = my_table)
predicted_y <- lm_results$fitted.values
residuals <- lm_results$residuals
```

To retrieve the regression's variance-covariance matrix we can use the function `vcov()`:

```
vcov(lm_results)
```

The command returns a matrix. Since we have two coefficients in this case, a 2×2 matrix is returned.

31.14 Export Results to Hard Drive

There are different methods for exporting regression results from R to other programs, if we for example are writing a report in Word and want to present our results there. A useful R package in this case is `stargazer`, which we can install and load as usual with the functions `install.packages()` and `library()`:

```
install.packages("stargazer")
library("stargazer")
```

In the stargazer package there is a function that is also called `stargazer()` which among other things can create nicer tables for presenting regression results. Here follows an example where we again estimate the two regression models $y = a + bx + u$ and $y = \beta_1 + \beta_2 x + \beta_3 z + v$. In the parentheses for `stargazer()` we define the option `type="text"` so that the table is displayed in readable format directly in RStudio's Console window. The result is shown in figure 31.10.

```
est_1 <- lm(y ~ x, data = my_table)
est_2 <- lm(y ~ x + z, data = my_table)
stargazer(est_1, est_2, type="text")
```

The package and function `stargazer()` have many options that we can use to influence the appearance of the results table. We can also export the results to different file types, for example `.xml` which can be read by Microsoft Word. Academic publications often use LaTeX to create results that follow certain rules, which `stargazer()` also has support for. See the package's and function's help section by using the commands:

```
help(package="stargazer")
help(stargazer)
```

```
> stargazer(est_1, est_2, type="text")

=======================================================
                          Dependent variable:
                  -------------------------------------
                                   y
                         (1)                (2)
-------------------------------------------------------
x                       0.500              0.284
                       (0.354)            (0.203)

z                                         -0.541
                                          (0.214)

Constant                1.000              2.892
                       (1.854)            (1.220)

-------------------------------------------------------
Observations              4                  4
R2                      0.500              0.932
Adjusted R2             0.250              0.797
Residual Std. Error  1.118 (df = 2)    0.581 (df = 1)
F Statistic         2.000 (df = 1; 2) 6.900 (df = 2; 1)
=======================================================
Note:                       *p<0.1; **p<0.05; ***p<0.01
```

Figure 31.10: Results from stargazer()

31.15 Create Plots

There are many functions in R that can be used to create graphics and charts. We will here introduce a few parts of the ggplot2 package, which is included in the tidyverse package. By loading tidyverse, ggplot2 is also loaded. In the ggplot2 package we can create charts using the function `ggplot()`. It works in the following way:

1. Specify data table with the columns that should be displayed in the chart.

2. Specify which columns in the table should be displayed, and define how the values from these columns should be illustrated in the chart. This is done with the function `aes()` (aesthetic mappings), which is specified in the parentheses for `ggplot()`.
3. Add one or more geom functions that describe the types of shapes and objects that should be drawn in the chart. Examples of geom functions: `geom_point()` draws points, `geom_line()` draws a line between the data points.

Here is a simple example where we create a simple time series chart. We use the pipe function `%>%` to pipe a small tibble table directly to the function `ggplot()`:

```
library(tidyverse)
tibble(time=2015:2022,
  my_variable = c(3,2,4,5,3,5,6,4)) %>%
ggplot(aes(x=time, y=my_variable)) +
geom_point() +
geom_line()
```

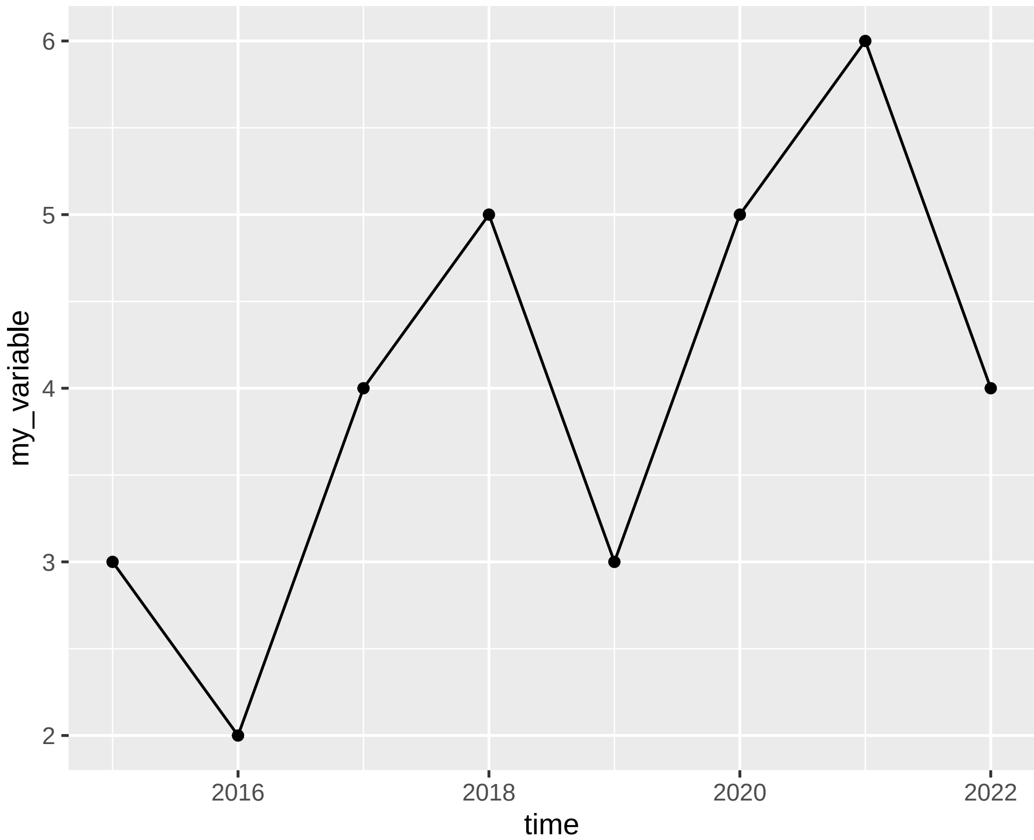


Figure 31.11: A graph with a line and dots

In the parentheses for the geom functions we can specify one or more arguments to change the appearance of the shapes that the function draws, for example: `color`, `shape`, `size`, `linewidth`, `linetype`, `fill`. In the functions' help sections there are descriptions of additional options and how we can use these to influence the chart's appearance. Below follows an example where we change color and shape of lines and dots:

```
tibble(time=2015:2022,
  my_variable = c(3,2,4,5,3,5,6,4)) %>%
ggplot(aes(x=time, y=my_variable)) +
geom_point(size=6, color="#F8766D", shape=3) +
geom_line(linetype="dashed", color="#00BFC4")
```

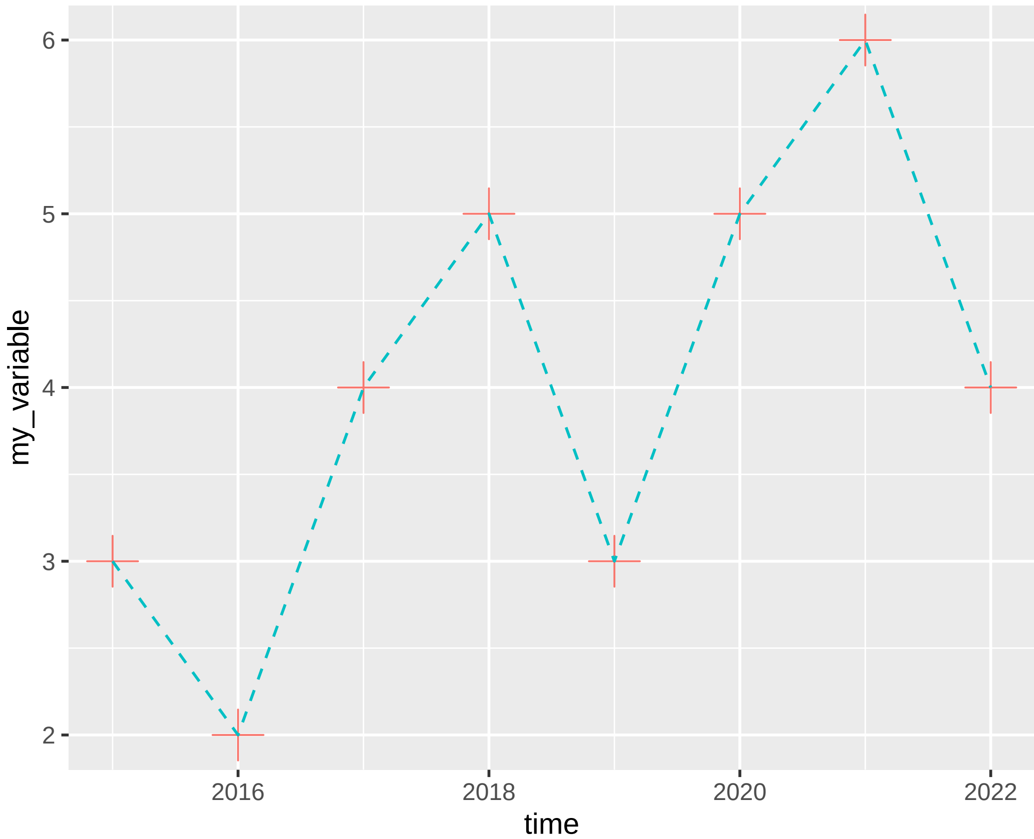


Figure 31.12: A graph with a line and dots (styled)

In the parentheses for `aes()` we can specify several arguments besides `x` and `y`. For example, we can use an aesthetic command, like `color` and define this as a variable, like `color=person`, where `person` is a variable in the table we use in the ggplot command. The unique values in the variable we refer to will define the aesthetic property, which in this case becomes the color of the data points that are drawn in the chart.

Below follows an example with three groups, where each group of observations is grouped in the variable `person`. To create the variable `person` we use the function `rep()` and create three values of each unique value: `"a"`, `"b"` and `"c"`. The table contains nine observations, three per group. By specifying `color=person` in the parentheses for `aes()`, each group will get its own color in the chart:

```
tibble(time=rep(c(1,2,3),3),
       person=c(rep("a",3), rep("b",3), rep("c",3)),
       value=c(2,4,3, 4,3,2.5, 3,2,4)) %>%
ggplot(aes(x=time, y=value, color=person)) + geom_line()
```

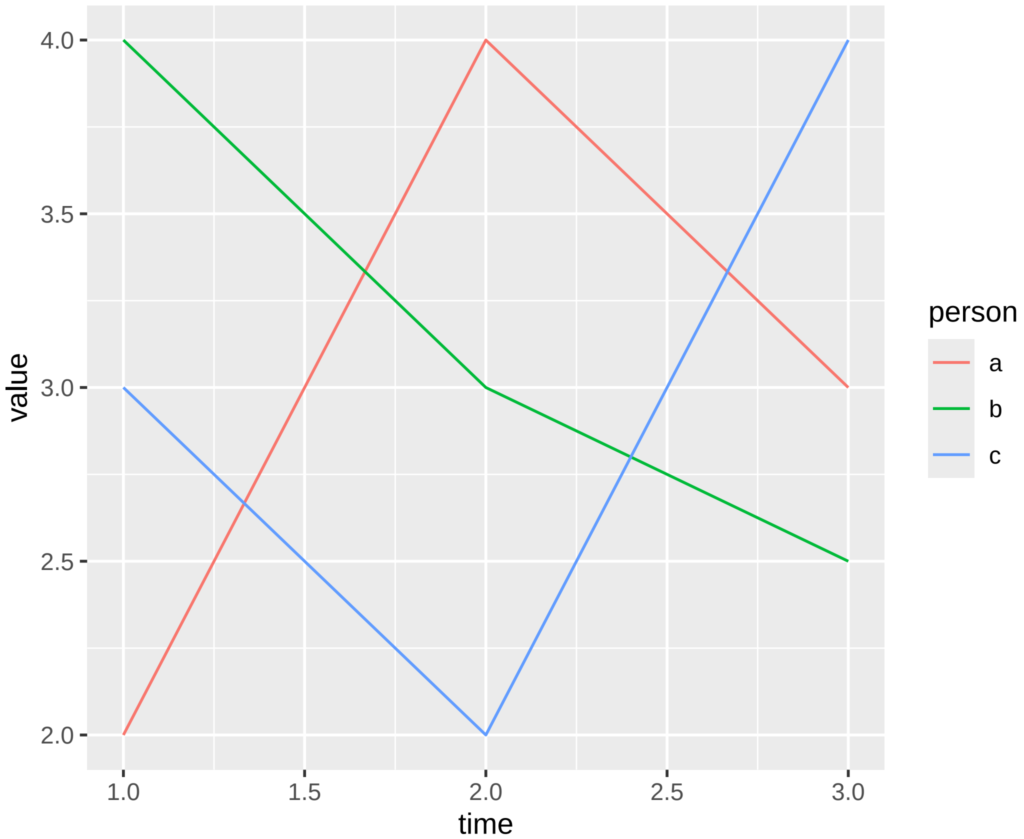


Figure 31.13: A graph with colored groups

To draw a regression line we can use `geom_smooth()`. In the parentheses we can among other things define the option `method`, which specifies which method we want to use to draw the regression line. Below follows an example where we use `method="lm"` where lm stands for linear model, which means that the line is estimated based on the least squares method.

Since we in the parentheses for `aes()` specified that we want to use variable `y` for the argument `y` and variable `x` for the argument `x`, `geom_smooth()` assumes that we want to draw the regression line based on an estimation of the regression model $y = a + bx + u$. In R this regression model can be written `y ~ x`. In the parentheses for `geom_smooth()` we also specify the option `se=TRUE`, which means that the confidence interval is drawn out. Read more in the help section for `geom_smooth()`:

```
tibble(x=c(3,4,6,7),
y=c(5,2,3,1)) %>%
ggplot(aes(x=x,y=y)) +
geom_point() +
geom_smooth(method="lm", se=TRUE)
```

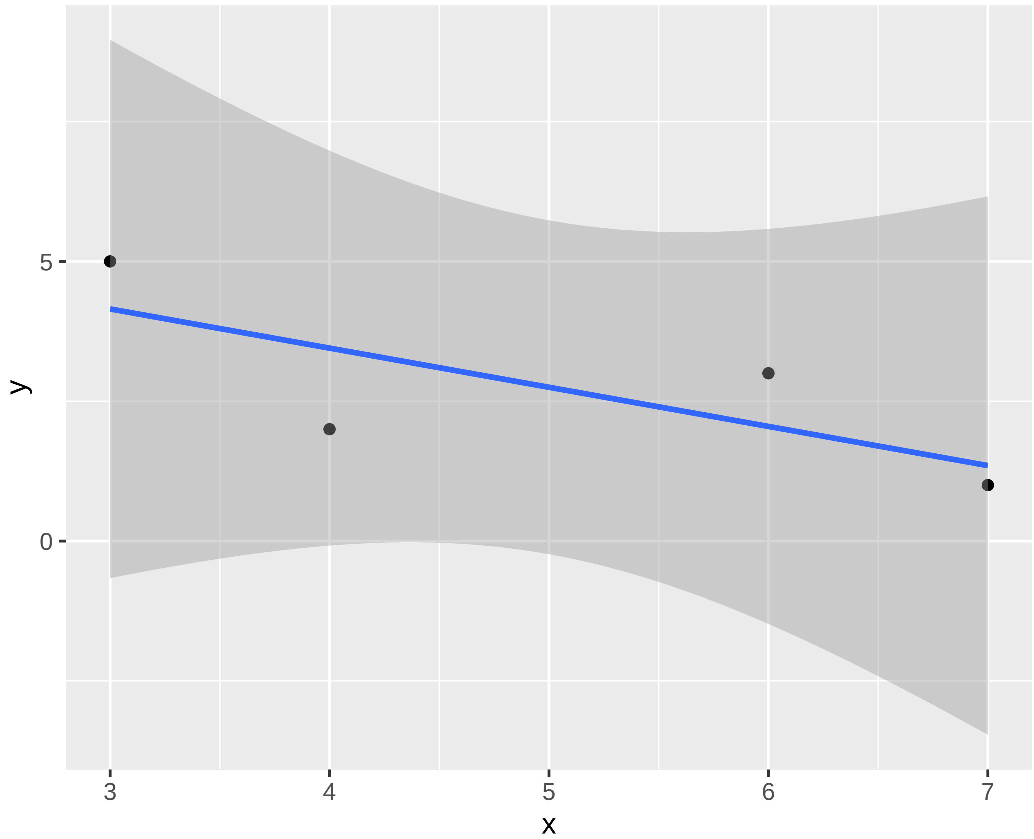


Figure 31.14: A graph with a regression line

To draw a histogram we can use the function `geom_histogram()`, which describes the frequency distribution of a variable. We then only need to specify one variable in the parentheses for `aes()`. Here follows an example where we create a variable that draws 300 random observations from the standard normal distribution and plot these in a histogram. We start by defining the random seed for our run, which we can do in R with the function `set.seed()`. In its parentheses we specify the value for the random seed. For the histogram we use the aesthetic argument `color` to define the color of the bars' edges, the option `fill` to define the color inside the bars and the option `alpha` to make the bars transparent:

```
# Define random seed
set.seed(1)
# Create variable based on normal distribution
tibble(x=rnorm(300)) %>%
# Create chart
ggplot(aes(x=x)) +
geom_histogram(alpha=.3, fill="#F8766D", color="black")
```

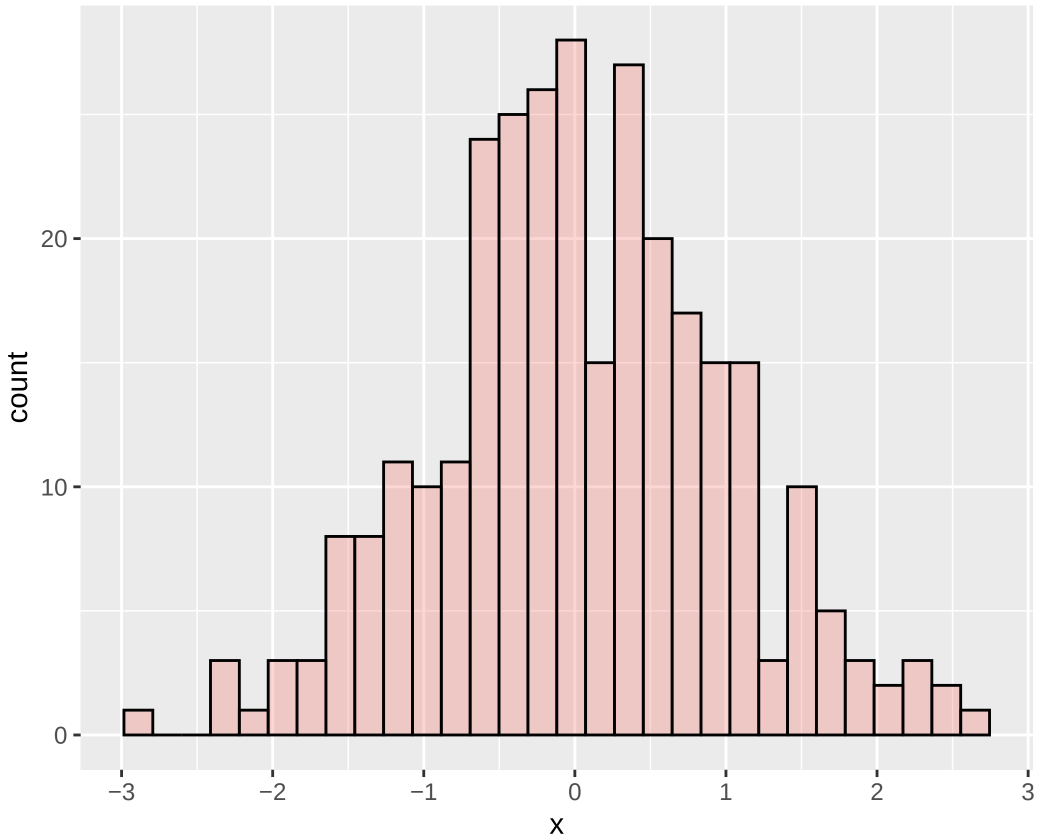


Figure 31.15: Histogram created with ggplot()

A histogram is a form of bar chart. To draw a bar chart generally we can use the function `geom_bar()`, which then requires that we specify at least two variables for the `y`- and `x`-axis. To draw mathematical functions we can use `geom_function()`. For this we don't need to use any data for `ggplot()`. We also don't need to use the function `aes()`. Instead we define all information in the parentheses for `geom_function()`.

Below follows an example where we first define the function using the R function called `function()`. We specify what we call the input variable, by writing `function(x)`. We then describe the mathematical function $y = x^2$ and then specify with the option `xlim` within which interval of x-values the mathematical function should be defined.

In the example below we use the function `labs()` which we can use to define titles. In the parentheses for `labs()` we define the option `x` to define the x-axis title, `y` to define the y-axis title, `title` to define the chart's title and we define the option `caption` to write a note below the chart. We also introduce here the function `theme()` which we can use to influence the chart's appearance in more detail. Read more about the many options available for `theme()` in the function's help section:

```
ggplot() +
geom_function(fun=function(x) x^2) +
labs(x="My x-variable",
     y="y = f(x)",
     title="My chart",
     subtitle="The subtitle here",
     caption="I drew this chart in R") +
theme(axis.title.y = element_text(angle=0),
      plot.caption = element_text(color="red"),
      plot.title= element_text(size=20, color="purple"))
```

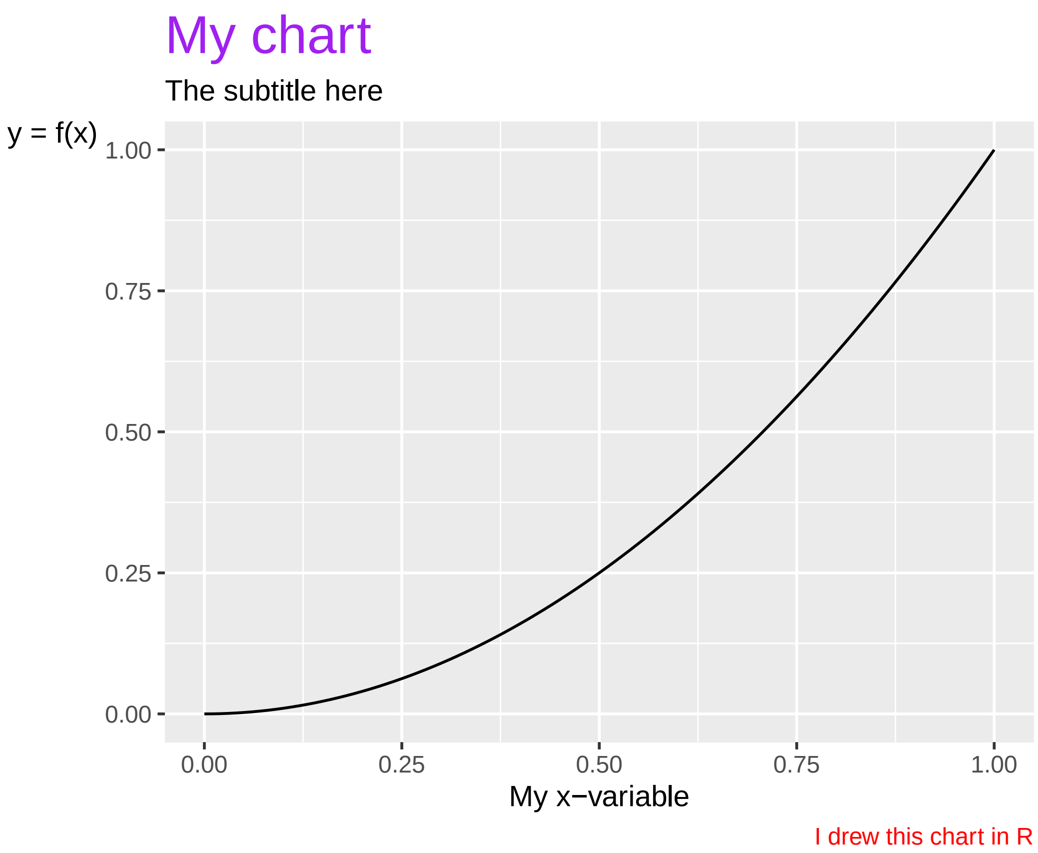


Figure 31.16: A graph with a mathematical function

Charts can also be saved in R objects by using `<-`. This can often be useful for several things, including for exporting charts to the hard drive in for example image files. To export charts we created with `ggplot()` to the hard drive we can use the function `ggsave()`. In the parentheses we specify with the help of the option `filename` the name of the file we want to save the chart as, for example `ggsave(filename="my_chart.png")`.

We can use different file extensions and thereby the file format, for example `.pdf` or `.png`. It is often helpful to also define height and width in the parentheses for `ggsave()` with the options `height` and `width`. See the function's help section. Here is an example where we first create a chart and save it in the R object `plot_1`. Then we use this object to export the chart in the object to a file on the hard drive using the function `ggsave()`.

```
plot_1 <- ggplot() +
geom_function(fun=function(x) x^3)
ggsave(plot_1, filename="my_chart.pdf", width=3, height=3)
```

Alongside the function `theme()` there are also special functions, such as `theme_bw()` and `theme_classic()` that we can use to change the appearance of the entire chart directly, without specifying any arguments. We can add free text in ggplot charts with for example the function `annotate()`. To add labels to data points the functions `geom_text()` and `geom_label()` are recommended. If we want to define intervals for the `y`- and `x`-axis we can use the functions `ylim()` and `xlim()`. If we want to control how the scales for any of the options in `aes()` are displayed we can also use a function such as `scale_x_continuous()`, `scale_fill_discrete()` or `scale_color_discrete()`. There isn't room here to describe these but you can read more about them in their respective help sections.

Saving charts in R objects can also be useful if we want to combine several charts in one and the same image. This can among other things be done with the help of the `patchwork` package. To install and activate it we first run the following commands:

```
install.packages("patchwork")
library("patchwork")
```

After we have loaded patchwork we can among other things use `+` and `/` to organize charts in one and the same image. Read more in the help section for the package by running the command `help(patchwork)`. Below follows an example:

```
library(patchwork)
# Create a chart
chart_1 <- ggplot() +
geom_function(fun=function(x) x^2) +
theme_bw()
# Create another chart
chart_2 <- ggplot() +
geom_function(fun=function(x) 1/x) +
theme_classic()
# Combine the charts with patchwork
chart_1 + chart_2
```

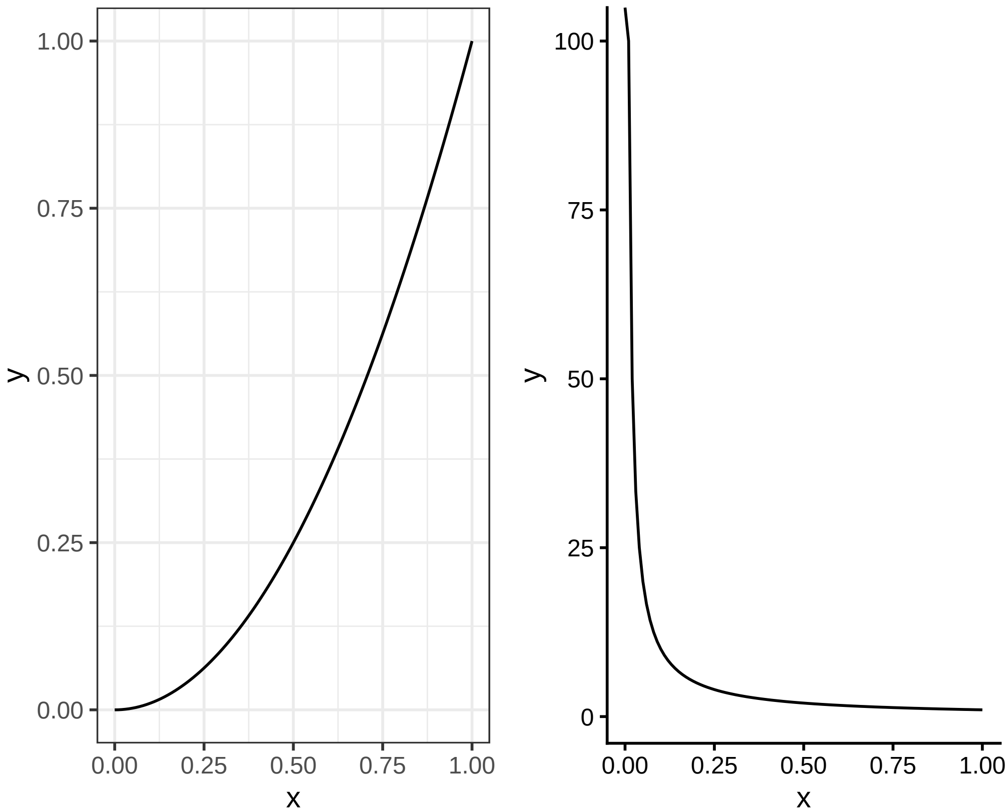


Figure 31.17: Graphs organized with patchwork

To save the combined charts in one file we can use `ggsave()`:

```
# Save the charts in one and the same file
(chart_1 + chart_2) %>%
ggsave("my_charts.pdf", width=4, height=2)
```

31.16 Iterations and Loops

There are many ways to work with iterations in R. We have already used certain forms of iterations in different examples above. For example to calculate the sum per group in a table using the function `group_by()` or to define colors and shapes in charts by combining options like `color` and `fill` in the parentheses for `aes()` with the name of a variable in our table. The function `group_by()` and other functions in among others the dplyr package can many times be combined to achieve unexpected results.

In this section we will introduce how we can build iterations and loops using the function `map()`. The function `map()` is for building loops and has two arguments. The first argument is called `.x` and is the input value that the function should use for each loop round. The input value can consist of a collection of values from the function `c()`, the elements in a vector, the columns in a table or something else. The second argument is `.f`, which defines what the function `map()` should do with the current value in each loop round. We can fill `.f` with a single simple operation or with a long chain of commands.

The function `map()` returns a form of data called a list. A list does not only refer to a collection of values (such as from the function `c()`) but is a special form for how data can be organized. Each element in a list can consist of different things. An element in a list can be a number or a text string but it can also be an entire table. A list can therefore be a collection of tables. We will not go through this in detail here but focus on how we can work with `map()` and tables, as well as get a table as a result of our loop in `map()`. Here is a simple example:

```
c(1,2,3) %>% map(log)
```

What happens in the command above is that the function `c()` sends the three values 1, 2 and 3 to `map()` which in turn uses the function `log()`. The first round the function takes the value 1 and returns the logarithm of this as the first element in the list that `map()` starts creating. The second round the function takes the value 2, creates the logarithm of this and returns this as the second element to the list from `map()`, and so on.

The entire command returns a list with the logarithmic values of 1, 2 and 3. To instead of a list object get a vector we can add the function `list_simplify()`:

```
c(1,2,3) %>% map(log) %>% list_simplify()
```

The example is of course completely unnecessary since we can instead take `log(c(1,2,3))`. Say now that we have the following table:

```
my_table <- tibble(
    y = c(3,2,5,4),
    x = c(3,4,6,7),
    z = c(1,4,0,1))
```

If we want to calculate a result per column, for example mean, we can for example use the following command with `map()`:

```
my_table %>% map(mean) %>% list_simplify()
```

Say now that we want to calculate several things at the same time and return a table. We can do this with the following command, which is explained below:

```
my_table %>% map(~{
tibble(
    mean = mean(.x),
    sd = sd(.x)) %>%
return()
}) %>%
list_rbind() %>%
mutate(variable = my_table %>% colnames())
```

The object `my_table` is sent to `map()`. The first round now `map()` uses the first column as input value `.x`. The second round `map()` uses the second column, and so on. In the parentheses for `map()` we write `~{}` and inside the curly brackets we write the commands we want to apply on each loop round. We can also write `.f=~{}` and get exactly the same result.

In this case we specify that we want to create a new tibble table on each round and in this create the variables `mean` and `sd`, which all use the values in `.x` as input values to calculate results. On each round a tibble table is now created with only one row (since `mean()` and `sd()` return only one constant each). We send the tibble table to the function `return()`, which controls what is returned on each loop round. The result becomes the same in this case if we delete `return()`.

After the parentheses for `map()` we send the result (a list) to the function `list_rbind()` which converts the list object to a table where each tibble table created in the loop rounds is converted to a common tibble table. Then we send this single table forward to the function `mutate()` to create a new variable, where we add the name of the three variables `x`, `y` and `z` from `my_table` using the function `colnames()`. The result is shown in figure 31.18.

```
# A tibble: 3 × 3
   mean    sd variable
  <dbl> <dbl> <chr>
1   3.5  1.29 y
2   5    1.83 x
3   1.5  1.73 z
```

Figure 31.18: Table created with map()

31.17 Example with GDP per Working Hours

In the Excel and Stata chapters we went through examples of how we can work with data on GDP per working hour. We will now do the same thing in R. Table 31.4 repeats the observations for real GDP per working hour, calculated in purchasing power adjusted millions of US dollars. To work with this example we need to import the data to R. One way to do this is the following command, where we save the table in the object `tcb_data`:

```
library(tidyverse)
tcb_data <- tibble(
country = c("Denmark","Finland","Norway","Sweden","United Kingdom","United States"),
gh1990 = c(62.6, 47, 68.1, 54.6, 50, 53.2),
gh2000 = c(76.6, 64.3, 89.3, 68.4, 63.1, 63.6),
gh2010 = c(84.2, 73.5, 95.9, 81.6, 70.4, 79.5),
gh2020 = c(95.9, 77, 101.1, 89.3, 73.4, 87.4))
```

Table 31.4: GDP per Working Hours. Source: The Conference Board. Real GDP purchasing power adjusted millions USD, divided by working hours per year.

country	**gh1990**	**gh2000**	**gh2010**	**gh2020**
Denmark	62.6	76.6	84.2	95.9
Finland	47	64.3	73.5	77
Norway	68.1	89.3	95.9	101.1
Sweden	54.6	68.4	81.6	89.3
United Kingdom	50	63.1	70.4	73.4
United States	53.2	63.6	79.5	87.4

Many work tasks in R are easier to perform with data that is organized in long format. We therefore start by pivoting the table to long format using the function `pivot_longer()`. To tidy up the table we also use in the command below the options `names_prefix` (which identifies that the columns gh1990...gh2020 begin with the letters gh), `values_to` (to name the new column with GDP per working hour) and `names_to` (to name the new variable with the years). After `pivot_longer()` we use the functions `mutate()` and `as.numeric()` to convert the new variable year to numbers instead of text strings:

```
tcb_data <- tcb_data %>%
pivot_longer(cols=gh1990:gh2020,
    names_prefix = "gh",
    values_to = "gdp_hour",
    names_to = "year") %>%
mutate(year = as.numeric(year))
```

Now the data is in more manageable format. Let us now calculate an index per country. We can do this using the functions `group_by()` and `mutate()`, for example like this:

```
tcb_data <- tcb_data %>%
group_by(country) %>%
mutate(gdp_index = gdp_hour/gdp_hour[1] * 100)
```

We have now saved the table in long format and added a new variable with a calculation of the indexed development of GDP per working hour per country. Let us look at the results in two charts. We start by creating a chart for GDP per working hour:

```
tcb_data %>% ggplot(
    aes(x=year, y=gdp_hour, color=country)) +
geom_line()
```

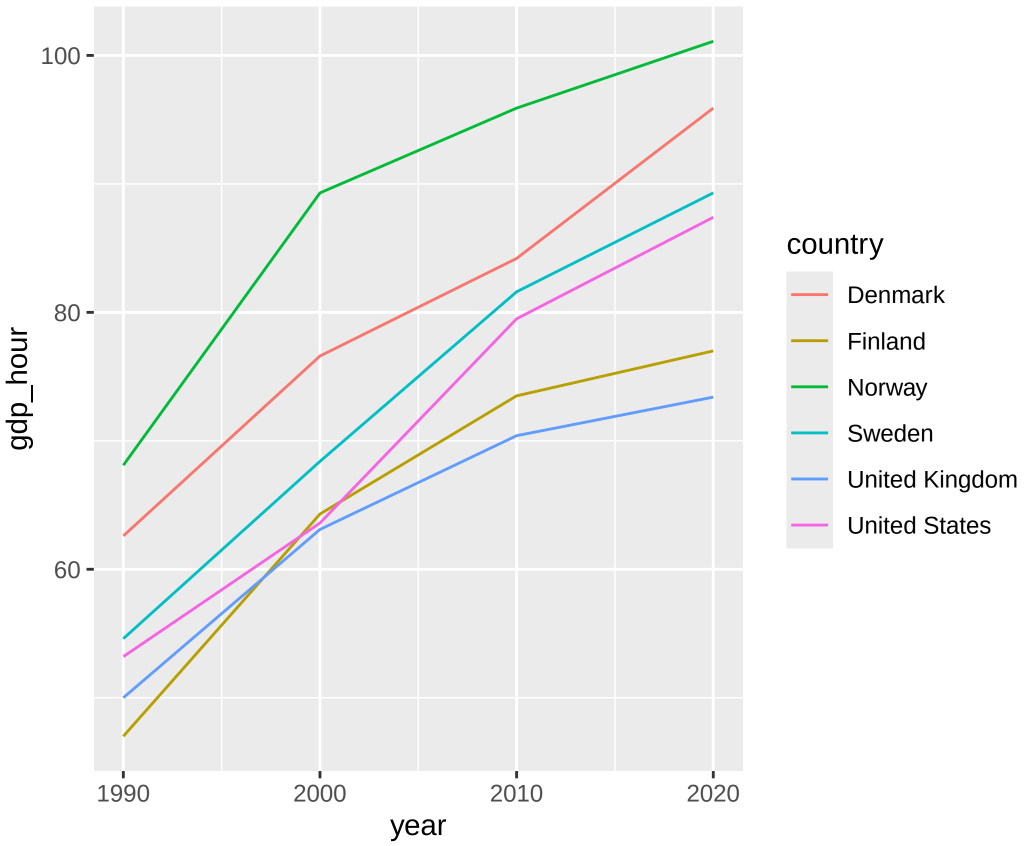


Figure 31.19: GDP per Working Hour

Then we can create a chart of the indexed development:

```
tcb_data %>% ggplot(
    aes(x=year, y=gdp_index, color=country)) +
geom_line()
```

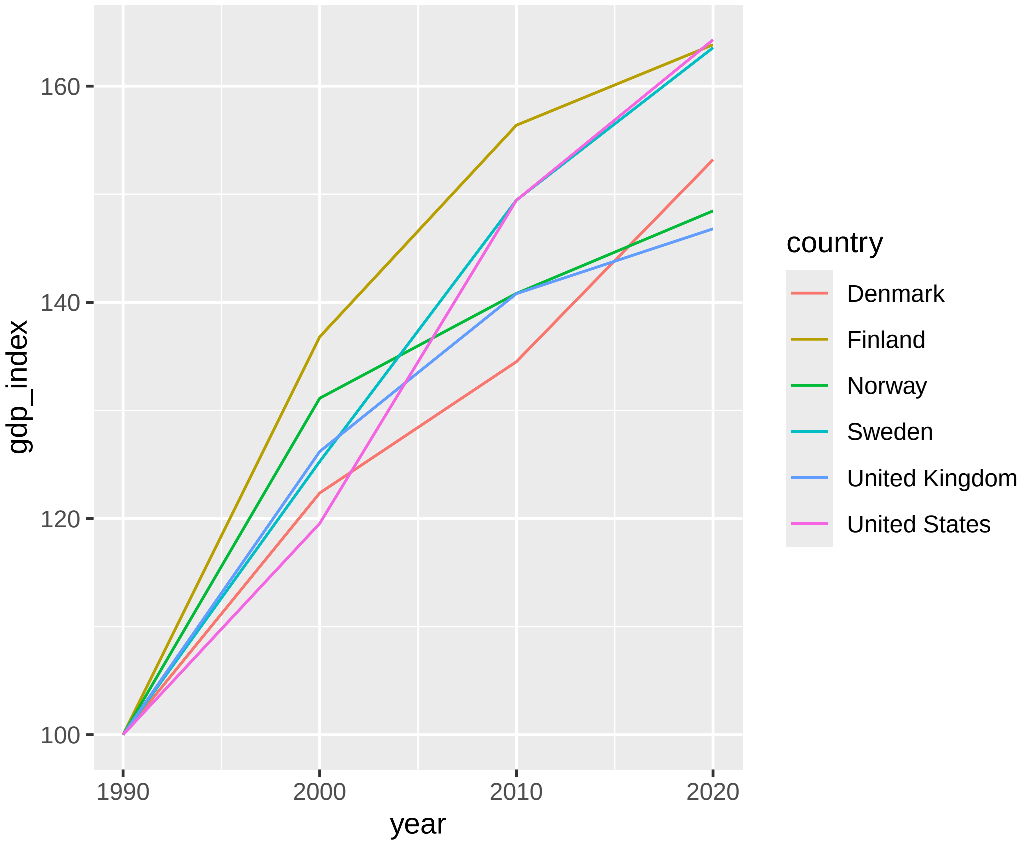


Figure 31.20: Indexed GDP per Working Hour

The upper chart shows GDP per working hour. The lower chart shows indexed development. Let us also estimate the following regression model for each country:

$$y_{it} = a_i + b_i t_{it} + u_{it}$$

where y_{it} is the natural logarithm of GDP per working hour, a_i is the y-intercept (a coefficient) for country i, b_i is the slope coefficient for country i with regard to variable t_{it} which is year in country i year t and u_{it} is the error term for country i year t. To estimate the regression model we can use the function `lm()`. One way to estimate the model one country at a time is to filter by country in the parentheses for `lm()`. For example like this:

```
lm_results <- lm(log(gdp_hour) ~ year,
data= tcb_data %>%
filter(country=="Denmark") )
```

Then we can for example return the estimated slope coefficients with the following command:

```
lm_results %>% coefficients()
```

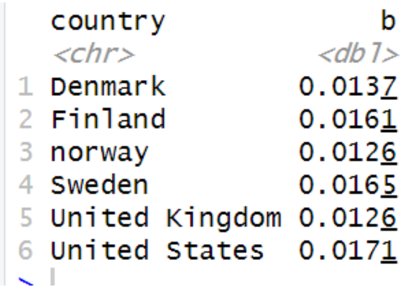
To only return the second coefficient, the slope coefficient $\hat{b}$, we can use the following command:

```
(lm_results %>% coefficients())[[2]]
```

There are several ways to avoid writing one call per country. We can for example create a loop using `map()`. Another way is to use `lm()` in the parentheses for `mutate()` and combine this with `group_by()`. Here follows an example. The code returns a table that is displayed in RStudio's Console window. An example of how it can look is shown in figure 31.21:

```
tcb_data %>%
group_by(country) %>%
mutate(b= (lm(log(gdp_hour) ~ year) %>%
```

```
    coefficients())[2]) %>%
distinct(country, b)
```



	country	b
	<chr>	<dbl>
1	Denmark	0.0137
2	Finland	0.0161
3	norway	0.0126
4	Sweden	0.0165
5	United Kingdom	0.0126
6	United States	0.0171

Figure 31.21: Table with estimated coefficients

31.18 Chapter Summary

- R can be used in the RStudio program. Functions in R are generally written with the function's name followed by parentheses, for example `mean()`. Generally all functions return information that we can save in R objects. To save something in an object we can use `<-`, for example: `result_1 <- mean(c(1,2,3))`. We can then refer to the saved object, such as: `result_1 * 4`.
- We can install new functions by downloading and installing R packages. For example: `install.packages("tidyverse")` and `library("tidyverse")`. The tidyverse package contains several R packages with many functions. Including pipes `%>%`.
- Tables are one of several ways that R can handle data. A useful type of table is tibble. In tidyverse the packages dplyr and tidyr are included which contain many good functions for working with tables. For example: `rename()`, `filter()`, `select()`, `mutate()`, `if_else()`, `case_when()`, `pivot_longer()`, `pivot_wider()`.
- Matrices can be saved as R objects, for example with the function `matrix()`. Calculations with matrices can be performed directly in the R script. For example: `A %*% B` means that we matrix multiply objects `A` and `B` which contain compatible matrices.
- Examples of functions for studying variation and covariation: `var()`, `cov()` and `cor()`. Examples of functions regarding probability: `dnorm()`, `pnorm()`, `qnorm()` and `rnorm()`.
- To estimate regression models with the least squares method we can use the function `lm()`. A regression model $Y = a + bX + U$ can in the parentheses for `lm()` be written `lm(y ~ x)`. We must also specify which data should be used. The results from `lm()` can be studied using `summary()`, `coefficients()` and `residuals()`. Result tables can be created with `stargazer()`.
- We can create charts with the ggplot2 package. The function `ggplot()` can be simplified and described in the following way: `ggplot(data = my_table, aes(y= y_variable, x= x_variable, color= color_variable)) + geom_point() + geom_line()`.
- We can create loops and iterations with `map()`, which has two input values: `.x` for the values that should be used in the loop and `.f` where we can describe what commands the loop should execute.

Chapter 32

Examples of Useful Data and Additional Tools

This chapter offers useful tips on freely available data sources and additional tools worth knowing about.

32.1 Examples of Free Data

For those writing a thesis, for example, it is useful to know where to find public statistics that can be downloaded for free. Large amounts of data are freely available online, organized in structured tables ready to download and use for analysis. Some data are easy to access; others require special permits, either because they contain sensitive information about individuals or companies, or because companies own the rights to the data.

Several well-known international databases provide free data for download. Some examples:

- Eurostat, the European Union's statistics portal: https://ec.europa.eu/eurostat
- OECD, Organisation for Economic Co-operation and Development, international organization and think tank. Publishes large amounts of statistics and reports: https://www.oecd.org and https://data.oecd.org
- World Bank database: https://data.worldbank.org
- ILO, International Labour Organization: https://ilostat.ilo.org
- IMF, International Monetary Fund: https://www.imf.org/en/Data
- Penn World Table: https://www.rug.nl/ggdc/productivity/pwt
- Federal Reserve Economic Data: https://fred.stlouisfed.org

In some cases there are also specially designed functions for connecting to databases directly from within the programs. For example, there are routines for connecting to the World Bank's open data directly from Stata and R. For Stata, go to the following link and search for Stata:

https://blogs.worldbank.org/opendata

For R, see the wbstats package, which has extensive information available on various websites. The package can be installed with `install.packages("wbstats")` and loaded with `library("wbstats")`.

32.2 Other Tools That Are Good to Know About

In this book we have covered three analysis tools. Beyond these, there are of course many other programs across different platforms that may be useful. This section gives some examples. The aim is to inspire and to remind you that staying open to new tools is always worthwhile. This list is not exhaustive. Which tool, function or application is best — or even relevant — depends entirely on the task at hand. Some have narrow applications but can be extremely valuable for specific projects.

Beyond the three tools covered in this book, many other programs offer similar functionality, each with their own specific strengths and weaknesses. There is also a large amount of literature and resources online for those who want to learn more. Here are some well-known examples:

- SPSS, Statistical Product and Service Solutions: https://www.ibm.com/products/spss-statistics
- SAS: https://www.sas.com
- EViews: https://eviews.com
- Minitab: https://www.minitab.com

Python is one of the world's most widely used programming languages, see https://www.python.org. It is used across a wide range of fields and can be applied to many different types of work. Within the social sciences it is used for data analysis and mathematical modelling, among other things. Another programming language that has grown in popularity is Julia: https://julialang.org.

Like R, both Python and Julia can be used in several different types of computer programs. In this book we used the RStudio program to work with R. In RStudio we can also use Python, for example by choosing in the menu File > New File > Python script.

An important part of all analysis work is being able to present results in a clear and effective way. For this purpose many use Microsoft Word or PowerPoint, which are widely installed alongside Excel. Word is a word processing program, but there are many other programs for editing text. If you are not familiar with any other word processing program, it can be valuable to explore alternatives that may work better for your specific purposes. Within academic work the typesetting system LaTeX is widely used; it is roughly like a programming language for word processing. LaTeX can be used in different programs, for example Texmaker or LyX. See the following links:

- LaTeX: https://www.latex-project.org/get
- Texmaker: https://www.xm1math.net/texmaker
- LyX: https://www.lyx.org/Download

To write and present material it is also possible to use the analysis programs themselves and write body text — introduction, analysis, presentation of results, discussion and conclusions — together with the actual programming code. This is called literate programming. In practice, this is most common in R. For example, there are several textbooks for R programming that are themselves written in R, see: https://bookdown.org/

When writing academic texts it is important that statements and information are supported by correctly cited sources. Special computer programs for reference management help to organize and insert citations in theses and academic articles. Some of the most well-known:

- Zotero: https://www.zotero.org
- Mendeley: https://www.mendeley.com
- EndNote: https://www.endnote.com

There is also a growing number of useful tools that can be run directly in the web browser. Two examples:

- posit Cloud: https://posit.cloud. Cloud service where you can work in RStudio.
- Overleaf: https://sv.overleaf.com. Tool where writers can collaborate to write in LaTeX.

Part V

Final Words

Part V: Final Words

This part of the book contains a final chapter with some common misconceptions, suggestions for further reading, and a brief closing word.

Chapter 33

Final Chapter

We are beginning to approach the end of this book but much remains for those who want to learn more.

33.1 Common Misconceptions

Analysis with mathematics and statistics often becomes abstract and it is then easy to confuse concepts. Here follow some reminders based on commonly occurring misconceptions.

Mathematics $\neq$ reality. In various examples we have used mathematics to describe theoretical relations and connections. Theoretical reasoning can sound so convincing that it is taken as strong arguments for how the world works. It is then important to remind oneself that mathematics and logic are not proof of how the world works, other than in very limited reasoning. Humans are complex, changeable and can function in many different ways in different situations.

Correlation is not causality. Causal relationships are something we read into what we observe. It cannot be proven mathematically. Even in a controlled experiment there can be room for interpretation and thereby more or less erroneous conclusions regarding causal relationships. The risks for erroneous interpretations also exist when we work with quasi-experiments.

Mean and variance. Let us assume that we have found a positive correlation between two variables, for example that higher income covaries with greater happiness. Covariation for groups does not necessarily mean that we can make statements about individual observations, or persons, with any great certainty. If the variance in the variables is sufficiently large in relation to the correlation then there will also be many low-income earners who are happier than many high-income earners.

There is always a risk that we are wrong. Suppose we estimate a correlation between the variables Y and X, where our statistical tests indicate that our null hypothesis (no correlation) is false at significance $\alpha = 0.01$, which feels safe. But there is still 1% probability that we reject the null hypothesis erroneously, a type 1 error. Furthermore, the power of our statistical test can be low. Other studies using other data or other methods could give other results that indicate that our calculations and/or our results and/or our conclusions were erroneous in some way, or that they maybe only apply to our specific case or some other problem.

Other factors can be important. Results that indicate that X has a causal effect on Y are not proof that some other phenomenon does not have an effect on Y. If we for example show that medicine A affects a disease symptom it does not mean that there does not exist some other medicine B that can also affect the same symptom.

Variables do not need to be normally distributed. For a regression model $Y = a + bX + U$ we often assume that the error term U follows a normal distribution. But the variables Y and X do not need to follow a normal distribution.

Results that are not statistically significant may still be interesting. When we work with regression analysis it is easy to believe that regression results without statistically significant results are uninteresting. But many times it is the statistically non-significant results that are the important ones, or at least as important. Say for example that there exists a widespread erroneous notion in society that

X affects Y. In that case a large amount of statistically non-significant results (that X does not seem to have any effect on Y) are very interesting.

33.2 Things That Perhaps Should Have Been Included in This Book

Here are some tips for those who want to read more.

Differential and difference equations are mathematical functions that describe how a variable can be a function of the change of the same variable. This can among other things be used to describe development over time. This type of mathematics can among other things be used to describe theoretical processes from one point in time to another. In empirical analysis, difference equations are used to study data that is sorted over time, within what is called time series econometrics.

Geometry is a part of mathematics that deals with distances and spaces, for example shape and size of different objects in a space. A particular part of geometry is trigonometry where we study angles and dimensions of triangles. In upper secondary school mathematics courses the trigonometric functions cos(), sin() and tan() are introduced. Within social science we can among other things use geometry to study cyclical phenomena, such as the economy's ups and downs. Geometry is also used within statistics and can therefore be valuable for those who want to deepen themselves within this.

Time series and panel data. In the examples we have primarily used cross-sectional data, observations that belong to one and the same point in time, for example how different countries a specific year. Data that is instead organized over time is called time series or time series data, for example GDP in a region or country over a collection of years. Data that consists of several time series for different groups is called panel data, for example GDP for several countries and years in the same table. Statistical analysis of time series and panel data brings particular challenges and sometimes requires other estimators than the least squares method. There is specific literature for this and at universities there are specific courses within time series econometrics.

Other statistical schools. The statistical method described in this book is based on what is called frequentist statistics, or classical statistics. There are also other approaches to how we should approach probability. Another popular approach is what is called Bayesian statistics, named after the mathematician Thomas Bayes (1702–1761). Bayesian statistics relates differently to knowledge and probability and emphasizes more subjective valuations.

Text analysis. In earlier sections we went through an example of how we can use correlation analysis for unstructured data, such as calculating to what extent the characters in Star Wars appear in the same films. The mathematics and statistics are basically the same regardless of what type of information we study. But different types of data sometimes require other methods and often encounter other types of challenges.

33.3 The Last Word

University departments differ in what they consider good science and which skills they value. In practice, most analytical and research work takes place outside academia. If you intend to work in analysis, a broad range of skills will serve you well: wide knowledge of the world, technical proficiency, and the ability to communicate clearly — in writing, through visuals, and in front of an audience. The ability to find and evaluate information, ideally in more than one language, is especially worth developing.

Finally, take a moment to appreciate what you have accomplished. There is always more to learn, and the work never truly ends. But you have now made it to the end of this book. Well done!